Lie Groups and Their Representation Theory

Two parts of fourteen lectures

Part I: Global structure and compact representations

Part II: Noncompact representations, geometry and physics

Draft course notes
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Studying this course

This is a self-reading course in two parts of fourteen lectures. The division into lectures is for convenience, so lectures differ in length. Study at your own pace. A lecture may need several passes. The exercise sheet is separate. Allow another two to four hours for its core problems.

The prerequisites are *Linear Algebra for Physics*, *Finite Groups and Their Representations*, *Lie Algebras and Their Representations*, and basic category language. We use their algebraic results throughout. Ordinary multivariable calculus and elementary ordinary differential equations are also assumed.

The course in one page

This course asks what a group determines that its Lie algebra does not. For example, $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ have isomorphic Lie algebras. The module V_1, of spin one half, is a representation of $\mathrm{SU}(2)$ and not of $\mathrm{SO}(3)$. The course also asks how the representations of a group describe the functions on it.

Part I treats compact groups. It can be read alone. Lectures 1 to 4 build Lie groups, the exponential map, quotients and covering groups. Lecture 4 decides which finite-dimensional representations of a Lie algebra come from a given group. Lectures 5 to 9 treat the structure of a compact group: the invariant integral, maximal tori and their lattices, and highest weights. Lecture 8 lists the irreducible representations of a compact connected group. Lectures 10 to 14 treat functions on a compact group. Lecture 10 proves the Peter–Weyl theorem, which contains Fourier series. Lectures 11 and 12 apply it to convolution and to functions on homogeneous spaces. Lecture 13 treats spin groups and projective representations. Lecture 14 recovers the group from its representations.

Part II removes compactness. Lectures 15 to 17 rebuild the tools of Part I: generators of unitary representations, Haar measure of infinite total mass, and induction. Lectures 18 and 19 treat the Heisenberg group and coadjoint orbits. Lectures 20 to 23 treat $\mathrm{SL}(2,\mathbb{R})$ and end with the list of its irreducible unitary representations. Lectures 24 and 25 treat the Lorentz and Poincaré groups. Lectures 26 and 27 return to compact groups and realise their representations on line bundles and on coadjoint orbits. Lecture 28 compares the function algebra of a group with its enveloping algebra, and it prepares for a later course on quantum groups.

Reading a lecture

Start with the objectives and outcomes. Work through the displayed calculations before reading the next paragraph. Then answer the self-checks without looking at their answers.

Attempt the core exercises before opening the worked solutions. Compare the reasoning as well as the final result. Some sheets include an optional extension. It is marked explicitly and is not needed for the next lecture.

The notes introduce the geometry and analysis needed at each stage. They do not replace courses on differential geometry, topology or functional analysis. Long existence theorems are stated with their hypotheses and a source. The main calculations are carried out here.

A cited theorem is marked as cited, by the word *cited* or a source in its title, or by a sentence next to it. Where a proof is only sketched, the text names the omitted step. The mark [C: source] gives the source of a cited statement, and the mark [I] means that a result comes from a prerequisite course. A cited theorem is an input to the argument that follows it. Longer checks that a source's hypotheses apply are gathered under *Sources and hypotheses* at the end of a lecture.

Conventions

Lie groups are finite-dimensional real Lie groups. Their manifolds are Hausdorff and second countable. Compactness and connectedness are stated when they are needed. Representations are continuous and act on complex vector spaces. A Hilbert-space unitary representation is strongly continuous. Inner products are antilinear in the first slot.

A real Lie algebra and its complexification are distinct objects. In particular, $\mathfrak{k} = \mathrm{Lie}(K)$ is real, while highest-weight theory applies to $\mathfrak{k}_{\mathbb{C}}$. For a torus with Lie algebra $\mathfrak{t}$, our period lattice is

$$\Lambda = \ker(\exp\colon \mathfrak{t} \to T).$$

We write a character as $\chi(\exp H) = e^{i\ell(H)}$. Thus its period condition is $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

The module V_m of SU(2) has dimension $m + 1$ and spin $m/2$. The operator of a group representation is $\Pi(g)$, and its differential is $d\Pi(X)$. When a basis is chosen, D denotes an operator matrix. Every change of convention is explained where it is used.

Books used in the notes

The reading lists give a route into the books. The following abbreviations are used in some lectures.

H Brian C. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, Springer, 2015. This is the main route for matrix groups, compact groups and integration of representations.

BD Theodor Bröcker and Tammo tom Dieck, *Representations of Compact Lie Groups*, Springer, 1985. This supplies the general compact-group theory and reconstruction.

S Mark R. Sepanski, *Compact Lie Groups*, Springer, 2007. This is a second route through compact structure and Peter–Weyl theory.

Knapp Beyond
Anthony W. Knapp, *Lie Groups Beyond an Introduction*, second edition. This supplies general Lie-group structure.

Knapp Representation Theory
Anthony W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. This supplies noncompact examples, smooth vectors and induced representations.

Kirillov Jr.
Alexander Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Cambridge University Press, 2008.

Kirillov Elements
A. A. Kirillov, *Elements of the Theory of Representations*, Springer, 1976.

Kirillov Orbit Method
A. A. Kirillov, *Lectures on the Orbit Method*, American Mathematical Society, 2004.

Warner
Frank W. Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Springer edition, 1983.

Further primary sources are named in the relevant lectures. Knapp's two books serve different purposes; their titles are distinguished throughout.

Part I

Global structure and compact representations

Lecture 1

Lie groups and their tangent algebras

Objectives. *A Lie group has both a group structure and a smooth geometry. We describe that geometry near the identity and recover its Lie algebra. The group* $\mathrm{SU}(2)$ *will give an example in which every step can be calculated.*

By the end of this lecture you will be able to:

- Identify the real dimension of a matrix Lie group.
- Find tangent spaces by differentiating a defining condition and exhibiting curves.
- Recover a matrix Lie bracket from left translations.
- Distinguish a real Lie algebra from its complexification.

The Lie algebras in the earlier course were complex. The tangent algebra of a real Lie group is a real Lie algebra. We will always state the field of scalars.

1.1 Smooth coordinates

A sphere cannot be described by one set of Euclidean coordinates without omitting a point or introducing a singularity. It can be described by several overlapping coordinate systems. A smooth manifold formalises this local description.

A topology specifies which subsets are open. A *neighbourhood* of a point is a set containing an open set around that point. A topological space is *Hausdorff* if distinct points have disjoint open neighbourhoods. It is *second countable* if its topology has a countable *basis*, namely a countable collection of open sets such that every open set is a union of some of them.

A map between topological spaces is *continuous* if the inverse image of every open set is open. A *homeomorphism* is a continuous bijection whose inverse is continuous.

Definition 1.1 (Smooth manifold)**.** A smooth real manifold of dimension d is a topological space M covered by coordinate charts

$$\varphi : U \longrightarrow V \subseteq \mathbb{R}^d.$$

Each chart has open domain $U \subseteq M$ and is a homeomorphism onto an open set V. Where the domains of two charts φ and ψ overlap, the coordinate change $\psi \circ \varphi^{-1}$ is infinitely differentiable. We require M to be Hausdorff and second countable.

A subset of a topological space has the *subspace topology*: its open sets are its intersections with open sets of the space. For subsets of Euclidean space we use this topology. For example, an open arc is open in the circle.

Subsets of Euclidean space are Hausdorff and second countable, so all the matrix groups in this lecture satisfy both conditions. Lecture 2 uses the Hausdorff condition for uniqueness of solutions of differential equations on a manifold. Lecture 3 uses second countability for homogeneous spaces. A smooth map between manifolds is a continuous map whose coordinate expressions are infinitely differentiable on their open domains. A *diffeomorphism* is a smooth bijection whose inverse is smooth.

Example 1.2 (A coordinate on the circle)**.** The unit circle is

$$S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

On the open right semicircle, use y as the coordinate. Its inverse chart is

$$y \longmapsto \left(\sqrt{1 - y^2}, y\right), \qquad -1 < y < 1.$$

The other three semicircles have similar charts. On the upper semicircle use x. On the open quarter circle where both charts apply, y runs over $(0, 1)$, and the coordinate change and its inverse are

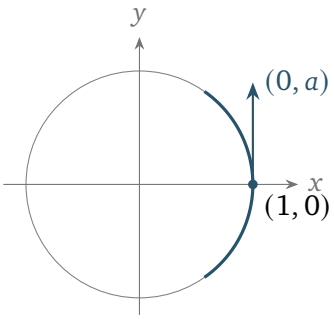
$$x = \sqrt{1 - y^2}, \qquad y = \sqrt{1 - x^2}.$$

These are smooth because $1 - y^2 > 0$ and $1 - x^2 > 0$ there. The other overlaps are similar. Thus the circle is a one-dimensional real manifold, although it lies in a two-dimensional ambient space.

The coordinate y can also describe a velocity. For a real number a, the curve on the circle

$$\gamma_a(t) = \left(\sqrt{1 - a^2 t^2}, at\right)$$

is defined near zero. It starts at $(1, 0)$ and has ambient derivative $(0, a)$. The derivative of its y-coordinate is a. Thus a one-dimensional tangent space can appear as a line inside a larger ambient space.



The coloured arc lies in the domain of one chart, in which y is the local coordinate. The tangent arrow is the velocity at $(1, 0)$ for positive a. Its drawn length is schematic.

1.2 Smooth groups

Product charts $\varphi \times \psi$ on $U \times V$ give $M \times N$ a smooth manifold structure of dimension $\dim M + \dim N$. A finite product of Hausdorff, second-countable spaces has both properties.

Definition 1.3 (Lie group)**.** A real Lie group is a group G that is also a smooth real manifold, with smooth multiplication and inversion:

$$G \times G \longrightarrow G, \quad (g,h) \longmapsto gh, \qquad G \longrightarrow G, \quad g \longmapsto g^{-1}.$$

A Lie group homomorphism is a smooth group homomorphism.

The additive group $\mathbb{R}^d$ is a Lie group. Any finite group with the discrete topology is also a Lie group, since each one-point subset is the domain of a chart modelled on $\mathbb{R}^0$. The tangent space of a finite group therefore has dimension zero. Hence a Lie algebra cannot distinguish finite groups.

Example 1.4 (General linear groups)**.** The group of invertible real matrices is

$$\mathrm{GL}_n(\mathbb{R}) = \{g \in M_{n\times n}(\mathbb{R}) : \det g \neq 0\}.$$

The determinant is continuous, so $\mathrm{GL}_n(\mathbb{R})$ is open in $\mathbb{R}^{n^2}$. Hence its dimension is n^2. The entries of a matrix product are polynomials in the entries of the factors, so matrix multiplication is smooth. Inversion is smooth because

$$g^{-1} = \frac{\operatorname{adj}(g)}{\det g}.$$

The same argument applies to $\mathrm{GL}_n(\mathbb{C})$. The real and imaginary parts of the entries give its coordinates, so its dimension as a real manifold is $2n^2$.

A *complex Lie group* has complex charts and holomorphic group operations. The group $\mathrm{GL}_n(\mathbb{C})$ has this extra structure. Its complex dimension is n^2. A group of complex matrices need not be a complex Lie group. Its defining conditions may involve complex conjugation.

For example, the unitary and special unitary groups are

$$\mathrm{U}(n) = \{U \in M_{n\times n}(\mathbb{C}) : U^*U = I\}, \qquad \mathrm{SU}(n) = \{U \in \mathrm{U}(n) : \det U = 1\}.$$

Here U^* denotes conjugate transpose. Their real Lie-group structures are a standard classical-group result, cited from Bröcker–tom Dieck, Chapter I, Section 1. We verify the manifold structure of $\mathrm{SU}(2)$ directly; Exercise Sheet 1 treats $\mathrm{U}(n)$.

A manifold is *connected* if it cannot be written as the union of two nonempty disjoint open subsets. For example, the orthogonal group is

$$\mathrm{O}(n) = \{g \in M_{n\times n}(\mathbb{R}) : g^{\mathsf{T}}g = I\}, \qquad n \geq 1.$$

Its determinant takes values in $\{1,-1\}$. Both values occur. The identity has determinant one, and $\operatorname{diag}(-1,1,\ldots,1)$ has determinant minus one. The inverse images of these two values are disjoint nonempty open subsets that cover the group. Thus $\mathrm{O}(n)$ is disconnected. Its subgroup with determinant one is $\mathrm{SO}(n)$.

1.3 Tangent vectors at the identity

Definition 1.5 (Tangent vector). Let $\varphi : U \to V$ be a chart of M with $p \in U$. A *smooth curve through* p is a smooth map γ from an open interval containing 0 to M with $\gamma(0) = p$. Two such curves γ and δ are *equivalent* when $(\varphi \circ \gamma)'(0) = (\varphi \circ \delta)'(0)$. A *tangent vector* at p is an equivalence class $[\gamma]$. The set of tangent vectors at p is denoted T_pM. The map $[\gamma] \mapsto (\varphi \circ \gamma)'(0)$ identifies T_pM with $\mathbb{R}^d$. Vector addition and real scalar multiplication on T_pM are transferred through this identification.

The class $[\gamma]$ is the *velocity* of γ, also written $\gamma'(0)$. The chain rule makes the relation between curves independent of the chart. The derivative of a coordinate change is linear, so the vector-space structure on T_pM is also independent of the chart.

A smooth map $f : M \to N$ sends tangent vectors to tangent vectors. If v is represented by γ, the image of v is the velocity of $f \circ \gamma$. This defines the *differential*

$$df_p : T_pM \longrightarrow T_{f(p)}N, \qquad df_p\big(\gamma'(0)\big) = (f \circ \gamma)'(0).$$

In charts this is the usual derivative, so it is real linear. The chain rule gives $d(f \circ h)_p = df_{h(p)} \circ dh_p$.

Our groups lie inside matrix spaces, and we calculate with matrix entries. An *embedded submanifold* is the notion that justifies this. It has the subspace topology. Near each of its points, a smooth change of ambient coordinates turns it into a coordinate subspace. For example, near $(1, 0)$ the coordinates $(r-1, \theta)$ turn the circle into the line $r-1 = 0$. We import the standard properties of an embedded submanifold. Its inclusion is smooth with injective differential. Products of embedded submanifolds are embedded in the corresponding product. Thus smooth ambient maps restrict smoothly to these submanifolds. A smooth ambient map whose image lies in an embedded submanifold is smooth as a map into that submanifold. The differential of the inclusion identifies its tangent vectors with ambient velocities.

We also use one result from multivariable calculus. Let $U \subseteq \mathbb{R}^N$ be open, and suppose a smooth map $F : U \to \mathbb{R}^r$ has surjective differential at every point of $F^{-1}(c)$. The *regular level-set theorem* says that $F^{-1}(c)$ is an embedded submanifold of U of dimension $N - r$, with the subspace topology. The tangent space of $F^{-1}(c)$ at a point p is $\ker dF_p$. The regular level-set theorem and the embedded-submanifold properties are cited from Warner, Chapter 1, sections “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem” and “Implicit Function Theorems” (Theorem 1.38). For example, the circle is $F^{-1}(1)$ with $F(x, y) = x^2 + y^2$. Here $dF_{(x,y)} = (2x, 2y)$ is nonzero on the circle, so the dimension is $2 - 1 = 1$.

The kernel clause of the regular level-set theorem now follows from the chain rule and rank–nullity. Put $P = F^{-1}(c)$ and let $i : P \to U$ be the inclusion. For $p \in P$, since $F \circ i$ is constant, $dF_p \circ di_p = 0$. Thus $di_p(T_pP) \subseteq \ker dF_p$. The inclusion is an embedding, so di_p is injective. Both spaces have dimension $N - r$, because dF_p is surjective. Hence they are equal.

For an embedded manifold in a matrix space, we compute velocities by differentiating the matrix entries of curves. The identity element of a group is denoted e, or I for a matrix group. For a Lie group G, the vector space of interest is

$$\mathfrak{g} = T_eG.$$

For $\mathrm{GL}_n(\mathbb{R})$, every matrix X is a tangent vector at I. Indeed, the curve $\gamma(t) = I + tX$ is invertible for sufficiently small t, and $\gamma'(0) = X$.

Example 1.6 (Tangent vectors to the orthogonal group)**.** Let $g(t)$ be a smooth curve in $\mathrm{O}(n)$ with $g(0) = I$. Put $X = g'(0)$. Differentiate the defining identity at zero by the product rule:

$$g(t)^{\mathsf{T}} g(t) = I \quad \Longrightarrow \quad X^{\mathsf{T}} + X = 0.$$

Thus every tangent vector at I is skew-symmetric. Differentiating a condition gives a necessary condition. We must also show that every such matrix occurs.

We could use the exponential series, which Lecture 2 develops. The Cayley curve needs no convergence argument. For a skew-symmetric matrix X, consider the Cayley curve

$$g_X(t) = \left(I + \frac{tX}{2}\right)\left(I - \frac{tX}{2}\right)^{-1}.$$

It is defined near zero. Set $A = I + tX/2$ and $B = I - tX/2$. Then $A^{\mathsf{T}} = B$, $B^{\mathsf{T}} = A$, and $AB = BA$. Therefore

$$g_X(t)^{\mathsf{T}} g_X(t) = A^{-1} B A B^{-1} = I.$$

The determinant is continuous, takes values in $\{1, -1\}$, and equals one at zero. It remains one near zero. Hence this curve lies in $\mathrm{SO}(n)$. The product rule gives

$$g_X'(0) = \frac{X}{2} + \frac{X}{2} = X.$$

Consequently both orthogonal groups have the same tangent space at I:

$$T_I \,\mathrm{SO}(n) = T_I \,\mathrm{O}(n) = \{X : X^{\mathsf{T}} = -X\}.$$

The example used that $\mathrm{O}(n)$ is a smooth manifold. To justify this assertion, apply the regular level-set theorem to $F(g) = g^{\mathsf{T}} g$, with target the real symmetric matrices. At an orthogonal matrix g,

$$dF_g(H) = H^{\mathsf{T}} g + g^{\mathsf{T}} H.$$

For any symmetric matrix S, choose $H = gS/2$. Since S is symmetric and g is orthogonal, $dF_g(H) = S$. Hence the surjectivity hypothesis holds. Subtracting the dimension of the space of real symmetric matrices from the ambient dimension gives

$$\dim_{\mathbb{R}} \mathrm{O}(n) = n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

The kernel clause gives the tangent space again: $dF_I(H) = H^{\mathsf{T}} + H$, so $T_I \,\mathrm{O}(n)$ consists of the skew-symmetric matrices. The Cayley curve showed the same thing by hand. The subgroup $\mathrm{SO}(n)$ is open in $\mathrm{O}(n)$, so it has the same local dimension.

Multiplication and inversion by transposition are restrictions of smooth matrix maps. By the embedded-submanifold properties, both orthogonal groups are Lie groups. Equal tangent spaces do not imply equal groups.

1.4 Translations and the bracket

A tangent space at one point is only a vector space, and it has no product. Vector fields do have a commutator, because they act on functions by differentiation. Left translation carries a tangent vector at e to a vector field on G. We therefore obtain a bracket on the tangent space. Lecture 2 shows that this bracket measures how far small group elements fail to commute.

Left multiplication by $g \in G$ is the smooth map

$$L_g : G \longrightarrow G, \qquad L_g(h) = gh.$$

Since L_g has inverse $L_{g^{-1}}$, the differential of L_g at e identifies tangent spaces:

$$(dL_g)_e : T_eG \longrightarrow T_gG.$$

For a matrix group this differential sends X to gX. For example, in $\mathrm{SO}(2)$ take the rotation and the tangent matrix

$$g = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Left translation gives the tangent vector at g:

$$gX = \begin{pmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix}.$$

This is the derivative of the rotation matrix with angle $\theta + t$ at $t = 0$.

A *vector field* assigns a tangent vector to each point. It is smooth if its components in each coordinate chart are smooth functions. The field determined by $X \in T_eG$ is

$$X^L(g) = (dL_g)_e X.$$

A vector field V on G is *left-invariant* if

$$(dL_h)_g\big(V(g)\big) = V(hg) \qquad \text{for all } g, h \in G.$$

The field X^L is left-invariant. Indeed, $L_h \circ L_g = L_{hg}$, so the chain rule gives $(dL_h)_g\big((dL_g)_e X\big) = (dL_{hg})_e X$.

Smooth vector fields act on smooth functions by directional differentiation. For a smooth vector field V and a smooth function f,

$$(Vf)(p) = df_p(V(p)).$$

Their commutator as differentiation operators is

$$[V, W]f = V(Wf) - W(Vf).$$

Theorem 1.7 (The tangent Lie algebra; cited). *The fields X^L obtained by left translation are smooth. The commutator of two smooth vector fields is again a vector field. The commutator of left-invariant fields is left-invariant. The value of this commutator at the identity defines a real Lie bracket on T_eG. For smooth vector fields V and W, in coordinates,*

$$[V, W](p) = DW_p(V(p)) - DV_p(W(p)).$$

For an embedded submanifold, this formula may be evaluated using smooth ambient extensions of tangent fields. The result is tangent to the submanifold.

The construction of the bracket from left-invariant fields is cited from Bröcker–tom Dieck, Chapter I, Section 2. The coordinate formula and the statement for embedded submanifolds are cited from Warner, Chapter 1, sections “Vector Fields” and “Submanifolds, Diffeomorphisms, and the Inverse Function Theorem”. The Jacobi identity is the commutator identity for the corresponding differentiation operators.

For an embedded matrix group, use the ambient extensions $X^L(g) = gX$ and $Y^L(g) = gY$. Since $Y^L(h) = hY$ is linear in h, its derivative in direction H is HY. Thus ambient calculus gives

$$D(Y^L)_g(gX) = gXY,$$
$$D(X^L)_g(gY) = gYX.$$

By the coordinate formula in the cited theorem, the vector-field bracket is the first of these derivatives minus the second. At the identity, this gives

$$[X, Y] = XY - YX.$$

This is the bracket studied in the earlier course. The cited submanifold compatibility shows that this ambient calculation is the intrinsic bracket on G. Hence $XY - YX$ belongs to T_IG. For skew-symmetric matrices, closure can also be seen directly:

$$(XY - YX)^{\mathsf{T}} = YX - XY = -(XY - YX).$$

The real tangent Lie algebra of $\mathrm{SO}(n)$ is denoted $\mathfrak{so}(n)$. For $\mathrm{GL}_n(\mathbb{C})$, viewed as a real group, the tangent algebra is all complex matrices with real scalars and the same commutator.

1.5 The group SU(2) and its real algebra

The two rows of a matrix in $\mathrm{SU}(2)$ are orthonormal. Write the first row as (α, β). The unit vectors orthogonal to this row are the multiples of $(-\overline{\beta}, \overline{\alpha})$ by complex numbers of modulus one. The determinant then equals that multiple, so determinant one forces it to be one. Hence every element of $\mathrm{SU}(2)$ has the form

$$U = \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Conversely, a matrix of the displayed form satisfies

$$\begin{pmatrix} \overline{\alpha} & -\beta \\ \overline{\beta} & \alpha \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ -\overline{\beta} & \overline{\alpha} \end{pmatrix} = \begin{pmatrix} |\alpha|^2 + |\beta|^2 & 0 \\ 0 & |\alpha|^2 + |\beta|^2 \end{pmatrix}.$$

The matrix U has determinant $|\alpha|^2 + |\beta|^2$. Thus this formula identifies $\mathrm{SU}(2)$ with the unit sphere S^3 in $\mathbb{C}^2 = \mathbb{R}^4$. The regular level-set theorem gives real dimension three. The displayed map from (α, β) to U is a real-linear injection from $\mathbb{R}^4$ into $M_{2\times 2}(\mathbb{C})$. It is a linear coordinate isomorphism onto its image, so it carries the embedded sphere to an embedded submanifold of the full matrix space. Multiplication and inversion are smooth restrictions of the matrix operations.

Proposition 1.8 (The tangent algebra of SU(2)). *The tangent Lie algebra of* $\mathrm{SU}(2)$ *is the real space*

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & y + iz \\ -y + iz & -ix \end{pmatrix} : x, y, z \in \mathbb{R} \right\},$$

with the matrix commutator.

Proof. First we prove the necessary conditions. Let $U(t)$ be a smooth curve in $\mathrm{SU}(2)$ with $U(0) = I$, and put $X = U'(0)$. Differentiating $U(t)^* U(t) = I$ at zero gives $X^* = -X$.

The determinant condition requires a separate derivative. Write $X = (x_{jk})$. To first order in t,

$$U(t) = \begin{pmatrix} 1 + tx_{11} + o(t) & tx_{12} + o(t) \\ tx_{21} + o(t) & 1 + tx_{22} + o(t) \end{pmatrix}.$$

Its determinant has expansion

$$\det U(t) = 1 + t(x_{11} + x_{22}) + o(t).$$

Here $o(t)$ denotes a quantity whose ratio to t tends to zero. The products of two off-diagonal entries have order t^2, so they do not contribute to the derivative. Differentiating $\det U(t) = 1$ now gives $\operatorname{tr} X = 0$.

A skew-Hermitian matrix has purely imaginary diagonal entries. Its entry below the diagonal is minus the conjugate of the entry above it. Since the trace is zero, the diagonal entries are ix and $-ix$ for some $x \in \mathbb{R}$. Writing the entry above the diagonal as $y + iz$ gives the displayed form.

Next we prove the converse. Every matrix of the displayed form is a real linear combination of

$$A = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Let $X = xA + yB + zC$ and $r^2 = x^2 + y^2 + z^2$. Direct multiplication gives

$$X^2 = -r^2 I, \qquad \det(I + tX) = 1 + t^2 r^2.$$

These identities provide a curve with the required velocity:

$$U_X(t) = \frac{I + tX}{\sqrt{1 + t^2 r^2}}.$$

The scalar factor appears squared in a 2×2 determinant. Hence the determinant of $U_X(t)$ is one. Since $X^* = -X$ and $X^2 = -r^2 I$,

$$U_X(t)^* U_X(t) = \frac{(I - tX)(I + tX)}{1 + t^2 r^2} = I.$$

Also $U_X(0) = I$ and $U_X'(0) = X$. Hence every matrix of the displayed form is tangent to $\mathrm{SU}(2)$ at I.

Finally, $\mathrm{SU}(2)$ is an embedded matrix group. By the calculation with ambient extensions in the previous section, its bracket is the matrix commutator. ∎

Each group has so far needed its own curve. Lecture 2 gives a single one, $\exp(tX)$.

The multiplication of the basis matrices starts with $AB = C$ and $BA = -C$. The next products confirm a cyclic pattern:

$$BC = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = A, \qquad CB = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -A.$$

Similarly, $CA = B$ and $AC = -B$. The bracket table is

$$[A, B] = 2C, \qquad [B, C] = 2A, \qquad [C, A] = 2B.$$

The scalars in this basis expansion are real. Multiplying A by i gives a Hermitian matrix, which is not in $\mathfrak{su}(2)$. The fact that its elements have complex entries does not make $\mathfrak{su}(2)$ a complex vector space. In physics, with the Pauli matrices $\sigma_1, \sigma_2, \sigma_3$ of *Linear Algebra for Physics*, one has $A = i\sigma_3$, $B = i\sigma_2$ and $C = i\sigma_1$.

The weight calculations of the earlier course use the matrices e and f given below. These matrices are not skew-Hermitian, so they are not in $\mathfrak{su}(2)$. To use them we must allow complex combinations of the real generators. We now relate this real algebra to the earlier complex algebra $\mathfrak{sl}_2(\mathbb{C})$.

Definition 1.9 (Complexification)**.** The complexification of a real Lie algebra $\mathfrak{k}$ is the complex vector space

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C},$$

with the complex-bilinear extension of its bracket. Thus $[X \otimes a, Y \otimes b] = [X, Y] \otimes ab$.

Let $\iota\colon \mathfrak{su}(2)_{\mathbb{C}} \to \mathfrak{sl}_2(\mathbb{C})$ be the complex-linear map given on pure tensors by $\iota(X \otimes a) = aX$. It extends the inclusion of $\mathfrak{su}(2)$ into complex matrices. Its values are traceless because the elements of $\mathfrak{su}(2)$ are traceless. The images of A, B, C span all traceless complex matrices, since

$$h = -iA, \qquad e = \frac{B - iC}{2}, \qquad f = -\frac{B + iC}{2}$$

give the three matrices

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

They are the usual basis matrices of $\mathfrak{sl}_2(\mathbb{C})$. These three matrices are complex linearly independent. The map ι preserves brackets by complex bilinearity. Its domain and image both have complex dimension three, so ι is a complex Lie algebra isomorphism:

$$\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}_2(\mathbb{C}).$$

The real algebra has dimension three over $\mathbb{R}$. Its complexification has dimension three over $\mathbb{C}$. The real algebra underlying $\mathfrak{sl}_2(\mathbb{C})$ has dimension six over $\mathbb{R}$; it is a different object.

Self-checks

1. Why is $\mathrm{GL}_n(\mathbb{C})$ open in $\mathbb{C}^{n^2}$ but $\mathrm{U}(n)$ is not?
2. Does differentiating $g(t)^{\mathsf{T}} g(t) = I$ prove that every skew-symmetric matrix is tangent to $\mathrm{SO}(n)$?
3. Is $\mathrm{diag}(1, -1)$ in $\mathfrak{su}(2)$? Is it in the complex span of $\mathfrak{su}(2)$?

Answers. (1) The determinant is continuous, so the set where it is nonzero is open in $\mathbb{C}^{n^2}$. For real ε with $0 < |\varepsilon| < 1$, a unitary matrix U satisfies

$$((1+\varepsilon)U)^*((1+\varepsilon)U) = (1+\varepsilon)^2 I \neq I.$$

These non-unitary matrices approach U, so no ambient ball about U lies in $\mathrm{U}(n)$. (2) No. Differentiation proves necessity. The Cayley curve proves sufficiency. (3) No. The matrix $\mathrm{diag}(1, -1)$ is traceless but Hermitian, so it is not skew-Hermitian. It does lie in the complex span, since it equals $-iA$.

Summary and reading

A Lie group has smooth multiplication and inversion. Its tangent space at the identity is a real Lie algebra. Left translation determines the bracket. For embedded matrix groups, the bracket is the matrix commutator. The algebra records local information. Distinct groups can have the same tangent algebra.

Read Hall, second edition, Sections 1.1–1.5 and 3.3–3.6, and Bröcker–tom Dieck, Chapter I, Sections 1–2. The regular level-set theorem and the embedded-submanifold properties are the geometric imports, cited from Warner, Chapter 1. Lecture 2 develops the exponential map and connects these tangent vectors to one-parameter subgroups.

Lecture 2

Exponential maps and local group structure

Objectives. *A tangent vector at the identity generates a one-parameter subgroup. The exponential map collects these subgroups into one map from the Lie algebra to the group. We ask how much of the multiplication is determined by the bracket. Near the identity, the bracket determines the whole multiplication. Far from the identity, it does not.*

By the end of this lecture you will be able to:

- Compute exponentials of rotation and boost generators and of elements of $\mathfrak{sl}_2(\mathbb{R})$.
- Identify continuous one-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$ with exponentials.
- Differentiate a homomorphism, and use $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ and $\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\,X}$.
- Compute the second-order BCH term and recognise non-exponential elements.

2.1 The matrix exponential

For a complex $n \times n$ matrix X, let $\|X\|$ be the operator norm for the standard Hermitian norm on $\mathbb{C}^n$. The operator norm satisfies $\|XY\| \le \|X\|\,\|Y\|$.

Definition 2.1 (Matrix exponential)**.** For $X \in M_{n\times n}(\mathbb{C})$, the exponential of X is

$$\exp X = e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

Since $\|X^k\| \le \|X\|^k$, the series is dominated by the scalar series of $e^{\|X\|}$. Hence the series converges absolutely, and uniformly on bounded sets. Each entry of $\exp X$ is an everywhere convergent power series in the entries of X. So exp is smooth, and its series may be differentiated term by term.

Proposition 2.2 (Properties of the matrix exponential)**.** *Let $X, Y \in M_{n\times n}(\mathbb{C})$ and $g \in \mathrm{GL}_n(\mathbb{C})$.*

(i) *If $XY = YX$, then $\exp(X+Y) = \exp X \exp Y$.*

(ii) *The matrix $\exp X$ is invertible, with inverse $\exp(-X)$.*

(iii) *The curve $t \mapsto \exp(tX)$ has derivative $X\exp(tX) = \exp(tX)X$.*

(iv) $g\exp(X)g^{-1} = \exp(gXg^{-1})$.

(v) $\det\exp X = e^{\mathrm{tr}\,X}$.

Proof. (i) Absolute convergence lets us multiply the series and collect terms of total degree m. The binomial theorem holds because $XY = YX$:

$$\exp X \exp Y = \sum_{m=0}^{\infty} \sum_{k+l=m} \frac{X^k Y^l}{k!\,l!} = \sum_{m=0}^{\infty} \frac{(X+Y)^m}{m!}.$$

(ii) Apply the product rule (i) to the commuting pair X and $-X$. (iii) Differentiate $\sum_k t^k X^k/k!$ term by term and factor out X on either side. (iv) We have $(gXg^{-1})^k = gX^k g^{-1}$. Conjugation by g is linear and continuous, so it commutes with the sum of the series. (v) Schur triangularisation gives $X = PTP^{-1}$ with T upper triangular. Let t_{jj} be the diagonal entries of T. Then $\exp T$ is upper triangular with diagonal entries $e^{t_{jj}}$. By the conjugation formula (iv),

$$\det \exp X = \det \exp T = \prod_j e^{t_{jj}} = e^{\operatorname{tr} T} = e^{\operatorname{tr} X}. \qquad \blacksquare$$

The hypothesis $XY = YX$ in the product rule, Proposition 2.2(i), cannot be dropped. Exercise 2.3 gives nilpotent matrices E and F with $\exp E \exp F \neq \exp(E+F)$. The Baker–Campbell–Hausdorff formula in § 2.5 shows that the bracket governs the correction.

2.2 One-parameter subgroups

Definition 2.3 (One-parameter subgroup)**.** A one-parameter subgroup of a Lie group G is a continuous homomorphism $\varphi\colon \mathbb{R} \to G$ from the additive group of real numbers.

For every matrix X, the curve $t \mapsto \exp(tX)$ is a one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$, by the product rule, Proposition 2.2(i). Conversely, continuity alone forces this form.

Theorem 2.4 (One-parameter subgroups of $\mathrm{GL}_n(\mathbb{C})$)**.** *Let $\varphi\colon \mathbb{R} \to \mathrm{GL}_n(\mathbb{C})$ be a one-parameter subgroup. Then φ is differentiable, and $\varphi(t) = \exp(tX)$ for the unique matrix $X = \varphi'(0)$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then X is a real matrix.*

Proof. We first show that φ is differentiable. Averaging a continuous function over a moving interval gives a differentiable function. The homomorphism law writes φ as its own moving average times a fixed matrix.

Step 1: differentiability. By continuity of φ, choose $\varepsilon > 0$ with $\|\varphi(s) - I\| \le \tfrac{1}{2}$ for $0 \le s \le \varepsilon$. Integrate entrywise and set

$$M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(s)\, ds.$$

Then $\|M - I\| \le \tfrac{1}{2}$, so the Neumann series $\sum_{k\ge 0} (I-M)^k$ inverts M. The homomorphism property gives

$$\varphi(t) M = \frac{1}{\varepsilon} \int_0^{\varepsilon} \varphi(t+s)\, ds = \frac{1}{\varepsilon} \int_t^{t+\varepsilon} \varphi(u)\, du.$$

The right side is continuously differentiable in t, by the fundamental theorem of calculus. Multiplication by the constant matrix M^{-1} preserves this property, so φ is continuously differentiable too.

Step 2: the equation. Put $X = \varphi'(0)$. Differentiate $\varphi(t+s) = \varphi(t)\varphi(s)$ in s at $s = 0$. This gives $\varphi'(t) = \varphi(t) X$.

Step 3: uniqueness. Let $\psi(t) = \varphi(t)\exp(-tX)$. By the Leibniz rule and the derivative formula, Proposition 2.2(iii),

$$\psi'(t) = \varphi(t)X\exp(-tX) - \varphi(t)X\exp(-tX) = 0.$$

Hence $\psi(t) = \psi(0) = I$, and $\varphi(t) = \exp(tX)$. Any Y with $\varphi(t) = \exp(tY)$ satisfies $Y = \varphi'(0) = X$. If φ takes values in $\mathrm{GL}_n(\mathbb{R})$, then its derivative $X = \varphi'(0)$ is a real matrix, as a limit of real difference quotients. ∎

Example 2.5 (Rotations)**.** The algebra $\mathfrak{so}(2)$ is spanned by

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad J^2 = -I.$$

Separating even and odd powers in the series gives

$$\exp(\theta J) = \cos\theta\, I + \sin\theta\, J = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The first column of an element of $\mathrm{SO}(2)$ is a unit vector, so it is $(\cos\theta, \sin\theta)$ for some θ. Orthogonality and determinant one then fix the second column. So $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ is onto. It is not injective, since $\exp(2\pi J) = I$.

Example 2.6 (Boosts)**.** In special relativity on a line, with the speed of light set to 1, an event has a time coordinate x_0 and a position x_1. Observers in uniform relative motion use coordinates related by linear maps preserving $x_0^2 - x_1^2$. With $\eta = \mathrm{diag}(1, -1)$, these maps form the group

$$\mathrm{O}(1,1) = \{g \in M_{2\times 2}(\mathbb{R}) : g^{\mathsf{T}}\eta g = \eta\}.$$

Let $g(t)$ be a smooth curve in $\mathrm{O}(1,1)$ with $g(0) = I$ and $g'(0) = X$. Differentiating $g(t)^{\mathsf{T}}\eta g(t) = \eta$ at $t = 0$ gives the necessary condition $X^{\mathsf{T}}\eta + \eta X = 0$. In terms of the entries of X,

$$X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad X^{\mathsf{T}}\eta + \eta X = \begin{pmatrix} 2a & b-c \\ b-c & -2d \end{pmatrix}.$$

The condition therefore forces $a = d = 0$ and $b = c$. So X is a multiple of

$$K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad K^2 = I.$$

The even and odd parts of the series now give hyperbolic functions:

$$B(u) = \exp(uK) = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix}.$$

Since $\cosh^2 u - \sinh^2 u = 1$, we get $B(u)^{\mathsf{T}}\eta B(u) = \eta$. So K is the velocity of a curve in the group, and the tangent algebra is exactly $\mathbb{R}K$.

The parameter u is the rapidity. The boost $B(u)$ sends the events $(\tau, 0)$ of an observer at rest to $(\tau\cosh u, \tau\sinh u)$, a motion with velocity $\tanh u$. Rapidities add, since $B(u)B(u') = B(u+u')$. Velocities do not add. Unlike the parametrisation of rotations, the map $u \mapsto B(u)$ is injective. Every boost has trace $2\cosh u \geq 2$.

2.3 The exponential map of a Lie group

So far our groups are matrix groups. Not every Lie group is one, and the reading of Lecture 4 cites an example. On a general Lie group, the exponential power series is not defined. Instead we use the left-invariant vector fields $X^L(g) = (dL_g)_e X$ of Lecture 1, with $X \in \mathfrak{g} = T_eG$.

Definition 2.7 (Integral curve). An integral curve of a vector field V on a manifold M is a smooth curve $\gamma : I \to M$, on an open interval I, with $\gamma'(t) = V(\gamma(t))$ for all $t \in I$.

In a chart, this condition is a first-order system of ODEs with smooth coefficients. The Picard–Lindelöf theorem of the ODE background gives local existence and uniqueness. Two integral curves with the same initial point agree on their common interval. The agreement set is open by local uniqueness, closed because limits in a Hausdorff manifold are unique, and nonempty. An interval is connected, so it is the whole interval.

Left translation maps integral curves of X^L to integral curves of X^L. For such a curve γ and $g \in G$, the chain rule and $L_g \circ L_h = L_{gh}$ give

$$(L_g \circ \gamma)'(t) = (dL_g)_{\gamma(t)}(dL_{\gamma(t)})_e X = (dL_{g\gamma(t)})_e X = X^L\big(g\gamma(t)\big).$$

Theorem 2.8 (Exponential map of a Lie group; cited). *Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $X \in \mathfrak{g}$. The integral curve γ_X of X^L with $\gamma_X(0) = e$ is defined on all of $\mathbb{R}$. The map*

$$\exp : \mathfrak{g} \longrightarrow G, \qquad \exp X = \gamma_X(1),$$

is smooth, and $\gamma_X(t) = \exp(tX)$ for all real t.

This is proved in Kirillov Jr., Chapter 3, in the section on the exponential map, and in Warner, Chapter 3. We sketch the algebraic part. The omitted, cited step is that γ_X exists for all t and that $\gamma_X(1)$ depends smoothly on X.

For fixed s, the curves $t \mapsto \gamma_X(s+t)$ and $t \mapsto \gamma_X(s)\gamma_X(t)$ are integral curves of X^L starting at $\gamma_X(s)$. By uniqueness,

$$\gamma_X(s+t) = \gamma_X(s)\gamma_X(t).$$

For a real number c, the curve $t \mapsto \gamma_X(ct)$ is the integral curve of $(cX)^L$ through e. So $\gamma_{cX}(t) = \gamma_X(ct)$. At $t = 1$ this gives $\exp(cX) = \gamma_X(c)$.

Let $G \subseteq \mathrm{GL}_n(\mathbb{C})$ be an embedded matrix Lie group, as in Lecture 1. Left translation is matrix multiplication, so γ_X solves $\gamma' = \gamma X$ with $\gamma(0) = I$ in the matrix space. Step 3 of the proof of the one-parameter subgroup theorem, Theorem 2.4, shows that $\exp(tX)$ is the only solution. Hence the two exponentials agree, and $\exp(tX) \in G$ for every $X \in \mathfrak{g}$.

Lemma 2.9 (Smooth one-parameter subgroups). *Let $\beta : \mathbb{R} \to G$ be a smooth homomorphism with $\beta'(0) = X$. Then $\beta(t) = \exp(tX)$ for all t.*

Proof. Differentiate $\beta(t+s) = L_{\beta(t)}\beta(s)$ in s at $s = 0$. This gives $\beta'(t) = X^L(\beta(t))$. A homomorphism sends the identity to the identity, so $\beta(0) = e$. Hence β and γ_X are both integral curves of X^L through e. By uniqueness of integral curves, they agree. ∎

Proposition 2.10 (Local invertibility). *The differential of* exp *at* 0 *is the identity of* $\mathfrak{g}$. *Consequently there are open sets* $U \ni 0$ *in* $\mathfrak{g}$ *and* $V \ni e$ *in* G *such that* exp *restricts to a diffeomorphism* $U \to V$. *We may take* $U = -U$.

Proof. Identify $T_0\mathfrak{g}$ with $\mathfrak{g}$. The line $t \mapsto tX$ has velocity X. Its image under exp is γ_X, whose velocity at 0 is $X^L(e) = X$. Hence $d\exp_0(X) = X$.

Read exp in a chart around e. The inverse function theorem of multivariable calculus, which we cite, says the following. A smooth map between open subsets of $\mathbb{R}^d$ with invertible derivative at a point restricts to a diffeomorphism between open neighbourhoods of that point and its image. This gives U and V. Replace U by $U \cap (-U)$, and replace V by the image of $U \cap (-U)$ under exp. Both sets remain open. ∎

Write $\log\colon V \to U$ for the inverse. Local invertibility, Proposition 2.10, is a local statement. Indeed, the rotation example, Example 2.5, shows that exp need not be injective on all of $\mathfrak{g}$.

Proposition 2.11 (Generation by exponentials). *If* G *is connected, every element of* G *is a finite product* $\exp X_1 \exp X_2 \cdots \exp X_k$.

Proof. Let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. Since $\exp(-X) = (\exp X)^{-1}$ and $U = -U$, we have $V^{-1} = V$. Let H be the set of finite products of elements of V. It is a subgroup. It is open, since $H = \bigcup_{h \in H} hV$. Its complement is a union of cosets gH, so it is open too. As $e \in H$ and G is connected, $H = G$. ∎

The proof uses only that V is open, contains e and satisfies $V^{-1} = V$.

A group that is not connected still has a connected part around e. The next definition describes it by paths.

Definition 2.12 (Identity component). The *identity component* G_0 of a Lie group G is the set of elements of G joined to e by a continuous path in G.

For example, let $G = \mathrm{O}(2)$. The determinant is continuous with values ± 1, so it is constant along a path. Hence $G_0 \subseteq \mathrm{SO}(2)$. Conversely, every element of $\mathrm{SO}(2)$ is $\exp(\theta J)$ by the rotation example, Example 2.5, and $t \mapsto \exp(t\theta J)$ joins it to I. So $G_0 = \mathrm{SO}(2)$.

Proposition 2.13 (Properties of the identity component). *Let* G *be a Lie group with Lie algebra* $\mathfrak{g}$.

(a) G_0 *is a normal subgroup of* G, *and it is both open and closed.*

(b) G_0 *has Lie algebra* $\mathfrak{g}$, *and it contains* $\exp(tX)$ *for all* $X \in \mathfrak{g}$ *and* $t \in \mathbb{R}$.

(c) G_0 *is connected, and it contains every connected subset of* G *through* e. *Hence* G *is connected exactly when* $G = G_0$.

Proof. (a) Let γ and δ be continuous paths from e to g and from e to h. Then $t \mapsto \gamma(t)\delta(t)$ joins e to gh, and $t \mapsto \gamma(t)^{-1}$ joins e to g^{-1}. For $x \in G$, the path $t \mapsto x\gamma(t)x^{-1}$ joins e to xgx^{-1}. So G_0 is a normal subgroup.

Next, let $V = \exp U$ be the neighbourhood given by local invertibility, Proposition 2.10. For $g \in G_0$ and $X \in U$, the path $t \mapsto g\exp(tX)$, $0 \le t \le 1$, joins g to $g\exp X$. Hence the open set

gV lies in G_0, so G_0 is open. Its complement is a union of cosets gG_0. These cosets are open, so G_0 is closed.

(b) An open subset containing e has the same tangent space at e as G. The curve $s \mapsto \exp(sX)$ joins e to $\exp(tX)$.

(c) Suppose $G_0 = O_1 \cup O_2$ with O_1 and O_2 disjoint, open in G_0 and nonempty, and with $e \in O_1$. Choose $g \in O_2$ and a path γ from e to g in G. Every point of this path is joined to e along γ, so γ lies in G_0. Then $\gamma^{-1}(O_1)$ and $\gamma^{-1}(O_2)$ would disconnect $[0,1]$. So G_0 is connected.

Let C be a connected subset of G containing e. By (a), $C \cap G_0$ is open and closed in C, and it is nonempty. So $C \subseteq G_0$. Thus G_0 is the connected component of e. Finally, if G is connected, then taking $C = G$ gives $G = G_0$. The converse is the first statement of this part. ∎

2.4 Conjugation and homomorphisms

Conjugation $C_g(h) = ghg^{-1}$ is a smooth automorphism of G fixing e. The adjoint action of g is the linear map $\mathrm{Ad}_g = (dC_g)_e \colon \mathfrak{g} \to \mathfrak{g}$, also written $\mathrm{Ad}(g)$. For a matrix group, differentiating $g\exp(tX)g^{-1}$ at $t = 0$ gives $\mathrm{Ad}_g X = gXg^{-1}$. Hence $gXg^{-1} \in \mathfrak{g}$.

Theorem 2.14 (Differentiating homomorphisms). *Let $\Phi \colon G \to H$ be a Lie group homomorphism and $\phi = d\Phi_e$.*

(i) $\Phi(\exp X) = \exp(\phi(X))$ *for all* $X \in \mathfrak{g}$.

(ii) $\exp(\mathrm{Ad}_g X) = g\exp(X)g^{-1}$ *for all* $g \in G$ *and* $X \in \mathfrak{g}$.

(iii) If G is connected, then Φ is determined by ϕ.

(iv) If G and H are matrix Lie groups, then $\phi([X,Y]) = [\phi(X), \phi(Y)]$.

Proof. (i) The curve $\beta(t) = \Phi(\exp tX)$ is a smooth homomorphism $\mathbb{R} \to H$ with velocity $\phi(X)$ at 0. Apply the lemma on smooth one-parameter subgroups, Lemma 2.9, in H and set $t = 1$.

(ii) Apply the intertwining formula (i) to $\Phi = C_g$, whose differential is Ad_g.

(iii) By generation by exponentials, Proposition 2.11, each element of G is a product $\exp X_1 \cdots \exp X_k$. By the intertwining formula (i), the image of this product under Φ is $\exp \phi(X_1) \cdots \exp \phi(X_k)$, which depends only on ϕ.

(iv) For $g \in G$ and $Y \in \mathfrak{g}$, the matrix $gYg^{-1} \in \mathfrak{g}$ is the velocity of $g\exp(tY)g^{-1}$ at $t = 0$. Differentiate $\Phi(g\exp(tY)g^{-1}) = \Phi(g)\Phi(\exp tY)\Phi(g)^{-1}$ at $t = 0$:

$$\phi(gYg^{-1}) = \Phi(g)\,\phi(Y)\,\Phi(g)^{-1}.$$

Put $g = \exp(sX)$ and use the intertwining formula (i) on the right:

$$\phi\big(e^{sX}Ye^{-sX}\big) = e^{s\phi(X)}\phi(Y)e^{-s\phi(X)}.$$

Differentiate at $s = 0$, using the derivative formula, Proposition 2.2(iii). Since ϕ is linear, it commutes with differentiation. The result is $\phi(XY - YX) = \phi(X)\phi(Y) - \phi(Y)\phi(X)$. This is the bracket identity $\phi([X,Y]) = [\phi(X), \phi(Y)]$. ∎

The bracket identity, Theorem 2.14(iv), holds for every Lie group homomorphism. The general proof uses the vector-field bracket of Lecture 1. We cite Kirillov Jr., Chapter 3.

The most important target is $H = \mathrm{GL}(V)$, for a finite-dimensional complex vector space V. A representation $\Pi\colon G \to \mathrm{GL}(V)$ has a differential $d\Pi$ at e, and the intertwining formula (i) reads $\Pi(\exp X) = \exp(d\Pi(X))$.

We now apply the theorem to the adjoint action itself. Here $\mathrm{GL}(\mathfrak{g})$ is the group of invertible linear maps of $\mathfrak{g}$, and $\mathrm{End}(\mathfrak{g})$ is its tangent space at the identity.

Proposition 2.15 (The adjoint representation)**.** *Let G be a Lie group with Lie algebra $\mathfrak{g}$. The map $\mathrm{Ad}\colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism. Write $\mathrm{ad}\colon \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ for its differential at e. Then, for all $X, Y \in \mathfrak{g}$,*

$$\mathrm{Ad}(\exp X) = e^{\mathrm{ad}\, X}, \qquad \mathrm{ad}\, X(Y) = [X, Y].$$

Proof. Since $C_{gh} = C_g \circ C_h$, the chain rule gives $\mathrm{Ad}_{gh} = \mathrm{Ad}_g\, \mathrm{Ad}_h$. The map $(g, h) \mapsto ghg^{-1}$ is smooth, so its derivative in h at $h = e$ depends smoothly on g. Hence Ad is a smooth homomorphism.

The intertwining formula, Theorem 2.14(i), with $\Phi = \mathrm{Ad}$ and $H = \mathrm{GL}(\mathfrak{g})$, gives $\mathrm{Ad}(\exp X) = \exp(\mathrm{ad}\, X)$. This is the exponential of the operator $\mathrm{ad}\, X$, which we write as $e^{\mathrm{ad}\, X}$.

By the definition of a differential, $\mathrm{ad}\, X(Y)$ is the derivative of $\mathrm{Ad}(\exp sX)Y$ at $s = 0$. For a matrix group, $\mathrm{Ad}(\exp sX)Y = e^{sX} Y e^{-sX}$, and the derivative formula, Proposition 2.2(iii), gives $\mathrm{ad}\, X(Y) = XY - YX = [X, Y]$. For a general Lie group we cite $\mathrm{ad}\, X(Y) = [X, Y]$ from Warner, Chapter 3, Section 3.46 and Proposition 3.47. ∎

For example, in $\mathrm{GL}_2(\mathbb{R})$ take

$$h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad [h, E] = 2E.$$

So $\mathrm{ad}\, h$ multiplies E by 2, and $e^{t\,\mathrm{ad}\, h} E = e^{2t} E$. Direct multiplication with $\exp(th) = \mathrm{diag}(e^t, e^{-t})$ gives the same result: $\exp(th) E \exp(-th) = e^{2t} E$. Exercise 2.5 computes the operator $\mathrm{ad}\, A$ on $\mathfrak{su}(2)$.

The adjoint representation gives the product rule for commuting elements of any Lie group.

Proposition 2.16 (Commuting exponentials)**.** *Let G be a Lie group, and let $X, Y \in \mathfrak{g}$ with $[X, Y] = 0$. Then $\exp(tX)$ and $\exp(sY)$ commute for all $s, t \in \mathbb{R}$, and*

$$\exp(X + Y) = \exp X \exp Y.$$

Proof. Since $\mathrm{ad}\, X(Y) = [X, Y] = 0$, we get $\mathrm{Ad}(\exp tX)Y = e^{t\,\mathrm{ad}\, X} Y = Y$. Apply Theorem 2.14(ii) with $g = \exp(tX)$ to the element sY:

$$\exp(tX) \exp(sY) \exp(-tX) = \exp\bigl(s\, \mathrm{Ad}(\exp tX) Y\bigr) = \exp(sY).$$

So $\exp(tX)$ and $\exp(sY)$ commute. Hence $\beta(t) = \exp(tX) \exp(tY)$ is a smooth homomorphism $\mathbb{R} \to G$.

Its velocity at 0 is $X+Y$. Indeed, let $m(g,h)=gh$. Since $m(g,e)=g$ and $m(e,h)=h$, the differential of m at (e,e) sends $(X,0)$ to X and $(0,Y)$ to Y. By linearity it sends (X,Y) to $X+Y$. The lemma on smooth one-parameter subgroups, Lemma 2.9, gives $\beta(t)=\exp\bigl(t(X+Y)\bigr)$. Now put $t=1$. ∎

The same differential shows that the velocity at 0 of a product of curves through e is the sum of their velocities.

2.5 The Baker–Campbell–Hausdorff formula

By local invertibility, Proposition 2.10, a product of exponentials of small elements has a logarithm in $\mathfrak{g}$. We compute its leading terms in $\mathrm{GL}_n(\mathbb{C})$. Write $O(t^k)$ for a quantity bounded in norm by a constant times $|t|^k$ for small $|t|$.

Proposition 2.17 (Second-order term)**.** *Let $X,Y\in M_{n\times n}(\mathbb{C})$. For small $|t|$, put $Z(t)=\log\bigl(\exp(tX)\exp(tY)\bigr)$. Then*

$$Z(t)=t(X+Y)+\frac{t^2}{2}[X,Y]+O(t^3).$$

Proof. The map Z is smooth with $Z(0)=0$. Taylor's theorem gives matrices A,B with $Z(t)=tA+t^2B+O(t^3)$. Hence $\|Z(t)\|=O(t)$, so the tail $\sum_{k\ge3}Z^k/k!$ has norm at most $\|Z\|^3e^{\|Z\|}=O(t^3)$. Therefore

$$\exp Z(t)=I+Z+\tfrac12Z^2+O(t^3)=I+tA+t^2\bigl(B+\tfrac12A^2\bigr)+O(t^3).$$

Multiplying the two exponential series to second order gives

$$\exp(tX)\exp(tY)=I+t(X+Y)+t^2\bigl(\tfrac12X^2+XY+\tfrac12Y^2\bigr)+O(t^3).$$

The two left sides are equal. If $tP+t^2Q=O(t^3)$, dividing by t and letting $t\to0$ gives $P=0$, and then $Q=0$. Comparing coefficients,

$$A=X+Y,\qquad B=\tfrac12X^2+XY+\tfrac12Y^2-\tfrac12(X+Y)^2=\tfrac12(XY-YX).$$ ∎

The second-order term also shows what the bracket means. Multiplying out the series gives

$$\exp(tX)\exp(tY)\exp(-tX)\exp(-tY)=I+t^2[X,Y]+O(t^3),$$

as Exercise 2.6 shows. So the bracket measures, to leading order, how far $\exp(tX)$ and $\exp(tY)$ fail to commute. On a general Lie group, the flow of X^L is right multiplication by $\exp(tX)$. So this product is the end point of the path from e that follows X, Y, $-X$ and $-Y$ in turn. Figure 2.1 draws this path.

Exercise 2.7 computes the third-order term. The general statement is the following.

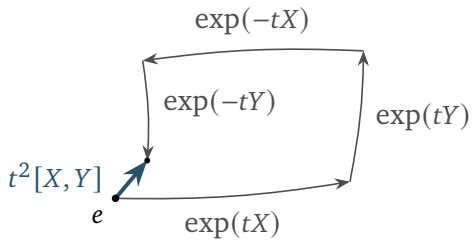


Figure 2.1: The path from e with steps $\exp(tX)$, $\exp(tY)$, $\exp(-tX)$ and $\exp(-tY)$, each multiplying on the right. The coloured arrow goes from e to the end point. In the coordinates given by log, this displacement is $t^2[X,Y] + O(t^3)$. The drawing is schematic.

Theorem 2.18 (Convergent BCH formula; cited)**.** *Let G be a Lie group with finite-dimensional real Lie algebra $\mathfrak{g}$. There is a neighbourhood W of 0 in $\mathfrak{g}$ with the following property. For $X, Y \in W$, the series grouped by total degree,*

$$H(X,Y) = X + Y + \tfrac{1}{2}[X,Y] + \tfrac{1}{12}\big([X,[X,Y]] + [Y,[Y,X]]\big) + \cdots,$$

converges in $\mathfrak{g}$, depends continuously on (X,Y), and satisfies $\exp X \exp Y = \exp H(X,Y)$. The degree-$m$ term is a combination of iterated brackets of m entries from X and Y, with universal rational coefficients independent of G.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves this convergent series for connected groups. The identity component reduces a general G to that case, by Proposition 2.13. The section on sources at the end of this lecture records what Hall, Knapp and Kirillov Jr. each cover.

The theorem answers the main question near the identity. As before, $\log\colon V \to U$ is the local inverse of exp on G.

Proposition 2.19 (Local determination)**.** *Let G and G' be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{g}'$. Write $\exp'$ and e' for the exponential and the identity of G'. Let $\psi\colon \mathfrak{g} \to \mathfrak{g}'$ be a homomorphism of real Lie algebras, and put $f = \exp' \circ \psi \circ \log$ on V. Then f is smooth and $df_e = \psi$. There is an open neighbourhood $V_1 \subseteq V$ of e such that, for all $a, b \in V_1$,*

$$ab \in V \quad \text{and} \quad f(ab) = f(a)f(b).$$

If ψ is an isomorphism, then f restricts to a diffeomorphism from an open neighbourhood of e onto an open neighbourhood of e'.

Proof. Since ψ is linear and preserves brackets, it maps each term of $H(X,Y)$ to the corresponding term of $H(\psi X, \psi Y)$. Since it is continuous, it maps the convergent sum to the sum.

Write W' for the BCH neighbourhood of G'. Choose V_1 so close to e that, for all $a, b \in V_1$, the elements $X = \log a$ and $Y = \log b$ satisfy three conditions:

(a) $X, Y \in W$;
(b) $\psi X, \psi Y \in W'$;
(c) $H(X,Y) \in U$, the set on which exp is inverted.

Continuity of log, ψ and H, with $H(0,0)=0$, makes this possible. Since $X, Y \in W$ by condition (a), $ab = \exp H(X,Y)$. Since $H(X,Y) \in U$ by condition (c), $ab \in V$ and $\log(ab) = H(X,Y)$. Therefore

$$f(ab) = \exp'\big(\psi H(X,Y)\big) = \exp'\big(H(\psi X, \psi Y)\big) = \exp'(\psi X)\exp'(\psi Y) = f(a)f(b).$$

The third equality uses condition (b), $\psi X, \psi Y \in W'$.

The map f is smooth, as a composite of smooth maps. We have $f(e) = \exp'(\psi(0)) = \exp'(0) = e'$. By local invertibility, Proposition 2.10, $d\exp_0$ and $d\exp'_0$ are identity maps, so $d\log_e$ is the identity too. The chain rule gives

$$df_e = d\exp'_0 \circ \psi \circ d\log_e = \psi.$$

If ψ is an isomorphism, it is invertible. The inverse function theorem then gives an open neighbourhood $V_0 \subseteq V$ of e that f maps diffeomorphically onto an open neighbourhood of e'. ∎

So isomorphic Lie algebras give the same multiplication near the identity. The multiplicative part of the proposition does not need ψ to be injective. Lecture 4 uses the same argument for a homomorphism that need not be injective.

The bracket does not determine the global group. For example, $\mathrm{O}(n)$ and $\mathrm{SO}(n)$ have the same Lie algebra. The next section shows a subtler effect.

2.6 Exponentials in $\mathrm{SL}(2,\mathbb{R})$

We now show that exp need not be onto, even on a connected group.

For $g \in \mathrm{SL}(2,\mathbb{R})$, expanding a 2×2 determinant gives

$$\det(g + tM) = 1 + t\,\mathrm{tr}(g^{-1}M) + t^2 \det(g^{-1}M).$$

Hence $d\det_g(M) = \mathrm{tr}(g^{-1}M)$. This differential is onto $\mathbb{R}$, since $d\det_g(g) = \mathrm{tr}\, I = 2$. By the regular level-set theorem of Lecture 1, $\mathrm{SL}(2,\mathbb{R})$ is an embedded matrix Lie group. Its algebra $\mathfrak{sl}_2(\mathbb{R})$ is the kernel of $d\det_I$, so it consists of the real traceless 2×2 matrices.

Let $X \in \mathfrak{sl}_2(\mathbb{R})$. Cayley–Hamilton gives $X^2 = \delta I$ with $\delta = -\det X$. Separating even and odd powers, as for rotations and boosts,

$$\exp X = c(\delta) I + s(\delta) X, \qquad c(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k)!}, \quad s(\delta) = \sum_{k\ge 0} \frac{\delta^k}{(2k+1)!}.$$

Comparison with the hyperbolic and circular series gives

$$c(\delta) = \begin{cases} \cosh\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \cos\sqrt{-\delta}, & \delta < 0, \end{cases} \qquad s(\delta) = \begin{cases} \sinh\sqrt{\delta}/\sqrt{\delta}, & \delta > 0, \\ 1, & \delta = 0, \\ \sin\sqrt{-\delta}/\sqrt{-\delta}, & \delta < 0. \end{cases}$$

Proposition 2.20 (A trace bound). *For every $X \in \mathfrak{sl}_2(\mathbb{R})$, $\operatorname{tr}\exp X = 2c(\delta) \geq -2$. Equality holds only when $\exp X = -I$. Consequently, an element of $\mathrm{SL}(2,\mathbb{R})$ is not an exponential if its trace is less than -2, or if its trace is -2 and it differs from $-I$.*

Proof. Since $\operatorname{tr} X = 0$, we have $\operatorname{tr}\exp X = 2c(\delta)$. For $\delta \geq 0$, $c(\delta) \geq 1$. For $\delta < 0$, $c(\delta)$ is a cosine, so it is at least -1.

Suppose $c(\delta) = -1$. Then $\delta < 0$, and $\sqrt{-\delta}$ is an odd multiple of π. Hence $\sin\sqrt{-\delta} = 0$ and $s(\delta) = 0$. Therefore $\exp X = c(\delta)I + s(\delta)X = -I$. ∎

Example 2.21 (A non-exponential). Let $a > 0$ with $a \neq 1$, and set

$$g_a = \begin{pmatrix} -a & 0 \\ 0 & -1/a \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}).$$

Its eigenvalues are negative and unequal, and

$$\operatorname{tr} g_a + 2 = -\left(a + \frac{1}{a} - 2\right) = -\frac{(a-1)^2}{a} < 0.$$

By the trace bound, Proposition 2.20, the element g_a is not $\exp X$ for any $X \in \mathfrak{sl}_2(\mathbb{R})$. For $a = 2$, the trace is $-5/2$.

The element g_a is a product of two exponentials. With $h = \operatorname{diag}(1,-1)$,

$$\exp(\pi J) = -I, \qquad \exp\big((\log a)h\big) = \begin{pmatrix} a & 0 \\ 0 & 1/a \end{pmatrix},$$

so $g_a = \exp(\pi J)\exp\big((\log a)h\big)$.

Exercise 2.4 proves that $\mathrm{SL}(2,\mathbb{R})$ is connected. So generation by exponentials, Proposition 2.11, applies, but one exponential does not suffice. Local invertibility, Proposition 2.10, is not contradicted. For a near 1, the element g_a is near $-I$, far from I. Lecture 6 shows that in a compact connected group every element is a single exponential.

Self-checks

1. Is $\exp\colon \mathfrak{so}(2) \to \mathrm{SO}(2)$ injective? Is $-I \in \mathrm{O}(1,1)$ of the form $\exp(uK)$?
2. Can a real matrix with negative determinant be $\exp X$ for a real matrix X? Does this contradict generation by exponentials, Proposition 2.11?
3. If $X, Y \in W$ and $[X,Y] = 0$, what is $H(X,Y)$? Is $\log(\exp X \exp Y) = X + Y$ for all commuting X, Y?

Answers. (1) No, since $\exp(2\pi J) = I = \exp 0$. Also no: $\operatorname{tr}(-I) = -2$, while $\operatorname{tr}\exp(uK) = 2\cosh u \geq 2$. (2) No: for real X, $\det\exp X = e^{\operatorname{tr} X} > 0$. There is no contradiction, because $\mathrm{GL}_n(\mathbb{R})$ is disconnected by the determinant sign. (3) Every later term is an iterated bracket, so $H(X,Y) = X + Y$. Still, take $G = \mathrm{SO}(2)$ and $X = Y = \pi J$. The product $\exp X \exp Y$ is I. For the local inverse $\log\colon V \to U$ of exp on $\mathrm{SO}(2)$, $\log I = 0 \neq 2\pi J$. Here $2\pi J \notin U$, since $\exp(2\pi J) = \exp 0$, $0 \in U$ and exp is injective on U.

Summary

Every continuous one-parameter subgroup of $\mathrm{GL}_n(\mathbb{C})$ is $t \mapsto \exp(tX)$. For a general Lie group, flows of left-invariant fields define exp, a local diffeomorphism at 0. Homomorphisms intertwine exponentials, and connected groups are generated by exponentials. The identity component is an open and closed normal subgroup with the same Lie algebra. The adjoint representation satisfies $\mathrm{Ad}(\exp X) = e^{\operatorname{ad} X}$ and $\operatorname{ad} X(Y) = [X, Y]$, and elements with $[X, Y] = 0$ satisfy $\exp(X + Y) = \exp X \exp Y$. Near the identity, the BCH series computes multiplication from the bracket. Globally, exp need not be injective or surjective, and the Lie algebra $\mathfrak{g}$ does not determine the components of G.

Sources and hypotheses

This section records what the sources prove for the convergent BCH formula, Theorem 2.18.

Hall, Sections 5.3–5.5, proves the integral form of the formula for matrix Lie groups and X, Y near 0. Hall, Section 5.6, computes the series only through degree three.

Knapp, *Lie Groups Beyond an Introduction*, Theorem B.22, proves the convergent series for connected groups. This suffices, because the identity component of G is open, has Lie algebra $\mathfrak{g}$ and contains each curve $t \mapsto \exp(tX)$. Knapp's proof reduces to matrices with $\|X\| + \|Y\| < \log 2$. There the degree-n term of his logarithm series (B.24) has norm at most $c_n(\|X\| + \|Y\|)^n$, where $-\log(2 - e^s) = \sum_n c_n s^n$ and $c_n \geq 0$. Hence the series converges uniformly for $\|X\| + \|Y\| \leq r < \log 2$, and H is continuous. Knapp's Proposition B.43 converts the rational coefficients of (B.24) into rational bracket coefficients. Kirillov Jr., Section 3.7, Theorem 3.37, states the same series.

Reading

Read Hall, second edition, Chapter 2 on the matrix exponential, Chapter 3 on matrix Lie algebras and homomorphisms, and Sections 5.3–5.7 on BCH and group versus algebra homomorphisms. Kirillov Jr., Chapter 3, treats the general exponential map and the Campbell–Hausdorff formula. Warner, Chapter 3, Sections 3.46–3.47, treats the adjoint representation on a general Lie group. Lecture 3 uses the exponential to study subgroups and quotients.

Lecture 3

Subgroups, quotients and homogeneous spaces

Objectives. *A subgroup of a Lie group need not be a submanifold, and a coset space need not be a manifold. Closedness of the subgroup makes the coset space a manifold. Normality is a separate condition, needed for a group law. We test both conditions on an irrational line in a torus and on two descriptions of the sphere S^2.*

By the end of this lecture you will be able to:

- State the closed-subgroup theorem and the subgroup–subalgebra correspondence with their hypotheses.
- Prove that an irrational line in the two-torus is dense and not embedded.
- Identify a homogeneous space with a coset space by computing a stabiliser.
- Compute tangent spaces and isotropy actions for $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$.

3.1 Closed and immersed subgroups

A subset S of a manifold M carries the subspace topology of Lecture 1: its open sets are the sets $U \cap S$ with U open in M. The subset S is *closed* if $M \setminus S$ is open. In a manifold, S is closed exactly when S contains the limit of each convergent sequence of its points.

The *closure* $\overline{S}$ consists of the points whose every open neighbourhood meets S. It is the smallest closed set containing S. The set S is *dense* if $\overline{S} = M$. By the definition of closure, S is dense exactly when S meets every nonempty open subset of M.

In a manifold, every point of $\overline{S}$ is a limit of a sequence in S. To obtain the sequence, choose points of S in chart balls of radii tending to zero.

We also need compactness, in the form of finite subcovers. Lecture 5 uses this form to prove strict positivity of the Haar integral.

Definition 3.1 (Compact space)**.** A topological space is *compact* if every cover by open sets has a finite subcover.

A continuous image of a compact space is compact. Indeed, the inverse images of an open cover of the image form an open cover of the space. A finite subfamily of them covers the space, so the corresponding open sets cover the image.

A subset of $\mathbb{R}^n$ is compact exactly when it is closed and bounded (Heine–Borel). For subsets of $\mathbb{R}^n$ this agrees with the definition by sequences in *Linear Algebra for Physics*. For a set of matrices, closed and bounded must be read in the matrix space. For example, the set $\{tI : 0 < t \leq 1\}$ is bounded and it is closed in $\mathrm{GL}_n(\mathbb{R})$. It is not compact, because its closure in the matrix space contains the zero matrix.

Definition 3.2 (Immersions and embeddings). A smooth map $f\colon N \to M$ is an *immersion* if df_p is injective for every $p \in N$. It is an *embedding* if it is an injective immersion and a homeomorphism from N onto $f(N)$ with the subspace topology.

The regular level sets of Lecture 1 are images of embeddings. An injective immersion can fail to be an embedding. The reason is that the topology carried over from N to $f(N)$ can be finer than the subspace topology, that is, it can have more open sets.

Definition 3.3 (Immersed and embedded subgroups). Let G be a Lie group. An *immersed subgroup* of G is a subgroup H with a Lie group structure for which the inclusion $\iota\colon H \to G$ is an immersion. It is an *embedded subgroup* if ι is an embedding.

The differential $d\iota_e$ identifies $\mathrm{Lie}(H)$ with a subspace $\mathfrak{h} \subseteq \mathfrak{g}$. By Lecture 2, the differential of a Lie group homomorphism preserves brackets. Hence $\mathfrak{h}$ is a real Lie subalgebra of $\mathfrak{g}$.

Theorem 3.4 (Closed-subgroup theorem; cited). *Let G be a Lie group and let $H \subseteq G$ be a subgroup that is a closed subset. Then H is an embedded submanifold of G, and with this structure H is a Lie group. Its Lie algebra is*

$$\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}.$$

The hypothesis concerns only the subset H. Smoothness of H is a conclusion. For example, $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ are closed in $\mathrm{GL}_n(\mathbb{C})$, because they are defined by continuous equations.

A related regularity theorem removes a hypothesis from homomorphisms.

Theorem 3.5 (Automatic smoothness; cited). *Every continuous homomorphism between finite-dimensional real Lie groups is smooth.*

See Bröcker–tom Dieck, Chapter I, Section 3, Proposition (3.12). In particular, finite-dimensional continuous representations have differentials. We use this fact from Lecture 4 onwards. In particular, by the lemma on smooth one-parameter subgroups in Lecture 2, every continuous one-parameter subgroup of a Lie group has the form $t \mapsto \exp(tX)$.

Theorem 3.6 (Subgroups and subalgebras; cited). *Let G be a Lie group and let $\mathfrak{h} \subseteq \mathfrak{g}$ be a real Lie subalgebra. There is exactly one connected immersed subgroup $H \subseteq G$ with $\mathrm{Lie}(H) = \mathfrak{h}$. As a set, H is the subgroup generated by $\exp(\mathfrak{h})$. Every connected immersed subgroup arises in this way from its own Lie algebra.*

Warner, Chapter 3, proves Theorems 3.4 and 3.6 for general Lie groups. Hall, Section 5.9, proves Theorem 3.6 for matrix Lie groups. Kirillov Jr., Chapters 2 and 3, states these two subgroup theorems.

The two theorems run in opposite directions. The first starts with a subset and concludes that it is a manifold. The second starts with a subalgebra and constructs a group. That group need not be closed or embedded, so the word “immersed” in Theorem 3.6 cannot be replaced by “embedded”. The next section gives an example of a connected immersed

subgroup that is neither closed nor embedded.

3.2 An irrational line in the two-torus

The two-torus $T^2 = \mathrm{U}(1) \times \mathrm{U}(1)$ is a compact abelian Lie group. Identify its Lie algebra $i\mathbb{R} \oplus i\mathbb{R}$ with $\mathbb{R}^2$ by $(2\pi i a, 2\pi i b) \mapsto (a, b)$. The exponential map then becomes the surjective homomorphism

$$E\colon \mathbb{R}^2 \longrightarrow T^2, \qquad E(a,b) = (e^{2\pi i a}, e^{2\pi i b}),$$

with kernel $\mathbb{Z}^2$. For a real number α, the subalgebra $\mathfrak{h}_\alpha = \mathbb{R}(1,\alpha)$ gives the homomorphism and image

$$\varphi_\alpha(t) = E(t,\alpha t) = (e^{2\pi i t}, e^{2\pi i \alpha t}), \qquad L_\alpha = \varphi_\alpha(\mathbb{R}).$$

The velocity of φ_α at t is the left translate of $(2\pi i, 2\pi i\alpha) \neq 0$, so φ_α is an immersion. For rational α the image is a closed circle subgroup (Exercise Sheet 3). Irrational α behaves differently.

We look at the line only where it crosses one vertical circle. At times $x + k$ with k an integer, the first coordinate stays at $e^{2\pi i x}$, and the second moves by α modulo one at each step. Figure 3.1 shows this. So density reduces to one claim: the multiples of α, taken modulo one, come close to every number.

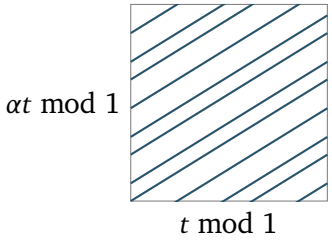


Figure 3.1: The line L_α for $0 \le t \le 8$ and $\alpha = (\sqrt{5}-1)/2$, drawn in the unit square with opposite sides identified. Each new unit-time piece is shifted upwards by α modulo one.

Lemma 3.7 (Small multiples). *Let α be irrational and $n \ge 1$ an integer. There are integers q and m with $1 \le q \le n$ and*

$$0 < |q\alpha - m| < \frac{1}{n}.$$

Proof. For $j = 0, 1, \ldots, n$, put $f_j = j\alpha - \lfloor j\alpha \rfloor \in [0,1)$. These $n+1$ numbers lie in the n intervals $[s/n, (s+1)/n)$. By the pigeonhole principle, two of them, f_{j_1} and f_{j_2} with $j_1 < j_2$, lie in one interval. Set $q = j_2 - j_1$ and $m = \lfloor j_2\alpha \rfloor - \lfloor j_1\alpha \rfloor$. Then $q\alpha - m = f_{j_2} - f_{j_1}$, which has absolute value less than $1/n$. It is nonzero, since otherwise $\alpha = m/q$ would be rational. ∎

Proposition 3.8 (The irrational line). *Let α be irrational. Then:*

(a) φ_α is injective;
(b) L_α is dense in T^2;
(c) $L_\alpha \neq T^2$, so L_α is not closed;
(d) φ_α is not a homeomorphism onto L_α with the subspace topology.

Proof. (a) If $\varphi_\alpha(t) = \varphi_\alpha(s)$, then $t - s$ and $\alpha(t - s)$ are integers. If $t \neq s$, then α is the ratio of these two integers, so it is rational. Hence $t = s$.

(b) Fix a point $(e^{2\pi i x}, e^{2\pi i y})$ and an integer $n \geq 1$. First, take q and m from the small-multiples lemma, Lemma 3.7, and put $\delta = q\alpha - m \neq 0$. Let k be an integer nearest to $(y - \alpha x)/\delta$. Multiplying by $|\delta|$,

$$|k\delta - (y - \alpha x)| \leq \tfrac{1}{2}|\delta|.$$

Next, put $t = x + kq$. Here kq is an integer, and $\alpha kq = k\delta + km$ with km an integer. Therefore

$$\varphi_\alpha(t) = \left(e^{2\pi i x},\, e^{2\pi i(\alpha x + k\delta)}\right).$$

Finally, compare coordinates. The first coordinates of $\varphi_\alpha(t)$ and the chosen point agree. A chord of the unit circle is no longer than its arc, so $|e^{2\pi i u} - e^{2\pi i v}| \leq 2\pi|u - v|$ for all $u \in \mathbb{R}$ and $v \in \mathbb{R}$. The second coordinates therefore differ by at most $\pi|\delta| < \pi/n$. As n is arbitrary, L_α meets every neighbourhood of the chosen point. Since the chosen point is arbitrary, L_α is dense.

(c) Suppose $\varphi_\alpha(t) = (1, -1)$. Then t is an integer r, and $r\alpha = m + \frac{1}{2}$ for an integer m. Here $r \neq 0$, so $\alpha = (2m + 1)/(2r)$ would be rational. Thus $(1, -1) \notin L_\alpha$. A dense closed subset is the whole space, so L_α is not closed.

(d) For each n, take q_n and m_n from the small-multiples lemma, Lemma 3.7. Then

$$\varphi_\alpha(q_n) = \left(1, e^{2\pi i(q_n\alpha - m_n)}\right) \longrightarrow (1, 1) = \varphi_\alpha(0),$$

but $q_n \geq 1$ for every n. If the inverse map $L_\alpha \to \mathbb{R}$ were continuous at $(1, 1)$, then q_n would converge to 0. This is impossible. Hence the inverse map is not continuous at $(1, 1)$, and φ_α is not a homeomorphism onto L_α. ∎

So points close together in the torus can be far apart along the line. Figure 3.2 shows this near the identity. This is why the subgroup is not embedded.

Give L_α the Lie group structure transported from $\mathbb{R}$ by the bijection φ_α. It is then the connected immersed subgroup with Lie algebra $\mathfrak{h}_\alpha$ given by the subgroup–subalgebra correspondence, Theorem 3.6. By the irrational-line proposition, Proposition 3.8, the map φ_α is not a homeomorphism onto L_α with the subspace topology, so this subgroup is not embedded. The same proposition shows that L_α is not closed, so the closed-subgroup theorem, Theorem 3.4, does not apply to it. Since L_α is dense by the same proposition, its closure is the closed subgroup T^2. The Lie algebra $\mathbb{R}^2$ of T^2 is strictly larger than $\mathfrak{h}_\alpha$.

3.3 Coset spaces, actions and quotient groups

Let H be a subgroup of a Lie group G. The *coset space* $G/H = \{gH : g \in G\}$ has the quotient map π given by $\pi(g) = gH$. Its *quotient topology* declares $V \subseteq G/H$ open when $\pi^{-1}(V)$ is open in G. The group G acts on G/H by left translation, $\ell_g(xH) = gxH$.

Closedness is necessary for G/H to be Hausdorff. In a Hausdorff space each point is closed, because every other point has an open neighbourhood avoiding it. So if G/H is Hausdorff, then $H = \pi^{-1}(eH)$ is closed. For irrational α, the space T^2/L_α is therefore not a manifold.

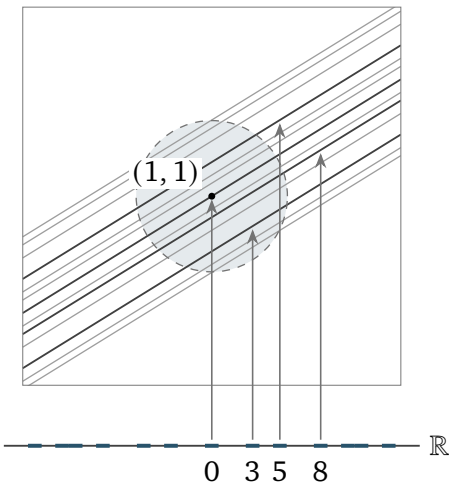


Figure 3.2: The square $[-\frac{1}{2}, \frac{1}{2}]^2$ is T^2, with opposite sides identified and the identity $(1,1)$ at the centre. Here $\alpha = (\sqrt{5}-1)/2$. The strands are the pieces of L_α for t in the heavy unit intervals around integers k of the line below. They are all the pieces with $|k| \le 13$ that meet the small dashed disc. The arrows show where the strands for $k = 0, 3, 5, 8$ come from.

Theorem 3.9 (Quotients by closed subgroups; cited)**.** *Let G be a Lie group and H a closed subgroup with Lie algebra $\mathfrak{h}$. Then:*

(1) G/H is Hausdorff and second countable. It has a unique smooth manifold structure for which π is a submersion*, that is, smooth with surjective differential at every point;*

(2) $\dim G/H = \dim G - \dim H$, and $d\pi_e$ induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_{eH}(G/H)$;

(3) the action $G \times G/H \to G/H$ is smooth;

(4) a map f from G/H to a manifold is smooth exactly when $f \circ \pi$ is smooth.

This theorem is in Bröcker–tom Dieck, Chapter I, Section 4, and in Warner, Chapter 3.

Definition 3.10 (Smooth actions)**.** A *smooth action* of G on a manifold M is a smooth map $G \times M \to M$, $(g,p) \mapsto g \cdot p$, with $e \cdot p = p$ and $g \cdot (g' \cdot p) = (gg') \cdot p$. The *stabiliser* of p is $G_p = \{g \in G : g \cdot p = p\}$, and the *orbit* of p is $G \cdot p$. The action is *transitive* if it has only one orbit. A manifold with a transitive smooth action is a *homogeneous space*.

Proposition 3.11 (Stabilisers and orbit maps)**.** *Let G act smoothly on M, fix $p \in M$, and put $H = G_p$.*

(a) H is a closed subgroup, and $G_{g\cdot p} = gHg^{-1}$.

(b) The orbit map $\beta(gH) = g \cdot p$ is a well-defined, injective, smooth map $G/H \to M$ with image $G \cdot p$. It is equivariant: $\beta(\ell_{g'}(gH)) = g' \cdot \beta(gH)$.

Proof. (a) Since $e \cdot p = p$, we have $e \in H$. If $g, g' \in H$, then $(gg') \cdot p = g \cdot (g' \cdot p) = p$ and $g^{-1} \cdot p = g^{-1} \cdot (g \cdot p) = p$. The point p is closed, because M is Hausdorff. So H is closed, as the preimage of p under the continuous map $g \mapsto g \cdot p$. Finally, $x \cdot (g \cdot p) = g \cdot p$ exactly when $(g^{-1}xg) \cdot p = p$, that is, when $x \in gHg^{-1}$.

(b) We have $g \cdot p = g' \cdot p$ exactly when $g^{-1}g' \in H$, that is, when $gH = g'H$. One direction gives

well-definedness, the other injectivity. Since H is closed by part (a), the quotient theorem, Theorem 3.9, applies to G/H. The composite $\beta \circ \pi(g) = g \cdot p$ is smooth, so β is smooth by the smoothness criterion of Theorem 3.9(4). Equivariance is the identity $(g'g) \cdot p = g' \cdot (g \cdot p)$. ∎

Theorem 3.12 (Homogeneous spaces; cited). *Let a Lie group G act smoothly and transitively on a manifold M, and let $p \in M$. Then the orbit map $\beta : G/G_p \to M$ is a G-equivariant diffeomorphism.*

The proof uses the second countability of G, which is part of the definition of a manifold in Lecture 1; see Warner, Chapter 3, or Kirillov Jr., Chapter 2.

We now compute T_pM from $\mathfrak{g}$. An element X of $\mathfrak{g}$ moves p with a velocity, namely the velocity of $t \mapsto \exp(tX) \cdot p$ at 0. For a transitive action, the elements with zero velocity form $\mathfrak{h}$, and the velocities fill T_pM. So $\mathfrak{g}/\mathfrak{h} \cong T_pM$. Next, the stabiliser H fixes p, so it acts on T_pM. This action tells the two sphere models below apart.

Two derivatives of the orbit map are used below. First, for $X \in \mathfrak{g}$, differentiate $\beta(\pi(\exp tX)) = \exp(tX) \cdot p$ at $t = 0$. This shows that $d\beta_{eH}(d\pi_e X)$ is the velocity of $t \mapsto \exp(tX) \cdot p$. This holds for every smooth action.

Second, assume now that the action is transitive. By the homogeneous-space theorem, Theorem 3.12, $d\beta_{eH}$ is an isomorphism $T_{eH}(G/H) \to T_pM$. Combining this with the tangent-space isomorphism of Theorem 3.9(2), we see that the composite $d\beta_{eH} \circ d\pi_e$ has kernel $\mathfrak{h}$ and induces an isomorphism $\mathfrak{g}/\mathfrak{h} \cong T_pM$.

As before, $H = G_p$. Let $k \in H$. The translation ℓ_k fixes eH. Its differential $d(\ell_k)_{eH}$ is called the *isotropy action* of k. The map $a_k : M \to M$, given by $a_k(p') = k \cdot p'$ for $p' \in M$, fixes p. Equivariance of β says $\beta \circ \ell_k = a_k \circ \beta$. The chain rule at eH therefore gives

$$d\beta_{eH} \circ d(\ell_k)_{eH} = d(a_k)_p \circ d\beta_{eH}.$$

So, transported by the isomorphism $d\beta_{eH}$, the isotropy action of k is the differential of a_k at p.

A coset space is not automatically a group.

Proposition 3.13 (Normality). *The rule $(xH)(yH) = xyH$ is well defined on G/H exactly when H is normal. It then makes G/H a group.*

Proof. Suppose the rule is well defined. Let $k \in H$ and $y \in G$. The representatives e and k of the coset H give the same product with yH, so $kyH = yH$. Hence $y^{-1}ky \in H$, and H is normal. Conversely, if H is normal, then for $k, k' \in H$,

$$xk\,yk' = xy\,(y^{-1}ky)\,k' \in xyH.$$

The group axioms are inherited from G. ∎

Theorem 3.14 (Quotient groups; cited). *Let H be a closed normal subgroup of a Lie group G. With the structure of Theorem 3.9, G/H is a Lie group and π is a homomorphism. The subalgebra $\mathfrak{h}$ is an ideal, and $\mathrm{Lie}(G/H) \cong \mathfrak{g}/\mathfrak{h}$ as Lie algebras.*

This theorem is in Warner, Chapter 3. Closedness of H makes G/H a manifold, and normality of H makes G/H a group. Since T^2 is abelian, L_α is normal, so T^2/L_α is a group. For irrational

α it is not a manifold. The two models of the sphere below are manifolds without a coset group law.

3.4 The sphere as SO(3)/SO(2)

The group $\mathrm{SO}(3)$ acts smoothly on $S^2 = \{v \in \mathbb{R}^3 : |v| = 1\}$ by matrix multiplication, with base point e_3. Let $J_1, J_2, J_3 \in \mathfrak{so}(3)$ be the matrices of Exercise Sheet 1, so that $J_k v = e_k \times v$. The rotations about the third axis are

$$R_\theta = \exp(\theta J_3) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Proposition 3.15 (Stabiliser of the north pole)**.** *The action of* $\mathrm{SO}(3)$ *on* S^2 *is transitive. The stabiliser of* e_3 *is* $H = \{R_\theta : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2)$.

Proof. Let $v \in S^2$. Extend v to an orthonormal basis (u, w, v) of $\mathbb{R}^3$. The matrix g with columns u, w, v is orthogonal and $ge_3 = v$. Replacing u by $-u$ if necessary, we get $\det g = 1$. Thus the action is transitive.

Now let $ge_3 = e_3$. The third column of g is e_3. The rows of g are also orthonormal, because $gg^{\mathsf{T}} = I$. The third row is a unit vector with third entry one, so its other entries vanish. Hence

$$g = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}, \qquad A^{\mathsf{T}} A = I, \qquad \det A = \det g = 1.$$

The first column of A is a unit vector, so it equals $(\cos\theta, \sin\theta)$ for some θ. The second column is a unit vector orthogonal to the first, and $\det A = 1$ selects $(-\sin\theta, \cos\theta)$. So $g = R_\theta$. Conversely, $R_\theta e_3 = e_3$. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $gH \mapsto ge_3$ is an equivariant diffeomorphism $\mathrm{SO}(3)/\mathrm{SO}(2) \to S^2$. The dimensions agree: $3 - 1 = 2$.

The sphere is the level set $|v|^2 = 1$, so $T_{e_3} S^2 = e_3^\perp$ by Lecture 1. By the first derivative computation of the previous section, $d\beta_{eH} \circ d\pi_e$ sends $X \in \mathfrak{so}(3)$ to the velocity of $\exp(tX)e_3$, which is Xe_3. On the basis,

$$J_1 e_3 = e_1 \times e_3 = -e_2, \qquad J_2 e_3 = e_2 \times e_3 = e_1, \qquad J_3 e_3 = 0.$$

So $X \mapsto Xe_3$ has kernel $\mathbb{R}J_3 = \mathrm{Lie}(H)$ and image $e_3^\perp$. It induces the isomorphism $\mathfrak{so}(3)/\mathbb{R}J_3 \cong T_{e_3} S^2$, which is $d\beta_{eH}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{R_\theta}(v) = R_\theta v$ is the restriction to S^2 of a linear map of $\mathbb{R}^3$. Its differential at e_3 is therefore R_θ restricted to $e_3^\perp$. In the coordinates (x, y) for $xe_1 + ye_2$, this is the rotation of the tangent plane by θ. Only $R_\theta = I$ acts trivially.

The stabiliser H is not normal. Take the rotation by $\pi/2$ about e_1,

$$g_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad g_0 e_3 = -e_2.$$

By the conjugation formula of Proposition 3.11(a), $g_0 H g_0^{-1}$ is the stabiliser of $-e_2$. But $R_{\pi/2}(-e_2) = e_1$. Hence $R_{\pi/2} \in H$ and $R_{\pi/2} \notin g_0 H g_0^{-1}$. By the normality proposition, Proposition 3.13, there is no coset multiplication on $\mathrm{SO}(3)/\mathrm{SO}(2)$.

3.5 The sphere as SU(2)/U(1)

Let V be the real vector space of traceless Hermitian 2×2 matrices. It has the coordinates

$$\xi(x,y,z) = \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}, \qquad -\det \xi(x,y,z) = x^2+y^2+z^2.$$

The group $\mathrm{SU}(2)$ acts on V by $U\cdot\xi = U\xi U^*$. The matrix $U\xi U^*$ is Hermitian. Since $U^* = U^{-1}$, it has the same trace and determinant as ξ. Thus $\mathrm{SU}(2)$ acts smoothly on $S = \{\xi \in V : \det\xi = -1\}$, which the coordinates identify with S^2. In physics, with the Pauli matrices of Lecture 1, one writes $\xi(x,y,z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, and S is the Bloch sphere of a two-level system. The base point is $N = \xi(0,0,1) = \mathrm{diag}(1,-1)$. With $A = \mathrm{diag}(i,-i)$ from Lecture 1, put $k_\theta = \exp(\theta A) = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$ and $T = \{k_\theta : \theta \in \mathbb{R}\} \cong \mathrm{U}(1)$.

Proposition 3.16 (Stabiliser in SU(2)). *The action of* $\mathrm{SU}(2)$ *on* S *is transitive, and the stabiliser of* N *is* T.

Proof. Let $\xi \in S$. Its characteristic polynomial is $s^2 - (\mathrm{tr}\,\xi)s + \det\xi = s^2 - 1$, so its eigenvalues are ± 1. By the spectral theorem, $\xi = WNW^*$ for a unitary W whose columns are unit eigenvectors of ξ for the eigenvalues 1 and -1. Since W is unitary, its determinant has modulus one. Choose $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $\lambda^2 = \overline{\det W}$. Then $U = \lambda W$ has determinant one, and $UNU^* = |\lambda|^2 WNW^* = \xi$.

For the stabiliser, $UNU^* = N$ is equivalent to $UN = NU$. Write U with entries a, b, c, d. Then

$$UN = \begin{pmatrix} a & -b \\ c & -d \end{pmatrix}, \qquad NU = \begin{pmatrix} a & b \\ -c & -d \end{pmatrix}.$$

Equality forces $b = c = 0$. A diagonal unitary matrix has $|a| = 1$, and $ad = 1$ gives $d = \overline{a}$. Thus $U = k_\theta$ for some θ. Conversely, k_θ commutes with N. ∎

By the homogeneous-space theorem, Theorem 3.12, the map $UT \mapsto UNU^*$ is an equivariant diffeomorphism $\mathrm{SU}(2)/T \to S \cong S^2$. Again the dimensions are $3 - 1 = 2$.

The map $d\beta_{eT} \circ d\pi_e$ sends $X \in \mathfrak{su}(2)$ to the velocity of $\exp(tX)N\exp(tX)^*$ at $t = 0$. By the product rule and since $X^* = -X$, this velocity is $XN + NX^* = [X, N]$. Use the basis A, B, C of $\mathfrak{su}(2)$ from Lecture 1, where

$$B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Direct multiplication gives $[A, N] = 0$ and

$$[B,N] = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = \xi(-2,0,0), \qquad [C,N] = \begin{pmatrix} 0 & -2i \\ 2i & 0 \end{pmatrix} = \xi(0,2,0).$$

The kernel of $X \mapsto [X, N]$ is $\mathbb{R}A = \mathrm{Lie}(T)$. The image of this map is $\{\xi(x, y, 0)\}$. In the coordinates, S is the unit sphere and N is its north pole, so this image is $T_N S$. Thus $X \mapsto [X, N]$ induces the isomorphism $\mathfrak{su}(2)/\mathbb{R}A \cong T_N S$, which is $d\beta_{eT}$ composed with the tangent-space isomorphism of Theorem 3.9(2).

For the isotropy action, the map $a_{k_\theta}(\xi) = k_\theta \xi k_\theta^*$ is the restriction to S of a linear map of V. Its differential at N is the same conjugation, acting on $T_N S$:

$$k_\theta\, \xi(x, y, 0)\, k_\theta^* = \begin{pmatrix} 0 & e^{2i\theta}(x - iy) \\ e^{-2i\theta}(x + iy) & 0 \end{pmatrix}.$$

Thus $x + iy \mapsto e^{-2i\theta}(x + iy)$, a rotation by -2θ. It is trivial exactly when $\theta \in \pi\mathbb{Z}$, that is, when $k_\theta = \pm I$. In physics, the element $k_{-\varphi/2} = \exp(-i\varphi\sigma_3/2)$ therefore rotates the tangent plane by φ, which is the half angle of spin one half.

The two models have the same space, circle stabilisers and tangent dimension. They differ in the isotropy action. Run each stabiliser circle once. Both R_θ and k_θ do this as θ runs over $[0, 2\pi]$. In the coordinates (x, y) used above, the tangent plane then turns once for SO(3), and twice in the opposite sense for SU(2).

The sense of rotation depends on the choice of parametrisation, because replacing k_θ by $k_{-\theta}$ reverses it. The invariant statement is the kernel of the isotropy action, which is trivial for SO(3) and $\{\pm I\}$ for SU(2). Moreover, $-I$ acts trivially on all of V, since $(-I)\xi(-I)^* = \xi$ for every $\xi \in V$. Lecture 4 uses the element $-I$ to relate the two groups.

The subgroup T is not normal in SU(2) (Exercise 3.6(ii)), so SU(2)/U(1) has no coset group law.

Self-checks

1. For which real α is L_α closed in T^2?
2. Is $(gH)(g'H) = gg'H$ a well-defined product on SO(3)/SO(2)?
3. Why is the stabiliser in SU(2) of every point of S a compact subgroup conjugate to T?

Answers. (1) Exactly for rational α. Let $\alpha = p/q$ with integers p and $q \geq 1$. Since q and $\alpha q = p$ are integers, $\varphi_\alpha(q) = (1, 1)$. So φ_α has period q, and $L_\alpha = \varphi_\alpha([0, q])$. This is a continuous image of a compact interval, hence compact in $\mathbb{C}^2$. It is therefore closed in $\mathbb{C}^2$, and so closed in T^2. For irrational α, the subgroup L_α is not closed by the irrational-line proposition, Proposition 3.8(c). (2) No. The stabiliser SO(2) is not normal, as g_0 shows. By the normality proposition, Proposition 3.13, the product is therefore not well defined. (3) By transitivity each point is $U \cdot N$. Its stabiliser is UTU^{-1} by the conjugation formula of Proposition 3.11(a). This is the image of the compact circle T under the continuous map $k \mapsto UkU^{-1}$, so it is compact.

Summary and reading

A closed subgroup of a Lie group is an embedded subgroup. A real Lie subalgebra determines a connected immersed subgroup, which need not be closed or embedded. For example,

the irrational line is dense and not embedded. For closed H, the coset space G/H is a manifold whose tangent space at eH is isomorphic to $\mathfrak{g}/\mathfrak{h}$. If a Lie group G acts smoothly and transitively on a manifold M, then M is equivariantly diffeomorphic to G/G_p for each $p \in M$. Normality is a further condition, needed for G/H to be a group. The sphere is diffeomorphic to both $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$. The two models have different isotropy actions.

Read Bröcker–tom Dieck, Chapter I, Section 4; Hall, Section 5.9; and Kirillov Jr., Chapter 2. Warner, Chapter 3, is the reference for the cited closed-subgroup, correspondence, quotient and homogeneous-space theorems. Lecture 4 studies the element $-I$ of $\mathrm{SU}(2)$ through covering groups.

Lecture 4

Covering groups, integration and descent

Objectives. *We ask when a finite-dimensional Lie algebra representation is the differential of a representation of a given connected Lie group. On a simply connected group it always is. On other connected groups we integrate on the universal covering group and then test a central kernel. For example, spin one half is a representation of* $\mathrm{SU}(2)$ *and not of* $\mathrm{SO}(3)$.

By the end of this lecture you will be able to:

- Describe the universal covering group by path classes and use the lifting theorem.
- Prove that a discrete normal subgroup of a connected group is central.
- State Lie's integration theorem and Lie III with their hypotheses, and explain path transport.
- Prove the descent criterion and apply it to $\mathbb{R} \to \mathrm{U}(1)$ and $\mathrm{SU}(2) \to \mathrm{SO}(3)$.

Lie groups are finite-dimensional, real, Hausdorff and second countable. From Lecture 1–Lecture 3 we use three facts. The map exp is a diffeomorphism near 0. A connected Lie group is generated by any neighbourhood of e. A continuous homomorphism Φ of Lie groups is smooth and satisfies $\Phi(\exp X) = \exp d\Phi(X)$. Hence two continuous homomorphisms on a connected Lie group with equal differentials are equal.

4.1 Loops, homotopy and covering groups

Definition 4.1. A *path* in a space X is a continuous map $\gamma\colon [0,1] \to X$. It is a *loop at* x if $\gamma(0) = \gamma(1) = x$. Paths γ_0, γ_1 from x to y are *homotopic* if there is a continuous map $F\colon [0,1]^2 \to X$ such that

$$F(0,t) = \gamma_0(t), \qquad F(1,t) = \gamma_1(t), \qquad F(s,0) = x, \qquad F(s,1) = y.$$

The space X is *simply connected* if it is path connected and every loop is homotopic to a constant loop.

A space is *path connected* if any two of its points are joined by a path. It is *locally path connected* if every neighbourhood of a point contains a path-connected neighbourhood of that point. Every manifold is locally path connected, and $[0,1]$ and $[0,1]^2$ are both path connected and locally path connected. A *topological group* is a group with a topology for which multiplication and inversion are continuous.

If σ starts where γ ends, the path $\gamma * \sigma$ traverses γ on $[0, \frac{1}{2}]$ and then σ on $[\frac{1}{2}, 1]$, each at double speed. Homotopy classes of loops at x form the *fundamental group* $\pi_1(X, x)$ under this product.

Definition 4.2. Let X and Y be path connected. A continuous surjection $p\colon Y \to X$ is a *covering map* if each point of X has an open neighbourhood U such that $p^{-1}(U)$ is a disjoint union of open sets, each mapped homeomorphically onto U by p.

The basic example is the map $p\colon \mathbb{R} \to \mathrm{U}(1)$ given by $p(t) = e^{it}$. Take a point $z \in \mathrm{U}(1)$ and a short open arc U around z. Then $p^{-1}(U)$ is a disjoint union of open intervals, spaced 2π apart, and p maps each interval homeomorphically onto U. So p is a covering map. Figure 4.1 draws the line as a helix, so that p becomes the vertical projection.

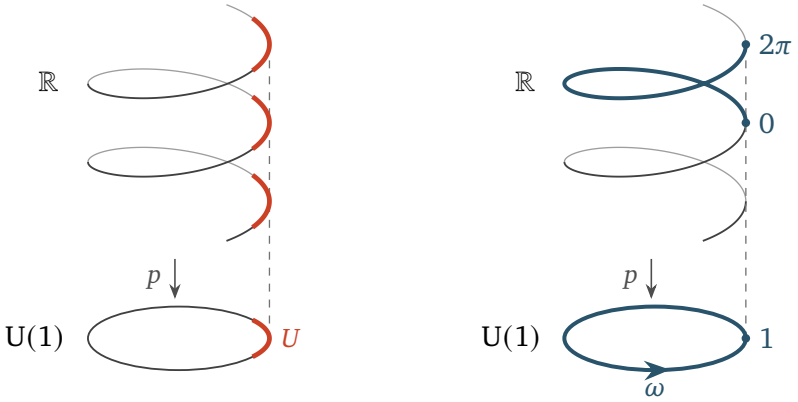


Figure 4.1: The map p projects the helix $\mathbb{R}$ vertically onto the circle $\mathrm{U}(1)$. In the first panel the coloured arcs of the helix lie over the arc U; only three of the infinitely many arcs over U are drawn. In the second panel the heavy path lies over the loop $\omega(t) = e^{2\pi i t}$ and runs from 0 to 2π, so it does not close.

Theorem 4.3 (Lifting theorem; cited). *Let $p\colon Y \to X$ be a covering map.*

(a) Let Z be path connected, locally path connected and simply connected. Let $f\colon Z \to X$ be continuous, and let $p(y_0) = f(z_0)$. Then there is a unique continuous $\tilde{f}\colon Z \to Y$ with $p \circ \tilde{f} = f$ and $\tilde{f}(z_0) = y_0$.

(b) Let γ_0, γ_1 be homotopic paths from x to x', and let $p(y_0) = x$. Then the lifts of γ_0 and γ_1 that start at y_0 have the same endpoint.

We cite this theorem from H §13.1 and S §1.2. The endpoint statement (b) follows from the lifting statement (a), applied to a homotopy F with $Z = [0,1]^2$ and $\tilde{F}(0,0) = y_0$. Each fibre of p is discrete. So if $\tilde{F}$ maps a connected set into one fibre, then $\tilde{F}$ maps that set to a single point.

The lift $\tilde{F}$ maps the edge $t = 0$ into the fibre over x, so $\tilde{F}$ maps this edge to y_0. By uniqueness of lifts in (a) with $Z = [0,1]$, the restrictions of $\tilde{F}$ to the edges $s = 0$ and $s = 1$ are the lifts of γ_0 and γ_1 from y_0.

The lift $\tilde{F}$ maps the edge $t = 1$ into the fibre $p^{-1}(x')$, so $\tilde{F}$ also maps this edge to a single point. Hence the two lifts have the same endpoint.

By the endpoint statement (b), a loop whose lift has distinct endpoints is not homotopic to a constant loop.

In the example of Figure 4.1, a path γ in $\mathrm{U}(1)$ starting at 1 lifts to a path $\tilde{\gamma}$ in $\mathbb{R}$ starting at 0. The endpoint $\tilde{\gamma}(1)$ is the *total angle* through which γ turns. The second panel shows a loop with total angle 2π.

Definition 4.4. A *universal covering group* of a connected Lie group G is a connected, simply connected Lie group $\tilde{G}$ with a smooth homomorphism $p\colon \tilde{G} \to G$ that is a covering map.

Path classes. Let G be connected. Let $\tilde{G}$ be the set of homotopy classes $[\gamma]$ of paths in G starting at e. Define

$$p[\gamma] = \gamma(1), \qquad [\gamma][\delta] = [\, t \mapsto \gamma(t)\delta(t)\,], \qquad [\gamma]^{-1} = [\, t \mapsto \gamma(t)^{-1}\,].$$

If F is a homotopy from γ to γ' and E one from δ to δ', then $(s,t) \mapsto F(s,t)E(s,t)$ is a homotopy from $\gamma\delta$ to $\gamma'\delta'$. So the product is well defined. The group axioms hold pointwise in t. The kernel of p consists of the classes of loops at e.

The pointwise product agrees with concatenation up to homotopy. Let c_x be the constant path at x. Replacing γ by $\gamma * c_{\gamma(1)}$ and δ by $c_e * \delta$ does not change their classes, since each is a reparametrisation. The pointwise product of the new paths is $\gamma * (\gamma(1)\delta)$, where $\gamma(1)\delta$ is the path $t \mapsto \gamma(1)\delta(t)$. For loops at e this is $\gamma * \delta$.

We compute this group for $G = \mathrm{U}(1)$. Two paths from 1 to the same point are homotopic exactly when they have the same total angle. If they are homotopic, the endpoint statement of the lifting theorem, Theorem 4.3(b), gives equal total angles. Conversely, if $\tilde{\gamma}_0(1) = \tilde{\gamma}_1(1)$, then

$$F(s,t) = e^{i((1-s)\tilde{\gamma}_0(t) + s\tilde{\gamma}_1(t))}$$

is a homotopy from γ_0 to γ_1, and both endpoints stay fixed. The pointwise product adds total angles, and every real number is the total angle of the path $t \mapsto e^{iat}$. So the path-class group of $\mathrm{U}(1)$ is $\mathbb{R}$ with addition. Fix θ with $e^{i\theta} = \gamma(1)$. The total angle of γ is $\theta + 2\pi k$ for an integer k. The class of γ records this integer, and the endpoint of γ does not.

It remains to give $\tilde{G}$ a topology. We sketch the construction. The details are cited after the theorem on smooth consequences, Theorem 4.5. Take a simply connected open neighbourhood U of $\gamma(1)$, for example a translate of a small exponential ball.

The basic open set $\mathcal{U}([\gamma], U)$ consists of the classes $[\gamma * \sigma]$, where σ is a path in U starting at $\gamma(1)$. The map p sends this set bijectively onto U, because paths in U with the same endpoints are homotopic in U.

The cited details show that p is a covering map, that the group operations are continuous and that $\tilde{G}$ is simply connected. Charts on G pulled back through p give the smooth structure on $\tilde{G}$.

An *isomorphism over G* from a universal covering group $p\colon \tilde{G} \to G$ to another one $p'\colon \tilde{G}' \to G$ is an isomorphism of Lie groups $\alpha\colon \tilde{G} \to \tilde{G}'$ with $p' \circ \alpha = p$.

Theorem 4.5 (Smooth consequences; cited)**.** *Let G be a connected Lie group.*

(a) The path-class construction gives a universal covering group $p\colon \tilde{G} \to G$, unique up to isomorphism over G. Its kernel is discrete and isomorphic to $\pi_1(G, e)$, and dp is an isomorphism.

> (b) *Let G' be a connected Lie group and $q\colon G' \to G$ a surjective smooth homomorphism with dq bijective. Then q is a covering map and $\ker q$ is discrete.*

We cite this from H §5.8 and from Knapp Beyond, Section I.11, *Covering groups*. Hall assumes that G is a matrix group, but $\tilde{G}$ need not be one.

> **Proposition 4.6.** *Let G be a connected topological group. Let N be a normal subgroup of G that is discrete in the subspace topology. Then N lies in the centre of G.*

Proof. Fix $n \in N$ and put $c(g) = gng^{-1}$. The map $c\colon G \to G$ is continuous, because multiplication and inversion are continuous. Normality gives $c(G) \subseteq N$. Since G is connected, $c(G)$ is connected. It contains $c(e) = n$.

Since N is discrete, every point of N is open in N. In particular, $\{n\}$ is open in N. Its complement $N \setminus \{n\}$ is a union of open singletons, so $\{n\}$ is also closed in N. Hence $\{n\}$ is open and closed in the subset $c(G)$ of N. Since $c(G)$ is connected and contains n, $c(G) = \{n\}$. In other words, $gng^{-1} = n$ for every $g \in G$. ∎

By the theorem on smooth consequences, Theorem 4.5(a), $\ker p$ is a discrete normal subgroup of the connected group $\tilde{G}$. By the proposition on discrete normal subgroups, Proposition 4.6, $\ker p$ is central. Hence $\pi_1(G, e)$, which is isomorphic to $\ker p$, is abelian. Exercise 4.1 shows that connectedness cannot be dropped.

4.2 Lie's integration theorem and Lie III

> **Theorem 4.7** (Lie's integration theorem; cited, with a sketch)**.** *Let G and H be Lie groups with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$. Assume that G is connected and simply connected. Let $\varphi\colon \mathfrak{g} \to \mathfrak{h}$ be a homomorphism of real Lie algebras. Then there is a unique smooth homomorphism $\Phi\colon G \to H$ with $d\Phi = \varphi$. It satisfies $\Phi(\exp X) = \exp \varphi(X)$.*

We cite this from H §5.7 for matrix groups. For general Lie groups we cite Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*. For representations take $H = \mathrm{GL}(V)$ with $\dim_{\mathbb{C}} V < \infty$.

Path transport. Near e the map we want is given by exp. To reach a far element g, we take a path in small steps and multiply the images of the steps in order. Two paths could give different answers, and simple connectivity excludes this. The existence proof has three steps and a conclusion. Only the conclusion uses simple connectivity.

Step 1: a local homomorphism. Choose a neighbourhood B of 0 on which exp is a diffeomorphism. Define $f(\exp X) = \exp \varphi(X)$ for $X \in B$. For small X, Y we have $\exp X \exp Y = \exp Z(X, Y)$. The Baker–Campbell–Hausdorff series is the series of iterated brackets

$$Z(X, Y) = X + Y + \frac{1}{2}[X, Y] + \cdots .$$

This is the series $H(X,Y)$ of Lecture 2. We write Z here because H is the target group. Since φ preserves brackets, it sends $Z(X,Y)$ to $Z(\varphi X, \varphi Y)$. Hence, for some neighbourhood W of e,

$$f(xy) = f(x)f(y) \qquad (x, y \in W). \tag{4.1}$$

Hall proves the series for matrix groups in H §§5.3–5.6. In general we cite Knapp Beyond, Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*.

Step 2: transport along a path. Let γ start at e. Choose $0 = t_0 < \cdots < t_n = 1$ with $\gamma(s)^{-1}\gamma(t) \in W$ whenever s, t lie in one interval $[t_{j-1}, t_j]$. Define

$$\Phi_\gamma = f\big(\gamma(t_0)^{-1}\gamma(t_1)\big) \cdots f\big(\gamma(t_{n-1})^{-1}\gamma(t_n)\big).$$

Inserting a point s into $[t_{j-1}, t_j]$ replaces one factor $f(ab)$ by $f(a)f(b)$, where $a = \gamma(t_{j-1})^{-1}\gamma(s)$ and $b = \gamma(s)^{-1}\gamma(t_j)$. By the local homomorphism property (4.1), applied to a and b, the product is unchanged. Any two such partitions have a common refinement. Hence Φ_γ depends only on γ, not on the partition.

Step 3: homotopy invariance. We sketch the topological step. Let F be a homotopy from γ to γ', both from e to g. Cut $[0,1]^2$ into squares so small that $F(q)^{-1}F(q') \in W$ for any two points q, q' of one square. Give an edge from q to q' the factor $f(F(q)^{-1}F(q'))$. The transport of an edge path is the product of its factors in order. Let one square have corners a, b, c, d in cyclic order, with images x_a, x_b, x_c, x_d. By the local homomorphism property (4.1),

$$f(x_a^{-1}x_b)\, f(x_b^{-1}x_c) = f(x_a^{-1}x_c) = f(x_a^{-1}x_d)\, f(x_d^{-1}x_c).$$

So the two routes $a \to b \to c$ and $a \to d \to c$ around one square have equal transport. Figure 4.2 shows such a move.

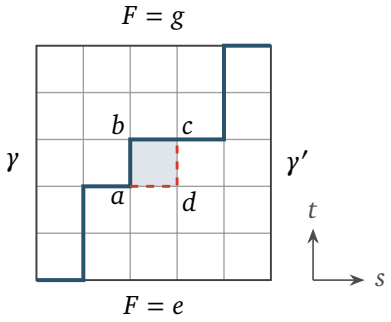


Figure 4.2: A homotopy F on a grid. The heavy edge path runs from $(0,0)$ to $(1,1)$. One move replaces its route $a \to b \to c$ by the dashed route $a \to d \to c$ across the shaded square.

Each edge path we use runs in the grid from $(0,0)$ to $(1,1)$. Start with the column $s = 0$ followed by the edge $t = 1$. End with the edge $t = 0$ followed by the column $s = 1$. Pass from the first to the last one square at a time. Each move swaps the routes $a \to b \to c$ and $a \to d \to c$ around one square, so it keeps the transport. The edges at $t = 0$ and $t = 1$ contribute $f(e) = e$, because F is constant there. By Step 2 the two columns give Φ_γ and $\Phi_{\gamma'}$. Hence $\Phi_\gamma = \Phi_{\gamma'}$. This step holds in every Lie group.

Conclusion. As G is simply connected, any two paths from e to g are homotopic. So $\Phi(g) = \Phi_\gamma$ does not depend on the path γ. The *increments* of a path δ are the elements $\delta(s)^{-1}\delta(t)$ that

enter the factors of Step 2. They do not change when δ is replaced by its left translate $g\delta$, given by $t \mapsto g\delta(t)$. So if δ runs from e to h, then

$$\Phi(gh) = \Phi_{\gamma * g\delta} = \Phi_\gamma \, \Phi_\delta = \Phi(g)\,\Phi(h).$$

Extending a path from e to g by a single short step gives $\Phi(g \exp X) = \Phi(g) \exp \varphi(X)$ for small X. Hence Φ is smooth and $d\Phi = \varphi$.

Without simple connectivity, Steps 1–3 still hold, but transport may depend on the homotopy class of the path. Self-check 2 gives an example.

Theorem 4.8 (Lie III; cited)**.** *Every finite-dimensional real Lie algebra is the Lie algebra of a connected, simply connected Lie group. This group is unique up to isomorphism.*

We cite existence from H §5.10. There Ado's theorem and H §5.9 give a connected Lie subgroup of a general linear group with the given Lie algebra. This subgroup need not be closed. The universal covering group of this subgroup is the required group. Uniqueness follows by integrating an isomorphism and its inverse with Lie's integration theorem, Theorem 4.7. The universal covering group of $\mathrm{SL}(2,\mathbb{R})$ is not isomorphic to any matrix Lie group (H §4.8). Hence Lie III needs abstract Lie groups.

4.3 Descent through a covering kernel

Theorem 4.9 (Descent)**.** *Let $p\colon \tilde{G} \to G$ be a surjective smooth homomorphism of Lie groups with dp bijective, and let $\Gamma = \ker p$. Let H be a topological group and $\tilde{\Phi}\colon \tilde{G} \to H$ a continuous homomorphism.*

(a) A homomorphism $\Phi\colon G \to H$ with $\Phi \circ p = \tilde{\Phi}$ exists if and only if $\Gamma \subseteq \ker \tilde{\Phi}$.

(b) Such Φ is unique and continuous. If H is a Lie group and $\tilde{\Phi}$ is smooth, then Φ is smooth and $d\Phi = d\tilde{\Phi} \circ (dp)^{-1}$.

Proof. Local sections. Since p is a homomorphism, left translations satisfy $p \circ L_x = L_{p(x)} \circ p$. Differentiating at e gives

$$dp_x \circ (dL_x)_e = (dL_{p(x)})_e \circ dp.$$

Both translations have invertible derivatives, and dp is bijective. Hence dp_x is bijective at every $x \in \tilde{G}$. By the inverse function theorem, each x has an open neighbourhood V_x that p maps diffeomorphically onto an open set. Let $s_x\colon p(V_x) \to V_x$ be the inverse of this restriction of p.

Necessity. If $\Phi \circ p = \tilde{\Phi}$ and $\gamma \in \Gamma$, then $\tilde{\Phi}(\gamma) = \Phi(e) = e$.

Sufficiency. Assume $\Gamma \subseteq \ker \tilde{\Phi}$. Let $g \in G$. Since p is surjective, we can write $g = p(x)$. Set $\Phi(g) = \tilde{\Phi}(x)$. If also $g = p(y)$, then $x^{-1}y \in \Gamma$ and

$$\tilde{\Phi}(y) = \tilde{\Phi}(x)\,\tilde{\Phi}(x^{-1}y) = \tilde{\Phi}(x).$$

Hence Φ is well defined and $\Phi \circ p = \tilde{\Phi}$. Moreover, Φ is a homomorphism, because

$$\Phi\big(p(x)p(y)\big) = \Phi\big(p(xy)\big) = \tilde{\Phi}(xy) = \tilde{\Phi}(x)\tilde{\Phi}(y).$$

Regularity. On $p(V_x)$ we have $\Phi = \Phi \circ p \circ s_x = \tilde{\Phi} \circ s_x$. This is continuous, and smooth when $\tilde{\Phi}$ is smooth. The open sets $p(V_x)$ cover G. Differentiating $\Phi \circ p = \tilde{\Phi}$ at e gives $d\Phi \circ dp = d\tilde{\Phi}$. Uniqueness holds because p is surjective. ∎

The proof uses only that H is a topological group. For example, the unitary group of a Hilbert space is a topological group for pointwise norm convergence. Lecture 19 uses this case.

Corollary 4.10. *Let G be a connected Lie group with universal covering group $p\colon \tilde{G} \to G$, and let $\Gamma = \ker p$. Let $\varphi\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation with $\dim_{\mathbb{C}} V < \infty$. Let $\tilde{\Pi}$ be the representation of $\tilde{G}$ with $d\tilde{\Pi} = \varphi \circ dp$. Then $\varphi = d\Pi$ for a representation Π of G if and only if $\tilde{\Pi}(\Gamma) = \{I\}$. In that case Π is unique and $\Pi \circ p = \tilde{\Pi}$.*

Proof. Since $\tilde{G}$ is simply connected, Lie's integration theorem, Theorem 4.7, gives $\tilde{\Pi}$, and $\tilde{\Pi}$ is unique. If Π exists, then $\Pi \circ p$ has differential $\varphi \circ dp$. Hence, by the uniqueness of $\tilde{\Pi}$, $\Pi \circ p = \tilde{\Pi}$, and $\tilde{\Pi}(\gamma) = \Pi(e) = I$ for $\gamma \in \Gamma$. Conversely, if $\tilde{\Pi}(\Gamma) = \{I\}$, then the Descent Theorem, Theorem 4.9, gives Π with $d\Pi = \varphi \circ dp \circ (dp)^{-1} = \varphi$. Uniqueness holds because G is connected. ∎

For disconnected groups the differential need not determine the homomorphism. For example, Exercise 4.6 gives two characters of $\mathrm{O}(2)$ with the same differential.

4.4 The line over the circle

Let the surjection $p\colon \mathbb{R} \to \mathrm{U}(1)$ be given by $p(t) = e^{it}$. Here $\operatorname{Lie} \mathrm{U}(1) = i\mathbb{R}$ and $\exp(i\theta) = e^{i\theta}$. Then

$$p(s+t) = p(s)\,p(t), \qquad dp(s) = \frac{d}{du}\Big|_{u=0} e^{ius} = is, \qquad \Gamma = \ker p = 2\pi\mathbb{Z}.$$

The line is simply connected, because $F(s,t) = (1-s)\gamma(t) + s\gamma(0)$ contracts any loop γ. By the theorem on smooth consequences, Theorem 4.5(b), p is a covering map, so it is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{U}(1), 1) \cong 2\pi\mathbb{Z} \cong \mathbb{Z}$.

Concretely, the loop $\omega(t) = e^{2\pi i t}$ lifts from 0 to $t \mapsto 2\pi t$, which ends at $2\pi \neq 0$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), ω is not homotopic to a constant loop.

A *scalar infinitesimal weight* on $\mathbb{R}$ is a number $\lambda \in \mathbb{C}$, viewed as the representation $\varphi_\lambda(s) = \lambda s$ on $\mathbb{C}$. It integrates to $\tilde{\chi}_\lambda(t) = e^{\lambda t}$.

The *period* of λ is $e^{2\pi\lambda} = \tilde{\chi}_\lambda(2\pi)$, the value at the generator 2π of Γ.

Say that λ *descends* if $\varphi_\lambda \circ (dp)^{-1}$ is the differential of a character of $\mathrm{U}(1)$. A *character* of $\mathrm{U}(1)$ here is a continuous homomorphism to the nonzero complex numbers. It is a one-dimensional representation, and its value is its trace. The descent criterion, Corollary 4.10, gives

$$\lambda \text{ descends} \iff e^{2\pi\lambda} = 1 \iff \lambda \in i\mathbb{Z}.$$

For $\lambda = in$ the descended character is $\chi(z) = z^n$, because $\chi(e^{it}) = e^{int}$. Its differential is $d\chi(i\theta) = in\theta$. In the course convention $d\chi = i\ell$ with $\ell(i\theta) = n\theta$. On $\Lambda = \ker\exp = 2\pi i\mathbb{Z}$ we get $\ell(2\pi ik) = 2\pi nk$, so $\ell(\Lambda) \subseteq 2\pi\mathbb{Z}$.

Success. For $\lambda = 3i$ the period is $e^{6\pi i} = 1$. The character $\chi(z) = z^3$ satisfies $\chi(p(t)) = e^{3it} = \tilde{\chi}_{3i}(t)$.

Failure. For $\lambda = i/2$ the period is $e^{i\pi} = -1$. Suppose a character χ had $d\chi(i\theta) = i\theta/2$. The corollary gives $\chi \circ p = \tilde{\chi}_{i/2}$, so

$$1 = \chi(1) = \chi\big(p(2\pi)\big) = \tilde{\chi}_{i/2}(2\pi) = -1.$$

This contradiction shows that $\lambda = i/2$ does not descend.

4.5 SU(2) over SO(3)

Let $\sigma_1, \sigma_2, \sigma_3$ be the *Pauli matrices*

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Put $X_j = -\frac{i}{2}\sigma_j$ for $j = 1, 2, 3$. Then $[X_1, X_2] = X_3$, with cyclic permutations.

Each element of SU(2) has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So SU(2) is homeomorphic to the unit sphere in $\mathbb{C}^2$, which is connected and simply connected (H §13.2).

The covering. The basis (X_1, X_2, X_3) is orthonormal for the form $\langle X, Y\rangle = -2\,\mathrm{tr}(XY)$. Each $\mathrm{Ad}(g)$, given by $\mathrm{Ad}(g)X = gXg^{-1}$, is a linear map of $\mathfrak{su}(2)$ to itself. It preserves this form, because the trace is invariant under conjugation. Hence the matrix of $\mathrm{Ad}(g)$ in the basis (X_1, X_2, X_3) lies in O(3). We identify $\mathrm{Ad}(g)$ with this matrix, so Ad maps SU(2) into O(3). Its image lies in SO(3), because $\det \circ \mathrm{Ad}$ is continuous on the connected group SU(2), takes values ± 1, and equals 1 at I.

The operator $\mathrm{ad}\, X_3$ of Lecture 2 sends $X_1 \mapsto X_2$, $X_2 \mapsto -X_1$ and $X_3 \mapsto 0$. Hence

$$\mathrm{Ad}(\exp\theta X_3) = e^{\theta\,\mathrm{ad}\,X_3} = R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The differential $d\,\mathrm{Ad} = \mathrm{ad}$ (Lecture 2) is injective, since $[X_1, X] = [X_2, X] = 0$ forces $X = 0$. By dimension it is bijective onto $\mathfrak{so}(3)$.

Lecture 3 let SU(2) act on the traceless Hermitian matrices V by $\xi \mapsto U\xi U^*$. This action is Ad. Indeed, $\xi(x, y, z) = x\sigma_1 + y\sigma_2 + z\sigma_3$, so $-\frac{i}{2}\xi(x, y, z) = xX_1 + yX_2 + zX_3$. This linear isomorphism from V to $\mathfrak{su}(2)$ commutes with conjugation by U. The basis of Lecture 1 is

$A = -2X_3$, $B = -2X_2$ and $C = -2X_1$. The factor $-\frac{1}{2}$ gives $[X_1, X_2] = X_3$ with no factor two. Hence $k_\theta = \exp(\theta A) = \exp(-2\theta X_3)$, and

$$\operatorname{Ad}(k_\theta) = R_3(-2\theta).$$

This is the rotation by -2θ found in Lecture 3. The matrix g above is the U of Lecture 1, with $a = \alpha$ and $b = -\bar{\beta}$. Exercise 1.3 chose a different isomorphism onto $\mathfrak{so}(3)$. It differs from this one by conjugation with an element of $\mathrm{SO}(3)$.

If $\operatorname{Ad}(g) = I$, then g commutes with σ_3 and σ_1. The first condition makes g diagonal. The second makes it scalar. So $g = zI$ with $z^2 = \det g = 1$, and $\ker \operatorname{Ad} = \{\pm I\}$.

The image of Ad contains exp of a neighbourhood of 0 in $\mathfrak{so}(3)$, because $d\operatorname{Ad}$ is bijective. Hence the image of Ad is an open subgroup of $\mathrm{SO}(3)$. It is also closed, because its complement is a union of open cosets.

Every rotation has the form $kR_3(\theta)k^{-1}$ with $k \in \mathrm{SO}(3)$. Indeed, let $g \in \mathrm{SO}(3)$. Then $\det(g - I) = \det g \det(I - g^{\mathsf{T}}) = \det(I - g) = -\det(g - I)$, so $\det(g - I) = 0$ and g fixes a unit vector u. By transitivity on S^2 (Lecture 3), there is $k \in \mathrm{SO}(3)$ with $ke_3 = u$. Then $k^{-1}gk$ fixes e_3, so it is some $R_3(\theta)$ by the stabiliser computation of Lecture 3. The path $t \mapsto kR_3(t\theta)k^{-1}$ joins this rotation to I. Hence $\mathrm{SO}(3)$ is connected. The image of Ad is open, closed and nonempty, so Ad is onto.

Let $\Gamma = \ker \operatorname{Ad}$. By the theorem on smooth consequences, Theorem 4.5(b), Ad is a covering map. Since $\mathrm{SU}(2)$ is connected and simply connected, Ad is a universal covering group. By the uniqueness and kernel statements of Theorem 4.5(a), $\pi_1(\mathrm{SO}(3), I) \cong \Gamma = \{\pm I\}$.

The loop $t \mapsto R_3(2\pi t)$ lifts to $t \mapsto \exp(2\pi t X_3) = \operatorname{diag}(e^{-i\pi t}, e^{i\pi t})$, which ends at $-I$. By the endpoint statement of the lifting theorem, Theorem 4.3(b), this loop is not homotopic to a constant loop.

Figure 4.3 shows this loop. The axis-angle form draws $\mathrm{SO}(3)$ as a ball. The point θn, with n a unit vector and $0 \le \theta \le \pi$, stands for the rotation by θ about n. Opposite points of the boundary are one rotation, since a half turn about n equals a half turn about $-n$.

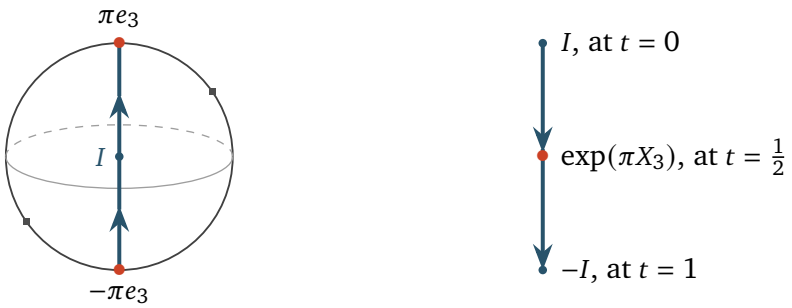


Figure 4.3: On the left, $\mathrm{SO}(3)$ as the ball of radius π. The coloured points are one rotation, and so are the square points. The heavy path is the loop $R_3(2\pi t)$: it rises from the centre to πe_3, reappears at $-\pi e_3$ and returns to the centre. On the right, the lift $t \mapsto \exp(2\pi t X_3)$ of this loop to $\mathrm{SU}(2)$ runs from I to $-I$, so it does not close.

Representations. Let V_m be the irreducible $\mathfrak{sl}(2,\mathbb{C})$-module of dimension $m+1$ and spin $m/2$. It has a basis $v_0,\dots,v_m$ with $\sigma_3 v_k = (m-2k)v_k$. Restriction to $\mathfrak{su}(2)$ gives φ_m, which integrates to Π_m on SU(2). Since $X_3 = -\frac{i}{2}\sigma_3$,

$$\Pi_m(\exp tX_3)v_k = e^{\lambda_k t}v_k, \qquad \lambda_k = -\tfrac{i}{2}(m-2k), \qquad \Pi_m(-I)v_k = e^{2\pi\lambda_k}v_k = (-1)^m v_k.$$

By the descent criterion, Corollary 4.10, $\varphi_m \circ (d\,\mathrm{Ad})^{-1}$ is the differential of a representation of SO(3) exactly when m is even. The homomorphism $t \mapsto R_3(t) = \mathrm{Ad}(\exp tX_3)$ from $\mathbb{R}$ has kernel $2\pi\mathbb{Z}$, and $\exp(2\pi X_3) = -I$. So the test is the scalar test of § 4.4 for each weight λ_k: its period $e^{2\pi\lambda_k}$ must equal 1.

Success: $m = 2$. Here $\Pi_2(-I) = I$. Hence there is a representation Π of SO(3) with $\Pi_2 = \Pi\circ\mathrm{Ad}$. This Π is equivalent to the defining representation. To see this, note that the matrices $d\,\mathrm{Ad}(X)$ also describe $\mathrm{ad}\,X$ on $\mathfrak{sl}(2,\mathbb{C})$ in the complex basis (X_1, X_2, X_3). This adjoint module has σ_3-weights $2, 0, -2$. Hence there is an invertible $E\colon \mathbb{C}^3 \to V_2$ with

$$E\,d\,\mathrm{Ad}(X)E^{-1} = \varphi_2(X).$$

The representation $g \mapsto E\,\mathrm{Ad}(g)E^{-1}$ of SU(2) has differential φ_2. By uniqueness on the connected group SU(2), it equals Π_2. Since Ad is onto, $\Pi(R) = ERE^{-1}$.

Failure: $m = 1$. Here $V_1 = \mathbb{C}^2$ and $\Pi_1(g) = g$. Suppose a representation Π of SO(3) had $d\Pi = \varphi_1 \circ (d\,\mathrm{Ad})^{-1}$. The corollary gives $\Pi \circ \mathrm{Ad} = \Pi_1$, so

$$I = \Pi\big(R_3(2\pi)\big) = \Pi\big(\mathrm{Ad}(-I)\big) = \Pi_1(-I) = -I.$$

Hence no representation of SO(3) has differential $\varphi_1 \circ (d\,\mathrm{Ad})^{-1}$.

For a spin-$\frac{1}{2}$ particle, $\mathbb{C}^2$ describes the internal spin degree of freedom. Rotations act on this space only up to sign. Lecture 13 treats such projective representations.

Self-checks

Check 1. Does the character $t \mapsto e^{i\sqrt{2}t}$ of $\mathbb{R}$ descend through $p(t) = e^{it}$ to U(1)?

Answer. No. Its period is $e^{2\pi i\sqrt{2}} \neq 1$, because $\sqrt{2} \notin \mathbb{Z}$.

Check 2. Take $G = \mathrm{U}(1)$ and $\varphi(i\theta) = i\theta/2$, so $f(e^{i\theta}) = e^{i\theta/2}$ for $|\theta| < \pi$. Use $W = \{e^{i\theta} : |\theta| < \pi/2\}$. Compute the transport around $\omega(t) = e^{2\pi i t}$. Which part of the path-transport argument fails?

Answer. The partition $t_j = j/5$ is admissible, since $2\pi/5 < \pi/2$. So $\Phi_\omega = f(e^{2\pi i/5})^5 = (e^{\pi i/5})^5 = -1$. The constant loop gives 1. Steps 1–3 remain valid. The Conclusion fails, because ω is not homotopic to the constant loop, by the lift computed in § 4.4. So $\Phi(1)$ is not well defined.

Check 3. Is φ_3 the differential of a representation of SU(2)? Is $\varphi_3 \circ (d\,\mathrm{Ad})^{-1}$ the differential of a representation of SO(3)?

Answer. Yes for SU(2), by Lie's integration theorem, Theorem 4.7. No for SO(3), since $\Pi_3(-I) = -I$. Equivalently, the weight $\lambda_0 = -\frac{3i}{2}$ has period $e^{-3\pi i} = -1$.

Summary

A connected Lie group G has a universal covering group $p\colon \tilde{G} \to G$. Its kernel $\Gamma \cong \pi_1(G, e)$ is discrete and central. Let φ be a representation of the Lie algebra of G on a finite-dimensional complex vector space. Then $\varphi \circ dp$ integrates to a representation $\tilde{\Pi}$ of $\tilde{G}$. The representation φ is the differential of a representation of G exactly when Γ acts trivially through $\tilde{\Pi}$. For $\mathbb{R} \to \mathrm{U}(1)$ this is the period condition $\lambda \in i\mathbb{Z}$. For $\mathrm{SU}(2) \to \mathrm{SO}(3)$ the condition holds exactly for even m, that is, for integer spin.

Reading

H is Hall, *Lie Groups, Lie Algebras, and Representations*, second edition, 2015. S is Sepanski, *Compact Lie Groups*, 2007. Knapp Beyond is Knapp, *Lie Groups Beyond an Introduction*, digital second edition, 2023.

- H §§4.7–4.8 and 5.7–5.10: group versus Lie algebra homomorphisms, universal covers, subgroups and subalgebras, Lie's third theorem, and a nonmatrix Lie group.
- H §§13.1–13.2: the fundamental group and the fundamental groups of the compact classical groups.
- S §§1.2, 4.2: topological background and the related material for compact groups.
- Knapp Beyond, Sections I.10–11, *Elementary theory of Lie groups* and *Covering groups*; Appendix B.4, Theorem B.22, *Campbell–Baker–Hausdorff formula*: general versions of the cited results.

Lecture 5

Haar Measure, Averaging and Orthogonality

Objectives. *For a finite group G, averaging with $|G|^{-1}\sum_{g\in G}$ proves unitarisability, complete reducibility and orthogonality of matrix coefficients. In this lecture we build an integral on a compact Lie group that replaces this average. It is linear and positive, has total mass 1, does not change when the integrand is shifted on either side, and is strictly positive on nonzero nonnegative functions. Sections 5.1 and 5.2 construct it. Sections 5.3 and 5.4 use no other property of it, and they record which property each argument uses.*

By the end of this lecture you will be able to:

- Integrate continuous functions on a compact manifold against a density.
- Construct the normalised Haar integral of a compact Lie group, and prove that it is right invariant and strictly positive.
- Average inner products and operators to prove unitarisability, complete reducibility and Schur orthogonality for finite-dimensional representations of compact Lie groups.
- Compute Haar integrals, an invariant-vector projector and a coefficient norm on $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Lie groups are real, finite-dimensional, Hausdorff and second countable. Representations are complex, finite-dimensional and continuous. Inner products are antilinear in the first slot.

5.1 Densities and integration on manifolds

Let M be a smooth n-manifold with $n \geq 1$. Charts and tangent spaces are as in Lecture 1. In a chart, the Euclidean integral $\int f\,du$ changes by a Jacobian factor when the chart changes. A density supplies a factor that cancels this change. We cite the routine analytic parts of this construction.

Definition 5.1. Let V be a real vector space of dimension n. A *density* on V is a function $\delta\colon V^n \to \mathbb{R}$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \quad \text{for all linear } T\colon V\to V \text{ and all } v_j \in V.$$

It is *positive* if it is positive on every basis.

A positive density assigns a volume to the box with edges $v_1, \dots, v_n$. The factor $|\det T|$ says how a linear map rescales volume.

Fix a basis $b = (b_j)$ and write $v_j = Tb_j$. Then $\delta(v_1, \dots, v_n) = |\det T|\,\delta(b_1, \dots, b_n)$, so δ is determined by one value. Conversely, let $[v]_b$ be the matrix whose jth column holds the coordinates of v_j. Then

$$\delta_b(v_1, \dots, v_n) = |\det [v]_b|$$

is a density, since $[Tv]_b = [T]_b[v]_b$. Hence the densities on V form the real line $\mathbb{R}\,\delta_b$, and $c\,\delta_b$ is positive exactly when $c > 0$.

In a chart x, let $\partial_{x_j}|_p \in T_pM$ be the velocity of the coordinate curve $t \mapsto x^{-1}(x(p) + te_j)$. These vectors form a basis of T_pM. A tangent vector with coordinate velocity (c_j) is $\sum_j c_j \partial_{x_j}|_p$.

Definition 5.2. A *continuous density* δ on M is a density δ_p on each T_pM such that, for every chart $x\colon U \to \Omega \subset \mathbb{R}^n$, the function ρ_x given by

$$\rho_x(x(p)) = \delta_p\big(\partial_{x_1}|_p, \dots, \partial_{x_n}|_p\big), \qquad p \in U,$$

is continuous on Ω.

Let $y\colon U' \to \Omega'$ be another chart, and put $\phi = y \circ x^{-1}$ on $x(U \cap U')$. The chain rule gives $\partial_{x_j} = \sum_i (\partial y_i/\partial x_j)\,\partial_{y_i}$. The defining property of a density, applied to this change of basis, gives

$$\rho_x(u) = |\det D\phi(u)|\,\rho_y(\phi(u)), \qquad u \in x(U \cap U'). \tag{5.1}$$

Let $f\colon M \to \mathbb{C}$ be continuous and vanish outside a compact subset of U. Define

$$\int_M f\,\delta = \int_\Omega f(x^{-1}(u))\,\rho_x(u)\,du. \tag{5.2}$$

If f vanishes outside a compact subset of $U \cap U'$, then the change-of-variables theorem and the transformation rule (5.1) show that the chart y gives the same value. For example, take spherical angles on S^2, with $p = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, and the density $\delta_p(v_1, v_2) = |\det[\,p\ v_1\ v_2\,]|$. Then the coefficient ρ_x is $\sin\theta$.

Now let M be compact. A finite *partition of unity* is a finite family of continuous functions $\chi_k \geq 0$ with $\sum_k \chi_k = 1$, each vanishing outside a compact subset of a chart domain U_k. We cite the existence of a finite partition of unity on M. For continuous f put

$$\int_M f\,\delta = \sum_k \int_M \chi_k f\,\delta.$$

Let (ψ_l) be a second partition of unity, with chart domains U_l'. Each $\chi_k\psi_l f$ vanishes outside a compact subset of $U_k \cap U_l'$, so its integral is the same in the charts on U_k and on U_l'. Since each partition sums to 1, linearity of the chart integral (5.2) shows that both sums equal $\sum_{k,l} \int_M \chi_k\psi_l f\,\delta$. Hence the integral is well defined. Moreover, it is linear, and it is nonnegative when $f \geq 0$ and δ is positive.

For a diffeomorphism $F\colon M \to M$, the *pullback* density is

$$(F^*\delta)_p(v_1, \dots, v_n) = \delta_{F(p)}(dF_p v_1, \dots, dF_p v_n).$$

The chain rule gives $(F_1 \circ F_2)^* = F_2^* F_1^*$.

Lemma 5.3. *$F^*\delta$ is a continuous density, positive if δ is. For $f \in C(M)$,*

$$\int_M (f \circ F)\, F^*\delta = \int_M f\, \delta.$$

Proof. Let x be a chart on U. Then $x \circ F$ is a chart on $F^{-1}(U)$, and dF sends its coordinate vectors to those of x. So the coefficient of $F^*\delta$ in the chart $x \circ F$ is the continuous function ρ_x. The transformation rule (5.1) then gives continuity in every chart. If f vanishes outside a compact subset of U, both integrals are the same Euclidean integral (5.2). For general f, apply this to each $\chi_k f$ and use the partition of unity $(\chi_k \circ F)$. ∎

In examples we integrate in one chart. Spherical angles on S^2 miss a half great circle, and we ignore it because a curve has zero area. The next lemma states this rule in general. Its proof is cited and uses Lebesgue theory.

One chart often covers M apart from a set contained in finitely many submanifolds of lower dimension. A set $Z \subset \mathbb{R}^n$ is *Lebesgue-null* if, for every $\varepsilon > 0$, countably many boxes of total volume less than ε cover Z. Changing a function on a Lebesgue-null set does not change its Lebesgue integral. We cite the following consequence of Lebesgue theory from Knapp Beyond, Section VIII.1. The integrability statement in the lemma, and the term *measurable* below, belong to that cited theory.

Lemma 5.4. *Let M be compact with a positive continuous density δ. Let $x : U \to \Omega$ be a chart such that $M \setminus U$ lies in a finite union of smooth submanifolds of dimension less than n. Then for $f \in C(M)$ the function $(f \circ x^{-1})\rho_x$ is Lebesgue integrable on Ω, and*

$$\int_M f\, \delta = \int_\Omega f(x^{-1}(u))\, \rho_x(u)\, du.$$

The lemma rests on two cited facts. A submanifold of dimension less than n has Lebesgue-null image in every chart. The change-of-variables formula holds for every nonnegative measurable function. For $f \geq 0$ one applies both facts to each $\chi_k f$, using the transformation rule (5.1), and sums over k. A general $f \in C(M)$ is a linear combination of four nonnegative continuous functions, namely the positive and negative parts of the real and imaginary parts of f.

5.2 The Haar integral

Let G be a compact Lie group with Lie algebra $\mathfrak{g} = T_eG$. If $\dim G = 0$, then G is discrete and compact, hence finite. Until the definition of the Haar integral, Definition 5.6, we assume $n = \dim G \geq 1$. Write $L_h(g) = hg$ and $R_h(g) = gh$. We choose a volume at e and carry it to every point by left translation, as Figure 5.1 shows. So fix a positive density δ_e on $\mathfrak{g}$ and define

$$\delta_g(v_1, \ldots, v_n) = \delta_e(dL_{g^{-1}}v_1, \ldots, dL_{g^{-1}}v_n), \qquad v_j \in T_gG. \tag{5.3}$$

Lemma 5.5. *δ is a positive continuous density, and $L_h^*\delta = \delta$ for all $h \in G$. Every continuous density δ' with $L_h^*\delta' = \delta'$ for all h is a constant multiple of δ.*

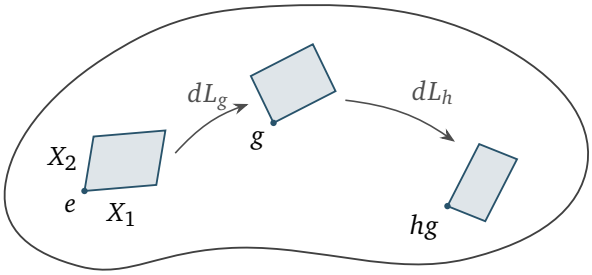


Figure 5.1: The box at e has edges X_1 and X_2. The boxes at g and hg are its images under dL_g and $dL_{hg} = dL_h \circ dL_g$. The density δ takes the same value on all three, although their shapes and areas in a chart may differ.

Proof. Each $dL_{g^{-1}} : T_gG \to \mathfrak{g}$ is a linear isomorphism, so δ_g is positive. Fix a basis (X_j) of $\mathfrak{g}$. The fields $X_j^L(g) = dL_g X_j$ are smooth and form a basis of each tangent space (Lecture 1). In a chart write $\partial_{x_i} = \sum_j a_{ji} X_j^L$. The matrix (a_{ji}) is the inverse of the coefficient matrix of the fields X_j^L in this chart. That coefficient matrix is continuous, hence so is (a_{ji}). Since $dL_{g^{-1}} X_j^L(g) = X_j$, we get $\rho_x = |\det(a_{ji})|\, \delta_e(X_1, \dots, X_n)$, which is continuous. With $v = (v_1, \dots, v_n)$,

$$(L_h^* \delta)_g(v) = \delta_{hg}(dL_h v) = \delta_e(dL_{(hg)^{-1}} dL_h v) = \delta_e(dL_{g^{-1}} v) = \delta_g(v).$$

Now let δ' be left invariant. Taking $h = g^{-1}$ gives $\delta'_g(v) = \delta'_e(dL_{g^{-1}} v)$. The densities on $\mathfrak{g}$ form a real line, so $\delta'_e = c\, \delta_e$ for a real constant c. Hence $\delta' = c\, \delta$. ∎

Definition 5.6. The *normalised Haar integral* of a compact Lie group G is the functional on $C(G)$ given by

$$\int_G f(g)\, dg = \begin{cases} \dfrac{\int_G f\, \delta}{\int_G 1\, \delta}, & \dim G \geq 1, \\[2ex] \dfrac{1}{|G|} \sum_{g \in G} f(g), & G \text{ finite}. \end{cases}$$

The denominator is positive because every ρ_x is positive. Rescaling δ_e does not change the quotient.

Theorem 5.7. *Let G be a compact Lie group, $f \in C(G)$ and $h \in G$.*

(a) The integral is linear, $\int_G f(g)\, dg \geq 0$ if $f \geq 0$, and $\int_G 1\, dg = 1$.

(b) Left invariance: $\int_G f(hg)\, dg = \int_G f(g)\, dg$.

(c) Right invariance: $\int_G f(gh)\, dg = \int_G f(g)\, dg$.

(d) Strict positivity: if $f \geq 0$ and $f(g_0) > 0$ for some g_0, then $\int_G f(g)\, dg > 0$.

Proof. For finite G all parts are clear, since translations permute G. Let $n \geq 1$. Linearity, positivity and normalisation are immediate. Left invariance follows from the pullback lemma, Lemma 5.3, with $F = L_h$, because $L_h^* \delta = \delta$ by the left-invariant density lemma, Lemma 5.5.

(c) By the pullback lemma, Lemma 5.3, $R_h^* \delta$ is a continuous density. Left and right translations commute, so

$$L_k^* R_h^* \delta = (R_h \circ L_k)^* \delta = (L_k \circ R_h)^* \delta = R_h^* L_k^* \delta = R_h^* \delta.$$

By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $R_h^*\delta = c(h)\,\delta$ for a real constant $c(h)$. We compute it at e. The map $L_{h^{-1}} \circ R_h$ is $g \mapsto h^{-1}gh$, whose differential at e is $\mathrm{Ad}(h^{-1})$ (Lecture 2). So for $X = (X_1, \dots, X_n)$ in $\mathfrak{g}$,

$$(R_h^*\delta)_e(X) = \delta_h(dR_h X) = \delta_e(dL_{h^{-1}} dR_h X) = \delta_e(\mathrm{Ad}(h^{-1})X) = j(h)^{-1}\delta_e(X),$$

where $j(h) = |\det \mathrm{Ad}(h)|$. Thus $R_h^*\delta = j(h)^{-1}\delta$. Since Ad is a continuous homomorphism, so is $j\colon G \to (0,\infty)$. Its image is a compact subgroup. If it contained some $t \neq 1$, the powers t^m, $m \in \mathbb{Z}$, would be unbounded. Hence $j = 1$ and $R_h^*\delta = \delta$. Now apply the pullback lemma, Lemma 5.3, with $F = R_h$.

(d) By continuity of f, there exist $c > 0$ and an open set $U \ni g_0$ with $f > c$ on U. The translates aU with $a \in G$ cover G, so compactness (Lecture 3) gives $G = a_1U \cup \cdots \cup a_NU$. Every $g \in G$ has some $a_i^{-1}g \in U$. Since $f > c$ on U and $f \geq 0$ everywhere, we get

$$\Phi(g) = \sum_{i=1}^{N} f(a_i^{-1}g) \geq c \qquad \text{for all } g \in G.$$

By linearity and left invariance, $N\int_G f(g)\,dg = \int_G \Phi(g)\,dg$. By positivity and normalisation, $\int_G \Phi(g)\,dg \geq c > 0$. Hence $\int_G f(g)\,dg > 0$. ∎

We cite uniqueness of the normalised Haar integral from BD I.5. Let $J\colon C(G) \to \mathbb{C}$ be linear with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all $f \in C(G)$ and $h \in G$. Then J is the normalised Haar integral. The left-invariant density lemma, Lemma 5.5, says that the left-invariant density is unique up to a constant factor. This theorem is stronger: it concerns any positive invariant functional on $C(G)$, which is not assumed to come from a density.

We also cite the Riesz representation theorem for positive linear functionals on $C(X)$, with X compact Hausdorff. It shows that the Haar integral is integration against a unique regular Borel probability measure. This is the *normalised Haar measure*. We integrate only continuous functions here, and this lecture does not use this measure. Lecture 9 uses Haar measure for null sets.

5.3 Averaging inner products and operators

Let W be a finite-dimensional complex vector space, and let $F\colon G \to W$ be continuous. We integrate F coordinatewise in a basis of W. The result does not depend on the basis. It satisfies $A\int_G F\,dg = \int_G AF\,dg$ for linear A. It also inherits left and right invariance from the Haar integral, Theorem 5.7.

> **Theorem 5.8** (Unitarisability). *Every representation (Π, V) of a compact Lie group G admits an inner product for which all $\Pi(g)$ are unitary.*

Proof. Choose any inner product $\langle\cdot,\cdot\rangle_0$ on V and set

$$\langle v, w\rangle = \int_G \langle \Pi(g)v, \Pi(g)w\rangle_0\,dg.$$

The integrand is continuous because Π is continuous. By linearity of the integral, the form is antilinear in v and linear in w. It is Hermitian, since the integral commutes with complex

conjugation. For $v \neq 0$, the function $g \mapsto \|\Pi(g)v\|_0^2$ is nonnegative and positive at e. Strict positivity gives $\langle v, v\rangle > 0$. Right invariance, applied to $g \mapsto \langle \Pi(g)v, \Pi(g)w\rangle_0$, gives

$$\langle \Pi(h)v, \Pi(h)w\rangle = \int_G \langle \Pi(gh)v, \Pi(gh)w\rangle_0 \, dg = \langle v, w\rangle. \qquad \blacksquare$$

Compactness cannot be dropped. The boost representation $u \mapsto B(u)$ of $\mathbb{R}$ from Lecture 2 has eigenvalues e^u and e^{-u}, so no inner product makes it unitary. The translation-invariant integral on $\mathbb{R}$ is $\int dt$, and it gives the constant 1 infinite value. The average in the proof is not available.

Corollary 5.9 (Complete reducibility)**.** *Every representation of a compact Lie group is a direct sum of irreducible subrepresentations.*

Proof. Let (Π, V) be a representation, and induct on $\dim V$. If V is irreducible, there is nothing to prove. Otherwise let U be a proper nonzero invariant subspace. Use the invariant inner product of unitarisability, Theorem 5.8. For $u \in U$ and $w \in U^\perp$,

$$\langle u, \Pi(g)w\rangle = \langle \Pi(g)^* u, w\rangle = \langle \Pi(g^{-1})u, w\rangle = 0.$$

So $U^\perp$ is invariant and $V = U \oplus U^\perp$. Apply induction to both summands. $\blacksquare$

The *invariant vectors* of a representation (Π, V) form the subspace

$$V^G = \{v \in V : \Pi(g)v = v \text{ for all } g \in G\}.$$

Proposition 5.10 (Invariant-vector projector)**.** *Let (Π, V) be a representation of a compact Lie group G and $P = \int_G \Pi(g)\, dg$. Then $\Pi(h)P = P = P\Pi(h)$ for all h, and P is a projection with image V^G. If all $\Pi(g)$ are unitary, P is the orthogonal projection onto V^G.*

Proof. Left invariance gives $\Pi(h)P = \int_G \Pi(hg)\, dg = P$. Right invariance gives $P\Pi(h) = P$. The first identity shows $\operatorname{im} P \subset V^G$. For $v \in V^G$ we have $Pv = \int_G v\, dg = v$. Hence $P^2 = P$ and $\operatorname{im} P = V^G$. Now let Π be unitary. As in the proof of complete reducibility, Corollary 5.9, the subspace $U = (V^G)^\perp$ is invariant. Since $\Pi(g)U \subset U$ for all g, also $PU \subset U \cap V^G = 0$. So $U \subset \ker P$. By rank–nullity, $\dim \ker P = \dim V - \dim V^G = \dim U$. Hence $\ker P = (V^G)^\perp$. $\blacksquare$

Let (σ, W) be a second representation. Then $\operatorname{Hom}(V, W)$ carries the representation in which g acts by $E \mapsto \sigma(g)E\Pi(g)^{-1}$. The invariant vectors of this representation are the intertwiners. By the invariant-vector projector, Proposition 5.10, applied to this representation, for every linear $E\colon V \to W$,

$$E^\natural = \int_G \sigma(g)E\Pi(g)^{-1}\, dg \in \operatorname{Hom}_G(V, W). \tag{5.4}$$

5.4 Orthogonality of matrix coefficients

Definition 5.11. Let Π act by unitary operators on a finite-dimensional inner product space V. The *matrix coefficient* of $v, w \in V$ is the function $c^{\Pi}_{v,w}(g) = \langle v, \Pi(g)w\rangle$ on G.

It is antilinear in v and linear in w. Let $D(g)$ be the matrix of $\Pi(g)$ in an orthonormal basis (e_i), so $\Pi(g)e_j = \sum_i D_{ij}(g)e_i$. Then $D_{ij} = c^{\Pi}_{e_i,e_j}$. On $C(G)$ we use

$$\langle f_1, f_2\rangle_{L^2} = \int_G \overline{f_1(g)}\, f_2(g)\, dg.$$

This inner product is also antilinear in the first slot. It is positive definite by strict positivity.

We import Schur's lemma from finite group theory. Its proof is algebraic and holds for any group. For irreducible finite-dimensional complex representations, $\operatorname{Hom}_G(V,W) = 0$ if $V \not\cong W$, and $\operatorname{Hom}_G(V,V) = \mathbb{C}\,\mathrm{id}$.

Theorem 5.12 (Schur orthogonality). *Let (Π, V) and (σ, W) be irreducible unitary representations of a compact Lie group G, with $d = \dim V$.*

(a) If $\Pi \not\cong \sigma$, then $\langle c^{\Pi}_{v,w}, c^{\sigma}_{v',w'}\rangle_{L^2} = 0$ for $v, w \in V$ and $v', w' \in W$.

(b) For $v, w, v', w' \in V$,

$$\int_G \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \Pi(g)w'\rangle\, dg = \frac{1}{d}\, \langle v', v\rangle\, \langle w, w'\rangle.$$

Proof. The key step writes the integrand as a coefficient of an averaged operator. In case (b) put $\sigma = \Pi$ and $W = V$. Define $E\colon V \to W$ by $Eu = \langle w, u\rangle\, w'$. Here $w \in V$ and $w' \in W$, so $E \in \operatorname{Hom}(V,W)$. Since $\Pi(g)^{-1} = \Pi(g)^{*}$,

$$\langle v', \sigma(g) E \Pi(g)^{-1} v\rangle = \langle w, \Pi(g)^{*} v\rangle \langle v', \sigma(g) w'\rangle = \overline{\langle v, \Pi(g)w\rangle}\, \langle v', \sigma(g)w'\rangle.$$

The map $u \mapsto \langle v', u\rangle$ is linear, so it commutes with the integral. Integrating over G therefore gives

$$\int_G \langle v', \sigma(g) E \Pi(g)^{-1} v\rangle\, dg = \langle v', E^{\natural} v\rangle,$$

with the averaged operator $E^{\natural}$ from (5.4). So the left sides in both parts equal $\langle v', E^{\natural} v\rangle$. In case (a), where $\Pi \not\cong \sigma$, Schur's lemma gives $E^{\natural} = 0$. In case (b), where $\sigma = \Pi$, Schur's lemma gives $E^{\natural} = \lambda\,\mathrm{id}$ for some $\lambda \in \mathbb{C}$. Trace is linear and conjugation invariant, so in an orthonormal basis (e_i)

$$d\lambda = \int_G \operatorname{tr}\!\big(\Pi(g) E \Pi(g)^{-1}\big)\, dg = \operatorname{tr} E = \sum_i \langle w, e_i\rangle \langle e_i, w'\rangle = \langle w, w'\rangle.$$

Hence $\langle v', E^{\natural} v\rangle = d^{-1} \langle w, w'\rangle \langle v', v\rangle$. ∎

Corollary 5.13. *Let (Π, V) and (σ, W) be as in Schur orthogonality, Theorem 5.12, with $d = \dim V$. Let D^{Π} and D^{σ} be their matrices in orthonormal bases of V and W. Then*

$$\int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\Pi}_{kl}(g)\, dg = \frac{\delta_{ik}\delta_{jl}}{d}, \qquad \int_G \overline{D^{\Pi}_{ij}(g)}\, D^{\sigma}_{kl}(g)\, dg = 0 \quad (\Pi \not\cong \sigma).$$

In particular $\|c^{\Pi}_{v,w}\|^2_{L^2} = \|v\|^2 \|w\|^2 / d$ for $v, w \in V$.

Proof. Take $v = e_i$, $w = e_j$, $v' = e_k$ and $w' = e_l$ in Schur orthogonality, Theorem 5.12. For the norm identity, take $v' = v$ and $w' = w$ in the relation for a single representation. ∎

In physics, for $G = \mathrm{SU}(2)$ and the module V_m, the entries in an orthonormal weight basis are the Wigner D-functions of spin $m/2$, with $d = m + 1$.

Put $j = i$ and $l = k$, and sum over i and k. The character $\chi_\Pi = \operatorname{tr}\Pi$ is the sum of the diagonal entries D^{Π}_{ii}. Hence the characters of inequivalent irreducible representations are orthonormal. For irreducible unitary Π and σ,

$$\int_G \overline{\chi_\Pi(g)}\,\chi_\sigma(g)\,dg = \begin{cases} 1, & \Pi \cong \sigma, \\ 0, & \Pi \not\cong \sigma. \end{cases}$$

The table records which properties of the integral each argument uses. There, “yes” means that the proof uses the property. Complete reducibility has the same column as unitarisability, since its proof uses only the invariant inner product.

Property	Unitarisability	Projector	Schur orthogonality
Linearity	yes	yes	yes
$\int_G 1\,dg = 1$	only via strict positivity	yes ($Pv = v$)	yes (trace step)
Left invariance	only via strict positivity	yes ($\Pi(h)P = P$)	yes ($E^\natural$ intertwines)
Right invariance	yes	yes ($P\Pi(h) = P$)	no
Strict positivity	yes	no	no

5.5 Examples: the groups U(1) and SU(2)

In each example the single-chart lemma, Lemma 5.4, lets us integrate over one chart.

Example 5.14 (The circle)**.** For $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ use the parametrisation $\theta \mapsto e^{i\theta}$, $0 < \theta < 2\pi$. Its inverse is a chart, whose domain misses only the point 1. At $z = e^{i\theta}$ we have $\partial_\theta = iz$ and $dL_{z^{-1}}(iz) = i$. Taking $\delta_e(i) = 1$ gives $\rho = 1$, so the integral of 1 against δ is 2π. Hence

$$\int_{\mathrm{U}(1)} f(z)\,dz = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta.$$

The characters $\chi_n(z) = z^n$, $n \in \mathbb{Z}$, are irreducible unitary representations, and

$$\int_{\mathrm{U}(1)} \overline{z^m}\,z^n\,dz = \frac{1}{2\pi}\int_0^{2\pi} e^{i(n-m)\theta}\,d\theta = \delta_{mn}.$$

Example 5.15 (SU(2))**.** Every $g \in \mathrm{SU}(2)$ has the form

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

So $g \mapsto (a, b)$ identifies $\mathrm{SU}(2)$ with $S^3 \subset \mathbb{C}^2 = \mathbb{R}^4$. This map and its inverse are restrictions of real linear maps, so the identification is a diffeomorphism. Differentiating $|\omega|^2 = 1$ gives $T_\omega S^3 \subset \omega^\perp$, and both have dimension 3. So $T_\omega S^3 = \omega^\perp$.

The first column of hg is $h(a,b)^T$. Thus L_h acts on S^3 by the unitary matrix h, which is orthogonal on $\mathbb{R}^4$. For $\omega \in S^3$ and $v_j \in \omega^\perp$ set

$$\nu_\omega(v_1,v_2,v_3) = \bigl|\det[\,\omega\ v_1\ v_2\ v_3\,]\bigr|,$$

a real 4×4 determinant. Extending $T\colon \omega^\perp \to \omega^\perp$ by $T\omega = \omega$ shows that ν_ω is a positive density. In a chart with inverse $u \mapsto \omega(u)$, the coefficient of ν is $|\det[\,\omega\ \partial_1\omega\ \partial_2\omega\ \partial_3\omega\,]|$, which is continuous.

For orthogonal A we get $\nu_{A\omega}(Av) = |\det A|\,\nu_\omega(v) = \nu_\omega(v)$. Hence $L_h^*\nu = \nu$. By the uniqueness part of the left-invariant density lemma, Lemma 5.5, $\nu = c\,\delta$ for the left-invariant density δ of (5.3), with $c > 0$. The density ν is the usual volume of the round sphere S^3, and left multiplication rotates this sphere. So the Haar integral on SU(2) is the uniform average over S^3.

Use Hopf coordinates $a = \cos\eta\, e^{i\xi_1}$ and $b = \sin\eta\, e^{i\xi_2}$, with $0 < \eta < \pi/2$ and $0 < \xi_1, \xi_2 < 2\pi$. This parametrisation is injective. The points it misses lie in the circles $\{a = 0\}$ and $\{b = 0\}$ and in the 2-spheres $\{\operatorname{Im} a = 0\}$ and $\{\operatorname{Im} b = 0\}$. In $\mathbb{R}^4$,

$$\begin{aligned}
\omega &= (\cos\eta\cos\xi_1,\ \cos\eta\sin\xi_1,\ \sin\eta\cos\xi_2,\ \sin\eta\sin\xi_2),\\
\partial_\eta\omega &= (-\sin\eta\cos\xi_1,\ -\sin\eta\sin\xi_1,\ \cos\eta\cos\xi_2,\ \cos\eta\sin\xi_2),\\
\partial_{\xi_1}\omega &= (-\cos\eta\sin\xi_1,\ \cos\eta\cos\xi_1,\ 0,\ 0),\\
\partial_{\xi_2}\omega &= (0,\ 0,\ -\sin\eta\sin\xi_2,\ \sin\eta\cos\xi_2).
\end{aligned}$$

These vectors are mutually orthogonal, with lengths $1, 1, \cos\eta, \sin\eta$. So the parametrisation is an injective immersion of an open box. By the inverse function theorem its inverse is a chart. The coefficient of ν in this chart is the product of these lengths, $\cos\eta\sin\eta$. So the integral of 1 against ν is $\frac{1}{2}(2\pi)^2 = 2\pi^2$, the volume of the unit 3-sphere. Therefore

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\!\!\int_0^{2\pi}\!\!\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2. \tag{5.5}$$

The monomial $a^p\bar a^q b^r \bar b^s$ carries the phase $e^{i(p-q)\xi_1}e^{i(r-s)\xi_2}$. So its integral vanishes unless $p = q$ and $r = s$. Also

$$\int_{\mathrm{SU}(2)} |a|^2\,dg = 2\int_0^{\pi/2}\cos^3\eta\sin\eta\,d\eta = 2\Bigl[-\frac{\cos^4\eta}{4}\Bigr]_0^{\pi/2} = \frac{1}{2}, \qquad \int_{\mathrm{SU}(2)} |b|^2\,dg = 1 - \frac{1}{2} = \frac{1}{2}.$$

A coefficient norm. The defining representation $V_1 = \mathbb{C}^2$ is irreducible of dimension 2. In the standard basis $D(g) = g$, so $D_{11} = a$ and $D_{12} = -\bar b$. The matrix-entry relations of Corollary 5.13 predict $\int |D_{11}|^2\,dg = \frac{1}{2}$ and $\int \overline{D_{11}}D_{12}\,dg = 0$. Both agree with the direct computation, since $\overline{D_{11}}D_{12} = -\bar a\bar b$ has a nonzero phase.

An invariant-vector projector. Let SU(2) act on $V_1 \otimes V_1$ by $g \otimes g$. Use the orthonormal basis $e_1\otimes e_1$, $e_1\otimes e_2$, $e_2\otimes e_1$, $e_2\otimes e_2$. The entry of $g\otimes g$ in row (i,k) and column (j,l) is $g_{ij}g_{kl}$. The entries are quadratic monomials in $a, \bar a, b, \bar b$. By the vanishing rule for monomials above, only the products $a\bar a$ and $b\bar b$ have nonzero integral. These products occur only in rows and columns $(1,2)$ and $(2,1)$:

$$g_{11}g_{22} = |a|^2, \qquad g_{12}g_{21} = -|b|^2, \qquad g_{21}g_{12} = -|b|^2, \qquad g_{22}g_{11} = |a|^2.$$

Let P be the averaging projector for this action. Writing D for matrices in this basis, we obtain

$$D(P) = \int_{\mathrm{SU}(2)} g \otimes g \, dg = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

With $\varepsilon = e_1 \otimes e_2 - e_2 \otimes e_1$ we get $Px = \frac{1}{2}\langle \varepsilon, x \rangle \, \varepsilon$. This is the orthogonal projection onto $\mathbb{C}\varepsilon$. In physics, ε is proportional to the singlet state of two spin-one-half particles. By the invariant-vector projector, Proposition 5.10, $(V_1 \otimes V_1)^{\mathrm{SU}(2)} = \mathbb{C}\varepsilon$.

Self-checks

1. Which property of the Haar integral makes the form in the proof of unitarisability, Theorem 5.8, positive definite? Which makes it invariant?

   Answer. Strict positivity gives $\langle v, v \rangle > 0$ for $v \neq 0$. Right invariance gives invariance. Left invariance and normalisation enter only through the proof of strict positivity.

2. Let Π be a nontrivial irreducible representation of a compact Lie group. What is $\int_G \Pi(g) \, dg$?

   Answer. It is 0. By the invariant-vector projector, Proposition 5.10, its image is the subrepresentation V^G. Irreducibility gives $V^G = 0$ or $V^G = V$. If $V^G = V$, then every $\Pi(g)$ is the identity, so every line in V is invariant. Irreducibility then forces $\dim V = 1$, so Π is the trivial representation, contrary to the hypothesis. Hence $V^G = 0$, and the integral is 0.

3. Explain $\int_{\mathrm{SU}(2)} a \, dg = 0$ in two ways.

   Answer. First, in Hopf coordinates the integrand carries the phase $e^{i\xi_1}$, so the integral over ξ_1 vanishes. Second, $a = D^{V_1}_{11}$, and the constant 1 is the matrix coefficient of the trivial representation V_0. Since $V_1 \not\cong V_0$, the relation for inequivalent representations in Corollary 5.13 gives $\int_G \overline{1} \, a \, dg = 0$.

Summary

On a compact Lie group G of positive dimension, left translation of one positive density on $\mathfrak{g}$ gives the Haar integral. Compactness forces $|\det \mathrm{Ad}| = 1$, which gives right invariance. A covering argument by translates gives strict positivity. With these properties, the finite-group proofs of unitarisability, complete reducibility and Schur orthogonality carry over with the sum replaced by the integral. By the matrix-entry relations, Corollary 5.13, the matrix entries of an irreducible unitary representation of dimension d in an orthonormal basis are pairwise L^2-orthogonal, and each has squared norm $1/d$. Completeness of matrix coefficients is the Peter–Weyl theorem of Lecture 10.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (BD), Sections I.5 and II.4.

- Sepanski, *Compact Lie Groups*, §1.4, §2.2 and §3.1.
- Knapp, *Lie Groups Beyond an Introduction* (Knapp Beyond), digital second edition, 2023, §VIII.1, for integration and sets of measure zero.

Lecture 6

Compact Structure and Maximal Tori

Objectives. *This lecture explains why conjugacy in a compact connected group can be studied on a torus. First, an averaged inner product splits the Lie algebra. Next, maximal tori meet every conjugacy class. Finally, a finite Weyl group records which elements of the torus are conjugate.*

By the end of this lecture you will be able to:

- Average an inner product and prove $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple.
- Prove that maximal tori are conjugate, explain the strategy of the torus theorem, and deduce that exp is surjective for compact connected groups.
- Compute $N_K(T)/T$ for $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ and match it with the algebraic Weyl group.
- Locate where these statements fail for $\mathrm{SL}(2, \mathbb{R})$.

Throughout, K is a compact Lie group with Lie algebra $\mathfrak{k}$. Its Haar probability measure dg is left and right invariant (Lecture 5). Connectedness of K is assumed only where a statement says so.

6.1 An invariant inner product and the splitting

Lemma 6.1. *There is an inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$ such that for all $g \in K$ and $X, Y, Z \in \mathfrak{k}$*

$$\langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle = \langle X, Y \rangle, \qquad \langle [Z, X], Y \rangle = -\langle X, [Z, Y] \rangle. \tag{6.1}$$

Proof. Choose any inner product $\langle \cdot, \cdot \rangle_0$ on $\mathfrak{k}$ and set

$$\langle X, Y \rangle = \int_K \langle \mathrm{Ad}(g)X, \mathrm{Ad}(g)Y \rangle_0 \, dg.$$

The integrand is continuous in g. So the form is well defined, symmetric and bilinear. For $X \neq 0$ the continuous function $g \mapsto \| \mathrm{Ad}(g)X \|_0^2$ is positive on the compact set K. It has a positive minimum m, so $\langle X, X \rangle \geq m > 0$. Right invariance of dg gives

$$\langle \mathrm{Ad}(h)X, \mathrm{Ad}(h)Y \rangle = \int_K \langle \mathrm{Ad}(gh)X, \mathrm{Ad}(gh)Y \rangle_0 \, dg = \langle X, Y \rangle.$$

For the second identity, recall $\mathrm{Ad}(\exp tZ) = e^{t \, \mathrm{ad} Z}$ from Lecture 2. By the first identity, the function $t \mapsto \langle e^{t \, \mathrm{ad} Z} X, e^{t \, \mathrm{ad} Z} Y \rangle$ is constant. By the product rule, its derivative at $t = 0$ is

$$0 = \langle [Z, X], Y \rangle + \langle X, [Z, Y] \rangle.$$

∎

In other words, every $\operatorname{ad} Z$ is skew-symmetric for $\langle \cdot, \cdot \rangle$. From now on we fix such an inner product.

Proposition 6.2. *Let $\mathfrak{z} = \mathfrak{z}(\mathfrak{k})$ be the centre and $\mathfrak{s} = [\mathfrak{k}, \mathfrak{k}]$.*

(a) $\mathfrak{s}^{\perp} = \mathfrak{z}$. Hence $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{s}$, an orthogonal sum of ideals.

(b) The Killing form B of $\mathfrak{s}$ is negative definite. Hence $\mathfrak{s}$ is semisimple.

Proof. (a) For $Z, X, Y \in \mathfrak{k}$, the skew-symmetry identity in (6.1) gives

$$\langle Z, [X, Y] \rangle = -\langle [X, Z], Y \rangle.$$

So $Z \perp \mathfrak{s}$ exactly when $[X, Z] \perp Y$ for all X, Y. This means $[X, Z] = 0$ for all X, that is, $Z \in \mathfrak{z}$. Both $\mathfrak{z}$ and $\mathfrak{s}$ are ideals. Since $\mathfrak{k}$ is finite-dimensional, $\mathfrak{k} = \mathfrak{s}^{\perp} \oplus \mathfrak{s} = \mathfrak{z} \oplus \mathfrak{s}$.

(b) First, $\mathfrak{z}(\mathfrak{s}) = 0$. An element of $\mathfrak{z}(\mathfrak{s})$ commutes with $\mathfrak{s}$ and with $\mathfrak{z}$, hence with $\mathfrak{k}$ by the splitting in (a). So it lies in $\mathfrak{z} \cap \mathfrak{s} = 0$.

Now let $X \in \mathfrak{s}$ and $A = \operatorname{ad} X|_{\mathfrak{s}}$. Since $\mathfrak{s}$ is an ideal, A maps $\mathfrak{s}$ to itself. The map A is skew-symmetric, since every $\operatorname{ad} X$ is. So in an orthonormal basis $e_1, \dots, e_m$ of $\mathfrak{s}$ the matrix of A satisfies $A^{\mathsf{T}} = -A$. Hence

$$B(X, X) = \operatorname{tr}(A^2) = -\operatorname{tr}(A^{\mathsf{T}} A) = -\sum_{i=1}^{m} \|Ae_i\|^2 \le 0.$$

Equality forces $A = 0$, so $X \in \mathfrak{z}(\mathfrak{s}) = 0$. Thus B is negative definite.

The complexification of B is the Killing form of $\mathfrak{s}_{\mathbb{C}}$, and it is nondegenerate. By the complex Cartan criterion from the Lie Algebras course, $\mathfrak{s}_{\mathbb{C}}$ is semisimple. A nonzero solvable real ideal of $\mathfrak{s}$ would complexify to a nonzero solvable ideal of $\mathfrak{s}_{\mathbb{C}}$. Hence $\mathfrak{s}$ is semisimple. ∎

For example, let $\mathfrak{k} = \mathfrak{u}(n)$. The form $-\operatorname{tr}(XY)$ satisfies both identities in (6.1) without averaging (Exercise 6.1). The proof of the splitting proposition, Proposition 6.2, uses only these identities, so it applies to this form. Here $\mathfrak{z} = i\mathbb{R}I$ and $\mathfrak{s} = \mathfrak{su}(n)$. The splitting of $X \in \mathfrak{u}(n)$ is

$$X = \tfrac{1}{n}(\operatorname{tr} X)\, I + \Big(X - \tfrac{1}{n}(\operatorname{tr} X)\, I\Big) \in i\mathbb{R}I \oplus \mathfrak{su}(n).$$

6.2 Maximal tori and their conjugacy

Definition 6.3. A *torus* is a compact connected abelian Lie group. A *maximal torus* of K is a torus subgroup $T \subseteq K$ not properly contained in another torus subgroup of K.

For example, the diagonal unitary matrices are a maximal torus of $\mathrm{U}(n)$, and the circle $T = \{k_\theta\}$ of Lecture 3 is a maximal torus of $\mathrm{SU}(2)$. § 6.5 proves this.

Let T be a torus with Lie algebra $\mathfrak{t}$. For $t \in T$ and $Y \in \mathfrak{t}$, differentiating $t \exp(sY) t^{-1} = \exp(sY)$ at $s = 0$ gives $\operatorname{Ad}(t)Y = Y$. Putting $t = \exp(rZ)$ with $Z \in \mathfrak{t}$ and differentiating at $r = 0$ gives $[Z, Y] = 0$. So $\mathfrak{t}$ is abelian, and $\exp(X + Y) = \exp X \exp Y$ on $\mathfrak{t}$ (Lecture 2).

Thus $\exp(\mathfrak{t})$ is a subgroup containing a neighbourhood of e (Lecture 2). Such a subgroup is open. It is also closed, because its complement is a union of open cosets. As T is connected, $\exp\colon \mathfrak{t} \to T$ is surjective.

We cite two further facts (H §11.1; S Ch. 5), and Lecture 7 proves the first:

$$T \cong \mathfrak{t}/\Lambda \text{ with } \Lambda = \ker \exp \text{ a lattice}; \qquad \overline{\exp(\mathbb{R}H)} = T \text{ for some } H \in \mathfrak{t}. \tag{6.2}$$

The second is Kronecker's theorem. We call such an H *generic*. Kronecker's theorem also gives $t_0 \in T$ whose powers are dense in T. For example, on the two-torus T^2 of Lecture 3 the irrational line L_α is dense, by the irrational-line proposition. So a nonzero element of $\mathfrak{t}$ tangent to L_α is generic.

Lemma 6.4. *A torus $T \subseteq K$ is maximal if and only if $\mathfrak{t}$ is a maximal abelian subalgebra of $\mathfrak{k}$. Maximal tori exist.*

Proof. Let $\mathfrak{t}$ be maximal abelian, and let $T \subseteq T'$ with T' a torus. Then $\mathfrak{t} \subseteq \mathfrak{t}'$ with $\mathfrak{t}'$ abelian, so $\mathfrak{t} = \mathfrak{t}'$. Hence $T' = \exp(\mathfrak{t}') = \exp(\mathfrak{t}) = T$.

Conversely, let T be maximal, and let $\mathfrak{a}$ be an abelian subalgebra with $\mathfrak{t} \subseteq \mathfrak{a}$. Since $\mathfrak{a}$ is abelian, $\exp(\mathfrak{a})$ is a subgroup, as for $\mathfrak{t}$ above. This subgroup is connected and abelian. Its closure A is a Lie group by the closed subgroup theorem (Lecture 3). The group A is compact, connected and abelian, so it is a torus containing T. Since T is maximal, $A = T$. Hence $\mathfrak{a} \subseteq \mathrm{Lie}(A) = \mathfrak{t}$.

For existence, take a maximal abelian subalgebra $\mathfrak{a}$. As in the second paragraph, $A = \overline{\exp(\mathfrak{a})}$ is a torus. Its Lie algebra is abelian and contains $\mathfrak{a}$, so it equals $\mathfrak{a}$. By the first paragraph, A is a maximal torus. ∎

Theorem 6.5 (Torus theorem; BD §IV.1; S Ch. 5; H §§11.2–11.5)**.** *Let K be compact and connected, and let T be a maximal torus.*

(a) Every $g \in K$ lies in xTx^{-1} for some $x \in K$.

(b) Every maximal torus of K equals xTx^{-1} for some $x \in K$.

Take $K = \mathrm{U}(n)$ with T the diagonal matrices. Then part (a) is the spectral theorem for unitary matrices (Self-check 2). Part (b) says that a maximal torus is diagonal after a change of orthonormal basis. So the theorem extends diagonalisation to every compact connected group.

Hall (H) proves this for compact matrix Lie groups, and Bröcker–tom Dieck (BD) for general compact connected Lie groups. Part (b), the conjugacy of maximal tori, and a Lie algebra form of part (a) have a short complete proof.

Proposition 6.6. *Let T be a maximal torus of K. For every $X \in \mathfrak{k}$ there is $x \in K$ with $\mathrm{Ad}(x)X \in \mathfrak{t}$. Moreover, any two maximal tori of K are conjugate.*

We rotate X by $\mathrm{Ad}(K)$ until it is as close as possible to a generic element $H \in \mathfrak{t}$, using the compactness of K. At the best position the first derivative vanishes, and this says that the rotated element commutes with H.

Proof. Choose a generic $H \in \mathfrak{t}$. Define

$$f \colon K \to \mathbb{R}, \qquad f(x) = \langle \mathrm{Ad}(x)X, H \rangle.$$

It is continuous, so it attains its maximum on the compact group K at some x_0. Put $X_0 = \mathrm{Ad}(x_0)X$ and fix $Y \in \mathfrak{k}$. By Lecture 2,

$$f\big(\exp(sY)x_0\big) = \langle \mathrm{Ad}(\exp sY)X_0, H\rangle = \langle e^{s\,\mathrm{ad}\,Y}X_0, H\rangle.$$

Since f is maximal at x_0, this function of s is maximal at $s = 0$. Hence its derivative $\langle [Y, X_0], H\rangle$ at $s = 0$ vanishes. Using the second identity in (6.1) twice, together with antisymmetry of the bracket, we get

$$0 = \langle [Y, X_0], H\rangle = -\langle X_0, [Y, H]\rangle = \langle X_0, [H, Y]\rangle = -\langle [H, X_0], Y\rangle.$$

Since Y is arbitrary, $[H, X_0] = 0$.

Next we pass from the single element H to all of $\mathfrak{t}$. This step uses genericity of H. Consider

$$S = \{t \in T : \mathrm{Ad}(t)X_0 = X_0\}.$$

It is a closed subgroup of T, since $t \mapsto \mathrm{Ad}(t)X_0$ is continuous. From $[H, X_0] = 0$, every term after the first in the exponential series vanishes. So

$$\mathrm{Ad}(\exp sH)X_0 = e^{s\,\mathrm{ad}\,H}X_0 = X_0 \qquad (s \in \mathbb{R}).$$

Thus S contains $\exp(\mathbb{R}H)$, and hence its closure T. Now let $Z \in \mathfrak{t}$. Differentiating $\mathrm{Ad}(\exp sZ)X_0 = X_0$ at $s = 0$ gives $[Z, X_0] = 0$. So $\mathfrak{t} + \mathbb{R}X_0$ is an abelian subalgebra containing $\mathfrak{t}$. By the characterisation of maximal tori in Lemma 6.4, $\mathfrak{t}$ is maximal abelian, so $X_0 \in \mathfrak{t}$.

Now let T' be a maximal torus with generic element H'. Choose x with $\mathrm{Ad}(x)H' \in \mathfrak{t}$. Then

$$x \exp(sH')x^{-1} = \exp\big(s\,\mathrm{Ad}(x)H'\big) \in T \qquad (s \in \mathbb{R}).$$

Conjugation by x is a homeomorphism and T is closed. Taking closures, $xT'x^{-1} \subseteq T$. The left side is a maximal torus, so equality holds. ∎

Definition 6.7. Let $T \subseteq K$ be a maximal torus. Its *normaliser* is $N_K(T) = \{n \in K : nTn^{-1} = T\}$. The *Weyl group* is $W(K, T) = N_K(T)/T$.

The group $W(K, T)$ acts on T by $nT \cdot t = ntn^{-1}$ and on $\mathfrak{t}$ by $\mathrm{Ad}(n)$. For $n \in N_K(T)$ and $H \in \mathfrak{t}$, the curve $s \mapsto n\exp(sH)n^{-1}$ lies in T. Differentiating it at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. Since $\mathrm{Ad}(n)$ is invertible and $\mathfrak{t}$ is finite-dimensional, $\mathrm{Ad}(n)$ maps $\mathfrak{t}$ onto itself. These actions are well defined on cosets because T is abelian.

Lemma 6.8. *Let T be a maximal torus of K and $N = N_K(T)$. Then the identity component N_0 equals T, and $W(K, T)$ is finite.*

Proof. The group N is closed in K, so it is a compact Lie group (Lecture 3). Let $n \in N$ and $H \in \mathfrak{t}$. Then

$$n\exp(sH)n^{-1} = \exp\big(s\,\mathrm{Ad}(n)H\big) \in T \qquad (s \in \mathbb{R}).$$

Differentiating at $s = 0$ gives $\mathrm{Ad}(n)H \in \mathfrak{t}$. If $\exp H = e$, the display gives $\exp(\mathrm{Ad}(n)H) = e$. So $A_n = \mathrm{Ad}(n)|_{\mathfrak{t}}$ maps the lattice Λ into itself.

First, N_0 centralises T. Choose a basis $\lambda_1, \dots, \lambda_r$ of $\mathfrak{t}$ inside Λ. Each map $n \mapsto A_n\lambda_i$ is continuous with values in the discrete set Λ. Since N_0 is connected, this map is constant on

N_0, equal to its value λ_i at e. So $A_n = I$ for $n \in N_0$. Then $n\exp(H)n^{-1} = \exp H$ for all $H \in \mathfrak{t}$. Since $\exp(\mathfrak{t}) = T$, every $n \in N_0$ commutes with every element of T.

Second, $\mathrm{Lie}(N) = \mathfrak{t}$. Let $X \in \mathrm{Lie}(N)$. Then $\exp(sX) \in N_0$ for all s. By the first step, $\mathrm{Ad}(\exp sX)H = H$ for $H \in \mathfrak{t}$. Differentiating at $s = 0$ gives $[X, H] = 0$. So $\mathfrak{t} + \mathbb{R}X$ is abelian, and $X \in \mathfrak{t}$ because $\mathfrak{t}$ is maximal abelian (Lemma 6.4). The connected group N_0 is generated by $\exp(\mathrm{Lie}\, N)$ (Lecture 2). Hence N_0 is generated by $\exp(\mathfrak{t}) = T$, so $N_0 = T$.

Finally, N_0 is open in N (Lecture 2). The cosets of N_0 form a cover of the compact group N by disjoint open sets. Hence there are finitely many cosets. Since $N_0 = T$, the group $W(K, T) = N/N_0$ is finite. ∎

6.3 An outline of the torus theorem via mapping degree

Since $x\exp(X)x^{-1} = \exp(\mathrm{Ad}(x)X)$, the conjugacy statement of Proposition 6.6 shows that every element of $\exp(\mathfrak{k})$ lies in a conjugate of T. Conversely, every conjugate of an element of $T = \exp(\mathfrak{t})$ is an exponential. So part (a) of the torus theorem, Theorem 6.5(a), is equivalent to surjectivity of exp. Surjectivity of exp needs a global argument. We outline the proof in Hall (H §§11.3–11.5).

For connected K define

$$\Phi\colon K/T \times T \to K, \qquad \Phi(xT, t) = xtx^{-1}.$$

This is well defined because T is abelian. The source $K/T \times T$ and the target K are compact connected manifolds of dimension $\dim K$ (Lecture 3).

We count how many times Φ covers a point, with signs. A point y is a *regular value* of a smooth map if the differential of the map is invertible at every point of the inverse image of y. For a smooth map between compact connected oriented manifolds of equal dimension, the signed count of preimages is the same at every regular value. A point outside the image would have count zero. So it is enough to show that the count is nonzero at one regular value. On a circle, an orientation is a direction of travel. The general definition is part of the cited theory.

Figure 6.1 shows the count on the circle, for the maps $f(x) = x + 0.9\sin 2x$ and $g(x) = 1.5 + \sin x$. A preimage counts $+1$ where the graph rises and -1 where it falls. The map f has count 1 at both levels drawn, and the map g has count 0 at both.

We now carry out the count.

1. Let $t_0 \in T$ be an element whose powers are dense in T, as after (6.2). Then

$$\Phi^{-1}(t_0) = \{(nT,\ n^{-1}t_0n) : nT \in W(K, T)\}.$$

   Each listed point maps to t_0. Conversely, let $xtx^{-1} = t_0$. Then $x^{-1}t_0^m x = t^m \in T$ for all $m \in \mathbb{Z}$. Taking closures, $x^{-1}Tx \subseteq T$. Both are maximal tori, so equality holds and $x \in N_K(T)$. The set is finite by the finiteness of the Weyl group, Lemma 6.8, which was proved without the torus theorem.

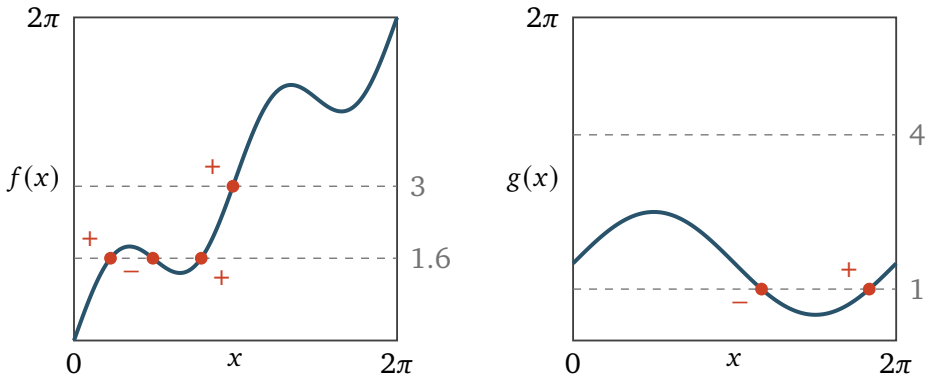


Figure 6.1: Two smooth maps of the circle, drawn on a square whose opposite sides are identified modulo 2π. The dashed lines are levels, numbered at the right, and the marked points are their preimages with the sign of the slope. The map f has signs $+, -, +$ at level 1.6 and $+$ at level 3. The map g has signs $-, +$ at level 1 and no preimage at level 4.

2. The quotient K/T is orientable. Fix orientations of K/T, T and K, and give $K/T \times T$ the product orientation. At each point of $\Phi^{-1}(t_0)$ the differential of Φ is invertible. Its determinant, computed in oriented bases, has the same sign at all of these points.
3. By step 2, t_0 is a regular value of Φ. A smooth map between compact connected oriented manifolds of equal dimension has an integer degree. It is the number of preimages of any regular value, each counted with the sign of the determinant of $d\Phi$ there.
4. So $\deg \Phi = \pm|W(K,T)| \neq 0$. A point outside the image would be a regular value with no preimages, giving degree 0. Hence Φ is surjective.

The omitted step is the degree theory in steps 2–4. It consists of the definition of degree, its independence of the regular value, the orientation of K/T, and the sign comparison at all preimages.

Step 2 rests on a local computation, shown here at (eT, t). By Lecture 3, $Y \mapsto \exp(Y)T$ identifies $\mathfrak{t}^\perp$ with the tangent space of K/T at eT. We identify the tangent spaces of T and of K at t with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation by t. So $Z \in \mathfrak{t}$ corresponds to the curve $s \mapsto t\exp(sZ)$. For $Y \in \mathfrak{t}^\perp$ and $Z \in \mathfrak{t}$,

$$t^{-1}\Phi\big(\exp(sY)T,\ t\exp(sZ)\big) = \exp\big(s\,\mathrm{Ad}(t^{-1})Y\big)\exp(sZ)\exp(-sY).$$

By the product rule, the derivative at $s = 0$ is $\mathrm{Ad}(t^{-1})Y - Y + Z$. The orthogonal map $\mathrm{Ad}(t^{-1})$ fixes $\mathfrak{t}$ pointwise, since T is abelian. Hence it preserves $\mathfrak{t}^\perp$. In these identifications, the differential is block diagonal in $\mathfrak{t}^\perp \oplus \mathfrak{t}$, with blocks $\mathrm{Ad}(t^{-1})|_{\mathfrak{t}^\perp} - I$ and I. Hence

$$\det d\Phi_{(eT,t)} = \det\big(\mathrm{Ad}(t^{-1}) - I\big)\big|_{\mathfrak{t}^\perp}. \tag{6.3}$$

Hall (H §11.5) shows that this determinant is positive at $t = t_0$, and that the sign is the same at the other preimages of t_0. We do not reproduce that part. The same determinant returns in Lecture 9, as the weight in the Weyl integration formula.

6.4 Surjectivity of exp and the Weyl group

Corollary 6.9. *If K is compact and connected, then $\exp\colon \mathfrak{k} \to K$ is surjective.*

Proof. Let $g \in K$. By the torus theorem, Theorem 6.5(a), $g = xtx^{-1}$ with $x \in K$ and $t \in T$. As $\exp\colon \mathfrak{t} \to T$ is surjective, $t = \exp H$ with $H \in \mathfrak{t}$. Hence

$$g = x \exp(H) x^{-1} = \exp\big(\mathrm{Ad}(x)H\big). \qquad \blacksquare$$

Connectedness is needed. The path $s \mapsto \exp(sX)$ joins e to $\exp X$, so $\exp(\mathfrak{k})$ lies in the identity component. For example, no reflection in $\mathrm{O}(2)$ is an exponential.

The *centraliser* of T is $Z_K(T) = \{g \in K : gt = tg \text{ for all } t \in T\}$. It contains T, since T is abelian.

Theorem 6.10 (H §§11.2–11.5; BD §§IV.1–2)**.** *Let K be compact and connected with maximal torus T. Then $Z_K(T) = T$. Two elements of T are conjugate in K if and only if they lie in one $W(K,T)$-orbit.*

Together with the torus theorem, Theorem 6.5(a), we obtain a bijection

$$T/W(K,T) \longrightarrow \{\text{conjugacy classes of } K\}, \qquad W(K,T)\cdot t \longmapsto \text{class of } t.$$

Surjectivity is the torus theorem. Injectivity is the orbit statement of Theorem 6.10. By Lemma 6.8, the group $W(K,T)$ is finite. So conjugacy in K reduces to the action of a finite group on T.

The Weyl group also has an algebraic description. For $H \in \mathfrak{t}$ the operator $\operatorname{ad} H$ is skew-symmetric for our fixed $\mathrm{Ad}(K)$-invariant inner product. So its complex-linear extension to $\mathfrak{k}_{\mathbb{C}}$ is diagonalisable with imaginary eigenvalues. These operators commute, because $\mathfrak{t}$ is abelian. So they have common eigenvectors. On a common eigenvector, the eigenvalue is i times a real number that depends linearly on H.

A *root* is a nonzero $\alpha \in \mathfrak{t}^*$ such that some nonzero $V \in \mathfrak{k}_{\mathbb{C}}$ satisfies

$$[H, V] = i\alpha(H)V \qquad (H \in \mathfrak{t}).$$

Let R be the set of roots. A root is real on $\mathfrak{t}$. Lectures 7, 8 and 9 keep this convention. The Lie Algebras course writes the same root as the complex-linear form $i\alpha$ on $\mathfrak{t}_{\mathbb{C}}$, and Lecture 7 gives a dictionary. When $\mathfrak{k}$ is semisimple, $\mathfrak{t}_{\mathbb{C}}$ is a Cartan subalgebra of $\mathfrak{k}_{\mathbb{C}}$ in the sense of that course. Lecture 8 proves this.

Define $\alpha^\sharp \in \mathfrak{t}$ by $\langle \alpha^\sharp, H\rangle = \alpha(H)$ for $H \in \mathfrak{t}$. The reflections and the *algebraic Weyl group* are

$$s_\alpha(H) = H - \frac{2\alpha(H)}{\langle \alpha^\sharp, \alpha^\sharp\rangle}\,\alpha^\sharp, \qquad W(R) = \langle s_\alpha : \alpha \in R\rangle \subseteq \mathrm{GL}(\mathfrak{t}).$$

Theorem 6.11 (H §11.7; BD Ch. V)**.** *Let K be compact and connected, with any $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$. Then $nT \mapsto \mathrm{Ad}(n)|_{\mathfrak{t}}$ is an isomorphism from $W(K,T)$ onto $W(R)$.*

6.5 Unitary groups and a noncompact example

Proposition 6.12. *Let $T \subseteq \mathrm{U}(n)$ be the diagonal unitary matrices. For $\sigma \in S_n$ let $P_\sigma e_j = e_{\sigma(j)}$. Then T is a maximal torus,*

$$N_{\mathrm{U}(n)}(T) = \{DP_\sigma : D \in T,\ \sigma \in S_n\},$$

and $DP_\sigma T \mapsto \sigma$ is an isomorphism $W(\mathrm{U}(n), T) \to S_n$. The action on T is

$$P_\sigma \operatorname{diag}(z_1, \dots, z_n) P_\sigma^{-1} = \operatorname{diag}\left(z_{\sigma^{-1}(1)}, \dots, z_{\sigma^{-1}(n)}\right). \tag{6.4}$$

Proof. The group $T \cong \mathrm{U}(1)^n$ is a torus. Fix $t_* = \operatorname{diag}(c_1, \dots, c_n) \in T$ with distinct entries. A matrix commuting with t_* preserves each eigenline $\mathbb{C}e_j$, so it is diagonal. Hence $Z_{\mathrm{U}(n)}(T) = T$. A torus $T' \supseteq T$ is abelian, so $T' \subseteq Z_{\mathrm{U}(n)}(T) = T$. Thus T is maximal.

Formula (6.4) holds for every $t = \operatorname{diag}(z_1, \dots, z_n) \in T$, because

$$P_\sigma t P_\sigma^{-1} e_{\sigma(j)} = P_\sigma t e_j = z_j e_{\sigma(j)}.$$

So each DP_σ normalises T. Conversely, let $g \in N_{\mathrm{U}(n)}(T)$ and $t' = g t_* g^{-1} \in T$. Then

$$t'(g e_j) = g t_* e_j = c_j\, g e_j.$$

The diagonal matrix t' has the distinct eigenvalues $c_1, \dots, c_n$. So its c_j-eigenspace is a coordinate line $\mathbb{C}e_{\sigma(j)}$. Distinct eigenvalues have distinct eigenlines, so σ is a permutation. Thus $g e_j = d_{\sigma(j)} e_{\sigma(j)}$ with $|d_{\sigma(j)}| = 1$. That is, $g = DP_\sigma$ with $D = \operatorname{diag}(d_1, \dots, d_n)$. Since $P_\sigma P_\tau = P_{\sigma\tau}$ and $P_\sigma D P_\sigma^{-1} \in T$, the map $DP_\sigma T \mapsto \sigma$ is a homomorphism. It is onto, since $P_\sigma \in \mathrm{U}(n)$. It is injective because $DP_\sigma \in T$ only when $\sigma = \mathrm{id}$. ∎

We now compare $W(\mathrm{U}(n), T)$ with the algebraic Weyl group, using the Ad-invariant form $-\operatorname{tr}(XY)$. On $\mathfrak{t} = \{i \operatorname{diag}(a_1, \dots, a_n) : a_j \in \mathbb{R}\}$ this form equals $\sum_j a_j b_j$. Here $\mathfrak{t}_{\mathbb{C}} = \mathfrak{gl}(n, \mathbb{C})$. Let E_{jk} be the matrix unit with 1 in row j and column k. For $H = i \operatorname{diag}(a)$ and $j \neq k$,

$$[H, E_{jk}] = i(a_j - a_k) E_{jk}.$$

Let ε_j be the linear form on $\mathfrak{t}$ with $\varepsilon_j\big(i \operatorname{diag}(a_1, \dots, a_n)\big) = a_j$. The roots are $\alpha_{jk} = \varepsilon_j - \varepsilon_k$ for $j \neq k$, with $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ and $\langle \alpha_{jk}^\sharp, \alpha_{jk}^\sharp \rangle = 2$. Then

$$s_{\alpha_{jk}}\big(i \operatorname{diag}(a)\big) = i \operatorname{diag}(a) - (a_j - a_k)\, i(E_{jj} - E_{kk}).$$

This reflection swaps a_j and a_k. By the permutation formula (6.4), the reflection equals $\operatorname{Ad}(P_{(jk)})|_{\mathfrak{t}}$. Transpositions generate S_n. So $W(R)$ is the image of $W(\mathrm{U}(n), T)$, as the algebraic description of the Weyl group in Theorem 6.11 predicts. Hence conjugacy classes of $\mathrm{U}(n)$ correspond to T/S_n. In other words, a unitary matrix is determined up to conjugacy by its eigenvalues with multiplicity.

Corollary 6.13. *Let $n \geq 2$ and $T_1 = T \cap \mathrm{SU}(n)$. Then T_1 is a maximal torus of $\mathrm{SU}(n)$. The normaliser $N_{\mathrm{SU}(n)}(T_1)$ consists of the DP_σ with $D \in T$ and $\det(DP_\sigma) = 1$, and $DP_\sigma T_1 \mapsto \sigma$ is an isomorphism $W(\mathrm{SU}(n), T_1) \to S_n$.*

Proof. The map $(z_1, \dots, z_{n-1}) \mapsto \operatorname{diag}(z_1, \dots, z_{n-1}, (z_1 \cdots z_{n-1})^{-1})$ shows $T_1 \cong \mathrm{U}(1)^{n-1}$. Let $0 < \epsilon < \pi/(n-1)$. The entries $c_j = e^{i\epsilon(2j-n-1)}$ are distinct, and their product is 1 since $\sum_j (2j-n-1) = 0$. The proof of Proposition 6.12, the normaliser computation for $\mathrm{U}(n)$, applies with this t_*. It shows that T_1 is maximal. It also shows that the normaliser consists of the DP_σ with

$$\det(DP_\sigma) = \det D \cdot \operatorname{sgn}\sigma = 1.$$

Every σ occurs, with $D = \operatorname{diag}(\operatorname{sgn}\sigma, 1, \dots, 1)$. The injectivity argument is unchanged. ∎

The roots match again. The Lie algebra $\mathfrak{t}_1$ of T_1 consists of the $i\operatorname{diag}(a)$ with $\sum_j a_j = 0$. Since $\mathfrak{su}(n)_{\mathbb{C}} = \mathfrak{sl}(n, \mathbb{C})$ is spanned by $\mathfrak{t}_{1,\mathbb{C}}$ and the E_{jk}, the roots are the restrictions of the α_{jk}. They are nonzero, because $\alpha_{jk}^\sharp = i(E_{jj} - E_{kk})$ lies in $\mathfrak{t}_1$ and $\alpha_{jk}(\alpha_{jk}^\sharp) = 2$. So $s_{\alpha_{jk}}$ still swaps a_j and a_k. Put $n_{(jk)} = \operatorname{diag}(-1, 1, \dots, 1)P_{(jk)}$, which lies in $N_{\mathrm{SU}(n)}(T_1)$. Diagonal matrices act trivially on $\mathfrak{t}$, so $\operatorname{Ad}(n_{(jk)})|_{\mathfrak{t}_1} = s_{\alpha_{jk}}$. Hence $W(R)$ is the image of $W(\mathrm{SU}(n), T_1) \cong S_n$.

For $\mathrm{SU}(2)$ the transposition is represented by

$$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \operatorname{diag}(-1, 1)P_{(12)}, \qquad w \operatorname{diag}(z, \bar{z}) w^{-1} = \operatorname{diag}(\bar{z}, z).$$

By the bijection between $T_1/W(\mathrm{SU}(2), T_1)$ and conjugacy classes, every conjugacy class of $\mathrm{SU}(2)$ contains exactly one matrix $\operatorname{diag}(e^{i\theta}, e^{-i\theta})$ with $\theta \in [0, \pi]$. In physics terms, choosing a maximal torus of $\mathrm{SU}(2)$ is like choosing a quantisation axis.

Now compare the connected noncompact group $\mathrm{SL}(2, \mathbb{R})$. Its connectedness was shown in Lecture 2 (Exercise 2.4). By the computation in Lecture 2, the element

$$\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$$

is not in $\exp(\mathfrak{sl}(2, \mathbb{R}))$. So the surjectivity statement of Corollary 6.9 fails for this connected noncompact group. The lines $\mathbb{R}\operatorname{diag}(1, -1)$ and $\mathfrak{so}(2)$ are both maximal abelian. Their nonzero elements have real and imaginary eigenvalues respectively, so they are not Ad-conjugate.

The proof of the conjugacy statement in Proposition 6.6 breaks at the invariant inner product of Lemma 6.1. For $H = \operatorname{diag}(1, -1)$ the operator $\operatorname{ad} H$ has eigenvalues $2, 0, -2$, but a skew-symmetric operator has imaginary eigenvalues. Averaging is unavailable because $\mathrm{SL}(2, \mathbb{R})$ is not compact. Exercise 6.4 gives the details.

6.6 Finite central quotients

The next lecture uses the following global form of the splitting.

Theorem 6.14 (BD §V.8). *Let K be compact and connected. Let $T_0 = Z(K)_0$ be the identity component of the centre. It is a torus with Lie algebra $\mathfrak{z}(\mathfrak{k})$. Let K_{sc} be the simply connected Lie group with Lie algebra $[\mathfrak{k}, \mathfrak{k}]$ (Lecture 4). Then K_{sc} is compact, and the inclusion $[\mathfrak{k}, \mathfrak{k}] \to \mathfrak{k}$ integrates to a homomorphism $\varphi \colon K_{\mathrm{sc}} \to K$. The map*

$$\mu \colon T_0 \times K_{\mathrm{sc}} \to K, \qquad \mu(z, y) = z\,\varphi(y),$$

is a surjective homomorphism with finite central kernel F. Hence $K \cong (T_0 \times K_{\mathrm{sc}})/F$.

Compactness of K_{sc} is Weyl's theorem and is part of the citation.

For example, for $K = \mathrm{U}(n)$ we have $T_0 = \{zI : |z| = 1\}$ and $K_{\mathrm{sc}} = \mathrm{SU}(n)$, which is simply connected (H Ch. 13). Here $\mu(z, A) = zA$. If $zA = I$, then $A = z^{-1}I$ and $\det A = z^{-n} = 1$. Hence

$$F = \{(\zeta, \zeta^{-1}I) : \zeta^n = 1\} \cong \mathbb{Z}/n, \qquad \mathrm{U}(n) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(n)\big)/F.$$

Exercise 6.5 checks surjectivity for $n = 2$.

Self-checks

1. Split
$$X = \begin{pmatrix} 3i & 1 \\ -1 & i \end{pmatrix} \in \mathfrak{u}(2)$$
along $\mathfrak{z} \oplus \mathfrak{su}(2)$.
Answer. The centre component is $\frac{1}{2}(\operatorname{tr} X)I = 2iI$. The remainder is skew-Hermitian with trace zero:
$$X - 2iI = \begin{pmatrix} i & 1 \\ -1 & -i \end{pmatrix}.$$
2. Let T be the diagonal torus of $\mathrm{U}(n)$. Without using the torus theorem, prove that every element of $\mathrm{U}(n)$ lies in a conjugate of T, as in Theorem 6.5(a).
Answer. A unitary U is normal. The spectral theorem for normal matrices gives a unitary V such that $V^{-1}UV$ is diagonal. The matrix $V^{-1}UV$ is also unitary, so it lies in T. Hence $U \in VTV^{-1}$.
3. Are $\operatorname{diag}(i, -i)$ and $\operatorname{diag}(-i, i)$ conjugate in $\mathrm{SU}(2)$? Are $\operatorname{diag}(i, -i)$ and I?
Answer. Yes. With w as above, $w \operatorname{diag}(i, -i)w^{-1} = \operatorname{diag}(-i, i)$. The second pair is not conjugate. Conjugate matrices have equal traces, but the traces of $\operatorname{diag}(i, -i)$ and I are 0 and 2.

Summary

An averaged inner product gives the central and semisimple splitting. Maximal tori are conjugate. For connected K, the cited global torus theorem makes exp surjective. Together with the cited orbit theorem for $W(K, T)$, it identifies the conjugacy classes of K with $T/W(K, T)$. The Weyl group is finite. For $\mathrm{U}(n)$ and $\mathrm{SU}(n)$ it is isomorphic to S_n. Compactness is essential, as the connected group $\mathrm{SL}(2, \mathbb{R})$ shows.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, §§11.1–11.5 and 11.7.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1–3 and V.8.
- S: Sepanski, *Compact Lie Groups*, Chapter 5.

Lecture 7

Lattices, centres and global forms

Objectives. *The groups* $\mathrm{SU}(2)$ *and* $\mathrm{SO}(3)$ *have isomorphic Lie algebras but are not isomorphic. This lecture asks which discrete data determine connected groups with the same Lie algebra. For compact groups, the answer is the kernel lattice* Λ *of a maximal torus, inside its Lie algebra* $\mathfrak{t}$*, read together with the root data.*

The lecture proves this for semisimple Lie algebras, using cited structure results. Different lattices can still give isomorphic groups, through automorphisms of the Lie algebra that preserve $\mathfrak{t}$*. When there is a central torus, the lecture shows how* Λ *determines a finite group* F*.*

By the end of this lecture you will be able to:

- Prove that a compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, with characters the ℓ satisfying $\ell(\Lambda) \subset 2\pi\mathbb{Z}$.
- Place Λ between the coroot and coweight lattices, and compute centres.
- Describe the connected compact groups (global forms) with a given semisimple Lie algebra $\mathfrak{k}$, and the groups $(T_0 \times K_{sc})/F$.
- Decide which representations descend to $\mathrm{SO}(3)$, $\mathrm{PSU}(3)$ and $\mathrm{U}(2)$.

The Lie groups denoted K or T in this lecture, with or without subscripts, are connected. The subgroups C, F and $Z(K)$ that appear later need not be connected. A Lie group and its identity component have the same Lie algebra, so we restrict to connected groups. Even connected groups with isomorphic Lie algebras need not be isomorphic, as $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ show.

7.1 Compact connected abelian groups

Definition 7.1. A *torus* is a Lie group isomorphic to $\mathrm{U}(1)^n$ for some $n \geq 0$. A *lattice* in a real vector space V is a subgroup $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_n$, where $v_1, \ldots, v_n$ is a basis of V.

Lemma 7.2 (Discrete subgroups)**.** *Every discrete subgroup of a finite-dimensional real vector space equals* $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_k$ *for some linearly independent* $v_1, \ldots, v_k$*.*

We cite this standard lemma from BD, Section I.3, Lemma (3.8).

Theorem 7.3. *Let* T *be a compact connected abelian Lie group with Lie algebra* $\mathfrak{t}$ *of dimension* n*. Then* $\exp\colon \mathfrak{t} \to T$ *is a surjective homomorphism, and* $\Lambda = \ker \exp$ *is a lattice*

in $\mathfrak{t}$. *For a basis* $v_1, \dots, v_n$ *of* Λ, *the map*

$$\Phi\colon \mathrm{U}(1)^n \to T, \qquad \Phi\big(e^{2\pi i x_1}, \dots, e^{2\pi i x_n}\big) = \exp(x_1 v_1 + \cdots + x_n v_n),$$

is an isomorphism of Lie groups.

Proof. Since T is abelian, so is $\mathfrak{t}$. Commuting elements satisfy $\exp(X+Y) = \exp X \exp Y$ (Lecture 2), so exp is a homomorphism. Choose an open set $U = -U$ containing 0 that exp maps diffeomorphically onto an open set (Lecture 2).

Surjectivity. The subgroup $\exp(\mathfrak{t})$ contains $g\exp(U)$ for each of its elements g, so it is open. Its complement is a union of open cosets, so it is also closed. Since T is connected, $\exp(\mathfrak{t}) = T$.

Discreteness. As exp is injective on U, we have $\Lambda \cap U = \{0\}$. If $H, H' \in \Lambda$ and $H' \in H + U$, then $H' - H \in \Lambda \cap U$, so $H' = H$. Hence Λ is discrete. The discrete-subgroup lemma, Lemma 7.2, gives independent generators $v_1, \dots, v_k$ of Λ.

Full rank. Suppose $k < n$. Choose a nonzero linear map $p\colon \mathfrak{t} \to \mathbb{R}$ with $p(v_1) = \cdots = p(v_k) = 0$, and define

$$f\colon T \to \mathbb{R}, \qquad f(\exp H) = p(H).$$

It is well defined because $p(\Lambda) = 0$. It is continuous: near $\exp H_1$ we have $f(\exp H_1 \exp Y) = p(H_1) + p(Y)$ for $Y \in U$, and $Y \mapsto \exp H_1 \exp Y$ is a homeomorphism of U onto a neighbourhood of $\exp H_1$. So $\mathbb{R} = p(\mathfrak{t}) = f(T)$ is compact, which is false. Hence $k = n$.

Isomorphism. Changing x_j by an integer changes $\sum_j x_j v_j$ by an element of Λ. So Φ is a well-defined homomorphism. It is continuous, because near any point it is the composite of a local inverse of $x \mapsto (e^{2\pi i x_j})_j$ with the continuous map $x \mapsto \exp(\sum_j x_j v_j)$. It is bijective because $\ker\exp = \bigoplus_j \mathbb{Z} v_j$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism. Continuous homomorphisms of Lie groups are smooth (Lecture 3), so Φ and Φ^{-1} are smooth. ∎

Lecture 6 defined a torus as a compact connected abelian Lie group. The theorem shows that such a group is a torus in the sense above, and the converse is clear. So the two definitions agree. The theorem also proves the statement $T \cong \mathfrak{t}/\Lambda$ cited in Lecture 6, and it uses nothing from that lecture.

7.2 Characters and cocharacters

Definition 7.4. A *character* of a torus T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. A *cocharacter* is a continuous homomorphism $\lambda\colon \mathrm{U}(1) \to T$. Under pointwise multiplication, the characters form a group $X^*(T)$ and the cocharacters form a group $X_*(T)$.

These characters are one-dimensional. Trace characters return in Lecture 9.

Theorem 7.5 (Torus character criterion). *Let* T *be a torus with* $\Lambda = \ker\exp$, *and let* $\ell \in \mathfrak{t}^*$. *There is a character* χ_ℓ *with*

$$\chi_\ell(\exp H) = e^{i\ell(H)} \qquad (H \in \mathfrak{t})$$

if and only if $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. *Every character equals* χ_ℓ *for exactly one such* ℓ, *and* $d\chi_\ell = i\ell$.

Hence $\ell \mapsto \chi_\ell$ is an isomorphism from $\{\ell \in \mathfrak{t}^* : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ onto $X^*(T)$.

Proof. Suppose $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. The homomorphism $H \mapsto e^{i\ell(H)}$ from $\mathfrak{t}$ to $\mathrm{U}(1)$ is constant on cosets of Λ. Since exp is onto with kernel Λ, it defines a homomorphism χ_ℓ on T. Continuity follows as for f above. Conversely, if χ_ℓ exists, then $1 = \chi_\ell(\exp H) = e^{i\ell(H)}$ for $H \in \Lambda$.

Now let χ be any character. It is smooth (Lecture 3). Its differential maps $\mathfrak{t}$ to $\operatorname{Lie}\mathrm{U}(1) = i\mathbb{R}$, so $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$. Naturality of exp (Lecture 2) gives

$$\chi(\exp H) = \exp\bigl(d\chi(H)\bigr) = e^{i\ell(H)}.$$

For $H \in \Lambda$ the left side is $\chi(1) = 1$, so $\ell(H) \in 2\pi\mathbb{Z}$. Thus $\chi = \chi_\ell$. The form ℓ is unique, because $i\ell(H)$ is the derivative of $s \mapsto \chi(\exp sH)$ at $s = 0$. Finally, $\chi_\ell\chi_{\ell'} = \chi_{\ell+\ell'}$ by the formula. ∎

For $H \in \Lambda$ define the cocharacter

$$\lambda_H(e^{i\theta}) = \exp\Bigl(\frac{\theta}{2\pi}H\Bigr).$$

Replacing θ by $\theta+2\pi$ multiplies the right side by $\exp H = 1$, so λ_H is well defined. Exercise 7.1 shows that $H \mapsto \lambda_H$ is an isomorphism $\Lambda \to X_*(T)$. Since $\chi_\ell(\lambda_H(e^{i\theta})) = e^{i\theta\ell(H)/2\pi}$,

$$\chi_\ell\bigl(\lambda_H(z)\bigr) = z^{\langle \ell, H\rangle}, \qquad \langle \ell, H\rangle = \frac{\ell(H)}{2\pi} \in \mathbb{Z}.$$

So $X^*(T)$ is the dual lattice of $X_*(T) \cong \Lambda$. In crystal physics terms, $X^*(T)$ is the reciprocal lattice of Λ, in the convention $k \cdot R \in 2\pi\mathbb{Z}$, and χ_ℓ is a plane wave that is periodic on Λ. From now on we identify $X^*(T)$ with $\{\ell : \ell(\Lambda) \subset 2\pi\mathbb{Z}\}$ and $X_*(T)$ with Λ.

A finite-dimensional representation of T is unitary for an averaged inner product (Lecture 5). The operators of the representation commute and are unitary, so they diagonalise simultaneously. Hence the representation space is a direct sum of lines on which $\exp H$ acts by $e^{i\ell(H)}$. These $\ell \in X^*(T)$ are the *weights* of the representation.

Example 7.6. Let $T \subset \mathrm{U}(n)$ be the diagonal torus. Then $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, \dots, i\theta_n) : \theta \in \mathbb{R}^n\}$ and $\Lambda = \{2\pi i \operatorname{diag}(k_1, \dots, k_n) : k \in \mathbb{Z}^n\}$. Put $\ell_m(\operatorname{diag}(i\theta)) = \sum_j m_j\theta_j$. Then $\ell_m \in X^*(T)$ if and only if $m \in \mathbb{Z}^n$, and

$$\chi_{\ell_m}\bigl(\operatorname{diag}(z_1, \dots, z_n)\bigr) = z_1^{m_1} \cdots z_n^{m_n}.$$

For $H = 2\pi i \operatorname{diag}(k)$ we get $\lambda_H(z) = \operatorname{diag}(z^{k_1}, \dots, z^{k_n})$ and $\langle \ell_m, H\rangle = \sum_j m_j k_j$.

7.3 Roots, coroots and the centre

Let K be a compact connected Lie group with maximal torus T (Lecture 6). We use the set $R \subset \mathfrak{t}^*$ of real roots from Lecture 6, and we import the root data. A root α has the root space $\mathfrak{g}_\alpha = \{X \in \mathfrak{k}_{\mathbb{C}} : [H, X] = i\alpha(H)X \text{ for } H \in \mathfrak{t}\}$, and

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \operatorname{Ad}(\exp H)X = e^{i\alpha(H)}X \quad (X \in \mathfrak{g}_\alpha,\ H \in \mathfrak{t}).$$

Each root α has a coroot H_α. It is the unique element

$$H_\alpha \in \mathfrak{t} \cap [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \qquad \text{with} \qquad \alpha(H_\alpha) = 2.$$

The condition $\alpha(H) = 2$ alone does not fix H once $\dim \mathfrak{t} \geq 2$, because then $\ker \alpha \neq 0$. There is a homomorphism $\varphi_\alpha \colon \mathrm{SU}(2) \to K$ with $d\varphi_\alpha(H_0) = H_\alpha$, where $H_0 = \mathrm{diag}(i, -i)$ is the matrix A of Lecture 1 and $-2X_3$ of Lecture 4. Its differential is the compact form of a root $\mathfrak{sl}_2$-triple (H, Chapter 11). It integrates because $\mathrm{SU}(2)$ is simply connected (Lecture 4).

The Lie Algebras course works with complex-linear forms. Its coroot h_α spans the line $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}]$, and $\mathfrak{t} \cap \mathbb{C}h_\alpha = \mathbb{R}H_\alpha$. The following table compares the two conventions, for $H \in \mathfrak{t}$.

Object	Lie Algebras course	Lectures 6 to 9
root	complex-linear $\mu = i\alpha$ on $\mathfrak{t}_\mathbb{C}$	real α on $\mathfrak{t}$
eigenvalue of $\mathrm{ad}\, H$ on $\mathfrak{g}_\alpha$	$\mu(H)$	$i\alpha(H)$
coroot	h_α, with $\mu(h_\alpha) = 2$	$H_\alpha = ih_\alpha$, with $\alpha(H_\alpha) = 2$

Definition 7.7. Define subgroups Q and P of $\mathfrak{t}^*$, and $Q^\vee$ and $P^\vee$ of $\mathfrak{t}$, by

$$Q = \mathbb{Z}R, \qquad P = \{\ell \in \mathfrak{t}^* : \ell(H_\alpha) \in \mathbb{Z} \text{ for all } \alpha\},$$
$$Q^\vee = \mathbb{Z}\{2\pi H_\alpha : \alpha \in R\}, \quad P^\vee = \{H \in \mathfrak{t} : \alpha(H) \in 2\pi\mathbb{Z} \text{ for all } \alpha\}.$$

A real form $\ell \in \mathfrak{t}^*$ is the complex weight $\mu = i\ell$ of the Lie Algebras course. With $h_\alpha = -iH_\alpha$ we get $\mu(h_\alpha) = \ell(H_\alpha)$. When $\mathfrak{k}$ is semisimple, P is the weight lattice of that course, written in real form.

The factors 2π in $Q^\vee$ and $P^\vee$ fit the convention $\Lambda = \ker \exp$, in which characters satisfy $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. Some texts use $\{H : \exp(2\pi H) = 1\}$ in place of Λ, and write $\chi(\exp 2\pi H) = e^{2\pi i \ell(H)}$. This divides the lattices Λ, $Q^\vee$ and $P^\vee$ in $\mathfrak{t}$ by 2π. The weight lattices $X^*(T)$, Q and P are unchanged.

For example, take $K = \mathrm{SU}(2)$, with $\mathfrak{t} = \mathbb{R}H_0$. Since $\exp(sH_0) = \mathrm{diag}(e^{is}, e^{-is})$, we have $\Lambda = 2\pi\mathbb{Z}H_0$. Since $[H_0, E_{12}] = 2iE_{12}$, the positive root has $\alpha(H_0) = 2$, so $H_\alpha = H_0$. Hence $Q^\vee = 2\pi\mathbb{Z}H_0$ and $P^\vee = \pi\mathbb{Z}H_0$. Identifying ℓ with $m = \ell(H_0)$, we get $Q = 2\mathbb{Z}$ and $P = \mathbb{Z}$.

The covering of Lecture 4 identifies the Lie algebra of $\mathrm{SO}(3)$ with $\mathfrak{su}(2)$. The element sH_0 then exponentiates to the identity of $\mathrm{SO}(3)$ exactly when $\exp(sH_0) = \pm I$ in $\mathrm{SU}(2)$. So $\mathrm{SO}(3)$ has $\Lambda = \pi\mathbb{Z}H_0$. These are the only two lattices between $Q^\vee$ and $P^\vee$. The torus character criterion gives $X^*(T) = \mathbb{Z}$ for $\mathrm{SU}(2)$ and $X^*(T) = 2\mathbb{Z}$ for $\mathrm{SO}(3)$. So the larger kernel lattice has the smaller character lattice. The table collects these values, and Fig. 7.1 draws them.

Lattice	Lives in	Defined by	$\mathrm{SU}(2)$	$\mathrm{SO}(3)$
$Q^\vee$	$\mathfrak{t}$	$\mathbb{Z}\{2\pi H_\alpha\}$	$2\pi\mathbb{Z}H_0$	$2\pi\mathbb{Z}H_0$
Λ	$\mathfrak{t}$	$\ker \exp$	$2\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
$P^\vee$	$\mathfrak{t}$	$\alpha(H) \in 2\pi\mathbb{Z}$ for all α	$\pi\mathbb{Z}H_0$	$\pi\mathbb{Z}H_0$
Q	$\mathfrak{t}^*$	$\mathbb{Z}R$	$m \in 2\mathbb{Z}$	$m \in 2\mathbb{Z}$
$X^*(T)$	$\mathfrak{t}^*$	$\ell(\Lambda) \subset 2\pi\mathbb{Z}$	$m \in \mathbb{Z}$	$m \in 2\mathbb{Z}$
P	$\mathfrak{t}^*$	$\ell(H_\alpha) \in \mathbb{Z}$ for all α	$m \in \mathbb{Z}$	$m \in \mathbb{Z}$

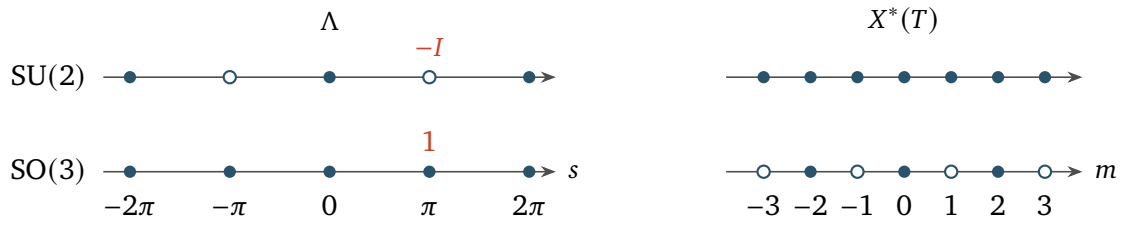


Figure 7.1: The kernel lattice Λ on the left, in the coordinate s of sH_0, and the character lattice $X^*(T)$ on the right, in the coordinate $m = \ell(H_0)$. Filled dots belong to the lattice. A value of m is filled exactly when $ms \in 2\pi\mathbb{Z}$ for every filled s in the same row. The element πH_0 exponentiates to $-I$ in SU(2) and to 1 in SO(3).

Proposition 7.8 (Lattice chain)**.**

$$Q^\vee \subset \Lambda \subset P^\vee, \qquad Q \subset X^*(T) \subset P.$$

Proof. Let $H \in \Lambda$ and $0 \neq X \in \mathfrak{g}_\alpha$. Then $X = \mathrm{Ad}(\exp H)X = e^{i\alpha(H)}X$, so $\alpha(H) \in 2\pi\mathbb{Z}$. Hence $\Lambda \subset P^\vee$. The same computation shows $\alpha \in X^*(T)$, so $Q \subset X^*(T)$. Next, naturality of exp gives

$$\exp(2\pi H_\alpha) = \varphi_\alpha\big(\exp(2\pi H_0)\big) = \varphi_\alpha\begin{pmatrix} e^{2\pi i} & 0 \\ 0 & e^{-2\pi i}\end{pmatrix} = \varphi_\alpha(I) = 1.$$

So $Q^\vee \subset \Lambda$. Finally, if $\ell \in X^*(T)$, then $\ell(2\pi H_\alpha) \in 2\pi\mathbb{Z}$, so $\ell(H_\alpha) \in \mathbb{Z}$ and $\ell \in P$. ∎

Theorem 7.9 (Centre)**.** *Let K be compact connected with maximal torus T. Then $Z(K) \subset T$, and* exp *restricts to a surjective homomorphism $P^\vee \to Z(K)$ with kernel Λ. Thus $Z(K) \cong P^\vee/\Lambda$.*

Proof. Let $z \in Z(K)$. By Lecture 6, $z = gtg^{-1}$ for some $g \in K$ and $t \in T$. Since z is central, $t = g^{-1}zg = z$, so $z \in T$.

Next we show that $g \in K$ is central if and only if $\mathrm{Ad}(g) = \mathrm{id}$. If g is central, differentiating $g\exp(sX)g^{-1} = \exp(sX)$ at $s = 0$ gives $\mathrm{Ad}(g) = \mathrm{id}$. If $\mathrm{Ad}(g) = \mathrm{id}$, then $g\exp(X)g^{-1} = \exp(\mathrm{Ad}(g)X) = \exp X$. The elements $\exp X$ generate the connected group K (Lecture 2), so g is central.

Now write $z = \exp H$ with $H \in \mathfrak{t}$, using the surjectivity of exp in the torus classification, Theorem 7.3. The operator $\mathrm{Ad}(\exp H) = e^{\mathrm{ad}\,H}$ is the identity on $\mathfrak{t}_\mathbb{C}$ and multiplication by $e^{i\alpha(H)}$ on $\mathfrak{g}_\alpha$. Hence, by the criterion just proved, $\exp H$ is central exactly when $H \in P^\vee$. The kernel of exp on $P^\vee$ is $\Lambda \cap P^\vee = \Lambda$. ∎

For the fundamental group we cite the following theorem.

Theorem 7.10 (Fundamental group; H, Chapter 13; BD, Section V.7)**.** *Let K be a compact connected Lie group with maximal torus T. Then*

$$\pi_1(K) \cong \Lambda/Q^\vee = X_*(T)/Q^\vee.$$

Call K *semisimple* if $\mathfrak{k}$ is semisimple. Then R spans $\mathfrak{t}^*$, and the simple roots form a basis of $\mathfrak{t}^*$. The simple coroots form a basis of $\mathfrak{t}$ (the Lie Algebras course, via $h_\alpha = -iH_\alpha$). So all four

subgroups are lattices, and the inclusions above have finite index. Weyl's theorem states that a compact connected semisimple group has finite fundamental group and compact universal cover (H, Chapter 13; Knapp, Chapter IV).

7.4 Global forms

Semisimple groups

By Weyl's theorem, the universal cover of our compact connected semisimple K is compact. Let K_{sc} be a compact, connected, simply connected, semisimple Lie group with Lie algebra $\mathfrak{k}$ and maximal torus T_{sc}. The fundamental group theorem, Theorem 7.10, gives $\Lambda_{sc} = Q^\vee$. The torus character criterion and the centre theorem, Theorems 7.5 and 7.9, then give

$$X^*(T_{sc}) = P, \qquad Z(K_{sc}) \cong P^\vee/Q^\vee.$$

Proposition 7.11 (Forms of a semisimple algebra). *(a) Every connected Lie group with Lie algebra $\mathfrak{k}$ is isomorphic to K_{sc}/C for a subgroup $C \subset Z(K_{sc})$.*

(b) For $K = K_{sc}/C$, the image T of T_{sc} is a maximal torus of K, with kernel lattice

$$\Lambda_C = \{H \in \mathfrak{t} : \exp_{sc} H \in C\}, \qquad \Lambda_C/Q^\vee \cong C.$$

The map $C \mapsto \Lambda_C$ is a bijection from subgroups of $Z(K_{sc})$ to lattices Λ with $Q^\vee \subset \Lambda \subset P^\vee$.

(c) $X^*(T) = \{\ell \in P : \chi_\ell(c) = 1 \text{ for all } c \in C\}$.

Proof. (a) By Lecture 4, such a group is isomorphic to K_{sc}/C with C discrete and normal. Such a subgroup C is central.

(b) First, T is a maximal torus with kernel lattice Λ_C. The covering map $p\colon K_{sc} \to K$ is a local isomorphism. So $\operatorname{Lie} T = \mathfrak{t}$ and $\exp_K = p \circ \exp_{sc}$. Hence $\exp_K H = 1$ exactly when $\exp_{sc} H \in C$. A larger torus would have a larger abelian Lie algebra, but $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$ (Lecture 6). So T is maximal.

Next, we prove $Q^\vee \subset \Lambda_C \subset P^\vee$ and $\Lambda_C/Q^\vee \cong C$. Since $C \subset Z(K_{sc})$, the centre theorem, Theorem 7.9, gives $Q^\vee \subset \Lambda_C \subset P^\vee$. The map $\exp_{sc}\colon \Lambda_C \to C$ is onto, since $C \subset T_{sc} = \exp_{sc}(\mathfrak{t})$. Its kernel is $Q^\vee$, so $\Lambda_C/Q^\vee \cong C$.

Conversely, let $Q^\vee \subset \Lambda \subset P^\vee$, and put $C = \exp_{sc}(\Lambda)$. By the centre theorem, Theorem 7.9, C is a subgroup of $Z(K_{sc})$. Clearly $\Lambda \subset \Lambda_C$. For the reverse inclusion, let $H \in \Lambda_C$. Then $\exp_{sc} H \in C = \exp_{sc}(\Lambda)$, so $\exp_{sc} H = \exp_{sc} H'$ for some $H' \in \Lambda$. Hence $H - H'$ lies in the kernel of $\exp_{sc}$ on $\mathfrak{t}$, which is $Q^\vee \subset \Lambda$. So $H \in \Lambda$, and $\Lambda_C = \Lambda$. We also showed $\exp_{sc}(\Lambda_C) = C$ for every C. So $C \mapsto \Lambda_C$ and $\Lambda \mapsto \exp_{sc}(\Lambda)$ are inverse to each other.

(c) A character of $T = T_{sc}/C$ is a character of T_{sc} that is trivial on C. ∎

Remark 7.12. Distinct lattices can give isomorphic groups. Let σ be an automorphism of $\mathfrak{k}$ with $\sigma(\mathfrak{t}) = \mathfrak{t}$. It integrates to an automorphism Σ of K_{sc} (Lecture 4). Since $\Sigma(\exp H) = \exp(\sigma H)$, we get $\Sigma(Z(K_{sc})) = Z(K_{sc})$ and $\Lambda_{\Sigma(C)} = \sigma(\Lambda_C)$. Hence $K_{sc}/C \cong K_{sc}/\Sigma(C)$. Exercise 7.5 gives an example. We do not discuss the converse.

The *simply connected form* is $C = 1$, with $\Lambda = Q^\vee$ and $X^*(T) = P$. The *adjoint form* is $K_{ad} = K_{sc}/Z(K_{sc})$, with $\Lambda = P^\vee$. Its centre $P^\vee/P^\vee$ is trivial by the centre theorem, Theorem 7.9. Write T_{ad} for the image of T_{sc} in K_{ad}. Every root is an integer combination of the simple roots α_i (the Lie Algebras course). So the lattice $P^\vee$ is defined by the conditions $\alpha_i(H) \in 2\pi\mathbb{Z}$ for the simple roots alone. Since the simple roots form a basis of $\mathfrak{t}^*$, we get $X^*(T_{ad}) = \mathbb{Z}\{\alpha_i\} = Q$.

Proposition 7.13 (Descent by central character)**.** *Let V be the irreducible representation of K_{sc} with dominant highest weight $\lambda \in P$, and let $C \subset Z(K_{sc})$. Then V descends to K_{sc}/C if and only if*

$$\chi_\lambda(c) = 1 \qquad \text{for all } c \in C.$$

The representation V exists by the highest-weight theorem of the Lie Algebras course and Lecture 4.

Proof. By Schur's lemma, a central element c acts on V by a scalar. On a highest weight vector this scalar is $\chi_\lambda(c)$. By Lecture 4, V descends to K_{sc}/C if and only if C acts trivially. ∎

Groups with a central torus

In U(2) the element $-I$ is both a scalar and an element of SU(2). A representation must act on it consistently. Exercise 4.3 did this by hand. We now state the condition for a general central torus.

If $\mathfrak{k}$ has nonzero centre, then $Z(K)$ is infinite and R does not span $\mathfrak{t}^*$. Moreover, $P^\vee$ contains the centre of $\mathfrak{k}$, so $P^\vee$ is not a lattice. The proposition on forms of a semisimple algebra, Proposition 7.11, does not apply. We use a structure theorem instead.

We recall the global structure theorem of Lecture 6, together with the splitting of that lecture.

Theorem 7.14 (Global structure; recalled from Lecture 6; BD, Section V.8; Knapp, Chapter IV)**.** *Let K be compact connected. Let T_0 be the identity component of $Z(K)$, with Lie algebra $\mathfrak{t}_0$. Let K_{sc} be the simply connected group with Lie algebra $[\mathfrak{k},\mathfrak{k}]$. Then T_0 is a torus, $\mathfrak{k} = \mathfrak{t}_0 \oplus [\mathfrak{k},\mathfrak{k}]$, the ideal $[\mathfrak{k},\mathfrak{k}]$ is semisimple, and K_{sc} is compact. Let $\varphi\colon K_{sc} \to K$ integrate the inclusion of $[\mathfrak{k},\mathfrak{k}]$. Then*

$$T_0 \times K_{sc} \to K, \qquad (t, g) \mapsto t\,\varphi(g),$$

is a surjective homomorphism with finite kernel F. Thus $K \cong (T_0 \times K_{sc})/F$.

Proposition 7.15. *In the global structure theorem, Theorem 7.14, let T_{sc} be a maximal torus of K_{sc} with Lie algebra $\mathfrak{t}_{sc}$, and let T be the image of $T_0 \times T_{sc}$. Identify $\mathfrak{t}^* = \mathfrak{t}_0^* \oplus \mathfrak{t}_{sc}^*$, and let $P_{sc} \subset \mathfrak{t}_{sc}^*$ be the weight lattice of K_{sc}. Then $T \cong (T_0 \times T_{sc})/F$ is a maximal torus of K, and*

$$X^*(T) = \big\{(\ell_0, \ell) \in X^*(T_0) \times P_{sc} : \ \chi_{\ell_0}(t)\,\chi_\ell(g) = 1 \text{ for all } (t, g) \in F\big\}.$$

Proof. A finite normal subgroup of a connected group is central (Lecture 4). So $F \subset T_0 \times Z(K_{sc}) \subset T_0 \times T_{sc}$ by the centre theorem, Theorem 7.9. Hence $T \cong (T_0 \times T_{sc})/F$. Its Lie algebra $\mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$ is maximal abelian in $\mathfrak{k}$, because $\mathfrak{t}_{sc}$ is maximal abelian in $[\mathfrak{k},\mathfrak{k}]$. So T is a maximal torus.

Each root space satisfies $\mathfrak{g}_\alpha = [\mathfrak{t}, \mathfrak{g}_\alpha] \subset [\mathfrak{k}, \mathfrak{k}]_{\mathbb{C}}$. So $H_\alpha \in \mathfrak{t} \cap [\mathfrak{k}, \mathfrak{k}] = \mathfrak{t}_{sc}$, and the coroots of K are those of K_{sc}. Thus $Q^\vee \subset \mathfrak{t}_{sc}$ is the kernel lattice of T_{sc}. Put

$$\Lambda_0 = \ker\bigl(\exp|_{\mathfrak{t}_0}\bigr) = \Lambda \cap \mathfrak{t}_0.$$

The torus $T_0 \times T_{sc}$ has kernel lattice $\Lambda_0 \oplus Q^\vee$. By the torus character criterion, Theorem 7.5, its characters are the pairs with $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$ and $\ell \in P_{sc}$. The characters of T are those trivial on F. ∎

In other words, (ℓ_0, ℓ) is a character of T exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product $\chi_{\ell_0}\chi_\ell$ is trivial on F.

The group F is determined by Λ. The map $T_0 \times T_{sc} \to T$ has differential the identity of $\mathfrak{t} = \mathfrak{t}_0 \oplus \mathfrak{t}_{sc}$. By the torus classification, Theorem 7.3,

$$T_0 \times T_{sc} \cong \mathfrak{t}/(\Lambda_0 \oplus Q^\vee), \qquad T \cong \mathfrak{t}/\Lambda.$$

The map is induced by the identity of $\mathfrak{t}$, so its kernel is

$$F \cong \Lambda/(\Lambda_0 \oplus Q^\vee).$$

7.5 Examples

SU(2) and SO(3)

We use the values found after the definition of the lattices, and drawn in Fig. 7.1. For SU(2) we have $Q^\vee = \Lambda$. With $m = \ell(H_0)$,

$$P = \mathbb{Z}, \qquad Q = 2\mathbb{Z}, \qquad P^\vee = \pi\mathbb{Z}H_0, \qquad Z(\mathrm{SU}(2)) = \exp(\pi\mathbb{Z}H_0) = \{\pm I\}.$$

The adjoint form is $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). It has $\Lambda = \pi\mathbb{Z}H_0$ and

$$X^*(T) = \{m : m\pi \in 2\pi\mathbb{Z}\} = 2\mathbb{Z} = Q, \qquad \pi_1(\mathrm{SO}(3)) \cong \pi\mathbb{Z}/2\pi\mathbb{Z} \cong \mathbb{Z}/2.$$

The module V_m has dimension $m+1$ and weights $m, m-2, \dots, -m$. Physicists call it the spin-$m/2$ representation. Since $-I = \exp(\pi H_0)$, the element $-I$ acts on V_m by $e^{i\pi m} = (-1)^m$. So V_m descends to SO(3) exactly when m is even.

SU(3) and PSU(3)

For the diagonal torus of SU(3),

$$\mathfrak{t} = \Bigl\{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum_j \theta_j = 0\Bigr\}, \qquad \Lambda = \Bigl\{2\pi i\,\mathrm{diag}(k) : k \in \mathbb{Z}^3,\ \sum_j k_j = 0\Bigr\}.$$

The roots are $\alpha_{jk}(H) = \theta_j - \theta_k$ for $j \neq k$, with coroots $H_{jk} = i(E_{jj} - E_{kk})$. Indeed $[E_{jk}, E_{kj}] = E_{jj} - E_{kk}$ and $\alpha_{jk}(H_{jk}) = 2$. Hence $Q^\vee = \Lambda$. Write $\ell_m(H) = \sum_j m_j\theta_j$ with $m \in \mathbb{R}^3$. Adding a multiple of $(1,1,1)$ to m does not change ℓ_m, since $\sum_j \theta_j = 0$. The condition $\ell_m(\Lambda) \subset 2\pi\mathbb{Z}$ means $m_j - m_k \in \mathbb{Z}$, and this is also the condition for P.

Such a form has a representative $m \in \mathbb{Z}^3$, obtained by subtracting $m_3(1,1,1)$. Two integer representatives differ by an element of $\mathbb{Z}(1,1,1)$, which changes $m_1+m_2+m_3$ by a multiple of 3. Using integer representatives,

$$X^*(T) = P \cong \mathbb{Z}^3/\mathbb{Z}(1,1,1),$$
$$Q = \{[m] : m \in \mathbb{Z}^3,\ m_1+m_2+m_3 \equiv 0 \bmod 3\}, \qquad P/Q \cong \mathbb{Z}/3.$$

An element of $P^\vee$ has $\theta_j = \theta + 2\pi k_j$ with $3\theta \in 2\pi\mathbb{Z}$. So $Z(\mathrm{SU}(3)) = \{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$. The adjoint form $\mathrm{PSU}(3) = \mathrm{SU}(3)/Z(\mathrm{SU}(3))$ has $X^*(T) = Q$, trivial centre and $\pi_1 \cong P^\vee/Q^\vee \cong \mathbb{Z}/3$.

Use the simple roots α_{12}, α_{23}, with simple coroots H_{12}, H_{23}. The fundamental weights are then $\varpi_1 = \ell_{(1,0,0)}$ and $\varpi_2 = \ell_{(1,1,0)}$. So the highest weight $a\varpi_1 + b\varpi_2$ is $\ell_{(a+b,b,0)}$. Write $V(a,b)$ for the irreducible module with this highest weight. Take

$$H_\omega = \mathrm{diag}\Big(\frac{2\pi i}{3}, \frac{2\pi i}{3}, -\frac{4\pi i}{3}\Big), \qquad \exp H_\omega = \mathrm{diag}(\omega, \omega, \omega^{-2}) = \omega I.$$

Then $\ell_{(a+b,b,0)}(H_\omega) = 2\pi(a+2b)/3$, so ωI acts on $V(a,b)$ by ω^{a+2b}. Thus $V(a,b)$ descends to PSU(3) if and only if $a \equiv b \bmod 3$. The adjoint module $V(1,1)$ descends. The module $\mathbb{C}^3 = V(1,0)$ does not. In physics terms, the number $a-b$ modulo 3 is the triality, the module $V(1,1)$ is the octet and $V(1,0)$ is the quark triplet.

Figure 7.2 draws the same data in the plane $\mathfrak{t}$ and in the dual plane $\mathfrak{t}^*$. In $\mathfrak{t}$ the kernel lattice of SU(3) is the smaller lattice $Q^\vee$, and that of PSU(3) is the larger lattice $P^\vee$. In $\mathfrak{t}^*$ the order reverses. The character lattice of SU(3) is the larger lattice P, and that of PSU(3) is the smaller lattice Q. The same three marks appear in both planes.

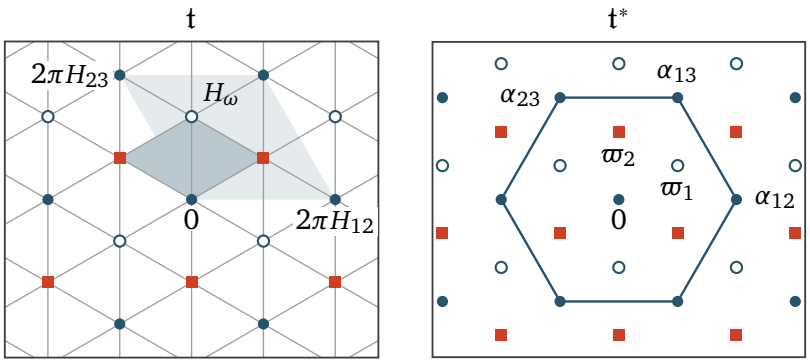


Figure 7.2: The plane $\mathfrak{t}$ of SU(3), with the form $-\mathrm{tr}(XY)$, on the left, and the dual plane $\mathfrak{t}^*$ on the right. Their scales differ. On the left, a disc, circle or square marks a point whose exponential is I, ωI or $\omega^2 I$. Discs form $Q^\vee$, and all marks form $P^\vee$. The thin lines are $\alpha_{jk}(H) \in 2\pi\mathbb{Z}$, and the shaded cells are cells of $Q^\vee$ and of $P^\vee$. On the right, a mark shows the scalar by which ωI acts on a weight: 1, ω or ω^2. Discs form Q, and all marks form P. The hexagon joins the roots, and $\alpha_{13} = \alpha_{12} + \alpha_{23} = \varpi_1 + \varpi_2$.

The group U(2)

Let $T_0 = \{zI : z \in \mathrm{U}(1)\}$ and $K_{sc} = \mathrm{SU}(2)$. The map

$$\Psi\colon\ \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2), \qquad \Psi(z,g) = zg,$$

is onto. Given u, choose z with $z^2 = \det u$. Then $z^{-1}u \in \mathrm{SU}(2)$. If $zg = I$, then $\det(z^{-1}I) = z^{-2} = 1$, so $z = \pm 1$. Hence

$$\mathrm{U}(2) \cong \big(\mathrm{U}(1) \times \mathrm{SU}(2)\big)/F, \qquad F = \{(1, I), (-1, -I)\}.$$

The lattice statements for semisimple groups fail for U(2) itself. In the semisimple case the roots span $\mathfrak{t}^*$, and Q, P, $Q^\vee$ and $P^\vee$ are lattices. By the computation for diagonal tori, Example 7.6, the diagonal torus of U(2) has $X^*(T) = \mathbb{Z}^2$. The roots are $\pm\alpha$ with $\alpha(\mathrm{diag}(i\theta)) = \theta_1 - \theta_2$, so $Q = \mathbb{Z}(1, -1)$ spans only a line. The set $P^\vee = \{\theta_1 - \theta_2 \in 2\pi\mathbb{Z}\}$ is not a lattice, and $P^\vee/\Lambda \cong \mathrm{U}(1)$ is the group of scalars.

The fundamental group theorem, Theorem 7.10, still applies. Since $[E_{12}, E_{21}] = \mathrm{diag}(1, -1)$, the coroot is $H_\alpha = i\,\mathrm{diag}(1, -1)$, and

$$\pi_1(\mathrm{U}(2)) \cong \mathbb{Z}^2/\mathbb{Z}(1, -1) \cong \mathbb{Z}, \qquad (k_1, k_2) \mapsto k_1 + k_2.$$

Here $\Lambda_0 = 2\pi i\mathbb{Z}I$, and $\Lambda_0 \oplus Q^\vee$ consists of the $2\pi i\,\mathrm{diag}(k_1, k_2)$ with $k_1 \equiv k_2 \bmod 2$. So $\Lambda/(\Lambda_0 \oplus Q^\vee) \cong \mathbb{Z}/2 \cong F$, as predicted.

Now we compute the central compatibility constraint. Recall from Lecture 4 that Π_m denotes the representation of SU(2) on V_m. The element $\mathrm{diag}(w, w^{-1})$, with $w \in \mathrm{U}(1)$, acts on a vector of weight j by w^j. For $k \in \mathbb{Z}$ and $m \geq 0$ define

$$\Pi_{k,m}(z, g) = z^k\,\Pi_m(g) \qquad \text{on } V_m.$$

The representation $\Pi_{k,m}$ is irreducible, already on SU(2). By Lecture 4 it descends to U(2) exactly when it is trivial on F. Since

$$\Pi_{k,m}(-1, -I) = (-1)^k(-1)^m\ \mathrm{id},$$

the constraint is

$$k \equiv m \pmod 2.$$

The descended representation is $\Pi(u) = z^k\Pi_m(z^{-1}u)$ with $z^2 = \det u$. The two choices of z give the same operator exactly because of the constraint. Exercise 7.3 shows that every irreducible representation of U(2) arises in this way.

The description of $X^*(T)$ for a central torus, Proposition 7.15, gives the same constraint. Here k is the exponent on scalars, and j runs over the weights of V_m. Put $\ell_0(isI) = ks$ and $\ell(sH_0) = js$. In T_0, the scalar $z = -1$ is the matrix $-I = \exp(i\pi I)$. So $(-1, -I) = (\exp(i\pi I), \exp(\pi H_0))$, and the pair (ℓ_0, ℓ) is a character of T exactly when $k + j$ is even. Every weight j of V_m satisfies $j \equiv m \bmod 2$, so this condition is the constraint found above.

Assume $k \equiv m \pmod 2$, so that $\Pi_{k,m}$ descends to U(2). Every weight j of V_m satisfies $j \equiv m \pmod 2$, so $k - j$ is even and the exponents below are integers. On $u = \mathrm{diag}(z_1, z_2)$ with $z^2 = z_1z_2$, a vector of weight j is multiplied by

$$z^k\left(\frac{z_1}{z}\right)^j = (z_1z_2)^{(k-j)/2}z_1^{\,j} = z_1^{(k+j)/2}\,z_2^{(k-j)/2}.$$

So $\Pi_{k,m}$ has highest weight $\left(\frac{k+m}{2}, \frac{k-m}{2}\right) \in \mathbb{Z}^2$. For example, $\mathbb{C}^2$ is $(k, m) = (1, 1)$, det is $(2, 0)$, and Ad on the complexification $\mathfrak{su}(2)_\mathbb{C}$ is $(0, 2)$.

Self-checks

1. Does the maximal torus of SO(3) have a character with $\ell(H_0) = 1$?

   Answer. No. Here $\Lambda = \pi\mathbb{Z}H_0$ and $\ell(\pi H_0) = \pi \notin 2\pi\mathbb{Z}$.

2. Compute $Z(\mathrm{PSU}(3))$ and $\pi_1(\mathrm{PSU}(3))$ from the lattices.

   Answer. Here $\Lambda = P^\vee$. So $Z = P^\vee/P^\vee = 1$ and $\pi_1 = P^\vee/Q^\vee \cong \mathbb{Z}/3$.

3. Is there a representation of U(2) on which scalars act by z^2 and SU(2) acts by V_1?

   Answer. No. The element $-I$ is the scalar -1, acting by $(-1)^2 = 1$. The element $-I$ also lies in SU(2), where it acts on V_1 by -1. The two actions disagree.

Summary

A compact connected abelian Lie group is $\mathfrak{t}/\Lambda$, and its characters are the ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. For compact connected K, the kernel lattice satisfies $Q^\vee \subset \Lambda \subset P^\vee$. The kernel lattice gives $Z(K) \cong P^\vee/\Lambda$ and, by a cited theorem, $\pi_1(K) \cong \Lambda/Q^\vee$.

For semisimple $\mathfrak{k}$, the connected forms correspond to lattices between $Q^\vee$ and $P^\vee$. Lattices related by an automorphism of $\mathfrak{k}$ preserving $\mathfrak{t}$ give isomorphic groups. With a central torus, write $K \cong (T_0 \times K_{sc})/F$, where $F \cong \Lambda/(\Lambda_0 \oplus Q^\vee)$. A pair (ℓ_0, ℓ) is then a character of the maximal torus exactly when $\ell_0(\Lambda_0) \subset 2\pi\mathbb{Z}$, $\ell \in P_{sc}$, and the product is trivial on F.

Lecture 8 uses $X^*(T)$ to state the classification of representations by highest weights.

Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations* (H), Chapter 13. T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (BD), Sections V.7–8. A. Knapp, *Lie Groups Beyond an Introduction*, Chapter IV. For the discrete-subgroup lemma, Lemma 7.2, see BD, Section I.3, Lemma (3.8).

Lecture 8

Highest weights of representations of compact groups

Objectives. *The Lie Algebras course classified finite-dimensional irreducible modules of semisimple algebras by dominant integral weights. This lecture adds the action of the centre of the Lie algebra and decides which of these modules are representations of a given compact connected Lie group K. These are exactly the modules whose highest weight is a character of a maximal torus.*

Lecture 7 decided, for a given highest-weight module, whether it descends. It did not show that every irreducible representation of K is obtained in this way, or that one is obtained only once. This lecture proves both.

By the end of this lecture you will be able to:

- Prove that highest weights give a bijection $\mathrm{Irr}(K) \to X^*(T)^+$, assuming cited structure results.
- Decide descent to K_{sc}/F by evaluating a highest weight λ on the elements $H \in \mathfrak{t}$ with $\exp H \in F$.
- Treat a central torus by an integral character and a finite compatibility condition.
- Explain why some modules of a reductive Lie algebra come from no compact group.

8.1 Torus characters and dominance

Throughout, K is a compact connected Lie group with Lie algebra $\mathfrak{k}$. Let T be a maximal torus with Lie algebra $\mathfrak{t}$ (Lecture 6). Put $\Lambda = \ker(\exp\colon \mathfrak{t} \to T)$. A *character* of T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. Its derivative has the form $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$, and

$$\chi(\exp H) = e^{i\ell(H)} \quad (H \in \mathfrak{t}), \qquad \ell(\Lambda) \subset 2\pi\mathbb{Z}. \tag{8.1}$$

Conversely, $\exp\colon \mathfrak{t} \to T$ is a surjective homomorphism with kernel Λ. So each ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$ defines a character. Let $X^*(T)$ be the lattice of these ℓ.

Let Π be a representation of K on V. By Lecture 5, V has a K-invariant inner product. So the operators $\Pi(t)$ with $t \in T$ are commuting unitary operators. Hence these operators have a common orthonormal eigenbasis. For a common eigenvector v, write $\Pi(t)v = c(t)v$. Then c

is a character of T, so $c(\exp H) = e^{i\ell(H)}$ for a unique $\ell \in X^*(T)$. Hence V is the direct sum of the weight spaces

$$V_\ell = \{v \in V : \ \Pi(\exp H)v = e^{i\ell(H)}v \ \text{ for all } H \in \mathfrak{t}\}, \qquad \ell \in X^*(T).$$

Differentiating $\Pi(\exp tH)v = e^{it\ell(H)}v$ at $t = 0$ gives $d\Pi(H)v = i\ell(H)v$ for $v \in V_\ell$. The complex-linear extension of $i\ell$ to $\mathfrak{t}_\mathbb{C}$ is a weight in the sense of the Lie Algebras course. We identify ℓ with this weight.

Roots are the nonzero weights of Ad on $\mathfrak{k}_\mathbb{C}$. Thus $[H, X] = i\alpha(H)X$ for $H \in \mathfrak{t}$ and X in the root space of α. In particular, roots lie in $X^*(T)$.

Fix positive roots R^+ as in [I]. Put

$$\mathbb{N}R^+ = \Big\{ \sum_{\alpha \in R^+} n_\alpha \alpha : \ n_\alpha \in \mathbb{Z}_{\geq 0} \Big\}, \qquad \ell' \preceq \ell \iff \ell - \ell' \in \mathbb{N}R^+.$$

There is $H_+ \in \mathfrak{t}$ with $\alpha(H_+) > 0$ for all $\alpha \in R^+$. So every nonzero element of $\mathbb{N}R^+$ is positive at H_+. Hence $\ell' \preceq \ell$ and $\ell \preceq \ell'$ force $\ell = \ell'$. A *highest weight* of Π is a weight ℓ of Π with $\ell' \preceq \ell$ for every weight ℓ' of Π. It is unique if it exists.

Write $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{k}_s$, with $\mathfrak{z}$ the centre and $\mathfrak{k}_s = [\mathfrak{k}, \mathfrak{k}]$ semisimple. Then $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ with $\mathfrak{t}_s = \mathfrak{t} \cap \mathfrak{k}_s$ (Lecture 6). Roots vanish on $\mathfrak{z}$, since $\operatorname{ad} Z = 0$ for $Z \in \mathfrak{z}$.

The highest-weight theorem of the Lie Algebras course is stated for a Cartan subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. We now check that $\mathfrak{t}_{s,\mathbb{C}}$ is one, so that the theorem applies here.

The algebra $\mathfrak{t}_s$ is maximal abelian in $\mathfrak{k}_s$. Indeed, let $\mathfrak{a} \supset \mathfrak{t}_s$ be an abelian subalgebra of $\mathfrak{k}_s$. Since $\mathfrak{z}$ is central, $\mathfrak{z} \oplus \mathfrak{a}$ is abelian and contains $\mathfrak{t}$. The Lie algebra of a maximal torus is maximal abelian (Lecture 6), so $\mathfrak{z} \oplus \mathfrak{a} = \mathfrak{t}$. Hence $\mathfrak{a} \subset \mathfrak{t} \cap \mathfrak{k}_s = \mathfrak{t}_s$.

The adjoint operators of elements of $\mathfrak{t}_s$ commute and are skew-adjoint, hence simultaneously diagonalisable over $\mathbb{C}$. So the complexification $\mathfrak{t}_{s,\mathbb{C}}$ is toral. It is also maximal toral. If $X \in \mathfrak{k}_s$ commutes with $\mathfrak{t}_s$, then $\mathfrak{t}_s + \mathbb{R}X$ is abelian, so $X \in \mathfrak{t}_s$. Now let $X, Y \in \mathfrak{k}_s$ with $[H, X + iY] = 0$ for all $H \in \mathfrak{t}_s$. The brackets $[H, X]$ and $[H, Y]$ lie in $\mathfrak{k}_s$, so both vanish, and $X, Y \in \mathfrak{t}_s$. Hence $\mathfrak{t}_{s,\mathbb{C}}$ is its own centraliser in $\mathfrak{k}_{s,\mathbb{C}}$. A toral subalgebra $\mathfrak{h} \supset \mathfrak{t}_{s,\mathbb{C}}$ is abelian [I], so $\mathfrak{h}$ lies in this centraliser and $\mathfrak{h} = \mathfrak{t}_{s,\mathbb{C}}$. Thus $\mathfrak{t}_{s,\mathbb{C}}$ is a maximal toral, or Cartan, subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. This is the Cartan subalgebra used in [I].

For a root α, [I] supplies a coroot $h_\alpha \in \mathfrak{t}_{s,\mathbb{C}}$. The corresponding root in the sense of the Lie Algebras course is the complex-linear extension of $i\alpha$. This extension takes the value 2 at h_α. By Lecture 7, the real coroot is $H_\alpha = ih_\alpha \in \mathfrak{t}_s$. Since $h_\alpha = -iH_\alpha$, the value of the complex-linear extension of $i\alpha$ at h_α equals $\alpha(H_\alpha)$. So

$$\alpha(H_\alpha) = 2, \qquad H_\alpha \in \mathfrak{t}_s.$$

In the same way, the complex-linear extension of $i\ell$ takes the value $\ell(H_\alpha)$ at h_α. So we put

$$\langle \ell, \alpha^\vee \rangle = \ell(H_\alpha), \qquad P = \{\lambda \in \mathfrak{t}_s^* : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \ \forall \alpha\},$$

$$X^*(T)^+ = \{\ell \in X^*(T) : \langle \ell, \alpha^\vee \rangle \geq 0 \ \forall \alpha \in R^+\}.$$

By Lecture 7, $2\pi H_\alpha \in \Lambda$. Hence every $\ell \in X^*(T)$ satisfies

$$\langle \ell, \alpha^\vee \rangle = \tfrac{1}{2\pi}\, \ell(2\pi H_\alpha) \in \mathbb{Z},$$

so $\ell|_{\mathfrak{t}_s} \in P$. Let Q be the root lattice, the lattice generated by the roots. Roots lie in $X^*(T)$, so for semisimple K we obtain $Q \subset X^*(T) \subset P$. If K is also simply connected, then Λ is generated as a lattice by the $2\pi H_\alpha$, and $X^*(T) = P$ (Lecture 7).

For SU(2), put $h = \operatorname{diag}(1, -1)$, so that ih is the matrix H_0 of Lecture 7. Then $\mathfrak{t} = \mathbb{R}\, ih$, with

$$\exp(\theta\, ih) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \qquad \Lambda = 2\pi\mathbb{Z}\, ih.$$

Since $[\theta\, ih, E_{12}] = 2i\theta E_{12}$, we get $\alpha(ih) = 2$ and $h_\alpha = h$. So $X^*(T) = P = \{\ell : \ell(ih) \in \mathbb{Z}\}$, and dominance says that $m = \ell(ih) \geq 0$. The classification theorem below says that this m labels V_m. For SO(3) only the even m remain.

8.2 The classification theorem

We prove the theorem by passing to a finite cover of K. First let V be irreducible for K. By Schur's lemma, the centre $\mathfrak{z}$ acts by a scalar $i\mu$. So V stays irreducible under $\mathfrak{k}_s$, and the highest-weight theorem of the Lie Algebras course gives a highest weight λ. Thus $\ell = \mu + \lambda$ is defined. A representation of a connected group is determined by its differential, so ℓ determines V.

For existence, start from ℓ. Integrate the module $V(\lambda)$ to the simply connected group K_s with Lie algebra $\mathfrak{k}_s$. Multiply by the character χ_μ of the central torus $Z_0 = \exp \mathfrak{z}$ given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. This gives a representation of the finite cover $Z_0 \times K_s$ of K. It descends if and only if the finite kernel F acts trivially. By the lifting lemma, Lemma 8.2, each element of F is the exponential of a point $H \in \Lambda$. It acts on a highest weight vector by $e^{i\ell(H)} = 1$, since $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. By Schur's lemma it then acts trivially.

We use the following structure inputs.

(S1) (Torus theorem) In a compact connected Lie group, every element lies in a maximal torus, and all maximal tori are conjugate (Lecture 6). The image of exp generates the group (Lecture 2). A compact connected abelian Lie group is the image of its Lie algebra under exp, and it is isomorphic to $\mathrm{U}(1)^n$ (Lecture 7). The Lie algebra of a maximal torus is a maximal abelian subalgebra (Lecture 6).

(S2) (Lecture 7; H Chapter 13; covering theorem) Let $Z_0 = \exp \mathfrak{z}$, a torus. Let K_s be the simply connected Lie group with Lie algebra $\mathfrak{k}_s$. Then K_s is compact, by Weyl's theorem on the fundamental group (Lecture 7). The inclusion $\mathfrak{k}_s \subset \mathfrak{k}$ integrates to $\iota\colon K_s \to K$. The map

$$p\colon \widetilde{K} = Z_0 \times K_s \to K, \qquad p(z, s) = z\, \iota(s),$$

is a surjective homomorphism with finite central kernel F, and $dp = \mathrm{id}_{\mathfrak{k}}$. Hence p is a covering homomorphism by Lecture 4. In the notation of Lecture 7, $Z_0 = T_0$, $K_s = K_{sc}$ and $\iota = \varphi$. The group $\widetilde{K}$ is a finite cover of K. It is not the universal cover when $\mathfrak{z} \neq 0$, because Z_0 is a torus. For this reason a character of Z_0 must satisfy an integrality condition.

(S3) (Lecture 4; integration and descent) Finite-dimensional representations of $\mathfrak{k}_s$ integrate to K_s. Let $\kappa\colon G_1 \to G_2$ be a covering homomorphism with kernel N. A representation $\widetilde{\Pi}$ of G_1 equals $\Pi \circ \kappa$ for a representation Π of G_2 if and only if $\widetilde{\Pi}(N) = \{I\}$.

(S4) ([I], highest-weight theorem) Finite-dimensional $\mathfrak{k}_{s,\mathbb{C}}$-modules are sums of weight spaces with weights in P. An irreducible such module has a unique highest weight λ. Its weight space is a line, every weight lies in $\lambda - \mathbb{N}R^+$, and $\langle \lambda, \alpha^\vee \rangle \geq 0$ for $\alpha \in R^+$. Each dominant $\lambda \in P$ is the highest weight of exactly one irreducible module $V(\lambda)$, up to isomorphism.

Lemma 8.1. *Let G be a compact connected Lie group with a maximal torus S and $\mathfrak{s} = \operatorname{Lie} S$. Every central element f of G equals $\exp H$ for some $H \in \mathfrak{s}$.*

Proof. By the torus theorem (S1), $f \in gSg^{-1}$ for some $g \in G$. As f is central, $f = g^{-1}fg \in S$. Since $\exp\colon \mathfrak{s} \to S$ is onto, $f = \exp H$ with $H \in \mathfrak{s}$. ∎

Lemma 8.2. *Assume the torus theorem (S1) and the covering theorem (S2). Then $\widetilde{T} = \exp_{\widetilde{K}} \mathfrak{t}$ is a maximal torus of $\widetilde{K}$. Every $f \in F$ equals $\exp_{\widetilde{K}} H$ for some $H \in \Lambda$.*

Proof. Since $dp = \mathrm{id}_{\mathfrak{k}}$, naturality of exp (Lecture 2) gives

$$p(\exp_{\widetilde{K}} X) = \exp_K X \qquad (X \in \mathfrak{k}). \tag{8.2}$$

Let T' be the identity component of the closed subgroup $p^{-1}(T)$. Its Lie algebra consists of the X with $\exp_K(tX) \in T$ for all t. This set is $\mathfrak{t}$. As $\mathfrak{t}$ is abelian, $\exp_{\widetilde{K}}|_{\mathfrak{t}}$ is a homomorphism. Its image is a subgroup of T' containing a neighbourhood of 1 (Lecture 2). So the image is open, hence also closed, in the connected group T'. Thus $\widetilde{T} = T'$.

So $\widetilde{T}$ is compact, connected and abelian, hence a torus. Let $S \supset \widetilde{T}$ be a torus. Then $\operatorname{Lie} S$ is abelian and contains $\mathfrak{t}$. Since $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$, $\operatorname{Lie} S = \mathfrak{t}$ and $S = \widetilde{T}$.

Finally, $f \in F$ is central in the compact connected group $\widetilde{K}$. By the central-element lemma, Lemma 8.1, $f = \exp_{\widetilde{K}} H$ with $H \in \mathfrak{t}$. By naturality (8.2), $\exp_K H = p(f) = 1$, so $H \in \Lambda$. ∎

Theorem 8.3. *Assume the structure inputs (S1)–(S4). Each irreducible representation of K has a unique highest weight. It lies in $X^*(T)^+$, and this gives a bijection $\operatorname{Irr}(K) \to X^*(T)^+$.*

Proof. Step 1: the highest weight. Let Π be irreducible on V, and let $Z \in \mathfrak{z}$. Since $[Z, X] = 0$ for every $X \in \mathfrak{k}$, the element $\exp tZ$ commutes with every $\exp X$. The elements $\exp X$ generate K by the torus theorem (S1), so $\exp tZ$ is central. Hence $\Pi(\exp tZ)$ commutes with $\Pi(K)$. Differentiating at $t = 0$, so does $d\Pi(Z)$. By Schur's lemma, $d\Pi(Z)$ is a scalar. The operator $d\Pi(Z)$ is skew-adjoint, so the scalar is imaginary. The scalar depends linearly on Z. Hence

$$d\Pi(Z) = i\mu(Z)I \qquad \text{for some } \mu \in \mathfrak{z}^*.$$

Let $W \subset V$ be a complex subspace invariant under $d\Pi(\mathfrak{k}_s)$. Then W is invariant under $d\Pi(\mathfrak{k})$, and so under every $\Pi(\exp X) = e^{d\Pi(X)}$. The elements $\exp X$ generate K by the torus theorem (S1), so W is K-invariant. Hence V is irreducible under $\mathfrak{k}_s$. The complexification $\mathfrak{k}_{s,\mathbb{C}}$ has the same invariant complex subspaces, so V is also irreducible under $\mathfrak{k}_{s,\mathbb{C}}$.

By the highest-weight theorem (S4), the $\mathfrak{k}_{s,\mathbb{C}}$-module V has a highest weight λ and a highest weight vector v. For $H = Z + H_s$ with $Z \in \mathfrak{z}$ and $H_s \in \mathfrak{t}_s$,

$$d\Pi(H)v = i\big(\mu(Z) + \lambda(H_s)\big)v = i\ell(H)v, \qquad \ell := \mu + \lambda \in \mathfrak{t}^*.$$

Exponentiating gives $\Pi(\exp H)v = e^{i\ell(H)}v$. So v is a common eigenvector of $\Pi(T)$, and $\ell \in X^*(T)$ by the description of characters in § 8.1.

Now let ℓ' be any T-weight of V. The restriction of ℓ' to $\mathfrak{z}$ equals μ. The restriction of ℓ' to $\mathfrak{t}_s$ lies in $\lambda - \mathbb{N}R^+$ by the highest-weight theorem (S4). Roots vanish on $\mathfrak{z}$, so $\ell - \ell' \in \mathbb{N}R^+$. Thus ℓ is the highest weight of Π. It is dominant, since $\langle \ell, \alpha^\vee \rangle = \langle \lambda, \alpha^\vee \rangle \geq 0$.

Step 2: injectivity. Let Π_1, Π_2 be irreducible with the same highest weight ℓ. They share μ and λ. By the uniqueness part of the highest-weight theorem (S4), there is an isomorphism $A\colon V_1 \to V_2$ of $\mathfrak{k}_{s,\mathbb{C}}$-modules. It also intertwines $d\Pi_j(Z) = i\mu(Z)I$. Hence for every $X \in \mathfrak{k}$

$$A\,\Pi_1(\exp X) = A\,e^{d\Pi_1(X)} = e^{d\Pi_2(X)}A = \Pi_2(\exp X)\,A.$$

The elements $\exp X$ generate K by the torus theorem (S1), so A is K-equivariant.

Step 3: existence. Let $\ell \in X^*(T)^+$, with $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. By the integrality computation in § 8.1, $\lambda \in P$, and λ is dominant. The module $V(\lambda)$ is only known to integrate to K_s. Since the map p has kernel F, we must show that F acts trivially. This step uses the hypothesis $\ell \in X^*(T)$, not only $\lambda \in P$.

The kernel $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$ of $\exp|_{\mathfrak{z}}$ lies in Λ. So $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, and

$$\chi_\mu(\exp Z) = e^{i\mu(Z)} \qquad (Z \in \mathfrak{z})$$

is a well-defined character of Z_0. By the highest-weight theorem (S4) and integration (S3), $V(\lambda)$ integrates to an irreducible representation Π_λ of K_s. Define on $V(\lambda)$

$$\widetilde{\Pi}(z,s) = \chi_\mu(z)\,\Pi_\lambda(s).$$

It is irreducible, because Π_λ already is. Let v be a highest weight vector. For $H = Z + H_s$ as above,

$$\widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\mu(Z)}e^{d\Pi_\lambda(H_s)}v = e^{i\mu(Z)}e^{i\lambda(H_s)}v = e^{i\ell(H)}v.$$

Let $f \in F$. By the lifting lemma, Lemma 8.2, $f = \exp_{\widetilde{K}} H$ with $H \in \Lambda$. As f is central, Schur's lemma gives $\widetilde{\Pi}(f) = cI$. Evaluating on v,

$$cv = \widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\ell(H)}v = v,$$

because $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. So F acts trivially. By the descent criterion (S3), $\widetilde{\Pi} = \Pi \circ p$ for a representation Π of K. It is irreducible, since p is onto.

It remains to identify the highest weight of Π. Since $dp = \mathrm{id}_{\mathfrak{k}}$, we have $d\Pi = d\widetilde{\Pi}$. So

$$d\Pi(Z) = i\mu(Z)I \quad (Z \in \mathfrak{z}), \qquad d\Pi|_{\mathfrak{k}_s} = d\Pi_\lambda.$$

Thus V is $V(\lambda)$ as an $\mathfrak{k}_{s,\mathbb{C}}$-module, with highest weight vector v. These are exactly the two data used in Step 1. Step 1 therefore gives the highest weight $\mu + \lambda = \ell$. ∎

Compactness gave the invariant inner product. This yields the weight decomposition of V. It also makes $d\Pi(Z)$ skew-adjoint, so μ is real. Compactness of $\widetilde{K}$ allowed the torus theorem (S1) inside the lifting lemma, Lemma 8.2. Schur's lemma needs only irreducibility. Connectedness entered through the torus theorem (S1). Connectedness is needed. The trivial character and det of $\mathrm{O}(2)$ have the same differential and differ on a reflection (Lecture 13).

H §§12.1–12.3 phrase the result with *analytically integral* elements. Hall's kernel lattice is our Λ divided by 2π. Hall obtains existence by analytic methods in Chapter 12. Our proof passes through the covering p.

8.3 Central characters and global forms

Let K_{sc} be compact, connected, simply connected and semisimple, with maximal torus T. Let F be a subgroup of its finite centre.

Corollary 8.4. *Assume the torus theorem (S1), integration and descent (S3), and the highest-weight theorem (S4). Let $\lambda \in P$ be dominant. Then $V(\lambda)$ descends to K_{sc}/F if and only if F acts trivially. Equivalently,*

$$\lambda(H) \in 2\pi\mathbb{Z} \qquad \text{whenever } H \in \mathfrak{t} \text{ and } \exp H \in F.$$

Proof. The first statement is the descent criterion (S3). Let Π be the representation of K_{sc} integrating $V(\lambda)$, which exists by integration (S3). Let $f \in F$. By the central-element lemma, Lemma 8.1, $f = \exp H$ with $H \in \mathfrak{t}$. By Schur's lemma, f acts by a scalar. Evaluating on a highest weight vector gives $\Pi(f) = e^{i\lambda(H)} I$. This is I exactly when $\lambda(H) \in 2\pi\mathbb{Z}$. ∎

Since $f \mapsto \Pi(f)$ is a homomorphism, it suffices to test generators of F.

Now allow a central torus. Here $F = \ker p$ again, as in the covering theorem (S2). Since F is central in $\widetilde{K}$, each $(z, s) \in F$ has s central in K_s.

We recall the notation of Step 3 in the proof of the classification theorem. The kernel of $\exp|_{\mathfrak{z}}$ is $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$. For $\mu \in \mathfrak{z}^*$ with $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, the character χ_μ of Z_0 is given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. For dominant $\lambda \in P$, the representation of K_s integrating $V(\lambda)$ is Π_λ.

Let Π be an irreducible representation of a compact group. By Schur's lemma, each central element acts by a scalar. The map sending a central element to its scalar is a homomorphism from the centre to $\mathbb{C}^\times$, the *central character* of Π. Let c_λ be the central character of Π_λ. So $c_\lambda(s)$ is the scalar by which an element s of the centre of K_s acts on $V(\lambda)$.

Corollary 8.5. *Assume the structure inputs (S1)–(S4). Let $\mu \in \mathfrak{z}^*$ and $\lambda \in \mathfrak{t}_s^*$ satisfy*

(a) $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, so χ_μ is a character of Z_0 (integral character);

(b) $\lambda \in P$ is dominant;

(c) $\chi_\mu(z)\, c_\lambda(s) = 1$ for all $(z, s) \in F$ (finite compatibility).

Then $\rho_{\mu,\lambda}(z, s) = \chi_\mu(z)\Pi_\lambda(s)$ descends to an irreducible representation of K. Every irreducible representation of K arises in this way from exactly one pair (μ, λ).

Proof. By the integrality condition (a), the dominance condition (b) and integration (S3), $\rho_{\mu,\lambda}$ is an irreducible representation of $\widetilde{K}$. Let $f = (z, s) \in F$. As noted before the corollary, s is central in K_s. So $\rho_{\mu,\lambda}(f) = \chi_\mu(z)c_\lambda(s)I$. By the descent criterion (S3), $\rho_{\mu,\lambda}$ descends if and only if the compatibility condition (c) holds. The descended representation is irreducible because p is onto. The computation in Step 3 of the proof of the classification theorem shows that the descended representation has highest weight $\mu + \lambda$.

Conversely, let Π be irreducible with highest weight ℓ. Put $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. Step 3 proves the integrality condition (a) and the dominance condition (b). It also gives Π' with $\Pi' \circ p = \rho_{\mu,\lambda}$ and highest weight ℓ. By the injectivity in Step 2, $\Pi \cong \Pi'$. Since $\Pi' \circ p$ is trivial on F, the compatibility condition (c) holds. The pair is unique because $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ recovers (μ, λ) from $\mu + \lambda$. ∎

This corollary and the classification theorem, Theorem 8.3, show that the integrality, dominance and compatibility conditions together are equivalent to $\mu + \lambda \in X^*(T)^+$. The compatibility condition is the description of $X^*(T)$ in Lecture 7. Indeed, a central s lies in a maximal torus of K_s by the central-element lemma, Lemma 8.1. Evaluating on a highest weight vector shows that $c_\lambda(s)$ is the value at s of the character χ_λ. The U(2) example, Example 8.9, checks this directly.

A module of the reductive algebra $\mathfrak{k}_{\mathbb{C}}$ can fail to come from a representation of K, because compactness constrains the action of the centre.

Proposition 8.6. *Let Π be a representation of K and $Z \in \mathfrak{z}$. Then $d\Pi(Z)$ is diagonalisable with eigenvalues in $i\mathbb{R}$. If $\exp Z = 1$, these eigenvalues lie in $2\pi i\mathbb{Z}$.*

Proof. With respect to an invariant inner product, $\Pi(\exp tZ)$ is unitary, so $d\Pi(Z)$ is skew-adjoint. Hence it is diagonalisable with eigenvalues $i\theta_j$, $\theta_j \in \mathbb{R}$. If $\exp Z = 1$, then $e^{d\Pi(Z)} = \Pi(1) = I$, so $e^{i\theta_j} = 1$. ∎

For $\mathfrak{k} = \mathfrak{u}(1) = i\mathbb{R}$, neither of the following modules comes from U(1):

$$\rho_1(it) = t\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \rho_2(it) = \tfrac{i}{2}t.$$

The first is not semisimple. The second has the wrong period: $\exp(2\pi i) = 1$, but $\rho_2(2\pi i) = \pi i \notin 2\pi i\mathbb{Z}$. Both integrate to the noncompact simply connected group $\mathbb{R}$, where no averaging is available. In physics terms, a compact U(1) symmetry has integer charges, and a noncompact $\mathbb{R}$ allows any charge.

The same happens for $\mathfrak{gl}(2,\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}(2,\mathbb{C})$. Letting I act on V_m by any $s \in \mathbb{C}$ gives a module. It comes from U(2) if and only if $s \in \mathbb{Z}$ and $s \equiv m \pmod 2$ (Exercise 8.5). On the semisimple part, Weyl's complete reducibility theorem already gives semisimplicity and weights in P. The remaining constraint is $X^*(T)$, which can be smaller.

8.4 Global forms in examples

Each irreducible representation of K is one dot of the lattice $X^*(T)$ in the dominant region. Passing to a quotient of K removes dots. Figure 8.1 draws this for SU(2) and for SU(3). The labels (a, b) are defined in Example 8.8, and the marks are those of Lecture 7.

Example 8.7 (SU(2) and SO(3))**.** We continue the computation at the end of § 8.1, with $h = \operatorname{diag}(1, -1)$. Thus $\langle \ell, \alpha^\vee \rangle = \ell(ih)$, and $Q = \{\ell : \ell(ih) \in 2\mathbb{Z}\}$. The highest weight of V_m has $\ell(ih) = m$. Now $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). The element $-I$ is central, and

$$\exp(\pi i h) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By the quotient criterion, Corollary 8.4, $-I$ acts on V_m by

$$e^{i\ell(\pi i h)} = e^{i\pi m} = (-1)^m.$$

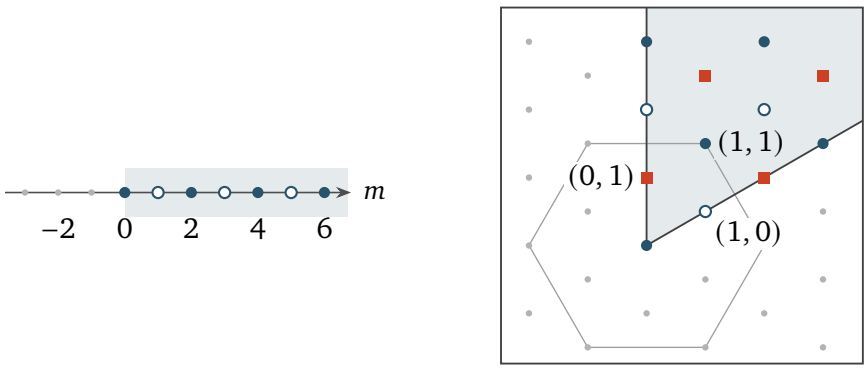


Figure 8.1: Characters of a maximal torus of SU(2), with coordinate $m = \ell(ih)$, on the left, and of SU(3) on the right, in the plane $\mathfrak{t}^*$. The shaded regions are the dominant regions, and the grey dots are not dominant. Filled dots in a shaded region are the highest weights of representations of SO(3) on the left and of PSU(3) on the right. On the right, a circle or a square marks the central value ω or ω^2. The thin hexagon joins the roots.

So V_m descends to SO(3) if and only if m is even. The irreducible representations of SO(3) have dimensions $1, 3, 5, \ldots$. In physics, V_m describes spin $m/2$. Hence the representations of integer spin descend to the rotation group SO(3). The representations of half-integer spin are representations of SU(2) that do not descend.

Let $r\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the covering. Its derivative is an isomorphism, and we identify the Lie algebras through it. Naturality of exp (Lecture 2) gives $\exp_{\mathrm{SO}(3)}(\theta\, ih) = r(\exp(\theta\, ih))$. This element is 1 if and only if $\exp(\theta\, ih) = \pm I$, that is, if and only if $\theta \in \pi\mathbb{Z}$. Hence the maximal torus $r(T)$ of SO(3) has $X^*(r(T)) = \{\ell : \pi\ell(ih) \in 2\pi\mathbb{Z}\} = Q$. This lattice has index 2 in P.

Example 8.8 (PSU(3))**.** Let $\mathfrak{t} = \{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum \theta_j = 0\}$ and $\varepsilon_j(H) = \theta_j$. The roots are $\varepsilon_j - \varepsilon_k$ with $j \neq k$, with root vectors E_{jk}. We take the positive roots $\varepsilon_j - \varepsilon_k$ with $j < k$. The simple roots $\varepsilon_1 - \varepsilon_2$ and $\varepsilon_2 - \varepsilon_3$ have the real coroots $H_{12} = i\,\mathrm{diag}(1, -1, 0)$ and $H_{23} = i\,\mathrm{diag}(0, 1, -1)$. Then

$$\varepsilon_1(H_{12}) = 1, \quad \varepsilon_1(H_{23}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{12}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{23}) = 1.$$

The fundamental weights ϖ_1 and ϖ_2 are dual to the simple coroots. Thus ϖ_1 takes the values 1 and 0 at H_{12} and H_{23}, and ϖ_2 takes the values 0 and 1. So $\varpi_1 = \varepsilon_1$ and $\varpi_2 = \varepsilon_1 + \varepsilon_2$. The label (a, b) denotes the highest weight

$$\ell = a\varpi_1 + b\varpi_2 = (a + b)\varepsilon_1 + b\varepsilon_2.$$

The centre is $\{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$, and

$$\omega I = \exp H_\omega, \qquad H_\omega = \tfrac{2\pi i}{3}\,\mathrm{diag}(1, 1, -2) \in \mathfrak{t}.$$

Then

$$\ell(H_\omega) = (a + b)\tfrac{2\pi}{3} + b\tfrac{2\pi}{3} = \tfrac{2\pi}{3}(a + 2b),$$

so ωI acts on the irreducible module with highest weight (a, b) by ω^{a+2b}. Hence this module descends to $\mathrm{PSU}(3) = \mathrm{SU}(3)/\langle \omega I \rangle$ if and only if $3 \mid a + 2b$. Since $(a + 2b) - (a - b) = 3b$, this means $a \equiv b \pmod 3$. The dimension formula $\frac{1}{2}(a + 1)(b + 1)(a + b + 2)$ [I] gives the following values.

(a,b)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$	$(0,3)$	$(2,2)$
dimension	3	3	6	8	10	10	27
ωI acts by	ω	ω^2	ω^2	1	1	1	1

The adjoint representation $(1,1)$ descends, since $\mathrm{Ad}(\omega I) = \mathrm{id}$.

Example 8.9 (U(2)). Let $p\colon \mathrm{U}(1)\times\mathrm{SU}(2)\to\mathrm{U}(2)$ be given by $p(z,g)=zg$. The map p is onto. Indeed, given u, choose z with $z^2=\det u$ and put $g=z^{-1}u$. For the kernel,

$$zg=I \implies g=z^{-1}I,\quad 1=\det g=z^{-2},\qquad \text{so}\qquad F=\{(1,I),(-1,-I)\}.$$

Here $\Lambda_{\mathfrak{z}}=2\pi i\mathbb{Z}\,I$, so the integral characters of $Z_0\cong\mathrm{U}(1)$ are $z\mapsto z^q$ with $q\in\mathbb{Z}$. Put $\Pi_{q,m}(z,g)=z^q\Pi_m(g)$. The compatibility condition (c) is

$$\Pi_{q,m}(-1,-I)=(-1)^q(-1)^m I=I \iff q+m \text{ is even.}$$

We cross-check with the classification theorem, Theorem 8.3. The torus $\mathrm{diag}(e^{i\theta_1},e^{i\theta_2})$ has $X^*(T)=\mathbb{Z}^2$, and (n_1,n_2) is dominant if and only if $n_1\ge n_2$. The torus element equals $p(e^{i\psi},\mathrm{diag}(e^{i\phi},e^{-i\phi}))$ when $\theta_1=\psi+\phi$ and $\theta_2=\psi-\phi$. On a highest weight vector of $\Pi_{q,m}$, this torus element acts by $e^{i(q\psi+m\phi)}$, while

$$n_1\theta_1+n_2\theta_2=(n_1+n_2)\psi+(n_1-n_2)\phi.$$

So $q=n_1+n_2$ and $m=n_1-n_2$. The numbers $n_{1,2}=(q\pm m)/2$ are integers exactly when $q+m$ is even. For example, $\mathbb{C}^2$ corresponds to $(q,m)=(1,1)$, and $\det$ corresponds to $(2,0)$.

8.5 Tensor products and consistency checks

Lemma 8.10. *Let f be central in K, acting by scalars c_1 on V_1 and c_2 on V_2. Then f acts by c_1c_2 on $V_1\otimes V_2$ and on each subrepresentation, and by c_1^{-1} on V_1^*.*

Proof. First, $(\Pi_1\otimes\Pi_2)(f)=c_1I\otimes c_2I=c_1c_2\,I$. Invariant subspaces inherit this scalar. On the dual, $\Pi_1^*(f)=\Pi_1(f^{-1})^T=c_1^{-1}I$. ∎

Hence the decompositions from [I] must respect central characters. We check this in the following cases.

- Clebsch–Gordan: $V_m\otimes V_n=\bigoplus_{k=0}^{\min(m,n)}V_{m+n-2k}$. Every summand has parity $m+n$, matching $(-1)^m(-1)^n$. In particular, $V_1\otimes V_1=V_2\oplus V_0$ is a sum of $\mathrm{SO}(3)$-representations, although V_1 is not an $\mathrm{SO}(3)$-representation.
- $\mathrm{SU}(3)$: $(1,0)\otimes(1,0)=(2,0)\oplus(0,1)$, with $9=6+3$ and $\omega\cdot\omega=\omega^2$ on both summands. Also $(1,0)\otimes(0,1)=(1,1)\oplus(0,0)$, with $9=8+1$ and $\omega\cdot\omega^2=1$.
- Duals: $V(a,b)^*\cong V(b,a)$ [I]. The lemma predicts $\omega^{b+2a}=\omega^{-(a+2b)}$. Indeed $(b+2a)+(a+2b)=3(a+b)$.
- Branching from $\mathrm{U}(2)$ to $\mathrm{SU}(2)$: $\Pi_{q,m}$ restricts to V_m. On the torus of $\mathrm{U}(2)$ it has the $m+1$ weights (n_1-j,n_2+j), $0\le j\le m$. Characters z^q multiply, so $\Pi_{1,1}\otimes\Pi_{1,1}=\Pi_{2,2}\oplus\Pi_{2,0}$. Both summands have $q+m$ even.

Self-checks

1. Does $V_3 \otimes V_5$ descend to $\mathrm{SO}(3)$? Decompose it.

   Answer. Yes. The element $-I$ acts by $(-1)^3(-1)^5 = 1$. By Clebsch–Gordan,

   $$V_3 \otimes V_5 = V_8 \oplus V_6 \oplus V_4 \oplus V_2, \qquad 9 + 7 + 5 + 3 = 24 = 4 \cdot 6.$$

2. Which irreducible representations of $\mathrm{PSU}(3)$ have dimension at most 10?

   Answer. The $\mathrm{SU}(3)$-irreducibles of dimension at most 10 are $(0,0)$, $(1,0)$, $(0,1)$, $(2,0)$, $(0,2)$, $(1,1)$, $(3,0)$, $(0,3)$. The next ones, such as $(2,1)$ and $(4,0)$, have dimension 15. The ones that descend to $\mathrm{PSU}(3)$ are those with $a \equiv b \pmod 3$, namely

   $$(0,0),\ (1,1),\ (3,0),\ (0,3),$$

   of dimensions $1, 8, 10, 10$.

3. Let $I \in \mathfrak{gl}(2,\mathbb{C})$ act by 1 on V_2. Does this module come from $\mathrm{U}(2)$?

   Answer. No. It would be $\Pi_{1,2}$, and $1 + 2$ is odd. Directly, compute $\Pi(-I)$ in two ways:

   $$-I = \exp(i\pi I) \implies \Pi(-I) = e^{i\pi \cdot 1} = -1, \qquad -I \in \mathrm{SU}(2) \implies \Pi(-I) = (-1)^2 = 1.$$

   These disagree.

Summary

Let K be a compact connected Lie group, and assume the cited structure inputs. An irreducible representation of K is determined by its highest weight. The possible highest weights are the dominant characters of a maximal torus T. For K_{sc}/F, test F on one highest weight vector. With a central torus, one adds an integral character of the central torus and a compatibility condition on F. In the examples, the descent conditions are m even for $\mathrm{SO}(3)$, $3 \mid a + 2b$ for $\mathrm{PSU}(3)$, and $q + m$ even for $\mathrm{U}(2)$.

Reading

H denotes Hall, *Lie Groups, Lie Algebras, and Representations*. BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

- H §§12.1–12.3: representations of compact groups, analytically integral elements, and the setting of the highest-weight classification for compact connected groups.
- H Chapter 13: fundamental groups of compact groups, and the comparison of algebraic and analytic integrality. This supports (S2).
- BD VI.2: the dominant weights of irreducible representations of compact connected Lie groups.
- Lie Algebras [I]: highest-weight theorem, Weyl's complete reducibility theorem, the dimension formula, Clebsch–Gordan, duals and $\mathfrak{sl}_3$ tensor products.

Lecture 9

Weyl integration and character calculations

Objectives. *A* class function *on a group is a function that is constant on conjugacy classes. On a compact connected group, a class function is determined by its values on a maximal torus. This lecture shows that the Haar integral of a class function can also be computed on the torus. The density is a product over root planes. A factor* $1/|W|$ *corrects for repeated conjugates. We then use the formula to check character norms and tensor-product multiplicities.*

By the end of this lecture you will be able to:

- State the Weyl integration formula with its measure conventions and its imported steps.
- Derive the density $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$ and the factor $1/|W|$.
- Compute class integrals on $\mathrm{SU}(2)$ and $\mathrm{SO}(3)$ and compare them.
- Check character norms and multiplicities with the Weyl character formula on T.

9.1 Measures, roots and the statement

Throughout, K is a compact connected Lie group and $T \subset K$ is a maximal torus. Their Lie algebras are $\mathfrak{t} \subset \mathfrak{k}$. By Lecture 6, every element of K is conjugate to an element of T. Also by Lecture 6, the group $W = N_K(T)/T$ is finite and acts on $X^*(T)$ as the Weyl group of the root system. Fix an $\mathrm{Ad}(K)$-invariant inner product on $\mathfrak{k}$ and set

$$\mathfrak{m} = \mathfrak{t}^{\perp} \subset \mathfrak{k}.$$

For $t \in T$ the map Ad_t is orthogonal and fixes $\mathfrak{t}$ pointwise. Hence Ad_t preserves $\mathfrak{m}$, and $\mathfrak{m} \cong \mathfrak{k}/\mathfrak{t}$ as representations of T.

We use three probability measures.

- dk is the normalised Haar measure on K (Lecture 5).
- dt is the normalised Haar measure on T.
- $d(gT)$ is the unique K-invariant probability measure on K/T.

The existence and uniqueness of $d(gT)$ are cited together with the theorem below. All three measures are given by smooth positive densities.

We keep the real weights and real roots of Lecture 6 and Lecture 7. For $\ell \in X^*(T)$, write t^ℓ for the value at t of the character of T with weight ℓ. Thus $t^\ell = e^{i\ell(H)}$ when $t = \exp H$ with $H \in \mathfrak{t}$, and $t^\ell t^{\ell'} = t^{\ell+\ell'}$.

With this convention, the root decomposition is

$$\mathfrak{k}_{\mathbb{C}} = \mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad \mathrm{Ad}_t E = t^\alpha E \quad (E \in \mathfrak{g}_\alpha,\ t \in T). \tag{9.1}$$

Each $\mathfrak{g}_\alpha$ is a complex line. Each root is a weight of the adjoint representation of K. So $t \mapsto t^\alpha$ is a character of T, that is, $\alpha \in X^*(T)$. If $t = \exp H$ with $H \in \mathfrak{t}$, then $t^\alpha = e^{i\alpha(H)}$ with $\alpha(H) \in \mathbb{R}$.

Fix a positive system R^+ and let ρ be half the sum of the positive roots. In general $\rho \notin X^*(T)$. § 9.4 shows how to avoid ρ as a function on T.

The Lie Algebras course states the Weyl character formula and the denominator identity for complex-linear weights on $\mathfrak{t}_{\mathbb{C}}$. In that course, a real weight ℓ appears as the complex weight $i\ell$, a root α as $i\alpha$, and ρ as $i\rho$. A complex weight $\mu = i\ell$ has the exponential $e^{\mu(H)} = e^{i\ell(H)} = t^\ell$ at $t = \exp H$. The map $\ell \mapsto i\ell$ is additive and commutes with W. So those identities keep their form when we write them with real exponents. They then hold in the ring $\mathbb{Z}[X^*(T)]$ of finite integer combinations of formal exponentials e^ℓ, where $e^\ell e^{\ell'} = e^{\ell+\ell'}$. Evaluating e^ℓ at t gives t^ℓ.

Definition 9.1. An element $t \in T$ is *regular* if $t^\alpha \neq 1$ for every $\alpha \in R$. The regular elements of T form T_{reg}. The regular elements of K form $K_{\mathrm{reg}} = \{gtg^{-1} : g \in K,\ t \in T_{\mathrm{reg}}\}$.

The conjugation map is

$$\psi \colon K/T \times T \to K, \qquad \psi(gT, t) = gtg^{-1}. \tag{9.2}$$

It is well defined because T is abelian. The domain and the codomain of ψ both have dimension $\dim K$.

Theorem 9.2 (Weyl integration formula)**.** *Let K, T, W and the measures be as above. Put*

$$J(t) = \det\bigl(1 - \mathrm{Ad}_t|_{\mathfrak{m}}\bigr) \qquad (t \in T).$$

Then $J \geq 0$, and for every continuous $f \colon K \to \mathbb{C}$,

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T J(t) \int_{K/T} f(gtg^{-1})\,d(gT)\,dt. \tag{9.3}$$

If f is a class function, then

$$\int_K f(k)\,dk = \frac{1}{|W|} \int_T f(t)\,J(t)\,dt. \tag{9.4}$$

The theorem is cited from H §11.6, BD IV.1 (1.11) and Knapp, Beyond, VIII.5. We prove the formula below from two imported analytic steps and the existence of $d(gT)$. The class-function version follows from the general formula (9.3). Indeed $f(gtg^{-1}) = f(t)$, and $d(gT)$ has total mass one.

We prove the formula by changing variables along ψ. Every element of K has the form gtg^{-1}. The Jacobian of ψ is a constant multiple of J, and the constant is fixed at the end. A regular

element has $|W|$ preimages, and this gives the factor $1/|W|$. The remaining elements form a set of Haar measure zero.

For example, take $K = \mathrm{SU}(2)$ and $t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})$. The space K/T is the sphere S of Lecture 3, and the coset gT corresponds to $\xi = g\,\mathrm{diag}(1,-1)g^{-1}$. Then

$$g t_\theta g^{-1} = \cos\theta\, I + i \sin\theta\, \xi.$$

The regular elements of T are the t_θ with $\theta \notin \pi\mathbb{Z}$, and K_{reg} is SU(2) without $\pm I$. So the remaining elements are two points. A regular element has the two preimages (ξ, θ) and $(-\xi, 2\pi - \theta)$.

9.2 The root-plane Jacobian and the factor $|W|$

Lemma 9.3 (Root planes). *For $\alpha \in R^+$ put $\mathfrak{m}_\alpha = \mathfrak{k} \cap (\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha})$. Then $\mathfrak{m} = \bigoplus_{\alpha \in R^+} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has a real basis consisting of X_α and Y_α, in which*

$$\mathrm{Ad}_t\,|_{\mathfrak{m}_\alpha} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \text{where } t^\alpha = e^{i\theta}.$$

Consequently, writing $t^\alpha = e^{i\theta_\alpha}$,

$$J(t) = \prod_{\alpha \in R^+} \left|1 - t^\alpha\right|^2 = \prod_{\alpha \in R} (1 - t^\alpha) = \prod_{\alpha \in R^+} 4 \sin^2 \frac{\theta_\alpha}{2}. \tag{9.5}$$

Proof. Let σ be complex conjugation on $\mathfrak{k}_{\mathbb{C}}$ with respect to $\mathfrak{k}$. It commutes with the real map Ad_t. For $E \in \mathfrak{g}_\alpha$ we get

$$\mathrm{Ad}_t(\sigma E) = \sigma(t^\alpha E) = \overline{t^\alpha}\,\sigma E = t^{-\alpha}\sigma E,$$

because $|t^\alpha| = 1$. So $\sigma\mathfrak{g}_\alpha = \mathfrak{g}_{-\alpha}$. Choose $E \neq 0$ in $\mathfrak{g}_\alpha$ and set

$$X_\alpha = E + \sigma E, \qquad Y_\alpha = i(E - \sigma E).$$

Both are fixed by σ, so they lie in $\mathfrak{k}$. An element $aE + b\,\sigma E$ is fixed by σ exactly when $b = \bar a$. If $a = x + iy$, then $aE + \bar a\,\sigma E = xX_\alpha + yY_\alpha$. Hence X_α and Y_α form a basis of $\mathfrak{m}_\alpha$. With $t^\alpha = e^{i\theta}$,

$$\begin{aligned} \mathrm{Ad}_t\, X_\alpha &= e^{i\theta}E + e^{-i\theta}\sigma E = \cos\theta\, X_\alpha + \sin\theta\, Y_\alpha, \\ \mathrm{Ad}_t\, Y_\alpha &= ie^{i\theta}E - ie^{-i\theta}\sigma E = -\sin\theta\, X_\alpha + \cos\theta\, Y_\alpha. \end{aligned}$$

Taking σ-fixed points in (9.1) gives $\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ is orthogonal to $\mathfrak{t}$. To see this, pick $H \in \mathfrak{t}$ with $\alpha(H) \neq 0$. Then $[H, X_\alpha] = \alpha(H)Y_\alpha$ and $[H, Y_\alpha] = -\alpha(H)X_\alpha$, so $\mathfrak{m}_\alpha \subset [\mathfrak{t}, \mathfrak{k}]$. Invariance of the inner product gives $\langle [H, X], H' \rangle = -\langle X, [H, H'] \rangle = 0$ for $H' \in \mathfrak{t}$. Counting dimensions, $\mathfrak{m} = \bigoplus_{\alpha>0} \mathfrak{m}_\alpha$.

For one plane,

$$\det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = |1 - e^{i\theta}|^2.$$

This equals $4\sin^2(\theta/2)$. Multiply over $\alpha \in R^+$. Finally $(1 - t^\alpha)(1 - t^{-\alpha}) = |1 - t^\alpha|^2$, since $t^{-\alpha} = \overline{t^\alpha}$. ∎

Thus $J \geq 0$, and $J(t) > 0$ exactly when $t \in T_{\mathrm{reg}}$. Also $J(t^{-1}) = J(t)$.

For $\mathrm{U}(n)$, with $t = \mathrm{diag}(z_1, \dots, z_n)$, the roots of Lecture 6 give the density $\prod_{j<k} |z_j - z_k|^2$. It vanishes when two eigenvalues coincide. So a Haar-random unitary matrix is unlikely to have two close eigenvalues.

Next we write the differential of ψ in translated coordinates. For $X \in \mathfrak{m}$ and $H \in \mathfrak{t}$ put

$$L_{(gT,t)}(X,H) = \frac{d}{ds}\Big|_{s=0} \psi(gT,t)^{-1}\,\psi\big(g\exp(sX)T,\ t\exp(sH)\big) \in \mathfrak{k}.$$

We identify the tangent space of K/T at gT with $\mathfrak{m}$ by $X \mapsto \frac{d}{ds}|_{s=0}\, g\exp(sX)T$. We identify the tangent spaces of T at t and of K at x with $\mathfrak{t}$ and $\mathfrak{k}$ by left translation. In these identifications, each invariant density is given at every point by its density at the base point. Let $\kappa > 0$ be the ratio of the base-point density of dk to the product of the base-point densities of $d(gT)$ and dt, all evaluated on orthonormal bases. Hence the Jacobian of ψ relative to $d(gT)\,dt$ and dk equals $\kappa\,|\det L_{(gT,t)}|$. Here the determinant is taken with respect to orthonormal bases of $\mathfrak{m} \oplus \mathfrak{t}$ and $\mathfrak{k}$.

The value $|\det L_{(gT,t)}|$ does not depend on the representative g of the coset gT. Indeed, replacing g by gs with $s \in T$ composes L with $\mathrm{Ad}_s|_{\mathfrak{m}}$. This map has determinant 1.

The next proposition is the Jacobian computation of Lecture 6, now at a general point. The map ψ is the map Φ there.

Proposition 9.4. *For all $g \in K$ and $t \in T$, $|\det L_{(gT,t)}| = J(t)$.*

Proof. Multiplying out,

$$\psi(gT,t)^{-1}\psi(ge^{sX}T, te^{sH}) = gt^{-1}e^{sX}t\,e^{sH}e^{-sX}g^{-1} = g\ \exp(s\,\mathrm{Ad}_{t^{-1}}X)\exp(sH)\exp(-sX)\,g^{-1}.$$

The velocity at $s = 0$ of a product of one-parameter subgroups is the sum of their velocities (Lecture 2). Hence

$$L_{(gT,t)}(X,H) = \mathrm{Ad}_g\big((\mathrm{Ad}_{t^{-1}} - 1)X + H\big).$$

Relative to $\mathfrak{k} = \mathfrak{m} \oplus \mathfrak{t}$ this is Ad_g composed with the block map

$$\begin{pmatrix} \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}} - 1 & 0 \\ 0 & 1_{\mathfrak{t}} \end{pmatrix}.$$

The map Ad_g is orthogonal, so $|\det \mathrm{Ad}_g| = 1$. Therefore $|\det L| = |\det(1 - \mathrm{Ad}_{t^{-1}}|_{\mathfrak{m}})| = J(t^{-1}) = J(t)$. ∎

If K is nonabelian, then $J(t)$ tends to zero as t approaches a central element. Indeed, every root character is 1 at that element. If K is a torus, the root product is empty and J is identically 1.

Proposition 9.5. *(a) For $t \in T$, $\dim Z_K(t) = \dim T + 2\,\#\{\alpha \in R^+ : t^\alpha = 1\}$. Hence $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.*

(b) If $t, t' \in T_{\mathrm{reg}}$ and $gtg^{-1} = t'$, then $g \in N_K(T)$.

(c) Every $x \in K_{\mathrm{reg}}$ has exactly $|W|$ preimages under ψ.

Proof. (a) The Lie algebra of the closed subgroup $Z_K(t)$ is $\{X \in \mathfrak{k} : \mathrm{Ad}_t X = X\}$ (Lecture 3). A plane rotation by θ fixes a nonzero vector only when $e^{i\theta} = 1$. By the root-plane lemma (Lemma 9.3), the fixed space of Ad_t is $\mathfrak{t} \oplus \bigoplus_{t^\alpha = 1} \mathfrak{m}_\alpha$. Each $\mathfrak{m}_\alpha$ has dimension 2, which gives the dimension formula. Centraliser dimensions are unchanged by conjugation. So if t is conjugate to a regular t', then t is regular.

Now let $(gT, t) \in \psi^{-1}(K_{\mathrm{reg}})$. Then gtg^{-1} is conjugate to some $t' \in T_{\mathrm{reg}}$, so t is conjugate to t', and hence $t \in T_{\mathrm{reg}}$. Conversely, $\psi(gT, t) \in K_{\mathrm{reg}}$ for every $t \in T_{\mathrm{reg}}$, by the definition of K_{reg}. These two inclusions give $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$.

(b) By the dimension formula in (a), the identity component $Z_K(t')_0$ has Lie algebra $\mathfrak{t}$. It contains the connected group T with the same Lie algebra, so $Z_K(t')_0 = T$ (Lecture 2). The torus gTg^{-1} is connected and centralises $gtg^{-1} = t'$. So $gTg^{-1} \subset T$. Then $\mathrm{Ad}_g \mathfrak{t} \subset \mathfrak{t}$, with equality by dimension. Hence $gTg^{-1} \supset \exp \mathfrak{t} = T$, and $g \in N_K(T)$.

(c) Let $x = \psi(gT, t)$ with t regular. Suppose $\psi(hT, s) = x$. By the description of $\psi^{-1}(K_{\mathrm{reg}})$ in (a), s is regular. Put $n = g^{-1}h$. Then $nsn^{-1} = t$. Since s and t are regular, the normaliser statement (b) gives $n \in N_K(T)$. Thus the preimages are the pairs $(gnT,\ n^{-1}tn)$ with $nT \in W$. Each such pair is a preimage. Different classes nT give different cosets gnT. So there are exactly $|W|$ preimages. ∎

Lemma 9.6. $\displaystyle\int_T J(t)\, dt = |W|.$

Proof. The Weyl denominator identity [I] holds in $\mathbb{Z}[X^*(T)]$:

$$\prod_{\alpha > 0} (1 - e^{-\alpha}) = \sum_{w \in W} \mathrm{sgn}(w)\, e^{w\rho - \rho}.$$

Here $\rho - w\rho$ is a sum of positive roots. So $w\rho - \rho \in X^*(T)$ even when $\rho \notin X^*(T)$. Evaluate both sides at t, and write $D(t)$ for the value of the left side. Then $|D(t)|^2 = J(t)$ by (9.5), so

$$J(t) = \sum_{w, w' \in W} \mathrm{sgn}(w)\, \mathrm{sgn}(w')\, t^{w\rho - w'\rho}.$$

Distinct characters of T are orthonormal in $L^2(T)$ (Lecture 5). Thus $\int_T t^\ell\, dt = 0$ for $\ell \neq 0$. Since ρ is regular [I], $w\rho = w'\rho$ only when $w = w'$. Hence only the diagonal terms survive, and the integral is $|W|$. ∎

The proof of the formula uses two imported analytic steps. The section on sources and hypotheses at the end of the lecture derives them from the cited results.

(C1) The set K_{reg} is open, and $K \setminus K_{\mathrm{reg}}$ has Haar measure zero. The null-set claim is cited from BD V.2, Proposition (2.6), together with Lee, Corollary 6.11.

(C2) Let M, N be compact manifolds of equal dimension. Give them finite measures defined by smooth positive densities. Let $\psi\colon M \to N$ be smooth and $U \subset N$ open. Assume $N \setminus U$ and $M \setminus \psi^{-1}(U)$ are null. Assume $\psi\colon \psi^{-1}(U) \to U$ is a surjective local diffeomorphism whose fibres all have d points. Then for continuous F on N,

$$\int_M F(\psi(x))\, J_\psi(x)\, dx = d \int_N F(y)\, dy,$$

where J_ψ is the Jacobian of ψ relative to the two densities. We derive this from Lee, Chapter 16, Propositions 16.40–16.42.

The second step is the change of variables for a map that covers its target d times. For $z \mapsto z^2$ on the circle, we have $d = 2$ and the Jacobian is 2, and

$$\frac{1}{2\pi}\int_0^{2\pi} F(e^{2i\theta})\,2\,d\theta = 2\cdot\frac{1}{2\pi}\int_0^{2\pi} F(e^{i\varphi})\,d\varphi.$$

Proof of the Weyl integration formula (Theorem 9.2) from (C1) and (C2). Null sets. The set $K\setminus K_{\mathrm{reg}}$ is null by the regular-set import (C1). The complement of $\psi^{-1}(K_{\mathrm{reg}})$ is $K/T\times(T\setminus T_{\mathrm{reg}})$. Now $T\setminus T_{\mathrm{reg}}$ is the union of the kernels T_α of $t\mapsto t^\alpha$. Each T_α has infinitely many cosets, because t^α maps the connected group T onto the circle. The cosets are disjoint and have equal Haar measure. Since T has finite measure, T_α is null.

Covering. On $K/T\times T_{\mathrm{reg}}$ the Jacobian is $\kappa J(t) > 0$, so ψ is a local diffeomorphism there. By the fibre count in Proposition 9.5, its fibres have $|W|$ points.

Change of variables. Both $K/T\times T$ and K are compact. The change-of-variables import (C2) gives

$$\kappa\int_T J(t)\int_{K/T} f(gtg^{-1})\,d(gT)\,dt = |W|\int_K f(k)\,dk.$$

Take $f = 1$. Then $\kappa\int_T J\,dt = |W|$. Since $\int_T J\,dt = |W|$ by Lemma 9.6, we get $\kappa = 1$. This is (9.3). ∎

9.3 The model: class integrals on $\mathrm{SU}(2)$

Let $K = \mathrm{SU}(2)$ with maximal torus $T = \{t_\theta\}$, where

$$t_\theta = \begin{pmatrix} e^{i\theta} & 0\\ 0 & e^{-i\theta}\end{pmatrix}, \qquad dt = \frac{d\theta}{2\pi}\ \text{ on } [0,2\pi).$$

Use the invariant inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$ on $\mathfrak{su}(2)$. The plane $\mathfrak{m}$ has basis

$$B = \begin{pmatrix} 0 & 1\\ -1 & 0\end{pmatrix}, \qquad C = \begin{pmatrix} 0 & i\\ i & 0\end{pmatrix}.$$

A direct multiplication gives

$$t_\theta B t_\theta^{-1} = \begin{pmatrix} 0 & e^{2i\theta}\\ -e^{-2i\theta} & 0\end{pmatrix} = \cos 2\theta\, B + \sin 2\theta\, C, \qquad t_\theta C t_\theta^{-1} = -\sin 2\theta\, B + \cos 2\theta\, C.$$

So $t_\theta^\alpha = e^{2i\theta}$ and

$$J(t_\theta) = |1 - e^{2i\theta}|^2 = 4\sin^2\theta.$$

The Weyl group has order 2. Its nontrivial element is represented by

$$n = \begin{pmatrix} 0 & -1\\ 1 & 0\end{pmatrix}, \qquad n t_\theta n^{-1} = t_{-\theta}.$$

Now (9.4) gives, for a continuous class function f,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{1}{\pi}\int_0^{2\pi} f(t_\theta)\sin^2\theta\,d\theta.$$

Since $f(t_{2\pi-\theta}) = f(t_{-\theta}) = f(t_\theta)$ and $\sin^2$ is symmetric about π,

$$\int_{\mathrm{SU}(2)} f\,dk = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta. \tag{9.6}$$

Every $g \in \mathrm{SU}(2)$ has eigenvalues $e^{\pm i\theta}$ for a unique $\theta \in [0,\pi]$, and $\operatorname{tr} g = 2\cos\theta$. So a class function is a function of $\theta \in [0,\pi]$. For $f = 1$ the right side is $\frac{2}{\pi}\cdot\frac{\pi}{2} = 1$.

Exercise 9.2 shows what this density means on S^3. For $0 < \theta < \pi$, the elements with trace $2\cos\theta$ form a 2-sphere of radius $\sin\theta$, the conjugacy class of t_θ. Its area is proportional to $\sin^2\theta$. At $\theta = 0$ and at $\theta = \pi$ the class is the single point I or $-I$. So $J(t)$ measures the size of the class of t. Elements near $\pm I$ are rare, since those classes are small.

Character norms. Let a representation V decompose as $\bigoplus_i m_i V_i$, with the V_i irreducible and pairwise inequivalent. This is possible by complete reducibility (Lecture 5). Orthonormality of characters (Lecture 5) gives $\int_K \chi_V\overline{\chi_i}\,dk = m_i$ and $\int_K |\chi_V|^2\,dk = \sum_i m_i^2$. So a character of norm one is irreducible.

By the Weyl character formula [I], the representation V_m of dimension $m+1$ has

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin(m+1)\theta}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Then (9.6) gives

$$\int_{\mathrm{SU}(2)} \chi_m\overline{\chi_n}\,dk = \frac{2}{\pi}\int_0^\pi \sin(m+1)\theta\,\sin(n+1)\theta\,d\theta = \delta_{mn}.$$

The factor $\sin^2\theta$ cancels both denominators. This confirms that each V_m is irreducible and that the representations V_m are pairwise inequivalent.

A tensor-product multiplicity. The multiplicity of V_n in V is $\int_K \chi_V\overline{\chi_n}\,dk$. Take $V = V_1^{\otimes 4}$, so $\chi_V(t_\theta) = (2\cos\theta)^4$. For $n = 0$ we need the Wallis integrals

$$\int_0^\pi \cos^4\theta\,d\theta = \frac{3\pi}{8}, \qquad \int_0^\pi \cos^6\theta\,d\theta = \frac{5\pi}{16}.$$

Write $\sin^2 = 1 - \cos^2$. Then

$$\int_0^\pi \cos^4\theta\sin^2\theta\,d\theta = \frac{3\pi}{8} - \frac{5\pi}{16} = \frac{\pi}{16}, \qquad \dim V^{\mathrm{SU}(2)} = \frac{2}{\pi}\cdot 16\cdot\frac{\pi}{16} = 2.$$

So the trivial representation V_0 occurs twice in $V_1^{\otimes 4}$. Exercise 9.3 finds the full decomposition from weights and checks the multiplicity of V_2 by integration.

Remark 9.7 (Indicator functions)**.** Probabilities of conjugation-invariant sets need discontinuous f. Let $c\colon K \to \mathbb{R}$ be a continuous class function and put $E = \{k \in K : c(k) \geq 0\}$. Then

$$\int_K 1_E\,dk = \frac{1}{|W|}\int_T 1_E(t)\,J(t)\,dt.$$

To prove this, note that the Haar integral on $C(K)$ extends to bounded measurable functions. This is the Haar measure mentioned in Lecture 5, and the same holds on T. We import one

fact about it, the dominated convergence theorem. If measurable functions f_n converge pointwise to f and $|f_n| \le g$ for an integrable g, then $\int f_n \to \int f$.

For $n \ge 1$ let $h_n : \mathbb{R} \to [0,1]$ be continuous with

$$h_n(x) = 1 \ \ (x \ge 0), \qquad h_n(x) = 0 \ \ (x \le -1/n), \qquad h_n \text{ linear on } [-1/n, 0].$$

Each $h_n \circ c$ is a continuous class function. For every k, the values $h_n(c(k))$ decrease to the indicator $1_E(k)$. All these functions are bounded by 1, and J is bounded on T. Applying dominated convergence to both sides of the class-function formula (9.4) gives the claim.

9.4 Characters on the torus and nonsimply connected groups

Let $\lambda \in X^*(T)$ be dominant. By the classification theorem of Lecture 8, K has an irreducible representation V_λ with highest weight λ. Write χ_λ for its character.

Since K is connected, V_λ is determined by its differential. The complex-linear extension of the differential makes V_λ an irreducible $\mathfrak{k}_{\mathbb{C}}$-module, with complex highest weight $i\lambda$. The centre of $\mathfrak{k}_{\mathbb{C}}$ acts by a scalar. Hence the semisimple Weyl character formula [I] applies to the derived algebra, with the central weight retained. Written with real exponents, as explained in § 9.1, the resulting formula is an identity in $\mathbb{Z}[X^*(T)]$. We use it in the form

$$\chi_\lambda(t)\, D(t) = \sum_{w \in W} \operatorname{sgn}(w)\, t^{w(\lambda+\rho)-\rho}, \qquad D(t) = \prod_{\alpha > 0} (1 - t^{-\alpha}). \tag{9.7}$$

Each exponent equals $w\lambda - (\rho - w\rho)$, so it lies in $X^*(T)$. Thus (9.7) is an identity of functions on T for every K. If $\rho \in X^*(T)$, multiplying by t^ρ gives the usual quotient form.

Proposition 9.8. *Let $\lambda, \mu \in X^*(T)$ be dominant. Then $\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \delta_{\lambda\mu}$.*

Proof. Since $|D|^2 = J$, (9.4) gives

$$\int_K \chi_\lambda \overline{\chi_\mu}\, dk = \frac{1}{|W|} \sum_{w,w'} \operatorname{sgn}(ww') \int_T t^{w(\lambda+\rho) - w'(\mu+\rho)}\, dt.$$

A term survives only if $w(\lambda+\rho) = w'(\mu+\rho)$. Both $\lambda+\rho$ and $\mu+\rho$ are strictly dominant. Each W-orbit meets the dominant chamber exactly once, and strictly dominant weights have trivial stabiliser [I]. So a term survives only if $\lambda = \mu$ and $w = w'$. There are $|W|$ such terms. ∎

This agrees with Schur orthogonality. It is a consistency check, not a new proof.

Comparison with SO(3). Let $T = \{R_\varphi\}$, where R_φ is the rotation through φ about the z-axis. Exercise 9.4 shows that Ad_{R_φ} rotates $\mathfrak{m}$ through φ. So $R_\varphi^\alpha = e^{i\varphi}$, $J(R_\varphi) = 4\sin^2(\varphi/2)$ and $|W| = 2$. Formal exponentiation of the weight $\rho = \alpha/2$ would give $e^{i\varphi/2}$. This is not a function on T, because φ and $\varphi + 2\pi$ give the same rotation.

The density J is still defined. Conjugation by the rotation $\mathrm{diag}(1,-1,-1)$ reverses the z-axis, so it maps R_φ to $R_{-\varphi} = R_{2\pi-\varphi}$. Hence a class function satisfies $f(R_{2\pi-\varphi}) = f(R_\varphi)$. The

factor $\sin^2\frac{\varphi}{2}$ is also unchanged under $\varphi \mapsto 2\pi - \varphi$, and $2\sin^2\frac{\varphi}{2} = 1 - \cos\varphi$. Therefore the class-function formula (9.4) gives

$$\int_{\mathrm{SO}(3)} f\,dk = \frac{1}{\pi}\int_0^{2\pi} f(R_\varphi)\sin^2\frac{\varphi}{2}\,d\varphi = \frac{1}{\pi}\int_0^{\pi} f(R_\varphi)(1-\cos\varphi)\,d\varphi. \tag{9.8}$$

We compare this with $\mathrm{SU}(2)$. The image of Haar measure under the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ is the Haar measure of $\mathrm{SO}(3)$, by uniqueness (Lecture 5). The element t_θ maps to a rotation through $\pm 2\theta$. So the substitution $\varphi = 2\theta$ turns (9.6) into the first form of (9.8). Exercise 9.4 shows that small rotations are rare.

The irreducible representations of $\mathrm{SO}(3)$ are the V_{2j} (Lecture 8). Here $\lambda = j\alpha$, and (9.7) reads

$$\chi_{2j}(R_\varphi)\,(1 - e^{-i\varphi}) = e^{ij\varphi} - e^{-i(j+1)\varphi}.$$

Hence

$$\int |\chi_{2j}|^2 = \frac{1}{2}\int_T |e^{ij\varphi} - e^{-i(j+1)\varphi}|^2 dt = \frac{1}{2}\cdot 2 = 1.$$

Self-checks

1. Find J for $\mathrm{U}(2)$ with $t = \mathrm{diag}(e^{ia}, e^{ib})$. Is $\rho \in X^*(T)$?
   Answer. There is one positive root, $t^\alpha = e^{i(a-b)}$. So $J = |e^{ia} - e^{ib}|^2 = 4\sin^2\frac{a-b}{2}$. Identify $X^*(T)$ with $\mathbb{Z}^2$, where (p,q) labels the weight ℓ with $t^\ell = e^{i(pa+qb)}$. Then $\alpha = (1,-1)$ and $\rho = (\frac{1}{2}, -\frac{1}{2}) \notin X^*(T)$.
2. What is the Haar probability that $g \in \mathrm{SU}(2)$ has $\operatorname{tr} g \geq 0$?
   Answer. The remark on indicator functions (Remark 9.7) with $c(g) = \operatorname{tr} g$ lets us use (9.6) for this indicator. The condition on the trace means $\theta \in [0, \pi/2]$. The probability is
   $$\frac{2}{\pi}\int_0^{\pi/2} \sin^2\theta\,d\theta = \frac{2}{\pi}\cdot\frac{\pi}{4} = \frac{1}{2}.$$
   This matches the symmetry $g \mapsto -g$.
3. Why can $\int_{\mathrm{SU}(2)} |g_{11}|^2 dk$ not be computed from (9.6) with $f(t_\theta) = |e^{i\theta}|^2$?
   Answer. The function $g \mapsto |g_{11}|^2$ is not a class function. Schur orthogonality (Lecture 5) gives the correct value $1/\dim V_1 = \frac{1}{2}$. Using $f(t_\theta) = 1$ in (9.6) would give the wrong value 1. To compute this integral with the Weyl integration formula, one must use the general form (9.3).

Summary

Let K be a compact connected Lie group with maximal torus T. Every regular element of K has exactly $|W|$ preimages under the conjugation map $K/T \times T \to K$. Relative to the normalised measures, the Jacobian of this map at (gT, t) is $\det(1 - \mathrm{Ad}_t)$ on $\mathfrak{k}/\mathfrak{t}$. This determinant is a product of root-plane factors $|1 - t^\alpha|^2$ over the positive roots α. The Weyl denominator identity fixes the constant. For $\mathrm{SU}(2)$ and a continuous class function f, the Haar integral of f is $\frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta$. With the character formula in the form (9.7), norms and multiplicities become Fourier calculations on T. This form never needs ρ to be a character.

Sources and hypotheses

The regular set, (C1). BD V.2, Proposition (2.6), covers $K \setminus K_{\mathrm{reg}}$ by the image of a compact manifold of dimension at most $\dim K - 3$. Such an image is null, because a smooth map from a manifold of lower dimension has a null image (Lee, Corollary 6.11). Haar measure has a smooth positive density, so a set that is null in charts has Haar measure zero. BD's regular elements agree with ours by the centraliser-dimension formula of Proposition 9.5.

Openness also follows from that proposition. The map ψ is onto (Lecture 6), and $\psi^{-1}(K_{\mathrm{reg}}) = K/T \times T_{\mathrm{reg}}$. Hence $K \setminus K_{\mathrm{reg}} = \psi\big(K/T \times (T \setminus T_{\mathrm{reg}})\big)$. The set $T \setminus T_{\mathrm{reg}}$ is a finite union of kernels of characters, so it is closed in T. So $K \setminus K_{\mathrm{reg}}$ is a continuous image of a compact set. Hence it is compact, and therefore closed in K.

The change of variables, (C2). The source is Lee, Chapter 16, Propositions 16.40–16.42. Proposition 16.40 is the absolute-Jacobian rule for pulling back a density in coordinates. Lee defines integrals of densities through charts and a partition of unity. Lee does not state (C2) itself, so we derive it from these results.

First, each point $y \in U$ has an open neighbourhood $O \subset U$ whose inverse image consists of d disjoint open sets, each mapped diffeomorphically onto O. To see this, let $x_1, \dots, x_d$ be the preimages of y. Choose disjoint open sets $V_i \ni x_i$ in $\psi^{-1}(U)$ that ψ maps diffeomorphically onto open sets $\psi(V_i)$. The set $C = M \setminus \bigcup_i V_i$ is closed in the compact manifold M, so it is compact. It contains no preimage of y. So $\psi(C)$ is compact, hence closed, and $y \notin \psi(C)$. Put $O = \bigcap_i \psi(V_i) \setminus \psi(C)$. This set is open and contains y. Every preimage of a point of O lies outside C, hence in some V_i. So the inverse image of O is the disjoint union of the sets $V_i \cap \psi^{-1}(O)$. Since $O \subset \psi(V_i)$, the map ψ sends each of these sets diffeomorphically onto O.

Thus each point of U has a neighbourhood that is evenly covered by d sheets, as in the definition of a covering map in Lecture 4. The domain need not be connected. For SU(2) it has two components, and each maps diffeomorphically onto K_{reg}.

Next, suppose F is supported in such a set O. Apply the change-of-variables rule on each of the d sets $V_i \cap \psi^{-1}(O)$ and add the results. A partition of unity on U subordinate to these sets O gives the general case: the partition is countable; sum the pieces by dominated convergence, since the partial sums are bounded by $|F \circ \psi| J_\psi$ and $|F|$. The null sets $N \setminus U$ and $M \setminus \psi^{-1}(U)$ do not contribute. BD I.5, (5.19), states the oriented version through the mapping degree.

Reading

- H: Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed., §11.6.
- BD: Bröcker–tom Dieck, *Representations of Compact Lie Groups*, IV.1, V.2 and VI.1.
- Knapp Beyond: *Lie Groups Beyond an Introduction*, VIII.5.
- Lee: *Introduction to Smooth Manifolds*, 2nd ed., Corollary 6.11, and Chapter 16, Propositions 16.40–16.42.

Lecture 10

Peter–Weyl and completeness

Objectives. *Lecture 5 proved that matrix coefficients of inequivalent irreducible representations are orthogonal. This lecture proves that finite-dimensional matrix coefficients span a dense subspace of the square-integrable functions on a compact Lie group. It then deduces that every compact Lie group is a closed subgroup of a unitary group.*

By the end of this lecture you will be able to:

- Average a rank-one projection to obtain a nonzero positive compact intertwiner, and use it to prove that the finite-dimensional invariant subspaces span a dense subspace in every strongly continuous unitary representation of a compact Lie group.
- Identify the finite-dimensional subspaces of $L^2(K)$ invariant under the right translations R with spaces of matrix coefficients, and state the Peter–Weyl theorem with multiplicities.
- Prove that a compact Lie group has a faithful finite-dimensional unitary representation.
- Compute the coefficient blocks of $\mathrm{U}(1)$ and $\mathrm{SU}(2)$.

Two results from earlier courses are special cases of the theorem of this lecture. For a finite group G, the regular module gives $\mathbb{C}[G] \cong \bigoplus_U \mathrm{End}(U)$, so that $\sum_U (\dim U)^2 = |G|$ (Finite Groups). For the circle $\mathrm{U}(1)$, a square-integrable function has a Fourier series in the functions $e^{in\theta}$. The Peter–Weyl theorem contains both. Each irreducible representation Π of a compact Lie group K gives a block of d_Π^2 matrix coefficients, and the span of these blocks is dense in $L^2(K)$. For $\mathrm{SU}(2)$ the blocks have $1, 4, 9, \dots$ functions. Compared with a finite group, the new difficulty is the infinite sum.

The proof has three parts. First we average a rank-one projection over the group. The result commutes with the group, and it is a compact operator. Next, its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant, so every unitary representation contains finite-dimensional pieces. Finally we apply this to $L^2(K)$, where a finite-dimensional invariant piece turns out to consist of matrix coefficients.

10.1 Analytic preliminaries

Throughout, K is a compact Lie group. We write $\int_K f(g)\,dg$ for its normalised Haar integral on $C(K)$. We import four facts from Lecture 5.

(H1) The integral is left and right invariant. It is invariant under $g \mapsto g^{-1}$, and $\int_K 1\,dg = 1$.

(H2) If $f \in C(K)$ is nonnegative and not identically zero, then $\int_K f\,dg > 0$.

(H3) A finite-dimensional continuous representation has an invariant inner product. It is then an orthogonal sum of irreducible subrepresentations.

(H4) Schur orthogonality, in the form (10.1) below.

We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5. It states that the two iterated Haar integrals agree.

Let (Π, V) be a finite-dimensional continuous unitary representation. Its *matrix coefficients* are the continuous functions

$$c^{\Pi}_{u,v}(x) = \langle u, \Pi(x)v\rangle, \qquad u, v \in V.$$

The *coefficient space* $\mathscr{R}(K)$ is the span of all such functions, for all such (Π, V). Let $\mathrm{Irr}(K)$ be a set of representatives of the isomorphism classes of irreducible finite-dimensional continuous unitary representations. Write $d_\Pi = \dim V_\Pi$. For $\Pi, \Pi' \in \mathrm{Irr}(K)$, Schur orthogonality (H4) reads

$$\int_K \overline{c^{\Pi}_{u,v}(x)}\, c^{\Pi'}_{u',v'}(x)\,dx = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{10.1}$$

Definition 10.1. Let $(H_i)_{i\in I}$ be mutually orthogonal closed subspaces of a Hilbert space H. Their *Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the closure of the algebraic sum $\bigoplus_i H_i$.

The elements of the Hilbert sum are the convergent series $\sum_i v_i$ with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$. For Hilbert spaces H_i not given inside one space, the *external Hilbert sum* $\widehat{\bigoplus}_i H_i$ is the space of families (v_i) with $v_i \in H_i$ and $\sum_i \|v_i\|^2 < \infty$.

A bounded operator on a Hilbert space H is *compact* if it is an operator-norm limit of finite-rank operators. A self-adjoint operator T on H is *positive* if $\langle v, Tv\rangle \geq 0$ for all $v \in H$.

We import the spectral theorem for compact self-adjoint operators from Linear Algebra for Physics, Weeks 13–14, in the following form. If T is compact and self-adjoint, then

$$H = \ker T \;\widehat{\oplus}\; \widehat{\bigoplus_{\mu\neq 0}} \ker(T - \mu).$$

Moreover, each $\ker(T-\mu)$ with $\mu \neq 0$ is finite-dimensional. If $T \neq 0$, then T has a nonzero eigenvalue.

Definition 10.2. The space $L^2(K)$ is the Hilbert completion of $C(K)$ for the sesquilinear form $\langle f, h\rangle = \int_K \overline{f(g)}\, h(g)\,dg$.

By positivity (H2), this sesquilinear form is an inner product on $C(K)$. Hence the map $C(K) \to L^2(K)$ is injective. An element of $L^2(K)$ is *continuous* if it lies in $C(K)$. Only continuous elements have values $f(x)$. For continuous functions, $\|f\|_2 \leq \|f\|_\infty = \sup_x |f(x)|$.

Definition 10.3. A *strongly continuous unitary representation* of K on a Hilbert space H is a homomorphism Π from K to the unitary group of H such that $g \mapsto \Pi(g)v$ is continuous for every $v \in H$.

We do not ask for continuity in operator norm. Already right translation on $L^2(\mathrm{U}(1))$ is not norm continuous (Exercise 10.2).

Lemma 10.4. *Let $F\colon K \to H$ be continuous. There is a unique vector $\int_K F(g)\,dg$ in H with*

$$\left\langle w, \int_K F(g)\,dg \right\rangle = \int_K \langle w, F(g)\rangle\,dg \qquad (w \in H).$$

Moreover $\|\int_K F\| \le \int_K \|F(g)\|\,dg$, and $A\int_K F = \int_K AF$ for bounded A. If each $F(g)$ lies in a closed subspace W, so does $\int_K F$. The integral is unchanged if $F(g)$ is replaced by $F(hg)$.

Proof. The scalar integrand is continuous. The map $w \mapsto \overline{\int_K \langle w, F\rangle}$ is linear with bound $\|w\| \int_K \|F\|$. The Riesz representation theorem for Hilbert spaces, from Linear Algebra for Physics, gives the vector and the norm bound. For bounded A, $\langle w, A\int F\rangle = \int \langle A^* w, F\rangle = \int \langle w, AF\rangle$. If $w \perp W$, the integrand vanishes, so $\int F \in W^{\perp\perp} = W$. Invariance follows from the left invariance in (H1). ∎

10.2 An averaged compact operator

Theorem 10.5. *Let Π be a strongly continuous unitary representation of K on $H \neq 0$.*

(a) The space H contains a nonzero finite-dimensional invariant subspace.

(b) The sum H_{fin} of all finite-dimensional invariant subspaces is dense in H.

We look for a nonzero compact self-adjoint intertwiner. An intertwiner has invariant eigenspaces. A compact self-adjoint operator has finite-dimensional eigenspaces for nonzero eigenvalues. Averaging a rank-one projection over K gives an intertwiner, and compactness of K keeps it a compact operator.

Proof. Fix a unit vector $u \in H$. Let E_u be the rank-one projection given by $E_u v = \langle u, v\rangle u$. Then $\Pi(g) E_u \Pi(g)^{-1} v = \langle \Pi(g)u, v\rangle \Pi(g)u$, which is continuous in g. Define the average

$$Tv = \int_K \langle \Pi(g)u, v\rangle\, \Pi(g)u\,dg. \tag{10.2}$$

Step 1: T is bounded, positive and nonzero. The vector-integral lemma, Lemma 10.4, gives $\|Tv\| \le \|v\|$ and

$$\langle w, Tv\rangle = \int_K \langle \Pi(g)u, v\rangle \langle w, \Pi(g)u\rangle\,dg.$$

Exchanging v, w and conjugating gives the same integral, so $T^* = T$. For $w = v$ the integrand is $|\langle \Pi(g)u, v\rangle|^2 \ge 0$, so T is positive. For $v = w = u$ this integrand is continuous with value 1 at e. By positivity (H2), $\langle u, Tu\rangle > 0$. Hence T is nonzero.

Step 2: T commutes with Π. For $h \in K$,

$$T\Pi(h)v = \int_K \langle \Pi(h^{-1}g)u, v\rangle\, \Pi(g)u\,dg = \int_K \langle \Pi(g)u, v\rangle\, \Pi(h)\Pi(g)u\,dg = \Pi(h)Tv.$$

The first equality uses unitarity. The second replaces g by hg. The third takes $A = \Pi(h)$ in the vector-integral lemma, Lemma 10.4.

Step 3: T is compact. The orbit $O = \{\Pi(g)u : g \in K\}$ is compact, since it is a continuous image of K. Let $\varepsilon > 0$. Since O is compact, finitely many open balls of radius ε with centres in O cover O.

Let P be the orthogonal projection onto the span of the centres. Put $x_g = \Pi(g)u$ and $y_g = Px_g$. Each x_g lies within ε of some centre, and y_g is the nearest point of the span of the centres to x_g. Hence $\|x_g - y_g\| < \varepsilon$ for all g. Now

$$\langle x_g, v\rangle\langle w, x_g\rangle - \langle y_g, v\rangle\langle w, y_g\rangle = \langle x_g - y_g, v\rangle\langle w, x_g\rangle + \langle y_g, v\rangle\langle w, x_g - y_g\rangle.$$

Because $\|x_g\| = 1$ and $\|y_g\| \le 1$, the Cauchy–Schwarz inequality shows that this difference has modulus at most $2\varepsilon\|v\|\|w\|$. On the other hand,

$$\langle w, PTPv\rangle = \langle Pw, TPv\rangle = \int_K \langle y_g, v\rangle\langle w, y_g\rangle\, dg.$$

Subtracting gives $|\langle w, (T - PTP)v\rangle| \le 2\varepsilon\|v\|\|w\|$, so $\|T - PTP\| \le 2\varepsilon$. The operator PTP has finite rank, since P has finite rank. Hence T is compact.

Step 4: proof of (a). By Steps 1 and 3, T is compact, self-adjoint and nonzero. Hence the spectral theorem gives an eigenvalue $\mu \neq 0$ of T with finite-dimensional eigenspace $H_\mu \neq 0$. If $Tv = \mu v$, then Step 2 gives $T\Pi(h)v = \Pi(h)Tv = \mu\Pi(h)v$. So H_μ is invariant.

Step 5: proof of (b). A sum of two finite-dimensional invariant subspaces is again one, so H_{fin} is a subspace. Its closure M is invariant. Hence $M^\perp$ is closed and invariant, since $\langle \Pi(g)w, m\rangle = \langle w, \Pi(g)^{-1}m\rangle$. Restricting Π gives a strongly continuous unitary representation on $M^\perp$. If $M^\perp \neq 0$, the existence statement (a), applied to $M^\perp$, gives a finite-dimensional invariant subspace $W \neq 0$ in $M^\perp$. But $W \subset H_{\mathrm{fin}} \subset M$. Thus $W \subset M \cap M^\perp = 0$, a contradiction. ∎

Corollary 10.6. *Let Π be a strongly continuous unitary representation of K on H. Suppose that Π is irreducible, meaning that $H \neq 0$ and that the only closed invariant subspaces are 0 and H. Then* $\dim H < \infty$.

Proof. Since $H \neq 0$, the existence statement, part (a) of Theorem 10.5, gives a nonzero finite-dimensional invariant subspace W. Finite-dimensional subspaces are closed, so $W = H$ by irreducibility. ∎

This corollary uses compactness. A noncompact Lie group can have irreducible unitary representations of infinite dimension, and Part II treats them.

10.3 The regular representation

We begin with a tool for continuous functions on K, which we use throughout this section.

Lemma 10.7. *(a) For $\varphi \in C(K)$ and $\varepsilon > 0$ there is an open $U \ni e$ with $|\varphi(yx) - \varphi(x)| < \varepsilon$ and $|\varphi(xy) - \varphi(x)| < \varepsilon$ for all $x \in K$, $y \in U$.*

(b) For each open $U \ni e$ there is a continuous $\varphi \ge 0$ vanishing outside U with $\varphi(e) > 0$ and $\int_K \varphi = 1$.

Proof. (a) The function $F(y,x) = |\varphi(yx) - \varphi(x)| + |\varphi(xy) - \varphi(x)|$ is continuous on $K \times K$ and vanishes on $\{e\} \times K$. For each x there are open sets $U_x \ni e$ and $N_x \ni x$ with $F < \varepsilon$ on $U_x \times N_x$. Finitely many $N_{x_1}, \dots, N_{x_r}$ cover K. Take $U = \bigcap_j U_{x_j}$.

(b) Fix a norm $|\cdot|$ on $\mathfrak{k}$, and let B_r be the open ball of radius r about 0. By Lecture 2, exp maps B_r homeomorphically onto an open neighbourhood of e for all sufficiently small $r > 0$. Shrink r so that $\exp(B_r) \subset U$. Put $\psi(\exp X) = \max(0, r/2 - |X|)$ on $\exp(B_r)$ and $\psi = 0$ elsewhere. Then ψ is continuous on $\exp(B_r)$.

The closed ball $\overline{B}_{r/2}$ is compact, so its image under exp is compact and hence closed in the Hausdorff space K. The function ψ vanishes on the open complement $K \setminus \exp(\overline{B}_{r/2})$. The two open sets $\exp(B_r)$ and $K \setminus \exp(\overline{B}_{r/2})$ cover K, so ψ is continuous on K. Since $\psi(e) = r/2 > 0$, positivity (H2) gives $\int \psi > 0$. Take $\varphi = \psi / \int \psi$. Then $\varphi(e) > 0$. ∎

For $f \in C(K)$ define

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

Lemma 10.8. *The translations L and R extend to commuting strongly continuous unitary representations on $L^2(K)$.*

Proof. We treat R. The proof for L is the same, except that it uses left invariance and replaces U below by $U \cap U^{-1}$.

First, R is a homomorphism on $C(K)$. For $f \in C(K)$ and $g, h, x \in K$,

$$(R_g R_h f)(x) = (R_h f)(xg) = f(xgh) = (R_{gh} f)(x).$$

By the right invariance in (H1), R_g preserves $\|\cdot\|_2$. So R_g extends to an isometry of $L^2(K)$ with inverse $R_{g^{-1}}$.

Next, we prove strong continuity at e. Let $f \in L^2(K)$ and $\varepsilon > 0$. Choose $f_0 \in C(K)$ with $\|f - f_0\|_2 < \varepsilon$. Uniform continuity, Lemma 10.7(a), gives an open $U \ni e$ with $\|R_g f_0 - f_0\|_2 \le \|R_g f_0 - f_0\|_\infty < \varepsilon$ for $g \in U$. For $g \in U$, the triangle inequality and the isometry property give

$$\|R_g f - f\|_2 \le \|R_g(f - f_0)\|_2 + \|R_g f_0 - f_0\|_2 + \|f_0 - f\|_2 < 3\varepsilon.$$

Continuity at g_0 follows from $R_g f - R_{g_0} f = R_{g_0}\big(R_{g_0^{-1}g} f - f\big)$, since R_{g_0} is an isometry.

Finally, for $f \in C(K)$ both $L_g R_h f$ and $R_h L_g f$ are the function $x \mapsto f(g^{-1}xh)$. The two bounded operators therefore agree on the dense subspace $C(K)$, so they commute on $L^2(K)$. ∎

For $\varphi \in C(K)$ and $f \in L^2(K)$ put

$$R_\varphi f = \int_K \varphi(g) R_g f \, dg.$$

Elements of $L^2(K)$ have no values at points, but a matrix coefficient must have them. Averaging removes this difficulty. The element $R_\varphi f$ is a weighted average of right translates of f, and it is continuous.

Lemma 10.9 (Smoothing). *For $\varphi \in C(K)$ and $f \in L^2(K)$, the element $R_\varphi f$ is continuous, and*

$$(R_\varphi f)(x) = \langle k_x, f \rangle, \qquad k_x(h) = \overline{\varphi(x^{-1}h)}.$$

Proof. Put $(Sf)(x) = \langle k_x, f\rangle$. Fix $\varepsilon > 0$. Choose U for φ by uniform continuity, Lemma 10.7(a). If $x' \in xU^{-1}$, then $y = x'^{-1}x \in U$. Writing $t = x^{-1}h$ gives $|k_x(h) - k_{x'}(h)| = |\varphi(t) - \varphi(yt)| < \varepsilon$. By the Cauchy–Schwarz inequality, $|Sf(x) - Sf(x')| \le \varepsilon\|f\|_2$. So $Sf \in C(K)$ and $\|Sf\|_2 \le \|Sf\|_\infty \le \|\varphi\|_\infty\|f\|_2$. For $f, w \in C(K)$, Fubini and the substitution $h = xg$ give

$$\langle w, R_\varphi f\rangle = \int_K \varphi(g) \int_K \overline{w(x)} f(xg)\,dx\,dg = \int_K \overline{w(x)} \int_K \varphi(x^{-1}h) f(h)\,dh\,dx = \langle w, Sf\rangle.$$

As $C(K)$ is dense, $R_\varphi f = Sf$ for $f \in C(K)$. Both operators are bounded, so they agree on $L^2(K)$. ∎

Proposition 10.10. *(a) Let $W \subset L^2(K)$ be a finite-dimensional R-invariant subspace. Then $W \subset C(K)$. There is $u_W \in W$ with $f(x) = \langle u_W, R_x f\rangle$ for all $f \in W$ and $x \in K$.*

(b) The coefficient space $\mathscr{R}(K)$ is the sum of all finite-dimensional R-invariant subspaces. It is dense in $L^2(K)$.

Proof. (a) Choose an orthonormal basis $f_1, \dots, f_n$ of W. Strong continuity gives an open $U \ni e$ with $\|R_g f_i - f_i\| < 1/(2n)$ for all i and $g \in U$. By Lemma 10.7(b), there is a continuous $\varphi \ge 0$ vanishing outside U with $\int_K \varphi = 1$. This normalisation gives $R_\varphi f - f = \int_K \varphi(g)\big(R_g f - f\big)\,dg$. For a unit vector $f = \sum a_i f_i$ we have $|a_i| \le 1$. Hence

$$\|R_\varphi f - f\| \le \sum_i |a_i| \int_K \varphi(g)\|R_g f_i - f_i\|\,dg \le n \cdot \frac{1}{2n} = \frac{1}{2}.$$

So $\|R_\varphi f\| \ge \tfrac12$ for every unit vector $f \in W$, and R_φ is injective on W. Since W is closed and R-invariant, the vector-integral lemma, Lemma 10.4, shows that R_φ maps W into W. As W is finite-dimensional, $R_\varphi W = W$. Every element of W is therefore of the form $R_\varphi f$, and the smoothing lemma, Lemma 10.9, gives $W \subset C(K)$.

Now evaluation $f \mapsto f(e)$ is a linear functional on W. The Riesz representation theorem in W gives u_W with $f(e) = \langle u_W, f\rangle$. For $f \in W$, $R_x f$ lies in W and is the function $h \mapsto f(hx)$. Hence $f(x) = (R_x f)(e) = \langle u_W, R_x f\rangle$.

(b) The restriction of R to W is a finite-dimensional continuous unitary representation. By the reproducing formula in (a), each $f \in W$ is a matrix coefficient of this representation. Hence $W \subset \mathscr{R}(K)$. Conversely, $R_h c^{\Pi}_{u,w} = c^{\Pi}_{u,\Pi(h)w}$, so each $c^{\Pi}_{u,v}$ lies in the finite-dimensional R-invariant space $\{c^{\Pi}_{u,w} : w \in V\}$. Density follows from the density statement Theorem 10.5(b), applied to the representation R of the regular-representation lemma, Lemma 10.8. ∎

10.4 The Peter–Weyl theorem

The group $K \times K$ acts on $L^2(K)$ by

$$((g,h)\cdot f)(x) = f(g^{-1}xh), \qquad \text{that is, } (g,h) \mapsto L_g R_h.$$

For $\Pi \in \mathrm{Irr}(K)$, give $\mathrm{End}(V_\Pi)$ the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^*B)$ and the action $(g,h)\cdot A = \Pi(h) A \Pi(g)^{-1}$.

Let $\overline{V}_\Pi$ be the conjugate space, with $\overline{\Pi}(g)\bar{u} = \overline{\Pi(g)u}$ and inner product $\langle \bar{u}, \bar{v}\rangle = \langle v, u\rangle$. For this inner product, $\overline{\Pi}$ is unitary. The representation $\overline{\Pi}$ is irreducible, since $W \mapsto \overline{W}$ matches the invariant subspaces of Π and $\overline{\Pi}$. The coefficients of $\overline{\Pi}$ are the conjugate functions:

$$c^{\overline{\Pi}}_{\bar{u},\bar{v}}(x) = \big\langle \bar{u}, \overline{\Pi(x)v}\big\rangle = \langle \Pi(x)v, u\rangle = \overline{c^{\Pi}_{u,v}(x)}.$$

The space $\overline{V}_\Pi$ is isomorphic to V_Π^* by $\bar{u} \mapsto \langle u, \cdot\rangle$. The map $\bar{u} \otimes v \mapsto v\langle u, \cdot\rangle$ identifies $\overline{V}_\Pi \otimes V_\Pi$ with $\mathrm{End}(V_\Pi)$. For $g, h \in K$, unitarity of $\Pi(g)$ gives $\Pi(h)v\langle u, \Pi(g)^{-1}\cdot\rangle = \Pi(h)v\langle \Pi(g)u, \cdot\rangle$. The left side is the action of (g,h) on $v\langle u,\cdot\rangle$. Hence the pair (g,h) acts on $\overline{V}_\Pi \otimes V_\Pi$ as $\overline{\Pi}(g) \otimes \Pi(h)$. Thus the first tensor factor carries the left action L.

In Dirac notation, $v\langle u,\cdot\rangle$ is the operator $|v\rangle\langle u|$, and $c^\Pi_{u,v}(x) = \langle u|\Pi(x)|v\rangle$ is a matrix element. Under (g,h) the ket moves by $\Pi(h)$ and the bra by $\Pi(g)$, and bras form the conjugate space $\overline{V}_\Pi$.

We define

$$\Phi_\Pi \colon \mathrm{End}(V_\Pi) \to C(K), \qquad \Phi_\Pi(A)(x) = \sqrt{d_\Pi}\ \mathrm{tr}\big(\Pi(x)A\big),$$

and let E_Π be its image. For $A = v\langle u,\cdot\rangle$ we get $\Phi_\Pi(A) = \sqrt{d_\Pi}\, c^\Pi_{u,v}$.

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π, and write $\Pi_{ij} = c^\Pi_{e_i,e_j}$. This is the entry of the matrix of $\Pi(x)$ in row i and column j. The operator $e_j\langle e_i,\cdot\rangle$ has matrix with a single entry 1, in row j and column i. Since $\mathrm{tr}\big(\Pi(x)e_j\langle e_i,\cdot\rangle\big) = \Pi_{ij}(x)$, the map Φ_Π sends this operator to $\sqrt{d_\Pi}\,\Pi_{ij}$. So a column of the matrix of A corresponds to a row of the array (Π_{ij}).

Theorem 10.11 (Peter–Weyl). *(a) Each Φ_Π is a $K \times K$-equivariant isometry from $(\mathrm{End}\, V_\Pi, \mathrm{HS})$ onto E_Π, and $E_\Pi \perp E_{\Pi'}$ for $\Pi \neq \Pi'$.*

(b) The coefficient space satisfies $\mathscr{R}(K) = \bigoplus_{\Pi\in\mathrm{Irr}(K)} E_\Pi$, an algebraic direct sum.

(c) The Hilbert completion of $\mathscr{R}(K)$ is $L^2(K) = \widehat{\bigoplus}_{\Pi\in\mathrm{Irr}(K)} E_\Pi$. The map $\bigoplus_\Pi \Phi_\Pi$ extends to a unitary $K \times K$-isomorphism

$$\widehat{\bigoplus_\Pi}\,\mathrm{End}(V_\Pi) \cong \widehat{\bigoplus_\Pi}\,\overline{V}_\Pi \otimes V_\Pi \longrightarrow L^2(K).$$

(d) Under R, $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$ and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = d_\Pi$. Under L, $E_\Pi \cong \overline{V}_\Pi^{\oplus d_\Pi}$, and again each irreducible occurs with multiplicity equal to its dimension.

The theorem says that functions on K are organised in square blocks, one for each irreducible representation. In an orthonormal basis, block Π holds the d_Π^2 matrix entries Π_{ij}, up to the factor $\sqrt{d_\Pi}$. Figure 10.1 draws these blocks as arrays of functions. Since $\Pi(xh) = \Pi(x)\Pi(h)$, we have $\Pi_{ij}(xh) = \sum_k \Pi_{ik}(x)\Pi_{kj}(h)$, so R_h keeps the row i. Since $\Pi(g^{-1}x) = \Pi(g^{-1})\Pi(x)$, we have $\Pi_{ij}(g^{-1}x) = \sum_k \Pi_{ik}(g^{-1})\Pi_{kj}(x)$, so L_g keeps the column j.

Proof. (a) For $A = v\langle u,\cdot\rangle$ and $B = v'\langle u',\cdot\rangle$ we have $A^*B = u\langle v, v'\rangle\langle u',\cdot\rangle$, so $\mathrm{tr}(A^*B) = \langle u', u\rangle\langle v, v'\rangle$. By Schur orthogonality (10.1),

$$\langle \Phi_\Pi(A), \Phi_\Pi(B)\rangle = d_\Pi \cdot d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle = \langle A, B\rangle_{\mathrm{HS}}.$$

Both sides are sesquilinear and rank-one operators span $\mathrm{End}(V_\Pi)$. The same argument gives $E_\Pi \perp E_{\Pi'}$. Equivariance is the calculation

$$\Phi_\Pi(A)(g^{-1}xh) = \sqrt{d_\Pi}\,\mathrm{tr}\big(\Pi(g)^{-1}\Pi(x)\Pi(h)A\big) = \Phi_\Pi\big(\Pi(h)A\Pi(g)^{-1}\big)(x).$$

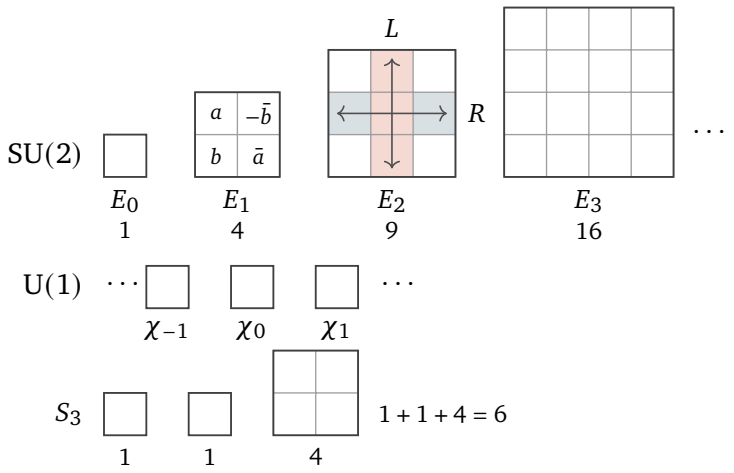


Figure 10.1: Blocks of functions in $L^2(K)$ for SU(2), U(1) and the finite group S_3. A cell is one entry Π_{ij}, with row i and column j, and a number below a block counts its cells. The SU(2) block E_1 shows the entries of the defining representation. In the block E_2, right translation R mixes the entries of the shaded row and left translation L mixes the entries of the shaded column.

(b) Let (σ, V) be a finite-dimensional continuous unitary representation. By complete reducibility (H3), $V = \bigoplus_j V_j$ orthogonally with each V_j irreducible. Write $u = \sum u_j$ and $v = \sum v_j$. Since $\sigma(x)v_k \in V_k$, we get $c^{\sigma}_{u,v} = \sum_j c^{\sigma_j}_{u_j,v_j}$.

Suppose that $\sigma_j \cong \Pi \in \mathrm{Irr}(K)$, and let S be an isomorphism. Since σ_j and Π are unitary, S^* is also an intertwiner. By Schur's lemma, $S^*S = \kappa \cdot 1$, and $\kappa > 0$ because S is injective. Hence $U = \kappa^{-1/2} S$ is a unitary intertwiner, and $c^{\sigma_j}_{u_j,v_j} = c^{\Pi}_{Uu_j,Uv_j} \in E_\Pi$. Thus $\mathscr{R}(K) = \sum_\Pi E_\Pi$. The sum is direct by the orthogonality in (a).

(c) Each E_Π is finite-dimensional, hence closed. By density of coefficients, Proposition 10.10(b), their sum is dense. So $L^2(K)$ is their Hilbert sum. The map $\bigoplus \Phi_\Pi$ is isometric with dense image, so it extends to a unitary. Equivariance passes to the closure by continuity.

(d) Under R alone, $A \mapsto \Pi(h)A$ acts on each column of A, so $E_\Pi \cong V_\Pi^{\oplus d_\Pi}$. Let $J : V_\Pi \to L^2(K)$ intertwine R. By the Hilbert-sum decomposition (c), $Jv = \sum_{\Pi'} P_{\Pi'} Jv$ in $L^2(K)$. The orthogonal projection $P_{\Pi'}$ onto $E_{\Pi'}$ commutes with R, because $E_{\Pi'}$ and $E_{\Pi'}^{\perp}$ are invariant. For $\Pi' \neq \Pi$, Schur's lemma gives $P_{\Pi'} J = 0$. So J maps into E_Π, and $\dim \mathrm{Hom}_K(V_\Pi, L^2(K)) = \dim \mathrm{Hom}_K(V_\Pi, V_\Pi^{\oplus d_\Pi}) = d_\Pi$.

Under L, $A \mapsto A\Pi(g)^{-1}$ acts on rows, giving $\overline{V}_\Pi^{\oplus d_\Pi}$. If $\overline{\Pi} \cong \overline{\Pi'}$, then conjugating the intertwiner gives $\Pi \cong \Pi'$. Also $\overline{\overline{\Pi}} = \Pi$. So conjugation induces a dimension-preserving bijection of the isomorphism classes in $\mathrm{Irr}(K)$. The projections $P_{\Pi'}$ also commute with L, so the same Hilbert-sum and Schur arguments apply. ∎

Theorem 10.12. *The coefficient space $\mathscr{R}(K)$ is dense in $C(K)$ for $\|\cdot\|_\infty$.*

Proof. The constant 1 is a coefficient of the trivial representation. Products are coefficients, since $c^{\Pi}_{u,v} c^{\Pi'}_{u',v'} = c^{\Pi\otimes\Pi'}_{u\otimes u', v\otimes v'}$. Conjugates are coefficients, since $\overline{c^{\Pi}_{u,v}} = c^{\overline{\Pi}}_{\bar{u},\bar{v}}$ for the inner product

on $\overline{V}_\Pi$ fixed above. Hence $\mathscr{R}(K)$ is a subalgebra of $C(K)$ that contains the constants and is closed under conjugation. It remains to prove that $\mathscr{R}(K)$ separates points.

Let $x \neq y$ and suppose $f(x) = f(y)$ for all $f \in \mathscr{R}(K)$. Since $L_g c^{\Pi}_{u,v} = c^{\Pi}_{\Pi(g)u,v}$, the space $\mathscr{R}(K)$ is L-invariant, so $f(g^{-1}x) = f(g^{-1}y)$ for all g. Put $s = x^{-1}y \neq e$ and $g = xt^{-1}$. Then $f(t) = f(ts)$ for all t. Hence $R_s f = f$ on $\mathscr{R}(K)$. Since $\mathscr{R}(K)$ is dense in $L^2(K)$ by Proposition 10.10(b), and R_s is bounded, $R_s = 1$ on $L^2(K)$.

Since K is Hausdorff, some open $U \ni e$ omits s. By Lemma 10.7(b), some continuous $\varphi \geq 0$ vanishes outside U and satisfies $\varphi(e) > 0 = \varphi(s)$. Then $(R_s\varphi)(e) = \varphi(s) \neq \varphi(e)$, so $R_s\varphi \neq \varphi$ in $C(K)$. Since $C(K) \to L^2(K)$ is injective, $R_s\varphi \neq \varphi$ in $L^2(K)$, which contradicts $R_s = 1$. Hence $\mathscr{R}(K)$ separates points.

We import the complex Stone–Weierstrass theorem. For compact Hausdorff X, it states that a subalgebra of $C(X)$ is uniformly dense if it contains the constants, is closed under conjugation and separates points. We apply it to $\mathscr{R}(K) \subset C(K)$. ∎

Separation of points has a structural consequence. A compact Lie group is defined abstractly, but it is always a group of matrices.

Theorem 10.13 (Faithful representation)**.** *A compact Lie group K has an injective finite-dimensional continuous unitary representation $\sigma\colon K \to \mathrm{U}(n)$. Its image is a closed subgroup of $\mathrm{U}(n)$, and σ is an isomorphism of Lie groups from K onto this image.*

Proof. We add summands to a representation until its kernel is trivial, and we show that this process stops.

Step 1: a smaller kernel. Let (σ, V) be a finite-dimensional continuous unitary representation with kernel N. For example, the trivial representation on $\mathbb{C}$ has kernel K. The kernel N is a closed subgroup of K, so it is a compact Lie group by the closed-subgroup theorem of Lecture 3. Suppose $N \neq \{e\}$, and choose $s \in N$ with $s \neq e$. By the proof of Theorem 10.12, the functions in $\mathscr{R}(K)$ separate e and s. These functions are spanned by matrix coefficients. So some finite-dimensional continuous unitary representation Π has a matrix coefficient with $c^{\Pi}_{u,v}(s) \neq c^{\Pi}_{u,v}(e)$. Hence $\Pi(s) \neq I$. The direct sum $\sigma \oplus \Pi$ has kernel $N' = N \cap \ker \Pi$. This is a closed subgroup of N, and $N' \neq N$ because $s \notin N'$.

Step 2: comparing N' with N. Since $N' \subseteq N$, we have $\dim N' \leq \dim N$. Suppose $\dim N' = \dim N$. The inclusion $N' \to N$ is an injective immersion between manifolds of equal dimension, so it is a local diffeomorphism. Hence N' is open in N. Let N_0 be the identity component of N. Then $N' \cap N_0$ is an open subgroup of the connected group N_0. An open subgroup is also closed, since its complement is a union of its cosets. So $N' \cap N_0 = N_0$, that is, $N_0 \subseteq N'$. The connected set N_0 is the largest connected subset of N containing e, so it is also the identity component of N'.

The components of N are the cosets of N_0. They are open and they cover the compact group N, so there are finitely many. Since $N_0 \subseteq N'$ and $N' \neq N$, the group N' consists of fewer of these cosets. So N' has fewer components than N.

Step 3: the process stops. By Step 2, each application of Step 1 either lowers the dimension of the kernel, or keeps the dimension and lowers the finite number of components. The dimension can decrease only finitely often. Between two decreases, the number of components decreases at each step. So after finitely many steps the kernel is $\{e\}$.

Step 4: the image. We now have an injective representation, and an orthonormal basis of its space gives $\sigma\colon K \to \mathrm{U}(n)$. The image $\sigma(K)$ is compact, so it is closed in $\mathrm{U}(n)$, and it is a Lie group by the closed-subgroup theorem. A continuous bijection from a compact space onto a Hausdorff space is a homeomorphism. So σ and its inverse are continuous homomorphisms between Lie groups. By the automatic smoothness theorem of Lecture 3, both are smooth. ∎

The proof applies to disconnected K. Step 3 uses that K is a Lie group. Compactness is also needed, since a covering group of a matrix group need not be a matrix group (Lecture 4).

10.5 Examples

Example 10.14 (The circle group)**.** Let $K = \mathrm{U}(1)$. Here $\int f = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$. Irreducible representations of an abelian group are one-dimensional by Schur's lemma. By Lecture 7 they are $\chi_n(e^{i\theta}) = e^{in\theta}$, $n \in \mathbb{Z}$. Then $d_{\chi_n} = 1$ and $E_n = \mathbb{C}\chi_n$. Hence

$$\mathscr{R}(\mathrm{U}(1)) = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}\chi_n, \qquad L^2(\mathrm{U}(1)) = \widehat{\bigoplus}_{n\in\mathbb{Z}} \mathbb{C}\chi_n.$$

Every $f \in L^2$ equals $\sum_n \hat{c}_n\chi_n$ with $\hat{c}_n = \langle \chi_n, f\rangle$ and $\|f\|_2^2 = \sum |\hat{c}_n|^2$. This is L^2 completeness of Fourier series.

For the $K \times K$ action,

$$((g,h)\cdot\chi_n)(x) = \chi_n(g^{-1}xh) = \overline{\chi_n(g)}\,\chi_n(h)\,\chi_n(x).$$

So $\mathbb{C}\chi_n \cong \overline{\chi}_n \otimes \chi_n$. Under L the line $\mathbb{C}\chi_n$ carries χ_{-n}. Under R it carries χ_n.

We can now see the operator (10.2) in this case. Let $\Pi = R$ on $L^2(\mathrm{U}(1))$, and let u be a unit vector with coefficients $a_n = \langle \chi_n, u\rangle$. Then $\langle \chi_k, R_{e^{it}}u\rangle = e^{ikt}a_k$. Hence

$$\langle \chi_k, T\chi_m\rangle = \frac{1}{2\pi}\int_0^{2\pi} e^{ikt}a_k\,\overline{e^{imt}a_m}\,dt = \delta_{km}|a_m|^2.$$

So $T\chi_m = |a_m|^2\chi_m$. Since $\sum_n |a_n|^2 = 1$, for each $\mu > 0$ only finitely many n have $|a_n|^2 = \mu$. So each eigenspace of T with $\mu \neq 0$ is finite-dimensional, as the spectral theorem says.

Example 10.15 (The group SU(2))**.** Let $K = \mathrm{SU}(2)$ and write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

By Lecture 8, $\mathrm{Irr}(\mathrm{SU}(2)) = \{V_m : m \geq 0\}$, where V_m has dimension $m+1$. Realise V_m as symmetric tensors in $(\mathbb{C}^2)^{\otimes m}$. Let $e_1^{m-k}e_2^k$ be the average of the $\binom{m}{k}$ words with $m-k$ letters e_1 and k letters e_2. The group acts by $g^{\otimes m}$, and the tensor inner product gives

$$\langle e_1^{m-k}e_2^k, e_1^{m-j}e_2^j\rangle = \delta_{jk}\binom{m}{k}^{-1}.$$

By the Peter–Weyl theorem, Theorem 10.11,

$$L^2(\mathrm{SU}(2)) = \widehat{\bigoplus}_{m\geq 0} E_m, \qquad \dim E_m = (m+1)^2 = 1, 4, 9, 16, \ldots$$

Under R, V_m occurs $m+1$ times. In physics, the matrix coefficients of V_m are called Wigner D-functions.

The block $m = 1$. For $V_1 = \mathbb{C}^2$, $c_{e_i,e_j}(g) = g_{ij}$. So $E_1 = \operatorname{span}\{a, -\bar{b}, b, \bar{a}\}$, and the functions $\sqrt{2}\, g_{ij}$ are orthonormal. In particular $\int |a|^2\, dg = \frac{1}{2}$. Under R, the row spans $\operatorname{span}\{a, -\bar{b}\}$ and $\operatorname{span}\{b, \bar{a}\}$ are copies of V_1.

The block m. Take the unit vector $u = e_1^m$. Since $ge_1 = ae_1 + be_2$ and the vectors $e_1^{m-k}e_2^k$ are orthogonal, only the $k = 0$ term of the binomial expansion has nonzero inner product with u:

$$c_{u,u}(g) = \left\langle e_1^m, (ae_1 + be_2)^m \right\rangle = a^m.$$

Then Schur orthogonality (10.1) gives $\int |a|^{2m}\, dg = 1/(m+1)$, matching $m = 1$.

Self-checks

1. Why is the operator (10.2) nonzero, although no invariant vector is known?

Answer. By the definition (10.2), $\langle u, Tu\rangle = \int |\langle \Pi(g)u, u\rangle|^2\, dg$. The integrand is continuous, nonnegative and equal to 1 at e. By positivity (H2), the integral is positive. Hence T is nonzero.

2. Give $\dim E_2 \subset L^2(\mathrm{SU}(2))$ and the multiplicity of V_2 under R.

Answer. The block has dimension $\dim E_2 = 9$. The representation V_2 occurs 3 times. Under L, $E_2 \cong \overline{V}_2^{\oplus 3}$.

3. Why does the coefficient proposition, Proposition 10.10, prove continuity before evaluating at e?

Answer. Only continuous elements of $L^2(K)$ have values. Evaluation is not even bounded on $\mathscr{R}(\mathrm{U}(1))$. For example, the function $f_N = (2N+1)^{-1/2} \sum_{|n|\le N} \chi_n$ has $\|f_N\|_2 = 1$ and $f_N(1) = \sqrt{2N+1}$.

Summary

Let K be a compact Lie group. In a strongly continuous unitary representation of K on $H \neq 0$, an averaged rank-one projection is a nonzero positive compact intertwiner. Its eigenspaces for nonzero eigenvalues are finite-dimensional and invariant. The finite-dimensional invariant subspaces span a dense subspace. In $L^2(K)$, smoothing identifies the finite-dimensional R-invariant subspaces with spaces of matrix coefficients. Schur orthogonality then gives the Peter–Weyl theorem. Under R, and also under L, each $\Pi \in \mathrm{Irr}(K)$ occurs in $L^2(K)$ with multiplicity d_Π. Density in $L^2(K)$ shows that $\mathscr{R}(K)$ separates points. By the Stone–Weierstrass theorem, $\mathscr{R}(K)$ is therefore uniformly dense in $C(K)$. Adding finitely many representations that separate points gives a faithful finite-dimensional unitary representation, so K is isomorphic as a Lie group to a closed subgroup of some $\mathrm{U}(n)$.

Reading

- Sepanski, *Compact Lie Groups*, §§3.2–3.3, including Theorem 3.25.
- Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, Sections 1–4.
- Etingof, MIT 18.755 lecture notes, Lectures 9–10.

Lecture 11

Fourier analysis and convolution

Objectives. *On the circle, a function has scalar Fourier coefficients. On a nonabelian compact group, the coefficient at an irreducible representation Π is an operator $\hat{f}(\Pi)$. We derive Plancherel from Peter–Weyl. We compute how convolution acts on these operators. We identify the convolution operators that commute with all translations.*

By the end of this lecture you will be able to:

- Compute $\hat{f}(\Pi)$ for matrix coefficients and derive the Plancherel theorem from Peter–Weyl.
- Prove $(f * h)^{\wedge} = \hat{h}\,\hat{f}$ and check this order by a direct $\mathrm{SU}(2)$ computation.
- Separate L^2 convergence of spectral partial sums from pointwise or uniform convergence.
- Characterise central convolution operators and write the isotypic projectors.

11.1 The operator-valued transform

Throughout, K is a compact Lie group with normalised Haar measure dg, so $\int_K dg = 1$. From Lecture 5 we import the following facts. The measure is invariant under left translation, right translation and inversion $g \mapsto g^{-1}$. A continuous function $\varphi \geq 0$ with $\int_K \varphi\, dg = 0$ vanishes. This property is strict positivity, which we call *positivity* below. We also import Fubini's theorem for continuous functions on $K \times K$ from Bröcker–tom Dieck, Section I.5, as in Lecture 10.

Every finite-dimensional representation is unitarisable and is a direct sum of irreducible ones. Finally, Schur orthogonality holds in the following form. For irreducible unitary representations Π, Π', vectors $u, v \in V_\Pi$ and vectors $u', v' \in V_{\Pi'}$,

$$\int_K \overline{\langle u, \Pi(g)v\rangle}\, \langle u', \Pi'(g)v'\rangle\, dg = \begin{cases} d_\Pi^{-1}\langle u', u\rangle\langle v, v'\rangle, & \Pi' = \Pi, \\ 0, & \Pi' \not\cong \Pi, \end{cases} \tag{11.1}$$

where $d_\Pi = \dim V_\Pi$.

From Lecture 10 we import two facts. The space $L^2(K)$ is the Hilbert completion of $C(K)$ with respect to the inner product $\langle \varphi, \psi\rangle = \int_K \overline{\varphi}\,\psi\, dg$. By the Peter–Weyl theorem, the span $\mathscr{R}(K)$ of matrix coefficients of finite-dimensional representations is dense in $L^2(K)$.

Fix a set $\widehat{K}$ containing exactly one irreducible unitary representation (Π, V_Π) from each equivalence class. On $\mathrm{End}(V_\Pi)$ we use the Hilbert–Schmidt inner product $\langle A, B\rangle_{\mathrm{HS}} = \mathrm{tr}(A^*B)$. For $u, v \in V_\Pi$, the operator uv^* is given by $w \mapsto \langle v, w\rangle u$.

Definition 11.1. Let $f \in C(K)$ and $\Pi \in \widehat{K}$. The *Fourier transform* of f at Π is

$$\hat{f}(\Pi) = \int_K f(g)\,\Pi(g^{-1})\,dg \in \mathrm{End}(V_\Pi),$$

integrated entry by entry in any basis. For $A \in \mathrm{End}(V_\Pi)$ the *matrix coefficient* c_A is the function given by $c_A(g) = \mathrm{tr}(A\,\Pi(g))$. The space of matrix coefficients of Π is $M_\Pi = \{c_A : A \in \mathrm{End}(V_\Pi)\}$.

For $K = \mathrm{U}(1)$ and $\Pi = \chi_n$ this is the classical coefficient $\frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})e^{-in\theta}\,d\theta$. The inverse g^{-1} in the definition gives the minus sign.

A rank-one operator gives the familiar coefficient $c_{uv^*}(g) = \langle v, \Pi(g)u\rangle$. Let Π' be equivalent to Π, say $\Pi'(g) = T\Pi(g)T^{-1}$ with T invertible. Then

$$\mathrm{tr}\big(A\,\Pi'(g)\big) = \mathrm{tr}\big(T^{-1}AT\,\Pi(g)\big) = c_{T^{-1}AT}(g).$$

Hence the matrix coefficients of Π' also lie in M_Π. So M_Π does not depend on the choice of representative. By complete reducibility, $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π.

These objects are those of Lecture 10 under other names. There $\widehat{K}$ is $\mathrm{Irr}(K)$, the space M_Π is E_Π, and $c_A = d_\Pi^{-1/2}\Phi_\Pi(A)$. The operator uv^* is $u\langle v, \cdot\rangle$, so the letters u and v are exchanged. The Peter–Weyl theorem already gives that the spaces M_Π are orthogonal and that $L^2(K)$ is their Hilbert sum. It also gives $\|f\|^2 = \sum_\Pi \|A_\Pi\|_{\mathrm{HS}}^2$ when $f = \sum_\Pi \Phi_\Pi(A_\Pi)$, and the coefficient lemma below gives $A_\Pi = \sqrt{d_\Pi}\,\hat{f}(\Pi)$. So what is new in this lecture is the formula for the projection and the operator form of the coefficient.

Lemma 11.2. *For $f \in C(K)$, $\Pi \in \widehat{K}$ and $B \in \mathrm{End}(V_\Pi)$,*

$$\langle c_B, f\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}}. \tag{11.2}$$

Proof. For every operator X we have $\overline{\mathrm{tr}\,X} = \mathrm{tr}\,X^*$. Since $\Pi(g)^* = \Pi(g^{-1})$,

$$\overline{c_B(g)} = \mathrm{tr}\big(\Pi(g)^*B^*\big) = \mathrm{tr}\big(B^*\,\Pi(g^{-1})\big).$$

Multiply by $f(g)$ and integrate. The trace is linear, so it passes through the entrywise integral. This gives $\mathrm{tr}(B^*\hat{f}(\Pi))$. ∎

We use (11.2) to extend the transform to $L^2(K)$. The left side of (11.2) is L^2-continuous in f. For fixed f, it is an antilinear functional of B. Hence the finite-dimensional Riesz theorem gives a unique operator $\hat{f}(\Pi)$ satisfying (11.2). For continuous f, this operator agrees with the integral of Definition 11.1, by the pairing lemma, Lemma 11.2.

Lemma 11.3. *Let $\Pi, \Pi' \in \widehat{K}$, $A \in \mathrm{End}(V_\Pi)$ and $B \in \mathrm{End}(V_{\Pi'})$. Then*

$$\langle c_B, c_A\rangle = \begin{cases} d_\Pi^{-1}\langle B, A\rangle_{\mathrm{HS}}, & \Pi' = \Pi, \\ 0, & \Pi' \neq \Pi. \end{cases} \tag{11.3}$$

Consequently $\hat{c}_A(\Pi) = d_\Pi^{-1}A$, and $\hat{c}_A(\Pi') = 0$ for $\Pi' \neq \Pi$.

Proof. Both sides are linear in A and antilinear in B. Rank-one operators span, so take $A = uv^*$ and $B = u'v'^*$. Distinct elements of $\widehat{K}$ are inequivalent. By (11.1),

$$\langle c_B, c_A\rangle = \int_K \overline{\langle v', \Pi'(g)u'\rangle}\,\langle v, \Pi(g)u\rangle\, dg = \delta_{\Pi\Pi'}\, d_\Pi^{-1}\langle v, v'\rangle\langle u', u\rangle.$$

On the other hand $(u'v'^*)^* = v'u'^*$, so

$$\langle B, A\rangle_{\mathrm{HS}} = \operatorname{tr}\big(v'u'^*uv^*\big) = \langle u', u\rangle\operatorname{tr}(v'v^*) = \langle u', u\rangle\langle v, v'\rangle.$$

This proves (11.3). By (11.2), $\langle B, \hat{c}_A(\Pi')\rangle_{\mathrm{HS}} = \langle c_B, c_A\rangle$ for every B. Comparing with (11.3) gives the transforms. ∎

11.2 Plancherel from Peter–Weyl

For $f \in L^2(K)$ and a finite set $F \subset \widehat{K}$, the *spectral partial sum* is

$$S_F f(g) = \sum_{\Pi\in F} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big).$$

Theorem 11.4 (Plancherel). *Let $f, h \in L^2(K)$.*

(a) The orthogonal projection of $L^2(K)$ onto M_Π is $P_\Pi f = d_\Pi\, c_{\hat{f}(\Pi)}$.

(b) For every $\varepsilon > 0$ there is a finite $F_0 \subset \widehat{K}$ with $\|f - S_F f\| < \varepsilon$ for every finite $F \supseteq F_0$.

(c) $\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$ and $\langle f, h\rangle = \sum_\Pi d_\Pi \langle \hat{f}(\Pi), \hat{h}(\Pi)\rangle_{\mathrm{HS}}$.

(d) If $\hat{f}(\Pi) = 0$ for every Π, then $f = 0$.

Proof. Step 1. By (11.3), $M_\Pi \perp M_{\Pi'}$ for $\Pi \neq \Pi'$, and $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Each M_Π is finite-dimensional, hence closed.

Step 2. Put $f_0 = d_\Pi c_{\hat{f}(\Pi)} \in M_\Pi$. For every $B \in \operatorname{End}(V_\Pi)$, (11.2) and (11.3) give

$$\langle c_B, f - f_0\rangle = \langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} - d_\Pi\, d_\Pi^{-1}\langle B, \hat{f}(\Pi)\rangle_{\mathrm{HS}} = 0.$$

So $f - f_0 \perp M_\Pi$, that is, $f_0 = P_\Pi f$. Step 1 gives $\|P_\Pi f\|^2 = d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$.

Step 3. This step uses the Peter–Weyl theorem. Let $M_F = \bigoplus_{\Pi\in F} M_\Pi$. By Steps 1 and 2, $S_F = \sum_{\Pi\in F} P_\Pi$ is the orthogonal projection onto M_F. Hence $S_F f$ is the point of M_F nearest to f. By the Peter–Weyl theorem, there is $\varphi \in \mathscr{R}(K)$ with $\|f - \varphi\| < \varepsilon$. Since $\mathscr{R}(K)$ is the algebraic sum of the spaces M_Π, there is a finite F_0 with $\varphi \in M_{F_0}$. If $F \supseteq F_0$, then $\varphi \in M_F$, so $\|f - S_F f\| \le \|f - \varphi\| < \varepsilon$.

Step 4. Pythagoras and Step 2 give

$$\|f\|^2 = \|S_F f\|^2 + \|f - S_F f\|^2, \qquad \|S_F f\|^2 = \sum_{\Pi\in F} d_\Pi\|\hat{f}(\Pi)\|_{\mathrm{HS}}^2.$$

Hence every finite partial sum of the series in (c) is at most $\|f\|^2$. By Step 3, the partial sum over F exceeds $\|f\|^2 - \varepsilon^2$ once $F \supseteq F_0$. A sum of nonnegative terms is the supremum of its finite partial sums, so the norm identity holds. Polarisation gives the inner product identity. Finally, if $\hat{f}(\Pi) = 0$ for every Π, then the norm identity gives $\|f\| = 0$. This is the uniqueness statement (d). ∎

On $\mathrm{U}(1)$ we have $d_\Pi = 1$, and the norm identity is Parseval's identity.

Corollary 11.5 (Inversion for finite coefficient sums)**.** *Let $f \in \mathscr{R}(K)$. Then $\hat f(\Pi) \neq 0$ for only finitely many Π, and for every $g \in K$*

$$f(g) = \sum_{\Pi \in \widehat{K}} d_\Pi \operatorname{tr}\big(\hat f(\Pi)\Pi(g)\big).$$

Proof. Write $f = \sum_{\Pi \in F} c_{A_\Pi}$ with F finite. By the coefficient lemma, Lemma 11.3, $\hat f(\Pi) = d_\Pi^{-1} A_\Pi$ for $\Pi \in F$ and $\hat f(\Pi) = 0$ otherwise. Substituting returns $\sum_{\Pi \in F} c_{A_\Pi}(g) = f(g)$. ∎

For general $f \in L^2(K)$ we only have the L^2 convergence of spectral partial sums in the Plancherel theorem, Theorem 11.4(b). This statement says nothing about the value of f at a single point.

11.3 Convolution reverses the order

For a finite group G, the product of the group algebra $\mathbb{C}[G]$ (Finite Groups) is

$$\Big(\sum_{y} f(y)y\Big)\Big(\sum_{z} h(z)z\Big) = \sum_{x}\Big(\sum_{y} f(y)h(y^{-1}x)\Big)x.$$

The next definition is the same formula, with the sum over y replaced by the normalised Haar integral. Finite Groups also proved $\mathbb{C}[G] \cong \bigoplus_U \operatorname{End}(U)$, by sending $\sum f(g)g$ to the operators $\sum f(g)U(g)$. Under the transform, a function becomes a family of matrices, one for each irreducible, and convolution multiplies these matrices one block at a time. The transform $\hat f$ uses $\Pi(g^{-1})$, so it reverses the order.

Definition 11.6. For $f, h \in C(K)$ the *convolution* $f * h$ is

$$(f * h)(x) = \int_K f(y)\, h(y^{-1}x)\, dy.$$

Fix x. The map $y \mapsto y^{-1}x$ is inversion followed by right translation, and both preserve dy. So $y \mapsto h(y^{-1}x)$ has L^2 norm $\|h\|$, and Cauchy–Schwarz gives $|(f * h)(x)| \le \|f\|\,\|h\|$.

The function $f * h$ is also continuous. By the uniform-continuity lemma of Lecture 10, h is uniformly continuous on the compact group K. In other words, for every $\varepsilon > 0$ there is a neighbourhood N of e with $|h(z) - h(zw)| < \varepsilon$ for all $z \in K$ and $w \in N$. Hence $|(f * h)(x) - (f * h)(xw)| \le \varepsilon \int_K |f|\, dy$ for $w \in N$.

Theorem 11.7 (Convolution theorem)**.** *For $f, h \in C(K)$ and $\Pi \in \widehat{K}$,*

$$(f * h)^{\wedge}(\Pi) = \hat h(\Pi)\, \hat f(\Pi).$$

Proof. Exchange the iterated integrals entrywise:

$$(f * h)^{\wedge}(\Pi) = \int_K f(y)\Big(\int_K h(y^{-1}x)\, \Pi(x^{-1})\, dx\Big)dy.$$

In the inner integral substitute $x = yz$. Left invariance gives $dx = dz$, and $x^{-1} = z^{-1}y^{-1}$. So the inner integral is

$$\int_K h(z)\,\Pi(z^{-1})\Pi(y^{-1})\,dz = \hat{h}(\Pi)\,\Pi(y^{-1}).$$

The factor $\Pi(y^{-1})$ coming from f stands on the right, because $x^{-1} = z^{-1}y^{-1}$. Integrating against $f(y)$ and using the definition of $\hat{f}(\Pi)$ gives $\hat{h}(\Pi)\hat{f}(\Pi)$. ∎

Some texts use $\tilde{f}(\Pi) = \int_K f(g)\Pi(g)\,dg$. The same computation then gives $(f * h)^{\sim} = \tilde{f}\,\tilde{h}$. Check the convention before importing formulas.

Corollary 11.8. *Let $A, B \in \mathrm{End}(V_\Pi)$. Then $c_A * c_B = d_\Pi^{-1} c_{BA}$. If instead $B \in \mathrm{End}(V_{\Pi'})$ with $\Pi' \neq \Pi$, then $c_A * c_B = 0$.*

Proof. By the coefficient lemma and the convolution theorem, Lemma 11.3 and Theorem 11.7, $(c_A * c_B)^\wedge(\Pi) = d_\Pi^{-2} BA$. This is the transform of $d_\Pi^{-1} c_{BA}$. At every other element of $\widehat{K}$ both transforms vanish. The difference φ of the two sides is continuous, and $\|\varphi\| = 0$ by the uniqueness statement of the Plancherel theorem, Theorem 11.4(d). Positivity applied to $|\varphi|^2$ gives $\varphi = 0$. In the second case one of $\hat{c}_A, \hat{c}_B$ vanishes at each element of $\widehat{K}$, so the same argument gives $c_A * c_B = 0$. ∎

For rank-one operators, $(u'v'^*)(uv^*) = \langle v', u\rangle\, u'v^*$. Hence the coefficient-convolution corollary, Corollary 11.8, gives

$$\langle v, \Pi(\cdot)u\rangle * \langle v', \Pi(\cdot)u'\rangle = d_\Pi^{-1}\langle v', u\rangle\, \langle v, \Pi(\cdot)u'\rangle.$$

Fix an orthonormal basis $e_1, \dots, e_{d_\Pi}$ of V_Π. Write $E_{ij} = e_i e_j^*$ for the matrix unit. Its matrix in this basis has 1 in row i and column j, and 0 elsewhere. By the last display, the matrix coefficients $D_{ij}(g) = \langle e_i, \Pi(g)e_j\rangle$ satisfy $D_{ij} * D_{kl} = d_\Pi^{-1}\delta_{jk} D_{il}$. So the functions multiply in the natural order. By the coefficient lemma, Lemma 11.3, their transforms are the transposed matrix units $d_\Pi^{-1} E_{ji}$, and these multiply in reverse.

Example 11.9 (Direct check on SU(2))**.** Write elements of SU(2) as

$$U = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Let Π be the defining representation on $\mathbb{C}^2$, with standard basis e_1, e_2, so $d_\Pi = 2$. Put $f(U) = U_{21} = b$ and $h(U) = U_{12} = -\bar{b}$. Since $\mathrm{tr}(E_{12}U) = U_{21}$ and $\mathrm{tr}(E_{21}U) = U_{12}$, we have $f = c_{E_{12}}$ and $h = c_{E_{21}}$. The convolution theorem predicts the transform of $f * h$. The coefficient-convolution corollary, Corollary 11.8, then gives the function, since $c_{E_{12}} * c_{E_{21}} = \tfrac{1}{2} c_{E_{21}E_{12}} = \tfrac{1}{2} c_{E_{22}}$. Hence

$$(f * h)^\wedge(\Pi) = \hat{h}\hat{f} = \tfrac{1}{4} E_{21}E_{12} = \tfrac{1}{4} E_{22}, \qquad \text{so} \quad (f * h)(U) = \tfrac{1}{2} U_{22} = \tfrac{1}{2}\bar{a}.$$

The opposite order would give the transform $\tfrac{1}{4}E_{11}$ and hence the function $\tfrac{1}{2}a$.

Now compute directly. Since $a = \langle e_1, Ue_1\rangle$ and $b = \langle e_2, Ue_1\rangle$, (11.1) gives

$$\int_{\mathrm{SU}(2)} |b|^2\,dU = \tfrac{1}{2}, \qquad \int_{\mathrm{SU}(2)} \bar{a}\,b\,dU = \tfrac{1}{2}\langle e_2, e_1\rangle = 0.$$

Let y have entries a, b and let x have entries α, β in the same pattern. Since y is unitary, $y^{-1} = y^{*}$. Hence

$$y^{-1}x = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix} \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} = \begin{pmatrix} \bar{a}\alpha + \bar{b}\beta & \bar{b}\bar{\alpha} - \bar{a}\bar{\beta} \\ -b\alpha + a\beta & a\bar{\alpha} + b\bar{\beta} \end{pmatrix}.$$

Since $h(y^{-1}x)$ is the $(1,2)$ entry of this matrix, we get

$$(f * h)(x) = \int b\,(\bar{b}\bar{\alpha} - \bar{a}\bar{\beta})\,dy = \bar{\alpha}\int |b|^2\,dy - \bar{\beta}\int \bar{a}\,b\,dy = \tfrac{1}{2}\bar{\alpha}.$$

This matches the prediction. Using $(y^{-1}x)_{21} = -b\alpha + a\beta$, the same method gives $(h * f)(x) = \tfrac{1}{2}\alpha$. So $f * h \neq h * f$.

11.4 Pointwise and uniform inversion

Lecture 10 also shows that $\mathscr{R}(K)$ is uniformly dense in $C(K)$. This density does not say that the approximants are the partial sums $S_F f$. Uniform convergence of $S_F f$ needs an extra hypothesis.

Proposition 11.10. *Let $f \in C(K)$. Assume*

$$\sum_{\Pi \in \widehat{K}} d_\Pi^{3/2}\, \|\hat{f}(\Pi)\|_{\mathrm{HS}} < \infty.$$

Then the series

$$\sum_{\Pi \in \widehat{K}} d_\Pi\, \operatorname{tr}\big(\hat{f}(\Pi)\,\Pi(g)\big)$$

converges absolutely and uniformly on K. Its sum is $f(g)$ for every $g \in K$.

Proof. We need two facts about the Hilbert–Schmidt norm. The first is Cauchy–Schwarz for $\langle \cdot, \cdot \rangle_{\mathrm{HS}}$, applied to X^{*} and Y. The second uses unitarity of $\Pi(g)$:

$$|\operatorname{tr}(XY)| \le \|X\|_{\mathrm{HS}}\,\|Y\|_{\mathrm{HS}}, \qquad \|\Pi(g)\|_{\mathrm{HS}}^2 = \operatorname{tr}\big(\Pi(g)^{*}\Pi(g)\big) = d_\Pi.$$

Hence, for every $g \in K$,

$$\big|d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(g)\big)\big| \le b_\Pi, \qquad b_\Pi = d_\Pi^{3/2}\|\hat{f}(\Pi)\|_{\mathrm{HS}}.$$

The numbers b_Π have finite sum by hypothesis. Hence, for each n, only finitely many b_Π exceed $1/n$, so only countably many terms are nonzero. By the Weierstrass test the series converges absolutely and uniformly to a continuous function Φ.

Explicitly, given $\varepsilon > 0$, choose a finite $F_1 \subset \widehat{K}$ with $\sum_{\Pi \notin F_1} b_\Pi < \varepsilon$. Then for every finite $F \supseteq F_1$,

$$\sup_{g \in K} \big|\Phi(g) - S_F f(g)\big| \le \sum_{\Pi \notin F} b_\Pi < \varepsilon.$$

Since dg has mass one, $\|\Phi - S_F f\| < \varepsilon$. Enlarge F to contain the set F_0 given for this ε by the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b). By the triangle inequality, $\|\Phi - f\| < 2\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $\|\Phi - f\| = 0$. Positivity applied to $|\Phi - f|^2$ then gives $\Phi = f$. ∎

Remark 11.11 (Cited results). (i) Let K be compact and connected, with maximal torus T of dimension r. Fix a Weyl-invariant inner product on $\mathfrak{t}^*$. By Lecture 8, each $\Pi \in \widehat{K}$ has a highest weight λ_Π. Distinct Π have distinct highest weights, and these lie in the weight lattice of T, a lattice of rank r. Now let $f \in C^\infty(K)$. Then for every N there is C_N with

$$\|\hat{f}(\Pi)\|_{\mathrm{HS}} \le C_N \, (1 + |\lambda_\Pi|)^{-N} \qquad \text{for all } \Pi \in \widehat{K}.$$

We cite this from M. Sugiura, *Fourier series of smooth functions on compact Lie groups*, Osaka J. Math. 8 (1971), 33–47. The proof uses the Casimir operator.

The Weyl dimension formula, imported from the prerequisite Lie algebras course, gives $d_\Pi \le C(1 + |\lambda_\Pi|)^q$, where q is the number of positive roots. A lattice of rank r has $O(R^r)$ points of norm at most R, so $\sum_\Pi (1 + |\lambda_\Pi|)^{-M} < \infty$ for $M > r$. Taking $N > \tfrac{3}{2}q + r$ gives the hypothesis of the uniform inversion proposition, Proposition 11.10. So the Fourier series of a smooth function on a compact connected Lie group converges absolutely and uniformly to it.

(ii) On $K = \mathrm{U}(1)$ there are continuous functions whose symmetric partial sums diverge at a point. This is du Bois-Reymond's classical example (Grafakos, *Classical Fourier Analysis*, Section 3.3.3, Proposition 3.3.5). So convergence at every point, and hence uniform convergence, can fail for continuous functions.

11.5 Central kernels and isotypic projectors

Convolution acts one block at a time. So we can read a function through the matrices $\hat{f}(\Pi)$. Figure 11.1 draws these matrices for SU(2), including two kernels that this section computes: $k(U) = |\operatorname{tr} U|^2$ of Example 11.15, and the projector kernel e_Π of Corollary 11.14 for $\Pi = V_2$. A central kernel turns out to have scalar blocks.

Left and right translation are given, as in Lecture 10, by $(L_g f)(x) = f(g^{-1}x)$ and $(R_g f)(x) = f(xg)$. The character of Π is $\chi_\Pi = \operatorname{tr} \circ \Pi = c_I$. Unitarity gives $\chi_\Pi(g^{-1}) = \overline{\chi_\Pi(g)}$.

Definition 11.12. A function $k \in C(K)$ is *central*, or a *class function* (Lecture 9), if $k(gxg^{-1}) = k(x)$ for all $g, x \in K$. The *convolution operator* of $k \in C(K)$ is the map $T_k \colon C(K) \to C(K)$ given by $T_k f = f * k$. The operator T_k is *central* if k is central.

Substituting $y = gy'$ in the convolution integral shows that $T_k L_g = L_g T_k$ for every k. The next theorem shows that commutation with right translations characterises the central kernels.

Theorem 11.13. *For $k \in C(K)$ the following are equivalent.*

(a) k is central.

(b) $T_k R_g = R_g T_k$ for every $g \in K$.

(c) Every $\hat{k}(\Pi)$ is a scalar multiple of the identity.

In this case $\hat{k}(\Pi) = \kappa_\Pi I$ with $\kappa_\Pi = d_\Pi^{-1} \langle \chi_\Pi, k \rangle$. For $f \in C(K)$ we have $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. Hence, for finite $F \subset \widehat{K}$, the sums $\sum_{\Pi \in F} \kappa_\Pi P_\Pi f$ converge to $f * k$ in L^2, in the sense of the convergence statement of the Plancherel theorem, Theorem 11.4(b).

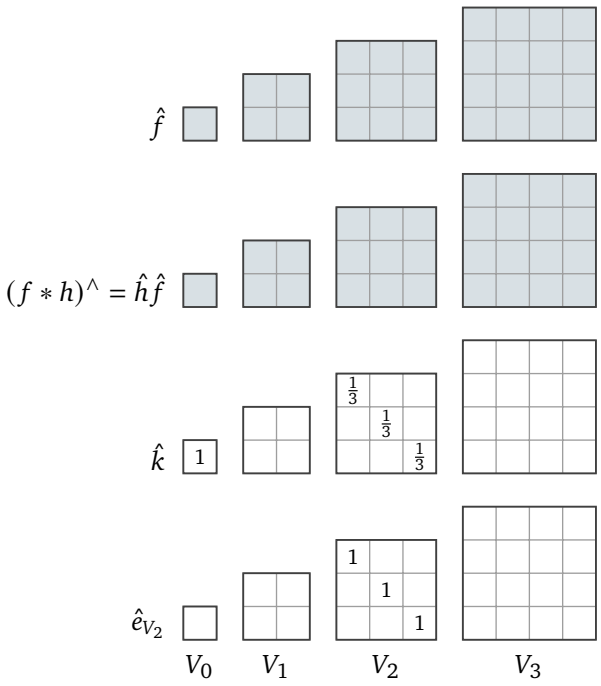


Figure 11.1: The operator blocks $\hat f(V_m)$ on SU(2), for $m = 0, 1, 2, 3$, with a shaded cell for a general entry and a blank cell for zero. The rows show a general f, a convolution $f * h$, the kernel $k(U) = |\operatorname{tr} U|^2$, and the projector kernel e_{V_2}. These are matrices of operators, not the arrays of functions of Lecture 10: since $\hat D_{ij} = d_\Pi^{-1} E_{ji}$, the two grids differ by a transpose.

Proof. *(a)⇔(c).* Fix g and put $k^g(x) = k(gxg^{-1})$. Substitute $x' = gxg^{-1}$, which preserves dx by two-sided invariance:

$$(k^g)^\wedge(\Pi) = \int_K k(x')\,\Pi(g^{-1}x'^{-1}g)\,dx' = \Pi(g)^{-1}\hat k(\Pi)\Pi(g).$$

If k is central, then $k^g = k$. Hence $\hat k(\Pi)$ commutes with every $\Pi(g)$, and by Schur's lemma it is a scalar multiple of the identity. Conversely, if every $\hat k(\Pi)$ is scalar, then k^g and k have equal transforms. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $k^g = k$.

(a)⇔(b). Right invariance with $y = y'g^{-1}$ gives

$$(T_k R_g f)(x) = \int_K f(y)\,k(gy^{-1}x)\,dy, \qquad (R_g T_k f)(x) = \int_K f(y)\,k(y^{-1}xg)\,dy.$$

If k is central, then $k(y^{-1}xg) = k(gy^{-1}x)$, which is (b). Conversely, assume (b) and set $x = e$. Put $\psi(y) = k(y^{-1}g) - k(gy^{-1})$. Then $\int_K f\psi\,dy = 0$ for all f. Taking $f = \bar\psi$ and using positivity gives $\psi = 0$. So $k(zg) = k(gz)$ for all z. Replacing z by zg^{-1} gives $k(z) = k(gzg^{-1})$.

Final claims. We have $\operatorname{tr}\hat k(\Pi) = \int_K k(g)\overline{\chi_\Pi(g)}\,dg = \langle \chi_\Pi, k\rangle$. Since $\hat k(\Pi) = \kappa_\Pi I$, this trace is $d_\Pi\kappa_\Pi$, which gives the formula for κ_Π. By the convolution theorem, Theorem 11.7, $(f * k)^\wedge(\Pi) = \kappa_\Pi \hat f(\Pi)$. Then the projection formula of the Plancherel theorem, Theorem 11.4(a), gives $P_\Pi(f * k) = \kappa_\Pi P_\Pi f$. So $S_F(f * k) = \sum_{\Pi\in F}\kappa_\Pi P_\Pi f$, and the L^2 convergence statement, Theorem 11.4(b), applied to $f * k$ gives the convergence. ∎

Corollary 11.14 (Isotypic projectors). *Let $e_\Pi = d_\Pi \chi_\Pi$. For $f \in C(K)$,*

$$P_\Pi f = f * e_\Pi = e_\Pi * f, \qquad (P_\Pi f)(x) = d_\Pi \int_K f(y)\, \chi_\Pi(y^{-1}x)\, dy.$$

Moreover $e_\Pi * e_{\Pi'} = \delta_{\Pi\Pi'} e_\Pi$.

Proof. First, characters are central, and $e_\Pi = d_\Pi c_I$. By the coefficient lemma, Lemma 11.3, $\hat{e}_\Pi(\Pi) = I$ and $\hat{e}_\Pi(\Pi') = 0$ for $\Pi' \neq \Pi$. So the central kernel theorem, Theorem 11.13, applies to $k = e_\Pi$ with $\kappa_{\Pi'} = \delta_{\Pi\Pi'}$. Its sums therefore equal $P_\Pi f$ once $\Pi \in F$, and they converge to $f * e_\Pi$ in L^2. Hence $\|f * e_\Pi - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f * e_\Pi = P_\Pi f$.

Next, by the convolution theorem, Theorem 11.7, the transforms of $f * e_\Pi$ and $e_\Pi * f$ agree, because $\hat{e}_\Pi$ is scalar at every element of $\widehat{K}$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), the difference of these functions is 0 in $L^2(K)$. Both functions are continuous, so positivity gives $f * e_\Pi = e_\Pi * f$.

Finally, the coefficient-convolution corollary, Corollary 11.8, with $A = B = d_\Pi I$ gives $e_\Pi * e_\Pi = e_\Pi$. Its second case gives $e_\Pi * e_{\Pi'} = 0$ for $\Pi' \neq \Pi$. ∎

As in the Finite Groups course, the kernels e_Π are orthogonal central idempotents. Substituting $y = xg$ in the formula of the corollary gives $P_\Pi f = d_\Pi \int_K \overline{\chi_\Pi(g)}\, R_g f\, dg$. The same formula projects in any finite-dimensional unitary representation σ of K: the operator $d_\Pi \int_K \overline{\chi_\Pi(g)}\, \sigma(g)\, dg$ is the orthogonal projection onto the Π-isotypic part. For trivial Π this is the invariant-vector projector of Lecture 5.

The Π-*isotypic part* of the right regular action R is the sum of all finite-dimensional R-invariant subspaces $W \subset C(K)$ with $R|_W \cong \Pi$.

We now identify M_Π as this isotypic part. By cyclicity of the trace, $c_A(xg) = \operatorname{tr}(\Pi(g)A\Pi(x))$. Hence $R_g c_A = c_{\Pi(g)A}$. The map $A \mapsto c_A$ is injective, because $\|c_A\|^2 = d_\Pi^{-1}\|A\|_{\mathrm{HS}}^2$. Let $e_1, \dots, e_{d_\Pi}$ be an orthonormal basis of V_Π. The map $A \mapsto \Pi(g)A$ sends each column Ae_j to $\Pi(g)Ae_j$. So R on M_Π is equivalent to $\Pi^{\oplus d_\Pi}$. The jth copy of Π consists of the functions c_A with $Ae_i = 0$ for $i \neq j$.

Now let $W \subset C(K)$ be finite-dimensional and R-invariant with $R|_W \cong \Pi$. Fix $\Pi' \neq \Pi$. The map $P_{\Pi'}|_W : W \to M_{\Pi'}$ intertwines R, since $P_{\Pi'} = T_{e_{\Pi'}}$ with $e_{\Pi'}$ central. Its image is 0 or isomorphic to Π, because W is irreducible. Every irreducible subrepresentation of $M_{\Pi'} \cong \Pi'^{\oplus d_{\Pi'}}$ projects nontrivially to some summand, so Schur's lemma makes it isomorphic to Π'. Hence $P_{\Pi'} W = 0$.

So for $f \in W$ we get $S_F f = P_\Pi f$ whenever $\Pi \in F$. By the L^2 convergence statement of the Plancherel theorem, Theorem 11.4(b), $\|f - P_\Pi f\| = 0$. Both functions are continuous, so positivity gives $f = P_\Pi f \in M_\Pi$. Hence every such W lies in M_Π. Conversely, M_Π is the sum of its d_Π copies of Π. Thus M_Π is the Π-isotypic part of R, and P_Π is the orthogonal projector onto it.

Example 11.15 (Central kernels on SU(2) and the circle). Let V_m be the irreducible representation of SU(2) of dimension $m+1$, and write P_m for P_{V_m}. Let $U \in \mathrm{SU}(2)$ have eigenvalues $e^{i\theta}$ and $e^{-i\theta}$ with $0 < \theta < \pi$. Then Lecture 9 gives $\chi_m(U) = \sin((m+1)\theta)/\sin\theta$. By the isotypic projector corollary, Corollary 11.14, $P_m f = (m+1)\, f * \chi_m$.

Take the kernel $k(U) = |\operatorname{tr} U|^2 = 4\cos^2\theta$. It is central, since the trace is invariant under conjugation. Since χ_1 is real and $V_1 \otimes V_1 \cong V_0 \oplus V_2$, we get $k = \chi_0 + \chi_2$. As a check, $\sin 3\theta / \sin\theta = 4\cos^2\theta - 1$.

Characters are orthonormal, so the central kernel theorem, Theorem 11.13, gives $\kappa_0 = 1$, $\kappa_2 = \frac{1}{3}$ and $\kappa_m = 0$ otherwise. Let $f \in C(\mathrm{SU}(2))$. For $F \supseteq \{V_0, V_2\}$, the sums in that theorem equal $P_0 f + \frac{1}{3} P_2 f$, and $P_0 f$ is the constant $\int_{\mathrm{SU}(2)} f\, dU$. These sums converge to $f * k$ in L^2. Both functions are continuous, so positivity gives

$$f * |\operatorname{tr}|^2 = \int_{\mathrm{SU}(2)} f\, dU + \tfrac{1}{3} P_2 f.$$

In particular $U_{11} * k = 0$, because $U_{11} = c_{E_{11}}$ lies in M_1. At $x = I$ this reads $\int a \cdot 4(\operatorname{Re} a)^2\, dy = 0$. Left translation by $-I$ preserves dy and changes the sign of the integrand. Hence this integral vanishes, as predicted.

On $K = \mathrm{U}(1)$ every $\Pi \in \widehat{K}$ is a character χ_n given by $\chi_n(z) = z^n$ for an integer n, with $d_{\chi_n} = 1$. Here $\hat{f}(\chi_n) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) e^{-in\theta} d\theta$ is the classical coefficient. We write $\hat{f}(n)$ for $\hat{f}(\chi_n)$ and P_n for P_{χ_n}. For $f, h \in C(\mathrm{U}(1))$, the convolution theorem, Theorem 11.7, reads $(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$. By the uniqueness statement of the Plancherel theorem, Theorem 11.4(d), and positivity, $f * h = h * f$. The group is abelian, so every function is central, and $P_n f = f * \chi_n = \hat{f}(n) z^n$.

Self-checks

1. On $\mathrm{U}(1)$, why is the reversal in the convolution theorem, Theorem 11.7, invisible?
   Answer. Each Π is one-dimensional. The transforms are numbers, and numbers commute.
2. Let $k(U) = U_{11}$ on $\mathrm{SU}(2)$. Is T_k central?
   Answer. No. Here $k = c_{E_{11}}$ for the defining representation Π, so $\hat{k}(\Pi) = \frac{1}{2} E_{11}$ is not a scalar multiple of the identity. Hence, by the central kernel theorem, Theorem 11.13, k is not central, and so T_k is not central. To check this directly, put

   $$w = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad t_\theta = \operatorname{diag}(e^{i\theta}, e^{-i\theta}).$$

   Then $k(t_\theta) = e^{i\theta}$, while $k(w t_\theta w^{-1}) = e^{-i\theta}$. These differ, for example, at $\theta = \pi/2$.
3. Compute $(\chi_\Pi * \chi_\Pi)(e)$ in two ways.
   Answer. By definition it is $\int_K \chi_\Pi(y)\chi_\Pi(y^{-1})\, dy = \int_K |\chi_\Pi|^2\, dy = 1$. By the identity $e_\Pi * e_\Pi = e_\Pi$ in the isotypic projector corollary, Corollary 11.14, $\chi_\Pi * \chi_\Pi = d_\Pi^{-1} \chi_\Pi$, whose value at e is 1.

Summary

The Fourier coefficient at Π is the operator $\hat{f}(\Pi) = \int f(g)\Pi(g^{-1})\, dg$. Peter–Weyl gives Plancherel. Hence $L^2(K)$ is the Hilbert direct sum of the spaces M_Π, and the orthogonal projection onto M_Π is $P_\Pi f = d_\Pi \operatorname{tr}(\hat{f}(\Pi)\Pi(\cdot))$. The algebraic sum of these spaces is $\mathscr{R}(K)$, which is dense in $L^2(K)$. The transform of a convolution is $(f * h)\hat{} = \hat{h}\hat{f}$. The inversion

formula holds exactly for finite coefficient sums. For general $f \in L^2(K)$, the spectral partial sums converge to f in L^2 norm. Uniform inversion needs an extra hypothesis, such as the summability condition of the uniform inversion proposition, Proposition 11.10. For compact connected K, smoothness suffices by the cited estimate in Remark 11.11. Central kernels are exactly those with scalar transforms. On $C(K)$, the isotypic projectors are $f \mapsto d_\Pi f * \chi_\Pi$.

The circle and a general compact Lie group compare as follows.

$\mathrm{U}(1)$	compact Lie group K
$\hat{f}(n) \in \mathbb{C}$	$\hat{f}(\Pi) \in \mathrm{End}(V_\Pi)$
$\|f\|^2 = \sum_n \|\hat{f}(n)\|^2$	$\|f\|^2 = \sum_\Pi d_\Pi \|\hat{f}(\Pi)\|_{\mathrm{HS}}^2$
$f = \sum_n \hat{f}(n) z^n$ (in L^2)	$f = \sum_\Pi d_\Pi \operatorname{tr}\big(\hat{f}(\Pi)\Pi(\cdot)\big)$ (in L^2)
$(f * h)\hat{}(n) = \hat{f}(n)\hat{h}(n)$	$(f * h)\hat{}(\Pi) = \hat{h}(\Pi)\hat{f}(\Pi)$

Reading

Sepanski, *Compact Lie Groups*, §3.4, for Fourier theory on compact groups. Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Chapter III, §§2–4, for representative functions and the Peter–Weyl theorem behind the Plancherel theorem, Theorem 11.4. Compare each source's transform and convolution conventions with Definitions 11.1 and 11.6 before using its formulas.

Lecture 12

Homogeneous spaces and compact induction

Objectives. *Let K be a compact Lie group and H a closed subgroup. Which irreducible representations of K occur in functions on K/H, and how often? We show that each irreducible representation V of K occurs $\dim V^H$ times. Induction from H extends this count to vector-valued functions.*

By the end of this lecture you will be able to:

- Construct the invariant integral on K/H and identify $L^2(K/H)$ with the right-H-invariant part of $L^2(K)$.
- Prove that each irreducible V occurs in $L^2(K/H)$ exactly $\dim V^H$ times.
- Define compact induction, describe an element of the induced representation in the tangent-field model on S^2, and prove Frobenius reciprocity with finite multiplicities.
- Decompose $L^2(S^2)$ under $\mathrm{SO}(3)$ and relate the summands to spherical harmonics.

Throughout, K is a compact Lie group, not necessarily connected, and $H \subseteq K$ is a closed subgroup. By Lecture 3, K/H is a compact manifold, and the projection $p\colon K \to K/H$, $p(x) = xH$, is continuous and open. The group K acts on K/H by $g \cdot xH = gxH$.

The Haar measures dx on K and dh on H have total mass 1. They are left, right and inversion invariant (Lecture 5). The space $L^2(K)$ is the completion of $C(K)$ for $\langle f_1, f_2\rangle = \int_K \overline{f_1}\, f_2\, dx$, and $C(K) \to L^2(K)$ is injective (Lecture 10).

We use the left and right translation representations

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

These commuting representations are unitary and strongly continuous (Lecture 10).

12.1 Functions and integration on K/H

By the definition of the quotient topology, a function φ on K/H is continuous exactly when $\varphi \circ p$ is continuous. A right-H-invariant function on K is constant on cosets. Hence the pullback p^* is a linear bijection

$$p^*\colon C(K/H) \to C(K)^H := \{f \in C(K) : f(xh) = f(x) \text{ for all } x \in K,\ h \in H\}, \qquad p^*\varphi = \varphi \circ p. \tag{12.1}$$

Definition 12.1. The *invariant integral* on K/H is the linear functional

$$\int_{K/H} \varphi(\xi)\, d\xi := \int_K \varphi(p(x))\, dx, \qquad \varphi \in C(K/H).$$

Proposition 12.2. *(a) The integral is positive,* $\int_{K/H} 1\, d\xi = 1$, *and* $\int_{K/H} \varphi(g^{-1}\xi)\, d\xi = \int_{K/H} \varphi(\xi)\, d\xi$ *for* $g \in K$.

(b) For $f \in C(K)$, *the* H*-average* $f^H(x) = \int_H f(xh)\, dh$ *lies in* $C(K)^H$, *and*

$$\int_K f(x)\, dx = \int_K f^H(x)\, dx = \int_{K/H} \Big(\int_H f(xh)\, dh \Big)\, d(xH).$$

Proof. (a) Positivity and normalisation hold for dx. Invariance follows from $\varphi(g^{-1}p(x)) = \varphi(p(g^{-1}x))$ and left invariance of dx.

(b) The map $(x, h) \mapsto f(xh)$ is uniformly continuous on the compact space $K \times H$. So f^H is continuous. For $k \in H$, left invariance of dh gives $f^H(xk) = \int_H f(xkh)\, dh = f^H(x)$. Fubini for continuous functions on $K \times H$ and right invariance of dx give

$$\int_K f^H(x)\, dx = \int_H \int_K f(xh)\, dx\, dh = \int_H \int_K f(x)\, dx\, dh = \int_K f(x)\, dx.$$

By the pullback bijection (12.1), the last expression in (b) is $\int_K f^H\, dx$ written on K/H. ∎

A positive normalised invariant integral I on K/H is unique. Indeed, the map $g \mapsto \varphi(g^{-1}\,\cdot\,)$ is continuous in the supremum norm. So averaging the identity $I(\varphi) = I(\varphi(g^{-1}\,\cdot\,))$ over K gives $I(\varphi) = I\big(\int_K \varphi(g^{-1}\,\cdot\,)\, dg\big)$. The function inside is constant, since K acts transitively on K/H, and its value is $\int_{K/H} \varphi\, d\xi$. In particular, for a maximal torus $H = T$ of a compact connected group K, this integral is the measure $d(gT)$ of Lecture 9.

Definition 12.3. The space $L^2(K/H)$ is the completion of $C(K/H)$ for $\langle \varphi, \psi \rangle = \int_{K/H} \overline{\varphi}\, \psi\, d\xi$. Also

$$L^2(K)^H = \{ f \in L^2(K) : R_h f = f \text{ for all } h \in H \}.$$

The form is definite since $\int_{K/H} |\varphi|^2\, d\xi = \|\varphi \circ p\|^2$ and $C(K) \to L^2(K)$ is injective. The same identity shows that p^* is isometric. So p^* extends to an isometry $L^2(K/H) \to L^2(K)$. The subspace $L^2(K)^H$ is closed because each R_h is bounded.

The operator of the next proposition replaces a function by its average over each coset. Averaging does not increase an L^2 norm, and it does not change a function that is constant on cosets. In the sense of the vector-integral lemma of Lecture 10, this operator is the integral $\int_H R_h\, dh$. So it is the projector onto invariants of Lecture 5, on an infinite-dimensional space.

Proposition 12.4. *There is a unique bounded operator* A_H *on* $L^2(K)$ *with* $A_H f = f^H$ *for* $f \in C(K)$. *It satisfies*

$$\langle f_1, A_H f_2 \rangle = \int_H \langle f_1, R_h f_2 \rangle\, dh \qquad (f_1, f_2 \in L^2(K)). \tag{12.2}$$

It is the orthogonal projection onto $L^2(K)^H$. *Consequently* p^* *maps* $L^2(K/H)$ *isometrically onto* $L^2(K)^H$.

Proof. Let $f \in C(K)$. Cauchy–Schwarz on H gives $|f^H(x)|^2 \le (|f|^2)^H(x)$. Integrate and apply the averaging identity of Proposition 12.2(b) to $|f|^2$:

$$\|f^H\|^2 \le \int_K (|f|^2)^H \, dx = \int_K |f|^2 \, dx = \|f\|^2.$$

Since $C(K)$ is dense, A_H exists uniquely, with $\|A_H\| \le 1$.

For continuous f_1, f_2, Fubini shows that both sides of (12.2) equal $\int_K \int_H \overline{f_1(x)} f_2(xh)\, dh\, dx$. Both sides are continuous in (f_1, f_2). On the right, the integrand is continuous in h by strong continuity and bounded by $\|f_1\|\,\|f_2\|$. So (12.2) holds on $L^2(K)$.

Now $R_h^* = R_{h^{-1}}$ and dh is inversion invariant. So the weak formula (12.2) gives $\langle f_1, A_H f_2\rangle = \langle A_H f_1, f_2\rangle$, and A_H is self-adjoint.

Next, A_H fixes $L^2(K)^H$. If $f \in L^2(K)^H$, then the weak formula gives $\langle f_1, A_H f\rangle = \langle f_1, f\rangle$ for all f_1, so $A_H f = f$.

Finally, the range of A_H lies in $L^2(K)^H$. Let $k \in H$. For continuous f, $A_H f = f^H$ lies in $C(K)^H$ by the description of the H-average in Proposition 12.2(b), so $R_k A_H f = A_H f$. Since $C(K)$ is dense and both sides are bounded in f, this identity holds for all f.

Hence A_H is a self-adjoint idempotent with range $L^2(K)^H$.

It remains to identify the image of p^*. Since p^* is an isometry, its image is the closure of $p^*(C(K/H)) = C(K)^H$. This closure lies in the closed subspace $L^2(K)^H$. Conversely, let $f \in L^2(K)^H$ and choose $f_n \in C(K)$ with $f_n \to f$. Then $f_n^H \in C(K)^H$ and $f_n^H = A_H f_n \to A_H f = f$. So the image is $L^2(K)^H$. ∎

Each L_g commutes with every R_h, so L_g preserves $L^2(K)^H$. Transported by p^*, the operator L_g acts on $C(K/H)$ by $\varphi \mapsto \varphi(g^{-1}\cdot)$. The resulting unitary representation of K on $L^2(K/H)$ is the *left regular representation*.

12.2 Multiplicities in $L^2(K/H)$

Let (Π, V) be a finite-dimensional unitary representation and $u, v \in V$. The *matrix coefficient* is $c_{u,v}(x) = \langle u, \Pi(x)v\rangle$. Unitarity gives

$$L_g c_{u,v} = c_{\Pi(g)u,\,v}, \qquad R_g c_{u,v} = c_{u,\,\Pi(g)v}. \tag{12.3}$$

For the first identity, $c_{u,v}(g^{-1}x) = \langle u, \Pi(g)^{-1}\Pi(x)v\rangle = \langle \Pi(g)u, \Pi(x)v\rangle$.

The map $u \mapsto c_{u,v}$ is antilinear. The *conjugate representation* $\overline{V}$ is the set V with scalar multiplication $t \cdot u = \bar{t}u$, the same operators $\Pi(g)$, and inner product $\langle u, u'\rangle_{\overline{V}} = \langle u', u\rangle$. It has the same invariant subspaces as V. So $\overline{V}$ is irreducible when V is, and $(\overline{V})^H = V^H$ as sets.

For each $\sigma \in \mathrm{Irr}(K)$ fix a unitary representative (Π_σ, V_σ) of dimension d_σ. Write $\bar\sigma$ for the class of $\overline{V_\sigma}$, and let $E_\sigma = \mathrm{span}\{c_{u,v} : u, v \in V_\sigma\} \subseteq C(K)$. We use two earlier results:

(PW1) Schur orthogonality (Lecture 5): $\langle c_{u,v}, c_{u',v'}\rangle = d_\sigma^{-1}\langle u', u\rangle\langle v, v'\rangle$ for $u, v, u', v' \in V_\sigma$.

(PW2) Peter–Weyl (Lecture 10): the E_σ are mutually orthogonal and $L^2(K) = \widehat{\bigoplus}_\sigma E_\sigma$.

By Schur orthogonality (PW1), the linear map $\overline{V_\sigma} \otimes V_\sigma \to E_\sigma$, $u \otimes v \mapsto c_{u,v}$, is $d_\sigma^{-1/2}$ times an isometry. So it is bijective. By the coefficient identities (12.3), it carries $\bar{\Pi}_\sigma \otimes 1$ to L and $1 \otimes \Pi_\sigma$ to R. Let Q_σ be the orthogonal projection onto E_σ. A unitary operator that preserves a finite-dimensional subspace also preserves its orthogonal complement. Hence Q_σ commutes with all L_g and R_g. This Q_σ is the projector onto the matrix coefficients of Π_σ, called P_{Π_σ} in Lecture 11.

For $\sigma \in \mathrm{Irr}(K)$ let $E_\sigma^H = \mathrm{span}\{c_{u,v} : u \in V_\sigma,\ v \in V_\sigma^H\}$, a subspace of E_σ.

We read the count from the matrix coefficient $c_{u,v}$. Right translation by H acts only on v, so $c_{u,v}$ is right-H-invariant when $v \in V_\sigma^H$. Left translation acts only on u, and u enters antilinearly. So the right-H-invariant part of E_σ is $\dim V_\sigma^H$ copies of $\overline{V_\sigma}$. Block σ therefore contributes to the multiplicity of $\bar{\sigma}$. Since $\dim V_\sigma^H = \dim V_{\bar{\sigma}}^H$, the count for σ is still $\dim V_\sigma^H$. For SO(3) every irreducible is self-conjugate, and no shift occurs.

Figure 12.1 draws this count in the arrays of functions of Lecture 10. In a block, the entry c_{e_i,e_j} has its second index on the side of R. So the condition $f(xh) = f(x)$ keeps the columns whose vector e_j lies in V_σ^H.

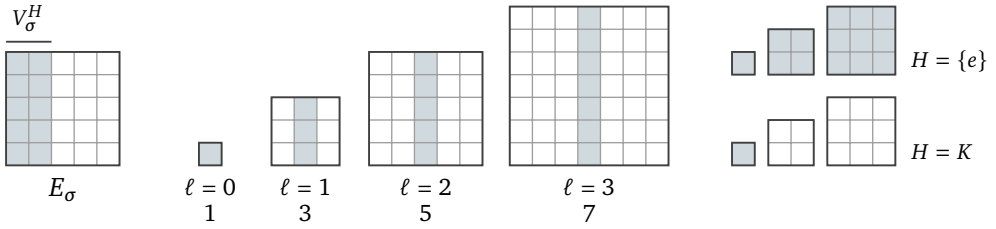


Figure 12.1: Blocks of functions in $L^2(K)$, as in Lecture 10, with the columns that remain on K/H shaded. From left to right: a general H, in a basis of V_σ whose first vectors span V_σ^H; the sphere SO(3)/SO(2), with blocks of side $2\ell + 1$ and one column each, of $2\ell + 1$ functions, for the weight-zero vector of § 12.4; and the extreme cases $H = \{e\}$ and $H = K$. Left translation mixes the entries of a shaded column.

Proposition 12.5. *Let $P_\sigma = \int_H \Pi_\sigma(h)\, dh$ be the orthogonal projection of V_σ onto V_σ^H (Lecture 5). Then $A_H c_{u,v} = c_{u,P_\sigma v}$. Moreover*

$$L^2(K/H) \cong L^2(K)^H = \overbrace{\bigoplus_{\sigma \in \mathrm{Irr}(K)}} E_\sigma^H, \qquad E_\sigma^H \cong \overline{V_\sigma} \otimes V_\sigma^H,$$

where K acts on $\overline{V_\sigma} \otimes V_\sigma^H$ by $\bar{\Pi}_\sigma \otimes 1$.

Proof. The finite-dimensional integral commutes with the functional $w \mapsto \langle u, \Pi_\sigma(x) w \rangle$. So

$$(A_H c_{u,v})(x) = \int_H \langle u, \Pi_\sigma(x)\Pi_\sigma(h) v \rangle\, dh = \left\langle u, \Pi_\sigma(x) \int_H \Pi_\sigma(h) v\, dh \right\rangle = c_{u,P_\sigma v}(x).$$

Thus $A_H(E_\sigma) = E_\sigma^H$, and A_H fixes E_σ^H. Hence $E_\sigma^H \subseteq L^2(K)^H$. Conversely, let $f \in L^2(K)^H$. By Peter–Weyl (PW2), $f = \sum_\sigma Q_\sigma f$ in L^2. Since A_H is bounded,

$$f = A_H f = \sum_\sigma A_H Q_\sigma f, \qquad A_H Q_\sigma f \in E_\sigma^H.$$

Since E_σ^H lies in E_σ, the spaces E_σ^H are mutually orthogonal by Peter–Weyl (PW2). This proves the decomposition. The isomorphism with $\overline{V_\sigma} \otimes V_\sigma^H$ is the restriction of the bijection above. ∎

Induction in § 12.3 uses vector-valued functions and evaluates them at the identity e. An L^2 class has no value at a point. The next lemma shows that equivariant images of V_σ consist of continuous functions, so these evaluations make sense.

Let W be a finite-dimensional Hilbert space with orthonormal basis $w_1, \dots, w_n$. Then $L^2(K) \otimes W$ is the Hilbert space of functions $\sum_j f_j w_j$ with $f_j \in L^2(K)$. It contains $C(K, W) = C(K) \otimes W$. Let K act by $L \otimes 1$.

Lemma 12.6. *Let $\sigma \in \operatorname{Irr}(K)$. Let $T : V_\sigma \to L^2(K) \otimes W$ be linear with $T\Pi_\sigma(g) = (L_g \otimes 1)T$ for all $g \in K$. Then $T(V_\sigma) \subseteq E_{\bar\sigma} \otimes W$. In particular, every vector in $T(V_\sigma)$ is a continuous function.*

Proof. Fix $\eta \in \operatorname{Irr}(K)$. The operator $Q_\eta \otimes 1$ commutes with $L_g \otimes 1$. So $(Q_\eta \otimes 1)T$ is a K-map from V_σ into

$$E_\eta \otimes W \cong \overline{V_\eta} \otimes (V_\eta \otimes W),$$

where K acts only on the first factor. This space is a direct sum of $d_\eta \dim W$ copies of $\overline{V_\eta}$. By Schur's lemma, $(Q_\eta \otimes 1)T = 0$ unless $V_\sigma \cong \overline{V_\eta}$, that is, unless $\eta = \bar\sigma$. Apply Peter–Weyl (PW2) in each coordinate. For $v \in V_\sigma$,

$$Tv = \sum_\eta (Q_\eta \otimes 1)Tv = (Q_{\bar\sigma} \otimes 1)Tv \in E_{\bar\sigma} \otimes W.$$ ∎

Thus the values $(Tv)(x)$ are defined for all $v \in V_\sigma$ and $x \in K$.

Let (E, Π) be a unitary representation of K and $\sigma \in \operatorname{Irr}(K)$. The *multiplicity* of σ in E is $m_\sigma(E) = \dim \operatorname{Hom}_K(V_\sigma, E)$. Here Hom_K consists of the linear maps T with $T\Pi_\sigma(g) = \Pi(g)T$. These maps are bounded because V_σ is finite-dimensional. The *σ-isotypic subspace* of E is the sum of the images of these maps.

Theorem 12.7. *For every $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2(K/H)\big) = \dim V_\sigma^H.$$

The σ-isotypic subspace is $E_{\bar\sigma}^H \subseteq C(K/H)$, of dimension $d_\sigma \dim V_\sigma^H$. The space $L^2(K/H)$ is the Hilbert sum of its isotypic subspaces.

Proof. Identify $L^2(K/H)$ with $L^2(K)^H$ through p^*, as in Proposition 12.4. Let $T : V_\sigma \to L^2(K)^H$ be a K-map. By Lemma 12.6 on equivariant images, with $W = \mathbb{C}$, $T(V_\sigma) \subseteq E_{\bar\sigma}$. A vector of $E_{\bar\sigma}$ fixed by A_H lies in $A_H(E_{\bar\sigma}) = E_{\bar\sigma}^H$. Hence

$$\operatorname{Hom}_K\big(V_\sigma, L^2(K/H)\big) = \operatorname{Hom}_K\big(V_\sigma, E_{\bar\sigma}^H\big).$$

Since $V_{\bar\sigma} \cong \overline{V_\sigma}$, we have $\overline{V_{\bar\sigma}} \cong V_\sigma$. So Proposition 12.5 on the decomposition of $L^2(K)^H$ gives

$$E_{\bar\sigma}^H \cong \overline{V_{\bar\sigma}} \otimes V_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H,$$

with K acting on the first factor only. By Schur's lemma,

$$\mathrm{Hom}_K\left(V_\sigma, V_\sigma \otimes V_{\bar\sigma}^H\right) \cong V_{\bar\sigma}^H.$$

A K-isomorphism $V_{\bar\sigma} \to \overline{V_\sigma}$ carries $V_{\bar\sigma}^H$ onto $(\overline{V_\sigma})^H$. This is the set V_σ^H with conjugated scalars, so its dimension is $\dim V_\sigma^H$. The images of the K-maps $V_\sigma \to E_{\bar\sigma}^H$ span $E_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H$. Finally, $\sigma \mapsto \bar\sigma$ is a bijection of $\mathrm{Irr}(K)$, so the decomposition in Proposition 12.5 gives the Hilbert sum. ∎

Thus the summand E_σ^H of Proposition 12.5 is indexed by σ but is $\bar\sigma$-isotypic. The count is unchanged because $\dim V_\sigma^H = \dim V_{\bar\sigma}^H$. Exercise 12.1 shows this label shift for $K = \mathrm{U}(1)$. Without any bar, the σ-isotypic subspace is spanned by the functions $x \mapsto \langle u, \Pi_\sigma(x)^{-1} v\rangle$ with $u \in V_\sigma^H$ and $v \in V_\sigma$.

12.3 Compact induction and Frobenius reciprocity

Let (τ, W) be a finite-dimensional unitary representation of H. Averaging over H gives every finite-dimensional representation of H an invariant inner product (Lecture 5).

A tangent vector field on S^2 is the model for what follows. Let $K = \mathrm{SO}(3)$, and let H be the subgroup of rotations $R_3(\theta)$ about e_3, as in Lecture 4. Let $x \in \mathrm{SO}(3)$. The columns xe_1 and xe_2 form an oriented orthonormal frame of the tangent plane at $p = xe_3$, and the coset xH is the set of all such frames at p. Let $F(x) \in \mathbb{C}^2$ hold the components of the field at p in this frame. Replacing x by $xR_3(\theta)$ rotates the frame through θ, and the components change by the inverse rotation. So

$$F(xh) = \tau(h)^{-1} F(x), \qquad h \in H,$$

where τ is the rotation action of H on $\mathbb{C}^2$. For example, the tangential part of a fixed vector $v \in \mathbb{R}^3$ has the components $F(x) = (xe_1 \cdot v,\ xe_2 \cdot v)$. Figure 12.2 shows one tangent vector in two frames.

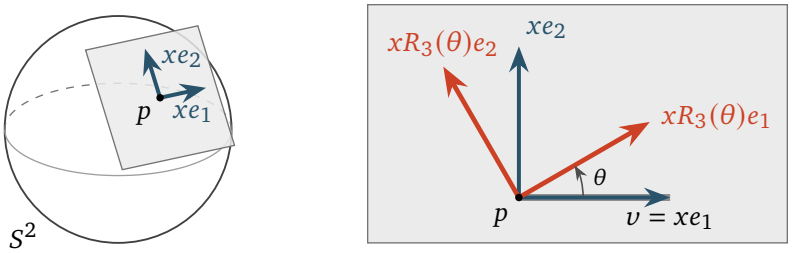


Figure 12.2: On the left, a point $p = xe_3$ of the sphere, with its tangent plane and the frame (xe_1, xe_2). On the right, the same plane seen from outside the sphere, with a second frame for $\theta = \pi/6$, turned anticlockwise. The tangent vector $v = xe_1$ has components $F(x) = (1, 0)$ in the first frame and $F(xR_3(\pi/6)) = (\sqrt{3}/2, -1/2)$ in the second frame.

Definition 12.8. The *induced representation* is

$$\mathrm{Ind}_H^K W = \{F \in C(K, W) : F(xh) = \tau(h)^{-1} F(x) \text{ for all } x \in K,\ h \in H\},$$

with K acting by $(L_g F)(x) = F(g^{-1}x)$.

Left translation preserves the covariance condition because left and right translations commute. For the trivial representation $W = \mathbb{C}$ of H, we recover $C(K)^H \cong C(K/H)$.

On a coset xH the function F is fixed by its value at x. So an element of $\operatorname{Ind}_H^K W$ assigns to each coset a vector of W, once a point of the coset is chosen. In the model above, with $W = \mathbb{C}^2$, the vector $F_1(x)xe_1 + F_2(x)xe_2$ does not depend on the point x of the coset. So $\operatorname{Ind}_H^K W$ is the space of continuous tangent fields on S^2 with complex components. Under this identification, L_g sends a field X to the rotated field $p \mapsto gX(g^{-1}p)$.

For finite K this is the induced representation of the Finite Groups course. The element $x \otimes w$ of $\mathbb{C}[K] \otimes_{\mathbb{C}[H]} W$ corresponds to the function F with $F(xh) = \tau(h)^{-1}w$ for $h \in H$ and $F = 0$ off xH.

Theorem 12.9 (Frobenius reciprocity, continuous form). *Let (Π, V) be a finite-dimensional representation of K. The map*

$$\Phi\colon \operatorname{Hom}_K\left(V, \operatorname{Ind}_H^K W\right) \to \operatorname{Hom}_H(V|_H, W), \qquad \Phi(T)v = (Tv)(e),$$

is a linear bijection. Its inverse sends A to $\Psi(A)$, where $(\Psi(A)v)(x) = A\big(\Pi(x)^{-1}v\big)$.

Proof. First, $\Phi(T)$ is H-equivariant. For $h \in H$, equivariance of T and covariance give

$$\Phi(T)\big(\Pi(h)v\big) = (L_h Tv)(e) = (Tv)(h^{-1}) = \tau(h^{-1})^{-1}(Tv)(e) = \tau(h)\,\Phi(T)v.$$

Next, $\Psi(A)v$ is continuous because $x \mapsto \Pi(x)^{-1}v$ is continuous. Since A is an H-map, $\Psi(A)v$ is covariant:

$$(\Psi(A)v)(xh) = A\big(\Pi(h)^{-1}\Pi(x)^{-1}v\big) = \tau(h)^{-1}A\big(\Pi(x)^{-1}v\big).$$

The map $\Psi(A)$ is a K-map, since $\Pi(x)^{-1}\Pi(g) = \Pi(g^{-1}x)^{-1}$ gives

$$(\Psi(A)\Pi(g)v)(x) = (\Psi(A)v)(g^{-1}x) = (L_g\Psi(A)v)(x).$$

Finally, $\Phi(\Psi(A))v = A(v)$. In the other order,

$$(\Psi(\Phi(T))v)(x) = \big(T\Pi(x^{-1})v\big)(e) = (L_{x^{-1}}Tv)(e) = (Tv)(x). \qquad \blacksquare$$

For finite K, the equality of dimensions is part of Frobenius reciprocity in the Finite Groups course. That course built its bijection on maps out of Ind. Here we build it on maps into Ind, where evaluation at e gives the map directly.

Now pass to L^2. For $h \in H$ define the operator $\widetilde{R}_h = R_h \otimes \tau(h)$ on $L^2(K) \otimes W$. These operators are unitary and commute with $L_g \otimes 1$. For continuous F, $(\widetilde{R}_h F)(x) = \tau(h)F(xh)$. Set

$$L^2\text{-}\operatorname{Ind}_H^K W = \{F \in L^2(K) \otimes W : \widetilde{R}_h F = F \text{ for all } h \in H\}.$$

This closed K-invariant subspace is a unitary representation of K. For the trivial representation $W = \mathbb{C}$, this subspace is $L^2(K)^H$.

Theorem 12.10 (finite multiplicities). *For $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2\text{-}\operatorname{Ind}_H^K W\big) = \dim \operatorname{Hom}_H(V_\sigma|_H, W) \le d_\sigma \dim W.$$

Proof. Let $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$ be a K-map. By Lemma 12.6 on equivariant images, each $F = Tv$ is continuous. Then $\widetilde{R}_h F$ and F are continuous and equal in L^2. Since $C(K)\otimes W \to L^2(K)\otimes W$ is injective, $\tau(h)F(xh) = F(x)$ for all x. So $F \in \operatorname{Ind}_H^K W$. Conversely, continuous covariant functions satisfy $\widetilde{R}_h F = F$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2\text{-}\operatorname{Ind}_H^K W\right) = \operatorname{Hom}_K\left(V_\sigma, \operatorname{Ind}_H^K W\right).$$

Frobenius reciprocity (Theorem 12.9) gives a linear bijection from this space onto $\operatorname{Hom}_H(V_\sigma|_H, W)$, so the dimensions agree. The bound holds because $\operatorname{Hom}_H(V_\sigma|_H, W)$ is a subspace of $\operatorname{Hom}(V_\sigma, W)$, which has dimension $d_\sigma \dim W$. ∎

Remark 12.11 (duals and covariance)**.** For $W = \mathbb{C}$ the count is $\dim \operatorname{Hom}_H(V_\sigma, \mathbb{C}) = \dim(V_\sigma^*)^H$. Since $\langle u, \Pi_\sigma(h)v\rangle = \langle \Pi_\sigma(h)^{-1}u, v\rangle$, the functional $\langle u, \cdot\rangle$ is H-invariant exactly when $u \in V_\sigma^H$. So $u \mapsto \langle u, \cdot\rangle$ is an antilinear bijection $V_\sigma^H \to (V_\sigma^*)^H$. Hence the count agrees with the multiplicity theorem (Theorem 12.7).

The inverse in $\tau(h)^{-1}$ makes the covariance condition compatible with products in H:

$$F(xh_1h_2) = \tau(h_2)^{-1}F(xh_1) = \tau(h_2)^{-1}\tau(h_1)^{-1}F(x) = \tau(h_1h_2)^{-1}F(x).$$

Remark 12.12 (scope)**.** For compact K, we obtained a finite invariant integral. The irreducible representations were finite-dimensional, and Peter–Weyl applied. We proved a count of finite multiplicities, not a general adjunction for Hilbert representations. For noncompact G, the space G/H may carry no invariant measure. Lecture 17 treats that case.

12.4 The sphere $\mathrm{SO}(3)/\mathrm{SO}(2)$

Let $K = \mathrm{SO}(3)$, let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$, and let, as in Lecture 4,

$$R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad H = \{R_3(\theta) : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2).$$

The subgroup H is the stabiliser of e_3. Every unit vector is the third column of some rotation, so the orbit map $q(x) = xe_3$ sends K onto S^2. By Lecture 3, q induces a K-equivariant diffeomorphism $K/H \to S^2$. So $F \mapsto F \circ q$ identifies $C(S^2)$ with $C(K)^H$, and $L_g(F\circ q) = (F\circ g^{-1})\circ q$. We define $L^2(S^2)$ through this identification. The invariant integral on S^2 is the usual area integral divided by 4π, because both are rotation-invariant and normalised, and such an integral is unique.

Let E be a finite-dimensional representation of H. An integer k is a *weight* of E if some nonzero $v \in E$ satisfies $R_3(\theta)v = e^{ik\theta}v$ for all $\theta \in \mathbb{R}$. These vectors and 0 form the *weight-k space*. Suppose E has a basis (v_j) with $R_3(\theta)v_j = e^{ik_j\theta}v_j$ for all θ. Then the weight-k space is spanned by the v_j with $k_j = k$. Indeed, a weight-k vector $\sum_j c_j v_j$ satisfies

$$c_j\left(e^{ik_j\theta} - e^{ik\theta}\right) = 0 \qquad \text{for all } \theta,$$

and this forces $c_j = 0$ when $k_j \neq k$.

From Lecture 8, the irreducibles of SO(3) are V_ℓ, $\ell = 0, 1, 2, \ldots$, of dimension $2\ell + 1$ and spin ℓ. In this section an unqualified V_ℓ means the SO(3)-module. The SU(2)-modules of Lecture 8 are always named as such. Choose the covering $\phi = \mathrm{Ad}$ in the ordered basis (X_1, X_2, X_3) of Lecture 4. The explicit calculation there gives

$$\phi\big(\mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2})\big) = R_3(-\theta), \qquad \ker\phi = \{\pm I\}.$$

The pullback $\Pi_\ell \circ \phi$ is the SU(2)-module $V_{2\ell}$. The diagonal element in this display acts on a basis of $V_{2\ell}$ by the scalars $e^{i(\ell-j)\theta}$, $0 \le j \le 2\ell$. Replacing θ by $-\theta$ therefore gives

$$V_\ell \text{ has a basis } (v_k)_{-\ell\le k\le\ell} \text{ with } \Pi_\ell(R_3(\theta))v_k = e^{ik\theta}v_k \text{ for all } \theta \in \mathbb{R}. \tag{12.4}$$

Theorem 12.13. *Let $\mathcal{Y}_\ell \subseteq L^2(S^2)$ be the V_ℓ-isotypic subspace. Then*

$$L^2(S^2) = \widehat{\bigoplus_{\ell\ge 0}} \mathcal{Y}_\ell, \qquad \mathcal{Y}_\ell \cong V_\ell, \qquad \mathcal{Y}_\ell \subseteq C(S^2).$$

Each integral-spin module occurs exactly once.

Proof. The space V_ℓ^H is the weight-zero space. By the weight basis (12.4) it is the line $\mathbb{C}v_0$, so $\dim V_\ell^H = 1$. Apply the multiplicity theorem (Theorem 12.7). ∎

The same weight basis counts tangent fields. In the tangent model, τ has the weights 1 and -1. By Theorem 12.10, V_ℓ occurs twice for $\ell \ge 1$ and not at all for $\ell = 0$. So S^2 carries no nonzero tangent field fixed by all rotations. This statement concerns invariant fields only, and it is not the hairy-ball theorem about a single continuous field.

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$, with $(L_g f)(v) = f(g^{-1}v)$. Let $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ and $\mathcal{H}_\ell = \{f \in \mathcal{P}_\ell : \Delta f = 0\}$. The chain rule gives $\Delta(f \circ g) = (\Delta f) \circ g$ for orthogonal g, so $\mathcal{H}_\ell$ is K-invariant. Restriction to S^2 is K-equivariant. Restriction is injective on $\mathcal{P}_\ell$, since $f(rv) = r^\ell f(v)$. Elements of $\mathcal{H}_\ell|_{S^2}$ are called *spherical harmonics* of degree ℓ.

Put $z_\pm = x \pm iy$. Since $R_3(\theta)^{-1} = R_3(-\theta)$,

$$(L_{R_3(\theta)} z_+)(x, y, z) = (\cos\theta\, x + \sin\theta\, y) + i(-\sin\theta\, x + \cos\theta\, y) = e^{-i\theta} z_+.$$

Similarly $L_{R_3(\theta)} z_- = e^{i\theta} z_-$ and $L_{R_3(\theta)} z = z$. The monomials $z_+^a z_-^b z^c$ with $a + b + c = \ell$ form a basis of $\mathcal{P}_\ell$, and for all θ,

$$L_{R_3(\theta)}\, z_+^a z_-^b z^c = e^{i(b-a)\theta}\, z_+^a z_-^b z^c, \qquad |b - a| \le \ell. \tag{12.5}$$

So every weight of $\mathcal{P}_\ell$ satisfies $|k| \le \ell$.

Proposition 12.14. *The space $\mathcal{Y}_\ell$ is spanned by the left translates of $f_\ell = z_+^\ell|_{S^2}$. Hence every element of $\mathcal{Y}_\ell$ is a spherical harmonic of degree ℓ.*

Proof. The polynomial z_+^ℓ is harmonic:

$$\partial_x^2 z_+^\ell + \partial_y^2 z_+^\ell = \ell(\ell-1)z_+^{\ell-2} + i^2\ell(\ell-1)z_+^{\ell-2} = 0.$$

Let U be the span of all $L_g f_\ell$. Then $U \subseteq \mathcal{H}_\ell|_{S^2}$ is finite-dimensional and K-invariant. Decompose U into irreducibles. A copy of V_j in U lies in the V_j-isotypic subspace $\mathcal{Y}_j$. Since $\mathcal{Y}_j \cong V_j$, this copy equals $\mathcal{Y}_j$. Thus $U = \bigoplus_{j \in S} \mathcal{Y}_j$ for a finite set S. Restriction is injective and equivariant, so the weights of U are weights of $\mathcal{P}_\ell$. Since V_j has weight j, this gives $S \subseteq \{0, \dots, \ell\}$.

Write $f_\ell = \sum_{j \in S} y_j$ with $y_j \in \mathcal{Y}_j$. Each $L_{R_3(\theta)}$ preserves each $\mathcal{Y}_j$, and $L_{R_3(\theta)} f_\ell = e^{-i\ell\theta} f_\ell$. Since the sum is direct, comparing components gives $L_{R_3(\theta)} y_j = e^{-i\ell\theta} y_j$ for all θ. For $j < \ell$ the weight $-\ell$ does not occur in V_j, so $y_j = 0$. Hence $f_\ell \in \mathcal{Y}_\ell$ and $U \subseteq \mathcal{Y}_\ell$. Since $U \neq 0$ and $\mathcal{Y}_\ell$ is irreducible, $U = \mathcal{Y}_\ell$. ∎

Figure 12.3 shades the harmonics of the next example by their sign.

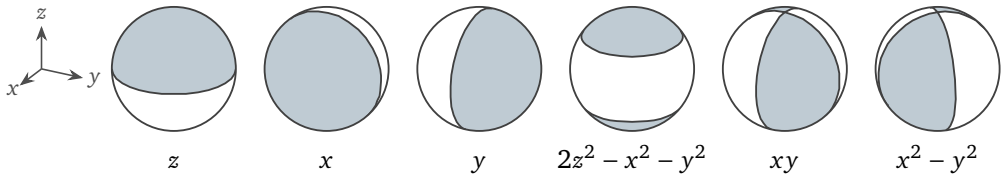


Figure 12.3: Real spherical harmonics of degrees one and two on the unit sphere, shaded where they are positive, with their zero sets drawn. The functions z and $2z^2 - x^2 - y^2$ depend on the latitude only, so they are fixed by H. The second of these vanishes on the latitudes $z = \pm 1/\sqrt{3}$.

Example 12.15 (degrees one and two)**.** We compute $\mathcal{Y}_1$ and $\mathcal{Y}_2$ and their H-fixed lines. Every linear polynomial is harmonic, so $\dim \mathcal{H}_1 = 3$. The map $\Delta\colon \mathcal{P}_2 \to \mathbb{C}$ is onto, since $\Delta(x^2) = 2$. So $\dim \mathcal{H}_2 = 6 - 1 = 5$.

By Proposition 12.14 on spherical harmonics, $\mathcal{Y}_\ell \subseteq \mathcal{H}_\ell|_{S^2}$. Restriction is injective, so $\mathcal{H}_\ell|_{S^2}$ has dimension $2\ell + 1$ for $\ell = 1, 2$. By the sphere theorem (Theorem 12.13), also $\dim \mathcal{Y}_\ell = 2\ell + 1$. So the inclusion is an equality for $\ell = 1, 2$:

$$\mathcal{Y}_1 = \operatorname{span}\{x, y, z\}|_{S^2}, \qquad \mathcal{Y}_2 = \operatorname{span}\{xy,\ yz,\ zx,\ x^2 - y^2,\ 2z^2 - x^2 - y^2\}|_{S^2}.$$

By the monomial weights (12.5), the weight-zero space of $\mathcal{P}_2$ is spanned by $z_+ z_- = x^2 + y^2$ and z^2. Since $\Delta\big(a(x^2 + y^2) + bz^2\big) = 4a + 2b$, its harmonic line is spanned by $2z^2 - x^2 - y^2$. This function spans $\mathcal{Y}_2^H$, the line predicted by $\dim V_2^H = 1$. Similarly z spans $\mathcal{Y}_1^H$. In atomic physics, $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are the angular parts of the p and d orbitals, and on S^2 the H-fixed function $2z^2 - x^2 - y^2 = 3z^2 - 1$ is a multiple of the Legendre polynomial $P_2(z)$.

Exercise 12.5 proves $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ for all ℓ. That exercise defines the Laplacian of the sphere and computes its eigenvalue on $\mathcal{H}_\ell$. Nothing in this lecture depends on it.

12.5 Self-checks

1. Let K be a finite group, regarded as a compact Lie group of dimension zero, and let $H \leq K$ be a subgroup. What does the multiplicity theorem (Theorem 12.7) say?

Answer. Here $L^2(K/H) = \mathbb{C}[K/H]$ is the permutation representation. An irreducible V occurs $\dim V^H$ times. This is finite-group Frobenius reciprocity for the trivial character of H.

2. Why not define $\Phi(T)v = (Tv)(e)$ directly for $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$?
   Answer. An L^2 class has no value at e. Lemma 12.6 on equivariant images first shows that T takes continuous values.
3. What are the multiplicities for $H = \{e\}$ and for $H = K$?
   Answer. For $H = \{e\}$, σ occurs d_σ times in $L^2(K)$. For $H = K$, a nontrivial irreducible has $V^K = 0$, so only the trivial representation occurs, once, as the constants.

Summary

The invariant integral on K/H is the Haar integral of pullbacks. The averaging operator A_H projects $L^2(K)$ onto $L^2(K)^H \cong L^2(K/H)$. Peter–Weyl then shows that each $\sigma \in \operatorname{Irr}(K)$ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times.

Let V be a finite-dimensional representation of K and W a finite-dimensional unitary representation of H. An element of $\operatorname{Ind}_H^K W$ is a function on K fixed on each coset by its value at one point, and on S^2 with $W = \mathbb{C}^2$ it is a tangent field with complex components, written in moving frames. Evaluation at e gives $\operatorname{Hom}_K(V, \operatorname{Ind}_H^K W) \cong \operatorname{Hom}_H(V|_H, W)$. Isotypic vectors in $L^2\text{-}\operatorname{Ind}_H^K W$ are continuous. So σ occurs in $L^2\text{-}\operatorname{Ind}_H^K W$ exactly $\dim \operatorname{Hom}_H(V_\sigma|_H, W)$ times.

On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ each $\mathrm{SO}(3)$-module V_ℓ occurs once in $L^2(S^2)$, realised by spherical harmonics of degree ℓ.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (1985), Section III.6: induction from closed subgroups and Frobenius reciprocity.
- Sepanski, *Compact Lie Groups* (2007), §2.3 and §§3.3–3.4: the assigned background route for this lecture and the sphere example.
- Knapp, *Lie Groups Beyond an Introduction*, digital second edition (2023), Section IX.2: induced representations of compact groups and Frobenius reciprocity.

Lecture 13

Spin groups and projective representations

Objectives. *A 2π rotation is the identity in* $\mathrm{SO}(3)$. *On an electron's state vector it acts by* -1. *We explain this sign using the double cover* $\mathrm{Spin}(n) \to \mathrm{SO}(n)$. *We then study actions on rays and their lifts to vectors.*

By the end of this lecture you will be able to:

- For $n \geq 2$, construct $\mathrm{Spin}(n)$ and prove that $\rho\colon\ \mathrm{Spin}(n) \to \mathrm{SO}(n)$ is surjective with kernel $\{\pm 1\}$.
- Identify $\mathrm{Spin}(3)$ with $\mathrm{SU}(2)$ and decide which V_m descend to $\mathrm{SO}(3)$.
- Define projective unitary representations, multipliers and central extensions, and lift finite-dimensional projective representations of $\mathrm{SO}(3)$ to $\mathrm{SU}(2)$.
- Use $\mathbb{R}^2$ to explain why a simply connected group can have a nontrivial multiplier.

13.1 Clifford algebras and spin groups

A rotation is a product of an even number of reflections. A reflection is determined by a unit vector u, up to the sign of u. We want an algebra in which conjugation by u gives the reflection in $u^{\perp}$ up to sign. For a product of an even number of unit vectors the signs cancel. The Clifford algebra has this property. For $n = 2$, Exercise 13.1 shows the result on a circle.

Fix $n \geq 2$ and an orthonormal basis $e_1, \dots, e_n$ of Euclidean $\mathbb{R}^n$. Recall from Lie Algebras that the *Clifford algebra* $Cl(n)$ is the real unital associative algebra generated by $\mathbb{R}^n$ subject to $v^2 = -|v|^2$. Replacing v by $u + v$ gives

$$uv + vu = -2\langle u, v\rangle \qquad (u, v \in \mathbb{R}^n). \tag{13.1}$$

This is the prerequisite convention $uv + vu = 2q(u, v)$ with the negative definite symmetric bilinear form $q(u, v) = -\langle u, v\rangle$.

The products $e_I = e_{i_1} \cdots e_{i_k}$ with $i_1 < \cdots < i_k$ form a basis of $Cl(n)$. The *even part* $Cl^0(n)$ is the span of the e_I with $|I|$ even. The *transpose* is the linear map $x \mapsto x^t$ with $(v_1 \cdots v_k)^t = v_k \cdots v_1$. The complexification $Cl(n)_{\mathbb{C}}$ has an irreducible *spinor module* S of dimension $2^{\lfloor n/2 \rfloor}$.

The units $Cl(n)^{\times}$ form an open set and a Lie group. A unit vector u has $u^2 = -1$ and $u^{-1} = -u$. Let $r_u(v) = v - 2\langle u, v\rangle u$ be reflection in $u^{\perp}$.

Definition 13.1. $\mathrm{Spin}(n)$ is the set of all products $u_1u_2\cdots u_{2k}$ with $k \geq 0$ and unit vectors $u_i \in \mathbb{R}^n$.

Lemma 13.2. *(a)* $\mathrm{Spin}(n)$ *is a subgroup of* $Cl^0(n)^\times$, *and* $x^{-1} = x^t$ *for* $x \in \mathrm{Spin}(n)$.
(b) For a unit vector u *and* $v \in \mathbb{R}^n$ *we have* $uvu^{-1} = -r_u(v)$.

Proof. (a) Even products of vectors lie in $Cl^0(n)$, and the set is closed under products. Let $x = u_1\cdots u_{2k}$, so $x^t = u_{2k}\cdots u_1$. In

$$x^t x = u_{2k}\cdots u_2\,(u_1u_1)\,u_2\cdots u_{2k}$$

the middle factor is the scalar -1. Next the two factors u_2 become adjacent and give $u_2u_2 = -1$, and so on. Hence $x^tx = (-1)^{2k} = 1$, and in the same way $xx^t = 1$. Since $x^t \in \mathrm{Spin}(n)$, the set is a subgroup.

(b) By the anticommutation relation (13.1), $uv = -vu - 2\langle u, v\rangle$. Using also $u^{-1} = -u$ and $u^2 = -1$, we get

$$uvu^{-1} = -uvu = (vu + 2\langle u, v\rangle)u = -v + 2\langle u, v\rangle u = -r_u(v). \qquad \blacksquare$$

13.2 The double covering of the rotation group

For $x \in \mathrm{Spin}(n)$ and $v \in \mathbb{R}^n$ put $\rho(x)v = xvx^{-1}$, an element of $Cl(n)$. Write $x = u_1\cdots u_{2k}$ with unit vectors $u_i \in \mathbb{R}^n$. Apply the conjugation formula of Lemma 13.2(b) once for each of the $2k$ factors of x. The $2k$ signs cancel, and we get

$$\rho(x)v = xvx^{-1} = r_{u_1}\circ\cdots\circ r_{u_{2k}}(v) \qquad (v \in \mathbb{R}^n). \tag{13.2}$$

So $\rho(x)$ maps $\mathbb{R}^n$ to itself, and it is a product of an even number of reflections. Hence $\rho(x) \in \mathrm{SO}(n)$, and ρ is a map from $\mathrm{Spin}(n)$ to $\mathrm{SO}(n)$.

The formula (13.2) explains the double cover. A reflection r_u does not change if we replace u by $-u$, but the element u of $Cl(n)$ does change. So x and $-x$ give the same rotation.

Two mirrors show this in a plane. Let e_a and e_b be orthonormal, let $u = \cos t\, e_a + \sin t\, e_b$, and put $x = -ue_a = u(-e_a)$. By (13.2), $\rho(x) = r_u \circ r_{e_a}$, so we reflect first in $e_a^\perp$ and then in $u^\perp$. These two mirrors meet at angle t. Example 13.5 below shows that the composition is the rotation through $2t$ from e_a towards e_b. Figure 13.1 draws the two reflections. In its second row, t runs from 0 to π. The mirror of u returns to its start, the rotation runs once around, and the product x runs from 1 to -1. At $t = \pi$ we have $u = -e_a$, with the same mirror and the same rotation, but with product -1.

Lemma 13.3 (Cartan–Dieudonné, weak form)**.** *Every* $g \in \mathrm{O}(n)$ *is a product of at most* n *reflections. If* $\det g = 1$, *the number of reflections is even.*

Proof. Induct on n. For $n = 1$ we have $\mathrm{O}(1) = \{1, r_{e_1}\}$. Let $n \geq 2$. If $ge_1 = e_1$, then g preserves $e_1^\perp \cong \mathbb{R}^{n-1}$. By induction $g|_{e_1^\perp}$ is a product of at most $n-1$ reflections of $e_1^\perp$, and each extends

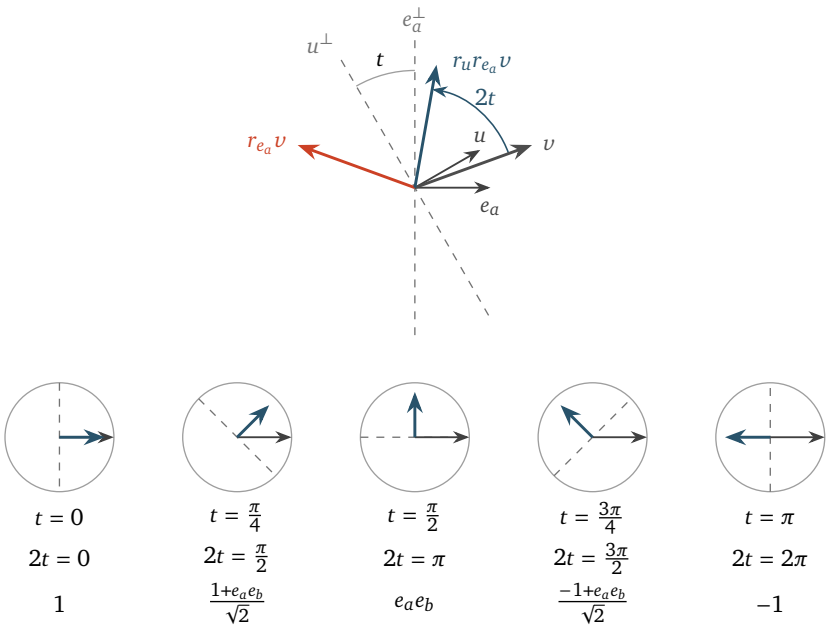


Figure 13.1: Two mirrors in the plane of e_a and e_b. The first panel has $t = \pi/6$: a vector v at angle $\pi/9$ goes to $r_{e_a} v$, and then to $r_u r_{e_a} v$, which is v turned through $2t$. The second row shows e_a, the vector u in colour, and the dashed mirror $u^\perp$, for five values of t. Beneath each are the rotation angle $2t$ and the product $x = -ue_a$.

to a reflection of $\mathbb{R}^n$ fixing e_1. If $a = ge_1 \neq e_1$, put $u = (a - e_1)/|a - e_1|$. Since $|a| = 1$, we have $\langle a - e_1, a\rangle = \frac{1}{2}|a - e_1|^2$. Therefore

$$r_u(a) = a - \frac{2\langle a - e_1, a\rangle}{|a - e_1|^2}(a - e_1) = e_1.$$

So $r_u g$ fixes e_1, and $g = r_u(r_u g)$ uses at most n reflections. Each reflection has determinant -1, which gives the parity. ∎

Theorem 13.4. *Let $n \geq 2$.*

(a) $\rho\colon \mathrm{Spin}(n) \to \mathrm{SO}(n)$ *is a continuous surjective homomorphism.*

(b) $\ker \rho = \{\pm 1\}$.

(c) $\mathrm{Spin}(n)$ *is a compact connected Lie group, and ρ is a two-sheeted covering homomorphism.*

(d) For $n \geq 3$, using the cited fact that $\pi_1(\mathrm{SO}(n))$ has order two, the group $\mathrm{Spin}(n)$ is simply connected. So $\mathrm{Spin}(n)$ is the universal covering group of $\mathrm{SO}(n)$.

Proof. (a) Clearly $\rho(xy) = \rho(x)\rho(y)$. Since $x^{-1} = x^t$, the entries of $\rho(x)$ are polynomials in the coordinates of x. Given $g \in \mathrm{SO}(n)$, write $g = r_{w_1} \cdots r_{w_{2k}}$ with $2k \leq n$ by the Cartan–Dieudonné lemma, Lemma 13.3. By the reflection formula (13.2), $y = w_1 \cdots w_{2k}$ satisfies $\rho(y) = g$.

(b) Note $-1 = e_1 e_1 \in \mathrm{Spin}(n)$, and ± 1 act trivially. Conversely, let $\rho(x) = \mathrm{id}$, so x commutes with every e_i. Fix i. Sorting the basis elements of x by whether they contain e_i, and moving e_i to the front, gives

$$x = a + e_i b,$$

where a, b are combinations of e_J with $i \notin J$, a even and b odd. By the anticommutation relation (13.1), e_i commutes with a and anticommutes with b. Hence

$$xe_i = e_i a - e_i e_i b, \qquad e_i x = e_i a + e_i e_i b.$$

Equality forces $2e_i^2 b = -2b = 0$. So no basis element containing e_i occurs in x. As i was arbitrary, $x = \lambda 1$ with $\lambda \in \mathbb{R}$. Then $1 = x^t x = \lambda^2$, so $\lambda = \pm 1$.

(c) Let $N = 2\lfloor n/2 \rfloor \geq 2$ and $x \in \mathrm{Spin}(n)$. The surjectivity proof in part (a) gives $y = w_1 \cdots w_{2k}$ with $2k \leq N$ and $\rho(y) = \rho(x)$. Append pairs $e_1(-e_1) = 1$ until there are N factors. By the kernel statement (b), $x = \pm y$, so we may replace w_1 by $-w_1$ to get $x = y$. Hence $\mathrm{Spin}(n)$ is the image of the compact connected space $(S^{n-1})^N$ under the continuous map $(w_1, \ldots, w_N) \mapsto w_1 \cdots w_N$. So $\mathrm{Spin}(n)$ is compact and connected.

Being compact, the group $\mathrm{Spin}(n)$ is closed in $Cl(n)^\times$, hence a Lie group by the closed subgroup theorem (Lecture 3).

The smooth transitive action of $\mathrm{Spin}(n)$ on $\mathrm{SO}(n)$ through ρ has stabiliser $\{\pm 1\}$. The homogeneous-space theorem (Lecture 3) identifies $\mathrm{Spin}(n)/\{\pm 1\}$ with $\mathrm{SO}(n)$. The quotient by the discrete subgroup $\{\pm 1\}$ is a local diffeomorphism, so $d\rho$ is an isomorphism. The covering criterion of Lecture 4 applies and gives two sheets.

(d) We cite that $\pi_1(\mathrm{SO}(n))$ has order two for $n \geq 3$ (Hall, Chapter 13). So the universal covering $q \colon \tilde{G} \to \mathrm{SO}(n)$ has kernel of order two. Integrate $(d\rho)^{-1} dq$ by the integration theorem of Lecture 4 to a homomorphism $\phi \colon \tilde{G} \to \mathrm{Spin}(n)$. The homomorphisms $\rho \circ \phi$ and q have the same differential on connected $\tilde{G}$, so they agree.

Since dq and $d\rho$ are isomorphisms, so is $d\phi$. Thus $\phi(\tilde{G})$ is an open, hence closed, subgroup, and ϕ is onto by connectedness. Now $\ker q = \phi^{-1}(\{\pm 1\})$ has order $2|\ker \phi|$, so $\ker \phi$ is trivial. A bijective homomorphism of Lie groups is an isomorphism (Lecture 3). ∎

For $n = 2$, Exercise 13.1 shows that $\mathrm{Spin}(2)$ is a circle and ρ is $z \mapsto z^2$. So $\mathrm{Spin}(2)$ is not simply connected.

Example 13.5 (Rotations in a coordinate plane)**.** Fix $a \neq b$ and put

$$x_{ab}(t) = \cos t + \sin t\, e_a e_b = -(\cos t\, e_a + \sin t\, e_b)\, e_a \in \mathrm{Spin}(n).$$

Since $(e_a e_b)^2 = -1$, we have $x_{ab}(t) = \exp(t\, e_a e_b)$, so $e_a e_b \in \mathrm{Lie}(\mathrm{Spin}(n))$. The vectors e_a, e_b anticommute with $e_a e_b$, so $e_a x_{ab}(t)^{-1} = x_{ab}(t) e_a$, and similarly for e_b. Every e_c with $c \neq a, b$ commutes with $e_a e_b$. Using $e_a e_b e_a = e_b$ and $e_a e_b e_b = -e_a$ we get

$$\rho(x_{ab}(t))e_a = x_{ab}(2t)e_a = \cos 2t\, e_a + \sin 2t\, e_b, \qquad \rho(x_{ab}(t))e_b = -\sin 2t\, e_a + \cos 2t\, e_b,$$

and $\rho(x_{ab}(t))e_c = e_c$. So $\rho(x_{ab}(t))$ is the rotation through $2t$ in the (e_a, e_b)-plane. For $t \in [0, \pi]$ the rotations run once around a closed loop. The lift runs from 1 to $x_{ab}(\pi) = -1$, so it is not closed.

Since $\mathrm{Spin}(n) \subset Cl(n)_{\mathbb{C}}$, the spinor module gives a representation Π_S of $\mathrm{Spin}(n)$. The element $-1 \in \mathrm{Spin}(n)$ is the scalar -1 of $Cl(n)$. Hence

$$\Pi_S(-1) = -\mathrm{id}_S, \qquad \rho(-1) = \mathrm{id}\,.$$

So Π_S is not of the form $\Pi' \circ \rho$. This gives the opening sign change.

13.3 Spin(3), SU(2) and descent

In $Cl^0(3)$ put $\mathbf{i} = e_2e_3$, $\mathbf{j} = e_3e_1$, $\mathbf{k} = e_1e_2$. For instance $\mathbf{ij} = e_2e_3e_3e_1 = -e_2e_1 = \mathbf{k}$. In the same way

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \qquad \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \quad \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \quad \mathbf{ki} = -\mathbf{ik} = \mathbf{j}.$$

So $Cl^0(3) = \operatorname{span}\{1, \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is the quaternion algebra $\mathbb{H}$. Since $\mathbf{i}^t = e_3e_2 = -\mathbf{i}$, the transpose is quaternion conjugation $q \mapsto \bar{q}$. For $q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}$ we have $q\bar{q} = |q|^2 = a^2 + b^2 + c^2 + d^2$. Let $\mathrm{Sp}(1) = \{q \in \mathbb{H} : |q| = 1\}$ be the group of unit quaternions.

Define a real linear map $\Phi\colon \mathbb{H} \to M_2(\mathbb{C})$ by

$$\Phi(1) = I, \quad \Phi(\mathbf{i}) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \Phi(\mathbf{j}) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \Phi(\mathbf{k}) = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}.$$

Direct multiplication shows that these matrices satisfy the quaternion relations. For example, $\Phi(\mathbf{i})\Phi(\mathbf{j}) = \Phi(\mathbf{k})$. Hence Φ is an algebra homomorphism, and

$$\Phi(a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}) = \begin{pmatrix} a + bi & -c - di \\ c - di & a - bi \end{pmatrix}, \qquad \det \Phi(q) = |q|^2. \tag{13.3}$$

Proposition 13.6. *(a)* $\mathrm{Spin}(3) = \mathrm{Sp}(1)$.
(b) Φ *restricts to an isomorphism of Lie groups* $\mathrm{Sp}(1) \to \mathrm{SU}(2)$.

Proof. (a) If $q \in \mathrm{Spin}(3)$, then $q\bar{q} = qq^t = 1$ by the inverse formula of Lemma 13.2(a), so $|q| = 1$. Conversely, let $|q| = 1$ and write $q = a + p$ with $p \in \operatorname{span}\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$. If $p = 0$, then $q = \pm 1 \in \mathrm{Spin}(3)$. Otherwise put $w = p/|p|$. Cross terms in w^2 cancel by anticommutation, so $w^2 = -1$. Choose θ with $\cos\theta = a$ and $\sin\theta = |p|$. Then

$$q = \cos\theta + \sin\theta\, w = \exp(\theta w).$$

By the plane-rotation example, Example 13.5, $\mathbf{i}, \mathbf{j}, \mathbf{k} \in \mathrm{Lie}(\mathrm{Spin}(3))$. The Lie algebra of a closed subgroup is a linear subspace, and exp maps it into the subgroup (Lecture 3). So $\theta w \in \mathrm{Lie}(\mathrm{Spin}(3))$ and $q \in \mathrm{Spin}(3)$.

(b) By the matrix formula (13.3), $\Phi(\bar{q}) = \Phi(q)^*$. So $\Phi(q)\Phi(q)^* = \Phi(q\bar{q}) = I$ for $|q| = 1$, and $\det\Phi(q) = 1$. Conversely, let $g \in \mathrm{SU}(2)$ have first column $(\alpha, \beta)^T$. The second column is a unit vector orthogonal to the first, and $\det g = 1$. This forces it to be $(-\bar{\beta}, \bar{\alpha})^T$. So $g = \Phi(q)$ with $\alpha = a + bi$ and $\beta = c - di$. Thus Φ is a bijective continuous homomorphism. Its inverse is linear, hence continuous. ∎

From now on we identify Spin(3) with SU(2) through Φ. Then $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ has kernel $\{\pm I\}$.

This ρ is the cover Ad of Lecture 4, up to a rotation of coordinates. Indeed, Exercise 13.2(a) shows that ρ is conjugation in the basis $(\mathbf{i}, \mathbf{j}, \mathbf{k})$, and in the basis (X_1, X_2, X_3) of Lecture 4 we have $\Phi(\mathbf{i}) = -2X_3$, $\Phi(\mathbf{j}) = 2X_2$ and $\Phi(\mathbf{k}) = 2X_1$. So $\rho = O\ \mathrm{Ad}\ O^{-1}$, where O is the rotation with $Oe_1 = e_3$, $Oe_2 = e_2$ and $Oe_3 = -e_1$. Hence the diagonal matrices rotate about e_1 here, and about the third axis in Lecture 4 and Lecture 12. A spinor module of $Cl(3)_{\mathbb{C}}$ is $\mathbb{C}^2$ with

e_j acting by $i\sigma_j$, for the Pauli matrices σ_j of Lecture 4. It sends $\mathbf{i}, \mathbf{j}, \mathbf{k}$ to $-i\sigma_1, -i\sigma_2, -i\sigma_3$, and with this isomorphism in place of Φ, the map ρ is exactly Ad.

For $m \geq 0$ let V_m be the space of homogeneous polynomials of degree m in $z = (z_1, z_2)$, with

$$(\Pi_m(g)f)(z) = f(g^{-1}z) \qquad (g \in \mathrm{SU}(2)).$$

The space V_m has dimension $m+1$ and spin $m/2$. Every finite-dimensional irreducible representation of SU(2) is isomorphic to exactly one V_m (Lecture 8). Fix an SU(2)-invariant inner product on each V_m (Lecture 5).

The spinor module of $Cl(3)_{\mathbb{C}}$ has dimension two. If its restriction to SU(2) were reducible, complete reducibility would make it $V_0 \oplus V_0$, since V_0 is the only one-dimensional irreducible. That restriction would be trivial, contrary to $\Pi_S(-1) = -\mathrm{id}_S$. Thus the restriction is isomorphic to V_1.

By the Descent Theorem of Lecture 4, a representation Π of SU(2) has the form $\Pi' \circ \rho$ exactly when $\Pi(\pm I) = \mathrm{id}$. The hypotheses hold by Theorem 13.4.

Proposition 13.7. *Π_m factors through $\rho\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ if and only if m is even.*

Proof. We have $\ker\rho = \{\pm I\}$ and $(\Pi_m(-I)f)(z) = f(-z) = (-1)^m f(z)$. Apply the Descent Theorem of Lecture 4. ∎

So the irreducible representations of SO(3) are the descents of the V_{2l}, of dimension $2l+1$. The spinor V_1 is not among them.

Example 13.8 (A full turn on V_m)**.** Put

$$u(\theta) = \Phi(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\mathbf{i}) = \mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2}).$$

Since $\mathbf{i} = e_2e_3$, the plane-rotation example, Example 13.5, shows that $\rho(u(\theta))$ is the rotation $R(\theta)$ through θ in the (e_2, e_3)-plane. On the monomial $f_k = z_1^{m-k}z_2^k$ we get

$$\Pi_m(u(\theta))f_k = (e^{-i\theta/2}z_1)^{m-k}(e^{i\theta/2}z_2)^k = e^{i(2k-m)\theta/2}f_k.$$

At $\theta = 2\pi$ every eigenvalue is $e^{i(2k-m)\pi} = (-1)^m$. Hence

$$R(2\pi) = \mathrm{id}, \qquad \Pi_m(u(2\pi)) = (-1)^m\,\mathrm{id}\,.$$

For odd m every vector changes sign. Each ray $\mathbb{C}f$ is unchanged, and so is every transition probability $|\langle f, h\rangle|^2/(\|f\|^2\|h\|^2)$. In physics, the sign is visible only relative to another part of a state, for example in the interference of a beam, one part of which has been turned through 2π.

Remark 13.9. Connectedness matters when a representation is recovered from its differential. The trivial character and det of O(2) both have differential 0 on $\mathfrak{so}(2)$, but they differ on a reflection. So for a disconnected group, $d\Pi$ need not determine Π (Exercise 13.4).

13.4 Projective unitary representations

In quantum mechanics a pure state is a ray $\mathbb{C}f$ in a Hilbert space, with $f \neq 0$. Transition probabilities depend only on rays. So a symmetry group acts on rays. Lifting this action to vectors may require choices of phase.

Let $\mathcal{H}$ be a complex Hilbert space of finite dimension $d \geq 1$, and let $\mathbb{T} = \{\lambda \in \mathbb{C} : |\lambda| = 1\}$.

Definition 13.10. The *projective unitary group* is $\mathrm{PU}(\mathcal{H}) = \mathrm{U}(\mathcal{H})/\mathbb{T}$ id with the quotient topology. We write $[U]$ for the class of U.

It is a Lie group (Lecture 3). It acts on rays by $[U] \cdot \mathbb{C}f = \mathbb{C}Uf$.

Definition 13.11. A *projective unitary representation* of a Lie group G on $\mathcal{H}$ is a continuous homomorphism $\bar{\pi} \colon G \to \mathrm{PU}(\mathcal{H})$.

We call $\bar{\pi}$ *irreducible* if 0 and $\mathcal{H}$ are the only subspaces invariant under every U with $[U] \in \bar{\pi}(G)$. A *lift* is any function $U \colon G \to \mathrm{U}(\mathcal{H})$ with $[U(g)] = \bar{\pi}(g)$. No continuity is assumed. Since $[U(g)U(h)] = [U(gh)]$, there is a function $c \colon G \times G \to \mathbb{T}$ with

$$U(g)U(h) = c(g,h)\,U(gh). \tag{13.4}$$

Definition 13.12. A *multiplier* of G is a function $c \colon G \times G \to \mathbb{T}$ such that

$$c(g,h)\,c(gh,k) = c(h,k)\,c(g,hk) \qquad (g,h,k \in G).$$

Multipliers c, c' are *equivalent* if $c'(g,h) = c(g,h)\,b(g)b(h)/b(gh)$ for some function $b \colon G \to \mathbb{T}$. A multiplier is *trivial* if it is equivalent to 1.

Lemma 13.13. *The function c in (13.4) is a multiplier. Its equivalence class does not depend on the lift. It is trivial if and only if $\bar{\pi}$ has a lift that is a homomorphism.*

Proof. Expand $U(g)U(h)U(k)$ in two ways:

$$(U(g)U(h))U(k) = c(g,h)c(gh,k)U(ghk), \qquad U(g)(U(h)U(k)) = c(h,k)c(g,hk)U(ghk).$$

This gives the identity. Any other lift is bU with $b \colon G \to \mathbb{T}$, and its multiplier is $c(g,h)\,b(g)b(h)/b(gh)$. So the class is well defined. The lift bU is a homomorphism exactly when this multiplier equals 1. ∎

Example 13.14 (Spin one half)**.** Let $\bar{\pi}_1$ be the action of $\mathrm{SO}(3)$ on rays of $\mathbb{C}^2$ of Corollary 13.18 below. For each rotation g choose one of its two preimages $s(g) \in \mathrm{SU}(2)$, and put $U(g) = \Pi_1(s(g))$. Since $s(g)s(h) = \pm s(gh)$ and $\Pi_1(-I) = -\mathrm{id}$, the multiplier c takes the values ± 1.

No choice of s is continuous. A continuous choice along the loop $t \mapsto \rho(x_{ab}(t))$, $t \in [0, \pi]$, would be a closed lift of this loop. Lifts are unique given their start (Lecture 4), and Example 13.5 shows that the lift starting at 1 runs to -1. Multiplying by -1 shows that the lift starting at -1 runs to 1. So no lift of the loop is closed.

Definition 13.15. A *central extension* of G by $\mathbb{T}$ is a Lie group E with a closed central subgroup $Z \cong \mathbb{T}$ and an isomorphism of Lie groups $E/Z \cong G$.

Proposition 13.16. *Let $\bar\pi$ be a projective unitary representation of G on $\mathcal{H}$. Then*

$$E_{\bar\pi} = \{(g, U) \in G \times \mathrm{U}(\mathcal{H}) : [U] = \bar\pi(g)\}$$

is a central extension of G by $Z = \{(1, \lambda\,\mathrm{id}) : \lambda \in \mathbb{T}\}$. The map $(g, U) \mapsto U$ is a unitary representation of $E_{\bar\pi}$ in which $(1, \lambda\,\mathrm{id})$ acts by λ.

Proof. The set $E_{\bar\pi}$ is a subgroup of $G \times \mathrm{U}(\mathcal{H})$. It is the preimage of the diagonal under the continuous map $(g, U) \mapsto (\bar\pi(g), [U])$ into the Hausdorff space $\mathrm{PU}(\mathcal{H})^2$. So it is closed, hence a Lie group (Lecture 3). Clearly Z is closed and central. The projection to G is surjective with kernel Z. The projection defines a smooth transitive action of $E_{\bar\pi}$ on G, with stabiliser Z. The homogeneous-space theorem of Lecture 3 therefore identifies $E_{\bar\pi}/Z$ smoothly with G. ∎

Conversely, a unitary representation of a central extension E in which Z acts by $\lambda \mapsto \lambda\,\mathrm{id}$ defines a projective representation of E/Z. In this sense, projective representations of G are ordinary representations of central extensions of G.

For $\bar\pi_1$ of Example 13.14, the map $(g, U) \mapsto U$ is an isomorphism $E_{\bar\pi_1} \to \mathrm{U}(2)$, with $Z = \mathbb{T}\,\mathrm{id}$, because $\bar\pi_1$ is a bijection $\mathrm{SO}(3) \to \mathrm{PU}(\mathbb{C}^2)$. So $\mathrm{U}(2)$ is a central extension of $\mathrm{SO}(3)$ by $\mathbb{T}$, and spin one half is an ordinary representation of it.

13.5 Lifting projective representations

Theorem 13.17. *Let G be a connected Lie group with universal covering homomorphism $q\colon \tilde G \to G$ (Lecture 4). Let $\bar\pi$ be a projective unitary representation of G on a finite-dimensional $\mathcal{H}$. Then there is a unique continuous homomorphism $U\colon \tilde G \to \mathrm{SU}(\mathcal{H})$ with $[U(x)] = \bar\pi(q(x))$.*

Proof. Let $d = \dim \mathcal{H}$. Every $V \in \mathrm{U}(\mathcal{H})$ equals λV_0 with $\lambda^d = \det V$ and $V_0 \in \mathrm{SU}(\mathcal{H})$. So the restriction $p\colon \mathrm{SU}(\mathcal{H}) \to \mathrm{PU}(\mathcal{H})$ of the quotient map is surjective. Its kernel $\{\lambda\,\mathrm{id} : \lambda^d = 1\}$ is finite, and $\mathrm{SU}(\mathcal{H})$ is connected.

The differential of p has kernel $\mathfrak{su}(d) \cap i\mathbb{R}I = 0$. Both $\mathfrak{su}(d)$ and the tangent algebra of $\mathrm{U}(d)/\mathbb{T}I$ have dimension $d^2 - 1$, so dp is an isomorphism. By the covering criterion of Lecture 4, p is a covering homomorphism. For $d = 1$ both groups are trivial.

Since $\tilde G$ is simply connected, the based lifting theorem of Lecture 4 gives a unique continuous lift U with $U(e) = I$ and $p \circ U = \bar\pi \circ q$. It remains to check that U is multiplicative. The two maps $(x, y) \mapsto U(xy)$ and $(x, y) \mapsto U(x)U(y)$ lift the same map on the connected space $\tilde G \times \tilde G$ and agree at (e, e). Uniqueness of covering lifts makes them equal. Thus U is a homomorphism. Any homomorphic lift sends e to I, so uniqueness of the based lift also gives the stated uniqueness. ∎

Corollary 13.18. *Let $\mathcal{H}$ be finite-dimensional.*

(a) For each projective unitary representation $\bar{\pi}$ of $\mathrm{SO}(3)$ on $\mathcal{H}$, there is a unitary representation U of $\mathrm{SU}(2)$ with $\bar{\pi}(\rho(x)) = [U(x)]$. Moreover $U(-I) = \pm\,\mathrm{id}$.

(b) If $\bar{\pi}$ is irreducible, then U is unitarily equivalent to Π_m with $m = \dim \mathcal{H} - 1$.

(c) For each $m \geq 0$, the rule $\bar{\pi}_m(\rho(x)) = [\Pi_m(x)]$ defines a projective unitary representation of $\mathrm{SO}(3)$. Its multiplier is trivial if and only if m is even.

Proof. (a) The group $\mathrm{SU}(2) \cong S^3$ is simply connected (Lecture 4). Apply the lifting theorem, Theorem 13.17. Since $[U(-I)] = \bar{\pi}(\mathrm{id})$, $U(-I)$ is a scalar. Its square is $U(I) = \mathrm{id}$, so the scalar is ± 1.

(b) Scalars preserve every subspace, so U is irreducible. By the classification, $U \cong \Pi_m$ with $m + 1 = \dim \mathcal{H}$. By Schur's lemma the isomorphism becomes unitary after rescaling (Lecture 5).

(c) If $\rho(x) = \rho(x')$, then $x' = \pm x$ and $\Pi_m(x') = \pm \Pi_m(x)$. So $\bar{\pi}_m$ is well defined. It is continuous by the Descent Theorem of Lecture 4, with target $\mathrm{PU}(V_m)$. For even m, the descent criterion Proposition 13.7 gives a lift that is a homomorphism.

Let m be odd. Put $A = \rho(\Phi(\mathbf{i}))$ and $B = \rho(\Phi(\mathbf{j}))$. Since $\mathbf{i} = x_{23}(\pi/2)$ and $\mathbf{j} = x_{31}(\pi/2)$, the plane-rotation example, Example 13.5, gives

$$A = \mathrm{diag}(1, -1, -1), \qquad B = \mathrm{diag}(-1, 1, -1), \qquad AB = BA.$$

Put $P = \Pi_m(\Phi(\mathbf{i}))$ and $Q = \Pi_m(\Phi(\mathbf{j}))$. Since $\mathbf{ij} = -\mathbf{ji}$ and $\Pi_m(-I) = -\,\mathrm{id}$,

$$PQ = \Pi_m(-I)\, QP = -QP.$$

Suppose $W\colon\ \mathrm{SO}(3) \to \mathrm{U}(V_m)$ is a homomorphism lifting $\bar{\pi}_m$, with no continuity assumed. Then $W(A) = \alpha P$ and $W(B) = \beta Q$ with $\alpha, \beta \in \mathbb{T}$. So $W(A)W(B) = -W(B)W(A)$. But $AB = BA$ gives $W(A)W(B) = W(B)W(A)$. Hence $2W(A)W(B) = 0$, which is impossible for invertible operators. By the triviality criterion of Lemma 13.13, the multiplier is nontrivial. ∎

So every finite-dimensional irreducible projective representation of $\mathrm{SO}(3)$ is equivalent to some $\bar{\pi}_m$. Here equivalence means that a unitary $T : V_m \to \mathcal{H}$ satisfies

$$\bar{\pi}(\rho(x)) = [T \Pi_m(x) T^{-1}] \qquad (x \in \mathrm{SU}(2)).$$

All spins $m/2$ occur, but only integer spins come from genuine representations. This is the mathematical form of the half-integer spin of the electron.

For separable Hilbert spaces we cite the following result.

Theorem 13.19 (Bargmann; cited)**.** *Let G be a connected Lie group with semisimple Lie algebra $\mathfrak{g}$ and universal covering $q\colon \tilde{G} \to G$. Let $\mathcal{H}$ be a separable Hilbert space, and let $\bar{\pi}$ be a homomorphism from G to $\mathrm{U}(\mathcal{H})/\mathbb{T}\,\mathrm{id}$. Assume that $g \mapsto |\langle f, U(g)h\rangle|$ is continuous for all $f, h \in \mathcal{H}$ and one (hence every) lift U. Then there is a strongly continuous unitary representation $\tilde{U}$ of $\tilde{G}$ with $[\tilde{U}(x)] = \bar{\pi}(q(x))$ for $x \in \tilde{G}$.*

Example 13.20 ($\mathbb{R}^2$ and the Heisenberg extension)**.** On $G = \mathbb{R}^2$ put

$$\omega(x, y) = x_1 y_2 - x_2 y_1, \qquad c(x, y) = e^{i\omega(x,y)/2}.$$

The exponent is bilinear, so c is a multiplier. The factor $b(x)b(y)/b(x+y)$ is symmetric in x, y. So the ratio

$$c(x, y)/c(y, x) = e^{i\omega(x,y)}$$

depends only on the class of c. The ratio is $e^i \neq 1$ for $x = (1, 0)$ and $y = (0, 1)$, so c is nontrivial. Yet $\mathbb{R}^2$ is its own universal cover.

On $E = \mathbb{R}^2 \times \mathbb{T}$ put $(x, \lambda)(y, \mu) = (x+y, \lambda\mu\, c(x, y))$. The multiplier identity gives associativity, and Exercise 13.5 shows that E is a group. Its multiplication is smooth. Projection onto $\mathbb{R}^2$ identifies the quotient $E/(\{0\} \times \mathbb{T})$ with $\mathbb{R}^2$, so E is a central extension. The commutator of (x, λ) and (y, μ) in E is $(0, e^{i\omega(x,y)})$.

This does not contradict the lifting theorem, Theorem 13.17. A lift on $\mathbb{C}^d$ with multiplier equivalent to c would satisfy $U(x)U(y) = e^{i\omega(x,y)}U(y)U(x)$. Taking determinants gives $e^{id\,\omega(x,y)} = 1$ for all x, y, which is false. So c occurs only on infinite-dimensional spaces (Lecture 18). In physics, this is the group form of the fact that $\operatorname{tr}[x, p] = 0$ rules out $[x, p] = i\hbar$ on a finite-dimensional space. Bargmann's theorem does not apply to $\mathbb{R}^2$, because its Lie algebra is abelian rather than semisimple.

Self-check questions

1. In Spin(4), what is $\rho(e_1e_2e_3e_4)$?
   Answer. The element $e_1e_2e_3e_4$ equals $x_{12}(\pi/2)\, x_{34}(\pi/2)$. By the plane-rotation example, Example 13.5, its image under ρ rotates through π in both coordinate planes, so $\rho(e_1e_2e_3e_4) = -\mathrm{id}$. The other lift of $-\mathrm{id}$ is $-e_1e_2e_3e_4$.
2. For odd m, does $\Pi_m(u(2\pi))$ change any transition probability?
   Answer. No. It equals $-\mathrm{id}$. Hence it fixes every ray, and $|\langle -f, -h\rangle| = |\langle f, h\rangle|$.
3. Is there a projective unitary representation of $\mathbb{R}^2$ on $\mathbb{C}^3$ with multiplier $e^{i\omega/2}$?
   Answer. No. The determinant argument of the Heisenberg example, Example 13.20, with $d = 3$ gives $e^{3i\omega(x,y)} = 1$ for all x, y, which is false.

Summary

The group $\mathrm{Spin}(n)$ consists of even products of unit vectors in $Cl(n)$. For $n \geq 2$, conjugation maps it onto $\mathrm{SO}(n)$ with kernel $\{\pm 1\}$. For $n \geq 3$, the cited value of $\pi_1(\mathrm{SO}(n))$ shows that it is the universal cover. Spinor modules send -1 to $-\mathrm{id}$ and do not factor through ρ.

For $n = 3$, $\mathrm{Spin}(3) = \mathrm{Sp}(1) \cong \mathrm{SU}(2)$, and V_m descends to $\mathrm{SO}(3)$ exactly for even m. The lift $u(2\pi)$ of a full turn acts on V_m by $(-1)^m$, which fixes every ray.

In finite dimensions, projective unitary representations of connected Lie groups lift to unitary representations of the universal cover. On separable Hilbert spaces, Bargmann's cited theorem gives strongly continuous lifts for connected groups with semisimple Lie algebra, under its continuity hypothesis. The multiplier on the simply connected group $\mathbb{R}^2$ shows that semisimplicity cannot be dropped.

Reading

- BD, Section I.6: Clifford algebras and Spin(n). Note the sign convention for v^2.
- Sepanski, Section 1.3: the Clifford-algebra construction of the double cover of SO(n).
- Kirillov, Elements, §14: projective representations and their multipliers.
- Hall, Chapter 13: fundamental groups of compact Lie groups, including SO(n).
- V. Bargmann, On unitary ray representations of continuous groups, Ann. of Math. 59 (1954).

Lecture 14

Representative functions and tensor reconstruction

Objectives. *This lecture asks what information about a compact Lie group K is contained in all its representations. We collect all matrix coefficients into one algebra $\mathscr{R}(K)$. The group law becomes three maps on this algebra. We then state how K is recovered, and what is recovered when the unitary structure is forgotten.*

By the end of this lecture you will be able to:

- Prove that $\mathscr{R}(K)$ is a $*$-algebra whose products come from tensor products and whose conjugation comes from conjugate representations.
- Define Δ, ε, S and prove the Hopf algebra identities, also in matrix form.
- State Tannaka–Krein reconstruction with the unitary fibre functor, and contrast it with the complex reconstruction of $K_{\mathbb{C}}$.
- Compute these structures for $\mathrm{U}(1)$ and $\mathrm{SU}(2)$, and explain why $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category when K is infinite.

A space can be replaced by the algebra of functions on it. A group has more: its product appears as a map on these functions. This description still makes sense for algebras that are not commutative, and Lecture 28 returns to them. Nothing here depends on that.

Throughout, K is a compact Lie group. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ has as objects the finite-dimensional continuous complex representations (Π_V, V). Its morphisms are the K-equivariant linear maps.

14.1 Matrix coefficients

Definition 14.1. Let V be an object of $\mathrm{Rep}_{\mathrm{fd}}(K)$, $\lambda \in V^*$ and $v \in V$. The *matrix coefficient* $c^V_{\lambda,v} \colon K \to \mathbb{C}$ is

$$c^V_{\lambda,v}(g) = \lambda\big(\Pi_V(g)v\big).$$

The space $\mathscr{R}(K)$ of *representative functions* is the linear span of all matrix coefficients of all objects V.

This is the coefficient space of Lecture 10, since every finite-dimensional continuous representation is unitarisable (Lecture 5).

Fix a basis (e_j) of V with dual basis (e^i). The matrix of $\Pi_V(g)$ in this basis is $D^V(g)$, with entries $D^V_{ij}(g) = e^i(\Pi_V(g)e_j)$. Expanding $\lambda = \sum_i \lambda(e_i)e^i$ and $v = \sum_j e^j(v)e_j$ gives

$$c^V_{\lambda,v} = \sum_{i,j} \lambda(e_i)\, e^j(v)\, D^V_{ij}.$$

So every coefficient of V is a linear combination of the D^V_{ij}. For a K-invariant inner product on V (Lecture 5) we may take $\lambda = \langle u, \cdot\rangle$, which gives $c^V_{\lambda,v}(g) = \langle u, \Pi_V(g)v\rangle$.

The *conjugate representation* $\overline{V}$ is the additive group V with scalar multiplication $t \cdot v = \bar{t}v$ and the same operators $\Pi_V(g)$. These operators are linear on $\overline{V}$:

$$\Pi_V(g)(t \cdot v) = \Pi_V(g)(\bar{t}v) = \bar{t}\,\Pi_V(g)v = t \cdot \Pi_V(g)v.$$

For $\lambda \in V^*$ put $\bar{\lambda}(v) = \overline{\lambda(v)}$. The functional $\bar{\lambda}$ is linear on $\overline{V}$:

$$\bar{\lambda}(t \cdot v) = \overline{\bar{t}\,\lambda(v)} = t\,\bar{\lambda}(v).$$

A basis (e_j) of V is also a basis of $\overline{V}$, with dual basis $(\overline{e^i})$. Hence

$$D^{\overline{V}}_{ij}(g) = \overline{e^i}\big(\Pi_V(g)e_j\big) = \overline{D^V_{ij}(g)}, \qquad \text{so} \qquad D^{\overline{V}}(g) = \overline{D^V(g)}.$$

Proposition 14.2. *Let $\lambda \in V^*$, $v \in V$, $\mu \in W^*$, $w \in W$ and $h \in K$. Then:*

(a) $c^V_{\lambda,v} + c^W_{\mu,w} = c^{V\oplus W}_{(\lambda,\mu),(v,w)}$;

(b) $c^V_{\lambda,v}\, c^W_{\mu,w} = c^{V\otimes W}_{\lambda\otimes\mu,\, v\otimes w}$;

(c) $\overline{c^V_{\lambda,v}} = c^{\overline{V}}_{\bar{\lambda},v}$;

(d) $g \mapsto c^V_{\lambda,v}(gh)$ *equals* $c^V_{\lambda,\Pi_V(h)v}$, *and* $g \mapsto c^V_{\lambda,v}(hg)$ *equals* $c^V_{\lambda\circ\Pi_V(h),v}$.

Hence $\mathscr{R}(K)$ is the set of matrix coefficients. It is a subalgebra of $C(K)$ containing the constants, closed under conjugation and translations.

Proof. Matrix coefficients are continuous because $g \mapsto \Pi_V(g)$ is continuous. The sum rule (a) holds because $\Pi_{V\oplus W}(g)$ acts componentwise. For the product rule (b), the tensor product representation gives

$$(\lambda \otimes \mu)\big(\Pi_V(g)v \otimes \Pi_W(g)w\big) = \lambda(\Pi_V(g)v)\,\mu(\Pi_W(g)w).$$

The conjugation rule (c) is the definition of $\bar{\lambda}$. The translation rule (d) uses $\Pi_V(gh) = \Pi_V(g)\Pi_V(h)$ and $\Pi_V(hg) = \Pi_V(h)\Pi_V(g)$.

Scalar multiples are coefficients, since $t\,c^V_{\lambda,v} = c^V_{t\lambda,v}$. By the sum rule (a), sums of coefficients are coefficients. So the span consists of coefficients. By the product, conjugation and translation rules (b)–(d), this set is closed under products, conjugation and translations. The constant 1 is $c^{\mathbb{C}}_{1,1}$ for the trivial representation. ∎

We use the Peter–Weyl theorem for compact Lie groups from Lecture 10. Choose one representative V for each class in $\mathrm{Irr}(K)$ and a basis of each. Then

$$\mathscr{R}(K) = \bigoplus_{[V]\in\mathrm{Irr}(K)} \operatorname{span}\{D^V_{ij}\},$$

an algebraic direct sum whose summand for V has dimension $(\dim V)^2$. Moreover $\mathscr{R}(K)$ is dense in $C(K)$ for the supremum norm and separates points.

14.2 The group law as structure maps

The group law is a map $K \times K \to K$. Pulling functions back along the group law produces functions of two variables. We first show that a tensor of functions is determined by the function of two variables that it defines.

Lemma 14.3. *Let A and B be linear spaces of functions on sets X and Y. The linear map from $A \otimes B$ to functions on $X \times Y$ with $f \otimes h \mapsto \big((x,y) \mapsto f(x)h(y)\big)$ is injective.*

Proof. Every tensor can be written $F = \sum_i f_i \otimes h_i$ with $h_1, \dots, h_n$ linearly independent. Suppose $\sum_i f_i(x)h_i(y) = 0$ for all x, y. Fix x. Then $\sum_i f_i(x)h_i$ is the zero function on Y. The numbers $f_i(x)$ are the coefficients of a vanishing combination of independent functions, so $f_i(x) = 0$ for all i. As x was arbitrary, each $f_i = 0$ and $F = 0$. ∎

Applying the lemma twice handles three factors. So we identify $\mathscr{R}(K)^{\otimes 2}$ and $\mathscr{R}(K)^{\otimes 3}$ with spaces of functions on K^2 and K^3. The identification is an algebra homomorphism for $(f \otimes h)(f' \otimes h') = ff' \otimes hh'$. Indeed both sides give the function $(x,y) \mapsto f(x)f'(x)h(y)h'(y)$.

Definition 14.4. A *Hopf algebra* is a unital algebra A with unital algebra homomorphisms $\Delta\colon A \to A \otimes A$, $\varepsilon\colon A \to \mathbb{C}$ and a linear map $S\colon A \to A$ such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta, \qquad (\varepsilon \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon)\Delta,$$
$$m(S \otimes \mathrm{id})\Delta = \eta\varepsilon = m(\mathrm{id} \otimes S)\Delta,$$

where m is multiplication and $\eta(t) = t1$. A *Hopf $*$-algebra* is a Hopf algebra with an antilinear map $*$ such that $f^{**} = f$, $(fh)^* = h^* f^*$, and Δ, ε preserve $*$. Here $(f \otimes h)^* = f^* \otimes h^*$ and $t^* = \bar{t}$ on $\mathbb{C}$.

For $f \in \mathscr{R}(K)$ and $g, h \in K$ define

$$\Delta f(g,h) = f(gh), \qquad \varepsilon f = f(e), \qquad Sf(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

For the coordinate function z on $\mathrm{U}(1)$ we have $\Delta z^n = z^n \otimes z^n$, since $(gh)^n = g^n h^n$. For a matrix entry we will find $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$, which is matrix multiplication. In general, a priori Δf is only a function on $K \times K$. The first task is to show that it is a tensor in $\mathscr{R}(K)^{\otimes 2}$.

Theorem 14.5. *These maps make $\mathscr{R}(K)$ a commutative Hopf $*$-algebra, and $S(f^*) = (Sf)^*$. For every V and basis (e_j),*

$$\Delta D^V_{ij} = \sum_k D^V_{ik} \otimes D^V_{kj}, \qquad \varepsilon D^V_{ij} = \delta_{ij}, \qquad SD^V_{ij}(g) = \big(D^V(g)^{-1}\big)_{ij}. \tag{14.1}$$

For an orthonormal basis of a K-invariant inner product, $SD^V_{ij} = (D^V_{ji})^*$.

Proof. Codomains. Insert the expansion $v = \sum_k e^k(v)e_k$ for the vector $v = \Pi_V(h)e_j$. This gives

$$D^V_{ij}(gh) = e^i\big(\Pi_V(g)\Pi_V(h)e_j\big) = \sum_k e^i(\Pi_V(g)e_k)\, e^k(\Pi_V(h)e_j).$$

So ΔD^V_{ij} is the function of the tensor $\sum_k D^V_{ik} \otimes D^V_{kj}$. This tensor is unique by the tensor injectivity lemma, Lemma 14.3. The same insertion for a general coefficient gives

$$\Delta c^V_{\lambda,v} = \sum_k c^V_{\lambda,e_k} \otimes c^V_{e^k,v}. \tag{14.2}$$

Hence Δ maps $\mathscr{R}(K)$ into $\mathscr{R}(K)^{\otimes 2}$. Let $\Pi_{V^*}(g)\lambda = \lambda \circ \Pi_V(g)^{-1}$ be the dual representation. Let $\hat{e}_j \in V^{**}$ be evaluation at e_j. Then

$$SD^V_{ij}(g) = e^i\big(\Pi_V(g)^{-1}e_j\big) = \hat{e}_j\big(\Pi_{V^*}(g)e^i\big) = c^{V^*}_{\hat{e}_j,e^i}(g).$$

So S preserves $\mathscr{R}(K)$. Also $\varepsilon D^V_{ij} = e^i(e_j) = \delta_{ij}$, and $f^* \in \mathscr{R}(K)$ by the conjugation rule in Proposition 14.2.

Algebra maps. As functions, $\Delta(ff')(g,h) = f(gh)f'(gh) = (\Delta f\,\Delta f')(g,h)$. The identification is an injective algebra homomorphism, so $\Delta(ff') = \Delta f\,\Delta f'$. Also $\Delta 1 = 1 \otimes 1$. Pointwise products show that ε and S are unital algebra homomorphisms.

Identities. Write $\Delta f = \sum_i f_i \otimes f'_i$, so $\sum_i f_i(g)f'_i(h) = f(gh)$ for $g,h \in K$. As functions of $(g,h,k') \in K^3$,

$$(\Delta \otimes \mathrm{id})\Delta f\,(g,h,k') = \textstyle\sum_i f_i(gh)f'_i(k') = f\big((gh)k'\big),$$
$$(\mathrm{id} \otimes \Delta)\Delta f\,(g,h,k') = \textstyle\sum_i f_i(g)f'_i(hk') = f\big(g(hk')\big).$$

These agree by associativity, so the tensors agree by the tensor injectivity lemma, Lemma 14.3. For the counit, $(\varepsilon \otimes \mathrm{id})\Delta f = \sum_i f_i(e)f'_i$ is the function $h \mapsto f(eh) = f(h)$. For the antipode,

$$m(S \otimes \mathrm{id})\Delta f\,(g) = \sum_i f_i(g^{-1})f'_i(g) = f(g^{-1}g) = f(e).$$

The mirror identities use $he = h$ and $gg^{-1} = e$. Each group axiom is used once.

$*$-structure. Conjugation is antilinear and involutive, and $\mathscr{R}(K)$ is commutative. The function $\Delta(f^*)(g,h) = \overline{f(gh)}$ is that of $(\Delta f)^*$. Also $\varepsilon(f^*) = \overline{f(e)}$ and $S(f^*)(g) = \overline{f(g^{-1})}$. Finally, if $D^V(g)$ is unitary then $(D^V(g)^{-1})_{ij} = \overline{D^V_{ji}(g)}$. ∎

On matrix coefficients the axioms become matrix identities. We write $D = D^V$ and evaluate each side at group elements:

$$\begin{aligned}
\text{coassociativity:} &\quad \textstyle\sum_{k,l} D_{ik}(g)D_{kl}(h)D_{lj}(k') = \big(D(g)D(h)D(k')\big)_{ij},\\
\text{counit:} &\quad \textstyle\sum_k \delta_{ik}D_{kj}(g) = \big(D(e)D(g)\big)_{ij} = D_{ij}(g),\\
\text{antipode:} &\quad \textstyle\sum_k (SD_{ik})(g)\,D_{kj}(g) = \big(D(g)^{-1}D(g)\big)_{ij} = \delta_{ij}.
\end{aligned}$$

So the three identities encode $D(ghk') = D(g)D(h)D(k')$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. In an orthonormal basis the antipode identity reads

$$\sum_k (D_{ki})^* D_{kj} = \delta_{ij}. \tag{14.3}$$

In other words, the columns of $D(g)$ are orthonormal. Hence S and $*$ together encode the unitarity of $D(g)$. Exercise 14.7 checks these identities on the spin 1 representation of SU(2).

14.3 Recovering K

Definition 14.6. A *character* of $\mathscr{R}(K)$ is a unital algebra homomorphism $\psi\colon \mathscr{R}(K) \to \mathbb{C}$. It is a *$*$-character* if $\psi(f^*) = \overline{\psi(f)}$ for all f.

This algebraic use of the word differs from the trace character of a representation and from the characters $T \to \mathrm{U}(1)$ of a torus T (Lecture 7).

For $g \in K$ the evaluation $\mathrm{ev}_g(f) = f(g)$ is a $*$-character.

Proposition 14.7. *The characters form a group $\mathcal{X}(K)$ with*

$$\psi_1\psi_2 = (\psi_1 \otimes \psi_2) \circ \Delta, \qquad 1 = \varepsilon, \qquad \psi^{-1} = \psi \circ S.$$

The $*$-characters form a subgroup $\mathcal{X}_*(K)$, and $g \mapsto \mathrm{ev}_g$ is an injective homomorphism $K \to \mathcal{X}_*(K)$.

Proof. Since Δ, $\psi_1 \otimes \psi_2$ and S are algebra homomorphisms, $\psi_1\psi_2$ and $\psi \circ S$ are characters. Applying $\psi_1 \otimes \psi_2 \otimes \psi_3$ to coassociativity gives $(\psi_1\psi_2)\psi_3 = \psi_1(\psi_2\psi_3)$. The counit identity gives $\varepsilon\psi = \psi\varepsilon = \psi$. Multiplicativity means $\psi \circ m = m_{\mathbb{C}} \circ (\psi \otimes \psi)$, so

$$(\psi \circ S)\psi = \psi \circ m \circ (S \otimes \mathrm{id}) \circ \Delta = \psi \circ \eta \circ \varepsilon = \varepsilon.$$

The mirror identity gives $\psi(\psi \circ S) = \varepsilon$. The relations $\Delta(f^*) = (\Delta f)^*$ and $S(f^*) = (Sf)^*$ show that products and inverses of $*$-characters are $*$-characters. Finally $(\mathrm{ev}_g \otimes \mathrm{ev}_h)\Delta f = f(gh)$. Injectivity holds because $\mathscr{R}(K)$ separates points. ∎

In other words, Δ, ε and S give the multiplication, the identity and the inverses in $\mathcal{X}(K)$. To recover K we must know that every $*$-character is an evaluation. We now describe the same question in categorical terms.

An element $g \in K$ gives one matrix $\Pi_V(g)$ on each representation V. These matrices are compatible with every intertwiner and with tensor products, and they are unitary for invariant inner products. We now ask whether every family of unitary matrices with these properties comes from an element of K.

Definition 14.8. The *fibre functor* $\omega\colon \mathrm{Rep}_{\mathrm{fd}}(K) \to \mathrm{Vect}_{\mathbb{C}}$ sends each object to its underlying space and each morphism to itself. A *tensor automorphism* of ω is a family $\theta = (\theta_V)$ with $\theta_V \in \mathrm{GL}(V)$ for each object V, such that

$$\theta_W T = T\theta_V \quad (T\colon V \to W \text{ a morphism}), \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

A tensor automorphism θ is *unitary* if θ_V is unitary for every K-invariant inner product on every V.

On $\mathrm{SU}(2)$, the decomposition $V_1 \otimes V_1 = V_2 \oplus V_0$ expresses θ_{V_2} through θ_{V_1}. So θ_{V_1} determines the family, as Example 14.12 shows.

Tensor automorphisms form a group $\mathrm{Aut}_{\otimes}(\omega)$ under $(\theta\theta')_V = \theta_V\theta'_V$, with identity (id_V) and inverse (θ_V^{-1}). Indeed, products of natural families are natural, and naturality $\theta_W T = T\theta_V$ gives $T\theta_V^{-1} = \theta_W^{-1}T$. Products and inverses also preserve the tensor and unit conditions.

Products and inverses of unitary operators are unitary, so the unitary tensor automorphisms form a subgroup $\mathrm{Aut}^{u}_{\otimes}(\omega)$.

The data of the unitary fibre functor are the following.

(i) The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ with its tensor product and unit object $\mathbb{C}$.
(ii) The functor ω, which remembers the underlying vector spaces and linear maps.
(iii) On each $\omega(V)$, the set of K-invariant inner products. On the algebra side these inner products correspond to the $*$-structure.

Each $g \in K$ gives the family with $\theta_V = \Pi_V(g)$. This family is natural because morphisms are equivariant. It is monoidal by the definition of $V \otimes W$. It is unitary because the inner products are K-invariant.

Tensor automorphisms and characters are related by the map $\theta \mapsto \psi_\theta$ given by

$$\psi_\theta\big(c^{V}_{\lambda,v}\big) = \lambda(\theta_V v). \tag{14.4}$$

Exercises 14.4 and 14.5 show that $\theta \mapsto \psi_\theta$ is a group isomorphism $\mathrm{Aut}_{\otimes}(\omega) \to \mathcal{X}(K)$. They also show that θ is unitary exactly when ψ_θ is a $*$-character. In any basis of V, the matrix of θ_V is $\big(\psi_\theta(D^{V}_{ij})\big)_{ij}$.

So a $*$-character ψ gives one operator θ_V on each V, determined by $\psi = \psi_\theta$. The operator θ_V is unitary for every K-invariant inner product on V. Hence its matrix in an orthonormal basis is unitary, while its matrix in a general basis need not be. The hard step of reconstruction is to find a single $g \in K$ whose operators $\Pi_V(g)$ equal the operators θ_V for all V at once. The $\mathrm{SU}(2)$ model, Example 14.12, finds g directly. In general we cite this step.

Theorem 14.9 (Tannaka–Krein duality; cited from BD III.7). *Let K be a compact Lie group. Every $*$-character of $\mathcal{R}(K)$ is ev_g for a unique $g \in K$. Equivalently, $g \mapsto (\Pi_V(g))_V$ is a group isomorphism $K \to \mathrm{Aut}^{u}_{\otimes}(\omega)$.*

The cases to keep in mind for the next theorem are $\mathrm{U}(1) \subset \mathbb{C}^{\times}$ and $\mathrm{SU}(2) \subset \mathrm{SL}(2,\mathbb{C})$, computed in the next section.

Theorem 14.10 (Complexification; cited from BD III.8). *Let K be a compact Lie group and put $K_{\mathbb{C}} = \mathcal{X}(K)$, with the topology of pointwise convergence on $\mathcal{R}(K)$. Then $K_{\mathbb{C}}$ carries a complex manifold structure that makes it a complex Lie group with Lie algebra $\mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \otimes_{\mathbb{R}} \mathbb{C}$. The map $g \mapsto \mathrm{ev}_g$ identifies K with a maximal compact subgroup of $K_{\mathbb{C}}$. Every object of $\mathrm{Rep}_{\mathrm{fd}}(K)$ extends uniquely to a holomorphic representation of $K_{\mathbb{C}}$. In algebraic language, $K_{\mathbb{C}}$ is a complex algebraic group with reductive identity component and coordinate ring $\mathcal{R}(K)$. For connected K, it is a connected complex reductive group.*

A map between complex manifolds is *holomorphic* if it is complex differentiable in charts. A *maximal compact subgroup* is a compact subgroup contained in no larger compact subgroup.

Thus plain complex tensor automorphisms recover $\mathrm{Aut}_{\otimes}(\omega) \cong K_{\mathbb{C}}$, not K. The unitary condition, equivalently the $*$-structure, cuts $K_{\mathbb{C}}$ down to K.

14.4 Two models

Example 14.11 (U(1)). The irreducible representations of U(1) are the lines $\mathbb{C}_n$, $n \in \mathbb{Z}$, on which z acts by z^n (characters of tori, Lecture 7). The functions z^n are orthonormal for Haar measure (Lecture 5), hence independent. So

$$\mathscr{R}(\mathrm{U}(1)) = \mathbb{C}[z, z^{-1}], \qquad \Delta z = z \otimes z, \quad \varepsilon z = 1, \quad Sz = z^{-1}, \quad z^* = z^{-1}.$$

A character ψ is determined by $w = \psi(z)$. Since $\psi(z)\psi(z^{-1}) = 1$, we have $w \neq 0$. Every $w \in \mathbb{C}^\times$ occurs, via $z^n \mapsto w^n$. The $*$-condition reads $\bar{w} = \psi(z^*) = w^{-1}$, that is $|w| = 1$. Thus $\mathcal{X} = \mathbb{C}^\times$ and $\mathcal{X}_* = \mathrm{U}(1)$.

The categorical computation agrees. Let $\theta \in \mathrm{Aut}_\otimes(\omega)$, and let w_n be the scalar $\theta_{\mathbb{C}_n}$. Every V is the sum of its weight spaces

$$V(n) = \{v \in V : \Pi_V(z)v = z^n v \text{ for all } z \in \mathrm{U}(1)\}.$$

For $v \in V(n)$, the map $T\colon \mathbb{C}_n \to V$, $1 \mapsto v$, is a morphism, so

$$\theta_V v = \theta_V T 1 = T\theta_{\mathbb{C}_n} 1 = w_n v.$$

The isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \to \mathbb{C}_{m+n}$, $1 \otimes 1 \mapsto 1$, gives $w_{m+n} = w_m w_n$, and $w_0 = 1$. Hence θ_V acts by w^n on $V(n)$, with $w = w_1 \in \mathbb{C}^\times$. Weight spaces are orthogonal for every invariant inner product, so θ is unitary exactly when $|w| = 1$. Conversely, each $w \in \mathbb{C}^\times$ occurs, through the character $z^n \mapsto w^n$ and Exercise 14.5. So $\mathrm{Aut}_\otimes(\omega) \cong \mathbb{C}^\times = \mathrm{U}(1)_\mathbb{C}$ and $\mathrm{Aut}^u_\otimes(\omega) \cong \mathrm{U}(1)$.

Example 14.12 (SU(2)). Define functions a, b, c, d on SU(2) by

$$g = \begin{pmatrix} a(g) & b(g) \\ c(g) & d(g) \end{pmatrix} = \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix}, \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Here a is the $(1,1)$ entry, as before, but b is now the $(1,2)$ entry. Earlier lectures used b for the $(2,1)$ entry. These functions are the coefficients D_{ij} of $\mathbb{C}^2$ in the standard basis, and

$$ad - bc = 1, \qquad a^* = d, \qquad b^* = -c.$$

Since $\det g = 1$, the inverse matrix is

$$g^{-1} = \begin{pmatrix} d(g) & -b(g) \\ -c(g) & a(g) \end{pmatrix}.$$

So the matrix formulas (14.1) give

$$\begin{aligned} \Delta a &= a \otimes a + b \otimes c, & \Delta b &= a \otimes b + b \otimes d, & Sa &= d, & Sb &= -b, \\ \Delta c &= c \otimes a + d \otimes c, & \Delta d &= c \otimes b + d \otimes d, & Sc &= -c, & Sd &= a. \end{aligned}$$

Also ε sends a, d to 1 and b, c to 0. One checks $SD_{ij} = D^*_{ji}$; for example $Sb = -b = c^*$.

The functions a, b, c, d generate $\mathscr{R}(\mathrm{SU}(2))$ as an algebra. Indeed every irreducible representation is $V_m \cong \mathrm{Sym}^m(\mathbb{C}^2)$ (Lecture 8). In it g sends $e_1^p e_2^{m-p}$ to $(ae_1 + ce_2)^p (be_1 + de_2)^{m-p}$. So each $D_{ij}^{V_m}$ is a polynomial of degree m in a, b, c, d. By the Peter–Weyl theorem these coefficients span $\mathscr{R}(\mathrm{SU}(2))$, which proves the claim.

Let ψ be a $*$-character. Collect its values on the generators in the matrix

$$A = \begin{pmatrix} \psi(a) & \psi(b) \\ \psi(c) & \psi(d) \end{pmatrix}.$$

The relations give

$$\psi(d) = \overline{\psi(a)}, \qquad \psi(c) = -\overline{\psi(b)}, \qquad |\psi(a)|^2 + |\psi(b)|^2 = \psi(ad - bc) = 1.$$

So $A \in \mathrm{SU}(2)$. The algebra maps ψ and ev_A agree on generators, so $\psi = \mathrm{ev}_A$. This verifies Tannaka–Krein duality, Theorem 14.9, for $\mathrm{SU}(2)$ directly.

Without the $*$-condition we only learn $\det A = 1$. Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ gives a character $\psi_A \colon p(a,b,c,d) \mapsto p(A)$ (Exercise 14.3). So $\mathcal{X}(\mathrm{SU}(2)) = \mathrm{SL}(2,\mathbb{C}) = \mathrm{SU}(2)_{\mathbb{C}}$. For $A = \mathrm{diag}(2, \frac{1}{2})$ we get $\psi_A(a) = 2$ but $\psi_A(a^*) = \psi_A(d) = \frac{1}{2} \neq \bar{2}$. The corresponding tensor automorphism has $\theta_{\mathbb{C}^2} = A$, which is not unitary. In physics, this matrix $\mathrm{diag}(2, \frac{1}{2})$ is a boost in the group $\mathrm{SL}(2,\mathbb{C})$, which covers $\mathrm{SO}^+(1,3)$ (Lecture 24).

14.5 The representation category

The Category Theory course studied fusion categories, such as $\mathrm{SU}(2)_k$, with finitely many simple objects. The Clebsch–Gordan rule has the same form. We show that an infinite compact Lie group falls outside this class, and that finiteness is the only axiom that fails.

We use the notions of a rigid and of a symmetric monoidal category from the prerequisite course. Rigidity requires duals, with evaluation and coevaluation maps that satisfy the zigzag identities. A symmetry is a natural isomorphism $\tau_{V,W} \colon V \otimes W \to W \otimes V$ with $\tau_{W,V}\tau_{V,W} = \mathrm{id}$, compatible with associativity.

Definition 14.13. A *fusion category* is a $\mathbb{C}$-linear semisimple rigid monoidal category with finite-dimensional morphism spaces, simple unit object, and finitely many isomorphism classes of simple objects.

The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is semisimple (Lecture 5) with simple unit $\mathbb{C}$ and finite-dimensional morphism spaces. It is rigid, with the dual representation and

$$\mathrm{ev}_V(\lambda \otimes v) = \lambda(v), \qquad \mathrm{coev}_V(1) = \sum_i e_i \otimes e^i.$$

It is symmetric with $\tau(v \otimes w) = w \otimes v$. Exercise 14.6 checks these claims.

Proposition 14.14. *If K is infinite, then $\mathrm{Irr}(K)$ is infinite and $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category. In particular, both conclusions hold when K is connected and nontrivial.*

Proof. We first show that $\mathscr{R}(K)$ is infinite-dimensional. Let $x_1, \dots, x_n \in K$ be distinct. Since $\mathscr{R}(K)$ separates points, for $i \neq j$ we can choose $f_{ij} \in \mathscr{R}(K)$ with $f_{ij}(x_i) \neq f_{ij}(x_j)$. Then

$$h_i = \prod_{j \neq i} \frac{f_{ij} - f_{ij}(x_j)}{f_{ij}(x_i) - f_{ij}(x_j)} \in \mathscr{R}(K).$$

At x_i every factor equals 1. At x_k with $k \neq i$ the factor $j = k$ vanishes. So $h_i(x_k) = \delta_{ik}$. A relation $\sum_i t_i h_i = 0$ evaluated at x_k gives $t_k = 0$. So $h_1, \dots, h_n$ are independent and $\dim \mathscr{R}(K) \geq n$ for every n. If $\mathrm{Irr}(K)$ were finite, Peter–Weyl would give

$$\dim \mathscr{R}(K) = \sum_{[V] \in \mathrm{Irr}(K)} (\dim V)^2 < \infty.$$

So $\mathrm{Irr}(K)$ is infinite. The simple objects of $\mathrm{Rep}_{\mathrm{fd}}(K)$ are the irreducible representations, and a fusion category has only finitely many isomorphism classes of simple objects. So $\mathrm{Rep}_{\mathrm{fd}}(K)$ is not a fusion category.

Finally, a finite Hausdorff space is discrete. Hence a connected Hausdorff space with at least two points is infinite, and the first part applies to connected nontrivial K. ∎

For a finite group the representation category is a fusion category (Exercise 14.2).

The algebra $\mathscr{R}(K)$ is commutative. Its coproduct is cocommutative only for abelian K. Indeed cocommutativity means

$$f(gh) = f(hg) \quad \text{for all } f \in \mathscr{R}(K),\ g, h \in K.$$

Since $\mathscr{R}(K)$ separates points, this forces $gh = hg$ for all $g, h \in K$.

A later course on quantum groups deforms this classical structure into noncommutative Hopf $*$-algebras. Lecture 28 returns to this comparison. Nothing in this lecture depends on that course.

Self-checks

1. In $\mathscr{R}(\mathrm{U}(1))$ let $f(z) = \mathrm{Re}\, z$. Compute Δf, εf and Sf.
   Answer. $f = \frac{1}{2}(z + z^{-1})$, so $\Delta f = \frac{1}{2}(z \otimes z + z^{-1} \otimes z^{-1})$, $\varepsilon f = 1$ and $Sf = f$. Note $\Delta f \neq f \otimes f$.
2. On $\mathrm{SU}(2)$ write $g \mapsto |g_{11}|^2$ in terms of a, b, c, d. Evaluate ψ_A on it for $A = \mathrm{diag}(2, \frac{1}{2})$.
   Answer. $|\alpha|^2 = aa^* = ad$, and $\psi_A(ad) = 2 \cdot \frac{1}{2} = 1$, while $|A_{11}|^2 = 4$. A non-$*$ character does not evaluate absolute values.
3. For $\mathrm{U}(1)$ let $\theta \in \mathrm{Aut}_{\otimes}(\omega)$ with $\theta_{\mathbb{C}_1} = 3$. Find $\theta_{\mathbb{C}_{-2}}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}}$. Is θ unitary?
   Answer. $\theta_{\mathbb{C}_{-2}} = \frac{1}{9}$ and $\theta_{\mathbb{C}_1 \oplus \mathbb{C}_{-1}} = \mathrm{diag}(3, \frac{1}{3})$. It is not unitary, since $|3| \neq 1$.

Summary

Matrix coefficients of all finite-dimensional continuous representations of the compact Lie group K form the commutative $*$-algebra $\mathscr{R}(K)$. Tensor products give its products and conjugate representations give its involution. The maps $\Delta f(g, h) = f(gh)$, $\varepsilon f = f(e)$ and $Sf(g) = f(g^{-1})$ make it a Hopf $*$-algebra. On coefficients they are $D(gh) = D(g)D(h)$, $D(e) = 1$ and $D(g^{-1}) = D(g)^{-1}$. The $*$-characters, or unitary tensor automorphisms of ω, recover K. All characters, or all tensor automorphisms, give $K_{\mathbb{C}}$. The category $\mathrm{Rep}_{\mathrm{fd}}(K)$ is rigid and symmetric. When K is infinite, it has infinitely many simple objects and is not a fusion category.

Part I in review

Part I asked what a group determines that its Lie algebra does not. A finite-dimensional representation of a Lie algebra integrates to the simply connected group, and it descends to a quotient exactly when the central kernel acts trivially (Lecture 4). This step does not use compactness. For a compact connected group the test reads from the lattice of a maximal torus, and the irreducible representations are labelled by dominant characters (Lecture 7 and Lecture 8). So SU(2) has every V_m, and SO(3) has only those with m even. The lift of a full spatial turn acts on V_m by $(-1)^m$. This sign fixes every ray, so Lecture 13 passes to projective representations.

Compactness gave the rest of Part I. The Haar integral has finite total mass. Averaging gives invariant inner products, complete reducibility and Schur orthogonality (Lecture 5). Averaging a rank-one projection gives a nonzero compact operator, and the Peter–Weyl theorem follows (Lecture 10). So every irreducible unitary representation is finite-dimensional, and $L^2(K)$ is a Hilbert sum of blocks of matrix coefficients. Lecture 11 and Lecture 12 read this as Fourier analysis and as a count of multiplicities on K/H. This lecture recovers K from its coefficient algebra, by a cited theorem.

A noncompact group has no normalised finite Haar integral, so these averaging arguments are no longer available in general. Part II starts from that point.

Reading

BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

BD III.1 for representative functions and their algebra structure. BD III.7 for Tannaka–Krein duality. BD III.8 for the complexification of compact Lie groups. The tensor-category terms are from the prerequisite course.

Part II

Noncompact representations, geometry and physics

Lecture 15

Strong Continuity and Infinitesimal Generators

Objectives. *Part II studies unitary representations on infinite-dimensional Hilbert spaces. Their derivatives may exist only on a proper subspace. This lecture fixes the continuity hypothesis, defines the derivative on smooth vectors, and uses Stone's theorem to identify the skew-adjoint generator of each one-parameter subgroup, together with its domain. Translations on $L^2(\mathbb{R})$ are the model throughout.*

By the end of this lecture you will be able to:

- Prove strong continuity of a homomorphism into the unitary group from continuity at the identity on a dense set.
- Define smooth vectors and $d\Pi$, and prove that each $d\Pi(X)$ is skew-symmetric on smooth vectors.
- State Stone's theorem with its hypotheses, and identify the closure of $d\Pi(X)$ with the generator.
- Locate explicit vectors relative to the domains for translations: Schwartz functions, a non-Schwartz vector in the domain, and a vector outside it. Prove that translations have no nonzero finite-dimensional invariant subspace.

15.1 From matrices to Hilbert spaces

Let $\Pi\colon G \to \mathrm{GL}(V)$ be a continuous representation on a finite-dimensional complex space V. Continuous homomorphisms of Lie groups are smooth (Lecture 3). So every $X \in \mathfrak{g}$ gives an operator

$$d\Pi(X) = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX) \in \mathrm{End}(V), \qquad \Pi(\exp tX) = e^{t\,d\Pi(X)}.$$

If Π is unitary, then $d\Pi(X)^* = -d\Pi(X)$. Every vector can be differentiated.

For compact G, the Peter–Weyl theorem (Lecture 10) decomposes each unitary representation into finite-dimensional subrepresentations. That theory rests on averaging. The Haar integral of a compact group has finite total mass, so Lecture 5 can average an inner product, and Lecture 10 can average an operator. A noncompact Lie group also has an invariant integral, but its total mass is infinite (Lecture 16). So it has no normalised finite Haar integral, and these averaging arguments are no longer available in general.

The theory changes in the following ways. First, a finite-dimensional representation need not preserve any inner product. For example, for $m \geq 1$ and $t \neq 0$, the matrix $\operatorname{diag}(e^t, e^{-t}) \in \mathrm{SL}(2,\mathbb{R})$ acts on the module V_m with the eigenvalue e^{mt}, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves more: every finite-dimensional unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. So a nontrivial unitary representation of this group is infinite-dimensional.

This is not a property of every noncompact group. The group $\mathbb{R}$ has the one-dimensional unitary characters $x \mapsto e^{icx}$. It shows another change. Its translation representation on $L^2(\mathbb{R})$ is unitary, and it has no nonzero finite-dimensional invariant subspace (Proposition 15.4 below). So a unitary representation need not contain any finite-dimensional piece.

Finally, on an infinite-dimensional space the derivative of a representation may be defined only on a proper subspace. We begin with this point, on translations. Lecture 16 and Lecture 17 then build Haar measure and induction without compactness. We apply these tools to the Heisenberg group, to $\mathrm{SL}(2,\mathbb{R})$, and to the Lorentz and Poincaré groups. Lecture 26 and Lecture 27 return to compact groups, and Lecture 28 returns to Lecture 14.

The basic example is the translation representation on $L^2(\mathbb{R})$, so we first construct that space.

Let $C_c(\mathbb{R})$ be the space of continuous functions $\mathbb{R} \to \mathbb{C}$ with compact support, with inner product

$$\langle f, g \rangle = \int_{\mathbb{R}} \overline{f(x)}\, g(x)\, dx.$$

Its Hilbert completion is $L^2(\mathbb{R})$.

We want the indicator $1_{[0,1]}$ as a vector of $L^2(\mathbb{R})$, because it will be our example of a vector with no derivative. As in Lecture 10, a general element of a completion has no pointwise values. So we say which functions define vectors. A *jump* of a function $f\colon \mathbb{R} \to \mathbb{C}$ is a point where f is discontinuous but has both one-sided limits. We call f *piecewise continuous* if it is continuous except at finitely many jumps in each bounded interval. If moreover the improper integral $\int_{\mathbb{R}} |f|^2\, dx$ is finite, we call f *square integrable*. Such a function f defines a vector of $L^2(\mathbb{R})$, and for two such functions f and $\tilde{f}$,

$$\|f\|^2 = \int_{\mathbb{R}} |f|^2\, dx, \qquad \langle f, \tilde{f} \rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx.$$

Changing f at finitely many points does not change the vector. § 15.6 constructs this vector and proves these formulas.

For $t \in \mathbb{R}$ and $f \in C_c(\mathbb{R})$ put

$$(\lambda(t)f)(x) = f(x - t).$$

Then $\lambda(s)\lambda(t) = \lambda(s+t)$, and $\|\lambda(t)f\| = \|f\|$ by substitution. So each $\lambda(t)$ extends to a unitary operator on $L^2(\mathbb{R})$ with inverse $\lambda(-t)$. On the piecewise continuous functions above, $\lambda(t)$ is still translation.

We identify the Lie algebra of $\mathbb{R}$ with $\mathbb{R}$. With this identification the exponential map $\mathbb{R} \to \mathbb{R}$ is the identity, so $d\lambda(1)f$ is the derivative of $t \mapsto \lambda(t)f$ at $t = 0$. Formally $\frac{d}{dt}\lambda(t)f\big|_{t=0} = -f'$. This formula has no meaning for a general vector. In the domain example, Example 15.16, the difference quotients of $1_{[0,1]}$ are unbounded. So $d\lambda(1)$ needs a stated domain.

15.2 Strong continuity

Definition 15.1. Let $\mathcal{H}$ be a complex Hilbert space and $\mathrm{U}(\mathcal{H})$ its group of unitary operators. A *unitary representation* of a Lie group G on $\mathcal{H}$ is a homomorphism $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ such that the map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is continuous for every $v \in \mathcal{H}$. This condition is *strong continuity*. The homomorphism is *norm continuous* if $g \mapsto \Pi(g)$ is continuous for the operator norm.

Lemma 15.2. *Let $\Pi\colon G \to \mathrm{U}(\mathcal{H})$ be a homomorphism. Let $D \subset \mathcal{H}$ be dense, and suppose $g \mapsto \Pi(g)w$ is continuous at e for every $w \in D$. Then Π is strongly continuous.*

Proof. Let $v \in \mathcal{H}$ and $w \in D$. Since $\Pi(g)$ is unitary,

$$\|\Pi(g)v - v\| \le \|\Pi(g)(v-w)\| + \|\Pi(g)w - w\| + \|w - v\| = 2\|v-w\| + \|\Pi(g)w - w\|.$$

Given $\varepsilon > 0$, choose w with $\|v - w\| < \varepsilon$. Then $\limsup_{g\to e} \|\Pi(g)v - v\| \le 2\varepsilon$. So $g \mapsto \Pi(g)v$ is continuous at e. At another point h we use

$$\|\Pi(g)v - \Pi(h)v\| = \|\Pi(h^{-1}g)v - v\|,$$

and $h^{-1}g \to e$ as $g \to h$. ∎

Proposition 15.3. *The translations λ form a unitary representation of $\mathbb{R}$ on $L^2(\mathbb{R})$.*

Proof. Only strong continuity remains. We apply the dense-set criterion, Lemma 15.2, with $D = C_c(\mathbb{R})$. Let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$. For $|t| \le 1$ the function $\lambda(t)f - f$ vanishes outside $[-R-1, R+1]$. Hence

$$\|\lambda(t)f - f\|^2 \le (2R+2) \sup_{x\in\mathbb{R}} |f(x-t) - f(x)|^2.$$

Since f is continuous with compact support, it is uniformly continuous. Hence the right side tends to 0 as $t \to 0$, and the dense-set criterion gives strong continuity. ∎

By the Peter–Weyl theorem (Lecture 10), every unitary representation of a compact group splits into finite-dimensional pieces. The next proposition shows that the unitary representation λ of the noncompact group $\mathbb{R}$ contains no such piece at all.

Proposition 15.4. *The representation λ has no nonzero finite-dimensional invariant subspace.*

Proof. First we show that matrix coefficients decay. Let $f, g \in C_c(\mathbb{R})$. For large $|t|$ the supports of g and $\lambda(t)f$ are disjoint, so $\langle g, \lambda(t)f\rangle = 0$. Now let $f, g \in L^2(\mathbb{R})$, and choose $f_0, g_0 \in C_c(\mathbb{R})$ close to f and g in norm. Since $\lambda(t)$ is unitary,

$$|\langle g, \lambda(t)f\rangle - \langle g_0, \lambda(t)f_0\rangle| \le \|g - g_0\|\,\|f\| + \|g_0\|\,\|f - f_0\|.$$

Hence $\langle g, \lambda(t)f\rangle \to 0$ as $|t| \to \infty$, for all $f, g \in L^2(\mathbb{R})$.

Suppose that $W \neq 0$ is a finite-dimensional invariant subspace, with orthonormal basis $e_1, \dots, e_k$. Let $f \in W$ be a unit vector. Then $\lambda(t)f$ is a unit vector of W, so

$$\sum_{j=1}^{k} |\langle e_j, \lambda(t) f\rangle|^2 = 1 \qquad \text{for every } t \in \mathbb{R}.$$

Each of the finitely many terms tends to 0 as $|t| \to \infty$. This is a contradiction. ∎

Norm continuity would be too strong a requirement, since translations are not norm continuous. Indeed, Exercise 15.1 shows that $\|\lambda(t) - I\| = 2$ for every $t \neq 0$. Strong continuity requires continuity separately for each vector. It does not require this continuity to be uniform over unit vectors.

Figure 15.1 shows the difference between a fixed vector and a vector chosen after t. For the fixed step $1_{[0,1]}$ and small $t > 0$, the translate differs from the step only on two slivers of total length $2t$. So $\|\lambda(t)1_{[0,1]} - 1_{[0,1]}\| \to 0$ as $t \to 0$. The comb of Exercise 15.1 depends on t, since its teeth have width t. Translation by t reverses its sign on the overlap, which keeps the ratio $\|\lambda(t)f - f\|/\|f\|$ away from 0 for every t. This does not contradict strong continuity, because strong continuity fixes the vector first.

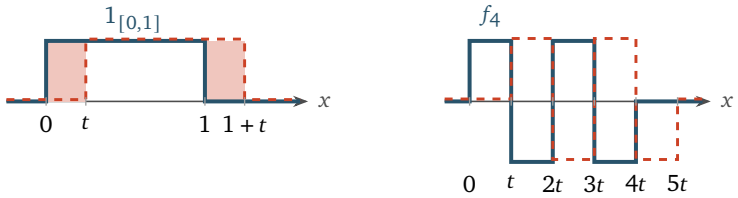


Figure 15.1: Two vectors, heavy, and their translates by t, dashed. The first panel shows the fixed step $1_{[0,1]}$, with the two slivers shaded. The second panel shows the comb f_4 of Exercise 15.1, whose teeth have width t.

15.3 Smooth vectors and the derived representation

A map from an open subset of $\mathbb{R}^n$ to $\mathcal{H}$ is *smooth* if all its partial derivatives of all orders exist as limits in the norm of $\mathcal{H}$ and are continuous. A map $G \to \mathcal{H}$ is smooth if it is smooth in every chart.

Definition 15.5. Let Π be a unitary representation of G on $\mathcal{H}$. A vector $v \in \mathcal{H}$ is *smooth* if $g \mapsto \Pi(g)v$ is a smooth map $G \to \mathcal{H}$. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ put

$$d\Pi(X)v = \frac{d}{dt}\Big|_{t=0} \Pi(\exp tX)v = \lim_{t \to 0} \frac{\Pi(\exp tX)v - v}{t}.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, since linear combinations of smooth maps $G \to \mathcal{H}$ are smooth. The limit defining $d\Pi(X)v$ exists because $t \mapsto \Pi(\exp tX)v$ is the smooth curve $t \mapsto \exp tX$ followed by the smooth map in the definition of a smooth vector.

Proposition 15.6. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$, and let $g \in G$, $X \in \mathfrak{g}$.*

(a) $\Pi(g)\mathcal{H}^\infty \subset \mathcal{H}^\infty$ *and* $d\Pi(X)\mathcal{H}^\infty \subset \mathcal{H}^\infty$.
(b) $\mathcal{H}^\infty$ *is dense in* $\mathcal{H}$.
(c) $X \mapsto d\Pi(X)$ *is a representation of* $\mathfrak{g}$ *on* $\mathcal{H}^\infty$, *and* $d\Pi(\mathrm{Ad}(g)X) = \Pi(g)\, d\Pi(X)\, \Pi(g)^{-1}$ *on* $\mathcal{H}^\infty$.
(d) *For* $v, w \in \mathcal{H}^\infty$ *we have* $\langle d\Pi(X)v, w\rangle = -\langle v, d\Pi(X)w\rangle$.

Proof. We cite Knapp, *Representation Theory of Semisimple Groups*, Sections III.3–4, for the invariance under $d\Pi(X)$ in (a), the density in (b), and the representation property and Ad formula in (c). We prove the rest.

For the invariance under $\Pi(g)$ in (a), let $v \in \mathcal{H}^\infty$, $h \in G$ and $F(g) = \Pi(g)v$. Then $\Pi(g)\Pi(h)v = F(gh)$. Right translation by h is smooth on G, so $g \mapsto F(gh)$ is smooth.

For the skew-symmetry in (d), unitarity gives

$$\langle \Pi(\exp tX)v, w\rangle = \langle v, \Pi(\exp(-tX))w\rangle.$$

Subtract $\langle v, w\rangle$ from both sides and divide by $t \neq 0$:

$$\left\langle \frac{\Pi(\exp tX)v - v}{t},\, w \right\rangle = -\left\langle v,\, \frac{\Pi(\exp(-tX))w - w}{-t} \right\rangle.$$

The inner product is continuous, so letting $t \to 0$ gives (d). ∎

The omitted density step in (b) is the hard one, so we sketch it. The sketch uses a vector-valued integral against a left Haar measure dg on G. For compact G, Haar measure is introduced in Lecture 5. For noncompact G it is introduced in Lecture 16, so until then we cite Knapp for this integral. For $\varphi \in C_c^\infty(G)$ and $v \in \mathcal{H}$ form the *Gårding vector*

$$\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(g)v\, dg.$$

Let $h \in G$. Since $\Pi(h)$ is bounded, it passes inside the integral. Left invariance of dg then allows the substitution $g \mapsto h^{-1}g$. It gives

$$\Pi(h)\Pi(\varphi)v = \int_G \varphi(g)\, \Pi(hg)v\, dg = \int_G \varphi(h^{-1}g)\, \Pi(g)v\, dg.$$

Now h appears only inside the smooth scalar function $(h, g) \mapsto \varphi(h^{-1}g)$. Differentiating under the integral in local coordinates of h shows that $h \mapsto \Pi(h)\Pi(\varphi)v$ is smooth on G. So $\Pi(\varphi)v$ is a smooth vector. If $\varphi_n \geq 0$, $\int_G \varphi_n\, dg = 1$ and the supports shrink to e, then

$$\|\Pi(\varphi_n)v - v\| \leq \sup_{g \in \operatorname{supp}\varphi_n} \|\Pi(g)v - v\| \longrightarrow 0$$

by strong continuity. Exercise 15.4 carries out the argument for $G = \mathbb{R}$ with Riemann integrals.

15.4 Unbounded operators and Stone's theorem

On $L^2(\mathbb{R})$ the operator $-d/dx$ is defined only on a subspace. We need language for such operators.

Definition 15.7. An *operator* A on $\mathcal{H}$ is a linear map $A\colon D(A) \to \mathcal{H}$ defined on a subspace $D(A) \subset \mathcal{H}$, its *domain*. It is *densely defined* if $D(A)$ is dense. We write $B \subset A$ if $D(B) \subset D(A)$ and $Bv = Av$ for $v \in D(B)$.

Definition 15.8. Let A be densely defined. The domain of the *adjoint* A^* is the set of $w \in \mathcal{H}$ for which some $u \in \mathcal{H}$ satisfies

$$\langle w, Av\rangle = \langle u, v\rangle \qquad \text{for all } v \in D(A),$$

and then $A^*w = u$. The operator A is *skew-symmetric* if $\langle Av, w\rangle = -\langle v, Aw\rangle$ for all $v, w \in D(A)$. It is *skew-adjoint* if $A^* = -A$, including $D(A^*) = D(A)$, and *self-adjoint* if $A^* = A$, including $D(A^*) = D(A)$.

Since $D(A)$ is dense, the vector u is unique. Skew-symmetry of A means exactly that $-A \subset A^*$.

Suppose $B \subset A$ and B is densely defined. Then $D(A) \supset D(B)$ is dense too, so both adjoints exist. In this case $A^* \subset B^*$. Indeed, the defining identity for A^*w holds in particular for $v \in D(B)$.

Lemma 15.9. *Let B be densely defined and skew-symmetric. If $v_n \in D(B)$, $v_n \to 0$ and $Bv_n \to u$, then $u = 0$.*

Proof. For every $w \in D(B)$, continuity of the inner product and skew-symmetry of B give

$$\langle w, u\rangle = \lim_n \langle w, Bv_n\rangle = -\lim_n \langle Bw, v_n\rangle = 0.$$

So u is orthogonal to the dense subspace $D(B)$, and $u = 0$. ∎

Definition 15.10. Let B be densely defined and skew-symmetric. The domain of its *closure* $\bar{B}$ is the set of $v \in \mathcal{H}$ for which there are $v_n \in D(B)$ with $v_n \to v$ and (Bv_n) convergent. Then $\bar{B}v = \lim_n Bv_n$.

The value $\bar{B}v$ does not depend on the sequence. Indeed, the difference of two such sequences satisfies the hypotheses of the limit lemma for skew-symmetric operators, Lemma 15.9.

Definition 15.11. Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a unitary representation. Its *generator* A has domain

$$D(A) = \left\{ v \in \mathcal{H} : \lim_{t \to 0} \frac{U(t)v - v}{t} \text{ exists in } \mathcal{H} \right\},$$

and Av is this limit.

Theorem 15.12 (Stone). *Let $U\colon \mathbb{R} \to \mathrm{U}(\mathcal{H})$ be a strongly continuous unitary representation with generator A.*

(a) A is densely defined and skew-adjoint. Moreover $U(t)D(A) \subset D(A)$ and $AU(t)v = U(t)Av$ for $v \in D(A)$.

(b) *Let $D \subset D(A)$ be a dense subspace with $U(t)D \subset D$ for all t. Then $A|_D$ is skew-symmetric, and its closure is A.*

(c) *Every skew-adjoint operator is the generator of exactly one strongly continuous unitary representation of $\mathbb{R}$, written $t \mapsto e^{tA}$.*

We cite most of this theorem and prove only the invariance statement in (a). The sources are Hall, *Quantum Theory for Mathematicians*, Section 10.2, and Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4. The section on sources and hypotheses at the end of this lecture matches their conventions with ours. For $v \in D(A)$, since $U(t)$ is bounded and commutes with $U(h)$,

$$\frac{U(h)U(t)v - U(t)v}{h} = U(t)\,\frac{U(h)v - v}{h} \longrightarrow U(t)Av \qquad (h \to 0).$$

So $U(t)v \in D(A)$ and $AU(t)v = U(t)Av$.

Our physical convention writes $U(t) = e^{-itH}$ with a self-adjoint operator H, so $H = iA$.

Corollary 15.13. *Let Π be a unitary representation of a Lie group G on $\mathcal{H}$. Let $X \in \mathfrak{g}$, and let B_X be the generator of $t \mapsto \Pi(\exp tX)$. Then $\mathcal{H}^\infty \subset D(B_X)$ and $B_X = d\Pi(X)$ on $\mathcal{H}^\infty$. The closure of $d\Pi(X)$ with domain $\mathcal{H}^\infty$ is the skew-adjoint operator B_X, and $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$.*

Proof. The map $t \mapsto \exp tX$ is a continuous homomorphism $\mathbb{R} \to G$. Composing it with Π gives a homomorphism $\mathbb{R} \to \mathrm{U}(\mathcal{H})$. For each v, the map $t \mapsto \Pi(\exp tX)v$ is a continuous map followed by the continuous map $g \mapsto \Pi(g)v$. So $t \mapsto \Pi(\exp tX)$ is a strongly continuous unitary representation of $\mathbb{R}$, and its generator B_X is defined.

The first two claims restate the definition of $d\Pi(X)$. By the invariance and density statements of Proposition 15.6, $\mathcal{H}^\infty$ is dense and invariant under every $\Pi(\exp tX)$. The dense invariant-domain criterion of Theorem 15.12(b), applied with $D = \mathcal{H}^\infty$, shows that the closure of $d\Pi(X)$ is B_X. The uniqueness statement of Theorem 15.12(c) then gives $\Pi(\exp tX) = e^{t\,\overline{d\Pi(X)}}$. ∎

The generator B_X can have a larger domain than $\mathcal{H}^\infty$. That domain can depend on X, and B_X need not preserve it. For example, Exercise 15.2 gives a vector g in the domain of the generator of translations whose image is outside that domain. Brackets $[d\Pi(X), d\Pi(Y)]$ need one subspace on which every $d\Pi(X)$ is defined and which every $d\Pi(X)$ preserves. By Proposition 15.6, the space $\mathcal{H}^\infty$ has both properties.

In the physical convention above, the *physical generator* of $t \mapsto \Pi(\exp tX)$ is $A_X = iB_X$, so $\Pi(\exp tX) = e^{-itA_X}$. Since B_X is skew-adjoint, A_X is self-adjoint. For translations and $X = 1 \in \mathbb{R}$, the generator B_1 is the operator that the next section calls A. The subscript matters: the physical generator $A_1 = iA$ is self-adjoint, while A is skew-adjoint. With $\hbar = 1$, the operator A_1 is the momentum operator P, formally $-i\,d/dx$.

15.5 Translations on $L^2(\mathbb{R})$

The *Schwartz space* $\mathcal{S}(\mathbb{R})$ is the space of smooth $f\colon \mathbb{R} \to \mathbb{C}$ such that

$$\sup_{x \in \mathbb{R}} |x^k f^{(j)}(x)| < \infty \qquad \text{for all } j, k \geq 0.$$

It contains $C_c^\infty(\mathbb{R})$ and $e^{-x^2/2}$. Let A be the generator of λ.

Lemma 15.14. *Let $f \in C^2(\mathbb{R})$ with f and f' square integrable. Suppose $|f''(y)| \le C(1+y^2)^{-1}$ for all y. Then $f \in D(A)$ and $Af = -f'$.*

Proof. Fix $0 < |h| \le 1$. Taylor's formula with integral remainder, for $s \mapsto f(x - sh)$ on $[0,1]$, gives

$$f(x-h) = f(x) - hf'(x) + h^2 \int_0^1 (1-s) f''(x-sh)\, ds.$$

So the error function is

$$q_h(x) = \frac{f(x-h) - f(x)}{h} + f'(x) = h \int_0^1 (1-s) f''(x-sh)\, ds,$$

and $|q_h(x)| \le \frac{|h|}{2} \sup_{|y-x|\le 1} |f''(y)|$. If $|y - x| \le 1$, then $x^2 \le (|y|+1)^2 \le 2y^2 + 2$, so $1 + x^2 \le 3(1+y^2)$. Therefore

$$|q_h(x)| \le \frac{3C|h|}{2(1+x^2)}, \qquad \|q_h\|^2 \le \frac{9C^2h^2}{4} \int_{\mathbb{R}} \frac{dx}{(1+x^2)^2} = \frac{9\pi C^2 h^2}{8}.$$

Thus $\left\| \frac{\lambda(h)f - f}{h} - (-f') \right\| = \|q_h\| \to 0$. ∎

The essential hypothesis is the weighted bound $|f''(y)| \le C(1+y^2)^{-1}$. The difference quotient in the proof converges pointwise for every C^1 function f. Pointwise convergence alone does not give convergence in norm. The weighted bound makes the error q_h at most $|h|$ times a fixed square-integrable function, so the norm of q_h is $O(|h|)$.

Proposition 15.15. *The space $\mathcal{S}(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, invariant under every $\lambda(t)$, and contained in $L^2(\mathbb{R})^\infty$. For $f \in \mathcal{S}(\mathbb{R})$ we have $d\lambda(1)f = -f'$. The closure of $B = -d/dx$ with domain $\mathcal{S}(\mathbb{R})$ is the skew-adjoint generator A.*

Proof. Invariance. Put $y = x - t$. Expanding $x^k = (y+t)^k$ bounds $|x^k f^{(j)}(x-t)|$ by a finite sum of terms $\binom{k}{i}|t|^{k-i}|y^i f^{(j)}(y)|$.

Smoothness. Each $f \in \mathcal{S}(\mathbb{R})$ satisfies the hypotheses of the Taylor-remainder lemma, Lemma 15.14. Indeed $(1+y^2)f$, $(1+y^2)f'$ and $(1+y^2)f''$ are bounded. So f and f' are square integrable, and $|f''| \le C(1+y^2)^{-1}$. In particular $\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})$. By the lemma, $t \mapsto \lambda(t)f$ is differentiable at 0 with derivative $-f'$. At any t_0,

$$\frac{\lambda(t_0+h)f - \lambda(t_0)f}{h} = \lambda(t_0)\, \frac{\lambda(h)f - f}{h} \longrightarrow \lambda(t_0)(-f'),$$

because $\lambda(t_0)$ is bounded. The derivative $t \mapsto \lambda(t)(-f')$ is continuous by the strong continuity of translations, Proposition 15.3. Since $-f' \in \mathcal{S}(\mathbb{R})$, we may repeat. The k-th derivative is $t \mapsto \lambda(t)\big((-1)^k f^{(k)}\big)$, so f is a smooth vector.

Density. Fix $\varphi \in C_c^\infty(\mathbb{R})$ with $\varphi \ge 0$, support in $[-1,1]$ and $\int \varphi = 1$. Put $\varphi_n(t) = n\varphi(nt)$. For $f \in C_c(\mathbb{R})$ vanishing outside $[-R,R]$, let

$$f_n(x) = \int_{\mathbb{R}} \varphi_n(x-y) f(y)\, dy = \int_{\mathbb{R}} \varphi_n(t) f(x-t)\, dt.$$

The first form shows $f_n \in C_c^\infty(\mathbb{R}) \subset \mathcal{S}(\mathbb{R})$, with support in $[-R-1, R+1]$. The second form gives

$$|f_n(x) - f(x)| \le \sup_{|t| \le 1/n} |f(x-t) - f(x)|,$$

which tends to 0 uniformly in x because f is uniformly continuous. Since f_n and f vanish outside $[-R-1, R+1]$, it follows that $f_n \to f$ in $L^2(\mathbb{R})$. Since $C_c(\mathbb{R})$ is dense in $L^2(\mathbb{R})$, so is $\mathcal{S}(\mathbb{R})$.

Closure. By the three steps above, $D = \mathcal{S}(\mathbb{R})$ is a dense invariant subspace of $D(A)$. So the dense invariant-domain criterion of Theorem 15.12(b) shows that the closure of B is A. ∎

The function f_n is the Gårding vector $\lambda(\varphi_n)f$ written pointwise. By analogy with the compact groups of Lecture 11, it is the convolution $\varphi_n * f$.

Example 15.16. The vectors w and h, and the operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$, show where the domains lie.

(i) Let $w(x) = (1+x^2)^{-1}$. Then

$$w'(x) = \frac{-2x}{(1+x^2)^2}, \qquad w''(x) = \frac{6x^2 - 2}{(1+x^2)^3}.$$

Since $|6x^2 - 2| \le 6(1+x^2)$, we get $|w''(x)| \le 6(1+x^2)^{-2} \le 6(1+x^2)^{-1}$. Both w and w' are square integrable. By the Taylor-remainder lemma, Lemma 15.14, $w \in D(A)$ and

$$(Aw)(x) = \frac{2x}{(1+x^2)^2}.$$

But $x^2 w(x) \to 1$, so $x^4 w(x)$ is unbounded and $w \notin \mathcal{S}(\mathbb{R})$.

(ii) The operator $B = -d/dx$ on $\mathcal{S}(\mathbb{R})$ is the restriction of $d\lambda(1)$ to $\mathcal{S}(\mathbb{R})$ by Proposition 15.15. So B is skew-symmetric by the skew-symmetry of $d\lambda(1)$ on smooth vectors, Proposition 15.6(d). It is not skew-adjoint. Indeed $B \subset A$, the domain $\mathcal{S}(\mathbb{R})$ is dense, and $A^* = -A$. So $-A = A^* \subset B^*$. Hence $w \in D(A) \subset D(B^*)$, while $w \notin D(B)$. So $D(B^*) \neq D(B)$, and therefore $B^* \neq -B$.

(iii) Let $h = 1_{[0,1]}$. For $0 < t \le 1$, up to finitely many points,

$$\lambda(t)h - h = 1_{[t,1+t]} - 1_{[0,1]} = 1_{(1,1+t]} - 1_{[0,t)}, \qquad \|\lambda(t)h - h\|^2 = 2t.$$

For $-1 \le t < 0$ the same computation gives $2|t|$. Therefore

$$\left\| \frac{\lambda(t)h - h}{t} \right\| = \sqrt{\frac{2}{|t|}} \longrightarrow \infty.$$

Convergent quotients would stay bounded near 0. So $h \notin D(A)$. Smooth vectors lie in $D(A)$ by Corollary 15.13, so also $h \notin L^2(\mathbb{R})^\infty$.

Together these give

$$\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})^\infty \subset D(A), \qquad w \in D(A) \setminus \mathcal{S}(\mathbb{R}), \qquad h \in L^2(\mathbb{R}) \setminus D(A).$$

Exercise 15.2 gives a tent function in $D(A)$ that is not a smooth vector. All derivatives of w satisfy the hypotheses of the Taylor-remainder lemma, Lemma 15.14, so w is a smooth vector that is not in $\mathcal{S}(\mathbb{R})$. So both inclusions are strict. The momentum operator $P = iA$ has domain $D(A)$, not $\mathcal{S}(\mathbb{R})$.

Remark 15.17. We cite Plancherel's theorem (Folland, *Real Analysis*, second edition, Chapter 8). The Fourier transform $\hat{f}(p) = (2\pi)^{-1/2} \int_{\mathbb{R}} f(x) e^{-ipx}\, dx$ extends to a unitary operator of $L^2(\mathbb{R})$, and it turns $\lambda(t)$ into multiplication by e^{-ipt}. As in Exercise 15.3, $D(A)$ is the set of f with $\int_{\mathbb{R}} p^2 |\hat{f}(p)|^2\, dp < \infty$, and P acts as multiplication by p. In quantum mechanics, $D(A)$ is the set of states for which P^2 has a finite expectation value.

This describes the three vectors above. The step $h = 1_{[0,1]}$ has $|\hat{h}(p)|^2 = 2\sin^2(p/2)/(\pi p^2)$, so $\int p^2 |\hat{h}|^2\, dp$ diverges. The tent g of Exercise 15.2 has $|\hat{g}(p)|^2 = 8\sin^4(p/2)/(\pi p^4)$, so $\int p^2 |\hat{g}|^2\, dp$ converges, while $\int p^4 |\hat{g}|^2\, dp$ diverges. The function w has $\hat{w}(p) = \sqrt{\pi/2}\, e^{-|p|}$, so all these integrals converge. Finally, $\|\lambda(t) - I\| = \sup_p |e^{-ipt} - 1| = 2$ for $t \neq 0$, which agrees with Exercise 15.1.

15.6 Piecewise continuous functions as vectors

We now construct the vector of a square integrable function, as stated in § 15.1. We first treat functions that vanish outside a bounded interval. Then we pass to square integrable functions by truncation.

Let f be piecewise continuous and vanish outside a bounded interval. Then f has finitely many jumps $x_1 < \cdots < x_J$, and the one-sided limits at these jumps make f bounded, say $|f| \le M$. Choose $\delta > 0$ so small that the intervals $[x_j - \delta, x_j + \delta]$ are disjoint. Let g_δ equal f outside these intervals, and let g_δ be linear on each of them, with the values of f at the two endpoints. Then $g_\delta \in C_c(\mathbb{R})$ and $|g_\delta| \le M$. So $|f - g_\delta| \le 2M$, and $f - g_\delta$ vanishes outside J intervals of length 2δ. Hence

$$\int_{\mathbb{R}} |f - g_\delta|^2\, dx \le 8JM^2\delta.$$

For piecewise continuous functions u and $\tilde{u}$ that vanish outside a bounded interval, write $\|u\|_2 = \big(\int_{\mathbb{R}} |u|^2\, dx\big)^{1/2}$. On $C_c(\mathbb{R})$ this is the norm of $L^2(\mathbb{R})$. The Cauchy–Schwarz inequality for Riemann integrals, $\int_{\mathbb{R}} |\bar{u}\tilde{u}|\, dx \le \|u\|_2 \|\tilde{u}\|_2$, gives the triangle inequality for $\|\cdot\|_2$.

Take $\delta_n \to 0$ and $g_n = g_{\delta_n}$, so that $\|f - g_n\|_2 \to 0$. Since $\|g_n - g_m\|_2 \le \|g_n - f\|_2 + \|f - g_m\|_2$, the sequence (g_n) is Cauchy in $L^2(\mathbb{R})$. We call its limit the vector f. By the same inequality, every sequence in $C_c(\mathbb{R})$ that approximates f in $\|\cdot\|_2$ has this limit. Sums and scalar multiples of approximating sequences approximate sums and scalar multiples, so the vector depends linearly on f. By the triangle inequality, the vector f has norm $\|f\| = \lim_n \|g_n\|_2 = \|f\|_2$.

Let $\tilde{f}$ be a second such function, approximated by $\tilde{g}_n \in C_c(\mathbb{R})$. By continuity of the inner product, $\langle f, \tilde{f}\rangle = \lim_n \int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx$. Also

$$\int_{\mathbb{R}} \overline{g_n}\, \tilde{g}_n\, dx - \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx = \int_{\mathbb{R}} \overline{(g_n - f)}\, \tilde{g}_n\, dx + \int_{\mathbb{R}} \bar{f}\, (\tilde{g}_n - \tilde{f})\, dx.$$

By the Cauchy–Schwarz inequality, the right side has absolute value at most $\|g_n - f\|_2 \|\tilde{g}_n\|_2 + \|f\|_2 \|\tilde{g}_n - \tilde{f}\|_2$. This tends to 0, because $\|\tilde{g}_n\|_2$ stays bounded. So $\langle f, \tilde{f}\rangle = \int_{\mathbb{R}} \bar{f}\tilde{f}\, dx$.

Now let f be square integrable. Each truncation $f \cdot 1_{[-n,n]}$ is piecewise continuous and vanishes outside $[-n, n]$, so it defines a vector. The difference of two truncations is $f \cdot$

$1_{\{n<|x|\le m\}}$. By linearity and the norm formula, the truncations form a Cauchy sequence, because

$$\left\| f\cdot 1_{[-m,m]} - f\cdot 1_{[-n,n]}\right\|^2 = \int_{n<|x|\le m} |f|^2\,dx \longrightarrow 0 \qquad (m>n\to\infty).$$

Their limit is the vector f. For square integrable f and $\tilde f$, letting $n\to\infty$ in the formulas for the truncations gives the two formulas of § 15.1. The second improper integral converges absolutely, since $\int_{-n}^{n}|\bar f\tilde f|\,dx \le \|f\|\,\|\tilde f\|$ by the Cauchy–Schwarz inequality.

Changing f at finitely many points does not change the vector. Indeed, the difference of the two functions is nonzero at only finitely many points, so its vector has norm 0.

15.7 Self-checks

1. Let $\dim\mathcal{H}<\infty$. What are $\mathcal{H}^\infty$ and $D(B_X)$?
   Answer. Both equal $\mathcal{H}$. The homomorphism $\Pi\colon G\to \mathrm{U}(\mathcal{H})$ is continuous, hence smooth (Lecture 3), so every vector is smooth. Since $d\Pi(X)$ is then defined on all of $\mathcal{H}$, its closure B_X also has domain $\mathcal{H}$.
2. Let $f(x)=xe^{-x^2/2}$. Compute Af and $\langle f, Af\rangle$.
   Answer. Since $f\in\mathcal{S}(\mathbb{R})$, the Schwartz-space proposition, Proposition 15.15, gives $Af=-f'=(x^2-1)e^{-x^2/2}$. Then $\langle f,Af\rangle=\int_{\mathbb{R}} x(x^2-1)e^{-x^2}\,dx=0$, because the integrand is odd. This is consistent with skew-symmetry, which forces $\langle f,Af\rangle$ to be purely imaginary.
3. Let D be the space of smooth functions with compact support in $(0,\infty)$. Then $D\subset D(A)$. Does the dense invariant-domain criterion of Theorem 15.12(b) apply to D?
   Answer. No. The space D is not dense, since every element is orthogonal to $1_{[-1,0]}\neq 0$. It is not invariant either. Given $t<0$, choose a nonzero $f\in D$ supported in $(0,|t|)$. Then $\lambda(t)f$ is supported in $(t,0)$, so $\lambda(t)f\notin D$.
4. Let $D=C_c^\infty(\mathbb{R}\setminus\{0\})$. This space is dense in $L^2(\mathbb{R})$ and contained in $D(A)$, but it is not invariant. Is the closure of $A|_D$ equal to A?
   Answer. No. Let g be a C^1 function with g and g' square integrable and $g(x)\to 0$ as $x\to-\infty$. Then $|g(0)|^2=\int_{-\infty}^{0}(|g|^2)'\,dx\le 2\|g\|\,\|g'\|$. Suppose that $v_n\in D$ with $v_n\to w$ and $Av_n\to Aw$, where $w=(1+x^2)^{-1}$. Take $g=w-v_n$. Since $v_n(0)=0$ and $w(0)=1$, we get $1\le 2\|w-v_n\|\,\|w'-v_n'\|\to 0$, which is false. So $w\in D(A)$ is not in the domain of the closure. The invariance hypothesis in the criterion cannot be dropped.

15.8 Summary

Unitary representations are required to be strongly continuous, not norm continuous. The translation representation on $L^2(\mathbb{R})$ has no nonzero finite-dimensional invariant subspace. Smooth vectors form a dense invariant subspace $\mathcal{H}^\infty$. On it, $d\Pi$ is a representation of $\mathfrak{g}$ by skew-symmetric operators. Stone's theorem attaches a skew-adjoint generator to each strongly continuous unitary representation of $\mathbb{R}$. The closure of $d\Pi(X)$ on $\mathcal{H}^\infty$ is this generator. For translations, $-d/dx$ on $\mathcal{S}(\mathbb{R})$ is skew-symmetric but not skew-adjoint. Its closure has the strictly larger domain $D(A)$, and $D(A)$ is still not all of $L^2(\mathbb{R})$.

Sources and hypotheses

Hall, *Quantum Theory for Mathematicians*, Section 10.2, Theorem 10.15, gives Stone's theorem. Hall writes $U(t) = e^{itH_{\text{Hall}}}$ with a self-adjoint operator H_{Hall}, so $H_{\text{Hall}} = -iA$. In our physical convention $H = iA$, so $H_{\text{Hall}} = -H$. Reed–Simon, *Methods of Modern Mathematical Physics I*, Section VIII.4, Theorems VIII.10–11, gives the dense invariant-domain criterion in Theorem 15.12(b).

15.9 Reading

- Hall, *Quantum Theory for Mathematicians* (2013): unbounded self-adjoint operators and Stone's theorem, Section 10.2.
- Reed–Simon, *Methods of Modern Mathematical Physics I: Functional Analysis*, Section VIII.4, especially Theorems VIII.10–11 on dense invariant domains.
- Knapp, *Representation Theory of Semisimple Groups* (2001), Sections III.3–4: smooth vectors, Gårding vectors and the action of $\mathfrak{g}$.
- Etingof, MIT 18.757 lecture notes, Lectures 1–7: an alternative introduction to continuous representations of Lie groups.

Lecture 16

Noncompact Haar Measure and the Modular Function

Objectives. *Lecture 5 integrated over compact groups. This lecture drops compactness. It asks which translation-invariant integrals a Lie group carries and how left and right integration differ. A noncompact Lie group has infinite Haar volume, so the normalised averaging of Lecture 5 is no longer available in general. Every Lie group has a left-invariant and a right-invariant density. With equal normalisations at the identity, the left density divided by the right density is the modular function. It controls right translation, inversion and the right regular representation.*

By the end of this lecture you will be able to:

- Construct the left and right Haar densities of a Lie group, prove that each is unique up to a positive factor, and prove that the total mass is infinite when the group is noncompact.
- Prove the right-translation and inversion identities for $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$.
- Compute both Haar densities and Δ for the affine group $ax + b$.
- Construct the left and right regular representations as strongly continuous unitary representations on $L^2(G)$, and state the integration rules of Part II with their hypotheses.

Lecture 5 built a left-invariant density on a compact Lie group by translating a density at e. That construction did not use compactness. Compactness entered later. It made the integral finite on every continuous function, and it allowed normalisation to total mass 1. It made the image of $j(h) = |\det \mathrm{Ad}(h)|$ a compact subgroup of $(0, \infty)$, so $j \equiv 1$. It gave strict positivity through a finite cover. Without compactness we integrate only compactly supported functions. Strict positivity comes from charts, and j survives as a homomorphism. The left Haar density μ_L below is the density δ of Lecture 5, and the modular function below is $\Delta = j^{-1}$.

Throughout, G is a Lie group of dimension n with Lie algebra $\mathfrak{g} = T_eG$. We write left translation, right translation, conjugation and inversion as

$$L_g(x) = gx, \qquad R_g(x) = xg, \qquad C_g = L_g \circ R_{g^{-1}}, \qquad \iota(x) = x^{-1}.$$

Then $R_g \circ R_h = R_{hg}$ and $d(C_g)_e = \mathrm{Ad}(g)$.

16.1 Densities and integration

Under a linear map T of $\mathbb{R}^n$, Lebesgue measure scales by $|\det T|$. A density imposes this rule on each tangent space.

Definition 16.1. Let V be a real vector space of dimension n. A *positive density* on V is a nonzero function $\delta\colon V^n \to [0,\infty)$ such that

$$\delta(Tv_1,\dots,Tv_n) = |\det T|\,\delta(v_1,\dots,v_n) \qquad (T \in \operatorname{End} V).$$

Let $e_1,\dots,e_n$ be a basis and $v_j = Te_j$. Then $\delta(v_1,\dots,v_n) = |\det T|\,\delta(e_1,\dots,e_n)$. So δ is fixed by the number $\delta(e_1,\dots,e_n)$, which is positive since $\delta \neq 0$. Two positive densities on V therefore differ by a positive factor.

Definition 16.2. A *smooth positive density* μ on G assigns a positive density μ_x on T_xG to each $x \in G$ such that, in every chart with coordinates y, the coefficient $m(y) = \mu_x(\partial_1,\dots,\partial_n)$ is smooth.

For a diffeomorphism F of G and a smooth function $\varphi > 0$ put

$$(F^*\mu)_x(v_1,\dots,v_n) = \mu_{F(x)}(dF_x v_1,\dots,dF_x v_n), \qquad (\varphi\mu)_x = \varphi(x)\,\mu_x.$$

The chain rule gives $(F\circ H)^* = H^*F^*$. The definitions give $F^*(\varphi\mu) = (\varphi\circ F)\,F^*\mu$.

Let $C_c(G)$ be the space of continuous functions $f\colon G \to \mathbb{C}$ with compact support.

If $\operatorname{supp} f$ lies in one chart, set $\int_G f\mu = \int_{\mathbb{R}^n} f(y)\,m(y)\,dy$. A change of chart multiplies the coefficient m by the absolute Jacobian determinant. Hence, by the change of variables formula, the integral is well defined.

For general $f \in C_c(G)$ we use a partition of unity, adapted from Lecture 5. Here a *partition of unity* on $\operatorname{supp} f$ is a finite family of smooth functions $\psi_i \geq 0$ on G, each with compact support in one chart, such that $\sum_i \psi_i = 1$ on $\operatorname{supp} f$. On a compact manifold, Lecture 5 requires the sum to equal 1 everywhere. Since f vanishes outside $\operatorname{supp} f$, we have $f = \sum_i \psi_i f$. We add the chart integrals of the functions $\psi_i f$.

We cite this construction and the following three facts. Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII, carries out the same construction with differential forms.

(I1) $f \mapsto \int_G f\mu$ is linear, and $\int_G f\mu > 0$ if $f \geq 0$ and $f \neq 0$.
(I2) $\int_G (f\circ F)\,F^*\mu = \int_G f\mu$ for every diffeomorphism F of G.
(I3) $\int_G f\,(\varphi\mu) = \int_G (f\varphi)\,\mu$.

In a chart, $F^*\mu$ has coefficient $m(F(y))\,|\det DF(y)|$. So (I2) is the change of variables formula.

16.2 Left and right Haar densities

Theorem 16.3 (Haar densities). *Fix a positive density μ_e on $\mathfrak{g}$.*

(a) There is exactly one smooth positive density μ_L with $(\mu_L)_e = \mu_e$ and $L_g^*\mu_L = \mu_L$ for all $g \in G$. Every left-invariant smooth positive density is a positive multiple of μ_L.

(b) There is exactly one smooth positive density μ_R with $(\mu_R)_e = \mu_e$ and $R_g^*\mu_R = \mu_R$ for all $g \in G$. Every right-invariant smooth positive density is a positive multiple of μ_R.

(c) Write $\int_G f(x)\,d_Lx$ for $\int_G f\mu_L$, and similarly d_Rx. For $f \in C_c(G)$ and $g \in G$,

$$\int_G f(gx)\,d_Lx = \int_G f(x)\,d_Lx, \qquad \int_G f(xg)\,d_Rx = \int_G f(x)\,d_Rx.$$

Proof. Uniqueness. Let ν be a left-invariant smooth positive density. Evaluate $L_{x^{-1}}^*\nu = \nu$ at x:

$$\nu_x(v_1,\dots,v_n) = \nu_e\big(d(L_{x^{-1}})_x v_1,\dots,d(L_{x^{-1}})_x v_n\big). \tag{16.1}$$

So ν is determined by ν_e. Two positive densities on $\mathfrak{g}$ differ by a positive factor. Thus $\nu_e = c\mu_e$ for some $c > 0$. Once μ_L is constructed below, the formula gives $\nu = c\mu_L$.

Existence. Define μ_L by (16.1) with $\nu_e = \mu_e$. Linear maps act entrywise on tuples $v = (v_1,\dots,v_n)$. Since $L_{(gx)^{-1}} \circ L_g = L_{x^{-1}}$, the chain rule gives

$$(L_g^*\mu_L)_x(v) = \mu_e\big(d(L_{(gx)^{-1}})_{gx}\,d(L_g)_x v\big) = \mu_e\big(d(L_{x^{-1}})_x v\big) = (\mu_L)_x(v).$$

Smoothness. This step uses the smooth group law. Choose a basis $X_1,\dots,X_n$ of $\mathfrak{g}$. The left-invariant fields $\tilde{X}_i(x) = d(L_x)_e X_i$ are smooth (Lecture 1). Since $L_{x^{-1}} \circ L_x = \mathrm{id}$, we have $d(L_{x^{-1}})_x \tilde{X}_i(x) = X_i$. Hence

$$(\mu_L)_x\big(\tilde{X}_1(x),\dots,\tilde{X}_n(x)\big) = \mu_e(X_1,\dots,X_n) =: c_0 > 0.$$

In a chart write $\tilde{X}_i = \sum_j P_{ji}(y)\,\partial_j$. The matrix $P(y)$ is smooth. It is invertible because $d(L_x)_e$ is an isomorphism. The density rule gives $c_0 = |\det P(y)|\,m(y)$. So $m = c_0\,|\det P|^{-1}$ is smooth.

Right translations. Use $\nu_x(v) = \nu_e(d(R_{x^{-1}})_x v)$. The key identity is $R_{(xg)^{-1}} \circ R_g = R_{x^{-1}}$, since $z\,g\,g^{-1}x^{-1} = zx^{-1}$. The fields $d(R_x)_e X_i$ replace $\tilde{X}_i$.

Integrals. Apply the change of variables (I2) with $F = L_g$ or $F = R_g$. ∎

On a compact group one normalises the total mass to 1, as in Lecture 5. For noncompact G this is impossible, as we now prove. Since the constant function 1 is not in $C_c(G)$, we state the result for compactly supported functions.

Proposition 16.4 (Infinite Haar volume). *Let G be a noncompact Lie group, and let $f \in C_c(G)$ with $f \geq 0$ and $f \neq 0$. There are elements $g_1, g_2, \dots$ of G such that the translates $f_k(x) = f(g_k^{-1}x)$ have pairwise disjoint supports. Hence $F_N = f_1 + \cdots + f_N$ lies in $C_c(G)$, it satisfies $0 \leq F_N \leq \sup f$, and*

$$\int_G F_N\,d_Lx = N\int_G f\,d_Lx, \qquad \int_G f\,d_Lx > 0.$$

Proof. We choose the elements one at a time. Let $S = \operatorname{supp} f$, a compact set, and take $g_1 = e$. Suppose $g_1, \dots, g_{m-1}$ are chosen. The set $\bigcup_{k<m} g_k S S^{-1}$ is compact, as a finite union of continuous images of $S \times S$. Since G is not compact, there is an element g_m outside it.

The support of f_k is $g_k S$. If $g_m S$ met $g_k S$ with $k < m$, then $g_m s = g_k s'$ for some $s, s' \in S$. That gives $g_m = g_k s' s^{-1} \in g_k S S^{-1}$, which contradicts the choice of g_m. So the supports are pairwise disjoint.

At each point at most one term of F_N is nonzero, so $0 \le F_N \le \sup f$. Left invariance in Theorem 16.3(c) gives $\int_G f_k \, d_L x = \int_G f \, d_L x$, and this value is positive by the positivity statement (I1). ∎

So the left Haar integral is unbounded on functions in $C_c(G)$ with values in $[0, \sup f]$, and the total mass of G is infinite. The right Haar integral behaves in the same way, with right translates. A noncompact Lie group therefore has no normalised finite Haar integral, and the averaging arguments of Lecture 5 are no longer available in general.

We use the same density μ_e for both μ_L and μ_R.

16.3 The modular function

Definition 16.5. The *modular function* of G is $\Delta(g) = |\det \operatorname{Ad}(g)|^{-1}$. The group G is *unimodular* if $\Delta \equiv 1$.

Since $\operatorname{Ad} \colon G \to \mathrm{GL}(\mathfrak{g})$ is a smooth homomorphism (Lecture 2), Δ is a smooth homomorphism into the multiplicative group $(0, \infty)$.

Theorem 16.6 (Integration identities)**.** *Let $f \in C_c(G)$ and $g \in G$.*

(i) $R_g^* \mu_L = \Delta(g)\, \mu_L$. *Hence* $\int_G f(xg)\, d_L x = \Delta(g)^{-1} \int_G f(x)\, d_L x$.

(ii) $\mu_R = \Delta^{-1} \mu_L$, *that is,* $\int_G f(x)\, d_R x = \int_G f(x)\, \Delta(x)^{-1}\, d_L x$.

(iii) $\int_G f(x^{-1})\, d_L x = \int_G f(x)\, d_R x$ *and* $\int_G f(x^{-1})\, \Delta(x^{-1})\, d_L x = \int_G f(x)\, d_L x$.

Proof. (i) Left and right translations commute. Hence

$$L_h^* R_g^* \mu_L = (R_g \circ L_h)^* \mu_L = (L_h \circ R_g)^* \mu_L = R_g^* L_h^* \mu_L = R_g^* \mu_L.$$

So $R_g^* \mu_L$ is left-invariant. By the uniqueness in Theorem 16.3(a), it equals $c(g) \mu_L$ for some $c(g) > 0$. We find $c(g)$ at e. Since $L_{g^{-1}} \circ R_g = C_{g^{-1}}$,

$$(R_g^* \mu_L)_e(v) = \mu_e\big(d(L_{g^{-1}})_g \, d(R_g)_e v\big) = \mu_e\big(\operatorname{Ad}(g^{-1}) v\big) = |\det \operatorname{Ad}(g)|^{-1} \mu_e(v).$$

Thus $c(g) = \Delta(g)$. Now the change of variables (I2) with $F = R_g$ and the multiplication rule (I3) give

$$\int_G f \mu_L = \int_G (f \circ R_g)\, R_g^* \mu_L = \Delta(g) \int_G f(xg)\, d_L x.$$

(ii) Use $F^*(\varphi \mu) = (\varphi \circ F) F^* \mu$, the right-translation identity (i) and the homomorphism property of Δ:

$$\big(R_g^* (\Delta^{-1} \mu_L)\big)_x = \Delta(xg)^{-1} \Delta(g)\, (\mu_L)_x = \Delta(x)^{-1} (\mu_L)_x.$$

So $\Delta^{-1}\mu_L$ is right-invariant with value μ_e at e. By uniqueness in Theorem 16.3(b) it equals μ_R. Then the multiplication rule (I3) gives the integral form.

(iii) First, $d\iota_e = -\mathrm{id}$. Indeed, the multiplication map $\mathrm{mult}\colon G\times G\to G$ satisfies $\mathrm{mult}(x,e)=x$ and $\mathrm{mult}(e,x)=x$ for $x\in G$. So $d(\mathrm{mult})_{(e,e)}(v,w)=v+w$ for $v,w\in\mathfrak{g}$. Differentiating $x\,\iota(x)=e$ at e gives $v+d\iota_e v=0$. Therefore

$$(\iota^*\mu_L)_e(v_1,\dots,v_n)=\mu_e(-v_1,\dots,-v_n)=|\det(-\mathrm{id})|\,\mu_e(v_1,\dots,v_n)=\mu_e(v_1,\dots,v_n).$$

Next, $(xg)^{-1}=g^{-1}x^{-1}$ says $\iota\circ R_g=L_{g^{-1}}\circ\iota$. Hence

$$R_g^*\iota^*\mu_L=(\iota\circ R_g)^*\mu_L=(L_{g^{-1}}\circ\iota)^*\mu_L=\iota^*L_{g^{-1}}^*\mu_L=\iota^*\mu_L.$$

By the uniqueness in Theorem 16.3(b), $\iota^*\mu_L=\mu_R$. Apply the change of variables (I2) with $F=\iota$ to $f\circ\iota$. This gives $\int_G f\,d_Rx=\int_G f(x^{-1})\,d_Lx$. For the second identity apply the first to $h=f\Delta\in C_c(G)$ and use the density relation (ii):

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\int_G h(x)\,d_Rx=\int_G f(x)\Delta(x)\Delta(x)^{-1}\,d_Lx.$$ ∎

Remark 16.7. The right-translation identity (i) is equivalent to $\int_G f(xg^{-1})\,d_Lx=\Delta(g)\int_G f\,d_Lx$. For Borel sets E, using the Radon measure of § 16.6, it reads $\mu_L(Eg)=\Delta(g)\mu_L(E)$. This is Folland's convention. Some texts call $|\det\mathrm{Ad}(g)|$ the modular function. When comparing sources, check which defining identity is used.

Corollary 16.8. *(a) Every compact Lie group is unimodular.*

(b) If G is connected, then $\Delta(\exp X)=e^{-\operatorname{tr}\operatorname{ad}X}$. So G is unimodular if and only if $\operatorname{tr}\operatorname{ad}X=0$ for all $X\in\mathfrak{g}$. This holds when $\mathfrak{g}$ is abelian, nilpotent or semisimple.

Proof. (a) Since G is compact and Δ is continuous, the image $\Delta(G)$ is a compact subgroup of $(0,\infty)$. If $t\neq 1$ lies in it, then so does the unbounded set $\{t^k:k\in\mathbb{Z}\}$. So $\Delta(G)=\{1\}$.

(b) We have $\mathrm{Ad}(\exp X)=e^{\operatorname{ad}X}$ (Lecture 2) and $\det e^A=e^{\operatorname{tr}A}$. This gives the formula. If all traces vanish, then $\Delta=1$ on $\exp\mathfrak{g}$. This set contains a neighbourhood U of e (Lecture 2). The subgroup H generated by U is open, since $hU\subset H$ for $h\in H$. It is closed, since its complement is a union of open cosets. By connectedness $H=G$, so $\Delta\equiv 1$. Conversely, $\Delta\equiv 1$ gives $e^{-t\operatorname{tr}\operatorname{ad}X}=1$ for all t, so $\operatorname{tr}\operatorname{ad}X=0$. For nilpotent $\mathfrak{g}$ each $\operatorname{ad}X$ is nilpotent. For semisimple $\mathfrak{g}$ we have $\mathfrak{g}=[\mathfrak{g},\mathfrak{g}]$ and $\operatorname{tr}\operatorname{ad}[X,Y]=\operatorname{tr}[\operatorname{ad}X,\operatorname{ad}Y]=0$. ∎

16.4 The affine group

The *$ax+b$ group* is

$$G=\left\{g(a,b)=\begin{pmatrix}a & b\\ 0 & 1\end{pmatrix}:\ a>0,\ b\in\mathbb{R}\right\}.$$

It acts on $\mathbb{R}$ by $x\mapsto ax+b$. The pair (a,b) is a global chart onto $(0,\infty)\times\mathbb{R}$. So integrals over G are ordinary double integrals. Matrix multiplication gives

$$g(a,b)\,g(a',b')=g(aa',\ ab'+b),\qquad g(a,b)^{-1}=g(a^{-1},-a^{-1}b).$$

A tangent vector at g is a matrix $\begin{pmatrix}\alpha & \beta\\ 0 & 0\end{pmatrix}$, which is $\alpha\partial_a+\beta\partial_b$ in the chart. The Lie algebra has basis

$$X=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix},\qquad Y=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix},\qquad [X,Y]=Y.$$

Translations are restrictions of linear maps on matrices. Hence $d(L_g)_eZ=gZ$ and $d(R_g)_eZ=Zg$. We fix the density μ_e on $\mathfrak{g}$ by $\mu_e(X,Y)=1$.

Left density. For $g=g(a,b)$,

$$gX=\begin{pmatrix}a & 0\\ 0 & 0\end{pmatrix}=a\,\partial_a,\qquad gY=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=a\,\partial_b.$$

So the matrix P from the smoothness step in the proof of Theorem 16.3 is

$$P=\begin{pmatrix}a & 0\\ 0 & a\end{pmatrix}.$$

Then $m_L=|\det P|^{-1}=a^{-2}$.

Right density. Similarly,

$$Xg=\begin{pmatrix}a & b\\ 0 & 0\end{pmatrix}=a\,\partial_a+b\,\partial_b,\qquad Yg=\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}=\partial_b,\qquad P=\begin{pmatrix}a & 0\\ b & 1\end{pmatrix}.$$

Then $m_R=a^{-1}$. In summary,

$$d_Lx=a^{-2}\,da\,db,\qquad d_Rx=a^{-1}\,da\,db.$$

Modular function. With $g^{-1}=\begin{pmatrix}a^{-1} & -a^{-1}b\\ 0 & 1\end{pmatrix}$ we compute

$$gXg^{-1}=\begin{pmatrix}1 & -b\\ 0 & 0\end{pmatrix}=X-bY,\qquad gYg^{-1}=\begin{pmatrix}0 & a\\ 0 & 0\end{pmatrix}=aY.$$

In the basis (X,Y),

$$\mathrm{Ad}(g)=\begin{pmatrix}1 & 0\\ -b & a\end{pmatrix},\qquad \det\mathrm{Ad}(g)=a,\qquad \Delta(g(a,b))=a^{-1}.$$

This matches the density relation in Theorem 16.6(ii): $\Delta^{-1}m_L=a\cdot a^{-2}=a^{-1}=m_R$. It also matches the trace formula in Corollary 16.8(b). Indeed $\operatorname{ad}X$ fixes Y and kills X, so $\operatorname{tr}\operatorname{ad}X=1$. And $\exp tX=g(e^t,0)$ has $\Delta=e^{-t}$. Hence the $ax+b$ group is not unimodular.

The connectedness hypothesis in Corollary 16.8(b) is needed. Let H be the subgroup of elements $g(a,b)$ with $a\in 2^{\mathbb{Z}}$ and $b\in\mathbb{R}$. It is closed, since $2^{\mathbb{Z}}$ is closed in $(0,\infty)$. Its Lie algebra $\mathbb{R}Y$ is abelian. Yet $\mathrm{Ad}(g)Y=gYg^{-1}=aY$ for $g=g(a,b)\in H$. So the modular function Δ_H of H satisfies $\Delta_H(g(a,b))=a^{-1}$, which is not identically 1.

Direct check of the right-translation identity in Theorem 16.6(i). Let $x=g(a',b')$ and $g=g(a,b)$. Then $xg=g(a'a,\ a'b+b')$. Substitute $u=a'a$, $v=a'b+b'$. Then $du\,dv=a\,da'\,db'$ and $a'^{-2}=a^2u^{-2}$. So

$$\int_G f(xg)\,d_Lx=\iint f(a'a,\ a'b+b')\,\frac{da'\,db'}{a'^2}=\iint f(u,v)\,\frac{a^2}{u^2}\,\frac{du\,dv}{a}=a\int_G f\,d_Lx.$$

This is $\Delta(g)^{-1}\int_G f\,d_Lx$.

Direct check of the second inversion identity in Theorem 16.6(iii). Here $x^{-1}=g(1/a,-b/a)$ and $\Delta(x^{-1})=a$. Substitute $u=1/a$, $v=-b/a$. The Jacobian determinant of $(a,b)\mapsto(u,v)$ is

$$\det\begin{pmatrix}-a^{-2} & 0\\ ba^{-2} & -a^{-1}\end{pmatrix}=a^{-3},\qquad\text{so}\qquad da\,db=u^{-3}\,du\,dv.$$

Since $a\cdot a^{-2}=a^{-1}=u$,

$$\int_G f(x^{-1})\Delta(x^{-1})\,d_Lx=\iint f\big(\tfrac{1}{a},-\tfrac{b}{a}\big)\,\frac{a\,da\,db}{a^2}=\iint f(u,v)\,u\,u^{-3}\,du\,dv=\int_G f\,d_Lx.$$

Translating a rectangle

In the $ax+b$ group the modular function can be computed from weighted areas, without differentiating a matrix. For a region E in the (a,b)-chart we write

$$\mu_L(E)=\iint_E a^{-2}\,da\,db$$

and call it the *left Haar measure* of E. Now let

$$E=\{(a,b):1\le a\le 2,\ 0\le b\le 1\}.$$

Its left Haar measure is

$$\mu_L(E)=\int_1^2\int_0^1 a^{-2}\,db\,da=\big[-a^{-1}\big]_1^2=\frac{1}{2}.$$

For $h=g(s,t)$, right translation sends (a,b) to $(as,b+at)$. Its Jacobian in (a,b) is s, while the new density coefficient is $(as)^{-2}$. Hence

$$\mu_L(Eh)=\int_E (as)^{-2}s\,da\,db=s^{-1}\mu_L(E)=\Delta(h)\mu_L(E).$$

The shear parameter t changes the shape of the region but not this scale factor. Left translation by h has Jacobian s^2 and density coefficient $(sa)^{-2}$, so it preserves $\mu_L(E)$.

An integrand $f(xh)$ involves a third region. The function $x\mapsto \mathbf{1}_E(xh)$ equals 1 exactly when $xh\in E$, so it is the indicator of Eh^{-1}. Figure 16.1 draws the three regions for $h=g(2,1)$. Here $hE=[2,4]\times[1,3]$, and Eh is the parallelogram with vertices $(2,1)$, $(4,2)$, $(4,3)$ and $(2,2)$. The region Eh^{-1} is the set of (a,b) with $\tfrac12\le a\le 1$ and $-a\le b\le 1-a$.

The weighted masses are

$$\mu_L(hE)=\tfrac12=\mu_L(E),\qquad \mu_L(Eh)=\tfrac14=\Delta(h)\mu_L(E),\qquad \mu_L(Eh^{-1})=1=\Delta(h)^{-1}\mu_L(E).$$

The inverse in the region Eh^{-1} explains why the integral identity in Theorem 16.6(i) has $\Delta(h)^{-1}$, whereas the region identity has $\Delta(h)$.

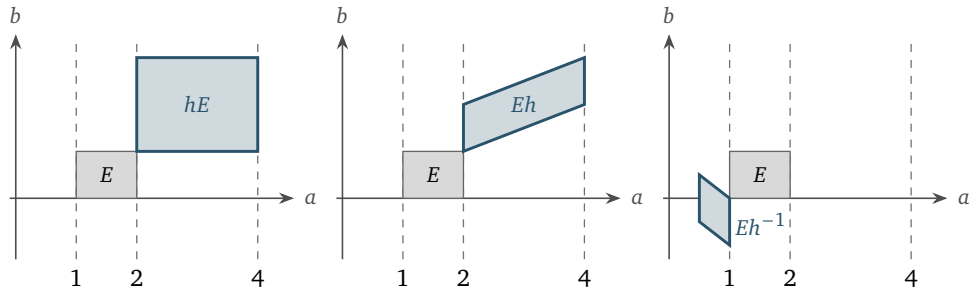


Figure 16.1: The rectangle E in the (a,b) half-plane, in grey, with the left translate hE, the right translate Eh and the region Eh^{-1} for $h = g(2,1)$. The measure of a region is the integral of $a^{-2}\,da\,db$, not its Euclidean area.

16.5 Regular representations

Definition 16.9. The space $L^2(G)$ is the Hilbert completion of $C_c(G)$ for the inner product

$$\langle f, h\rangle = \int_G \overline{f(x)}\, h(x)\, d_L x.$$

By the positivity statement (I1), this inner product is positive definite. For $f \in C_c(G)$ define

$$(\lambda(g)f)(x) = f(g^{-1}x), \qquad (\rho(g)f)(x) = \Delta(g)^{1/2} f(xg).$$

Lemma 16.10. *Let $f \in C_c(G)$ and $\varepsilon > 0$. There is a neighbourhood $V = V^{-1}$ of e with $|f(yx) - f(x)| < \varepsilon$ and $|f(xy) - f(x)| < \varepsilon$ for all $x \in G$ and $y \in V$.*

Proof. We treat $f(yx)$. The other case is symmetric. If U_L works for $f(yx)$ and U_R for $f(xy)$, put $V' = U_L \cap U_R$ and $V = V' \cap V'^{-1}$.

Fix a compact neighbourhood W of e. Choose a decreasing neighbourhood basis $V_1 \supset V_2 \supset \cdots$ of e inside W. Suppose no V_k works. Then there are $y_k \in V_k$ and x_k with $|f(y_k x_k) - f(x_k)| \geq \varepsilon$. One of x_k and $y_k x_k$ lies in $K = \operatorname{supp} f$. So x_k lies in the compact set $K \cup W^{-1}K$. Since G is second countable and Hausdorff, compact subsets are sequentially compact. So a subsequence has $x_k \to x$. As $y_k \to e$, also $y_k x_k \to x$. Continuity gives $f(y_k x_k) - f(x_k) \to 0$, which is a contradiction. ∎

Proposition 16.11. *The maps λ and ρ extend to strongly continuous unitary representations of G on $L^2(G)$, and $\lambda(g)\rho(h) = \rho(h)\lambda(g)$ for all $g, h \in G$.*

Proof. Both are homomorphisms on $C_c(G)$. For example,

$$\rho(g)\rho(h)f(x) = \Delta(g)^{1/2}\Delta(h)^{1/2} f(xgh) = \rho(gh)f(x).$$

Both operators preserve norms. By the left invariance in Theorem 16.3(c) and the right-translation identity in Theorem 16.6(i), applied to $|f|^2$,

$$\|\lambda(g)f\|^2 = \int_G |f(g^{-1}x)|^2\, d_L x = \|f\|^2, \qquad \|\rho(g)f\|^2 = \Delta(g)\int_G |f(xg)|^2\, d_L x = \|f\|^2.$$

Each operator is a bijection of $C_c(G)$, with inverse $\lambda(g^{-1})$ or $\rho(g^{-1})$. An isometric bijection of a dense subspace extends to an isometry with closed, dense range, hence to a unitary operator. The homomorphism property passes to the extensions by continuity. The operators commute, by the computation

$$\lambda(g)\rho(h)f(x) = \Delta(h)^{1/2} f(g^{-1}xh) = \rho(h)\lambda(g)f(x).$$

Strong continuity. Fix $f \in C_c(G)$ with support K, and a compact neighbourhood $W = W^{-1}$ of e. For $g \in W$, the functions $\lambda(g)f - f$ and $f(\cdot\, g) - f$ vanish outside $WK \cup KW$. A finite sum of chart bump functions gives $\varphi \in C_c(G)$ with $\varphi \geq 0$ and $\varphi \geq 1$ on $WK \cup KW$. Fix $\varepsilon > 0$. Take the neighbourhood supplied by the uniform-continuity statement of Lemma 16.10 and intersect it with W to obtain $V \subset W$. For $g \in V$,

$$\|\lambda(g)f - f\|^2 \leq \varepsilon^2 \int_G \varphi \, d_L x.$$

The function φ was fixed before ε, and $\varepsilon > 0$ is arbitrary. Hence $\lambda(g)f \to f$ as $g \to e$. For ρ use

$$\|\rho(g)f - f\| \leq \Delta(g)^{1/2} \, \|f(\cdot\, g) - f\| + |\Delta(g)^{1/2} - 1| \, \|f\|$$

and the continuity of Δ. The squared norm $\|f(\cdot\, g) - f\|^2$ obeys the same bound as $\|\lambda(g)f - f\|^2$, by the second estimate of Lemma 16.10. Hence $\rho(g)f \to f$ as $g \to e$.

Thus λ and ρ are continuous at e on the dense subspace $C_c(G)$. The dense-set criterion for strong continuity in Lecture 15 extends this to every vector of $L^2(G)$ and every point of G. ∎

Without the factor $\Delta(g)^{1/2}$, right translation multiplies squared norms by $\Delta(g)^{-1}$. On the $ax + b$ group the factor $\Delta(g)^{-1}$ equals a for $g = g(a, b)$. Exercise 16.5 shows that λ and ρ are unitarily equivalent.

We test this on the rectangle E of the $ax + b$ group, with $h = g(s, t)$ as in § 16.4. For $f \in C_c(G)$, the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, gives $\|f(\cdot\, h)\|_2^2 = \Delta(h)^{-1}\|f\|_2^2 = s\,\|f\|_2^2$.

The indicator $\mathbf{1}_E$ of E is not continuous, so it is not in $C_c(G)$. It defines a vector of $L^2(G)$, as follows. For $0 < \delta \leq \frac{1}{2}$ let $\varphi_\delta(a, b) = u_\delta(a)v_\delta(b)$ be a product of trapezoid functions. The function u_δ equals 1 on $[1, 2]$, vanishes outside $[1 - \delta, 2 + \delta]$, and is linear in between. The function v_δ is built in the same way on $[0, 1]$. Then $\varphi_\delta \in C_c(G)$ and $0 \leq \varphi_\delta \leq 1$. It equals $\mathbf{1}_E$ except on a region of Euclidean area $4\delta + 4\delta^2$, and $a^{-2} \leq 4$ on this region. Hence $\int_G |\mathbf{1}_E - \varphi_\delta|^2 \, d_L x \leq 4(4\delta + 4\delta^2) \to 0$. So the functions φ_δ form a Cauchy family in $L^2(G)$, whose limit is the vector $\mathbf{1}_E$, with $\|\mathbf{1}_E\|_2^2 = \mu_L(E)$. Applying the identity above to φ_δ and passing to the limit gives

$$\|\mathbf{1}_E(\,\cdot\, h)\|_2^2 = \mu_L(Eh^{-1}) = s\,\mu_L(E).$$

Multiplication by $\Delta(h)^{1/2} = s^{-1/2}$ restores the norm in both cases.

16.6 Locally compact groups

A topological group is *locally compact* if e has a compact neighbourhood. We cite the following result (Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2). Let G be a locally compact Hausdorff group. A linear functional I on $C_c(G)$ is *positive* if $I(f) \geq 0$ whenever $f \geq 0$. There is a nonzero positive linear functional I on $C_c(G)$ with $I(f(g\,\cdot)) = I(f)$ for all $f \in C_c(G)$ and $g \in G$. It is unique up to a positive factor. There is a unique continuous homomorphism $\Delta\colon G \to (0,\infty)$ satisfying $I(f(\cdot\, g^{-1})) = \Delta(g) I(f)$.

A general locally compact group has no normalisation at e that compares left and right integrals. The cited result instead contains two identities. They replace the density relation in Theorem 16.6(ii) and the inversion identities in part (iii). First, the functional $f \mapsto I(f\Delta^{-1})$ is right-invariant. Second, I takes the same value on $f \in C_c(G)$ and on the function $x \mapsto f(x^{-1})\Delta(x^{-1})$.

The Riesz representation theorem realises I as a Radon measure. We cite Folland, *Real Analysis*, second edition, Theorem 7.2 in §7.1.

For a Lie group, uniqueness makes I a multiple of $f \mapsto \int_G f\, d_L x$. So the two modular functions agree. Since $C_c(G)$ is dense in L^2 of a Radon measure (Folland, *Real Analysis*, Proposition 7.9 in §7.2), our $L^2(G)$ is the usual one, in the sense made precise below.

Integration rules for Part II

We now collect the measure theory used in the rest of Part II. We cite it from Folland, *Real Analysis*, second edition, Chapters 2, 6 and 7, and we do not prove it. Let X be a Lie group, a quotient G/H of a Lie group by a closed subgroup, or an open subset of $\mathbb{R}^n$. Such a space is locally compact and second countable, and it is a countable union of compact sets. We state the rules only for these spaces. General locally compact groups need further hypotheses in the product theorems.

A *Borel set* of X is a member of the σ-algebra generated by the open sets. A function $f\colon X \to \mathbb{C}$ is *measurable* if the preimage of every open set is a Borel set. A *positive Radon measure* μ on X assigns a value in $[0,\infty]$ to each Borel set. It is countably additive and finite on compact sets. On our spaces $\mu(E)$ is the infimum of $\mu(U)$ over open $U \supset E$, and the supremum of $\mu(K)$ over compact $K \subset E$. A *null set* is a Borel set of measure 0, or a subset of one. A property holds *almost everywhere* if it fails only on a null set.

The word "positive" says only that the values are nonnegative. The measure μ has *full support* if $\mu(U) > 0$ for every nonempty open set U. Equivalently, $\int_X \varphi\, d\mu > 0$ for every nonzero $\varphi \in C_c(X)$ with $\varphi \geq 0$. For example, the point mass at e is a positive Radon measure without full support. Haar measures have full support by the positivity statement (I1), and so do the quotient measures of Lecture 17. Lecture 25 also uses positive measures without full support.

Suppose that μ has full support. Then $\int_X \bar{\varphi}\psi\, d\mu$ is an inner product on $C_c(X)$, and its Hilbert completion is the usual space $L^2(X,\mu)$. Its vectors are the measurable f with $\int_X |f|^2\, d\mu < \infty$, and two functions define one vector when they agree almost everywhere. On $\mathbb{R}$, the elements of the completion $L^2(\mathbb{R})$ of Lecture 15 are therefore classes of functions of this kind.

A measurable f is *integrable* if $\int_X |f|\,d\mu < \infty$. Let μ and ν be positive Radon measures on such spaces X and Y. We use the following rules.

- *Tonelli.* If $F\colon X \times Y \to [0,\infty]$ is measurable, then the two iterated integrals of F are equal. Their common value may be infinite.
- *Fubini.* If $F\colon X \times Y \to \mathbb{C}$ is measurable and one iterated integral of $|F|$ is finite, then the two iterated integrals of F exist, are finite and are equal. We use Tonelli to check the hypothesis.
- *Dominated convergence.* Let f_j be measurable with $f_j \to f$ almost everywhere. Suppose there is one integrable function g with $|f_j| \le g$ almost everywhere for all j. Then f is integrable and $\int_X f_j\,d\mu \to \int_X f\,d\mu$.
- *Differentiation under the integral.* Let $I \subset \mathbb{R}$ be an open interval. Let $F(s,\cdot)$ be integrable for each $s \in I$, and let $s \mapsto F(s,x)$ be differentiable for each $x \in X$. Suppose there is one integrable function g with $|\partial_s F(s,x)| \le g(x)$ for all $s \in I$ and $x \in X$. Then $\frac{d}{ds}\int_X F(s,x)\,d\mu(x) = \int_X \partial_s F(s,x)\,d\mu(x)$. Indeed, the mean value inequality bounds the difference quotients by g, and dominated convergence applies to them.
- *Change of variables.* Let $\Phi\colon U \to V$ be a C^1 diffeomorphism of open subsets of $\mathbb{R}^n$, and let $f\colon V \to \mathbb{C}$ be integrable, or measurable with values in $[0,\infty]$. Then $\int_V f(y)\,dy = \int_U f(\Phi(x))\,|\det D\Phi(x)|\,dx$.

The common bound in dominated convergence cannot be dropped. On $\mathbb{R}$ the functions $j\,\mathbf{1}_{(0,1/j)}$ tend to 0 everywhere, but each has integral 1. For the differentiation rule, take $F(s,t) = e^{-t^2+ist}$ on $\mathbb{R}$. Then $|\partial_s F(s,t)| = |t|e^{-t^2}$, which is integrable and does not depend on s. So $\int_{\mathbb{R}} e^{-t^2+ist}\,dt$ may be differentiated under the integral.

For Tonelli and Fubini, here is the bound that Lecture 18 uses. Let $f, h \in L^2(\mathbb{R})$ and $\lambda \neq 0$. Tonelli lets us integrate over t first. Cauchy–Schwarz and translation invariance bound the inner integral by $\|h\|\,\|f\|$. The two Gaussian integrals are $2\sqrt{\pi}$ and $2\sqrt{\pi}/|\lambda|$. Hence

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\,|f(t+a)|\,dt\,da\,db \le \frac{4\pi}{|\lambda|}\,\|h\|\,\|f\|.$$

This finite bound is the hypothesis of Fubini for the same integrand without absolute values.

Self-checks

1. In the $ax+b$ group let $g = g(2,5)$. Suppose $\int_G f\,d_L x = 7$. Find $\Delta(g)$ and $\int_G f(xg)\,d_L x$.

   Answer. $\Delta(g) = 1/2$. By the right-translation identity in Theorem 16.6(i), the integral is $\Delta(g)^{-1} \cdot 7 = 14$.

2. Suppose some left-invariant smooth positive density on G is also right-invariant. Must G be unimodular?

   Answer. Yes. By the uniqueness in Theorem 16.3(a), the density is $c\mu_L$ for some $c > 0$, so $R_g^*\mu_L = \mu_L$ for all $g \in G$. By the right-translation identity in Theorem 16.6(i), this means $\Delta(g) = 1$ for all $g \in G$.

3. On the $ax+b$ group, by what factor does $f \mapsto f(\cdot\, g(3,1))$ multiply $\|f\|^2$ in $L^2(G)$?

   Answer. By the right-translation identity in Theorem 16.6(i), applied to $|f|^2$, the factor is $\Delta(g(3,1))^{-1} = 3$. Hence the map is not unitary without the factor $\Delta^{1/2}$.

Summary

Translating a fixed positive density on $\mathfrak{g}$ gives the left and right Haar densities. Each is unique up to a positive factor. On a noncompact Lie group the total mass is infinite, so no normalised finite Haar integral exists. Right translation scales μ_L by $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. When μ_L and μ_R have the same value at e, we have $\mu_R = \Delta^{-1}\mu_L$, and inversion exchanges μ_L and μ_R. For the $ax + b$ group, $d_L x = a^{-2}\,da\,db$, $d_R x = a^{-1}\,da\,db$ and $\Delta = a^{-1}$. The factor $\Delta^{1/2}$ makes right translation unitary on $L^2(G)$. Haar measures and quotient measures are positive Radon measures with full support, and the cited integration rules of § 16.6 hold in the second-countable spaces of Part II.

The main groups of Part II are unimodular: $\mathbb{R}^n$, the Heisenberg group, $\mathrm{SL}(2,\mathbb{R})$, and the Lorentz and Poincaré groups. Their closed subgroups often are not. For example, the upper triangular subgroup of $\mathrm{SL}(2,\mathbb{R})$ with positive diagonal is an $ax + b$ group (Exercise 20.6). Lecture 17 compares Δ_G with Δ_H to build unitary representations.

Reading

- Knapp, *Lie Groups Beyond an Introduction*, Chapter VIII “Integration”, sections “Differential Forms and Measure Zero” and “Haar Measure for Lie Groups”. Compare its normalisation of the modular function with ours.
- Kirillov, *Elements of the Theory of Representations*, §9, from the course reference route. Check its modular-function convention.
- Folland, *A Course in Abstract Harmonic Analysis*, Chapter 2, sections “Haar Measure” and “The Modular Function”, for general locally compact groups.

Lecture 17

Unitary induction

Objectives. *Lecture 12 constructed induced representations of compact groups using an invariant measure on K/H. For a noncompact group G, the quotient G/H may carry no invariant measure. We replace it by a* quasi-invariant *measure, which translation by an element of G multiplies by a positive continuous function. We then correct the action by a square-root factor. The model is the $ax+b$ group, with representations induced from characters of its translation subgroup.*

By the end of this lecture you will be able to:

- State the quotient-measure theorem and derive the quasi-invariance factor and its cocycle identity.
- Define the normalised induced representation and prove that it is unitary.
- Compute the $ax+b$ representations induced from translation characters in two measure pictures.
- Explain why the unitary equivalence class of the induced representation depends only on the scaling orbit of the frequency.

17.1 Equivariant functions

Throughout, G is a Lie group, H is a closed subgroup and $q(x) = xH$ is the quotient map $G \to G/H$. By Lecture 3, G/H is a manifold on which G acts by $g \cdot xH = gxH$. The map q is continuous and open. Let (σ, V) be a unitary representation of H on a Hilbert space V.

Definition 17.1. An *equivariant function of type σ* is a continuous map $f : G \to V$ with

$$f(xh) = \sigma(h)^{-1} f(x) \qquad (x \in G,\ h \in H). \tag{17.1}$$

Let $\mathcal{F}_c(\sigma)$ be the space of those f for which $q(\operatorname{supp} f)$ is compact.

This is the convention of Lecture 12. Let $s : U \to G$ be a continuous section of q over an open set $U \subseteq G/H$. Then $f(s(p)h) = \sigma(h)^{-1} f(s(p))$. So over U the function f is just the ordinary function $f \circ s : U \to V$. Continuous local sections exist near every point; they are part of the construction of G/H in Lecture 3. Geometrically f is a section of a vector bundle over G/H. We use only the function description.

Since $\sigma(h)$ is unitary, $n_f(xH) = \|f(x)\|^2$ is well defined. It is continuous on G/H because q is open, and its support is compact.

We check that $\mathcal{F}_c(\sigma)$ is not zero when $V \neq 0$. Averaging over H turns compactly supported continuous functions $G \to V$ into elements of $\mathcal{F}_c(\sigma)$. Fix a left Haar measure dh on H. For continuous compactly supported $F\colon G \to V$ define $P_\sigma F$ by

$$P_\sigma F(x) = \int_H \sigma(h) F(xh)\, dh.$$

The integral is defined weakly by the Riesz argument of Lecture 10, applied to a continuous Hilbert-space-valued function with compact support. For fixed x, the integrand vanishes outside $H \cap x^{-1} \operatorname{supp} F$.

Left invariance of dh under $h' = h_0 h$ gives

$$P_\sigma F(xh_0) = \int_H \sigma(h_0^{-1}h') F(xh')\, dh' = \sigma(h_0)^{-1} P_\sigma F(x).$$

For x in a compact neighbourhood, all the integrands have support in one compact subset of H. Uniform continuity there makes $P_\sigma F$ continuous. Also $q(\operatorname{supp} P_\sigma F) \subseteq q(\operatorname{supp} F)$, which is compact. Thus $P_\sigma F \in \mathcal{F}_c(\sigma)$.

This element can be nonzero. Let $v \in V$ be nonzero. By strong continuity of σ, we can choose $\beta \in C_c(G)$ with $\beta \geq 0$ and $\beta(e) > 0$, supported so near e that $\operatorname{Re}\langle v, \sigma(h)v\rangle > 0$ for $h \in H \cap \operatorname{supp}\beta$. For $F(x) = \beta(x)v$ the number $\langle v, P_\sigma F(e)\rangle = \int_H \beta(h)\langle v, \sigma(h)v\rangle\, dh$ has positive real part. So $\mathcal{F}_c(\sigma) \neq 0$.

17.2 Quotient measures

Rescale a function on the line: $u \mapsto u((t - b_0)/a_0)$ with $a_0 > 0$. This multiplies $\|u\|^2$ by a_0, because dt is not invariant under dilation. Dividing by $a_0^{1/2}$ repairs the norm. This lecture carries out that repair for any quotient G/H.

In this lecture a *measure* on a locally compact space X means a positive Radon measure with full support, in the terms of Lecture 16. Equivalently, it is a positive linear functional $\varphi \mapsto \int_X \varphi\, d\mu$ on $C_c(X)$ which is strictly positive on nonzero $\varphi \geq 0$. Haar measures have this property. Then $L^2(X, \mu)$ is the completion of $C_c(X)$ for $\langle \varphi, \psi\rangle = \int_X \bar{\varphi}\psi\, d\mu$.

Fix left Haar measures dx on G and dh on H. Modular functions follow Lecture 16:

$$\int_G F(xg)\, dx = \Delta_G(g)^{-1} \int_G F(x)\, dx,$$

and likewise for Δ_H. For $F \in C_c(G)$ define $F^H \in C_c(G/H)$ by

$$F^H(xH) = \int_H F(xh)\, dh. \tag{17.2}$$

It is well defined by left invariance of dh.

Lemma 17.2. *If a measure μ on G/H satisfies $\int_G F\, dx = \int_{G/H} F^H\, d\mu$ for all $F \in C_c(G)$, then $\Delta_G(h) = \Delta_H(h)$ for all $h \in H$.*

Proof. Fix $h_0 \in H$ and apply the identity to $F_{h_0}(x) = F(xh_0)$. The left side becomes $\Delta_G(h_0)^{-1}\int_G F\,dx$. On the right,

$$(F_{h_0})^H(xH) = \int_H F(xhh_0)\,dh = \Delta_H(h_0)^{-1}F^H(xH).$$

Hence $\Delta_G(h_0)^{-1}\int_G F\,dx = \Delta_H(h_0)^{-1}\int_G F\,dx$. Choose $F \geq 0$ nonzero, so that $\int_G F\,dx > 0$. Dividing both sides by $\int_G F\,dx$ gives $\Delta_G(h_0)^{-1} = \Delta_H(h_0)^{-1}$, so $\Delta_G(h_0) = \Delta_H(h_0)$. ∎

For the $ax+b$ group and its scaling subgroup A, we will see $\Delta_G(a,0) = 1/a$ while $\Delta_A = 1$. Hence, by the modular compatibility lemma, Lemma 17.2, G/A carries no such measure. Instead, we insert a weight function on G into the integral over G.

The weight must transform in a definite way. Suppose that a continuous function $\rho > 0$ and a measure μ satisfy $\int_G F\rho\,dx = \int_{G/H} F^H\,d\mu$ for all $F \in C_c(G)$. Apply this identity to $F_{h_0}(x) = F(xh_0)$. As in the proof of Lemma 17.2, the right side is multiplied by $\Delta_H(h_0)^{-1}$. On the left, the modular identity applied to $x \mapsto F(x)\rho(xh_0^{-1})$ gives

$$\int_G F(xh_0)\rho(x)\,dx = \Delta_G(h_0)^{-1}\int_G F(x)\rho(xh_0^{-1})\,dx.$$

Since F is arbitrary, $\Delta_G(h_0)^{-1}\rho(xh_0^{-1}) = \Delta_H(h_0)^{-1}\rho(x)$ for all $x \in G$. Replacing x by xh_0 gives the transformation law in part (a) of the next theorem.

Theorem 17.3 (Quotient measures; cited). *Let G be a Lie group and H a closed subgroup.*

(a) There is a continuous function $\rho\colon G \to (0,\infty)$ with

$$\rho(xh) = \frac{\Delta_H(h)}{\Delta_G(h)}\,\rho(x) \qquad (x \in G,\ h \in H). \tag{17.3}$$

(b) The map $F \mapsto F^H$ sends $C_c(G)$ onto $C_c(G/H)$. If $\varphi \geq 0$, a lift F with $F^H = \varphi$ can be chosen with $F \geq 0$.

(c) For each ρ as in (a) there is a unique measure μ_ρ on G/H with

$$\int_G F(x)\rho(x)\,dx = \int_{G/H} F^H\,d\mu_\rho \qquad (F \in C_c(G)). \tag{17.4}$$

For a general closed subgroup, see G. B. Folland, *A Course in Abstract Harmonic Analysis*, second edition (2016), Chapter 2, “Measures on Homogeneous Spaces”, cited below as Folland, Abstract Harmonic Analysis. Its modular convention agrees with ours. Uniqueness of μ_ρ follows from the onto property of the averaging map. A. A. Kirillov, *Elements of the Theory of Representations* (1976), Section 9.1, cited below as Kirillov, Elements, treats quasi-invariant measures and the modular criterion. Its conventions must be translated before use.

A continuous positive function ρ with the transformation law (17.3) is called a *rho-function*. If $\Delta_G|_H = \Delta_H$ we may take $\rho = 1$. In this lecture ρ denotes a rho-function, and λ below denotes its cocycle. They are not the regular representations of Lecture 16.

Proposition 17.4. *Let ρ be a rho-function and $\mu = \mu_\rho$. Define λ by $\lambda(g, xH) = \rho(gx)/\rho(x)$. Then λ is well defined. For $g, g_1, g_2 \in G$, $p \in G/H$ and $\varphi \in C_c(G/H)$,*

$$\lambda(g_1 g_2, p) = \lambda(g_1, g_2 p)\,\lambda(g_2, p), \qquad \lambda(e, p) = 1, \tag{17.5}$$

$$\int_{G/H} \varphi(g^{-1}p)\,d\mu(p) = \int_{G/H} \varphi(p)\,\lambda(g, p)\,d\mu(p). \tag{17.6}$$

Proof. By the transformation law (17.3), replacing x by xh multiplies $\rho(gx)$ and $\rho(x)$ by the same factor. So λ is well defined. The function $(g, x) \mapsto \rho(gx)/\rho(x)$ is continuous and constant on fibres of $\mathrm{id} \times q$. Since q is open, $\mathrm{id} \times q$ is an open surjection, hence a quotient map. Thus λ is jointly continuous in (g, p). For $p = xH$,

$$\lambda(g_1 g_2, p) = \frac{\rho(g_1 g_2 x)}{\rho(g_2 x)} \cdot \frac{\rho(g_2 x)}{\rho(x)} = \lambda(g_1, g_2 p)\,\lambda(g_2, p).$$

Also $\lambda(e, p) = \rho(x)/\rho(x) = 1$.

For the quasi-invariance identity (17.6), use the onto property of averaging, Theorem 17.3(b), to write $\varphi = F^H$ with $F \in C_c(G)$. The function $x \mapsto F(g^{-1}x)$ has average $\varphi(g^{-1}\,\cdot)$. By the weighted quotient integral (17.4) and left invariance of dx, with $y = g^{-1}x$,

$$\int \varphi(g^{-1}p)\,d\mu(p) = \int_G F(g^{-1}x)\rho(x)\,dx = \int_G F(y)\rho(gy)\,dy = \int_G F(y)\,\lambda(g, yH)\,\rho(y)\,dy.$$

Since $\lambda(g, \cdot)$ is constant on cosets, $y \mapsto F(y)\lambda(g, yH)$ has average $\varphi\,\lambda(g, \cdot)$. Applying the weighted quotient integral (17.4) again gives the right side of (17.6). ∎

We say μ is *quasi-invariant*: translation by g changes it by the positive continuous factor $\lambda(g, \cdot)$. If $\rho = 1$, then $\lambda = 1$ and μ is G-invariant.

17.3 The normalised induced representation

The substitution $p = gq$ in the quasi-invariance identity (17.6) gives $d\mu(gq) = \lambda(g, q)\,d\mu(q)$. So $\lambda(g, \cdot)$ is the Jacobian of $q \mapsto gq$ relative to μ, and the cocycle identity (17.5) is the chain rule. As an analogy, think of $\|f\|^2\,d\mu$ as a probability density. Then an amplitude carries the square root of the Jacobian.

Fix a rho-function ρ, and let μ and λ be as in the quasi-invariance proposition, Proposition 17.4. For $f_1, f_2 \in \mathcal{F}_c(\sigma)$ put

$$\langle f_1, f_2 \rangle = \int_{G/H} \langle f_1(x), f_2(x) \rangle\,d\mu(xH), \qquad \|f\|^2 = \int_{G/H} n_f\,d\mu.$$

The integrand is well defined because σ is unitary. If $\|f\| = 0$, then $n_f = 0$ by strict positivity, so $f = 0$.

Theorem 17.5. *For $f \in \mathcal{F}_c(\sigma)$ and $g \in G$ define*

$$(\Pi(g)f)(x) = \left(\frac{\rho(g^{-1}x)}{\rho(x)}\right)^{1/2} f(g^{-1}x) = \lambda(g^{-1}, xH)^{1/2} f(g^{-1}x). \tag{17.7}$$

Then (a) $\Pi(g)f \in \mathcal{F}_c(\sigma)$; (b) $\Pi(e) = I$ *and* $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$; (c) $\|\Pi(g)f\| = \|f\|$.

Proof. (a) The factor depends only on xH. So replacing x by xh only changes $f(g^{-1}x)$ into $\sigma(h)^{-1}f(g^{-1}x)$. Also $q(\operatorname{supp}\Pi(g)f) = g\cdot q(\operatorname{supp} f)$ is compact.

(b) Let $p = xH$. Applying the normalised action (17.7) twice,

$$(\Pi(g_1)\Pi(g_2)f)(x) = \lambda(g_1^{-1},p)^{1/2}\,\lambda(g_2^{-1},g_1^{-1}p)^{1/2}\,f(g_2^{-1}g_1^{-1}x).$$

The cocycle identity for the pair g_2^{-1}, g_1^{-1} turns the product of factors into $\lambda((g_1g_2)^{-1},p)^{1/2}$. Also $\lambda(e,p) = 1$ gives $\Pi(e) = I$.

(c) Translation by g changes μ by the factor $\lambda(g,\cdot)$. The square-root factor in $\Pi(g)$ is chosen to cancel this change. We have $n_{\Pi(g)f}(p) = \lambda(g^{-1},p)\,n_f(g^{-1}p)$. Put $\varphi(p) = n_f(p)\,\lambda(g^{-1},gp)$, so that $\varphi(g^{-1}p) = n_{\Pi(g)f}(p)$. Then the quasi-invariance identity (17.6) gives

$$\|\Pi(g)f\|^2 = \int \varphi(g^{-1}p)\,d\mu = \int n_f(p)\,\lambda(g^{-1},gp)\,\lambda(g,p)\,d\mu(p) = \int n_f\,d\mu.$$

The last step is the cocycle identity (17.5) with $g_1 = g^{-1}$ and $g_2 = g$. ∎

By the representation law and the isometry property in the normalised-induction theorem, Theorem 17.5, each $\Pi(g)$ is an isometry of $\mathcal{F}_c(\sigma)$ with inverse $\Pi(g^{-1})$. So $\Pi(g)$ extends to a unitary operator on the completion. The representation law persists by continuity.

Definition 17.6. The *normalised induced representation* $\operatorname{Ind}_H^G\sigma$ is the completion of $\mathcal{F}_c(\sigma)$, for the inner product defined by the chosen rho-function ρ, together with these unitary operators $\Pi(g)$.

Proposition 17.7. *The induced representation* $\operatorname{Ind}_H^G\sigma$ *is strongly continuous.*

Proof. First take $f \in \mathcal{F}_c(\sigma)$ and let $C = q(\operatorname{supp} f)$, a compact set. Choose a compact neighbourhood B of e in G. For $g \in B$, the quotient supports of both $\Pi(g)f$ and f lie in the compact set

$$S = B\cdot C \subset G/H.$$

It suffices to show that their difference tends uniformly to zero in fibre norm on S. This norm is independent of the representative in G because σ is unitary.

By compactness of S and the manifold structure of G/H, we can cover S by finitely many compact subsets S_i. Each S_i lies in an open set with a continuous local section s_i of q. The lifted sets $s_i(S_i)$ are compact. At these representatives,

$$(\Pi(g)f - f)(s_i(p)) = \lambda(g^{-1},p)^{1/2}f(g^{-1}s_i(p)) - f(s_i(p)).$$

The function $(g,p)\mapsto\lambda(g^{-1},p)$ is jointly continuous and equals one at $g = e$. Also $(g,p)\mapsto f(g^{-1}s_i(p))$ is jointly continuous as a Hilbert-space-valued function. On the compact sets $B\times S_i$, this continuity is uniform. Thus the displayed difference tends uniformly to zero for $p\in S_i$ as $g\to e$. There are only finitely many S_i, so the same holds on S.

Choose $\varphi\in C_c(G/H)$ with $\varphi\ge 0$ and $\varphi\ge 1$ on S. If the uniform bound is ε, then

$$\|\Pi(g)f - f\|^2 \le \varepsilon^2\int_{G/H}\varphi\,d\mu.$$

This proves that $\Pi(g)f\to f$ in the norm of the completion as $g\to e$, for each $f\in\mathcal{F}_c(\sigma)$.

For a vector v in the completion, choose f with $\|v - f\| < \varepsilon$. Unitarity gives

$$\|\Pi(g)v - v\| \le 2\varepsilon + \|\Pi(g)f - f\|.$$

Hence continuity holds at e for every v. By the representation law and unitarity, continuity at e implies continuity at every g_0. ∎

Remark 17.8. Without the square root, the map $f \mapsto f(g^{-1}\cdot)$ still preserves $\mathcal{F}_c(\sigma)$ and satisfies the representation law. By the quasi-invariance identity (17.6), this map sends $\|f\|^2$ to $\int n_f\,\lambda(g,\cdot)\,d\mu$. So in general it is not unitary; in the $ax+b$ example it multiplies $\|\psi\|^2$ by a_0 (Proposition 17.9(a)). For compact G and finite-dimensional σ, we take $\rho = 1$ and recover the construction of Lecture 12. Different rho-functions give unitarily equivalent representations (Exercise 17.3). Finite-group Frobenius reciprocity does not transfer verbatim. For example, the representation Π_0 of § 17.5 is induced from the trivial character of the translation subgroup N. A character of the $ax+b$ group that is trivial on N would have a nonzero N-map to that trivial character, yet no such character occurs in Π_0. We do not use reciprocity.

17.4 Representations of the $ax + b$ group induced from translations

Let $G = \{(a, b) : a > 0,\ b \in \mathbb{R}\}$ with

$$(a,b)(a',b') = (aa', b + ab'), \qquad (a,b)^{-1} = (a^{-1}, -a^{-1}b), \qquad (a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}.$$

It acts on $\mathbb{R}$ by $t \mapsto at + b$. By Lecture 16, $dx = a^{-2}\,da\,db$ is a left Haar measure and $\Delta_G(a,b) = 1/a$. Let $N = \{n_c = (1,c)\}$ be the translations and $A = \{(a,0)\}$ the scalings. Then $N \cong \mathbb{R}$ has Haar measure dc and $\Delta_N = 1$. For a *frequency* $\xi \in \mathbb{R}$, the map $\chi_\xi(n_c) = e^{i\xi c}$ is a unitary character of N.

The quotient G/N carries an invariant measure, as we see below. We treat it first because its points carry the orbit of the frequency. The affine line G/A of the opening preview has no invariant measure, and it comes at the end of § 17.5.

17.4.1 The quotient and two measures

Since $(a,b)n_c = (a, b + ac)$, the coset $(a,b)N$ is $\{a\} \times \mathbb{R}$. So $(a,b)N \mapsto a$ identifies G/N with $(0,\infty)$, and $s(a) = (a,0)$ is a global section. The action is $(a_0,b_0)\cdot a = a_0 a$.

Since $\Delta_G(n_c) = 1 = \Delta_N(n_c)$, the transformation law (17.3) says $\rho(xn) = \rho(x)$. Thus rho-functions are positive continuous functions of a alone. The substitution $b = ac$ gives

$$F^N(a) = \int_{\mathbb{R}} F(a, ac)\,dc = \frac{1}{a}\int_{\mathbb{R}} F(a,b)\,db.$$

Hence for $\rho = \rho(a)$,

$$\int_G F\rho\,dx = \int_0^\infty \Big(\int_{\mathbb{R}} F(a,b)\,db\Big)\frac{\rho(a)\,da}{a^2} = \int_0^\infty F^N(a)\,\rho(a)\,\frac{da}{a}.$$

By the uniqueness in the quotient-measure theorem, Theorem 17.3, $d\mu_\rho = \rho(a)\,da/a$.

We use two choices. The rho-function $\rho_1 = 1$ gives the invariant measure, and the rho-function $\rho_{\mathrm{Leb}}(a,b) = a$ gives Lebesgue measure:

$$d\mu_{\rho_1} = \frac{da}{a}, \qquad d\mu_{\rho_{\mathrm{Leb}}} = da.$$

For the Lebesgue choice, take $g = (a_0, b_0)$ and $x = (a, 0)$. Then $gx = (a_0 a, b_0)$, so $\lambda_{\mathrm{Leb}}(g, a) = a_0$. In this case the quasi-invariance identity (17.6) is the substitution rule $\int_0^\infty \varphi(a/a_0)\, da = a_0 \int_0^\infty \varphi(u)\, du$. The cocycle identity reads $a_1 a_2 = a_1 \cdot a_2$, because $g_1 g_2$ has first entry $a_1 a_2$.

17.4.2 Equivariant functions and the action

For $\sigma = \chi_\xi$, the equivariance condition (17.1) says $f(a, b + ac) = e^{-i\xi c} f(a, b)$. Take $b = 0$ and then $c = b/a$:

$$f(a,b) = e^{-i\xi b/a}\varphi(a), \qquad \varphi = f \circ s. \tag{17.8}$$

Conversely, for $\varphi \in C_c((0,\infty))$ this f lies in $\mathcal{F}_c(\chi_\xi)$, since $e^{-i\xi(b+ac)/a} = e^{-i\xi c} e^{-i\xi b/a}$. So $f \mapsto \varphi$ is a bijection onto $C_c((0,\infty))$. Moreover $n_f = |\varphi|^2$ and $\|f\|^2 = \int_0^\infty |\varphi|^2 \rho(a)\, da/a$.

Figure 17.1 draws such a function. Left translation by n_c sends (a, b) to $(a, b + c)$. So on the coset over a, the function f oscillates in b with frequency ξ/a. In the coset parameter $c = b/a$ the frequency is ξ.

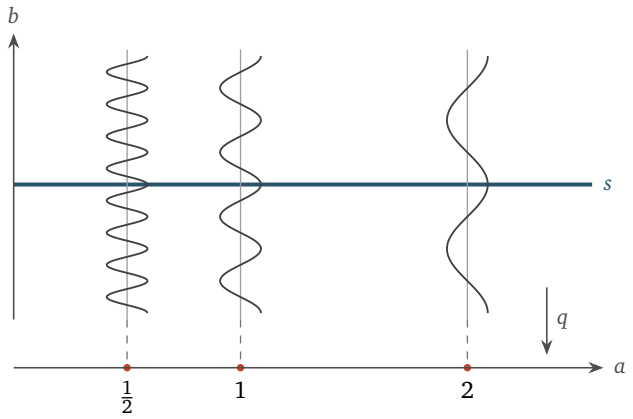


Figure 17.1: The real part of an equivariant function $f(a,b) = e^{-i\xi b/a}\varphi(a)$ with $\xi = 2\pi$, drawn as a horizontal offset on the three cosets of N over $a = \frac{1}{2}, 1, 2$. The heavy line is the section s, and q sends each vertical coset to its point of the base $(0,\infty)$.

Let $g = (a_0, b_0)$. Then $g^{-1}(a, 0) = (a/a_0,\ -b_0/a_0)$, and the equivariant form (17.8) gives

$$f(g^{-1}(a,0)) = \exp\Big(-i\xi\,\frac{-b_0/a_0}{a/a_0}\Big)\varphi\Big(\frac{a}{a_0}\Big) = e^{i\xi b_0/a}\,\varphi\Big(\frac{a}{a_0}\Big).$$

The factor at $x = (a, 0)$ is $(\rho(a/a_0)/\rho(a))^{1/2}$. This yields two pictures:

$$(\Pi_\xi(a_0,b_0)\varphi)(a) = e^{i\xi b_0/a}\,\varphi(a/a_0) \qquad \text{on } L^2((0,\infty), da/a), \tag{17.9}$$

$$(\Pi^{\mathrm{Leb}}_\xi(a_0,b_0)\psi)(a) = a_0^{-1/2} e^{i\xi b_0/a}\,\psi(a/a_0) \qquad \text{on } L^2((0,\infty), da). \tag{17.10}$$

The normalised-induction theorem, Theorem 17.5, already gives unitarity. We now verify unitarity and the representation law directly in this model.

Proposition 17.9. *Let $g = (a_0, b_0)$, $g_j = (a_j, b_j)$ and $\varphi, \psi \in C_c((0,\infty))$.*

(a) $\|\Pi^{\mathrm{Leb}}_\xi(g)\psi\| = \|\psi\|$, *while* $\psi \mapsto e^{i\xi b_0/a}\psi(a/a_0)$ *multiplies* $\|\psi\|^2$ *by* a_0.

(b) $\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2) = \Pi^{\mathrm{Leb}}_\xi(g_1 g_2)$.

(c) $(W\varphi)(a) = a^{-1/2}\varphi(a)$ *extends to a unitary map* $L^2(da/a) \to L^2(da)$ *with* $W\Pi_\xi(g) = \Pi^{\mathrm{Leb}}_\xi(g)W$.

Proof. (a) Substitute $a = a_0 u$:

$$\int_0^\infty a_0^{-1}|\psi(a/a_0)|^2\,da = \int_0^\infty |\psi(u)|^2\,du.$$

Without the factor a_0^{-1} the result is $a_0\|\psi\|^2$.

(b) Since $g_1 g_2 = (a_1 a_2,\ b_1 + a_1 b_2)$,

$$\begin{aligned}(\Pi^{\mathrm{Leb}}_\xi(g_1)\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a) &= a_1^{-1/2}e^{i\xi b_1/a}\,(\Pi^{\mathrm{Leb}}_\xi(g_2)\psi)(a/a_1)\\ &= a_1^{-1/2}a_2^{-1/2}\,e^{i\xi b_1/a}e^{i\xi a_1 b_2/a}\,\psi\Big(\frac{a}{a_1 a_2}\Big)\\ &= (a_1a_2)^{-1/2}e^{i\xi(b_1+a_1b_2)/a}\,\psi\Big(\frac{a}{a_1a_2}\Big) = (\Pi^{\mathrm{Leb}}_\xi(g_1g_2)\psi)(a).\end{aligned}$$

The square roots combine because λ_{Leb} is multiplicative; this is the cocycle identity. The phases combine into the second entry of $g_1 g_2$; this is where equivariance enters.

(c) We have $\int |a^{-1/2}\varphi|^2\,da = \int |\varphi|^2\,da/a$, and W has inverse multiplication by $a^{1/2}$. Finally,

$$(\Pi^{\mathrm{Leb}}_\xi(g)W\varphi)(a) = a_0^{-1/2}e^{i\xi b_0/a}\Big(\frac{a}{a_0}\Big)^{-1/2}\varphi\Big(\frac{a}{a_0}\Big) = a^{-1/2}e^{i\xi b_0/a}\varphi\Big(\frac{a}{a_0}\Big) = (W\Pi_\xi(g)\varphi)(a). \qquad \blacksquare$$

Note that W is multiplication by $(\rho_1/\rho_{\mathrm{Leb}})^{1/2}$.

17.5 The scaling orbit of a frequency

On translations, the first picture (17.9) gives $(\Pi_\xi(n_c)\varphi)(a) = e^{i(\xi/a)c}\varphi(a)$. So at the point $a \in G/N$ the group N acts by the character $\chi_{\xi/a}$. Let A act on characters of N by $(t\cdot\chi)(n) = \chi(t^{-1}nt)$. For $t = (a, 0)$,

$$t^{-1}n_c t = (a^{-1}, a^{-1}c)(a,0) = (1, c/a), \qquad \text{so} \qquad t\cdot\chi_\xi = \chi_{\xi/a}.$$

Thus the character at aN is $s(a)\cdot\chi_\xi$. As a runs through $(0,\infty)$, the frequency ξ/a fills the orbit $\mathbb{R}_{>0}\,\xi$. There are three orbits: $(0,\infty)$, $(-\infty, 0)$ and $\{0\}$.

Proposition 17.10. *Let $\kappa > 0$. The operator $(U\varphi)(a) = \varphi(a/\kappa)$ is unitary on $L^2((0,\infty), da/a)$, and $U\Pi_\xi(g) = \Pi_{\kappa\xi}(g)U$ for all $g \in G$.*

Proof. The substitution $a = \kappa u$ preserves da/a, so U is isometric on $C_c((0,\infty))$. Its inverse is $\varphi \mapsto \varphi(\kappa\,\cdot)$. For $g = (a_0, b_0)$,

$$(U\Pi_\xi(g)\varphi)(a) = (\Pi_\xi(g)\varphi)(a/\kappa) = e^{i\xi b_0\kappa/a}\,\varphi\Big(\frac{a}{\kappa a_0}\Big),$$
$$(\Pi_{\kappa\xi}(g)U\varphi)(a) = e^{i\kappa\xi b_0/a}\,(U\varphi)(a/a_0) = e^{i\kappa\xi b_0/a}\,\varphi\Big(\frac{a}{a_0\kappa}\Big).$$

The two expressions agree on the dense subspace $C_c((0,\infty))$ of $L^2((0,\infty), da/a)$. Both sides are bounded operators, so they agree on the whole space. ∎

Thus, by Proposition 17.10, the unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. We write $\Pi_+ = \Pi_1$ and $\Pi_- = \Pi_{-1}$. By Exercise 17.5, the three representations Π_+, Π_- and Π_0 are pairwise inequivalent.

In Π_0 we have $(\Pi_0(a_0, b_0)\varphi)(a) = \varphi(a/a_0)$. So N acts trivially. The representation Π_0 is the left regular representation of $(0,\infty) \cong G/N$, composed with the quotient map $G \to G/N$.

We do not prove irreducibility here. Mackey's theorem, cited in Lecture 25, applies with $n = 1$ and $H = (0,\infty)$. It shows that Π_+ and Π_- are irreducible. It also shows that, together with the characters $(a, b) \mapsto a^{is}$ with $s \in \mathbb{R}$, they are all the irreducible unitary representations of G, up to unitary equivalence. The representation Π_0 is not irreducible. By Exercise 17.4(a) it is the translation representation on $L^2(\mathbb{R})$, composed with $(a_0, b_0) \mapsto \log a_0$. By the proposition on finite-dimensional invariant subspaces in Lecture 15, it has no invariant line. An irreducible unitary representation of the abelian group $(0,\infty)$ is one-dimensional, by the Schur lemma that Lecture 18 cites. So Π_0 is reducible.

In physics this is the pattern of Wigner's classification. Translations act by phases, the remaining group moves the frequency along an orbit, and the representation lives on that orbit. Lecture 25 carries this out for the Poincaré group. Nothing in this lecture depends on it.

A quotient with no invariant Radon measure

For comparison, give the scaling subgroup $A = \{(a, 0) : a > 0\}$ the Haar measure da/a and induce its trivial character. Identify G/A with the line by $(a, b)A \mapsto b$. Left multiplication gives the familiar affine action

$$(a_0, b_0) \cdot t = b_0 + a_0 t.$$

Lebesgue measure dt is quasi-invariant, but a dilation multiplies it by a_0. Any invariant Radon measure would be translation invariant. By Haar uniqueness on $\mathbb{R}$, it would be a constant multiple of dt. Dilation invariance then forces this constant to be zero.

Here $\Delta_A = 1$ and $\Delta_G(a, 0) = 1/a$, so a rho-function is $\rho(a, b) = a$. The quotient identity follows directly:

$$\int_G F(a,b)\rho(a,b)\,\frac{da\,db}{a^2} = \int_{\mathbb{R}} \left(\int_0^\infty F(a,t)\,\frac{da}{a} \right) dt.$$

Since $(1, t)(a, 0) = (a, t)$ and A carries the Haar measure da/a, the inner integral is the coset average $F^A((1,t)A)$ of (17.2) with $H = A$. By the uniqueness in the quotient-measure theorem, Theorem 17.3, dt is exactly the measure μ_ρ supplied by this rho-function.

The square-root factor in the normalised action (17.7) becomes $a_0^{-1/2}$, and the induced action is

$$(\Pi(a_0, b_0)u)(t) = a_0^{-1/2} u\left(\frac{t - b_0}{a_0}\right).$$

To verify its norm, set $r = (t - b_0)/a_0$. Then $dt = a_0\, dr$, so

$$\|\Pi(a_0, b_0)u\|_2^2 = a_0^{-1} \int_{\mathbb{R}} |u((t - b_0)/a_0)|^2 dt = \int_{\mathbb{R}} |u(r)|^2 dr.$$

Without the square root, the squared norm would be multiplied by a_0. This quotient makes the measure correction visible in one variable. In the earlier translation-subgroup model, the quotient was the positive scaling line and an invariant quotient measure was available. The subgroup, not just the ambient group, controls the answer. Exercise 17.6 shows that this representation is unitarily equivalent to $\Pi_1 \oplus \Pi_{-1}$.

Self-checks

1. In the $ax + b$ group, derive the transformation law of a rho-function for $H = A$ and check that $\rho(a, b) = a$ satisfies it. Is $\rho = 1$ possible?
   Answer. For $h = (t, 0)$ we have $\Delta_A(h)/\Delta_G(h) = 1/(1/t) = t$, so we need $\rho(xh) = t\rho(x)$. Since $(a, b)(t, 0) = (at, b)$, the function $\rho(a, b) = a$ works. The choice $\rho = 1$ fails for $t \neq 1$.
2. Is $\rho(a, b) = a + b^2$ a rho-function for $H = N$?
   Answer. No. A rho-function for N must satisfy $\rho(xn) = \rho(x)$. But $\rho((a, 0)n_1) = \rho(a, a) = a + a^2 \neq a = \rho(a, 0)$.
3. Give $\Pi_\xi(g)^*$ for $g = (a_0, b_0)$ in the first picture (17.9).
   Answer. Since Π_ξ is unitary, the adjoint is $\Pi_\xi(g^{-1})$, with $g^{-1} = (a_0^{-1}, -a_0^{-1}b_0)$. Hence

   $$(\Pi_\xi(g)^*\varphi)(a) = e^{-i\xi b_0/(a_0 a)}\varphi(a_0 a).$$

   Substituting $a = a_0 u$ in $\langle \Pi_\xi(g)\varphi, \psi\rangle$ confirms this formula.

Summary

A rho-function gives a quasi-invariant measure μ_ρ on G/H. Its translation factor $\lambda(g, xH) = \rho(gx)/\rho(x)$ is a cocycle. The action $f \mapsto \lambda(g^{-1}, \cdot)^{1/2} f(g^{-1}\,\cdot)$ on equivariant functions is unitary, because the cocycle cancels the change of measure. For the representation Π_ξ of the $ax + b$ group induced from the character χ_ξ of the translation subgroup N, the point $a \in G/N$ carries the frequency ξ/a. The unitary equivalence class of Π_ξ depends only on the scaling orbit of ξ. By Mackey's theorem, cited in Lecture 25, Π_+ and Π_- are irreducible, while Π_0 is not.

Reading

Folland, Abstract Harmonic Analysis, Chapter 2, "Measures on Homogeneous Spaces": quotient measures and rho-functions for a general closed subgroup. Kirillov, Elements,

Section 9.1: quasi-invariant measures on homogeneous spaces. Kirillov, Elements, Section 13: models of induced representations. A. W. Knapp, *Representation Theory of Semisimple Groups* (2001), Sections VII.1–VII.2: pictures of representations induced from a parabolic subgroup of a linear connected reductive group.

Lecture 18

Heisenberg group and the Weyl relations

Objectives. *The Heisenberg group is the simplest noncommutative nilpotent Lie group. This lecture builds its Schrödinger representation on $L^2(\mathbb{R})$ from translations and modulations, and proves the Weyl relations, which describe how translations and modulations commute up to a phase. On this representation, the bracket $[X, Y] = Z$ becomes the commutation relation $[Q, P] = i\hbar$ of quantum mechanics. We then treat the Stone–von Neumann theorem. It says that a fixed nonzero central character, which records the scalars by which the centre acts, determines one irreducible unitary representation up to unitary equivalence. We prove the irreducibility of the Schrödinger representation from an explicit rank-one operator, and we cite the uniqueness. For central character zero, the irreducibles are one-dimensional characters, and the translation representation on $L^2(\mathbb{R})$ contains none of them.*

By the end of this lecture you will be able to:

- Compute products, inverses, commutators, the exponential map and Ad in the Heisenberg group.
- Prove that the Schrödinger formula is a strongly continuous unitary representation satisfying the Weyl relations, and compute its derivative on $C_c^\infty(\mathbb{R})$.
- State the Stone–von Neumann theorem with its hypotheses, and turn a Weyl pair into a representation to which it applies.
- Compute a Gaussian matrix coefficient, use it to prove that Π_λ is irreducible for $\lambda \neq 0$, and contrast nonzero and zero central character.

18.1 The group law

Definition 18.1. The *Heisenberg group* H is the set of real matrices

$$(a, b, c) := \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}, \qquad a, b, c \in \mathbb{R},$$

with matrix multiplication.

Multiplying the matrices gives

$$(a, b, c)(a', b', c') = (a + a',\ b + b',\ c + c' + ab'). \tag{18.1}$$

The identity is $(0,0,0)$, and

$$(a,b,c)^{-1} = (-a,-b,-c+ab).$$

Indeed, the third coordinate of $(a,b,c)(-a,-b,-c+ab)$ is $c-c+ab-ab=0$. The group H is a closed subgroup of $\mathrm{GL}_3(\mathbb{R})$, hence a Lie group by the closed-subgroup theorem of Lecture 3. The coordinates identify H with $\mathbb{R}^3$, so H is connected and simply connected.

Swapping the factors in the group law (18.1) changes only the last coordinate. For $g=(a,b,c)$ and $g'=(a',b',c')$,

$$gg' = (0,0,ab'-a'b)\,g'g. \tag{18.2}$$

By the group law (18.1), each $(0,0,d)$ commutes with every element. So the commutation formula (18.2) shows that g is central exactly when $ab'-a'b=0$ for all a',b', that is, when $a=b=0$. The centre is therefore

$$Z(H) = \{(0,0,c) : c\in\mathbb{R}\} \cong \mathbb{R},$$

and $(a,b,c)Z(H)\mapsto(a,b)$ identifies $H/Z(H)$ with $\mathbb{R}^2$.

The Lie algebra $\mathfrak{h}$ has basis $X=E_{12}$, $Y=E_{23}$, $Z=E_{13}$, where E_{ij} are matrix units. Then

$$[X,Y]=Z, \qquad [X,Z]=[Y,Z]=0.$$

Let $W=xX+yY+zZ$. Multiplying matrix units gives $W^2=xyZ$ and $W^3=0$. So the exponential series of W stops after the quadratic term:

$$\exp(xX+yY+zZ) = I+W+\tfrac12 W^2 = \big(x,\ y,\ z+\tfrac12 xy\big). \tag{18.3}$$

So $\exp\colon\mathfrak{h}\to H$ is a diffeomorphism with inverse $(a,b,c)\mapsto aX+bY+(c-\frac12 ab)Z$. In particular $\exp(sX)=(s,0,0)$, $\exp(sY)=(0,s,0)$ and $\exp(sZ)=(0,0,s)$.

For the adjoint action we show one representative step. By the group law (18.1),

$$g\exp(sX)g^{-1} = (a+s,b,c)(-a,-b,-c+ab) = (s,0,-sb) = \exp\big(s(X-bZ)\big).$$

Thus $\mathrm{Ad}(g)X = X-bZ$. In the same way $\mathrm{Ad}(g)Y = Y+aZ$ and $\mathrm{Ad}(g)Z=Z$. In the ordered basis (X,Y,Z),

$$D_{\mathrm{Ad}}(g) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -b & a & 1 \end{pmatrix}, \qquad \det D_{\mathrm{Ad}}(g) = 1.$$

With the convention $\Delta(g)=|\det\mathrm{Ad}(g)|^{-1}$ of Lecture 16, H is unimodular. Directly, left and right translations are polynomial maps of $\mathbb{R}^3$ with triangular Jacobian matrices and ones on the diagonal. So Lebesgue measure $da\,db\,dc$ is a left and right Haar measure.

18.2 The Schrödinger representation

We use $L^2(\mathbb{R})$ with $\langle f,h\rangle = \int_{\mathbb{R}} \overline{f(t)}\,h(t)\,dt$. We use the integration rules stated at the end of Lecture 16, in its section on locally compact groups: the identification of $L^2(\mathbb{R})$ with classes of functions, Tonelli and Fubini, dominated convergence and change of variables. To differentiate under an integral, we apply dominated convergence to the difference

quotients, using a common integrable bound for the derivative. We also use two standard facts. First, Lebesgue integrals are invariant under $t \mapsto t + a$. Second, the space $C_c(\mathbb{R})$ of continuous compactly supported functions, and its subspace $C_c^\infty(\mathbb{R})$ of smooth ones, are dense in $L^2(\mathbb{R})$.

Definition 18.2. Fix $\lambda \in \mathbb{R}$. For $g = (a, b, c) \in H$ and $f \in L^2(\mathbb{R})$, the *Schrödinger operator* $\Pi_\lambda(g)$ is defined by

$$(\Pi_\lambda(a, b, c)f)(t) = e^{i\lambda(c+bt)} f(t + a). \tag{18.4}$$

The *translations* $U(a)$ and *modulations* $V_\lambda(b)$ are defined by

$$(U(a)f)(t) = f(t + a), \qquad (V_\lambda(b)f)(t) = e^{i\lambda bt} f(t).$$

By the definitions, the Schrödinger operators factor as

$$\Pi_\lambda(a, b, c) = e^{i\lambda c}\, V_\lambda(b)\, U(a).$$

This matches the factorisation of group elements

$$(a, b, c) = (0, 0, c)(0, b, 0)(a, 0, 0).$$

Our $U(a)$ is the translation by $-a$ in the convention of Lecture 15, and $V_1(b)$ is the modulation $M(b)$ of Exercise 15.5. So at $\lambda = 1$ the Weyl relations below are the relation of Exercise 15.5(d).

Proposition 18.3. *For every $\lambda \in \mathbb{R}$, Π_λ is a strongly continuous unitary representation of H on $L^2(\mathbb{R})$, and $\Pi_\lambda(0, 0, c) = e^{i\lambda c} I$. The* Weyl relations *hold:*

$$U(a)V_\lambda(b) = e^{i\lambda ab} V_\lambda(b)U(a), \qquad a, b \in \mathbb{R}. \tag{18.5}$$

Proof. Homomorphism. Let $g = (a, b, c)$ and $g' = (a', b', c')$. Applying the defining formula (18.4) twice,

$$\begin{aligned}(\Pi_\lambda(g)\Pi_\lambda(g')f)(t) &= e^{i\lambda(c+bt)}(\Pi_\lambda(g')f)(t + a)\\ &= e^{i\lambda(c+bt)} e^{i\lambda(c'+b'(t+a))} f(t + a + a')\\ &= e^{i\lambda(c+c'+ab'+(b+b')t)} f(t + a + a').\end{aligned}$$

By the group law (18.1), this is $(\Pi_\lambda(gg')f)(t)$. Also $\Pi_\lambda(0, 0, 0) = I$.

Unitarity. Since $|e^{i\lambda(c+bt)}| = 1$, translation invariance gives $\|\Pi_\lambda(g)f\|^2 = \int |f(t+a)|^2 dt = \|f\|^2$. Each $\Pi_\lambda(g)$ is an isometry with inverse $\Pi_\lambda(g^{-1})$, hence unitary.

Centre and Weyl relations. The first claim is the defining formula (18.4) with $a = b = 0$. For the second,

$$(U(a)V_\lambda(b)f)(t) = (V_\lambda(b)f)(t + a) = e^{i\lambda b(t+a)} f(t + a) = e^{i\lambda ab}(V_\lambda(b)U(a)f)(t).$$

Strong continuity. This is the only analytic step. First let $f \in C_c(\mathbb{R})$ vanish outside $[-R, R]$, and let $|a| \le 1$. Then $\Pi_\lambda(g)f - f$ vanishes outside $[-R-1, R+1]$. On that interval, $|e^{i\theta} - 1| \le |\theta|$ gives

$$|(\Pi_\lambda(g)f)(t) - f(t)| \le |\lambda|\big(|c| + (R + 1)|b|\big)\|f\|_\infty + \omega_f(|a|),$$

where $\omega_f(\delta) = \sup\{|f(t+s)-f(t)| : t \in \mathbb{R},\ |s| \le \delta\}$. Uniform continuity of f gives $\omega_f(\delta) \to 0$ as $\delta \to 0$. Integrating the square over an interval of length $2R+2$,

$$\|\Pi_\lambda(g)f - f\| \le (2R+2)^{1/2}\Big(|\lambda|\big(|c| + (R+1)|b|\big)\|f\|_\infty + \omega_f(|a|)\Big) \longrightarrow 0 \quad (g \to e).$$

Now let $f \in L^2(\mathbb{R})$ and $\varepsilon > 0$. Choose $h \in C_c(\mathbb{R})$ with $\|f-h\| < \varepsilon$. Since $\Pi_\lambda(g)$ is unitary,

$$\|\Pi_\lambda(g)f - f\| \le \|\Pi_\lambda(g)(f-h)\| + \|\Pi_\lambda(g)h - h\| + \|h - f\| < 2\varepsilon + \|\Pi_\lambda(g)h - h\|.$$

So $\limsup_{g\to e} \|\Pi_\lambda(g)f - f\| \le 2\varepsilon$ for every ε. Finally, $\|\Pi_\lambda(g)f - \Pi_\lambda(g_0)f\| = \|\Pi_\lambda(g_0^{-1}g)f - f\|$, and $g_0^{-1}g \to e$ as $g \to g_0$. ∎

Combining the commutation formula (18.2) with the scalar action of the centre in Proposition 18.3 gives

$$\Pi_\lambda(g)\Pi_\lambda(g') = e^{i\lambda(ab'-a'b)}\,\Pi_\lambda(g')\Pi_\lambda(g).$$

For $\lambda \neq 0$ the operators need not commute. For $\lambda = 0$ they commute.

This answers the question left open by the Heisenberg multiplier example of Lecture 13. There $\omega(x,y) = x_1y_2 - x_2y_1$ on $\mathbb{R}^2$ and $c(x,y) = e^{i\omega(x,y)/2}$. For $x \in \mathbb{R}^2$ put $U_x = \Pi_1(\exp(x_1X + x_2Y))$. The exponential formula (18.3) and the group law give

$$U_xU_y = e^{i\omega(x,y)/2}\,U_{x+y}.$$

So the multiplier c is realised on $L^2(\mathbb{R})$. By Exercise 18.4(d), Π_1 has no nonzero finite-dimensional invariant subspace, as that example requires. The extension $E = \mathbb{R}^2 \times \mathbb{T}$ of Exercise 13.5 is the quotient H/Γ with $\Gamma = \{(0,0,2\pi k) : k \in \mathbb{Z}\}$, through the map $(a,b,c) \mapsto ((a,b), e^{i(c-ab/2)})$.

The exponent $ab' - a'b$ is a signed area in the plane $H/Z(H)$, as Figure 18.1 shows. Multiply on the right by $(a,0,0)$ and then by $(0,b,0)$. This route ends at (a,b,ab). The other route ends at $(a,b,0)$. The central coordinates differ by the area ab of the rectangle, and this is the phase in the Weyl relations. Similarly, the four steps g, g', g^{-1}, g'^{-1} form a closed loop in the plane, and by (18.2) they end at $(0,0,ab'-a'b)$. The loop runs counterclockwise when $ab' - a'b > 0$. The operator products in the figure name these group products. As operators they act on a vector from the right, so the order of action is the reverse of the order of the route.

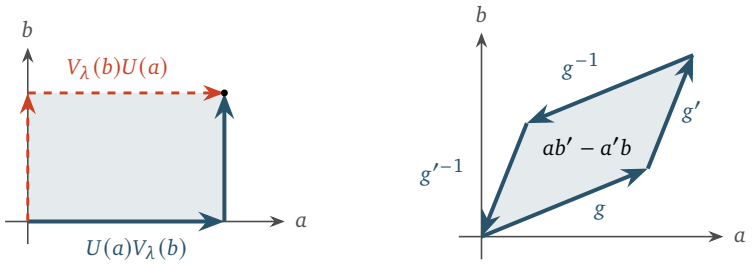


Figure 18.1: Routes in the plane $H/Z(H)$, with each step a multiplication on the right. In the first panel, two routes join e to the same point of the plane, and the rectangle lies between them. In the second panel, the loop of a commutator encloses the parallelogram on (a,b) and (a',b'), and its signed area is $ab' - a'b$.

Remark 18.4. Let $A = \{(0,b,c)\}$, a closed abelian normal subgroup, with character $\chi_\lambda(0,b,c) = e^{i\lambda c}$. Exercise 18.5 shows that the representation unitarily induced from χ_λ, as in Lecture 17, is equivalent to Π_λ.

Next we differentiate. Write $\mathcal{D} = C_c^\infty(\mathbb{R})$ and tf for the function $t \mapsto tf(t)$.

Proposition 18.5. *Each $\Pi_\lambda(g)$ maps $\mathcal{D}$ into $\mathcal{D}$. For $f \in \mathcal{D}$ and $W \in \{X, Y, Z\}$, the limit*

$$d\Pi_\lambda(W)f = \lim_{s\to 0} s^{-1}(\Pi_\lambda(\exp sW)f - f)$$

exists in $L^2(\mathbb{R})$, and

$$d\Pi_\lambda(X)f = f', \qquad d\Pi_\lambda(Y)f = i\lambda t f, \qquad d\Pi_\lambda(Z)f = i\lambda f.$$

On $\mathcal{D}$ we have $[d\Pi_\lambda(X), d\Pi_\lambda(Y)] = d\Pi_\lambda(Z) = d\Pi_\lambda([X,Y])$.

Proof. Translation and multiplication by the smooth function $e^{i\lambda(c+bt)}$ preserve smoothness and compact support. Let f vanish outside $[-R,R]$ and let $0 < |s| \le 1$. For X, Taylor's theorem gives

$$\left|\frac{f(t+s)-f(t)}{s} - f'(t)\right| \le \tfrac{1}{2}|s|\,\|f''\|_\infty,$$

and the left side vanishes outside $[-R-1, R+1]$. Its L^2 norm is at most $(2R+2)^{1/2}\tfrac{1}{2}|s|\|f''\|_\infty \to 0$. For Y, the bound $|e^{i\theta} - 1 - i\theta| \le \theta^2/2$ gives

$$\left|\frac{e^{i\lambda st}-1}{s} - i\lambda t\right| |f(t)| \le \tfrac{1}{2}\lambda^2|s|R^2|f(t)|,$$

whose L^2 norm tends to 0. For Z, $(e^{i\lambda s}-1)/s \to i\lambda$. Finally,

$$[d\Pi_\lambda(X), d\Pi_\lambda(Y)]f = (i\lambda t f)' - i\lambda t f' = i\lambda f = d\Pi_\lambda(Z)f. \qquad \blacksquare$$

Remark 18.6. Since Π_λ is strongly continuous (Proposition 18.3), $s \mapsto \Pi_\lambda(\exp sW)$ is a strongly continuous one-parameter unitary group. Its generator B_W, as defined in Lecture 15, has domain exactly the vectors f at which $s \mapsto \Pi_\lambda(\exp sW)f$ is differentiable at $s = 0$ in $L^2(\mathbb{R})$. Stone's theorem (Lecture 15) shows that B_W is skew-adjoint. So the physical generator $A_W = iB_W$ is self-adjoint and satisfies $\Pi_\lambda(\exp sW) = \exp(-isA_W)$. The derivative formulas of Proposition 18.5 show that $\mathcal{D}$ lies in the domain of A_W and that on $\mathcal{D}$

$$A_X f = if', \qquad A_Y f = -\lambda t f, \qquad A_Z f = -\lambda f.$$

Since $\mathcal{D}$ is dense and invariant under each $\Pi_\lambda(\exp sW)$, the dense invariant-domain criterion of Stone's theorem in Lecture 15 shows that B_W is the closure of $d\Pi_\lambda(W)$ on $\mathcal{D}$.

In quantum mechanics a particle on a line has state space $L^2(\mathbb{R})$. Take $\lambda = 1/\hbar$, where $\hbar > 0$ is the reduced Planck constant. Put $P = -\hbar A_X$ (momentum) and $Q = -\hbar A_Y$ (position). The minus sign in P is there because $U(a)$ is the translation by $-a$. On $\mathcal{D}$, $Qf = tf$, $Pf = -i\hbar f'$ and $[Q,P] = i\hbar$.

In terms of these operators, $U(a) = \exp(iaP/\hbar)$ and $V_\lambda(b) = \exp(ibQ/\hbar)$. So the Weyl relations (18.5) are the bounded, exponentiated form of the commutation relation.

We never use the commutation relation as a hypothesis. A commutation relation of unbounded operators on a dense domain does not by itself imply the Weyl relations. For example, on $L^2(0,1)$ take $Qf = tf$ and $P = -i\hbar\, d/dt$ with periodic boundary conditions. Both operators are self-adjoint, and $[Q,P] = i\hbar$ on $C_c^\infty(0,1)$. Here $\exp(iP/\hbar) = I$, since P has the eigenvectors $e^{2\pi i n t}$ with eigenvalues $2\pi\hbar n$. The Weyl relation at $a = 1$ would then give $I = e^{ib/\hbar} I$ for every $b \in \mathbb{R}$, which is false (Hall, *Quantum Theory for Mathematicians*, Section 14.1). Exercise 18.3 shows that no bounded operators on a nonzero Hilbert space satisfy $[Q,P] = i\hbar I$.

18.3 The Stone–von Neumann theorem

Definition 18.7. A unitary representation ρ of H on a Hilbert space $\mathcal{H}$ has *central character* $\lambda \in \mathbb{R}$ if $\rho(0,0,c) = e^{i\lambda c} I$ for all $c \in \mathbb{R}$.

By the centre statement in Proposition 18.3, the representation Π_λ has central character λ. Now let ρ be an irreducible unitary representation of H. Schur's lemma for unitary representations (Kirillov, *Lectures on the Orbit Method*, 2004, Appendix V) makes each central operator $\rho(0,0,c)$ a scalar. These scalars form a continuous homomorphism $\mathbb{R} \to \mathrm{U}(1)$, hence have the form $e^{i\lambda c}$. So every irreducible unitary representation has a central character. Reducible ones need not, for example $\Pi_1 \oplus \Pi_2$.

Lemma 18.8. *Let σ and τ be strongly continuous one-parameter unitary groups on a Hilbert space $\mathcal{H}$, and let $\lambda \in \mathbb{R}$. Suppose $\sigma(a)\tau(b) = e^{i\lambda ab}\tau(b)\sigma(a)$ for all $a, b \in \mathbb{R}$. Then $\rho(a,b,c) = e^{i\lambda c}\tau(b)\sigma(a)$ defines a strongly continuous unitary representation ρ of H on $\mathcal{H}$ with central character λ.*

Proof. Move $\sigma(a)$ past $\tau(b')$ using the hypothesis:

$$\rho(g)\rho(g') = e^{i\lambda(c+c')}\tau(b)\sigma(a)\tau(b')\sigma(a') = e^{i\lambda(c+c'+ab')}\tau(b+b')\sigma(a+a') = \rho(gg').$$

Each $\rho(g)$ is unitary, and $\rho(0,0,c) = e^{i\lambda c} I$. For continuity at (a_0,b_0,c_0), let $f \in \mathcal{H}$ and put $h = \sigma(a_0)f$. Then

$$\|\tau(b)\sigma(a)f - \tau(b_0)\sigma(a_0)f\| \le \|\sigma(a)f - \sigma(a_0)f\| + \|\tau(b)h - \tau(b_0)h\| \to 0,$$

and the scalar factor $e^{i\lambda c}$ is continuous. ∎

We call a pair (σ,τ) satisfying the hypotheses of Lemma 18.8 a *Weyl pair* with parameter λ. By the Weyl relations (18.5) and the strong continuity in Proposition 18.3, the translations and modulations form a Weyl pair (U, V_λ) with parameter λ. Two Weyl pairs are *equivalent* when a single unitary operator from the first space to the second conjugates each group of the first pair into the corresponding group of the second.

Theorem 18.9 (Stone–von Neumann)**.** *Fix $\lambda \neq 0$.*

(a) The Schrödinger representation Π_λ is irreducible.

(b) Let ρ be a strongly continuous unitary representation of H on a nonzero separable Hilbert space $\mathcal{H}$ with central character λ. Then $\mathcal{H}$ is a Hilbert direct sum of mutually orthogonal closed ρ-invariant subspaces. On each summand ρ is unitarily equivalent to Π_λ.

We prove part (a) in § 18.4, after a kernel computation that does not use this theorem. We cite part (b), the uniqueness statement, from Kirillov, *Lectures on the Orbit Method*, Chapter 2. The proof of (a) does not prove (b).

The Hilbert space of an irreducible representation of H is automatically separable. By irreducibility, every nonzero vector is cyclic. By strong continuity, the closed span of the orbit of such a vector is already generated by its orbit under a countable dense subset of H. This closed span is the whole space, and it is separable.

The uniqueness statement in the Stone–von Neumann theorem, Theorem 18.9, has the following consequences.

- Every irreducible unitary representation of H with central character $\lambda \neq 0$ is unitarily equivalent to Π_λ.
- By the lemma on Weyl pairs, Lemma 18.8, every Weyl pair with parameter $\lambda \neq 0$ on a nonzero separable space is equivalent to a direct sum of copies of (U, V_λ). In particular, every Weyl pair with parameter $1/\hbar$ on a nonzero separable space is unitarily equivalent to a direct sum of copies of $(\exp(iaP/\hbar), \exp(ibQ/\hbar))$.
- Distinct values of λ give inequivalent representations Π_λ (Exercise 18.6(c) on inequivalence).

In quantum mechanics, the first two consequences say that every irreducible Weyl pair with parameter $1/\hbar$ is the Schrödinger pair, which is the precise form of the statement that matrix mechanics and wave mechanics agree.

A standard proof of (b), going back to von Neumann, turns on one step. Let ρ be a representation as in the uniqueness statement (b) of Theorem 18.9. For $f, h \in \mathcal{H}$, consider the sesquilinear form

$$B(f,h) = \frac{|\lambda|}{2\pi} \int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \, \big\langle f, \rho(\exp(aX + bY))h \big\rangle \, da\, db.$$

The Gaussian weight has integral $4\pi/|\lambda|$, so $|B(f,h)| \leq 2\|f\|\|h\|$. Hence the Riesz representation theorem for bounded sesquilinear forms gives a bounded operator E on $\mathcal{H}$ with $\|E\| \leq 2$ and $B(f,h) = \langle f, Eh\rangle$ for all $f, h \in \mathcal{H}$.

That proof shows, from the Weyl relations and strong continuity, that E is a nonzero orthogonal projection. Each unit vector $v \in E\mathcal{H}$ then generates one copy of Π_λ. In that copy, v corresponds to the Gaussian of the next section, § 18.4. The weight is that Gaussian's matrix coefficient, given by (18.6) at $g = \exp(aX + bY)$.

18.4 Gaussian matrix coefficients and central character zero

Let $\varphi(t) = \pi^{-1/4} e^{-t^2/2}$. Since $\int e^{-t^2} dt = \sqrt{\pi}$, $\|\varphi\| = 1$. We compute $\langle \varphi, \Pi_\lambda(g)\varphi\rangle$ for $g = (a, b, c)$. We use the calculus fact

$$\int_{\mathbb{R}} e^{-(t+w)^2} dt = \sqrt{\pi} \qquad (w \in \mathbb{C}).$$

For real w this is a translation. Write $w = u + iv$. The v-derivative of the integrand is $i\,\frac{d}{dt} e^{-(t+w)^2}$, which integrates to 0. Differentiation under the integral is justified on $|v| \leq V$

by the common integrable bound $2(|t+u|+V)e^{V^2}e^{-(t+u)^2}$ for the derivative. So the integral does not depend on v.

By the defining formula (18.4), and since φ is real,

$$\langle \varphi, \Pi_\lambda(g)\varphi\rangle = \pi^{-1/2} e^{i\lambda c} \int_{\mathbb{R}} e^{-t^2/2-(t+a)^2/2+i\lambda b t}\, dt.$$

Complete the square with $w = \tfrac{1}{2}(a - i\lambda b)$:

$$-\tfrac{t^2}{2} - \tfrac{(t+a)^2}{2} + i\lambda b t = -t^2 - (a - i\lambda b)t - \tfrac{a^2}{2} = -(t+w)^2 + w^2 - \tfrac{a^2}{2},$$

$$w^2 - \tfrac{a^2}{2} = \tfrac{1}{4}\big(a^2 - 2i\lambda ab - \lambda^2 b^2\big) - \tfrac{a^2}{2} = -\tfrac{1}{4}\big(a^2 + \lambda^2 b^2\big) - \tfrac{i\lambda ab}{2}.$$

The integral is $\sqrt{\pi}$ times $e^{w^2 - a^2/2}$. Therefore

$$\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle = e^{i\lambda(c - ab/2)}\, e^{-(a^2+\lambda^2 b^2)/4}. \tag{18.6}$$

We make two checks. First, at $g = e$ the value is $1 = \|\varphi\|^2$. Second, replacing g by $g^{-1} = (-a, -b, -c + ab)$ conjugates the value, as unitarity requires.

By the exponential formula (18.3), $c - ab/2$ is the Z-coordinate of $\log g$, where log is the inverse of the exponential map. Hence at $\exp(xX + yY + zZ)$ the coefficient is $e^{i\lambda z} e^{-(x^2+\lambda^2 y^2)/4}$.

The modulus depends only on the image (a, b) in $H/Z(H) \cong \mathbb{R}^2$. For $\lambda \neq 0$ it decays in every direction, and

$$\int_{\mathbb{R}^2} \big|\langle \varphi, \Pi_\lambda(a,b,c)\varphi\rangle\big|^2 da\, db = \int e^{-a^2/2} da \int e^{-\lambda^2 b^2/2} db = \sqrt{2\pi} \cdot \frac{\sqrt{2\pi}}{|\lambda|} = \frac{2\pi}{|\lambda|}.$$

Its reciprocal $|\lambda|/(2\pi)$ is the normalising constant in the form B above.

Compare Schur orthogonality (Lecture 5). For a unit vector v in an irreducible unitary representation of dimension d of a compact group, $\int |\langle v, \Pi(g)v\rangle|^2\, dg = 1/d$, and $d \int \overline{\langle v, \Pi(g)v\rangle}\, \Pi(g)\, dg$ is the orthogonal projection onto $\mathbb{C}v$. The form B has the same shape on $H/Z(H)$, with φ in place of v. The number $|\lambda|/(2\pi)$ takes the place of d. Only the shape can be compared, because the measure here is $da\, db$ and not a normalised integral.

The operator E defined after the Stone–von Neumann theorem can be computed explicitly in this model, for $\lambda \neq 0$. The computation does not use that theorem. Write E_λ for that operator with $\rho = \Pi_\lambda$. Since

$$\Pi_\lambda(\exp(aX + bY))f(t) = e^{i\lambda(bt + ab/2)} f(t + a),$$

we integrate first over b. For $f, h \in C_c(\mathbb{R})$, Tonelli, Cauchy–Schwarz in t and translation invariance give, as at the end of Lecture 16,

$$\int_{\mathbb{R}^2} e^{-(a^2+\lambda^2 b^2)/4} \int_{\mathbb{R}} |h(t)|\, |f(t+a)|\, dt\, da\, db \le \frac{4\pi}{|\lambda|} \|f\| \|h\|.$$

Thus Fubini's theorem identifies the following kernel calculation with the form B defining E_λ. The Gaussian Fourier integral, obtained by the same square completion as above, gives

$$\int_{\mathbb{R}} e^{-\lambda^2 b^2/4} e^{i\lambda b(t + a/2)}\, db = \frac{2\sqrt{\pi}}{|\lambda|} e^{-(t+a/2)^2}.$$

Consequently, for $f \in C_c(\mathbb{R})$,

$$(E_\lambda f)(t) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} e^{-a^2/4-(t+a/2)^2} f(t+a)\, da.$$

Set $u = t + a$. The exponent simplifies to

$$-\frac{(u-t)^2}{4} - \frac{(t+u)^2}{4} = -\frac{t^2+u^2}{2}.$$

We therefore obtain

$$(E_\lambda f)(t) = \varphi(t) \int_{\mathbb{R}} \overline{\varphi(u)} f(u)\, du = \varphi(t)\langle \varphi, f\rangle.$$

Both operators are bounded, so density extends the identity to all $L^2(\mathbb{R})$. Thus E_λ is exactly the orthogonal projection onto the Gaussian line. For a general representation ρ with central character $\lambda \neq 0$, proving that the operator E is a nonzero projection is the key step of the standard proof of the uniqueness statement. Here its rank-one integral kernel is visible. In quantum mechanics, with $\lambda = \hbar = 1$, the function φ is the ground state of the oscillator $\frac{1}{2}(P^2 + Q^2)$, the vector $\Pi_1(g)\varphi$ is a coherent state, and E_1 is the projection on the ground state.

We can now prove part (a) of the theorem. The rank of E_λ forces every closed invariant subspace to contain φ or to be orthogonal to it, and the orbit of φ spans a dense subspace.

Proof of Theorem 18.9(a). Fix $\lambda \neq 0$, and let $W \subset L^2(\mathbb{R})$ be a closed invariant subspace.

The orbit of φ spans a dense subspace. Let $f \in L^2(\mathbb{R})$ be orthogonal to $\Pi_\lambda(g)\varphi$ for all $g \in H$. Taking $g = (0, b, 0)$ gives

$$\int_{\mathbb{R}} \overline{f(t)}\, \varphi(t)\, e^{i\lambda b t}\, dt = 0 \qquad \text{for every } b \in \mathbb{R}.$$

By the Cauchy–Schwarz inequality, $u = \bar{f}\varphi$ is integrable. We cite the injectivity of the Fourier transform on $L^1(\mathbb{R})$: if u is integrable and $\int_{\mathbb{R}} u(t)e^{ist}\, dt = 0$ for every $s \in \mathbb{R}$, then $u = 0$ almost everywhere (Folland, *Real Analysis*, second edition, Chapter 8). Since $\lambda \neq 0$, the number $s = \lambda b$ runs over $\mathbb{R}$. So $\bar{f}\varphi = 0$ almost everywhere. Since $\varphi > 0$ everywhere, $f = 0$.

The operator E_λ commutes with the projection onto W. Since $\Pi_\lambda(g)^* = \Pi_\lambda(g^{-1})$, the subspace $W^\perp$ is also invariant. Let $f \in W$ and $h \in W^\perp$. Then $\langle h, \Pi_\lambda(g)f\rangle = 0$ for all $g \in H$. The form defining E_λ therefore gives $\langle h, E_\lambda f\rangle = 0$. Hence $E_\lambda W \subset W$. The same argument with W and $W^\perp$ exchanged gives $E_\lambda W^\perp \subset W^\perp$. So E_λ commutes with the orthogonal projection P_W onto W.

Conclusion. Since $E_\lambda \varphi = \varphi$, we get $E_\lambda P_W \varphi = P_W E_\lambda \varphi = P_W \varphi$. So $P_W\varphi$ lies in the range $\mathbb{C}\varphi$ of E_λ, and $P_W\varphi = \kappa\varphi$ for some $\kappa \in \mathbb{C}$. Since $P_W^2 = P_W$, we have $\kappa = 0$ or $\kappa = 1$. If $\kappa = 1$, then $\varphi \in W$, and the first step gives $W = L^2(\mathbb{R})$. If $\kappa = 0$, then $\varphi \in W^\perp$, and the first step gives $W^\perp = L^2(\mathbb{R})$, so $W = 0$. ∎

Now set $\lambda = 0$. The defining formula (18.4) becomes pure translation, $(\Pi_0(a,b,c)f)(t) = f(t+a)$. The modulations become trivial, and the Gaussian coefficient (18.6) becomes $e^{-a^2/4}$. This does not decay in b, and the integral above diverges.

The irreducible unitary representations of H with central character 0 factor through $H/Z(H) \cong \mathbb{R}^2$. By Schur's lemma (Kirillov, Appendix V), irreducible unitary representations of an abelian group are one-dimensional. Hence the irreducible unitary representations of H with central character 0 are the characters

$$\chi_{\alpha,\beta}(a,b,c) = e^{i(\alpha a+\beta b)}, \qquad \alpha,\beta \in \mathbb{R}.$$

Kirillov's Chapter 2 records this together with the Stone–von Neumann theorem, Theorem 18.9, as the full list of irreducibles. Lecture 19 explains the list through coadjoint orbits.

Proposition 18.10. *Π_0 has no one-dimensional invariant subspace. Hence it has no irreducible closed invariant subspace.*

Proof. Suppose $f \neq 0$ spans an invariant line. Then $U(a)f = \mu(a)f$ with $|\mu(a)| = 1$. For each integer n,

$$\int_n^{n+1} |f(t)|^2 dt = \int_0^1 |f(t+n)|^2 dt = \int_0^1 |\mu(n)f(t)|^2 dt = \int_0^1 |f(t)|^2 dt.$$

Summing over $n \in \mathbb{Z}$ gives $\|f\|^2 = \sum_n \int_0^1 |f|^2$. This is finite only if $\int_0^1 |f|^2 = 0$, and then $\|f\| = 0$, a contradiction. An irreducible closed invariant subspace would carry an irreducible representation with central character 0. By the classification above it would be a line. ∎

The pieces one expects in Π_0 are the plane waves $e^{i\alpha t}$, on which $U(a)$ acts by $e^{i\alpha a}$. They are not square integrable. The Fourier transform of Lecture 15 turns $U(a)$ into multiplication by $e^{i\alpha a}$ on $L^2(\mathbb{R}, d\alpha)$. So Π_0 is an integral of the characters $\chi_{\alpha,0}$ rather than a sum. This is called a *direct integral*, and we do not develop it.

The following table compares central character $\lambda \neq 0$ with central character 0.

	central character $\lambda \neq 0$	central character 0
irreducibles	one class, Π_λ	$\chi_{\alpha,\beta}$, $\alpha,\beta \in \mathbb{R}$
dimension	infinite	one
the $L^2(\mathbb{R})$ model	Π_λ itself, irreducible	Π_0 has no irreducible piece
Gaussian coefficient	decays on $H/Z(H)$	constant in b

Self-checks

1. Compute gg' and $g'g$ for $g = (1,2,3)$ and $g' = (4,5,6)$, and check the commutation formula (18.2).
   Answer. $gg' = (5, 7, 3+6+5) = (5,7,14)$ and $g'g = (5,7,9+8) = (5,7,17)$. Here $ab' - a'b = 5 - 8 = -3$, and $(0,0,-3)(5,7,17) = (5,7,14)$.
2. Why are Π_2 and Π_3 inequivalent?
   Answer. An intertwiner T satisfies $Te^{2ic} = e^{3ic}T$ for all c. At $c = \pi$ this reads $T = -T$, so $T = 0$.
3. Compute $\langle \varphi, \Pi_1(2,2,2)\varphi \rangle$ and $\langle \varphi, \Pi_0(0,b,0)\varphi \rangle$.
   Answer. By the Gaussian coefficient (18.6), $c - ab/2 = 0$ and $(a^2+b^2)/4 = 2$, so the first is e^{-2}. The second is 1 for every b.

Summary

The Heisenberg group has centre $Z(H) \cong \mathbb{R}$, and every commutator lies in it. Translations and modulations on $L^2(\mathbb{R})$ form the Schrödinger representation Π_λ, which satisfies the Weyl relations. For $\lambda \neq 0$ the Gaussian form computes an explicit rank-one projection, and this proves that Π_λ is irreducible. The uniqueness statement of the Stone–von Neumann theorem, cited, says that for $\lambda \neq 0$ every strongly continuous unitary representation on a nonzero separable space with central character λ is a Hilbert direct sum of copies of Π_λ, up to unitary equivalence. At $\lambda = 0$ the irreducibles are the characters $\chi_{\alpha,\beta}$, and the translation representation Π_0 contains none of them.

Reading

- Kirillov, *Lectures on the Orbit Method* (2004), Chapter 2: the Heisenberg group, its irreducible unitary representations and the Stone–von Neumann theorem.
- Kirillov, *Lectures on the Orbit Method*, Appendix V: background on unitary representations, including Schur's lemma and induced representations.
- Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Chapter 1: the Heisenberg group among the examples of matrix Lie groups.

Lecture 19

Coadjoint orbits and the nilpotent orbit method

Objectives. *A representation acts on vectors. The coadjoint action of a Lie group instead acts on linear functionals on its Lie algebra. Each orbit of this action has a natural symplectic form. For connected simply connected nilpotent groups, these orbits label all irreducible strongly continuous unitary representations up to unitary equivalence. We prove the geometric construction and work through the Heisenberg example. The general classification is an explicit cited theorem.*

By the end of this lecture you will be able to:

- Compute a coadjoint action and its stabilisers.
- Construct the orbit symplectic form and check its signs, using the rules for 1-forms, the exterior derivative and pullback.
- State the nilpotent orbit theorem with its global hypotheses.
- Choose a polarisation for a functional on a two-dimensional Heisenberg orbit and recover the Schrödinger representation by induction.

19.1 The dual action

Throughout, G is a finite-dimensional real Lie group with Lie algebra $\mathfrak{g}$. Its real dual is $\mathfrak{g}^* = \mathrm{Hom}_{\mathbb{R}}(\mathfrak{g}, \mathbb{R})$. We pair a functional ℓ and a vector X by $\ell(X)$. This dual is different from the complex dual of a representation space.

We first say why this dual appears. For $G = \mathbb{R}^n$, a unitary character has the form $x \mapsto e^{i\ell(x)}$ with $\ell \in \mathfrak{g}^*$. So $\mathfrak{g}^*$ labels the irreducible unitary representations of $\mathbb{R}^n$. A noncommutative group acts on $\mathfrak{g}^*$. The guiding idea of this lecture is that two functionals on one orbit should label one representation. In Lecture 17, the scaling subgroup of the $ax + b$ group moved the frequency ξ to ξ/a, and the class of Π_ξ depended only on the orbit.

Definition 19.1. The *coadjoint action* of G on $\mathfrak{g}^*$ is

$$(g \cdot \ell)(X) = \ell(\mathrm{Ad}(g^{-1})X).$$

The *coadjoint orbit* through ℓ is $O_\ell = \{g \cdot \ell : g \in G\}$. Its stabiliser is $G_\ell = \{g : g \cdot \ell = \ell\}$.

The inverse is needed for the action law. Indeed,

$$(g \cdot (h \cdot \ell))(X) = \ell(\mathrm{Ad}(h^{-1})\,\mathrm{Ad}(g^{-1})X) = \ell(\mathrm{Ad}((gh)^{-1})X) = ((gh) \cdot \ell)(X).$$

The stabiliser is closed by continuity. It is therefore an embedded Lie subgroup by the closed-subgroup theorem of Lecture 3. Differentiating the action gives

$$(\mathrm{ad}^*(X)\ell)(Y) = -\ell([X,Y]), \qquad \mathfrak{g}_\ell = \{X : \ell([X,Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}. \tag{19.1}$$

The identity $\exp(tX)\cdot\ell = e^{t\,\mathrm{ad}^*(X)}\ell$ shows that $\exp(tX) \in G_\ell$ for every t exactly when $\mathrm{ad}^*(X)\ell = 0$. Thus $\mathrm{Lie}(G_\ell) = \mathfrak{g}_\ell$. For example, a central X lies in every $\mathfrak{g}_\ell$. Central directions therefore do not generate motion on an orbit.

We give $\mathcal{O}_\ell$ the smooth structure of G/G_ℓ. This uses the homogeneous-space theorem from Lecture 3. The natural map

$$G/G_\ell \longrightarrow \mathfrak{g}^*, \qquad gG_\ell \longmapsto g\cdot\ell$$

is injective and an immersion. At the identity coset, its derivative is induced by $X \mapsto \mathrm{ad}^*(X)\ell$, whose kernel is precisely $\mathfrak{g}_\ell$. At other points, equivariance gives the same rank. The orbit dimension is consequently $\dim\mathfrak{g} - \dim\mathfrak{g}_\ell$.

For a general Lie group, an orbit need not be an embedded submanifold of $\mathfrak{g}^*$. All tangent-space statements below use the quotient smooth structure. The Heisenberg orbits will be ordinary affine planes and points, so this distinction disappears there.

Write

$$\xi_X(\eta) = \left.\frac{d}{dt}\right|_0 \exp(tX)\cdot\eta = \mathrm{ad}^*(X)\eta, \qquad \eta \in \mathcal{O}_\ell.$$

The vectors $\xi_X(\eta)$ span the tangent space of $\mathcal{O}_\ell$ at η. Two choices of X give the same vector exactly when their difference belongs to $\mathfrak{g}_\eta$.

A first example: the Heisenberg group

Let H be the Heisenberg group of Lecture 18, with Lie algebra $\mathfrak{h}$ spanned by X, Y and Z. We use exactly its matrix coordinates:

$$(a,b,c)(a',b',c') = (a+a', b+b', c+c'+ab'), \qquad [X,Y] = Z.$$

In particular, c is a matrix entry. It is not the central coefficient in the logarithm, which is $c - ab/2$.

Write $\ell = (p,q,\lambda)$ when $\ell(X) = p$, $\ell(Y) = q$ and $\ell(Z) = \lambda$. The adjoint formulas of Lecture 18 are

$$\mathrm{Ad}(a,b,c)X = X - bZ, \qquad \mathrm{Ad}(a,b,c)Y = Y + aZ, \qquad \mathrm{Ad}(a,b,c)Z = Z.$$

The inverse of $g = (a,b,c)$ has first coordinates $(-a,-b)$. Hence $\mathrm{Ad}(g^{-1})X = X + bZ$ and $\mathrm{Ad}(g^{-1})Y = Y - aZ$. Applying ℓ to these vectors gives

$$(a,b,c)\cdot(p,q,\lambda) = (p+\lambda b, q-\lambda a, \lambda). \tag{19.2}$$

For $\lambda \neq 0$, this reaches every point of the plane with fixed last coordinate. Its stabiliser is the centre. For $\lambda = 0$, every point is fixed. Thus the orbit list consists of planes for nonzero λ and individual points $(p,q,0)$.

On a plane, p and q are coordinates. Differentiating the action formula (19.2) gives

$$\xi_X = -\lambda\partial_q, \qquad \xi_Y = \lambda\partial_p, \qquad \xi_Z = 0.$$

Figure 19.1 draws the orbits and these fields.

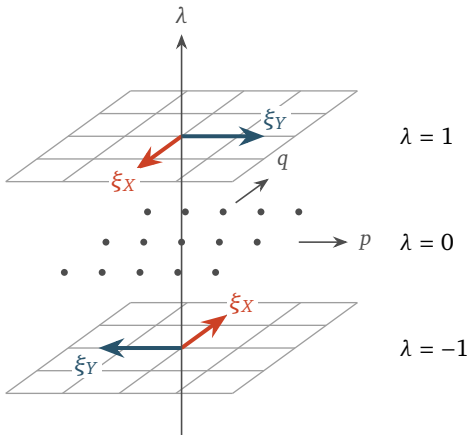


Figure 19.1: Coadjoint orbits of the Heisenberg group in the coordinates (p, q, λ), with λ upward. Each plane $\lambda = \pm 1$ is one orbit, and each dot at $\lambda = 0$ is one orbit. The arrows are the fields ξ_X and ξ_Y at one point of each plane.

19.2 A symplectic form on each orbit

We use the form of this section as follows. First, it shows that orbits have even dimension. Next, at ℓ it is the form B_ℓ on $\mathfrak{g}/\mathfrak{g}_\ell$ below, which decides what a polarisation is. Closedness is not used in this lecture. Lecture 27 needs it for a cited theorem.

We need a small amount of differential-form language. A smooth 2-form ω assigns an alternating bilinear form ω_x to every tangent space, smoothly in local coordinates. On $\mathbb{R}^2$, for example,

$$(dp \wedge dq)((u_1, u_2), (v_1, v_2)) = u_1 v_2 - u_2 v_1.$$

A form ω is *nondegenerate* if, at every point x, the only tangent vector u at x with $\omega_x(u, v) = 0$ for every tangent vector v at x is $u = 0$.

In coordinates $x_1, \dots, x_d$, put $\omega_{ij} = \omega(\partial_i, \partial_j) = -\omega_{ji}$. Then $\omega = \frac{1}{2} \sum_{i,j} \omega_{ij}\, dx_i \wedge dx_j$. The form ω is *closed* when

$$\partial_i \omega_{jk} + \partial_j \omega_{ki} + \partial_k \omega_{ij} = 0 \quad \text{for all } i, j, k.$$

Equivalently, closedness means $d\omega = 0$. A closed nondegenerate 2-form is called a *symplectic form*. These are coordinate-independent conditions. We import this elementary differential-form calculus, including pullbacks and the exterior derivative d, from Warner, Chapter 2. We use the following rules from it.

A smooth 1-*form* α assigns a linear functional α_x on the tangent space to every point x, smoothly in local coordinates. Here dx_j is the 1-form with $dx_j(\partial_k) = \delta_{jk}$. So in coordinates $\alpha = \sum_j a_j\, dx_j$ with $a_j = \alpha(\partial_j)$. The differential of a smooth function f is the 1-form $df = \sum_j (\partial_j f)\, dx_j$. The *wedge* of two 1-forms is the 2-form given by

$$(\alpha \wedge \beta)(u, v) = \alpha(u)\beta(v) - \alpha(v)\beta(u).$$

In particular $\beta \wedge \alpha = -\alpha \wedge \beta$. The exterior derivative of 1-forms is linear, and it satisfies $d(f\, dg) = df \wedge dg$ and $d(dg) = 0$ for smooth functions f and g. In coordinates,

$$\alpha = \sum_j a_j\, dx_j, \qquad d\alpha = \sum_{i<j} (\partial_i a_j - \partial_j a_i)\, dx_i \wedge dx_j.$$

A smooth map F has a *pullback*, given on 1-forms and 2-forms by

$$(F^*\alpha)_x(u) = \alpha_{F(x)}(dF_x u), \qquad (F^*\omega)_x(u,v) = \omega_{F(x)}(dF_x u, dF_x v).$$

It satisfies $F^*(df) = d(f\circ F)$, $F^*(f\alpha) = (f\circ F)F^*\alpha$, $F^*(\alpha\wedge\beta) = F^*\alpha\wedge F^*\beta$ and $d(F^*\alpha) = F^*(d\alpha)$.

For example, on $\mathbb{R}^2$ we have $d(y\,dx) = dy\wedge dx = -dx\wedge dy$. The map $F(s,t) = (s+t, st)$ into the (p,q)-plane gives

$$F^*(dp\wedge dq) = (ds+dt)\wedge(t\,ds + s\,dt) = (s-t)\,ds\wedge dt.$$

The proof below uses two further identities, which it states.

Theorem 19.2 (Orbit symplectic form)**.** *Every coadjoint orbit has a G-invariant symplectic form ω defined by*

$$\omega_\eta(\xi_X(\eta), \xi_Y(\eta)) = \eta([X,Y]). \tag{19.3}$$

This is the Kirillov–Kostant–Souriau form, with the sign convention of the infinitesimal action (19.1).

Proof. Well-definedness. If X changes by $A\in\mathfrak{g}_\eta$, then $\eta([A,Y]) = 0$. Changing Y has the same effect by antisymmetry. Thus the defining formula (19.3) depends only on the two tangent vectors. It is alternating and bilinear.

Nondegeneracy. If $\omega_\eta(\xi_X,\xi_Y) = 0$ for every Y, then $X\in\mathfrak{g}_\eta$ by the stabiliser formula (19.1). Hence $\xi_X(\eta) = 0$.

Smoothness. The fields ξ_X are smooth and span every tangent space. Choose a local frame from these fields. The defining formula (19.3) gives smooth coefficients in that frame, so ω is smooth.

Invariance. For $g\in G$, let g_* be the differential of the map $\eta\mapsto g\cdot\eta$ on the orbit. Since $g\exp(tX) = \exp(t\,\mathrm{Ad}(g)X)g$, differentiating $g\cdot(\exp(tX)\cdot\eta)$ at $t=0$ gives

$$g_*\xi_X(\eta) = \xi_{\mathrm{Ad}(g)X}(g\cdot\eta).$$

Using that $\mathrm{Ad}(g)$ preserves brackets, we get

$$\begin{aligned}\omega_{g\cdot\eta}(g_*\xi_X, g_*\xi_Y) &= (g\cdot\eta)([\mathrm{Ad}(g)X, \mathrm{Ad}(g)Y])\\ &= \eta([X,Y]).\end{aligned}$$

So the form is invariant.

Closedness. Let $\Phi: G\to O_\ell$ be the orbit map, given by $\Phi(g) = g\cdot\ell$. Put $\beta = \Phi^*\omega$. Invariance makes β a left-invariant form on G, and $\beta_e(X,Y) = \ell([X,Y])$. Exterior differentiation commutes with pullback. For a left-invariant 2-form, the differential at e is

$$(d\beta)_e(X,Y,Z) = -\beta_e([X,Y],Z) + \beta_e([X,Z],Y) - \beta_e([Y,Z],X).$$

We have now stated the two identities imported from Warner, Chapter 2. They are the commutation of d with pullback and the preceding formula for a left-invariant 2-form. Substitution gives

$$(d\beta)_e(X,Y,Z) = -\ell([[X,Y],Z]) + \ell([[X,Z],Y]) - \ell([[Y,Z],X]) = 0$$

by the Jacobi identity. Left invariance gives $d\beta = 0$ everywhere. Therefore $\Phi^*(d\omega) = 0$. Since Φ is a submersion, every tuple of tangent vectors on the orbit lifts to G. This forces $d\omega = 0$. ∎

A nondegenerate alternating form exists only in even dimension. To see this, let A be the matrix of such a form in a basis. Then $A^T = -A$. If the size d of A were odd, then transposition invariance of the determinant would give $\det A = \det(-A) = -\det A$. Hence $\det A = 0$, and the form would be degenerate. In particular, every coadjoint orbit has even dimension. This is a useful first check on a proposed calculation.

Reading the form through linear functions

There is a useful way to remember the sign in the defining formula (19.3). For $X \in \mathfrak{g}$, restrict the linear function $\mu_X(\eta) = \eta(X)$ to an orbit. Its derivative in a tangent direction is

$$(d\mu_X)_\eta(\xi_Y) = (\mathrm{ad}^*(Y)\eta)(X) = -\eta([Y,X]) = \eta([X,Y]).$$

Thus $d\mu_X = \iota_{\xi_X}\omega$, where contraction means $(\iota_V\omega)(W) = \omega(V,W)$. This identifies the infinitesimal action using only the orbit form and the linear functions.

For a smooth real function f, nondegeneracy determines a unique tangent vector field X_f by $\iota_{X_f}\omega = df$. It is called the Hamiltonian vector field in this convention. The preceding calculation says $X_{\mu_X} = \xi_X$. Consequently the bracket

$$\{f,h\} = \omega(X_f, X_h)$$

satisfies $\{\mu_X, \mu_Y\} = \mu_{[X,Y]}$. We only need this identity for linear functions. It says that the Lie bracket is still visible after passing to an orbit.

A central element $W \in \mathfrak{g}$ has $\xi_W = 0$. Thus $d\mu_W = \iota_{\xi_W}\omega = 0$, so μ_W is locally constant on each orbit. It is constant when G is connected. Such a constant can still be the value of a bracket of linear functions. For the Heisenberg basis X, Y and Z of the first example in § 19.1, the identity above gives $\{\mu_X, \mu_Y\} = \mu_Z$, and μ_Z takes one constant value on each orbit.

A change of convention for Hamiltonian vector fields would change some of these signs. Here the definitions, rather than a remembered physics formula, determine them. This calculation concerns finite-dimensional smooth manifolds. It does not differentiate an unbounded operator or assert a quantisation rule.

19.3 Heisenberg planes and points

On a plane, the fields $\xi_X = -\lambda\partial_q$ and $\xi_Y = \lambda\partial_p$ of the first example in § 19.1 span the tangent spaces, and $\xi_Z = 0$. The defining value of the form is $\omega(\xi_X, \xi_Y) = \ell(Z) = \lambda$. Since $(dp \wedge dq)(\xi_X, \xi_Y) = \lambda^2$, we obtain

$$\omega_\lambda = \frac{1}{\lambda}\, dp \wedge dq. \tag{19.4}$$

The coefficient is $1/\lambda$, not λ, because the vectors ξ_X and ξ_Y already carry a factor λ.

By Exercise 19.2, the orbit map pulls ω_λ back to $\lambda\, da \wedge db$ on H. On the vectors (a,b) and (a',b') its value is $\lambda(ab' - a'b)$. This is the exponent of the commutator phase in Lecture 18, where $ab' - a'b$ is drawn as a signed area.

Point orbits have zero-dimensional tangent spaces. The zero form on such a tangent space is nondegenerate, since its kernel is the zero space. The orbit method will associate $(p, q, 0)$ with the one-dimensional character

$$\chi_{p,q}(a, b, c) = e^{i(pa+qb)}.$$

These are different representations for different (p, q). The trivial central character, with $\lambda = 0$, does not mean that the representation is trivial. Only $(0, 0, 0)$ gives the trivial character.

19.4 Polarisations and the orbit theorem

A Lie algebra is *nilpotent* if its lower central series terminates:

$$\mathfrak{g}_1 = \mathfrak{g}, \qquad \mathfrak{g}_{r+1} = [\mathfrak{g}, \mathfrak{g}_r], \qquad \mathfrak{g}_s = 0 \text{ for some } s.$$

For the Heisenberg algebra, $\mathfrak{g}_2 = \mathbb{R}Z$ and $\mathfrak{g}_3 = 0$. Nilpotence is stronger than solvability.

We import a global fact: for a connected simply connected nilpotent group, $\exp : \mathfrak{g} \to G$ is a diffeomorphism, and BCH is a finite polynomial. Every Lie subalgebra $\mathfrak{m}$ gives a closed subgroup $M = \exp \mathfrak{m}$. Indeed, finite BCH keeps products in $\mathfrak{m}$, and the exponential diffeomorphism takes the closed vector subspace $\mathfrak{m}$ to a closed subset. See Knapp, *Lie Groups Beyond an Introduction*, Section I.16.

Definition 19.3. A *real polarisation* for $\ell \in \mathfrak{g}^*$ is a Lie subalgebra $\mathfrak{m} \subset \mathfrak{g}$ such that

$$\ell([\mathfrak{m}, \mathfrak{m}]) = 0, \qquad \dim \mathfrak{m} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

A subspace $A \subseteq \mathfrak{g}$ is called *isotropic* for ℓ when

$$\ell([A, A]) = 0.$$

A real polarisation is thus an isotropic Lie subalgebra of the displayed dimension. This dimension says that the image of $\mathfrak{m}$ in $\mathfrak{g}/\mathfrak{g}_\ell$ is as large as the image of an isotropic subspace can be. In fact, the two displayed conditions force $\mathfrak{g}_\ell \subset \mathfrak{m}$, because an isotropic subspace of maximal dimension contains $\mathfrak{g}_\ell$. The subsection on the dimension of a polarisation below proves both statements.

For such a subalgebra, put

$$\chi_\ell(\exp W) = e^{i\ell(W)}, \qquad W \in \mathfrak{m}.$$

This is well-defined because exp is bijective. It is a character of M. Every BCH correction is a bracket of elements of $\mathfrak{m}$, so ℓ kills it and $\chi_\ell(\exp U \exp V) = \chi_\ell(\exp U)\chi_\ell(\exp V)$.

For example, take the Heisenberg algebra and $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Then $\mathfrak{g}_\ell = \mathbb{R}Z$, so a polarisation has dimension two. Both $\operatorname{span}\{Y, Z\}$ and $\operatorname{span}\{X, Z\}$ are polarisations. All of $\mathfrak{h}$ is too large: $\ell([X, Y]) = \lambda \neq 0$, so the formula above does not define a character of H. The line $\mathbb{R}Z$ is isotropic but too small. The representation $\operatorname{Ind}_{Z(H)}^{H} \chi_\lambda$ of Lecture 17 acts on functions of two variables. Right translations of H preserve its space, since $Z(H)$ is central,

and they are unitary and commute with the action. So it is not irreducible. By the Stone–von Neumann theorem of Lecture 18, it is a direct sum of copies of Π_λ.

Theorem 19.4 (Nilpotent orbit theorem; cited)**.** *Let G be connected, simply connected and nilpotent. Every $\ell \in \mathfrak{g}^*$ has a real polarisation. The unitary induced representation*

$$\Pi_\ell = \operatorname{Ind}_{\exp \mathfrak{m}}^{G} \chi_\ell$$

is irreducible. Its equivalence class does not depend on the polarisation. Two such representations are equivalent exactly when their functionals lie on the same coadjoint orbit. Every irreducible strongly continuous unitary representation of G is unitarily equivalent to some Π_ℓ.

This is Kirillov's theorem, imported from *Lectures on the Orbit Method*, Chapter 3. Induction means unitary induction with the quotient measure convention of Lecture 17. The statement concerns complex Hilbert representations, not just finite-dimensional modules.

The conclusion can fail without these hypotheses. A central quotient can restrict which representations descend (Exercise 19.3). For SU(2) the coadjoint orbits are the origin and the spheres of every radius, as Lecture 27 computes. These form a continuum, while the irreducible representations V_m are labelled by integers. Apart from the trivial group, no compact group and no semisimple group satisfies this theorem's hypotheses. We will meet related orbit constructions in Lecture 27, with a different integrality condition and a different representation model.

The hypotheses are sufficient, not necessary. A solvable group need not be nilpotent. The $ax + b$ group is not nilpotent, yet its orbit list in Exercise 19.1 matches Π_+, Π_- and the characters of Lecture 17 (Kirillov, *Lectures on the Orbit Method*, Chapter 4).

The dimension of a polarisation

We now isolate the linear algebra behind a polarisation. Define the alternating form

$$B_\ell(X, Y) = \ell([X, Y])$$

on $\mathfrak{g}$. Its kernel is $\mathfrak{g}_\ell$. Suppose $\dim \mathcal{O}_\ell = 2d$ and $\dim \mathfrak{g}_\ell = k$. Then $\dim \mathfrak{g} = 2d + k$. The form B_ℓ induces a nondegenerate alternating form on the quotient $\mathfrak{g}/\mathfrak{g}_\ell$, which has dimension $2d$. Under the isomorphism $\mathfrak{g}/\mathfrak{g}_\ell \to T_\ell \mathcal{O}_\ell$ given by $X \mapsto \xi_X(\ell)$, this induced form is ω_ℓ. A vector space with a nondegenerate alternating form is called a *symplectic vector space*.

Let L be a subspace of this quotient. Its *orthogonal* $L^\perp$ consists of the vectors whose induced pairing with every vector of L is zero. The subspace L is *isotropic* when the induced form vanishes on pairs of vectors of L, that is, when $L \subseteq L^\perp$. Nondegeneracy gives $\dim L^\perp = 2d - \dim L$, so an isotropic L has $\dim L \leq d$. Let $A \subset \mathfrak{g}$ be an isotropic subspace for ℓ. Its image in the quotient is isotropic, so this image has dimension at most d. Therefore

$$\dim A = \dim(A \cap \mathfrak{g}_\ell) + \dim\big((A + \mathfrak{g}_\ell)/\mathfrak{g}_\ell\big) \leq k + d.$$

Equality forces $\mathfrak{g}_\ell \subset A$. It also forces the image of A to be a d-dimensional isotropic subspace. The dimension in the definition of a polarisation is exactly $k + d$.

There is one extra demand: A must be a Lie subalgebra. A maximal isotropic vector subspace need not have this property. The existence of a suitable subalgebra is a substantial part of the nilpotent orbit theorem. The definition alone does not prove existence.

This count also predicts the number of variables in an induced model. If $\mathfrak{m}$ is a polarisation, then

$$\dim(G/M) = \dim \mathfrak{g} - \dim \mathfrak{m} = d = \tfrac{1}{2} \dim O_\ell .$$

The Heisenberg plane is two-dimensional, but its representation acts on functions of one real variable. The number of these variables is half the orbit dimension. The orbit does not single out which variables to use. A polarisation makes that choice.

19.5 Induction recovers the Schrödinger formula

Take $\ell = (0, 0, \lambda)$ with $\lambda \neq 0$. Its stabiliser algebra is $\mathbb{R}Z$, so a polarisation must have dimension two. Choose

$$\mathfrak{m} = \operatorname{span}\{Y, Z\}, \qquad M = \{(0, b, c)\}, \qquad \chi_\ell(0, b, c) = e^{i\lambda c}.$$

The subalgebra $\mathfrak{m}$ is abelian and two-dimensional, so it satisfies both polarisation conditions. A section of $H/M \cong \mathbb{R}$ is $s(t) = (t, 0, 0)$. The quotient measure is dt, invariant under the induced translation action.

An induced function satisfies $f(gm) = \chi_\ell(m)^{-1} f(g)$. Write $\varphi(t) = f(s(t))$. Since

$$(a, b, c) = s(a)(0, b, c - ab),$$

its values are

$$f(a, b, c) = e^{-i\lambda(c-ab)} \varphi(a).$$

Left translation acts by $(\Psi_\lambda(g) f)(h) = f(g^{-1}h)$. For $g = (a, b, c)$,

$$g^{-1} s(t) = (t - a, -b, -c + ab).$$

Insert this into the preceding covariance formula. Its central expression becomes $-c + ab - (t - a)(-b) = -c + bt$. Therefore

$$(\Psi_\lambda(a, b, c)\varphi)(t) = e^{i\lambda(c-bt)} \varphi(t - a). \tag{19.5}$$

The operator is unitary: the phase has absolute value one and translation preserves dt. The general unitary induction construction of Lecture 17 supplies the representation law and strong continuity. Both can also be checked from this formula.

Exercise 18.5 carries out the same computation. The sign of the translation differs from the Schrödinger formula of Lecture 18, but it gives an equivalent model. Let $R\varphi(x) = \varphi(-x)$. Then $R^{-1} = R$, and

$$(R\Psi_\lambda(a, b, c)R^{-1}u)(x) = e^{i\lambda(c+bx)} u(x + a) = \Pi_\lambda(a, b, c)u(x).$$

Thus the orbit theorem reproduces the Schrödinger representation with the precise convention already used. The plane's parameter λ determines its nonzero central character.

For a point $(p, q, 0)$, the polarisation is all of $\mathfrak{h}$. Induction from H to itself gives $\chi_{p,q}$. The plane and point calculations therefore recover the complete Heisenberg classification from Lecture 18. The general orbit theorem extends that pattern to every connected simply connected nilpotent group.

The polarisation chooses the variable of the model. On a plane, the group M moves ℓ along lines in the direction of ξ_Y, that is, along ∂_p. By the action formula (19.2), the orbit of $\ell = (0, 0, \lambda)$ has $q = -\lambda a$. So the variable $t = a$ of the model is the transverse coordinate q, up to a factor. The second polarisation $\mathfrak{m}' = \operatorname{span}\{X, Z\}$ moves ℓ along ξ_X, that is, along ∂_q. Its section $s(u) = (0, u, 0)$ gives a model on functions of the transverse coordinate $p = \lambda u$. Figure 19.2 shows both choices. The orbit theorem says that the two models are equivalent, and here the equivalence is a Fourier transform.

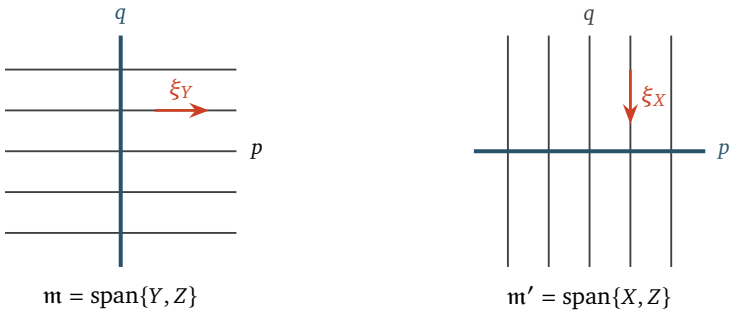


Figure 19.2: Two polarisations on the plane $\lambda = 1$. The parallel lines are the orbits of the subgroup $\exp \mathfrak{m}$ or $\exp \mathfrak{m}'$ on the plane, and the arrow gives their direction. The heavy line is the transverse coordinate, which is the variable of the corresponding induced model up to a factor.

What the Heisenberg orbit list explains

The orbit calculation separates two phenomena. For $\lambda \neq 0$, changing p or q stays within one plane. It cannot change the equivalence class of the representation. For $\lambda = 0$, each point is an orbit of its own, so p and q both survive as representation labels. The discontinuous change in orbit dimension matches the change from infinite-dimensional representations to characters.

One can check the zero-charge conclusion without the general orbit theorem. In an irreducible unitary representation, every central group element acts by a scalar, by the unitary Schur lemma used in Lecture 18. If that scalar character is trivial, the representation factors through $H/Z(H) \cong \mathbb{R}^2$. An irreducible unitary representation of this abelian group is one-dimensional. Its continuous characters are $e^{i(pa+qb)}$, obtained by applying the one-parameter character classification to each coordinate.

For nonzero central character, the Stone–von Neumann theorem gives uniqueness directly. The orbit theorem recovers that result and explains its geometric label. It also shows what to try for a larger nilpotent group: calculate ad^*, find orbit invariants, determine the stabiliser dimension, choose a real polarisation for ℓ, and induce its character. The last exercise carries out the geometric steps in one four-dimensional example.

Remark 19.5. This remark gives a physics dictionary. Compare Lecture 18 with $\lambda = 1/\hbar$. Since $\chi_\ell(\exp W) = e^{i\ell(W)}$, the value $\ell(W)$ corresponds to the operator $-i\,d\Pi(W) = -A_W$. Thus p corresponds to $P/\hbar$ and q to $Q/\hbar$, and $\omega_\lambda = \hbar^{-1} dP \wedge dQ$ is phase-space area in units of $\hbar$. This is a correspondence of labels, not an equality of operators.

Self-checks

1. Why must the stabiliser algebra contain the centre?

Answer. If X is central, then $[X, Y] = 0$ for every Y. The stabiliser formula (19.1) puts X in $\mathfrak{g}_\ell$ for every ℓ.

2. Does the formula $\omega_\lambda = \lambda\, dp \wedge dq$ have the right normalisation?

Answer. No. Its value on $(-\lambda\partial_q, \lambda\partial_p)$ would be λ^3. The required value is λ, so the coefficient is $1/\lambda$.

3. Can one apply the orbit theorem directly to a Heisenberg group with circle centre?

Answer. No. That group is not simply connected. Apply the theorem to its simply connected cover. Then use the Descent theorem of Lecture 4. Only representations trivial on the covering kernel descend.

Summary

The coadjoint action turns $\mathfrak{g}^*$ into a union of even-dimensional symplectic orbits. For connected simply connected nilpotent groups, a polarisation converts an orbit into an irreducible induced representation. For the Heisenberg group, planes give Schrödinger representations and points give characters. The cited classification is exact in this setting. Its hypotheses are part of the result.

Reading

- Kirillov, *Lectures on the Orbit Method*, Chapters 2–3: Heisenberg representations, polarisations and the nilpotent correspondence.
- Kirillov, *Lectures on the Orbit Method*, Chapter 4: the correspondence for exponential groups, which include the $ax + b$ group.
- Knapp, *Lie Groups Beyond an Introduction*, Section I.16: simply connected nilpotent groups.
- Warner, *Foundations of Differentiable Manifolds and Lie Groups*, Chapter 2: differential forms and exterior differentiation. The lecture uses only the rules stated before the orbit form theorem and the two identities stated in its proof.

Lecture 20

Real Forms and Noncompact Structure

Objectives. *A connected real semisimple group with finite centre has a product decomposition into a compact subgroup, an abelian subgroup and a nilpotent subgroup. The orbit theorem of Lecture 19 concerns nilpotent groups and does not apply to such a group, so we start from its structure. This lecture compares compact and split real forms and states the Cartan and Iwasawa decompositions. It proves both decompositions for* $\mathrm{SL}(2,\mathbb{R})$*, where they give coordinates on the upper half-plane.*

By the end of this lecture you will be able to:

- Distinguish the compact and split examples of real forms using their Killing forms and Cartan involutions.
- State the Cartan and Iwasawa decompositions with all their hypotheses.
- Compute $g = kan$ and $g = k \exp X$ for an explicit matrix in $\mathrm{SL}(2,\mathbb{R})$.
- Read both decompositions as coordinates on the upper half-plane.

20.1 Real forms and Cartan involutions

Throughout this section $\mathfrak{g}_{\mathbb{C}}$ is a complex semisimple Lie algebra with Killing form

$$\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \ \operatorname{ad} Y).$$

We cite Knapp, *Lie Groups Beyond an Introduction*, digital second edition 2023, as Knapp Beyond.

Definition 20.1. A *real form* of $\mathfrak{g}_{\mathbb{C}}$ is a real Lie subalgebra $\mathfrak{g} \subset \mathfrak{g}_{\mathbb{C}}$ with $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \oplus i\mathfrak{g}$ as real vector spaces.

A real basis of $\mathfrak{g}$ is then a complex basis of $\mathfrak{g}_{\mathbb{C}}$. For $X \in \mathfrak{g}$, the matrix of $\operatorname{ad} X$ in this basis is real. It is the same over $\mathbb{R}$ and over $\mathbb{C}$. Hence the Killing form of $\mathfrak{g}$ is the restriction $\kappa|_{\mathfrak{g}}$. It is real and nondegenerate. Its signature is unchanged by real isomorphisms.

Definition 20.2. A real form $\mathfrak{g}$ is *compact* if $\kappa|_{\mathfrak{g}}$ is negative definite. It is *split* if it contains an abelian subalgebra $\mathfrak{h}_0$ with the following two properties. Every $\operatorname{ad} H$ with $H \in \mathfrak{h}_0$ is diagonalisable over $\mathbb{R}$. The sum $\mathfrak{h}_0 \oplus i\mathfrak{h}_0$ is a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$.

Every complex semisimple Lie algebra has both types. The Lie Algebras course does not treat real forms, so we cite this fact from Knapp Beyond, Section VI.1 (Theorem 6.6, Corollary 6.10 and Theorem 6.11).

The construction starts from a *Chevalley basis*. This basis consists of the coroots H_α and of root vectors E_α with $[E_\alpha, E_{-\alpha}] = H_\alpha$. Its structure constants are real, and the linear map given by $E_\alpha \mapsto -E_{-\alpha}$ and $H_\alpha \mapsto -H_\alpha$ is an automorphism. Knapp scales these vectors by real factors, which does not change the real spans below. The real span of the Chevalley basis is a split form, with $\mathfrak{h}_0$ the real span of the H_α. The elements iH_α, $E_\alpha - E_{-\alpha}$ and $i(E_\alpha + E_{-\alpha})$ span a compact form. For $\mathfrak{sl}(2,\mathbb{C})$, with $E_\alpha = E$, $E_{-\alpha} = F$ and $H_\alpha = H$, these elements are the matrices A, B and C of the next example.

Example 20.3 proves both claims for $\mathfrak{sl}(2,\mathbb{C})$, and Proposition 20.5 proves them for $\mathfrak{sl}(n,\mathbb{C})$. The general case stays cited.

The name “compact” refers to compact groups. By Lecture 6, the Lie algebra of a compact group has an Ad-invariant inner product, so each $\operatorname{ad} X$ is skew and

$$\kappa(X,X) = \operatorname{tr}\left((\operatorname{ad} X)^2\right) \le 0.$$

So a real form whose Killing form takes a positive value is not the Lie algebra of a compact group.

Example 20.3 (The two forms of $\mathfrak{sl}(2,\mathbb{C})$)**.** On $\mathfrak{sl}(2,\mathbb{C})$ one has $\kappa(X,Y) = 4\operatorname{tr}(XY)$. Take

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $H^2 = I$, $(E+F)^2 = I$ and $(E-F)^2 = -I$. The products

$$H(E+F) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad H(E-F) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (E-F)(E+F) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are traceless. In the basis $(H, E+F, E-F)$ of $\mathfrak{sl}(2,\mathbb{R})$ the Gram matrix of κ is therefore $\operatorname{diag}(8,8,-8)$. The signature is $(2,1)$. The real form $\mathfrak{sl}(2,\mathbb{R})$ is split with $\mathfrak{h}_0 = \mathbb{R}H$, because $\operatorname{ad} H$ has eigenvalues $2, 0, -2$ on E, H, F.

The course basis of $\mathfrak{su}(2)$ is

$$A = iH, \qquad B = E - F, \qquad C = i(E+F).$$

Thus $\kappa(A,A) = i^2 \cdot 8 = -8$, $\kappa(C,C) = -8$ and $\kappa(B,B) = -8$. The cross terms are multiples of the vanishing ones above. The Gram matrix is $-8I$, so $\mathfrak{su}(2)$ is compact. The signatures $(2,1)$ and $(0,3)$ differ, so the two real forms are not isomorphic.

The two real forms in the example share the line $\mathbb{R}B$. The split form adds the plane $\operatorname{span}(H, E+F)$. The compact form adds the plane $i\operatorname{span}(H, E+F)$, spanned by A and C. Thus

$$\mathfrak{sl}(2,\mathbb{R}) = \mathbb{R}B \oplus \operatorname{span}(H, E+F), \qquad \mathfrak{su}(2) = \mathbb{R}B \oplus i\operatorname{span}(H, E+F).$$

A Cartan involution makes this relation precise. Let $\mathfrak{g}$ now be any real semisimple Lie algebra. Write $\kappa(X,Y) = \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ for the Killing form of $\mathfrak{g}$ itself. For a real form of $\mathfrak{g}_{\mathbb{C}}$, this is the restriction $\kappa|_{\mathfrak{g}}$ described above.

Definition 20.4. A *Cartan involution* of $\mathfrak{g}$ is a Lie algebra automorphism θ with $\theta^2 = \mathrm{id}$ such that

$$\kappa_\theta(X,Y) = -\kappa(X,\theta Y)$$

is positive definite on $\mathfrak{g}$.

The form κ_θ is symmetric. Since θ is an automorphism, $[\theta X, Z] = \theta[X, \theta^{-1}Z]$ for all $Z \in \mathfrak{g}$. So $\operatorname{ad}(\theta X) = \theta \operatorname{ad} X\, \theta^{-1}$. Hence $\kappa(\theta X, \theta Y) = \operatorname{tr}(\theta \operatorname{ad} X \operatorname{ad} Y\, \theta^{-1}) = \kappa(X, Y)$. Using $\theta^2 = \mathrm{id}$ and the symmetry of κ, we get

$$\kappa(X, \theta Y) = \kappa(\theta X, \theta^2 Y) = \kappa(\theta X, Y) = \kappa(Y, \theta X).$$

Thus $\kappa_\theta(X, Y) = \kappa_\theta(Y, X)$. So κ_θ is an inner product on $\mathfrak{g}$, as the spectral theorem in § 20.2 requires.

Write $\mathfrak{k}$ and $\mathfrak{p}$ for the $(+1)$- and (-1)-eigenspaces of θ. Since $\theta^2 = \mathrm{id}$, we have $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. Because θ is an automorphism,

$$[\mathfrak{k}, \mathfrak{k}] \subset \mathfrak{k}, \qquad [\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}, \qquad [\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}.$$

For instance, $X, Y \in \mathfrak{p}$ give $\theta[X, Y] = [-X, -Y] = [X, Y]$. We cite from Knapp Beyond, Section VI.2, that every real semisimple Lie algebra has a Cartan involution, unique up to inner automorphism.

Proposition 20.5. *Let $\mathfrak{g} = \mathfrak{sl}(n, \mathbb{R})$ and $\theta(X) = -X^{T}$.*

(a) θ is a Cartan involution. Here $\mathfrak{k} = \mathfrak{so}(n)$, and $\mathfrak{p}$ consists of the real symmetric traceless matrices.

(b) $\mathfrak{k} \oplus i\mathfrak{p} = \mathfrak{su}(n)$, and $\mathfrak{su}(n)$ is a compact real form of $\mathfrak{sl}(n, \mathbb{C})$.

(c) The traceless diagonal matrices form a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. Each $\operatorname{ad} H$, $H \in \mathfrak{a}$, is diagonalisable over $\mathbb{R}$. Hence $\mathfrak{sl}(n, \mathbb{R})$ is split.

Proof. (a) Clearly $\theta^2 = \mathrm{id}$. It is an automorphism because

$$[\theta X, \theta Y] = X^{T}Y^{T} - Y^{T}X^{T} = (YX - XY)^{T} = \theta[X, Y].$$

On $\mathfrak{sl}(n, \mathbb{C})$ we import $\kappa(X, Y) = 2n \operatorname{tr}(XY)$. Then

$$\kappa_\theta(X, X) = 2n \operatorname{tr}(XX^{T}) = 2n \sum_{i,j} X_{ij}^2 > 0 \qquad (X \neq 0).$$

The eigenspace descriptions are immediate.

(b) Write $Z \in \mathfrak{gl}(n, \mathbb{C})$ as $Z = X + iY$ with X, Y real. Then $Z^{*} = X^{T} - iY^{T}$. So $Z^{*} = -Z$ exactly when $X^{T} = -X$ and $Y^{T} = Y$. Also $\operatorname{tr} Z = 0$ exactly when $\operatorname{tr} Y = 0$, since $\operatorname{tr} X = 0$ already. Hence $\mathfrak{su}(n) = \mathfrak{k} \oplus i\mathfrak{p}$. It is a real form because

$$\mathfrak{sl}(n, \mathbb{C}) = (\mathfrak{k} \oplus \mathfrak{p}) \oplus i(\mathfrak{k} \oplus \mathfrak{p}) = (\mathfrak{k} \oplus i\mathfrak{p}) \oplus i(\mathfrak{k} \oplus i\mathfrak{p}).$$

For nonzero $Z \in \mathfrak{su}(n)$, we get $\kappa(Z, Z) = 2n \operatorname{tr}(Z^2) = -2n \operatorname{tr}(Z^{*}Z) < 0$.

(c) Let $H = \operatorname{diag}(h_1, \dots, h_n)$ and let E_{ij} be the matrix units. Then

$$[H, E_{ij}] = (h_i - h_j) E_{ij}.$$

So $\operatorname{ad} H$ is diagonal, with real eigenvalues, in the basis of all E_{ij} with $i \neq j$ together with a basis of $\mathfrak{a}$. For maximality, choose $H \in \mathfrak{a}$ with distinct entries. If $Y \in \mathfrak{p}$ commutes with H, then $(h_i - h_j) Y_{ij} = 0$, so Y is diagonal. Finally, $\mathfrak{a} \oplus i\mathfrak{a}$ is the diagonal Cartan subalgebra of $\mathfrak{sl}(n, \mathbb{C})$. ∎

The recipe $\mathfrak{k} \oplus i\mathfrak{p}$ from the compact-form statement in Proposition 20.5 works in general. If θ is a Cartan involution of $\mathfrak{g}$, then $\mathfrak{k} \oplus i\mathfrak{p}$ is a compact real form of the complexification $\mathfrak{g}_{\mathbb{C}}$ of $\mathfrak{g}$, the complex Lie algebra in which $\mathfrak{g}$ is a real form. This general fact is cited from Knapp Beyond, Sections VI.1–VI.2. For $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ this recipe returns the decomposition of $\mathfrak{su}(2)$ displayed above.

20.2 Restricted roots and the global decompositions

Fix a real semisimple Lie algebra $\mathfrak{g}$ with Cartan involution θ, and write $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$.

Lemma 20.6. *For $X \in \mathfrak{p}$, the operator $\operatorname{ad} X$ is symmetric for κ_θ.*

Proof. Since θ is an automorphism and $\theta X = -X$, we have $[X, \theta Z] = -\theta[X,Z]$. Invariance of κ gives $\kappa([X,Y],W) = -\kappa(Y,[X,W])$. Hence

$$\kappa_\theta([X,Y],Z) = \kappa(Y,[X,\theta Z]) = -\kappa(Y,\theta[X,Z]) = \kappa_\theta(Y,[X,Z]).$$

∎

Choose a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$. The operators $\operatorname{ad} H$, $H \in \mathfrak{a}$, commute and are κ_θ-symmetric. By the spectral theorem they are simultaneously diagonalisable over $\mathbb{R}$.

Definition 20.7. For $\lambda \in \mathfrak{a}^*$ put

$$\mathfrak{g}_\lambda = \{X \in \mathfrak{g} : [H,X] = \lambda(H)X \text{ for all } H \in \mathfrak{a}\}.$$

A *restricted root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. The *real rank* of $\mathfrak{g}$ is $\dim \mathfrak{a}$.

The name refers to the roots of the complexification $\mathfrak{g}_{\mathbb{C}}$. For a Cartan subalgebra of $\mathfrak{g}_{\mathbb{C}}$ containing $\mathfrak{a}$, each restricted root is the restriction to $\mathfrak{a}$ of a root of $\mathfrak{g}_{\mathbb{C}}$. For example, § 20.3 takes $\mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ and $\mathfrak{a} = \mathbb{R}H$, and finds the restricted roots $\pm\lambda_1$ with $\lambda_1(H) = 2$.

Fix $H_0 \in \mathfrak{a}$ with $\lambda(H_0) \neq 0$ for the finitely many restricted roots. Such an H_0 exists, because $\mathfrak{a}$ is not a union of finitely many hyperplanes $\ker\lambda$. Call λ positive if $\lambda(H_0) > 0$, and set

$$\mathfrak{n} = \bigoplus_{\lambda(H_0)>0} \mathfrak{g}_\lambda.$$

The Jacobi identity gives $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$. Hence $\mathfrak{n}$ is a subalgebra. Each $\operatorname{ad} X$ with $X \in \mathfrak{n}$ is nilpotent, because it strictly raises the value at H_0.

For $\mathfrak{su}(n)$ the identity is a Cartan involution, since $\kappa_{\mathrm{id}} = -\kappa$ is positive definite. The two forms of $\mathfrak{sl}(n,\mathbb{C})$ compare as follows.

Real form	θ	$\mathfrak{k}$	$\mathfrak{p}$	real rank
$\mathfrak{su}(n)$, compact	id	$\mathfrak{su}(n)$	0	0
$\mathfrak{sl}(n,\mathbb{R})$, split	$X \mapsto -X^{T}$	$\mathfrak{so}(n)$	symmetric traceless	$n-1$

For $\mathfrak{sl}(n,\mathbb{R})$, with $H_0 = \operatorname{diag}(n-1, n-3, \dots, 1-n)$, the algebra $\mathfrak{n}$ is the strictly upper triangular matrices (Exercise 20.2).

The real rank in the table does not depend on the choices of θ and $\mathfrak{a}$. We cite this from Knapp Beyond, Chapter VI. First fix θ. Any two maximal abelian subspaces of $\mathfrak{p}$ are then conjugate by a finite product of automorphisms $e^{\operatorname{ad} Z}$ with $Z \in \mathfrak{k}$. This conjugacy is a statement about the Lie algebra $\mathfrak{g}$ alone.

Next, let θ' be a second Cartan involution of $\mathfrak{g}$, with (-1)-eigenspace $\mathfrak{p}'$. By the uniqueness statement in § 20.1, $\theta' = \varphi\theta\varphi^{-1}$ for an inner automorphism φ. If $\theta X = -X$, then $\theta'(\varphi X) = \varphi\theta X = -\varphi X$. Symmetrically, $\theta = \varphi^{-1}\theta'\varphi$, so $\varphi^{-1}(\mathfrak{p}') \subseteq \mathfrak{p}$. Thus φ maps $\mathfrak{p}$ onto $\mathfrak{p}'$. Since φ is a Lie algebra automorphism, it maps maximal abelian subspaces of $\mathfrak{p}$ to maximal abelian subspaces of $\mathfrak{p}'$. Hence $\dim \mathfrak{a}$ is unchanged.

Theorem 20.8 (Cartan and Iwasawa decompositions; cited). *Let G be a connected semisimple Lie group with finite centre. Let θ be a Cartan involution of $\mathfrak{g}$, with $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$, and let K be the connected subgroup with Lie algebra $\mathfrak{k}$. Choose $\mathfrak{a}$ and $\mathfrak{n}$ as above, and put $A = \exp \mathfrak{a}$ and $N = \exp \mathfrak{n}$. Then:*

(a) K is compact, and $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, is a diffeomorphism;

(b) A and N are closed subgroups, and $K \times A \times N \to G$, $(k, a, n) \mapsto kan$, is a diffeomorphism.

For proofs see Knapp Beyond, Sections VI.3 and VI.4, or P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18. The finite-centre hypothesis ensures that K is compact.

For $\mathrm{SL}(n, \mathbb{R})$ these decompositions are matrix factorisations. The form $G = KAN$ is the QR decomposition $g = k(an)$, where an is upper triangular with positive diagonal. Gram–Schmidt produces it (Exercise 20.2). The form $G = K \exp \mathfrak{p}$ writes $g = kP$ with $P = \exp X$ symmetric and positive definite. This is the polar decomposition of g, and $g^{T}g = P^{2} = \exp 2X$ determines P. In § 20.3 we prove both forms for $\mathrm{SL}(2, \mathbb{R})$, and we also obtain the singular value decomposition $G = KAK$.

A real form keeps the finite-dimensional complex modules of $\mathfrak{g}_{\mathbb{C}}$, by restriction and complex-linear extension. It does not keep their unitarity. For example, $\mathrm{SL}(2, \mathbb{R})$ acts on $V_m = \operatorname{Sym}^m \mathbb{C}^2$. For $m \geq 1$ and $t \neq 0$ the matrix $\operatorname{diag}(e^{t}, e^{-t})$ has the eigenvalue e^{mt} on V_m, and $|e^{mt}| \neq 1$. So no inner product makes V_m unitary for $\mathrm{SL}(2, \mathbb{R})$, whereas $\mathrm{SU}(2)$ has one by Lecture 5. To build unitary representations of $\mathrm{SL}(2, \mathbb{R})$ we use infinite-dimensional spaces, with K in Lecture 21 and KAN in Lecture 22.

In physics language, the adjoint action identifies $\mathfrak{sl}(2, \mathbb{R})$ with $\mathfrak{so}(2, 1)$. Then $\mathfrak{k}$ is the rotation generator and $\mathfrak{p}$ consists of the boosts, and $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$ says that the commutator of two boosts is a rotation (Lecture 24).

20.3 The rank-one model $\mathrm{SL}(2, \mathbb{R})$

Let $G = \mathrm{SL}(2, \mathbb{R})$ and $\theta(X) = -X^{T}$. By the description of $\mathfrak{sl}(n, \mathbb{R})$ in Proposition 20.5, $\mathfrak{k} = \mathbb{R}(E - F)$ and $\mathfrak{p} = \operatorname{span}(H, E + F)$. Take $\mathfrak{a} = \mathbb{R}H$. The restricted roots are $\pm\lambda_1$ with $\lambda_1(H) = 2$, and $\mathfrak{g}_{\lambda_1} = \mathbb{R}E$ and $\mathfrak{g}_{-\lambda_1} = \mathbb{R}F$. Hence the real rank is one. With $H_0 = H$ we get $\mathfrak{n} = \mathbb{R}E$. The subgroups K, A and N consist of the matrices

$$k_\varphi = \begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^{t} & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with $K = \mathrm{SO}(2) = \{k_\varphi\}$, $A = \{a_t\}$ and $N = \{n_x\}$. We now prove the conclusions of the Cartan and Iwasawa theorem, Theorem 20.8, directly for G, without using that theorem.

Proposition 20.9 (Iwasawa decomposition of $\mathrm{SL}(2,\mathbb{R})$). *The map $K\times A\times N \to G$, $(k,a,n)\mapsto kan$, is a diffeomorphism. Explicitly, for*

$$g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in G, \qquad r = \sqrt{\alpha^2+\gamma^2},$$

one has $g = k_\varphi a_t n_x$ with

$$\cos\varphi = \frac{\alpha}{r}, \qquad \sin\varphi = \frac{\gamma}{r}, \qquad e^t = r, \qquad x = \frac{\alpha\beta+\gamma\delta}{r^2}.$$

Proof. The first column of g is nonzero, so $r > 0$. Let $k = k_\varphi$ with φ as stated. This is Gram–Schmidt applied to the columns of g. Using $\alpha\delta - \beta\gamma = 1$,

$$k^{-1}g = \frac{1}{r}\begin{pmatrix} \alpha & \gamma \\ -\gamma & \alpha \end{pmatrix}\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & \dfrac{\alpha\beta+\gamma\delta}{r} \\ 0 & \dfrac{1}{r} \end{pmatrix}.$$

On the other hand,

$$a_t n_x = \begin{pmatrix} e^t & e^t x \\ 0 & e^{-t} \end{pmatrix}.$$

These agree for $e^t = r$ and $x = (\alpha\beta+\gamma\delta)/r^2$. This proves existence.

For uniqueness, suppose $kan = k'a'n'$. Then $k'^{-1}k = a'n'(an)^{-1}$ is orthogonal and upper triangular with positive diagonal. Its first column is $(p,0)^T$ with $p>0$ and unit length, so $p=1$. The second column is a unit vector orthogonal to $(1,0)^T$ with positive second entry, so it is $(0,1)^T$. Hence $k = k'$ and $an = a'n'$. Comparing diagonals gives $a = a'$, and then $n = n'$.

The multiplication map is smooth. Its inverse is given by the formulas above, which are smooth in the entries of g because $r > 0$. ∎

In particular G is connected and diffeomorphic to a circle times $\mathbb{R}^2$. So $\mathrm{SL}(2,\mathbb{R})$ satisfies the hypotheses of the Cartan and Iwasawa theorem, Theorem 20.8, with centre $\{\pm I\}$.

For the Cartan decomposition we need the exponential on $\mathfrak{p}$. Define the entire functions

$$C(u) = \sum_{k\ge 0}\frac{u^k}{(2k)!}, \qquad S(u) = \sum_{k\ge 0}\frac{u^k}{(2k+1)!}.$$

For $u = s^2 \ge 0$ they are $\cosh s$ and $\sinh s/s$, with the latter interpreted as 1 at $s=0$. We use $u = s^2$ rather than $s = \sqrt{-\det Y}$, because s is not smooth at $Y = 0$. Let $\mathcal{P}$ be the set of symmetric positive definite 2×2 matrices of determinant 1.

Lemma 20.10. *For $Y \in \mathfrak{p}$, set $u = -\det Y \ge 0$. Then $Y^2 = uI$ and*

$$\exp Y = C(u)I + S(u)Y.$$

The map $\exp\colon \mathfrak{p} \to \mathcal{P}$ *is a bijection. Its inverse is*

$$L(Q) = \frac{Q - cI}{\mathcal{S}(C^{-1}(c))}, \qquad c = \tfrac{1}{2}\operatorname{tr} Q,$$

which is smooth on a neighbourhood of $\mathcal{P}$ *among symmetric matrices.*

Proof. Write $Y = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}$. Then $Y^2 = (p^2 + q^2)I = uI$. Splitting the exponential series into even and odd powers gives the formula. The matrix $\exp Y = (\exp \frac{Y}{2})^2$ is the square of an invertible symmetric matrix. Hence $\exp Y$ is positive definite. Moreover, $\det \exp Y = e^{\operatorname{tr} Y} = 1$.

Every coefficient of the series for C' is positive, so $C' > 0$ on $(0, \infty)$. Also, $C'(0) = \frac{1}{2}$. So $C' > 0$ on some interval $(-\varepsilon, \infty)$. Thus C is a strictly increasing bijection $[0, \infty) \to [1, \infty)$, with smooth inverse near $[1, \infty)$. Also $\mathcal{S} > 0$ near $[0, \infty)$.

For injectivity, let $Q = \exp Y$. Then $\operatorname{tr} Q = 2C(u)$, which determines u because C is strictly increasing on $[0, \infty)$. The formula for $\exp Y$ then gives $Y = (Q - C(u)I)/\mathcal{S}(u) = L(Q)$.

For surjectivity, let $Q \in \mathcal{P}$ have eigenvalues $\mu, \mu^{-1} > 0$. Then $c = \frac{1}{2}(\mu + \mu^{-1}) \geq 1$. Put $u = C^{-1}(c)$ and $Y = L(Q)$. This Y is symmetric and traceless. Cayley–Hamilton gives $Q^2 = 2cQ - I$, so $(Q - cI)^2 = (c^2 - 1)I$. Since $u\,\mathcal{S}(u)^2 = C(u)^2 - 1 = c^2 - 1$, we get $Y^2 = uI$. Therefore

$$\exp Y = C(u)I + \mathcal{S}(u)Y = cI + (Q - cI) = Q. \qquad \blacksquare$$

Proposition 20.11 (Cartan decomposition of $\mathrm{SL}(2, \mathbb{R})$)**.** *The map* $K \times \mathfrak{p} \to G$, $(k, X) \mapsto k \exp X$, *is a diffeomorphism. Its inverse is*

$$g \mapsto \bigl(g \exp(-X),\, X\bigr), \qquad X = \tfrac{1}{2} L(g^T g).$$

Proof. Suppose first that $g = k \exp X$ with $k \in K$. Since X is symmetric, so is $\exp X$. Also k is orthogonal. Hence

$$g^T g = \exp(X) k^T k \exp(X) = \exp(2X).$$

By the exponential lemma, Lemma 20.10, $2X = L(g^T g)$, so X and then k are forced. Conversely, given g, the matrix $g^T g$ lies in $\mathcal{P}$. Define X and $k = g \exp(-X)$ as stated. Then

$$k^T k = \exp(-X) \exp(2X) \exp(-X) = I, \qquad \det k = 1,$$

so $k \in K$. The map and its inverse are smooth by the exponential lemma, Lemma 20.10. $\blacksquare$

Corollary 20.12 ($G = KAK$)**.** *Every* $g \in G$ *can be written as* $g = k_1 a_t k_2$ *with* $k_1, k_2 \in K$ *and* $t \geq 0$*, and* t *is unique. The number* e^{2t} *is the larger eigenvalue of* $g^T g$*. So* e^t *is the larger singular value of* g*.*

Proof. Write $g = k \exp X$. The symmetric traceless matrix X has eigenvalues $\pm t$ with $t \geq 0$. Take an orthonormal eigenbasis, ordered $(+t, -t)$, and negate one vector if needed. This gives $k' \in K$ with $X = k'(tH)k'^{-1}$. Then $\exp X = k' a_t k'^{-1}$ and $g = (kk') a_t k'^{-1}$. If $g = k_1 a_t k_2$, then $g^T g = k_2^{-1} a_{2t} k_2$ has eigenvalues $e^{\pm 2t}$, which fixes t. $\blacksquare$

Example 20.13. We compute the Iwasawa, Cartan and KAK data of the lower triangular unipotent matrix

$$g = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The Cartan data involve the golden ratio $\tau = \frac{1+\sqrt{5}}{2}$. Recall $\tau + \tau^{-1} = \sqrt{5}$ and $\tau - \tau^{-1} = 1$.

Iwasawa. Here $r = \sqrt{2}$, so $k = k_{\pi/4}$, $e^t = \sqrt{2}$ and $x = \frac{1}{2}$. Check:

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{\sqrt{2}}{2} \\ 0 & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Cartan. Here

$$Q = g^{T} g = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

The half-trace of Q is $c = \frac{3}{2}$. The equation $C(u) = \cosh\sqrt{u} = \frac{3}{2}$ gives $\sqrt{u} = 2\log\tau$, since $\tau^2 + \tau^{-2} = 3$. Also $\sinh\sqrt{u} = \frac{\sqrt{5}}{2}$. Hence

$$X = \frac{1}{2} L(Q) = \frac{2\log\tau}{\sqrt{5}} \begin{pmatrix} \frac{1}{2} & 1 \\ 1 & -\frac{1}{2} \end{pmatrix} = \frac{\log\tau}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}.$$

Now $-\det X = (\log\tau)^2$, so the exponential lemma, Lemma 20.10, gives

$$\exp X = \frac{\sqrt{5}}{2} I + \frac{1}{2\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Then

$$k = g \exp(-X) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \in \mathrm{SO}(2).$$

The eigenvalues of X are $\pm\log\tau$. Hence, as in the proof of the KAK corollary, Corollary 20.12, the KAK parameter of g is $t = \log\tau$.

20.4 The upper half-plane

Let $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$. For g with entries $\alpha, \beta, \gamma, \delta$ as in the Iwasawa decomposition, Proposition 20.9, put

$$g \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}.$$

Lemma 20.14. *For $z \in \mathbb{H}$ one has $\gamma z + \delta \neq 0$ and*

$$\operatorname{Im}(g \cdot z) = \frac{\operatorname{Im} z}{|\gamma z + \delta|^2}.$$

This defines an action of G on $\mathbb{H}$. The stabiliser of i is K, and $-I$ acts trivially.

Proof. If $\gamma \neq 0$, the root of $\gamma z + \delta$ is real. If $\gamma = 0$, then $\delta \neq 0$. Next,

$$g \cdot z = \frac{(\alpha z + \beta)(\gamma \bar{z} + \delta)}{|\gamma z + \delta|^2},$$

and the numerator has imaginary part $(\alpha\delta - \beta\gamma)\operatorname{Im} z = \operatorname{Im} z$.

For the action law, write $[v] = v_1/v_2$ for $v \in \mathbb{C}^2$ with $v_2 \neq 0$. Then $g \cdot z = [g(z,1)^T]$, and $[\lambda v] = [v]$ for $\lambda \neq 0$. If h has bottom row (γ', δ'), then $h(z,1)^T = (\gamma' z + \delta')(h \cdot z, 1)^T$. Hence

$$(gh) \cdot z = [gh(z,1)^T] = [g(h \cdot z, 1)^T] = g \cdot (h \cdot z).$$

If $g \cdot i = i$, then $\alpha i + \beta = -\gamma + \delta i$. So $\alpha = \delta$ and $\beta = -\gamma$, with $\alpha^2 + \gamma^2 = 1$. That is, $g = k_\varphi$. Conversely, $k_\varphi \cdot i = ie^{i\varphi}/e^{i\varphi} = i$. Finally, $(-I) \cdot z = z$. ∎

Inverting $g^{-1} = kan$ gives $g = n^{-1}a^{-1}k^{-1}$. So the Iwasawa decomposition, Proposition 20.9, also yields a unique decomposition $G = NAK$.

Proposition 20.15. *Write $g = n_x a_t k$. Then*

$$g \cdot i = x + ie^{2t}, \qquad x = \frac{\alpha\gamma + \beta\delta}{\gamma^2 + \delta^2}, \qquad e^{2t} = \frac{1}{\gamma^2 + \delta^2}.$$

The map $n_x a_t \mapsto n_x a_t \cdot i$ is a diffeomorphism $NA \to \mathbb{H}$. The map $gK \mapsto g \cdot i$ is a bijection $G/K \to \mathbb{H}$.

These numbers x and t are not those of the Iwasawa decomposition, Proposition 20.9, of g. From $g = n_x a_t k$ we get $g^{-1} = k^{-1}a_{-t}n_{-x}$. So they are the negatives of the parameters of g^{-1} in $G = KAN$.

Proof. We have $a_t \cdot z = e^{2t}z$ and $n_x \cdot z = z + x$. Since $k \cdot i = i$, this gives $n_x a_t k \cdot i = x + ie^{2t}$. The stated formulas for x and e^{2t} follow by comparing real and imaginary parts in the direct computation

$$g \cdot i = \frac{(\alpha i + \beta)(\delta - \gamma i)}{\gamma^2 + \delta^2} = \frac{\alpha\gamma + \beta\delta + i}{\gamma^2 + \delta^2}.$$

The map $(x,t) \mapsto x + ie^{2t}$ has smooth inverse $z \mapsto (\operatorname{Re} z, \frac{1}{2}\log \operatorname{Im} z)$. Finally, $g \cdot i = g' \cdot i$ exactly when $g^{-1}g' \in K$, by the stabiliser statement of Lemma 20.14. ∎

The action of G on $\mathbb{H}$ is smooth and transitive. So, by the homogeneous-space theorem of Lecture 3, the bijection $G/K \to \mathbb{H}$ is a diffeomorphism.

So Iwasawa coordinates are Cartesian coordinates (x,y) with $y = e^{2t}$. Here N translates horizontally and A dilates from 0. The subgroup NA acts by the affine maps $z \mapsto e^{2t}z + x$; Exercise 20.6 identifies it with the affine group of Lecture 16. Under this identification the left Haar measure $dy\,db/y^2$ of Lecture 16 becomes the measure $dx\,dy/y^2$ on $\mathbb{H}$.

The KAK form describes $\mathbb{H}$ by K-orbits. If $g \cdot i = z$ and $g = k_1 a_t k_2$, then $z = k_1 \cdot (ie^{2t})$. The number $t \geq 0$ depends only on z, because replacing g by gk does not change the eigenvalues of $g^T g$. So $\mathbb{H}$ is the disjoint union of the K-orbits of the points ie^{2t}, $t \geq 0$. Exercise 20.5 shows that these orbits are Euclidean circles.

Figure 20.1 draws both descriptions. It marks the point $g \cdot i = \frac{1}{2}(1+i)$ for the matrix g of Example 20.13. This point has Iwasawa coordinates $x = \frac{1}{2}$ and $e^{2t} = \frac{1}{2}$. Its KAK parameter

is $t = \log \tau$, so it lies on the K-orbit of $i\tau^2$. The derivative of $z \mapsto k_\varphi \cdot z$ at i is $e^{-2i\varphi}$, so k_φ with $\varphi > 0$ turns clockwise around i.

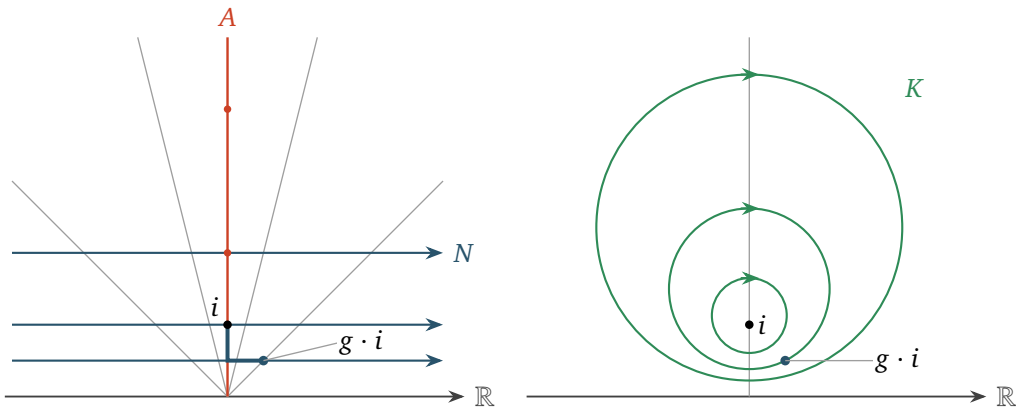


Figure 20.1: Coordinates on $\mathbb{H}$, with the point $g \cdot i$ of the lower triangular example. The first panel shows N-orbits (horizontal lines), A-orbits (rays from 0, with the imaginary axis through i) and a heavy path from i along A and then along N. The second panel shows K-orbits (circles, and the point i itself), with arrows for k_φ, $\varphi > 0$.

20.5 Self-checks

1. What are the real ranks of $\mathfrak{sl}(n, \mathbb{R})$ and $\mathfrak{su}(n)$? What do A and N become for $\mathrm{SU}(n)$?
   Answer. The real rank of $\mathfrak{sl}(n, \mathbb{R})$ is $n - 1$, and the real rank of $\mathfrak{su}(n)$ is 0. For $\mathrm{SU}(n)$, $\mathfrak{p} = 0$, so $A = N = \{I\}$ and the Iwasawa decomposition reads $G = K$.

2. Find the Iwasawa decomposition of $g = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}$.
   Answer. Here $r = 1$, $\cos\varphi = 0$ and $\sin\varphi = -1$, so $k = k_{-\pi/2}$ and $a = I$. Also $x = (0 \cdot 1 + (-1) \cdot 2)/1 = -2$. Indeed
   $$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}.$$

3. Where does $-I$ sit in $G = KAN$, and what does it do on $\mathbb{H}$?
   Answer. In $G = KAN$, the element $-I$ has factors $k = k_\pi$ and $a = n = I$. The element $-I$ acts trivially on $\mathbb{H}$. Hence the action factors through $G/\{\pm I\}$.

Summary

A Cartan involution splits $\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$. The compact form $\mathfrak{k} \oplus i\mathfrak{p}$ of $\mathfrak{g}_{\mathbb{C}}$ has real rank zero. The split form $\mathfrak{sl}(n, \mathbb{R})$ has real rank $n - 1$. For connected semisimple groups with finite centre, the cited Cartan and Iwasawa theorem gives $G = K \exp \mathfrak{p}$ and $G = KAN$. For $\mathrm{SL}(2, \mathbb{R})$ we proved both decompositions, with explicit inverses. On $\mathbb{H}$, identified with G/K, the subgroup NA gives Cartesian coordinates. The KAK form describes $\mathbb{H}$ as the disjoint union of the K-orbits of the points ie^{2t} with $t \geq 0$.

Reading

- Knapp Beyond, Chapter VI, Sections 1–4: compact real forms, the Cartan decomposition on the Lie algebra and group levels, and the Iwasawa decomposition.
- P. Etingof, *Lie Groups and Lie Algebras*, MIT 18.755 lecture notes, Lectures 15 and 18: a second route through real forms and these decompositions.
- Lecture 22 uses $G = KAN$ to construct the principal series.

Lecture 21

Smooth Vectors, K-Types and $(\mathfrak{g}, K)$-Modules

Objectives. *A representation of a noncompact group usually acts on an infinite-dimensional Hilbert space. We use its derivatives on smooth vectors. These operators are in general unbounded, and the space of smooth vectors has no convenient basis. Let K be a compact subgroup. If $\Pi|_K$ is unitary, then Lecture 10 splits the Hilbert space into a Hilbert sum of blocks, one for each K-type. So we pass to the smooth vectors that are finite under K. On these vectors $\mathfrak{g}$ and K act together, subject to compatibility conditions. For $\mathrm{SL}(2,\mathbb{R})$ with $K = \mathrm{SO}(2)$, the study of a representation becomes a raising and lowering calculation.*

By the end of this lecture you will be able to:

- Define smooth vectors, K-finite vectors, K-types and admissibility.
- Prove that the K-finite smooth vectors of a strongly continuous representation on a Hilbert space form a $(\mathfrak{g}, K)$-module.
- Use the compatibility conditions to show that E and F shift the circle K-types of $\mathrm{SL}(2,\mathbb{R})$ by ± 2.
- State which global questions the $(\mathfrak{g}, K)$-module leaves open, and which cited theorems address them.

21.1 Smooth vectors and K-finite vectors

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$. The subgroup $K \subseteq G$ is compact, with Lie algebra $\mathfrak{k} \subseteq \mathfrak{g}$. The representation Π of G on a Hilbert space $\mathcal{H}$ is strongly continuous. Here each operator $\Pi(g)$ is bounded, linear and invertible. The map $g \mapsto \Pi(g)v$ from G to $\mathcal{H}$ is continuous for every $v \in \mathcal{H}$. Unitarity is assumed only where stated.

Definition 21.1. A vector $v \in \mathcal{H}$ is *smooth* if the orbit map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is smooth. This means that in every chart all iterated partial derivatives exist as norm limits and are continuous. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ we set

$$d\Pi(X)v = \left.\frac{d}{dt}\right|_{t=0} \Pi(\exp tX)v.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, because a linear combination of smooth orbit maps is smooth.

For unitary representations, the following facts were imported in Lecture 15. We now also import the density and invariance facts (F1)–(F2) for strongly continuous representations by bounded operators. A source for this more general smooth-vector theorem is Knapp RT, Sections III.3–4. Here Knapp RT is A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (2001).

(F1) *Density* (Gårding). The space $\mathcal{H}^\infty$ is dense in $\mathcal{H}$.

(F2) *Invariance.* Each operator $d\Pi(X)$ maps $\mathcal{H}^\infty$ into itself, and $X \mapsto d\Pi(X)$ is a Lie algebra representation of $\mathfrak{g}$ on $\mathcal{H}^\infty$.

(F3) *Skew-symmetry.* If Π is unitary, then $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for all $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$.

We extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$ by $d\Pi(X + iY) = d\Pi(X) + i\, d\Pi(Y)$ for $X, Y \in \mathfrak{g}$. By the invariance property (F2), this is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{H}^\infty$.

One elementary fact is proved here. If $v \in \mathcal{H}^\infty$ and $h \in G$, then $\Pi(h)v \in \mathcal{H}^\infty$. Indeed, the orbit map of $\Pi(h)v$ is $g \mapsto \Pi(gh)v$. This is the orbit map of v composed with the smooth right translation $g \mapsto gh$.

Definition 21.2. A vector $v \in \mathcal{H}$ is *K-finite* if $\operatorname{span}\{\Pi(k)v : k \in K\}$ is finite-dimensional. We write $\mathcal{H}_K$ for the set of K-finite vectors.

The set $\mathcal{H}_K$ is a subspace, because for $a, b \in \mathbb{C}$ the span of the vectors $\Pi(k)(av + bw)$, $k \in K$, lies in $\operatorname{span}\Pi(K)v + \operatorname{span}\Pi(K)w$.

Let $\widehat{K}$ be the set of equivalence classes of irreducible continuous representations of K. Each class δ has a finite-dimensional representative space V_δ (Lecture 10). Let $\operatorname{Hom}_K(V_\delta, \mathcal{H})$ be the space of linear maps $V_\delta \to \mathcal{H}$ that commute with K.

Definition 21.3. The *multiplicity* of $\delta \in \widehat{K}$ in Π is $m_\delta(\Pi) = \dim \operatorname{Hom}_K(V_\delta, \mathcal{H})$. The class δ is a *K-type* of Π if $m_\delta(\Pi) \geq 1$. The representation Π is *admissible* if $m_\delta(\Pi) < \infty$ for every $\delta \in \widehat{K}$.

Let $v \in \mathcal{H}_K$ and $W_v = \operatorname{span}\Pi(K)v$. Then W_v is finite-dimensional and K-invariant. All norms on W_v are equivalent, so strong continuity makes K act continuously on W_v. By complete reducibility (Lecture 5), W_v is a direct sum of irreducible subspaces. Hence every K-finite vector is a finite sum of vectors from K-types.

For the circle we write k_θ, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, with $k_\theta k_\phi = k_{\theta+\phi}$. Its irreducible continuous representations are the characters $\chi_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 7). We label the K-type χ_n by n.

Example 21.4. Let $G = K = \mathrm{SO}(2)$ and $\mathcal{H} = L^2(\mathbb{R}/2\pi\mathbb{Z})$, with

$$\langle f, h\rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{f(x)}\, h(x)\, dx, \qquad (\Pi(k_\theta)f)(x) = f(x + \theta).$$

This representation is unitary and strongly continuous. The functions $e_n(x) = e^{inx}$ form an orthonormal basis (Lecture 10). Put $\mathcal{H}(n) = \{v : \Pi(k_\theta)v = e^{in\theta}v \text{ for all } \theta\}$. Then $e_n \in \mathcal{H}(n)$.

We show that $\mathcal{H}(n) = \mathbb{C}e_n$. Let $a_m(v) = \langle e_m, v\rangle$. For continuous v, the substitution $y = x + \theta$ and periodicity give

$$a_m(\Pi(k_\theta)v) = \frac{1}{2\pi}\int_0^{2\pi} e^{-im(y-\theta)}v(y)\,dy = e^{im\theta}a_m(v).$$

Both sides are continuous linear functionals of v, so density extends the identity to all L^2 vectors. If $v \in \mathcal{H}(n)$, the left side is also $e^{in\theta}a_m(v)$. For $m \neq n$ take $\theta = \pi/(m-n)$. Then $e^{im\theta} = -e^{in\theta}$, so $a_m(v) = 0$. Hence $v = a_n(v)e_n$.

So $\mathcal{H}_K$ is the space of trigonometric polynomials $f = \sum_{|n|\le N} c_n e_n$. Each such f is smooth, because its orbit map

$$\theta \mapsto \Pi(k_\theta)f = \sum_{|n|\le N} c_n\, e^{in\theta}\, e_n$$

is a finite sum of smooth scalar functions times fixed vectors. Thus every K-finite vector is smooth. Every multiplicity is 1, so Π is admissible. The space $\mathcal{H}_K$ is dense but proper. For example, $\sum_n 2^{-|n|}e_n$ is not K-finite.

A K-finite vector need not be smooth. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by translation, and let $K = \{0\}$. Every vector is K-finite. The indicator function of $[0,1]$ is not smooth (Exercise 21.5). The domain example of Lecture 15 computes the same difference quotients. So we impose both conditions. We write $\mathcal{H}_K^\infty$ for the space of smooth K-finite vectors:

$$\mathcal{H}_K^\infty = \mathcal{H}^\infty \cap \mathcal{H}_K.$$

21.2 The compatibility conditions

Definition 21.5. A $(\mathfrak{g}, K)$-*module* is a complex vector space V with a Lie algebra representation of $\mathfrak{g}$ on V, written $v \mapsto X \cdot v$, and a group homomorphism $K \to \mathrm{GL}(V)$, written $v \mapsto k \cdot v$, such that:

(a) (*finiteness condition*) every $v \in V$ lies in a finite-dimensional K-invariant subspace W on which K acts continuously;

(b) (*derivative condition*) for $X \in \mathfrak{k}$ and $v \in V$, with W as in the finiteness condition (a),

$$X \cdot v = \left.\frac{d}{dt}\right|_{t=0} \exp(tX)\cdot v \qquad \text{(derivative computed in } W\text{)};$$

(c) (*covariance condition*) for $k \in K$, $X \in \mathfrak{g}$ and $v \in V$,

$$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v).$$

A *morphism* of $(\mathfrak{g}, K)$-modules is a linear map that commutes with both actions. A $(\mathfrak{g}, K)$-module V is *admissible* if $\dim \mathrm{Hom}_K(V_\delta, V) < \infty$ for every $\delta \in \widehat{K}$.

The derivative in the derivative condition (b) does not depend on the choice of W. Indeed, two choices W and W' both lie in the finite-dimensional space $W + W'$, and all norms on $W + W'$ are equivalent.

The $\mathfrak{g}$-action extends complex-linearly to $\mathfrak{g}_\mathbb{C}$. Both sides of the covariance condition (c) are complex-linear in X, so the covariance condition also holds for $X \in \mathfrak{g}_\mathbb{C}$.

The derivative condition (b) involves only elements $\exp(tX)$, which lie in the identity component of K. Hence, if K is disconnected, the K-action carries information that $\mathfrak{k}$ cannot detect (Exercise 21.7).

The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. In § 21.3 it gives the shifts of the circle K-types for $\mathrm{SL}(2,\mathbb{R})$.

Proposition 21.6. *Let Π be a strongly continuous representation of G on $\mathcal{H}$, and let $K \subseteq G$ be compact. Then $\mathcal{H}_K^\infty$ is invariant under $\Pi(k)$ for $k \in K$ and under $d\Pi(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. With these actions $\mathcal{H}_K^\infty$ is a $(\mathfrak{g}, K)$-module. Let Π' be a strongly continuous representation of G on a Hilbert space $\mathcal{H}'$. Let $T : \mathcal{H} \to \mathcal{H}'$ be bounded and linear, with $T\Pi(g) = \Pi'(g)T$ for all $g \in G$. Then T maps $\mathcal{H}_K^\infty$ into $\mathcal{H}_K'^\infty$, and its restriction is a morphism.*

Proof. Step 1: the K-action. Let $v \in \mathcal{H}_K^\infty$ and $k \in K$. Then $\Pi(k)v$ is smooth by the right-translation argument in § 21.1. Since $Kk = K$,

$$\operatorname{span} \Pi(K)\Pi(k)v = \operatorname{span} \Pi(Kk)v = \operatorname{span} \Pi(K)v,$$

which is finite-dimensional. So $\Pi(k)v \in \mathcal{H}_K^\infty$. In particular, $W_v = \operatorname{span} \Pi(K)v$ is contained in $\mathcal{H}_K^\infty$.

Step 2: the covariance identity. Let $k \in K$, $X \in \mathfrak{g}$ and $w \in \mathcal{H}^\infty$. We claim

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w. \tag{21.1}$$

By the conjugation formula for exponentials (Lecture 2), $k \exp(tX)k^{-1} = \exp(t\,\mathrm{Ad}(k)X)$. Hence, for $t \neq 0$,

$$\Pi(k)\Big[\frac{\Pi(\exp tX)w - w}{t}\Big] = \frac{\Pi(\exp(t\,\mathrm{Ad}(k)X))\,\Pi(k)w - \Pi(k)w}{t}.$$

Let $t \to 0$. On the left, the bracket tends to $d\Pi(X)w$, and $\Pi(k)$ is bounded. So the left side tends to $\Pi(k)\, d\Pi(X)w$. On the right, $\Pi(k)w$ is smooth, so the quotient tends to $d\Pi(\mathrm{Ad}(k)X)\Pi(k)w$.

Step 3: $\mathfrak{g}$ preserves K-finiteness. This is the main step. The difficulty is that $d\Pi(X)$ does not commute with $\Pi(k)$. So the span of $\Pi(K)\, d\Pi(X)v$ is not obviously finite-dimensional. The covariance identity (21.1) replaces commutation. It moves X to $\mathrm{Ad}(k)X$, which still lies in the finite-dimensional space $\mathfrak{g}$.

Let $v \in \mathcal{H}_K^\infty$ and $W = W_v$. By Step 1, $W \subseteq \mathcal{H}^\infty$, so $d\Pi(X)w$ is defined for $w \in W$. Choose bases $X_1, \dots, X_d$ of $\mathfrak{g}$ and $w_1, \dots, w_r$ of W. Put

$$U = \operatorname{span}\{d\Pi(X_i)w_j : 1 \le i \le d,\ 1 \le j \le r\}.$$

Then $\dim U \le dr$. Let $X \in \mathfrak{g}$, $w \in W$ and $k \in K$. Expand

$$\mathrm{Ad}(k)X = \sum_{i=1}^{d} a_i(k)X_i, \qquad \Pi(k)w = \sum_{j=1}^{r} b_j(k)w_j,$$

with scalars $a_i(k)$ and $b_j(k)$. The second expansion uses $\Pi(k)w \in W$. By the covariance identity (21.1) and linearity of $d\Pi$ in both arguments,

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w = \sum_{i=1}^{d}\sum_{j=1}^{r} a_i(k)\, b_j(k)\, d\Pi(X_i)w_j \in U.$$

In particular $\Pi(K)\, d\Pi(X)v \subseteq U$. So $d\Pi(X)v$ is K-finite. It is smooth by the invariance property (F2). Complex combinations stay in the subspace $\mathcal{H}_K^\infty$, so the same holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

Step 4: the axioms. By the invariance property (F2) and Step 3, $d\Pi$ is a representation of $\mathfrak{g}$ on $\mathcal{H}_K^\infty$. The finiteness condition (a) holds with $W = W_v$, by Step 1 and the continuity of the K-action on W_v shown in § 21.1. For the derivative condition (b), let $X \in \mathfrak{k}$. The curve $t \mapsto \Pi(\exp tX)v$ lies in W_v. Its derivative in $\mathcal{H}$ is $d\Pi(X)v$. A finite-dimensional subspace is closed, and W_v carries the restricted norm. So the difference quotients converge in W_v to $d\Pi(X)v$. The covariance condition (c) is the covariance identity (21.1).

Step 5: intertwiners. Let $v \in \mathcal{H}_K^\infty$. The orbit map $g \mapsto \Pi'(g)Tv = T\Pi(g)v$ is smooth, because T is bounded and linear. Also $\operatorname{span} \Pi'(K)Tv = T(\operatorname{span} \Pi(K)v)$ is finite-dimensional. Differentiating $T\Pi(\exp tX)v = \Pi'(\exp tX)Tv$ at $t = 0$ gives $T\, d\Pi(X)v = d\Pi'(X)Tv$. Commutation with K is the hypothesis. ∎

Definition 21.7. The $(\mathfrak{g}, K)$-*module of* Π is $\mathcal{H}_K^\infty$ with the actions given in the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6. Two representations are *infinitesimally equivalent* if their $(\mathfrak{g}, K)$-modules are isomorphic.

21.3 Circle K-types for $\mathrm{SL}(2, \mathbb{R})$

Now $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. Put

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

The matrix k_θ here is $k_{-\theta}$ in the notation of Lecture 20. Then $\mathfrak{k} = \mathbb{R}W$, and $\theta \mapsto k_\theta$ identifies $\mathbb{R}/2\pi\mathbb{Z}$ with K. So K is connected.

Let V be a $(\mathfrak{g}, K)$-module. For $n \in \mathbb{Z}$ put

$$V(n) = \{v \in V : k_\theta \cdot v = e^{in\theta} v \text{ for all } \theta\}.$$

By the finiteness condition (a) and complete reducibility (Lecture 5), every $v \in V$ is a finite sum of vectors from the spaces $V(n)$. The sum is direct, because the functions $e^{in\theta}$ are linearly independent. So $V = \bigoplus_{n\in\mathbb{Z}} V(n)$, and $\dim V(n)$ is the multiplicity of the K-type n.

Let $H = \operatorname{diag}(1, -1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then H, S, W is a basis of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{R})$. Multiplying matrices gives

$$[H, S] = 2W, \qquad [W, H] = -2S, \qquad [W, S] = 2H.$$

In $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$ we set

$$H' = -iW = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}(H + iS) = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}(H - iS) = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}.$$

The real brackets give

$$\begin{aligned} [H', E] &= -\tfrac{i}{2}\big([W, H] + i[W, S]\big) = -\tfrac{i}{2}(-2S + 2iH) = H + iS = 2E, \\ [H', F] &= -\tfrac{i}{2}\big([W, H] - i[W, S]\big) = -\tfrac{i}{2}(-2S - 2iH) = -(H - iS) = -2F, \\ [E, F] &= \tfrac{1}{4}\big(-i[H, S] + i[S, H]\big) = -\tfrac{i}{2}[H, S] = -iW = H'. \end{aligned}$$

Thus (H', E, F) is an $\mathfrak{sl}_2$-triple. The $\mathfrak{sl}_2$ theory of the prerequisite Lie Algebras course applies to it. The element H' spans $\mathfrak{k}_{\mathbb{C}}$, whereas $H \notin \mathfrak{k}$. Also $F = \overline{E}$, where $\overline{X + iY} = X - iY$ for $X, Y \in \mathfrak{g}$.

These E and F are not the real matrices of Lecture 20. Let $u_\pm = (1, \pm i)^T$, and let Q be the matrix with columns u_+ and u_-. Then

$$H' = Q \operatorname{diag}(1, -1) Q^{-1}, \qquad E = Q \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} Q^{-1}, \qquad F = Q \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} Q^{-1}.$$

The vectors $u_\pm$ are eigenvectors of K, as Example 21.10 shows. In the K-eigenbasis (u_+, u_-), the matrices of (H', E, F) are the standard triple of Lecture 20. Up to a scalar, Q^{-1} is the matrix c of Exercise 20.1.

Lemma 21.8. *For every $\theta \in \mathbb{R}$, the complex-linear extension of $\operatorname{Ad}(k_\theta)$ to $\mathfrak{g}_{\mathbb{C}}$ satisfies $\operatorname{Ad}(k_\theta)H' = H'$, $\operatorname{Ad}(k_\theta)E = e^{2i\theta}E$ and $\operatorname{Ad}(k_\theta)F = e^{-2i\theta}F$.*

Proof. Since $W = iH'$, we get

$$\operatorname{ad}(W)E = i[H', E] = 2iE, \qquad \operatorname{ad}(W)F = i[H', F] = -2iF, \qquad \operatorname{ad}(W)H' = 0.$$

By the relation between Ad and ad in Lecture 2, $\operatorname{Ad}(\exp \theta W) = e^{\theta \operatorname{ad} W}$, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. On an eigenvector of $\operatorname{ad} W$ with eigenvalue μ this operator is multiplication by $e^{\theta\mu}$. The eigenvalues 0, $2i$ and $-2i$ of $\operatorname{ad} W$ on H', E and F give the three formulas. ∎

Exercise 21.1 checks the adjoint lemma by matrix multiplication.

Proposition 21.9. *Let V be a $(\mathfrak{g}, K)$-module for $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. If $v \in V(n)$, then*

$$H' \cdot v = nv, \qquad E \cdot v \in V(n+2), \qquad F \cdot v \in V(n-2).$$

Proof. Let $v \in V(n)$. Since $\exp(\theta W) \cdot v = e^{in\theta} v$, the derivative condition (b) with $X = W$ gives

$$W \cdot v = \frac{d}{d\theta}\Big|_{\theta=0} e^{in\theta} v = in\, v, \qquad \text{so} \qquad H' \cdot v = -i\, W \cdot v = nv.$$

By the covariance condition (c), extended to $X \in \mathfrak{g}_{\mathbb{C}}$, and the adjoint lemma, Lemma 21.8,

$$k_\theta \cdot (E \cdot v) = (\operatorname{Ad}(k_\theta)E) \cdot (k_\theta \cdot v) = e^{2i\theta} E \cdot (e^{in\theta} v) = e^{i(n+2)\theta}\, E \cdot v.$$

So $E \cdot v \in V(n+2)$. The argument for F is the same with $e^{-2i\theta}$, and gives $F \cdot v \in V(n-2)$. ∎

In this proof the shift by 2 comes from the covariance condition (c). For this connected group K, the shift also follows from the derivative condition (b) and the bracket $[H', E] = 2E$. Indeed, this bracket and $H' \cdot v = nv$ show that H' multiplies $E \cdot v$ by $n + 2$. The operator H' acts on each summand $V(m)$ of $V = \bigoplus_{m \in \mathbb{Z}} V(m)$ by m, so $E \cdot v \in V(n+2)$. The bracket $[H', F] = -2F$ gives $F \cdot v \in V(n-2)$ in the same way.

Let Ω be the element

$$\Omega = H'^2 + 2EF + 2FE$$

of the universal enveloping algebra $U(\mathfrak{g}_{\mathbb{C}})$. It is the Casimir element of the triple. By the prerequisite Lie Algebras course, it is central in $U(\mathfrak{g}_{\mathbb{C}})$. Substituting $EF = FE + H'$ gives

$$\Omega = H'^2 + 2H' + 4FE, \qquad \Omega \cdot v = n(n+2)v + 4F \cdot E \cdot v \quad (v \in V(n)). \tag{21.2}$$

By the shift proposition, Proposition 21.9, $F \cdot E \cdot v \in V(n)$. So Ω preserves each $V(n)$.

Example 21.10. The group $\mathrm{SL}(2,\mathbb{R})$ acts on $\mathbb{C}^2$ by matrix multiplication, and hence on $V_m = \mathrm{Sym}^m \mathbb{C}^2$, of dimension $m+1$. The orbit maps are polynomial in the matrix entries, so every vector is smooth. Since V_m is finite-dimensional, every vector is also K-finite. Hence V_m is its own $(\mathfrak{g}, K)$-module. The Lie algebra acts on products by the Leibniz rule. For $u_\pm = (1, \pm i)^T$ we have

$$k_\theta u_+ = \begin{pmatrix} \cos\theta + i\sin\theta \\ -\sin\theta + i\cos\theta \end{pmatrix} = e^{i\theta} u_+, \qquad k_\theta u_- = e^{-i\theta} u_-.$$

Direct multiplication (Exercise 21.2) gives $Eu_- = u_+$, $Eu_+ = 0$, $Fu_+ = u_-$ and $Fu_- = 0$. Let $w_j = u_+^j u_-^{m-j}$ for $0 \le j \le m$. The group acts on each factor, so

$$k_\theta w_j = (e^{i\theta} u_+)^j (e^{-i\theta} u_-)^{m-j} = e^{i(2j-m)\theta} w_j.$$

Hence the K-types of V_m are $-m, -m+2, \dots, m$, each with multiplicity one. The Leibniz rule gives

$$E \cdot w_j = (m-j) w_{j+1} \quad (0 \le j < m), \qquad F \cdot w_j = j w_{j-1} \quad (0 < j \le m).$$

The endpoint actions are $Ew_m = 0$ and $Fw_0 = 0$, in agreement with the shift proposition, Proposition 21.9. Since $E \cdot w_m = 0$ and $w_m \in V(m)$, the Casimir formula (21.2) gives $\Omega \cdot w_m = m(m+2) w_m$. The K-types are the familiar $\mathfrak{sl}_2$-weights, read with respect to the compact element H' rather than the diagonal element H.

For $m \ge 1$ no inner product makes this module unitary. Indeed, $\mathrm{diag}(e^t, e^{-t})$ has the eigenvalue e^{mt} on V_m, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. For this reason we allow nonunitary Π in this lecture.

If Π is unitary and Ω acts on $\mathcal{H}_K^\infty$ by a scalar λ, the skew-symmetry (F3) turns the Casimir formula (21.2) into formulas for $\|d\Pi(E)v\|^2$ and $\|d\Pi(F)v\|^2$. Exercise 21.4 derives them. Here $\langle d\Pi(E)u, w\rangle = -\langle u, d\Pi(F)w\rangle$ for smooth vectors u and w, because $F = \overline{E}$. In a unitary representation of $\mathrm{SU}(2)$ the same triple gives the sign +, because $F = E^*$ as matrices. In the physics notation $n = 2j$, the value $n(n+2)$ in (21.2) is $4j(j+1)$.

Exercise 21.3 builds infinite-dimensional $(\mathfrak{g}, K)$-modules from the same raising and lowering pattern. Its module V_λ has a basis v_n, $n \in 2\mathbb{Z}$, with $E \cdot v_n = v_{n+2}$ and $F \cdot v_n = \frac{1}{4}\big(\lambda - (n-2)n\big) v_{n-2}$. Figure 21.1 compares it with the module V_4 of Example 21.10. The shift proposition, Proposition 21.9, says that every arrow joins neighbouring dots of one parity. A missing arrow means that the operator gives zero. In the figure, generic means that $\lambda + 1$ is not the square of an odd integer. Exercise 21.3 shows that V_λ is then irreducible.

21.4 What the $(\mathfrak{g}, K)$-module does not record

A $(\mathfrak{g}, K)$-module has no topology. It does not record the norm or the action of elements of G outside K. We cite theorems that relate it back to representations. None of them is proved or used later in this lecture.

Admissibility. A *linear connected reductive group* is a closed connected group of real or complex matrices that is stable under conjugate transpose (Knapp RT, Section I.1). Examples are

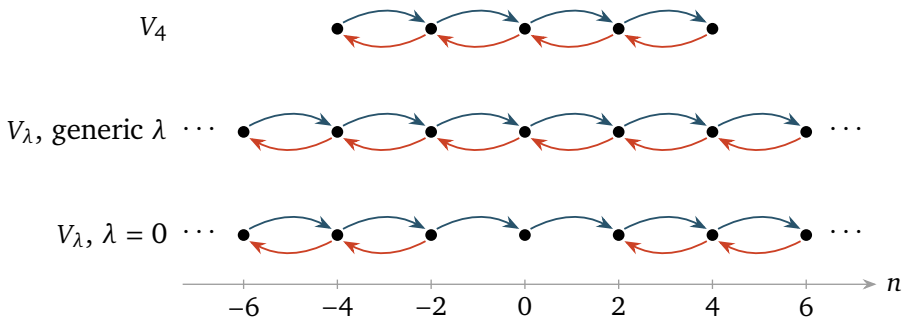


Figure 21.1: Ladders of K-types. Each dot is a one-dimensional space $V(n)$. Upper arrows show E and lower arrows show F. The rows show V_4, the module V_λ of Exercise 21.3 for generic λ, and the same family at $\lambda = 0$, where F vanishes on v_0 and v_2.

$\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(p,q)$. This class is not the class of Lecture 20, as the note at the end of this lecture explains.

A *maximal compact subgroup* of a group is a compact subgroup contained in no strictly larger compact subgroup. For a linear connected reductive group $G \subseteq \mathrm{GL}(n,\mathbb{C})$, the subgroup $K = G \cap \mathrm{U}(n)$ is a maximal compact subgroup (Knapp RT, Section I.1, Proposition 1.2). For $G = \mathrm{SL}(2,\mathbb{R})$ it is $\mathrm{SO}(2)$. If G is also semisimple, then $\theta(X) = -X^*$ is a Cartan involution of $\mathfrak{g}$, and this K is the group K of the Cartan and Iwasawa theorem of Lecture 20.

Let G be a linear connected reductive group and $K = G \cap \mathrm{U}(n)$. The following two statements are in Knapp RT, Chapter VIII, and the note at the end of this lecture gives their numbers.

(i) If a strongly continuous Hilbert representation Π is admissible and $\Pi|_K$ is unitary, then every K-finite vector is smooth, so $\mathcal{H}_K^\infty = \mathcal{H}_K$.

(ii) (Harish-Chandra) Every irreducible unitary representation of G is admissible.

The circle example, Example 21.4, shows the conclusion of the smoothness statement (i) directly in the case $G = K$.

Globalization. Passing from a $(\mathfrak{g}, K)$-module to a continuous group representation is called globalization. A Harish-Chandra module is an admissible $(\mathfrak{g}, K)$-module finitely generated over $U(\mathfrak{g}_\mathbb{C})$. A positive definite Hermitian form on a Harish-Chandra module is invariant if the K-action is unitary and the $\mathfrak{g}$-action is skew-symmetric for real $X \in \mathfrak{g}$.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, an irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization. It is unique up to unitary equivalence. This theorem is cited from P. Etingof's MIT 18.757 notes, Theorem 7.5 and Proposition 7.1. The theorem supplies an action of G on a Hilbert space completing the module, together with its strong continuity. A completion of the vector space alone supplies neither.

One $(\mathfrak{g}, K)$-module can arise from completions of different kinds. Take $G = K = \mathrm{SO}(2)$ as in the circle example, Example 21.4. Translation also acts continuously on the Banach space $C(\mathbb{R}/2\pi\mathbb{Z})$ with the supremum norm, which is not a Hilbert space. With the same definitions, the smooth K-finite vectors of $C(\mathbb{R}/2\pi\mathbb{Z})$ are again the trigonometric polynomials.

So $L^2(\mathbb{R}/2\pi\mathbb{Z})$ and $C(\mathbb{R}/2\pi\mathbb{Z})$ have the same $(\mathfrak{g}, K)$-module. So the module does not record the norm.

This common module has infinitely many distinct circle types and is not finitely generated over $U(\mathfrak{k}_{\mathbb{C}})$. Thus this example is outside the irreducible Harish-Chandra theorem. The more general Casselman–Wallach globalization theorem belongs to further study.

Integration. The identities among $d\Pi(E)$, $d\Pi(F)$ and $d\Pi(H')$ were valid because Π was given. Formal commutation relations among unbounded operators do not by themselves produce a group representation. For the modules of Exercise 21.3, the question of which ones come from unitary representations is taken up in Lecture 22 and Lecture 23.

21.5 Self-checks

Check 1. In Step 3 of the proof of the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6, why must all of W_v consist of smooth vectors?

Answer. The space U is spanned by vectors $d\Pi(X_i)w_j$ with $w_j \in W_v$. These are defined only for smooth w_j. Step 1 shows that each $\Pi(k)v$ is smooth.

Check 2. What are the K-types of Ad on $\mathfrak{g}_{\mathbb{C}}$? Which V_m has the same K-types?

Answer. By the adjoint lemma, Lemma 21.8, the elements E, H' and F have K-types 2, 0, -2, each with multiplicity one. The module V_2 has the same K-types.

Check 3. Could the eigenvalues of $H = \operatorname{diag}(1, -1)$ serve as K-type labels?

Answer. No. The derivative condition (b) relates the K-action only to elements of $\mathfrak{k}_{\mathbb{C}} = \mathbb{C}H'$. Since $H \notin \mathfrak{k}_{\mathbb{C}}$, the action of H is not determined by the K-action. In the V_m example, Example 21.10, the element $H = E + F$ moves $w_j \in V(n)$, where $n = 2j - m$, into $V(n+2) \oplus V(n-2)$.

Summary

Let K be a compact subgroup of G. For a strongly continuous representation of G on a Hilbert space, the smooth K-finite vectors carry an action of $\mathfrak{g}$ and of K. The derivative condition (b) relates the two actions on $\mathfrak{k}$. The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. The proof that $\mathfrak{g}$ preserves K-finiteness uses the covariance condition.

For the circle, the K-finite vectors are the trigonometric polynomials. For $\mathrm{SL}(2, \mathbb{R})$ with $K = \mathrm{SO}(2)$, the covariance condition (c) shows that E and F shift the circle K-types by 2 and -2, respectively. The derivative condition (b) shows that H' acts on the K-type n by multiplication by n. For this connected K, the brackets $[H', E] = 2E$ and $[H', F] = -2F$ then give the same shifts.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, three results are cited. Irreducible unitary representations are admissible. In an admissible strongly continuous Hilbert representation Π with $\Pi|_K$ unitary, K-finite vectors are smooth. An irreducible Harish-Chandra module

with an invariant positive definite Hermitian form has an irreducible unitary globalization, unique up to unitary equivalence.

Sources and hypotheses

The class of groups. Knapp RT, Section I.1, defines linear connected reductive groups, and calls such a group with finite centre linear connected semisimple. Its Proposition 1.2 states that K is compact, connected and maximal compact, and that $K \times \mathfrak{p} \to G$ is a diffeomorphism. This class differs from the class of Lecture 20, which consists of connected semisimple Lie groups with finite centre. For example, the identity component $\mathrm{GL}(n, \mathbb{R})_0$ lies in the present class and is not semisimple, whereas a nonlinear finite cover of $\mathrm{SL}(2, \mathbb{R})$ lies only in the class of Lecture 20. The group $\mathrm{SL}(2, \mathbb{R})$ lies in both.

Admissibility. The smoothness statement (i) is Knapp RT, Proposition 8.5. The definition immediately after the proof of Theorem 8.1 includes the unitarity of $\Pi|_K$ and finite multiplicities, which are the hypotheses of the smoothness statement. The admissibility statement (ii) is Knapp RT, Theorem 8.1.

Globalization. In the notes of Etingof, the unitary globalization is Theorem 7.5, printed p. 38, and its uniqueness is Proposition 7.1, printed p. 37.

Reading

- Knapp RT, Section I.1: linear connected reductive groups.
- Knapp RT, Sections III.3–4: smooth vectors and the facts (F1)–(F3).
- Knapp RT, Chapter VIII: admissible representations and Harish-Chandra's admissibility theorem.
- P. Etingof, MIT 18.757 *Representations of Lie Groups*, lecture notes, Lectures 2 and 5–7: a second account of this material.
- Etingof, Lecture 7, Theorem 7.5 (printed p. 38) and Proposition 7.1 (printed p. 37): the unitary globalization result stated here and its uniqueness. For further study, Wallach, *Real Reductive Groups II*, develops Casselman–Wallach globalization.

Lecture 22

The principal series of $\mathrm{SL}(2,\mathbb{R})$

Objectives. *This lecture builds the principal series of $G = \mathrm{SL}(2,\mathbb{R})$. The principal series is induced from characters of the upper triangular subgroup. We realise it on functions on a circle. We prove unitarity when the parameter is purely imaginary. Then we compute how the Lie algebra moves vectors between K-types. Finally we use this ladder to decide which of these representations are irreducible.*

By the end of this lecture you will be able to:

- State the inducing data (ε, ν) and the role of the modular function.
- Pass between induced functions, homogeneous functions on the plane and the circle model.
- Prove that $\Pi_{\varepsilon,\nu}$ is unitary for $\nu \in i\mathbb{R}$.
- Compute the ladder of K-types, and prove that $\Pi_{\varepsilon,\nu}$ has a proper closed invariant subspace exactly at the points where the ladder breaks.

22.1 Inducing data

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$. We use the matrices

$$k_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad a_t = \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Let $K = \{k_\theta\}$, $A = \{a_t\}$, $N = \{n_x\}$ and $M = \{\pm I\}$. The group M is the centre of G.

Definition 22.1. The *minimal parabolic subgroup* of G is the group $P = MAN$ of upper triangular matrices in G.

The subgroup P is the stabiliser of the line $\mathbb{R}(1,0)^{\mathsf{T}}$. So G/P is the set of lines through the origin of $\mathbb{R}^2$. This set is a circle, and it is compact. We induce from P for this reason: the functions below live on a circle, and their integrals over it are finite.

Each $p \in P$ has a unique form $p = ma_tn_x$. Its diagonal determines m and t; its upper-right entry then determines x. The subgroup N is normal in P, so $ma_tn_x \mapsto ma_t$ is a homomorphism. Note that $K \cap P = M$.

The tangent algebra of P has basis $H = \mathrm{diag}(1,-1)$ and $E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Direct multiplication gives

$$\mathrm{Ad}(a_t)H = H, \qquad \mathrm{Ad}(a_t)E = e^{2t}E, \qquad \mathrm{Ad}(n_x)H = H - 2xE, \qquad \mathrm{Ad}(n_x)E = E,$$

and $\mathrm{Ad}(-I) = \mathrm{id}$. Since Ad and det are multiplicative, these formulas give $|\det \mathrm{Ad}(ma_tn_x)| = e^{2t}$. We use the modular convention of Lecture 16, namely

$$\int f(xg)\,dx = \Delta(g)^{-1} \int f(x)\,dx.$$

With this convention $\Delta(g) = |\det \mathrm{Ad}(g)|^{-1}$. Therefore

$$\Delta_P(ma_tn_x) = e^{-2t}. \tag{22.1}$$

Up to the centre, P is the $ax + b$ group of Lecture 16 and Lecture 17 with $a = e^{2t}$, by Exercise 22.2. So this is its modular function $\Delta = 1/a$.

The group G itself is unimodular because it is connected with semisimple Lie algebra, by Lecture 16.

Definition 22.2. Let $\varepsilon \in \{0, 1\}$ and $\nu \in \mathbb{C}$. The *inducing character* $\chi_{\varepsilon,\nu} \colon P \to \mathbb{C}^\times$ is

$$\chi_{\varepsilon,\nu}(ma_tn_x) = \sigma_\varepsilon(m)\, e^{\nu t}, \qquad \sigma_\varepsilon(\pm I) = (\pm 1)^\varepsilon.$$

It is a continuous character, since it factors through $P \to MA$. It is unitary exactly when $\nu \in i\mathbb{R}$.

Definition 22.3. The space $I_{\varepsilon,\nu}$ consists of the continuous $f \colon G \to \mathbb{C}$ with

$$f(gp) = \Delta_P(p)^{1/2} \chi_{\varepsilon,\nu}(p)^{-1} f(g) \qquad (g \in G,\ p \in P).$$

For $g \in G$, the operator $\Pi_{\varepsilon,\nu}(g)$ on $I_{\varepsilon,\nu}$ is given by $(\Pi_{\varepsilon,\nu}(g)f)(h) = f(g^{-1}h)$ for $h \in G$.

By the modular formula (22.1), the defining condition reads

$$f(g\, ma_tn_x) = \sigma_\varepsilon(m)\, e^{-(\nu+1)t} f(g). \tag{22.2}$$

Left and right translations commute, so $\Pi_{\varepsilon,\nu}(g)$ preserves $I_{\varepsilon,\nu}$. Also $\Pi(g_1g_2) = \Pi(g_1)\Pi(g_2)$. We drop subscripts when (ε, ν) is fixed.

Remark 22.4. The factor $\Delta_P^{1/2}$ gives another form of the normalisation in Lecture 17. In Lecture 17 the square root appears in the action.

Let q be the rho-function of Lecture 17. It is a positive continuous function on G with $q(gp) = \Delta_P(p)q(g)$ for $g \in G$ and $p \in P$. Let f satisfy the unitary covariance condition of Lecture 17, and set $F(g) = \sqrt{q(g)}f(g)$. By the transformation law of q, the square root moves into the covariance condition. The action becomes plain left translation. For $\nu \in i\mathbb{R}$ the resulting condition is

$$|F(gp)|^2 = \Delta_P(p)|F(g)|^2.$$

This condition allows integration over G/P. Below we give the circle integral and prove its invariance directly. Nothing else from Lecture 17 is used.

The shift $e^{-t} = \Delta_P(a_t)^{1/2}$ is often written $e^{-\rho(tH)}$. Here ρ is the linear functional on $\mathbb{R}H$ with $\rho(H) = 1$, half the eigenvalue 2 of $\mathrm{ad}\, H$ on E. It is the analogue of the ρ of Lecture 9.

22.2 The plane model and the circle model

Let $\mathbf{e}_1 = (1,0)^{\mathsf{T}}$ and $u_\theta = (\cos\theta, \sin\theta)^{\mathsf{T}} = k_\theta \mathbf{e}_1$. Multiplying g on the right by n_x leaves the first column of g unchanged and adds a multiple of it to the second column. So a function with $f(gn_x) = f(g)$ depends only on the first column $g\mathbf{e}_1$, which can be any nonzero vector.

Lemma 22.5. *The map $g \mapsto g\mathbf{e}_1$ induces a homeomorphism $G/N \to \mathbb{R}^2 \setminus \{0\}$.*

Proof. If $g\mathbf{e}_1 = \mathbf{e}_1$, the first column of g is $\mathbf{e}_1$, and $\det g = 1$ forces $g \in N$. So the fibres of $g \mapsto g\mathbf{e}_1$ are the cosets gN. The map is onto, with continuous section

$$s(v) = \begin{pmatrix} v_1 & -v_2/|v|^2 \\ v_2 & v_1/|v|^2 \end{pmatrix}, \qquad s(v)\mathbf{e}_1 = v, \qquad \det s(v) = 1.$$

Thus $v \mapsto s(v)N$ is a continuous inverse. ∎

Proposition 22.6. *The rule $F(g\mathbf{e}_1) = f(g)$ is a bijection from $I_{\varepsilon,\nu}$ onto the continuous functions F on $\mathbb{R}^2 \setminus \{0\}$ satisfying*

$$F(cv) = \operatorname{sgn}(c)^{\varepsilon}\,|c|^{-(\nu+1)} F(v) \qquad (c \in \mathbb{R}^\times). \tag{22.3}$$

Under it, $\Pi_{\varepsilon,\nu}(g)$ becomes $F \mapsto F(g^{-1}\cdot)$.

Proof. Taking $p = n_x$ in the covariance law (22.2) shows $f(gn_x) = f(g)$. By the plane identification (Lemma 22.5), F is well defined, and $F = f \circ s$ is continuous. For $c = \pm e^t$ we have $g(\pm a_t)\mathbf{e}_1 = c\,g\mathbf{e}_1$. So the covariance law with $p = \pm a_t$ gives the homogeneity rule (22.3).

Conversely, given F, put $f(g) = F(g\mathbf{e}_1)$. Since $g\,m a_t n_x \mathbf{e}_1 = \pm e^t g\mathbf{e}_1$ for $m = \pm I$, the homogeneity rule gives the covariance law. Finally, $(\Pi(g)f)(h) = f(g^{-1}h) = F(g^{-1}h\mathbf{e}_1)$ for $h \in G$. So $\Pi(g)$ becomes $F \mapsto F(g^{-1}\cdot)$. ∎

A homogeneous function F is determined by its values on the unit circle. Put $\varphi(\theta) = F(u_\theta) = f(k_\theta)$. Then

$$F(ru_\theta) = r^{-(\nu+1)}\varphi(\theta) \quad (r > 0), \qquad \varphi(\theta+\pi) = (-1)^{\varepsilon}\varphi(\theta).$$

The second rule is the homogeneity rule (22.3) with $c = -1$. Let C_ε be the space of continuous 2π-periodic functions with this parity. For every ν, restriction to K identifies $I_{\varepsilon,\nu}$ with C_ε. This identification is the *compact picture*. In the compact picture the space C_ε does not depend on ν.

Restriction to K loses no data because of the Iwasawa decomposition of Lecture 20. It gives $G = KAN$, so $G = KP$. Every $g \in G$ has the form kp, and the defining condition gives $f(kp) = \Delta_P(p)^{1/2}\chi_{\varepsilon,\nu}(p)^{-1} f(k)$. So f is determined by its values on K.

Definition 22.7. For $h \in G$, the map τ_h of $\mathbb{R}/2\pi\mathbb{Z}$ is defined by $hu_\theta = |hu_\theta|\,u_{\tau_h(\theta)}$.

Apply the plane model (Proposition 22.6) with $g^{-1}u_\theta = |g^{-1}u_\theta|\,u_{\tau_{g^{-1}}(\theta)}$. This gives the circle model:

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = |g^{-1}u_\theta|^{-(\nu+1)}\,\varphi\big(\tau_{g^{-1}}(\theta)\big). \tag{22.4}$$

The parameter ν enters only through the scalar factor.

Figure 22.1 draws τ_h for $h = a_{1/2}$. The matrix h turns the unit circle into an ellipse, and each point of the ellipse is projected back to the circle along its ray. Directions crowd towards the horizontal axis.

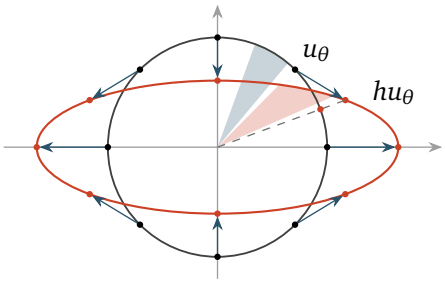


Figure 22.1: The map τ_h for $h = a_{1/2}$. Black dots are the points u_θ at multiples of $\pi/4$, and arrows lead to their images on the ellipse. The dashed ray projects $hu_{\pi/4}$ back to the circle at the angle $\tau_h(\pi/4)$. The thin sector of the disk and its image in the ellipse have equal area.

Example 22.8. For rotations $k_\alpha^{-1} u_\theta = u_{\theta-\alpha}$, so

$$(\Pi(k_\alpha)\varphi)(\theta) = \varphi(\theta - \alpha), \qquad \Pi(-I)\varphi = (-1)^\varepsilon \varphi.$$

The second formula uses $-I = k_\pi$. Thus the centre M acts by σ_ε for every ν. We call ε the central parity.

For the diagonal subgroup, $a_s^{-1} u_\theta = (e^{-s}\cos\theta,\ e^{s}\sin\theta)^{\mathsf{T}}$. Hence

$$r_s(\theta)^2 := |a_s^{-1} u_\theta|^2 = e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta, \qquad \tan \tau_{a_s^{-1}}(\theta) = e^{2s}\tan\theta.$$

When $\varepsilon = 0$ the constant function $e_0 = 1$ lies in C_0, and $\Pi_{0,\nu}(a_s)e_0 = r_s^{-(\nu+1)}$.

22.3 Unitarity on the imaginary axis

Let $\mathcal{H}_\varepsilon$ be the closure of C_ε in $L^2(\mathbb{R}/2\pi\mathbb{Z})$, with inner product

$$\langle \varphi, \psi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \overline{\varphi(\theta)}\, \psi(\theta)\, d\theta.$$

By Fourier analysis on the circle (Lecture 10), $\mathcal{H}_\varepsilon$ is the closed span of the functions $e_n(\theta) = e^{in\theta}$ with $n \equiv \varepsilon \pmod 2$. The key step towards unitarity is a change of variables on the circle.

Lemma 22.9. *Each τ_h is an orientation-preserving diffeomorphism. Moreover $\tau_{h_1 h_2} = \tau_{h_1} \circ \tau_{h_2}$ and*

$$\tau_h'(\theta) = |hu_\theta|^{-2}.$$

Proof. Linear maps commute with positive scaling, so $h_1 h_2 u_\theta$ is a positive multiple of $h_1 u_{\tau_{h_2}(\theta)}$. This gives the composition law, and $\tau_{h^{-1}}$ inverts τ_h. Locally $\tau_h(\theta)$ is a smooth branch of the argument of the nonvanishing smooth curve hu_θ. So τ_h is smooth.

For the derivative, write $r = |hu_\theta|$ and $\tau = \tau_h(\theta)$. Differentiate $hu_\theta = r\,u_\tau$ in θ:

$$hu'_\theta = r'u_\tau + r\tau' u'_\tau, \qquad u'_\theta = (-\sin\theta, \cos\theta)^{\mathsf{T}}.$$

Now compute the determinant of the pair (hu_θ, hu'_θ) in two ways. By multiplicativity of the determinant it equals $\det h \cdot \det(u_\theta, u'_\theta) = 1$. By the first display it also equals

$$\det(ru_\tau,\ r'u_\tau + r\tau' u'_\tau) = r^2\tau' \det(u_\tau, u'_\tau) = r^2\tau'.$$

Hence $\tau' = r^{-2} > 0$. The inverse $\tau_{h^{-1}}$ is smooth by the same argument. ∎

Figure 22.1 shows the reason. A thin sector of angle $d\theta$ in the unit disk has area $\frac{1}{2}d\theta$. Its image under h is, to first order, a sector of radius r and angle $d\tau$, of area $\frac{1}{2}r^2 d\tau$. Since $\det h = 1$, the matrix h preserves area, so $d\theta = r^2 d\tau$.

As a check, take $h = a_s^{-1}$ and differentiate $\tan\tau = e^{2s}\tan\theta$ from the diagonal example (Example 22.8):

$$\tau' = \frac{e^{2s}\sec^2\theta}{1 + e^{4s}\tan^2\theta} = \frac{e^{2s}}{\cos^2\theta + e^{4s}\sin^2\theta} = \frac{1}{e^{-2s}\cos^2\theta + e^{2s}\sin^2\theta} = \frac{1}{r_s(\theta)^2}.$$

Proposition 22.10. *For $\nu \in \mathbb{C}$, $\varphi, \psi \in C_\varepsilon$ and $g \in G$,*

$$\langle \Pi_{\varepsilon,-\bar\nu}(g)\varphi,\ \Pi_{\varepsilon,\nu}(g)\psi\rangle = \langle \varphi, \psi\rangle.$$

Proof. Put $h = g^{-1}$, $r(\theta) = |hu_\theta|$ and $\theta' = \tau_h(\theta)$. Since $r > 0$, we have $\overline{r^{-(-\bar\nu+1)}} = r^{\nu-1}$. By the circle-model formula (22.4), the integrand is

$$\overline{(\Pi_{\varepsilon,-\bar\nu}(g)\varphi)(\theta)}\,(\Pi_{\varepsilon,\nu}(g)\psi)(\theta) = r^{\nu-1}r^{-\nu-1}\,\overline{\varphi(\theta')}\psi(\theta') = \overline{\varphi(\theta')}\psi(\theta')\,r(\theta)^{-2}.$$

By the Jacobian lemma (Lemma 22.9), $r(\theta)^{-2}\,d\theta = d\theta'$. As θ runs once around the circle, so does θ', in the same direction. So the integral equals $\int_0^{2\pi} \overline{\varphi(\theta')}\psi(\theta')\,d\theta'$. After division by 2π, this is $\langle\varphi,\psi\rangle$. ∎

The Jacobian lemma rewrites the circle formula. With $\tau = \tau_{g^{-1}}$ we have $|g^{-1}u_\theta|^{-2} = \tau'(\theta)$, so

$$(\Pi_{\varepsilon,\nu}(g)\varphi)(\theta) = \tau'(\theta)^{(\nu+1)/2}\,\varphi(\tau(\theta)).$$

For $\nu = 0$ the factor is the square root of the Jacobian, and it keeps $|\varphi|^2 d\theta$ fixed when the circle is stretched. This is the square root of Lecture 17. For $\nu \in i\mathbb{R}$ the factor gains a phase of modulus one. For $\nu = -1$ the factor is 1, so φ transforms as a function. For $\nu = 1$ the factor is the Jacobian, so φ transforms as a density and its integral is invariant. Without $\Delta_P^{1/2}$, the two powers of r in the proof above would multiply to 1, and the Jacobian would be left over.

Theorem 22.11. *Let $\nu \in i\mathbb{R}$. Each $\Pi_{\varepsilon,\nu}(g)$ extends uniquely from C_ε to a unitary operator on $\mathcal{H}_\varepsilon$. The map $g \mapsto \Pi_{\varepsilon,\nu}(g)$ is a strongly continuous unitary representation of G.*

Proof. For $\nu \in i\mathbb{R}$ we have $-\bar{\nu} = \nu$. So the pairing identity (Proposition 22.10) says that $\Pi(g)$ preserves norms on C_ε. As C_ε is dense, $\Pi(g)$ extends uniquely to an isometry of $\mathcal{H}_\varepsilon$.

The relation $\Pi(g_1 g_2) = \Pi(g_1)\Pi(g_2)$ holds on C_ε by the plane model (Proposition 22.6). By continuity it holds on $\mathcal{H}_\varepsilon$. So $\Pi(g^{-1})$ inverts $\Pi(g)$. An invertible isometry is unitary.

For strong continuity, fix $\varphi \in C_\varepsilon$ and $g_0 \in G$. By the circle-model formula (22.4), the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is continuous. It is uniformly continuous on $U \times [0, 2\pi]$ for a compact neighbourhood U of g_0. Hence $\sup_\theta |(\Pi(g)\varphi - \Pi(g_0)\varphi)(\theta)| \to 0$ as $g \to g_0$. The L^2 norm is at most this supremum.

For general $\psi \in \mathcal{H}_\varepsilon$ and $\delta > 0$, choose $\varphi \in C_\varepsilon$ with $\|\psi - \varphi\| < \delta$. Since every $\Pi(g)$ is unitary,

$$\|\Pi(g)\psi - \Pi(g_0)\psi\| \le 2\delta + \|\Pi(g)\varphi - \Pi(g_0)\varphi\|,$$

and the right side is below 3δ for g near g_0. ∎

Example 22.12. Unitarity can fail off the imaginary axis. For example, take $\varepsilon = 0$ and $\nu \in \mathbb{R}$. By the diagonal example (Example 22.8),

$$\|\Pi_{0,\nu}(a_s)e_0\|^2 = \frac{1}{2\pi}\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^{\nu+1}}, \qquad A = e^{-2s},\ B = e^{2s}.$$

For $A, B > 0$, the substitution $u = \tan\theta$ on each quarter period gives

$$\int_0^{2\pi} \frac{d\theta}{A\cos^2\theta + B\sin^2\theta} = 4\int_0^{\infty} \frac{du}{A + Bu^2} = \frac{2\pi}{\sqrt{AB}}.$$

Applying $-\partial_A - \partial_B$ to this identity gives

$$\int_0^{2\pi} \frac{d\theta}{(A\cos^2\theta + B\sin^2\theta)^2} = \frac{\pi(A+B)}{(AB)^{3/2}}.$$

Here $AB = 1$. For $\nu = 0$ the norm is 1, as the unitarity theorem (Theorem 22.11) predicts. For $\nu = 1$ the squared norm is $\frac{1}{2}(e^{2s} + e^{-2s}) = \cosh 2s$, which exceeds 1 for $s \neq 0$. Exercise 22.6 gives the general skew-symmetry test.

This shows only that the L^2 norm is not preserved. For $\varepsilon = 0$ and $0 < |\nu| < 1$, Lecture 23 uses a different inner product, which is preserved.

For general $\nu \in \mathbb{C}$ the substitution in the proof of the pairing identity (Proposition 22.10) still gives

$$\|\Pi_{\varepsilon,\nu}(g)\varphi\|^2 \le \max_\theta |g^{-1}u_\theta|^{-2\operatorname{Re}\nu}\,\|\varphi\|^2.$$

So each $\Pi_{\varepsilon,\nu}(g)$ is bounded, locally uniformly in g. The proof of the unitarity theorem (Theorem 22.11) then gives a strongly continuous representation on $\mathcal{H}_\varepsilon$. In its density step, 2δ is replaced by $2C\delta$, where C bounds $\|\Pi_{\varepsilon,\nu}(g)\|$ on a compact neighbourhood of g_0. By the pairing identity (Proposition 22.10), the adjoint is $\Pi_{\varepsilon,\nu}(g)^* = \Pi_{\varepsilon,-\bar{\nu}}(g^{-1})$.

22.4 K-types and the Lie algebra action

By the rotation example (Example 22.8), $\Pi(k_\alpha)e_n = e^{-in\alpha}e_n$. So the character $k_\alpha \mapsto e^{-in\alpha}$ of K occurs exactly once when $n \equiv \varepsilon \pmod 2$, and not at all otherwise. This matches the centre: $k_\pi = -I$ acts on e_n by $(-1)^n = (-1)^\varepsilon$. Let

$$V_{\varepsilon,\nu} = \operatorname{span}\{e_n : n \equiv \varepsilon \pmod 2\} \qquad \text{(finite sums)}.$$

For $\nu \in i\mathbb{R}$ this is the space of K-finite vectors of Lecture 21.

For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g} = \mathfrak{sl}(2,\mathbb{R})$ put

$$d\Pi(X)\varphi = \frac{d}{ds}\Big|_{s=0} \Pi_{\varepsilon,\nu}(\exp sX)\varphi.$$

By the circle-model formula (22.4), the function $(s,\theta) \mapsto (\Pi(\exp sX)\varphi)(\theta)$ is smooth. So the difference quotients converge uniformly in θ. Hence they converge in $\mathcal{H}_\varepsilon$.

More generally, the function $(g,\theta) \mapsto (\Pi(g)\varphi)(\theta)$ is jointly smooth. On compact sets in a group chart, the difference quotients for its derivatives converge uniformly in θ. Hence φ is a smooth vector in the Hilbert representation, in the sense used in Lecture 21.

For $\nu \in i\mathbb{R}$ this derivative agrees with the derivative on smooth vectors in Lecture 15. We extend $d\Pi$ complex linearly to $\mathfrak{g}_\mathbb{C}$.

Lemma 22.13. *For smooth $\varphi \in C_\varepsilon$ and $X \in \mathfrak{g}$, using the dot product of $\mathbb{R}^2$,*

$$(d\Pi(X)\varphi)(\theta) = (\nu+1)\,(u_\theta \cdot Xu_\theta)\,\varphi(\theta) - \det(u_\theta, Xu_\theta)\,\varphi'(\theta).$$

Proof. Let $v(s) = \exp(-sX)u_\theta = r(s)u_{\tau(s)}$. Then $r(0)=1$, $\tau(0)=\theta$ and $v'(0) = -Xu_\theta$. As in the proof of the Jacobian lemma (Lemma 22.9), $\det(v,v') = r^2\tau'$, so $\tau'(0) = -\det(u_\theta, Xu_\theta)$. Also $(r^2)' = 2\,v\cdot v'$, so $r'(0) = -u_\theta \cdot Xu_\theta$. Now differentiate $r(s)^{-(\nu+1)}\varphi(\tau(s))$ at $s=0$. The product rule with these values of $r'(0)$ and $\tau'(0)$ gives the stated formula. ∎

We use the basis

$$W = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

with $\exp(\alpha W) = k_\alpha$. We have $Wu_\theta = u_\theta'$, $Hu_\theta = (\cos\theta, -\sin\theta)^{\mathsf T}$ and $Qu_\theta = (\sin\theta, \cos\theta)^{\mathsf T}$. The derivative formula (Lemma 22.13) gives the table:

X	$u_\theta \cdot Xu_\theta$	$\det(u_\theta, Xu_\theta)$	$d\Pi(X)\varphi$
W	0	1	$-\varphi'$
H	$\cos 2\theta$	$-\sin 2\theta$	$(\nu+1)\cos 2\theta\,\varphi + \sin 2\theta\,\varphi'$
Q	$\sin 2\theta$	$\cos 2\theta$	$(\nu+1)\sin 2\theta\,\varphi - \cos 2\theta\,\varphi'$

Set $X_\pm = H \pm iQ \in \mathfrak{g}_\mathbb{C}$. Since $\sin 2\theta \mp i\cos 2\theta = \mp i e^{\pm 2i\theta}$,

$$d\Pi(X_\pm)\varphi = e^{\pm 2i\theta}\big((\nu+1)\varphi \mp i\varphi'\big).$$

The element W used here is the negative of the W in Lecture 21, so k_α here is $k_{-\alpha}$ there. In the notation of Lecture 21, $E = X_+/2$, $F = X_-/2$ and $H' = iW$. Thus the Fourier vector e_n has type n in that lecture's parametrisation, even though its character here is $e^{-in\alpha}$. These E and F are not the real matrices of § 22.1, where E is upper triangular.

Proposition 22.14. *For $n \equiv \varepsilon \pmod 2$,*

$$d\Pi(W)e_n = -in\,e_n, \qquad d\Pi(X_+)e_n = (\nu+1+n)\,e_{n+2}, \qquad d\Pi(X_-)e_n = (\nu+1-n)\,e_{n-2}.$$

In the letters of Lecture 21, $d\Pi(E)e_n = \tfrac{1}{2}(\nu+1+n)e_{n+2}$ and $d\Pi(F)e_n = \tfrac{1}{2}(\nu+1-n)e_{n-2}$.

Proof. Insert $e'_n = in\,e_n$ into the table row for W and into the display for $d\Pi(X_\pm)$. For example, $(\nu+1) - i(in) = \nu+1+n$. ∎

The ladder formulas also show that $d\Pi$ respects brackets, for every $\nu \in \mathbb{C}$. On e_n the eigenvalues of $d\Pi(W)$ give

$$[d\Pi(W), d\Pi(X_\pm)]e_n = \big(-i(n\pm 2) + in\big)(\nu+1\pm n)\,e_{n\pm 2} = \mp 2i\,d\Pi(X_\pm)e_n.$$

The products of the two ladder operators are

$$d\Pi(X_\mp)d\Pi(X_\pm)e_n = (\nu+1\pm n)(\nu-1\mp n)\,e_n = \big(\nu^2 - (1\pm n)^2\big)e_n.$$

Subtracting the two cases gives

$$[d\Pi(X_+), d\Pi(X_-)]e_n = \big((1+n)^2 - (1-n)^2\big)e_n = 4n\,e_n = 4i\,d\Pi(W)e_n.$$

In $\mathfrak{g}_{\mathbb{C}}$ the brackets $[W,H] = 2Q$, $[W,Q] = -2H$ and $[H,Q] = -2W$ give $[W, X_\pm] = \mp 2iX_\pm$ and $[X_+, X_-] = 4iW$ (Exercise 22.4). The elements W, X_+ and X_- span $\mathfrak{g}_{\mathbb{C}}$, and $d\Pi$ is complex linear. Hence $d\Pi([Y,Z]) = [d\Pi(Y), d\Pi(Z)]$ for all $Y, Z \in \mathfrak{g}_{\mathbb{C}}$. So $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $V_{\varepsilon,\nu}$ for every $\nu \in \mathbb{C}$.

Proposition 22.15. *A subspace $U \subseteq V_{\varepsilon,\nu}$ is stable under $\Pi(K)$ and $d\Pi(\mathfrak{g})$ exactly when $U = \operatorname{span}\{e_n : n \in S\}$ for a set S such that the following two conditions hold. If $n \in S$ and $\nu+1+n \neq 0$, then $n+2 \in S$. If $n \in S$ and $\nu+1-n \neq 0$, then $n-2 \in S$. Hence the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is irreducible if and only if ν is not an integer with $\nu \equiv \varepsilon + 1 \pmod 2$.*

Proof. Let U be K-stable and let $u = \sum c_n e_n \in U$ be a finite sum. The e_n with $c_n \neq 0$ span a finite-dimensional K-stable space F. The continuous curve $\alpha \mapsto e^{in\alpha}\Pi(k_\alpha)u$ lies in the closed subspace $U \cap F$ of F. Hence

$$\frac{1}{2\pi}\int_0^{2\pi} e^{in\alpha}\,\Pi(k_\alpha)u\,d\alpha = c_n e_n$$

lies in U. Thus U is spanned by the e_n it contains. Since H and Q are linear combinations of $X_\pm$, and $d\Pi(W)$ preserves every span of the e_n, stability under $d\Pi(\mathfrak{g})$ means stability under $d\Pi(X_\pm)$. By the ladder formulas (Proposition 22.14), this is the stated property of S.

If no coefficient $\nu+1\pm n$ vanishes, every nonempty admissible S contains all $n \equiv \varepsilon$. If $\nu+1+n_0 = 0$, then $S = \{n \le n_0\}$ is admissible and proper. If $\nu+1-n_0 = 0$, use $S = \{n \ge n_0\}$. Finally, $\nu = -(n_0+1)$ or $\nu = n_0 - 1$ with $n_0 \equiv \varepsilon$ means exactly that $\nu \in \mathbb{Z}$ and $\nu \equiv \varepsilon + 1 \pmod 2$. ∎

Figure 22.2 shows the criterion at four parameters.

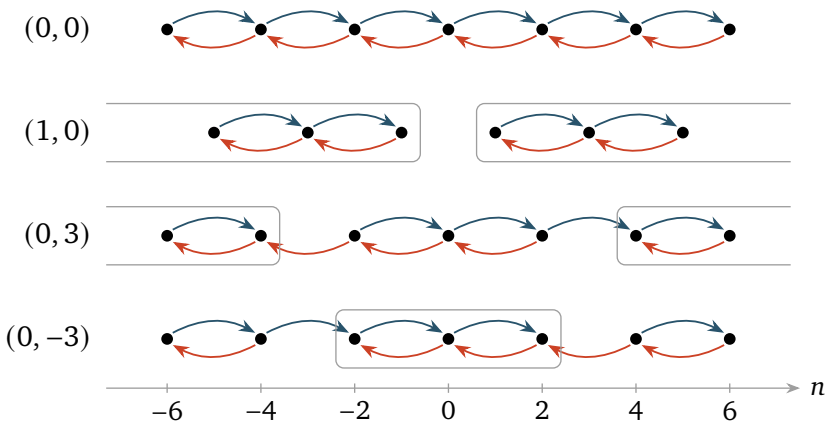


Figure 22.2: The ladder of $V_{\varepsilon,\nu}$, with rows labelled (ε,ν). Each dot is the one-dimensional space $\mathbb{C}e_n$; the rows continue in both directions. Upper arrows show $d\Pi(X_+)$ and lower arrows show $d\Pi(X_-)$. An arrow is missing where its coefficient vanishes. An invariant set of dots is one that no arrow leaves, and the boxes mark such sets.

22.5 Closed invariant subspaces

We now pass from the module $V_{\varepsilon,\nu}$ to the Hilbert representation on $\mathcal{H}_\varepsilon$. A $(\mathfrak{g},K)$-submodule is an algebraic object and a closed invariant subspace is an analytic one, so the submodule criterion does not settle the question alone. In general the passage between them is Harish-Chandra's correspondence for admissible representations. For this family we prove what we need directly. First, a closed invariant subspace contains Fourier vectors. Next, a half-ladder has an invariant closed span.

Lemma 22.16. *Let $Y \neq 0$ be a closed subspace of $\mathcal{H}_\varepsilon$ invariant under $\Pi_{\varepsilon,\nu}(G)$. If $v \in Y$ and $\langle e_k, v\rangle \neq 0$, then $e_k \in Y$. If $e_k \in Y$, then $d\Pi(X_\pm)e_k \in Y$.*

Proof. By the rotation example (Example 22.8), the Fourier coefficient of $\Pi(k_\alpha)v$ at m is $e^{-im\alpha}\langle e_m, v\rangle$. So the integral

$$\frac{1}{2\pi}\int_0^{2\pi} e^{ik\alpha}\,\Pi(k_\alpha)v\,d\alpha = \langle e_k, v\rangle e_k$$

has Fourier coefficient $\langle e_k, v\rangle$ at k and 0 elsewhere. It is a norm limit of Riemann sums of vectors in Y, because the integrand is continuous. Since Y is closed, $e_k \in Y$.

The vector e_k is smooth, as shown before the derivative formula (Lemma 22.13). For $X \in \mathfrak{g}$, the quotient $t^{-1}(\Pi(\exp tX)e_k - e_k)$ lies in Y and converges in norm to $d\Pi(X)e_k$. So $d\Pi(X)e_k \in Y$. Complex combinations give $d\Pi(X_\pm)e_k \in Y$. ∎

Let $n \geq 1$ be an integer and let $\varepsilon \equiv n \pmod 2$. At $\nu = n-1$ the ladder shows the half-ladder $k \geq n$. We define the closed span

$$H_n^+ = \overline{\operatorname{span}}\{e_{n+2m} : m \geq 0\} \subset \mathcal{H}_\varepsilon,$$

and likewise $H_n^- = \overline{\operatorname{span}}\{e_{-n-2m} : m \geq 0\}$. The ladder alone does not show that H_n^+ is invariant under the group, since $\Pi(g)$ is not a polynomial in $d\Pi$. We use power series in $z = e^{2i\theta}$. For a polynomial f put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = e_{n+2m}.$$

We use polynomials because a general vector of H_n^+ need not extend to a function holomorphic near the closed unit disk.

We also fix the complex matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$. For $g \in G$ with $g^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiplying out gives

$$\gamma g^{-1} \gamma^{-1} = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{22.5}$$

The determinant gives $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. So $|\beta| < |\alpha|$.

Proposition 22.17. *Let $n \geq 1$, $\varepsilon \equiv n \pmod 2$, $g \in G$, and let f be a polynomial. With $z = e^{2i\theta}$ and α, β as in (22.5),*

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = e^{in\theta}\,(\bar\beta z + \bar\alpha)^{-n}\, f\Big(\frac{\alpha z + \beta}{\bar\beta z + \bar\alpha}\Big). \tag{22.6}$$

Proof. For a real vector v we have $\gamma v = (v_1 + iv_2,\ v_1 - iv_2)^{\mathsf T}$. Hence $\gamma u_\theta = e^{-i\theta}(z, 1)^{\mathsf T}$. Write $g^{-1} u_\theta = r u_\tau$, with $r = |g^{-1} u_\theta|$ and $\tau = \tau_{g^{-1}}(\theta)$. Apply γ to both sides. Since $\gamma g^{-1} = (\gamma g^{-1} \gamma^{-1})\gamma$, this gives

$$r e^{-i\tau} \begin{pmatrix} e^{2i\tau} \\ 1 \end{pmatrix} = e^{-i\theta} \begin{pmatrix} \alpha z + \beta \\ \bar\beta z + \bar\alpha \end{pmatrix}.$$

The second entries give $\bar\beta z + \bar\alpha = r e^{i(\theta - \tau)}$. Dividing the first entries by the second gives $e^{2i\tau} = (\alpha z + \beta)/(\bar\beta z + \bar\alpha)$. Now the circle-model formula (22.4) with $\nu + 1 = n$ gives

$$(\Pi_{\varepsilon,n-1}(g) U_n f)(\theta) = r^{-n} e^{in\tau} f(e^{2i\tau}) = e^{in\theta}\big(r e^{i(\theta-\tau)}\big)^{-n} f(e^{2i\tau}),$$

which is the stated formula. ∎

Proposition 22.18. *Let $n \geq 1$ and $\varepsilon \equiv n \pmod 2$. The subspaces H_n^+ and H_n^- are invariant under $\Pi_{\varepsilon,n-1}$. Their orthogonal complements in $\mathcal{H}_\varepsilon$ are invariant under $\Pi_{\varepsilon,-(n-1)}$.*

Proof. The half H_n^+. Let $g \in G$, and let f be a polynomial. Since $|\beta/\alpha| < 1$, the geometric series

$$(\bar\beta w + \bar\alpha)^{-1} = \bar\alpha^{-1} \sum_{l \geq 0} (-\bar\beta w/\bar\alpha)^l$$

converges absolutely and uniformly on $|w| \leq 1$, and so do its powers. Therefore the function of w obtained from the right side of (22.6) has a power series $\sum_{l \geq 0} b_l w^l$ that converges uniformly on $|w| = 1$. Setting $w = e^{2i\theta}$, we see that $\Pi_{\varepsilon,n-1}(g) U_n f$ is a uniform limit, hence an L^2 limit, of finite combinations of the e_{n+2l}. So it lies in H_n^+. The vectors $U_n f$ span a dense subspace of H_n^+, and $\Pi_{\varepsilon,n-1}(g)$ is bounded. Hence $\Pi_{\varepsilon,n-1}(g) H_n^+ \subset H_n^+$ for every g.

The half H_n^-. For real ν the factor in (22.4) is real, so complex conjugation commutes with $\Pi_{\varepsilon,\nu}(g)$. Conjugation carries H_n^+ onto H_n^-. So H_n^- is invariant at $\nu = n - 1$.

The complements. By the adjoint formula after the pairing identity (Proposition 22.10), $\Pi_{\varepsilon,-(n-1)}(g)^* = \Pi_{\varepsilon,n-1}(g^{-1})$. Let $v \perp H_n^{\pm}$ and $u \in H_n^{\pm}$. Then

$$\langle u, \Pi_{\varepsilon,-(n-1)}(g)v\rangle = \langle \Pi_{\varepsilon,n-1}(g^{-1})u, v\rangle = 0.$$

■

Theorem 22.19 (Irreducibility; the two equivalences are cited)**.** *Let $\nu \in \mathbb{C}$. The representation $\Pi_{\varepsilon,\nu}$ on $\mathcal{H}_\varepsilon$ has no closed invariant subspace other than 0 and $\mathcal{H}_\varepsilon$ exactly when $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$. For $\nu \in i\mathbb{R}$, all $\Pi_{\varepsilon,\nu}$ are irreducible except $\Pi_{1,0}$. The space $\mathcal{H}_1$ of $\Pi_{1,0}$ is the orthogonal sum of the closed spans of $\{e_n : n > 0\}$ and $\{e_n : n < 0\}$, and each summand is irreducible.*
Moreover, for $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$ we have $V_{\varepsilon,\nu} \cong V_{\varepsilon,-\nu}$ as $(\mathfrak{g}, K)$-modules. For $\nu \in i\mathbb{R}$, the representations $\Pi_{\varepsilon,\nu}$ and $\Pi_{\varepsilon,-\nu}$ are unitarily equivalent. These two equivalences are cited from Knapp, Representation Theory of Semisimple Groups*, §II.5, and Etingof's 18.757 notes, Lecture 19.*

Proof. We prove the first paragraph.

Irreducibility. Let $\nu \notin \varepsilon + 1 + 2\mathbb{Z}$, and let $W \neq 0$ be a closed invariant subspace. A nonzero vector of W has a nonzero Fourier coefficient, so $e_k \in W$ for some k, by Lemma 22.16. No coefficient $\nu + 1 \pm n$ with $n \equiv \varepsilon$ vanishes. So the ladder formulas (Proposition 22.14) and Lemma 22.16 put every e_n with $n \equiv \varepsilon$ in W. Hence $W = \mathcal{H}_\varepsilon$.

Reducibility. Let $\nu \in \varepsilon + 1 + 2\mathbb{Z}$. Then $\nu = n - 1$ or $\nu = -(n-1)$ for an integer $n \geq 1$ with $n \equiv \varepsilon \pmod 2$. By Proposition 22.18, H_n^+ or its orthogonal complement is a closed invariant subspace in the two cases. Each is nonzero and proper.

The imaginary axis. A number $\nu \in i\mathbb{R}$ lies in $\varepsilon + 1 + 2\mathbb{Z}$ only for $\nu = 0$ and $\varepsilon = 1$. So the exception is $\Pi_{1,0}$.

The point $(1,0)$. Take $n = 1$. Then H_1^+ and H_1^- are invariant, mutually orthogonal, and together they span $\mathcal{H}_1$. Let $W \neq 0$ be a closed invariant subspace of H_1^+. A nonzero vector of W has a nonzero coefficient at some odd $k \geq 1$, so $e_k \in W$. For odd $n \geq 1$ we have $1 + n \neq 0$, and for odd $n \geq 3$ we have $1 - n \neq 0$. So the ladder puts every e_n with odd $n \geq 1$ in W. Hence $W = H_1^+$. Complex conjugation gives the same result for H_1^-. ■

The pieces at the reducibility points return in Lecture 23.

The reducible module at $(\varepsilon, \nu) = (1, 0)$

The simplest reducibility point is $(\varepsilon, \nu) = (1, 0)$. The Fourier labels are the odd integers, and the ladder formulas become

$$d\Pi(X_+)e_n = (1+n)e_{n+2}, \qquad d\Pi(X_-)e_n = (1-n)e_{n-2}.$$

At $n = -1$, the upward coefficient vanishes. At $n = 1$, the downward coefficient vanishes. Thus the space $V_{1,0}$ is the direct sum

$$V_{1,0} = \operatorname{span}\{e_1, e_3, e_5, \ldots\} \;\oplus\; \operatorname{span}\{e_{-1}, e_{-3}, e_{-5}, \ldots\}.$$

Each summand is preserved by both ladder operators and by K. Within either chain, the remaining connecting coefficients are nonzero. The weight-projection argument in the

proof of the submodule criterion (Proposition 22.15) shows that each chain is algebraically irreducible.

The irreducibility theorem (Theorem 22.19) gives the two closed irreducible summands. Their group invariance is Proposition 22.18, which goes beyond the algebraic calculation above. Lecture 23 identifies them as limits of discrete series and explains their norms. The full odd ladder breaks at the gap between -1 and 1.

For comparison, $(\varepsilon,\nu)=(0,0)$ has only even labels. Neither equation $1+n=0$ nor $1-n=0$ has an even solution. Its algebraic module is therefore irreducible.

We compare the two modules through the Casimir element $\Omega = H^2+Q^2-W^2$ of $U(\mathfrak{g}_{\mathbb{C}})$. Exercise 22.4 shows $\Omega = X_+X_- - 2iW - W^2$ in $U(\mathfrak{g}_{\mathbb{C}})$. Since $d\Pi$ is a representation of $\mathfrak{g}_{\mathbb{C}}$, it extends to $U(\mathfrak{g}_{\mathbb{C}})$, and this identity holds for the operators. By the ladder formulas (Proposition 22.14), the three terms act on e_n by $\nu^2-(1-n)^2$, $-2n$ and n^2. Hence Ω acts on $V_{\varepsilon,\nu}$ by the scalar

$$\nu^2-(1-n)^2-2n+n^2=\nu^2-1.$$

By the dictionary of Lecture 21 above, $2EF+2FE=H^2+Q^2$ and $H'^2=-W^2$. So this Ω is the Casimir element $H'^2+2EF+2FE$ of Lecture 21.

Lecture 21 found $\Omega = m(m+2)$ on V_m. This equals ν^2-1 at $\nu=\pm(m+1)$. Indeed, take $\nu=-(m+1)$ with ε of the parity of m. For a function F on the plane, the homogeneity rule (22.3) then reads $F(cv)=c^mF(v)$. The homogeneous polynomials of degree m on $\mathbb{R}^2$ form a finite-dimensional invariant space. On the circle they span the e_n with $|n|\le m$ and $n\equiv m \pmod 2$.

At $(\varepsilon,\nu)=(1,0)$ and at $(0,0)$ this scalar is -1. The two modules nevertheless have different central parity and different reducibility. A Casimir value alone is not a classification label.

22.6 Self-checks

1. What is $\Pi_{\varepsilon,\nu}(-I)$?
   Answer. $(-I)^{-1}u_\theta = u_{\theta+\pi}$ has length 1. So $(\Pi(-I)\varphi)(\theta)=\varphi(\theta+\pi)=(-1)^\varepsilon\varphi(\theta)$. The centre acts by σ_ε for every ν.
2. For which (ε,ν) is e_0 fixed by G?
   Answer. We need $e_0\in C_\varepsilon$, so $\varepsilon=0$. We also need $d\Pi(X_+)e_0=(\nu+1)e_2=0$, so $\nu=-1$. Conversely, for $\nu=-1$ the factor in the circle-model formula (22.4) is 1, so $\Pi(g)e_0=e_0$.
3. Why is $V_{1,0}$ reducible?
   Answer. Here $\nu+1+n=0$ at $n=-1$ and $\nu+1-n=0$ at $n=1$. So $d\Pi(X_+)e_{-1}=0$ and $d\Pi(X_-)e_1=0$. The sets $\{n\ge 1\}$ and $\{n\le -1\}$ are admissible, and $V_{1,0}$ is the direct sum of their spans.

22.7 Summary

The principal series $\Pi_{\varepsilon,\nu}$ is induced from the character $\sigma_\varepsilon\otimes e^{\nu t}$ of $P=MAN$, shifted by $\Delta_P^{1/2}$. Restriction to K identifies every $I_{\varepsilon,\nu}$ with the continuous parity-ε functions on the circle. The group acts on these functions by the circle-model formula (22.4). The identity $\tau'_h=|hu_\theta|^{-2}$

makes the pairing of ν with $-\bar{\nu}$ invariant. Hence, for $\nu \in i\mathbb{R}$, the operators $\Pi_{\varepsilon,\nu}(g)$ extend to a strongly continuous unitary representation on $\mathcal{H}_\varepsilon$.

Each K-type occurring in $V_{\varepsilon,\nu}$ occurs once and is spanned by one e_n with $n \equiv \varepsilon \pmod 2$. The elements $X_\pm$ move n by ± 2 with coefficients $\nu + 1 \pm n$. We proved that the $(\mathfrak{g}, K)$-module $V_{\varepsilon,\nu}$ is reducible exactly at integers ν of parity opposite to ε. We also proved the same criterion for closed invariant subspaces of $\mathcal{H}_\varepsilon$. Fourier projection and the ladder give irreducibility away from these points. At $\nu = n - 1$ the power series of (22.6) show that the closed span of the half-ladder $k \geq n$ is invariant, and at $\nu = -(n-1)$ its orthogonal complement is. At $(\varepsilon, \nu) = (1, 0)$ the two half-ladders split $\mathcal{H}_1$ into irreducible summands. The module equivalence of $\pm\nu$ and its unitary version stay cited.

22.8 Further reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (Princeton, 2001), §II.5 for $\mathrm{SL}(2,\mathbb{R})$, and §§VII.1–2 for induction from parabolic subgroups in general.
- P. Etingof, lecture notes for MIT 18.757, *Representations of Lie Groups*, Lectures 9 and 19.

Lecture 23

Discrete series and the rank one unitary picture

Objectives. *We realise the holomorphic discrete series of* $\mathrm{SL}(2,\mathbb{R})$ *on weighted spaces of holomorphic functions on the disk. We then compare it with the finite-dimensional and principal series modules, and a map* U_n *identifies its polynomials with a half-ladder of the principal series. Finally we state the full list of irreducible unitary representations, and we prove that a finite-dimensional unitary representation is trivial.*

By the end of this lecture you will be able to:

- Prove that Π_n is an irreducible unitary representation on the weighted disk space $\mathcal{A}_n$ for each integer $n \geq 2$.
- Compute its K-types $n, n{+}2, \dots$ and the action of H', E, F, and identify its polynomials with a half-ladder of the principal series by the map U_n.
- Decide which lowest weights λ occur in irreducible unitary representations of $\mathrm{SL}(2,\mathbb{R})$, and when such a representation factors through $\mathrm{PSL}(2,\mathbb{R})$.
- State Bargmann's classification, derive the positivity bounds behind it, and prove that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial.

23.1 The disk model

Lecture 22 found half-ladders at $\nu = n-1$. For $n \geq 2$ the L^2 norm of Lecture 22 is not invariant there, so a half-ladder needs a new norm. A half-ladder has one sign of K-type. Power series in z have the same shape, because rotation multiplies z^m by a power of $e^{2i\theta}$. So we look for holomorphic functions on a disk.

Let $G = \mathrm{SL}(2,\mathbb{R})$. We keep the conventions of Lecture 21:

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}, \qquad K = \{k_\theta\}.$$

Thus $k_{2\pi} = I$ and $k_\pi = -I$. With $H_0 = \mathrm{diag}(1,-1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ put

$$H' = -iW, \qquad E = \tfrac{1}{2}(H_0 + iS), \qquad F = \tfrac{1}{2}(H_0 - iS) = \overline{E}.$$

Then $[H', E] = 2E$, $[H', F] = -2F$ and $[E, F] = H'$. In a representation Π, a vector v has *K-type k* if $\Pi(k_\theta)v = e^{ik\theta}v$ for all θ. If such a v is smooth, then $d\Pi(H')v = kv$.

The disk carries an action of

$$\mathrm{SU}(1,1) = \left\{ \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix} : |\alpha|^2 - |\beta|^2 = 1 \right\}.$$

Fix the Cayley matrix $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and set $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G$, multiplying out gives

$$\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar{\beta} & \bar{\alpha} \end{pmatrix}, \qquad \alpha = \tfrac{1}{2}\big(a + d + i(c - b)\big), \qquad \beta = \tfrac{1}{2}\big(a - d + i(b + c)\big). \tag{23.1}$$

Then $|\alpha|^2 - |\beta|^2 = ad - bc = 1$. Solving (23.1) for a, b, c, d shows that φ is an isomorphism of Lie groups from G onto $\mathrm{SU}(1,1)$. Also

$$\varphi(k_\theta) = \mathrm{diag}(e^{-i\theta}, e^{i\theta}), \qquad \varphi(-I) = -I. \tag{23.2}$$

Conjugation by γ is complex linear on all 2×2 matrices. A direct multiplication gives

$$\gamma H' \gamma^{-1} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma E \gamma^{-1} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad \gamma F \gamma^{-1} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \tag{23.3}$$

The matrix $\bar{\gamma}$, which is the matrix c of Exercise 20.1, gives the triple in the order of Lecture 21. Our γ reverses it, so that Π_n below has positive K-types.

A function on an open subset of the complex plane is *holomorphic* if it is complex differentiable at every point.

We import three results from elementary complex analysis. The first is the local power-series theorem. A holomorphic function on a disk has a Taylor series converging uniformly on every smaller closed disk. The second concerns power series. A power series about the centre of a disk that converges at every point of the disk defines a holomorphic function there. The third follows from the Cauchy–Riemann equations. The real Jacobian of a holomorphic map is the squared modulus of its complex derivative. These are the analytic inputs used below. See MIT 18.112, the lectures on Cauchy's formula, power series and the Cauchy–Riemann equations.

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ with closure $\overline{D}$. For $h \in \mathrm{SU}(1,1)$ as above and $z \in \overline{D}$ put

$$h \cdot z = \frac{\alpha z + \beta}{\bar{\beta} z + \bar{\alpha}}, \qquad j(h, z) = \bar{\beta} z + \bar{\alpha}.$$

Lemma 23.1. *Let $h, h_1, h_2 \in \mathrm{SU}(1,1)$ and $z \in \overline{D}$.*

(a) $|j(h,z)| \geq |\alpha| - |\beta| > 0$ *and* $|j(h,z)|^2 - |\alpha z + \beta|^2 = 1 - |z|^2$.

(b) h *maps* D *into* D *and* $\overline{D}$ *into* $\overline{D}$, $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$ *and* $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$.

(c) $d(h \cdot z)/dz = j(h,z)^{-2}$.

Proof. (a) Use $|z| \leq 1$ and $|\alpha|^2 = 1 + |\beta|^2$. Expanding both squares, the cross terms cancel and leave $(|\alpha|^2 - |\beta|^2)(1 - |z|^2)$.

(b) Dividing (a) by $|j(h,z)|^2$ gives

$$1 - |h \cdot z|^2 = \frac{1 - |z|^2}{|j(h,z)|^2}. \tag{23.4}$$

Hence $|h \cdot z| < 1$ whenever $|z| < 1$, so h maps D into D. Also, $1 - |h \cdot z|^2 \geq 0$ when $|z| \leq 1$, with equality exactly when $|z| = 1$. So h maps $\overline{D}$ into $\overline{D}$.

Next,

$$j(h_1, h_2 \cdot z)\, j(h_2, z) = \bar{\beta}_1(\alpha_2 z + \beta_2) + \bar{\alpha}_1(\bar{\beta}_2 z + \bar{\alpha}_2).$$

The right side is the bottom row of $h_1 h_2$ applied to $(z,1)^T$, so it equals $j(h_1 h_2, z)$. In the same way, the top row of $h_1 h_2$ applied to $(z,1)^T$ is the numerator of $(h_1 h_2) \cdot z$. It equals $j(h_2, z)$ times the numerator of $h_1 \cdot (h_2 \cdot z)$. Dividing the two rows cancels $j(h_2, z)$ and gives $(h_1 h_2) \cdot z = h_1 \cdot (h_2 \cdot z)$.

(c) By the quotient rule the numerator is $\alpha(\bar{\beta} z + \bar{\alpha}) - \bar{\beta}(\alpha z + \beta) = 1$. ∎

Before we choose a norm, we look for an invariant area. By the identity (23.4) and Lemma 23.1(c), the area form $dA/(1 - |z|^2)^2$ on D is invariant under $\mathrm{SU}(1,1)$. This is the *hyperbolic area*. The matrix $\bar{\gamma}$ moves the upper half-plane $\mathbb{H}$ of Lecture 20 to D, with i going to 0, and this form becomes $\frac{1}{4} y^{-2} dx\, dy$. So it is a constant multiple of the invariant measure of Lecture 20, which is the left Haar measure of Lecture 16. Figure 23.1 shows this chart. Our γ uses the lower half-plane instead, with $-i$ going to 0, so that the K-types of Π_n are positive.

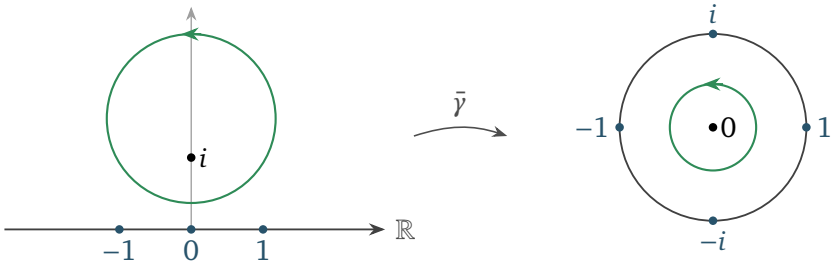


Figure 23.1: The chart $z \mapsto \bar{\gamma} \cdot z$. The points -1, 0 and 1 of the real line go to i, -1 and $-i$ on the boundary circle, and the point at infinity goes to 1. The point i goes to 0, and the K-orbit of ie goes to a centred circle. Arrows show k_θ for $\theta > 0$ in the parametrisation of this lecture.

Now take f and $\Pi_n(h) f$ as in (23.5) below. At $z = h \cdot w$ the value $|\Pi_n(h) f|^2$ gains the factor $|j(h,w)|^{2n}$, and $(1 - |z|^2)^n$ gains the factor $|j(h,w)|^{-2n}$ by (23.4). These cancel. So the natural norm integrates $|f|^2 (1 - |z|^2)^n$ against the hyperbolic area. This is why the power below is $n - 2$.

Definition 23.2. Let $n \geq 2$ be an integer and $dA = dx\, dy$. For f holomorphic on D put

$$\|f\|_n^2 = \frac{n-1}{\pi} \lim_{r \to 1} \int_{|z| \leq r} |f(z)|^2 (1 - |z|^2)^{n-2}\, dA(z) \in [0, \infty].$$

The *weighted disk space* $\mathcal{A}_n$ is the space of holomorphic f on D with $\|f\|_n < \infty$.

The limit exists because the integrand is nonnegative. For the same reason it equals the supremum of the integrals over compact regions in D with smooth boundary.

Lemma 23.3. *If $f(z)=\sum_{m\ge 0} a_m z^m$ is holomorphic on D, then*

$$\|f\|_n^2=\sum_{m\ge 0} c_m|a_m|^2,\qquad c_m=\|z^m\|_n^2=\frac{m!\,(n-1)!}{(n+m-1)!}=\binom{n+m-1}{m}^{-1}.$$

So $\mathcal{A}_n$ is a Hilbert space with orthonormal basis $e_m=c_m^{-1/2}z^m$. Moreover $\|g\|_n\le \sup_D|g|$ for bounded holomorphic g.

Proof. On $|z|=s<1$ the series converges uniformly. Orthogonality of the functions $e^{im\theta}$ gives

$$\int_0^{2\pi}|f(se^{i\theta})|^2\,d\theta=2\pi\sum_m |a_m|^2 s^{2m}.$$

Integration in polar coordinates then gives

$$\frac{n-1}{\pi}\int_{|z|\le r}|f|^2(1-|z|^2)^{n-2}\,dA=\sum_m |a_m|^2\,(n-1)\int_0^r 2s^{2m+1}(1-s^2)^{n-2}\,ds.$$

The terms are nonnegative and increase with r, so we may let $r\to 1$ termwise. With $t=s^2$ and repeated integration by parts,

$$(n-1)\int_0^1 t^m(1-t)^{n-2}\,dt=(n-1)\frac{m!\,(n-2)!}{(n+m-1)!}=c_m.$$

Conversely, let $(a_m)_{m\ge 0}$ be complex numbers with $C=\sum_m c_m|a_m|^2<\infty$. Then $|a_m|^2\le C\binom{n+m-1}{m}\le C(m+1)^{n-1}$, so $\sum a_m z^m$ converges on D. By the power-series import, its sum f is holomorphic on D. The formula above then gives $\|f\|_n^2=C$. Hence $f\mapsto (c_m^{1/2}a_m)_m$ is an isometry of $\mathcal{A}_n$ onto ℓ^2. This gives completeness, the inner product by polarisation, and the basis. Finally $\|g\|_n^2\le \sup|g|^2\,\|1\|_n^2=\sup|g|^2$, since $c_0=1$. ∎

Theorem 23.4. *For $h\in \mathrm{SU}(1,1)$ and $f\in\mathcal{A}_n$ define*

$$(\Pi_n(h)f)(z)=j(h^{-1},z)^{-n}\,f(h^{-1}\cdot z). \tag{23.5}$$

Then $g\mapsto \Pi_n(\varphi(g))$ is a strongly continuous unitary representation of G on $\mathcal{A}_n$. We write $\Pi_n(g)$ for $\Pi_n(\varphi(g))$.

Proof. Homomorphism. Apply the cocycle identity of Lemma 23.1(b) to h_2^{-1} and h_1^{-1}:

$$(\Pi_n(h_1)\Pi_n(h_2)f)(z)=j(h_1^{-1},z)^{-n}\,j(h_2^{-1},h_1^{-1}\cdot z)^{-n}f(h_2^{-1}h_1^{-1}\cdot z)=(\Pi_n(h_1h_2)f)(z).$$

Isometry. Substitute $z=h\cdot w$. The cocycle identity with $h^{-1}h=I$ gives $j(h^{-1},h\cdot w)=j(h,w)^{-1}$. With the identity (23.4) for $1-|h\cdot w|^2$ and the derivative formula of Lemma 23.1(c) we get

$$\begin{aligned}&|(\Pi_n(h)f)(h\cdot w)|^2(1-|h\cdot w|^2)^{n-2}\left|\frac{d(h\cdot w)}{dw}\right|^2\\&\quad=|j(h,w)|^{2n}|f(w)|^2\frac{(1-|w|^2)^{n-2}}{|j(h,w)|^{2n-4}}\frac{1}{|j(h,w)|^4}.\end{aligned}$$

The powers of $|j(h,w)|$ cancel, leaving $|f(w)|^2(1-|w|^2)^{n-2}$.

For a holomorphic map the real Jacobian determinant is $|d(h\cdot w)/dw|^2$, by the Cauchy–Riemann equations. So on each compact region $R\subset D$, the change of variables formula equates the weighted integral of $|\Pi_n(h)f|^2$ over $h\cdot R$ with that of $|f|^2$ over R.

The map $w\mapsto h\cdot w$ is a bijection of D with inverse $w\mapsto h^{-1}\cdot w$. So the regions $h\cdot R$ also run over all compact regions. Taking suprema gives $\|\Pi_n(h)f\|_n=\|f\|_n$. Thus $\Pi_n(h)$ is an isometry with inverse $\Pi_n(h^{-1})$, hence unitary.

Continuity. Let f,g be polynomials. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), the function $\overline{g(z)}(\Pi_n(h)f)(z)(1-|z|^2)^{n-2}$ is continuous on $\mathrm{SU}(1,1)\times\overline{D}$. Here $n\geq 2$ is used at the boundary. It is uniformly continuous on $U\times\overline{D}$ for a compact neighbourhood U of I. So $h\mapsto\langle g,\Pi_n(h)f\rangle_n$ is continuous. Polynomials are dense and $\|\Pi_n(h)\|=1$, so an $\varepsilon/3$ argument extends this to all f,g. Then

$$\|\Pi_n(h)f-f\|_n^2=2\|f\|_n^2-2\,\mathrm{Re}\langle f,\Pi_n(h)f\rangle_n\to 0\qquad(h\to I).$$

Unitarity moves this to every other point. ∎

Remark 23.5. Put $\delta=\mathrm{diag}(1,-1)$. Since $\bar\gamma=\gamma\delta$, the matrix $\bar\gamma$ in place of γ gives $\Pi_n^-(g)=\Pi_n(\delta g\delta^{-1})$. Conjugation by δ preserves G, although $\delta\notin G$. Since $\delta k_\theta\delta^{-1}=k_{-\theta}$, the K-types of Π_n^- are the negatives of those of Π_n. Which one is called holomorphic depends on the Cayley matrix and the sign of the K-angle. We label them by K-types.

23.2 K-types and Lie algebra operators

Proposition 23.6. *For all θ and $m\geq 0$,*

$$\Pi_n(k_\theta)z^m=e^{i(n+2m)\theta}z^m.$$

Hence the K-types of Π_n are $n,n+2,n+4,\dots$, each with multiplicity one, and the K-finite vectors are the polynomials.

Proof. By the formula (23.2), the element $h=\varphi(k_\theta)$ has $h^{-1}=\mathrm{diag}(e^{i\theta},e^{-i\theta})$. So $h^{-1}\cdot z=e^{2i\theta}z$ and $j(h^{-1},z)=e^{-i\theta}$. Now (23.5) gives $e^{in\theta}(e^{2i\theta}z)^m$.

For an integer k define

$$P_k f=\frac{1}{2\pi}\int_0^{2\pi}e^{-ik\theta}\,\Pi_n(k_\theta)f\,d\theta,$$

a Riemann integral of a continuous $\mathcal{A}_n$-valued function. Then $P_k e_m=e_m$ if $k=n+2m$, and $P_k e_m=0$ otherwise. Being bounded, P_k is the orthogonal projection onto $\mathbb{C}e_m$, or zero. Its range is the space of vectors of K-type k.

If v is K-finite, its orbit spans a finite-dimensional space U. Every $P_k v$ lies in U, as a limit of Riemann sums there. The nonzero $P_k v$ are orthogonal, so only finitely many occur. The basis expansion of v is therefore a polynomial. Conversely, every polynomial is a finite sum of K-types. ∎

Lemma 23.7. *Polynomials are smooth vectors for Π_n. For $Y=\begin{pmatrix}p&q\\r&s\end{pmatrix}\in\mathfrak{su}(1,1)$ and a*

polynomial f,

$$d\Pi_n(Y)f = n(rz+s)f + \big(rz^2 + (s-p)z - q\big)f'. \tag{23.6}$$

Proof. By the lower bound for $|j(h,z)|$ in Lemma 23.1(a), $(h,z) \mapsto (\Pi_n(h)f)(z)$ is smooth on $\mathrm{SU}(1,1) \times \overline{D}$. By Taylor's theorem on compact sets, difference quotients in h of every order converge uniformly on $\overline{D}$. By the supremum bound in Lemma 23.3, they converge in $\mathcal{A}_n$. So f is a smooth vector in the sense of Lecture 21. Moreover $d\Pi_n(Y)f$ is the pointwise t-derivative at 0 of $(\Pi_n(h_t)f)(z)$, where $h_t = \exp(tY)$. Since $h_t^{-1} = I - tY + O(t^2)$, uniformly in $z \in \overline{D}$,

$$j(h_t^{-1}, z) = 1 - t(rz+s) + O(t^2), \qquad h_t^{-1} \cdot z = \frac{(1-tp)z - tq}{1 - t(rz+s)} = z + t\big(rz^2 + (s-p)z - q\big) + O(t^2).$$

So $j(h_t^{-1}, z)^{-n} = 1 + nt(rz+s) + O(t^2)$, and the product and chain rules give (23.6). ∎

Proposition 23.8. *Extend $d\Pi_n$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. Then*

$$\begin{aligned} d\Pi_n(H')z^m &= (n+2m)z^m, \\ d\Pi_n(E)z^m &= (n+m)z^{m+1}, \\ d\Pi_n(F)z^m &= -m\,z^{m-1} \quad (m \geq 1), \qquad d\Pi_n(F)1 = 0. \end{aligned}$$

The Casimir element $\Omega = H'^2 + 2EF + 2FE$ acts on every polynomial by $n(n-2)$.

Proof. The right side of (23.6) is complex linear in (p,q,r,s). So it also computes $d\Pi_n$ on the matrices in (23.3). For H' we have $p=-1$, $s=1$, giving $nf + 2zf'$. For E we have $r=1$, giving $nzf + z^2f'$. For F we have $q=1$, giving $-f'$. Apply these to z^m. Since $[E,F] = H'$, we have $\Omega = H'^2 + 2H' + 4FE$. On z^m its eigenvalue is

$$(n+2m)^2 + 2(n+2m) - 4(n+m)(m+1) = n^2 - 2n.$$ ∎

For a unitary representation Π and $X \in \mathfrak{g}$, the operator $d\Pi(X)$ is skew-symmetric on smooth vectors. Extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. The matrices H_0, S and W are real, and the inner product is antilinear in the first slot. Moving $d\Pi(E)$ to the second slot therefore negates the real generators H_0 and S. It also turns the coefficient i of S into $-i$, which produces $-d\Pi(F)$. For $H' = -iW$ the two sign changes cancel. Hence smooth vectors u and v satisfy

$$\langle d\Pi(E)u, v\rangle = -\langle u, d\Pi(F)v\rangle, \qquad \langle d\Pi(H')u, v\rangle = \langle u, d\Pi(H')v\rangle. \tag{23.7}$$

In the model, $u = z^m$ and $v = z^{m+1}$ give $(n+m)c_{m+1}$ on the left and $(m+1)c_m$ on the right. These agree by the formula for c_m in Lemma 23.3. Conversely, distinct K-types are orthogonal, so (23.7) forces $c_{m+1}/c_m = (m+1)/(n+m)$. Up to a scalar, the weighted norm is the only such inner product on polynomials.

Theorem 23.9. *Π_n is irreducible for every integer $n \geq 2$.*

Proof. Let $V \neq 0$ be a closed invariant subspace and $0 \neq v \in V$. Each P_k maps V into V, since its Riemann sums do and V is closed. As $P_{n+2m}v = \langle e_m, v\rangle e_m$, some e_m lies in V. For $X \in \mathfrak{g}$, $d\Pi_n(X)e_m$ is the norm limit of $t^{-1}(\Pi_n(\exp tX)e_m - e_m) \in V$, so it lies in V. Hence $d\Pi_n(E)e_m$ and $d\Pi_n(F)e_m$ lie in V. By the operator formulas of Proposition 23.8, E sends z^m to $(n+m)z^{m+1}$ with $n+m \neq 0$. For $m \geq 1$, F sends z^m to $-mz^{m-1}$ with $m \neq 0$. So every z^k lies in V. Polynomials are dense, so $V = \mathcal{A}_n$. ∎

An irreducible unitary representation belongs to the *discrete series* if its matrix coefficients are square integrable on G. The family Π_n, $n \geq 2$, is the holomorphic discrete series in our labelling. Its square integrability is cited from Knapp's §VI.1; Exercise 23.6 checks the coefficient of e_0. The end of § 23.4 explains the word discrete.

23.3 Descent conditions

Proposition 23.10. *Let Π be a strongly continuous representation of G on a Hilbert space. Let $v \neq 0$ have K-type k.*

(a) $k \in \mathbb{Z}$. In particular, for $\lambda \notin \mathbb{Z}$ no such Π has a lowest-weight module with weights $\lambda, \lambda+2, \dots$ as its K-finite part.

(b) $\Pi(-I)v = (-1)^k v$.

(c) $\Pi_n(-I) = (-1)^n \,\mathrm{id}$. Hence Π_n factors through $\mathrm{PSL}(2,\mathbb{R}) = G/\{\pm I\}$ if and only if n is even.

Proof. (a) Since $k_{2\pi} = I$, we have $v = \Pi(k_{2\pi})v = e^{2\pi i k}v$. As $v \neq 0$, this forces $e^{2\pi i k} = 1$, so $k \in \mathbb{Z}$. A lowest-weight K-finite part with weights $\lambda, \lambda+2, \dots$ contains a vector of K-type λ, so it requires $\lambda \in \mathbb{Z}$.

(b) Since $k_\pi = -I$, the definition of K-type at $\theta = \pi$ gives $\Pi(-I)v = e^{i\pi k}v$. This equals $(-1)^k v$ because $k \in \mathbb{Z}$.

(c) By the formula (23.2) for $\varphi(-I)$, $(-I)\cdot z = z$ and $j(-I, z) = -1$, so $\Pi_n(-I)f = (-1)^{-n}f$. The quotient map $G \to \mathrm{PSL}(2,\mathbb{R})$ is onto with kernel $\{\pm I\}$. So Π_n factors exactly when $\Pi_n(-I) = \mathrm{id}$. ∎

The value of $\Pi_n(-I)$ in Proposition 23.10(c) agrees with the sign rule of Proposition 23.10(b). Indeed, every K-type $n+2m$ of Π_n has the parity of n. The integrality argument of Proposition 23.10(a) uses the period 2π of K in G. On the universal covering group the lift of K is a copy of $\mathbb{R}$, so (a) gives no condition there. We do not study that group.

Remark 23.11. The ratio forced by (23.7) still makes sense for $n = 1$. It gives $c_m = 1$, so the norm is $\sum |a_m|^2$. No positive measure μ on D gives this norm by area integration. It would satisfy $\int_D 1\, d\mu = \int_D |z|^2 d\mu = 1$, although $1 - |z|^2$ is strictly positive on D. This is impossible. So lowest weight 1 lies outside the family $\mathcal{A}_n$. It appears below as a limit of discrete series.

23.4 Comparison and the unitary picture

Let V be a $(\mathfrak{g}, K)$-module of G (Lecture 21) on which Ω acts by a scalar c. Write v_k for a vector of K-type k. Since $\Omega = H'^2 + 2H' + 4FE = H'^2 - 2H' + 4EF$,

$$4FEv_k = \big(c - k(k+2)\big)v_k, \qquad 4EFv_k = \big(c - k(k-2)\big)v_k. \tag{23.8}$$

Example 23.12. Fix $n \geq 2$ and $c = n(n-2) = (n-1)^2 - 1$. By (23.8), $FEv_k = 0$ exactly when $k \in \{-n, n-2\}$.

(i) In Π_n the K-types are $n, n+2, \dots$. F kills z^0, and E is injective.

(ii) Put $d = n-2$ and replace n by $-d$ in (23.5). Call the resulting operator $\Pi_{-d}(h)$. It sends f to $j(h^{-1}, z)^d f(h^{-1} \cdot z)$ and preserves the polynomials of degree at most d. The computation of (23.6) gives $d\Pi(E)z^m = (m-d)z^{m+1}$, which kills z^d. The K-types are $-d, \dots, d$. On the lowest vector $v_{-d} = z^0$, we have $Fv_{-d} = 0$. Thus (23.8) gives $c = (-d)(-d-2) = d(d+2) = n(n-2)$. Exercise 23.3 gives the polynomial matrices.

(iii) The principal series of Lecture 22 has every integer of one parity as a K-type, once each. Take the principal series module $V_{\varepsilon,\nu}$ of Lecture 22 with $\nu = n-1$, and with ε chosen so that its K-types have the parity of n. By Lecture 22, Exercise 22.4, Ω acts on it by $\nu^2 - 1 = n(n-2)$. At $k = n-2$, (23.8) says that the coefficient of $E\colon v_{n-2} \mapsto v_n$ times that of $F\colon v_n \mapsto v_{n-2}$ vanishes. The same holds between $-n$ and $2-n$.
The ladder of Lecture 22 says which arrow vanishes. At $\nu = n-1$ the arrows E from $-n$ and F from n vanish. So the half-ladders $k \geq n$ and $k \leq -n$ are submodules, and the finite segment $2-n \leq k \leq n-2$ is what remains when both are removed. At $\nu = -(n-1)$ the arrows E from $n-2$ and F from $2-n$ vanish, so the finite segment is a submodule. In both cases the three pieces have the K-types and Casimir value of Π_n^-, of the finite-dimensional module Π_{-d} with $d = n-2$, and of Π_n.

We now identify the piece $k \geq n$ with Π_n. In this lecture e_m is the basis of $\mathcal{A}_n$, so we write $\mathsf{e}_k(\theta) = e^{ik\theta}$ for the Fourier vectors of Lecture 22. For $n \geq 1$ and f holomorphic on a neighbourhood of $\overline{D}$, put

$$(U_n f)(\theta) = e^{in\theta} f(e^{2i\theta}), \qquad \text{so that} \qquad U_n z^m = \mathsf{e}_{n+2m}.$$

This is the map U_n of Lecture 22, with the same formula. Formula (23.5) also makes sense for $n = 1$ on such functions.

Proposition 23.13. *Let $n \geq 1$ and let $\varepsilon \equiv n \pmod 2$. For a polynomial f and $g \in G$, the function $\Pi_n(g)f$ is holomorphic on a neighbourhood of $\overline{D}$, and*

$$U_n \Pi_n(g) f = \Pi_{\varepsilon,n-1}(g) U_n f.$$

Proof. Put $h = \varphi(g)^{-1} = \gamma g^{-1} \gamma^{-1}$. By (23.5), $(\Pi_n(g)f)(z) = j(h,z)^{-n} f(h \cdot z)$. By Lemma 23.1(a), $j(h, \cdot)$ has no zero on $\overline{D}$, and its only possible zero lies outside. So $\Pi_n(g)f$ is holomorphic near $\overline{D}$.

Let $z = e^{2i\theta}$ and $g^{-1}u_\theta = r u_\tau$, with $u_\theta = (\cos\theta, \sin\theta)^T$ as in Lecture 22. Then $\gamma u_\theta = e^{-i\theta}(z, 1)^T$. Applying $\gamma g^{-1} = h\gamma$ to u_θ gives $h \cdot z = e^{2i\tau}$ and $j(h,z) = re^{i(\theta-\tau)}$, as in the half-ladder formula of Lecture 22. Hence

$$e^{in\theta} j(h,z)^{-n} f(h \cdot z) = r^{-n} e^{in\tau} f(e^{2i\tau}) = (\Pi_{\varepsilon,n-1}(g) U_n f)(\theta). \qquad \blacksquare$$

This identity uses only matrices, so the opposite sign of k_θ in this lecture does not enter. It says that the polynomials of Π_n are the piece $k \geq n$ of $V_{\varepsilon,n-1}$. The ladders agree too: for $\nu = n-1$ and $k = n+2m$ we have $\frac{1}{2}(\nu+1+k) = n+m$ and $\frac{1}{2}(\nu+1-k) = -m$, which are the coefficients of Proposition 23.8. The boundary point $z = e^{2i\theta}$ is the line $\mathbb{R}u_\theta$ of Lecture 22.

Only the norm differs. The squared disk norm of z^m is c_m, and the squared circle norm of e_{n+2m} is 1. So for $n \geq 2$ the map U_n is not an isometry. Also a general vector of $\mathcal{A}_n$ need not have boundary values, so we do not define U_n on all of $\mathcal{A}_n$. At $n = 1$ we have $c_m = 1$, and the two norms agree.

Corollary 23.14. *Let $\mathcal{A}_1$ be the completion of the polynomials in the norm $\|\sum a_m z^m\|^2 = \sum |a_m|^2$. Then U_1 extends to a unitary map from $\mathcal{A}_1$ onto the closed span H_1^+ of the vectors $\mathtt{e}_1, \mathtt{e}_3, \dots$. The operators $\Pi_1(g) = U_1^{-1}\Pi_{1,0}(g)U_1$ form an irreducible strongly continuous unitary representation of G on $\mathcal{A}_1$, and on polynomials they are given by (23.5) with $n = 1$. Put $\Pi_1^-(g) = \Pi_1(\delta g \delta^{-1})$. Then $\Pi_{1,0}$ is the orthogonal sum of a copy of Π_1 on H_1^+ and a copy of Π_1^- on the closed span of $\mathtt{e}_{-1}, \mathtt{e}_{-3}, \dots$.*

Proof. By Parseval, $\|U_1 f\|^2 = \sum |a_m|^2$, and the vectors $\mathtt{e}_{1+2m}$ are an orthonormal basis of H_1^+. So U_1 extends to a unitary map onto H_1^+. The representation $\Pi_{1,0}$ is unitary and strongly continuous, since $\nu = 0$ is imaginary. By the irreducibility theorem of Lecture 22, H_1^+ is invariant and irreducible. These properties pass to Π_1 through U_1. A function holomorphic near $\overline{D}$ has a Taylor series that converges in $\mathcal{A}_1$, so the extended U_1 agrees on it with the formula above. Hence, on polynomials, Proposition 23.13 gives the stated formula for $\Pi_1(g)$.

For the second half, put $(R\varphi)(\theta) = \varphi(-\theta)$. This is unitary and carries H_1^+ onto the negative half. Since $\delta u_\theta = u_{-\theta}$, the circle formula of Lecture 22 gives $\Pi_{1,0}(\delta g \delta^{-1}) = R\,\Pi_{1,0}(g)R$. Hence $RU_1\Pi_1^-(g) = \Pi_{1,0}(g)RU_1$. ∎

We now compare the sign in the two real forms of $\mathfrak{sl}(2,\mathbb{C})$. In a unitary representation of SU(2) the same triple has, in physics notation, $H' = 2J_3$, $E = J_+$ and $F = J_-$, and now $\langle Eu, v\rangle = \langle u, Fv\rangle$. Then this sign and (23.8) give $\|Ev_k\|^2 = \frac{1}{4}\big(c - k(k+2)\big)\|v_k\|^2$. For $c = 4j(j+1)$ and $k = 2m$, this is $j(j+1) - m(m+1)$ times $\|v_k\|^2$. For large k the right side is negative, so the ladder must stop. For $\mathrm{SL}(2,\mathbb{R})$ the relation (23.7) has a minus sign, and the sign of the norm formula is reversed. So a ladder need not stop.

Proposition 23.15. *Let V be an irreducible $(\mathfrak{g}, K)$-module of G. Assume each K-type has multiplicity at most one and Ω acts by c. Suppose V has a positive definite inner product satisfying (23.7) for the module action. If k and $k+2$ are K-types, then*

$$\|Ev_k\|^2 = \tfrac{1}{4}\big(k(k+2) - c\big)\|v_k\|^2 > 0. \tag{23.9}$$

Consequently:

(i) if the K-types are $\lambda, \lambda+2, \lambda+4, \dots$, then $\lambda \geq 1$ is an integer and $c = \lambda(\lambda-2)$;

(ii) if the K-types are $2\mathbb{Z}$, then $c < 0$;

(iii) if the K-types are $2\mathbb{Z}+1$, then $c < -1$.

Proof. By (23.7) and (23.8), $\|Ev_k\|^2 = -\langle v_k, FEv_k\rangle = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Since $\|Ev_k\|^2$ and $\|v_k\|^2$ are real and $v_k \neq 0$, this identity shows that c is real.

Next we show that $Ev_k \neq 0$, arguing by contradiction. Suppose $Ev_k = 0$. For each parity, the sum of the K-type spaces of that parity is a submodule. By irreducibility, all K-types have the parity of k. Since E raises and F lowers K-types by 2, the span of the v_j with $j \leq k$ is invariant. It is nonzero and misses v_{k+2}, which contradicts irreducibility. So (23.9) is strict.

For the lowest-weight pattern (i), $Fv_\lambda = 0$, so (23.8) gives $c = \lambda(\lambda-2)$. Then (23.9) at $k = \lambda$ reads $\|Ev_\lambda\|^2 = \lambda\|v_\lambda\|^2$, so $\lambda > 0$. The circle action on the $(\mathfrak{g}, K)$-module satisfies $k_{2\pi} = I$. Thus $v_\lambda = e^{2\pi i\lambda}v_\lambda$, which makes λ an integer. For the two-sided patterns (ii) and (iii), write $k(k+2) = (k+1)^2 - 1$. Its minimum is 0 over even k and -1 over odd k. ∎

Figure 23.2 shows formula (23.9). At $c = 8$ the dots on the line are $k = 2$ and $k = -4$, where the step vanishes. Between them the even dots $k = -2$ and $k = 0$ are the steps of the finite-dimensional module of K-types $-2, 0, 2$. For $\mathrm{SL}(2,\mathbb{R})$ these steps have negative squared norm. For $\mathrm{SU}(2)$ they are exactly the steps with positive squared norm.

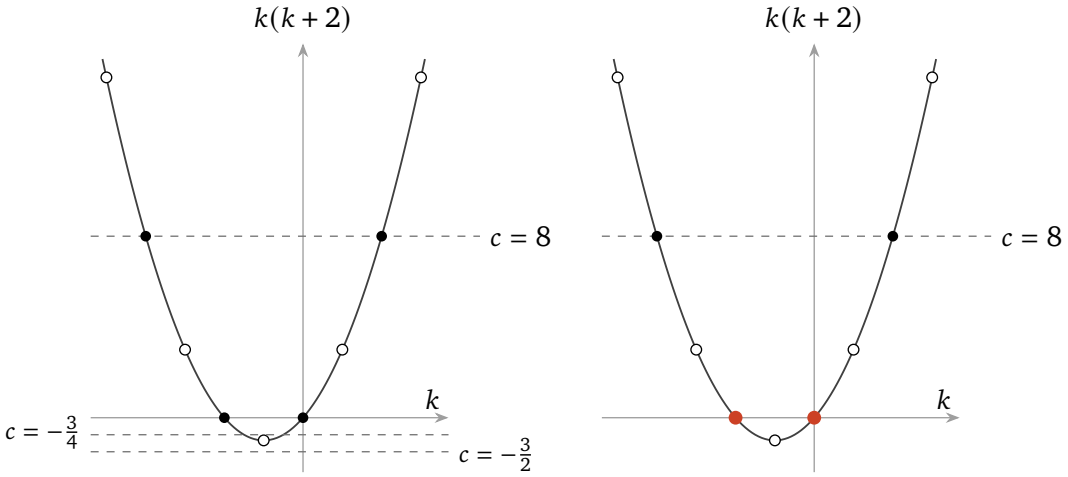


Figure 23.2: Formula (23.9) for $\mathrm{SL}(2,\mathbb{R})$ in the first panel and its $\mathrm{SU}(2)$ analogue in the second. Filled dots are even k and hollow dots are odd k. In the first panel a nonzero step from k to $k+2$ needs its dot strictly above the line $y = c$; a dot on the line is a zero step. In the second panel the step needs its dot strictly below the line, and the coloured dots are the steps of the module with K-types $-2, 0, 2$.

Twisting by δ from Remark 23.5 interchanges E, F, sends H' to $-H'$ and preserves (23.7). Thus the lowest-weight case (i) also gives $c = \lambda(\lambda - 2)$ with integer $\lambda \geq 1$ for highest-weight patterns $-\lambda, -\lambda - 2, \ldots$.

Theorem 23.16 (Bargmann; cited). *Every irreducible unitary representation of G is equivalent to exactly one entry of the following list. Each entry is irreducible and unitary, and is determined by its K-types and its Casimir value c. Here Π_1 and Π_1^- are the representations of Corollary 23.14, whose unitarity and irreducibility are proved there.*

Family	*K-types*	c	ν
trivial	$\{0\}$	0	± 1
even principal series	$2\mathbb{Z}$	$c \leq -1$	$i\mathbb{R}$
odd principal series	$2\mathbb{Z}+1$	$c < -1$	$i\mathbb{R} \setminus \{0\}$
complementary series	$2\mathbb{Z}$	$-1 < c < 0$	$(-1,1) \setminus \{0\}$
Π_n, $n \geq 1$	$n, n+2, \ldots$	$n(n-2)$	$\pm(n-1)$
Π_n^-, $n \geq 1$	$-n, -n-2, \ldots$	$n(n-2)$	$\pm(n-1)$

For $n \geq 2$ the last two rows are the discrete series. For $n = 1$ they are the limits of discrete series, realised with the norm of Remark 23.11. Their sum is the odd principal series at $c = -1$.

The last column gives the parameters of Lecture 22 with $c = \nu^2 - 1$. In the principal and complementary rows the entry has the module $V_{\varepsilon,\nu}$, with $\varepsilon = 0$ in the even rows and $\varepsilon = 1$ in the odd row, and ν and $-\nu$ give the same module. In the other rows the entry is only one

piece of a reducible module $V_{\varepsilon,\nu}$. So a value of ν can carry several entries of the list.

Proposition 23.15 proves the necessity of the c-ranges for these K-type patterns. Call a $(\mathfrak{g}, K)$-module *even* if its K-types are $2\mathbb{Z}$, and *odd* if they are $2\mathbb{Z}+1$. An irreducible even module with a positive definite inner product satisfying (23.7) needs $c < 0$. The part $c \le -1$ is the unitary principal series of Lecture 22. The interval $-1 < c < 0$ is the complementary series. The inner product of the complementary series is fixed by (23.9), not by the L^2 norm of Lecture 22 (Exercise 23.5). An irreducible odd module with such an inner product needs $c < -1$, so there is no odd complementary series. At $c = -1$ the odd principal series module has $FEv_{-1} = 0$ and splits.

Figure 23.3 sets the list against c. We use c and not ν because of the last remark on the table. At $c = -1$ the odd principal series splits into Π_1 and Π_1^-. At $c = 0$ the representations Π_2, Π_2^- and the trivial representation are distinct.

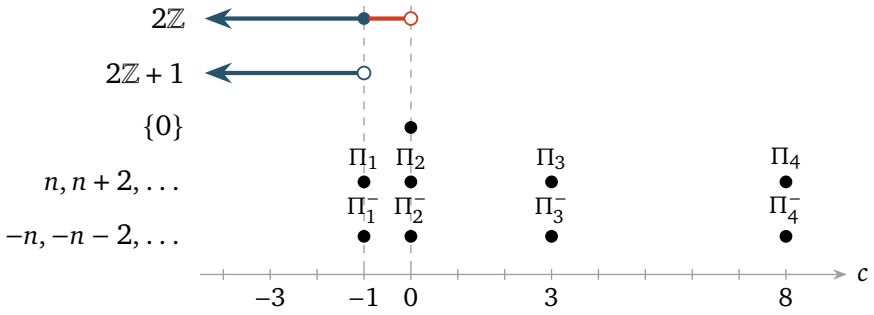


Figure 23.3: Bargmann's list against the Casimir value c, with one row for each pattern of K-types. In the top row the long segment $c \le -1$ is the even principal series, and the short segment $-1 < c < 0$ is the complementary series. The second row is the odd principal series, the third is the trivial representation, and the last two rows show the first few Π_n and Π_n^-.

The cited parts are the converse and the exhaustion. They use two results of Harish-Chandra (compare Lecture 21). First, every irreducible unitary representation is admissible. Its K-finite vectors form an irreducible $(\mathfrak{g}, K)$-module with multiplicity-one K-types on which Ω is scalar. Second, each module above with an invariant positive definite Hermitian form integrates to an irreducible unitary representation of G. Etingof's MIT 18.757 notes treat admissibility in Lecture 5 and unitary globalisation in Theorem 7.5.

We can now explain the word discrete. For a compact group, the Plancherel theorem of Lecture 11 writes the squared norm of a function as a sum over the irreducible representations. For G the corresponding formula has a sum and integrals. Let h be smooth with compact support on G, and put $\Pi(h) = \int_G h(g)\Pi(g)\,dg$. For a suitable normalisation of Haar measure,

$$h(I) = \int_{\mathbb{R}} \operatorname{tr} \Pi_{0,i\nu}(h)\, \nu \tanh \frac{\pi\nu}{2}\, d\nu + \int_{\mathbb{R}} \operatorname{tr} \Pi_{1,i\nu}(h)\, \nu \coth \frac{\pi\nu}{2}\, d\nu + \sum_{n \ge 2} 4(n-1) \operatorname{tr}\bigl(\Pi_n(h) + \Pi_n^-(h)\bigr).$$

We cite this inversion formula from Knapp, §II.7, formula (2.25), and we do not use it. Knapp proves it as Theorem 11.6 in §XI.3 and credits it to Harish-Chandra (1952). The Plancherel formula for G follows from it. The discrete series enters through a discrete sum, and the

unitary principal series enters through integrals over $\nu = i\nu$. Neither the complementary series nor $\Pi_1^{\pm}$ appears. The number $n-1$ is the one in Exercise 23.6, where the coefficient of e_0 has squared norm $C/(n-1)$.

Finally we return to finite dimensions. The next statement does not use Bargmann's theorem, and it needs no irreducibility hypothesis.

Corollary 23.17. *Every finite-dimensional continuous unitary representation Π of G is trivial.*

Proof. By the theorem of Lecture 3 on continuous homomorphisms, Π is smooth. The complex-linear extension of $d\Pi$ makes the space V a finite-dimensional module of $\mathfrak{sl}(2,\mathbb{C})$. The real element $H_0 = \operatorname{diag}(1,-1)$ is the diagonal element of the triple of Lecture 20. So complete reducibility and the highest-weight calculation of the prerequisite Lie Algebras course make $d\Pi(H_0)$ diagonalisable with integer eigenvalues. Unitarity makes the same operator skew-Hermitian, with imaginary eigenvalues. Therefore $d\Pi(H_0) = 0$.

The kernel of the complexified differential is an ideal of $\mathfrak{sl}(2,\mathbb{C})$ containing the nonzero element H_0. Since $\mathfrak{sl}(2,\mathbb{C})$ is simple, this kernel is everything, so $d\Pi = 0$. Since G is connected, it is generated by exponentials (Lecture 2), and $\Pi(\exp X) = \exp d\Pi(X) = I$. Hence Π is trivial. ∎

So a nontrivial unitary representation of G is infinite-dimensional. Exercise 23.3 checks this directly for the finite-dimensional modules Π_{-d} of Example 23.12(ii).

23.5 Self-checks

1. For $n = 3$, compute c_1 and c_2. Check (23.7) for $u = z$ and $v = z^2$.

   Answer. $c_1 = 1!\,2!/3! = 1/3$ and $c_2 = 2!\,2!/4! = 1/6$. The left side is $(3+1)c_2 = 2/3$. The right side is $-\langle z, -2z\rangle = 2c_1 = 2/3$.

2. Does Π_5 factor through $\mathrm{PSL}(2,\mathbb{R})$? Compute $\Pi_5(k_{\pi/2})z^m$.

   Answer. No, since $\Pi_5(-I) = -\mathrm{id}$. By the K-type formula of Proposition 23.6, $\Pi_5(k_{\pi/2})z^m = i^{5+2m}z^m = (-1)^m\, i\, z^m$.

3. Is there an irreducible unitary representation with K-types $1, 3, 5, \ldots$ and $c = -\frac{1}{2}$?

   Answer. No. By the cited Harish-Chandra facts after Theorem 23.16, the K-finite vectors of such a representation satisfy the hypotheses of Proposition 23.15. Its lowest-weight case (i) shows that the lowest K-type 1 forces $c = 1\cdot(1-2) = -1$.

23.6 Summary

For each integer $n \geq 2$, formula (23.5) defines an irreducible unitary representation Π_n on $\mathcal{A}_n$. Its K-types are $n, n+2, \ldots$, each once. The operator E raises the K-type by 2 and F lowers it by 2. The Casimir element Ω acts by $n(n-2)$. The relation $k_{2\pi} = I$ forces integral K-types. The representation Π_n descends to $\mathrm{PSL}(2,\mathbb{R})$ exactly for even n. In an irreducible unitary module with K-types $\lambda, \lambda+2, \ldots$, the lowest weight is an integer $\lambda \geq 1$, and Ω acts

by $\lambda(\lambda-2)$. Positivity of $\|Ev_k\|^2$ explains the shape of Bargmann's cited list, including the complementary series.

The map U_n carries the polynomials of Π_n onto the half-ladder $k \geq n$ of the principal series at $\nu = n-1$, and only the norm differs. For $n=1$ the norms agree, and this constructs Π_1 as a summand of the odd principal series at $\nu = 0$. Every finite-dimensional continuous unitary representation of G is trivial. We also cited the inversion formula for G, whose discrete sum runs over the discrete series.

23.7 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001 edition. §VI.1 treats the holomorphic discrete series of $\mathrm{SU}(1,1)$. §§II.5–II.6 treat the unitary representations of $\mathrm{SL}(2,\mathbb{R})$. §II.7 states the inversion formula cited above, and §XI.3 proves it.
- MIT 18.112, *Functions of a Complex Variable*, Fall 2008: Cauchy's formula, power series and the Cauchy–Riemann equations.
- P. Etingof, *Representations of Lie Groups*, MIT 18.757 lecture notes, Fall 2023. Lecture 5 treats admissibility, and Theorem 7.5 is the unitary globalisation theorem.
- Harish-Chandra, Plancherel formula for the 2×2 real unimodular group, *Proc. Nat. Acad. Sci. USA* 38 (1952), 337–342.
- V. Bargmann, Irreducible unitary representations of the Lorentz group, *Ann. of Math.* 48 (1947), 568–640, for the original classification.

Lecture 24

Lorentz groups and finite-dimensional representations

Objectives. *We build the double cover* $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ *from Hermitian* 2×2 *matrices. We then classify the finite-dimensional irreducible continuous complex representations of both groups by complexifying the real Lie algebra.*

By the end of this lecture you will be able to:

- Prove that $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ is a surjective covering homomorphism with kernel $\{\pm I\}$, and identify $\mathrm{SL}(2,\mathbb{R})$ as the double cover of $\mathrm{SO}^+(1,2)$ inside it.
- Identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$.
- Label the irreducible finite-dimensional continuous complex representations of $\mathrm{SL}(2,\mathbb{C})$ by (j_L, j_R) and decide when they descend to $\mathrm{SO}^+(1,3)$.
- Recognise Weyl spinors and four-vectors in this list.

In special relativity an event has coordinates $x = (x_0, \vec{x})$, with time x_0 and position $\vec{x} \in \mathbb{R}^3$. We use units in which light has speed 1. Inertial observers are related by linear maps that preserve the Minkowski form below. Relativistic fields take values in finite-dimensional representations of the Lorentz group or its double cover $\mathrm{SL}(2,\mathbb{C})$.

24.1 Minkowski space and the Lorentz group

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$. The *Minkowski form* is

$$\langle x, y\rangle = x_0 y_0 - \vec{x}\cdot\vec{y} = x^{T}\eta y, \qquad \eta = \mathrm{diag}(1,-1,-1,-1).$$

Definition 24.1. The *Lorentz group* is $\mathrm{O}(1,3) = \{\Lambda \in \mathrm{GL}(4,\mathbb{R}) : \Lambda^{T}\eta\Lambda = \eta\}$. The *future cone* is $T^+ = \{x \in \mathbb{R}^4 : x_0 > |\vec{x}|\}$.

Let $\Lambda \in \mathrm{O}(1,3)$. Then $\det\Lambda = \pm 1$ and $\Lambda^{-1} = \eta\Lambda^{T}\eta$, so $(\Lambda^{-1})_{00} = \Lambda_{00}$. Since $\langle \Lambda e_0, \Lambda e_0\rangle = 1$,

$$\Lambda_{00}^2 = 1 + \Lambda_{10}^2 + \Lambda_{20}^2 + \Lambda_{30}^2 \geq 1.$$

So either $\Lambda_{00} \geq 1$ or $\Lambda_{00} \leq -1$.

Lemma 24.2. *Let* $\Lambda \in \mathrm{O}(1,3)$. *Then* $\Lambda_{00} \geq 1$ *if and only if* $\Lambda T^+ \subseteq T^+$. *Consequently the elements with* $\Lambda_{00} \geq 1$ *form a subgroup* $\mathrm{O}^+(1,3)$.

Proof. Suppose $\Lambda_{00} \geq 1$ and let $x \in T^+$. Put $w = \Lambda^{-1} e_0$. Then $\langle w, w\rangle = 1$ and $w_0 = (\Lambda^{-1})_{00} = \Lambda_{00} \geq 1$, so $w_0 > |\vec{w}|$. Since Λ preserves the form,

$$(\Lambda x)_0 = \langle e_0, \Lambda x\rangle = \langle w, x\rangle = w_0 x_0 - \vec{w}\cdot\vec{x} \geq w_0 x_0 - |\vec{w}|\,|\vec{x}| > 0.$$

The last step uses $w_0 > |\vec{w}|$ and $x_0 > |\vec{x}|$. Also $\langle \Lambda x, \Lambda x\rangle = \langle x, x\rangle > 0$. These two facts give $\Lambda x \in T^+$. Conversely, if $\Lambda T^+ \subseteq T^+$, then $\Lambda e_0 \in T^+$. So $\Lambda_{00} > 0$, hence $\Lambda_{00} \geq 1$. The cone condition is preserved by products, and $(\Lambda^{-1})_{00} = \Lambda_{00}$. So these elements form a subgroup. ∎

Definition 24.3. The *proper orthochronous Lorentz group* is

$$\mathrm{SO}^+(1,3) = \{\Lambda \in \mathrm{O}(1,3) : \det\Lambda = 1,\ \Lambda_{00} \geq 1\}.$$

By the cone lemma, Lemma 24.2, it is a group. It is closed in $\mathrm{GL}(4,\mathbb{R})$, hence a Lie group by the closed-subgroup theorem (Lecture 3). By the two dichotomies above, for $\det\Lambda$ and for Λ_{00}, it is also open in $\mathrm{O}(1,3)$. So it has the Lie algebra of $\mathrm{O}(1,3)$:

$$\mathfrak{so}(1,3) = \{Y \in \mathfrak{gl}(4,\mathbb{R}) : Y^T\eta + \eta Y = 0\}.$$

The map $Y \mapsto \eta Y$ is a bijection onto the antisymmetric matrices, so $\dim\mathfrak{so}(1,3) = 6$.

A basis consists of the *rotation generators* J_k and the *boost generators* K_k, for $k = 1,2,3$. Here $J_k e_0 = 0$, and J_k acts on $\vec{x}$ by $e_k \times \vec{x}$. Moreover $K_k e_0 = e_k$ and $K_k e_k = e_0$, and K_k maps the other two basis vectors to zero. For example,

$$J_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad K_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Then $R_3(\theta) = \exp(\theta J_3)$ rotates the (x_1, x_2)-plane by the positive angle θ. The *boost* $B_3(t) = \exp(tK_3)$ fixes x_1, x_2 and acts by

$$x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3).$$

The quantity $x_0^2 - x_3^2 = (x_0 + x_3)(x_0 - x_3)$ does not change, so a point moves along a hyperbola. Figure 24.1 shows this motion in the plane $x_1 = x_2 = 0$.

Parity $\mathrm{diag}(1,-1,-1,-1)$ and time reversal $\mathrm{diag}(-1,1,1,1)$ lie in $\mathrm{O}(1,3)$ but outside the open subgroup $\mathrm{SO}^+(1,3)$. The Lie algebra $\mathfrak{so}(1,3)$ carries no information about these elements. We study only $\mathrm{SO}^+(1,3)$.

24.2 The double cover

Lecture 4 built the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ as the adjoint action on $\mathfrak{su}(2)$. It also identified this with the action $\xi \mapsto U\xi U^*$ on traceless Hermitian matrices. We now allow all Hermitian matrices and all of $\mathrm{SL}(2,\mathbb{C})$, with the action $X \mapsto AXA^*$. The same construction gives the Lorentz group.

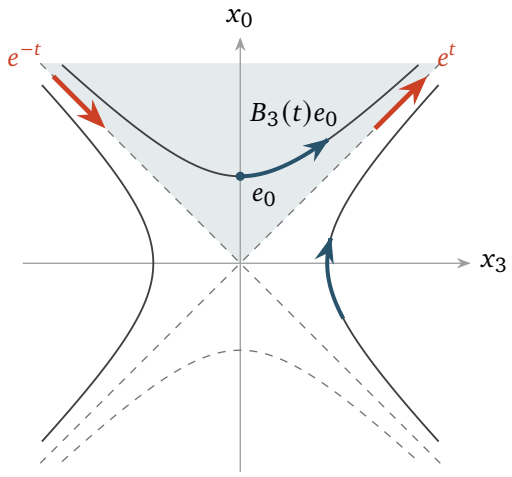


Figure 24.1: The boost $B_3(t)$ in the plane $x_1 = x_2 = 0$. The dashed diagonals are the null lines $x_0 = \pm x_3$, and the shaded wedge is the slice of T^+. The solid curves are the hyperbolas $x_0^2 - x_3^2 = \pm 1$. The coloured arrows show that $B_3(t)$ stretches the diagonal $x_0 = x_3$ by e^t and shrinks the diagonal $x_0 = -x_3$ by e^{-t}. The heavy arcs show the motion of two points for one fixed $t > 0$: the point e_0, which is the lowest point of the upper branch, and a point with $x_0^2 - x_3^2 = -1$. The dashed lower branch is the slice of the sheet on which Λe_0 lies when $\Lambda_{00} \le -1$.

Let $\sigma_0 = I$ and

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These form a real basis of the space $\mathcal{H}$ of Hermitian 2×2 matrices. For $x \in \mathbb{R}^4$ put

$$X(x) = \sum_{\mu=0}^{3} x_\mu \sigma_\mu = \begin{pmatrix} x_0 + x_3 & x_1 - i x_2 \\ x_1 + i x_2 & x_0 - x_3 \end{pmatrix}.$$

Direct expansion gives

$$\det X(x) = \langle x, x \rangle, \qquad x_0 = \tfrac{1}{2} \operatorname{tr} X(x).$$

For $A \in \mathrm{SL}(2, \mathbb{C})$ the matrix $AX(x)A^*$ is again Hermitian. Define $\varphi(A) \in \mathrm{GL}(4, \mathbb{R})$ by

$$X(\varphi(A)x) = A\, X(x)\, A^*. \tag{24.1}$$

The entries of $\varphi(A)$ are real polynomials in the real and imaginary parts of the entries of A. Hence φ is smooth.

Example 24.4 (Diagonal elements). Let $A = \operatorname{diag}(a, a^{-1})$ with $a \in \mathbb{C}^\times$. Then

$$AX(x)A^* = \begin{pmatrix} |a|^2 (x_0 + x_3) & (a/\bar{a})(x_1 - i x_2) \\ (\bar{a}/a)(x_1 + i x_2) & |a|^{-2}(x_0 - x_3) \end{pmatrix}.$$

For $a = e^{-i\theta/2}$, the coordinates x_0, x_3 are fixed and $x_1 + i x_2 \mapsto e^{i\theta}(x_1 + i x_2)$. For $a = e^{t/2}$, we get $x_0 \pm x_3 \mapsto e^{\pm t}(x_0 \pm x_3)$, with x_1, x_2 fixed. Put $X_k = -\frac{i}{2}\sigma_k$. Then

$$\varphi(\exp \theta X_3) = R_3(\theta), \qquad \varphi(\exp t i X_3) = B_3(t).$$

Differentiating at 0 gives $d\varphi(X_3) = J_3$ and $d\varphi(iX_3) = K_3$. Also $\operatorname{diag}(e^{i\theta}, e^{-i\theta}) = \exp(-2\theta X_3)$ maps to $R_3(-2\theta)$, as in the convention for $\mathrm{SU}(2) \to \mathrm{SO}(3)$ in Lecture 4. Finally, $\exp(2\pi X_3) = -I$ maps to $R_3(2\pi) = I$.

Let $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ denote $\mathfrak{sl}(2,\mathbb{C})$ regarded as a six-dimensional real Lie algebra.

Theorem 24.5. *The map φ is a surjective homomorphism $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its differential $d\varphi : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \to \mathfrak{so}(1,3)$ is an isomorphism of real Lie algebras, and φ is a two-sheeted covering map.*

Proof. Step 1: a homomorphism into $\mathrm{O}^+(1,3)$. Since $(AB)X(AB)^* = A(BXB^*)A^*$, we have $\varphi(AB) = \varphi(A)\varphi(B)$. Since $\det A = 1$, $\det(AXA^*) = \det X$. So $\varphi(A)$ preserves $\langle x, x\rangle$. By polarisation it preserves $\langle x, y\rangle$, so $\varphi(A) \in \mathrm{O}(1,3)$. Its 00 entry is the time component of AIA^*:

$$\varphi(A)_{00} = \tfrac{1}{2}\operatorname{tr}(AA^*) = \tfrac{1}{2}\textstyle\sum_{i,j} |A_{ij}|^2 > 0.$$

By the dichotomy above, $\varphi(A)_{00} \geq 1$.

Step 2: determinant one. We show that $\mathrm{SL}(2,\mathbb{C})$ is path connected. Let $A \in \mathrm{SL}(2,\mathbb{C})$. Let $P = (A^*A)^{1/2}$ be the positive definite square root, and $U = AP^{-1}$. Then $U^*U = I$ and $\det P = |\det A| = 1$, so $U \in \mathrm{SU}(2)$. Diagonalise U and P by the spectral theorem. Since both have determinant 1, we can write $U = \exp Y_1$ and $P = \exp Y_2$ with Y_1 skew-Hermitian, Y_2 Hermitian, and both traceless. Then $t \mapsto \exp(tY_1)\exp(tY_2)$, for $t \in [0,1]$, is a path in $\mathrm{SL}(2,\mathbb{C})$ from I to A. Along it, $\det\varphi$ is continuous with values in $\{\pm 1\}$, and it starts at 1. So $\det\varphi(A) = 1$.

Step 3: the kernel. Suppose $\varphi(A) = I$. Then $AXA^* = X$ for all $X \in \mathcal{H}$. Taking $X = I$ gives $AA^* = I$, so $AXA^{-1} = X$ for all $X \in \mathcal{H}$. Every complex matrix is $X_1 + iX_2$ with $X_1, X_2 \in \mathcal{H}$. So A commutes with all of $M_2(\mathbb{C})$, hence $A = cI$. Then $\det A = 1$ gives $c = \pm 1$. Conversely, $\pm I$ lie in the kernel.

Step 4: surjectivity. This is the main step. Let $\Lambda \in \mathrm{SO}^+(1,3)$ and $v = \Lambda e_0$. First we remove the boost part. The matrix $V = X(v)$ satisfies

$$\det V = \langle v, v\rangle = 1, \qquad \operatorname{tr} V = 2\Lambda_{00} > 0.$$

Its real eigenvalues have product 1 and positive sum, so both are positive. Thus $V = W\operatorname{diag}(\lambda, \lambda^{-1})W^*$ with W unitary and $\lambda > 0$. Put $P = W\operatorname{diag}(\lambda^{1/2}, \lambda^{-1/2})W^*$. Then $P \in \mathrm{SL}(2,\mathbb{C})$, $P^* = P$ and $P^2 = V$. Hence $X(\varphi(P)e_0) = PIP^* = V$, that is, $\varphi(P)e_0 = v$.

Next we treat the rotation part. Put

$$\Lambda' = \varphi(P)^{-1}\Lambda.$$

This element lies in $\mathrm{SO}^+(1,3)$ and fixes e_0. So it preserves $e_0^{\perp} = \operatorname{span}(e_1, e_2, e_3)$, where $-\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Hence $\Lambda' = \operatorname{diag}(1, R)$ with $R \in \mathrm{SO}(3)$. Let $n = Re_3$, a unit vector, and set $N = \sum_k n_k\sigma_k$. This Hermitian matrix satisfies

$$\operatorname{tr} N = 0, \qquad \det N = -|n|^2 = -1,$$

so the eigenvalues of N are ± 1. By the spectral theorem, $N = W'\sigma_3 W'^*$ with W' unitary. We correct the determinant of W' by replacing W' with

$$W'\operatorname{diag}(1, \overline{\det W'}).$$

Diagonal matrices commute with σ_3, so this replacement preserves the diagonalisation of N. The corrected W' belongs to $\mathrm{SU}(2)$.

Then $\varphi(W')e_0 = e_0$, since $W'W'^* = I$. Also $\varphi(W')e_3 = n$, since $W'\sigma_3 W'^* = N$. So $\varphi(W')^{-1}\Lambda'$ fixes e_0 and e_3. It acts on $\operatorname{span}(e_1, e_2)$ by an element of $\mathrm{SO}(2)$, so it equals $R_3(\theta)$ for some θ. By the rotation formula of Example 24.4,

$$\Lambda = \varphi(P)\,\varphi(W')\,R_3(\theta) = \varphi\big(PW' \exp(\theta X_3)\big).$$

In particular, φ maps $\mathrm{SU}(2)$ onto the rotations $\operatorname{diag}(1, R)$.

Step 5: the differential and the covering. Differentiate (24.1) along $\exp(tY)$, using $\exp(tY)^* = \exp(tY^*)$:

$$X(d\varphi(Y)x) = YX(x) + X(x)Y^*, \qquad Y \in \mathfrak{sl}(2,\mathbb{C}). \tag{24.2}$$

Suppose $d\varphi(Y) = 0$. Taking $X = I$ gives $Y^* = -Y$. Then $YX - XY = 0$ for all $X \in \mathcal{H}$, so Y is scalar, as in Step 3. Since $\operatorname{tr} Y = 0$, we get $Y = 0$. Thus $d\varphi$ is injective. Both algebras have real dimension 6, so $d\varphi$ is a linear bijection. The differential of a Lie group homomorphism preserves brackets, so $d\varphi$ is an isomorphism of real Lie algebras.

By Steps 2 and 4, $\mathrm{SO}^+(1,3) = \varphi(\mathrm{SL}(2,\mathbb{C}))$ is connected. A continuous surjective homomorphism of connected Lie groups with bijective differential is a covering map (Lecture 4). Its fibres are cosets of $\ker\varphi = \{\pm I\}$, so it has two sheets. ∎

Formula (24.2) gives $d\varphi(X_k) = J_k$ and $d\varphi(iX_k) = K_k$ for every k. The case $k = 3$ is the diagonal computation of Example 24.4, Exercise 24.1 treats $k = 1$, and the case $k = 2$ is identical. The brackets $[X_i, X_j] = \epsilon_{ijk}X_k$ then transfer through $d\varphi$:

$$[J_i, J_j] = \epsilon_{ijk}J_k, \qquad [J_i, K_j] = \epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\epsilon_{ijk}J_k.$$

Steps 2 and 4 of the proof are the Cartan decomposition of Lecture 20, for the real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ with $\mathfrak{k} = \mathfrak{su}(2)$ and $\mathfrak{p} = i\,\mathfrak{su}(2)$. Step 2 writes $A = UP$, with $U \in \mathrm{SU}(2)$ and $P = \exp Y_2$ for a traceless Hermitian Y_2. So $\varphi(A) = \varphi(U)\varphi(P)$, which is a boost followed by a rotation. Step 4 writes the same kind of element in the other order, as a rotation followed by a boost.

The same construction one dimension down

The construction also works for the real subgroup $\mathrm{SL}(2,\mathbb{R})$ of $\mathrm{SL}(2,\mathbb{C})$. It then relates $\mathrm{SL}(2,\mathbb{R})$ to a Lorentz group with two space coordinates. Let $\mathbb{R}^{1,2}$ be the subspace $\{x \in \mathbb{R}^4 : x_2 = 0\}$, with the form $x_0^2 - x_1^2 - x_3^2$. Let $\mathrm{SO}^+(1,2)$ be the group of linear maps of $\mathbb{R}^{1,2}$ that preserve this form, have determinant one and satisfy $\Lambda_{00} \geq 1$.

> **Proposition 24.6.** *Each $\varphi(A)$ with $A \in \mathrm{SL}(2,\mathbb{R})$ fixes e_2 and preserves $\mathbb{R}^{1,2}$. Restriction to $\mathbb{R}^{1,2}$ gives a surjective homomorphism $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$ with kernel $\{\pm I\}$.*

Proof. Let A be real, so $A^* = A^T$. If $x_2 = 0$, then $X(x)$ is real symmetric, and so is $AX(x)A^T$. Hence $\varphi(A)$ preserves $\mathbb{R}^{1,2}$. Next, $X(e_2) = \sigma_2$, and $A\sigma_2 A^T = (\det A)\sigma_2 = \sigma_2$, so $\varphi(A)$ fixes e_2. Thus $\varphi(A)$ is the identity on $\mathbb{R}e_2$ and its restriction on $\mathbb{R}^{1,2}$. By Theorem 24.5, this restriction preserves the form, has determinant one and has $\Lambda_{00} \geq 1$. If the restriction is the identity, then $\varphi(A) = I$, so $A = \pm I$.

For surjectivity we repeat Step 4 with real matrices. Let $\Lambda \in \mathrm{SO}^+(1,2)$ and $v = \Lambda e_0$. The matrix $V = X(v)$ is real symmetric with positive eigenvalues and determinant one. A real orthogonal diagonalisation gives a real positive definite P with $\det P = 1$ and $P^2 = V$, so $\varphi(P)e_0 = v$. Then $\varphi(P)^{-1}\Lambda$ fixes e_0, so it acts on the (x_3, x_1)-plane by an element of $\mathrm{SO}(2)$. Direct multiplication shows that the rotation matrix with first column $(\cos\theta, \sin\theta)^T$ fixes e_0 and rotates the (x_3, x_1)-plane through the angle 2θ. So this element is $\varphi(k)$ for a rotation matrix k, and $\Lambda = \varphi(Pk)$ on $\mathbb{R}^{1,2}$. ∎

So Lecture 21, Lecture 22 and Lecture 23 studied the double cover of the Lorentz group of two space coordinates and one time. This is why Bargmann's paper, cited in Lecture 23, has the title *Irreducible unitary representations of the Lorentz group*.

The earlier models of that block appear here as well. The real matrices $\frac{1}{2}\sigma_3$ and $\frac{1}{2}\sigma_1$ map to the boost generators K_3 and K_1. The stabiliser of e_0 in $\mathrm{SL}(2,\mathbb{R})$ is $\{A : AA^T = I\}$, which is the subgroup K of rotation matrices. The proof above shows that the group acts transitively on the upper sheet $x_0^2 - x_1^2 - x_3^2 = 1$ with $x_0 > 0$. So this sheet is $\mathrm{SL}(2,\mathbb{R})/K$. As a homogeneous space it is the half-plane of Lecture 20 and the disk of Lecture 23. Finally, a real line $\mathbb{R}v$ in $\mathbb{R}^2$ goes to the matrix vv^T. This matrix has determinant zero and trace $|v|^2 > 0$, so it is a future light-like vector, and $A(vv^T)A^T = (Av)(Av)^T$. Thus the circle of real lines in Lecture 22 is the circle of light rays in $\mathbb{R}^{1,2}$.

24.3 Complexifying the Lorentz algebra

This section rests on one change of basis. In the complexified Lorentz algebra $\mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$, the elements $\frac{1}{2}(J_k - iK_k)$ obey the brackets of $\mathfrak{su}(2)$. So do the elements $\frac{1}{2}(J_k + iK_k)$, and the two sets commute. We now derive this.

By the double-cover theorem, Theorem 24.5, $\mathfrak{so}(1,3) \cong \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$. A complex representation of this real algebra need not be complex linear. For example, $Y \mapsto \bar{Y}$ is a real Lie algebra automorphism that is not complex linear. Complexification treats both kinds together.

Let $\mathfrak{g}$ be a real Lie algebra and V a finite-dimensional complex space. A real Lie algebra map $\rho : \mathfrak{g} \to \mathfrak{gl}(V)$ extends uniquely to a complex-linear Lie algebra map $\rho_{\mathbb{C}}(Y \otimes z) = z\rho(Y)$ on $\mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$. A complex subspace is ρ-invariant if and only if it is $\rho_{\mathbb{C}}$-invariant. A linear map intertwines two such maps ρ if and only if it intertwines their extensions.

The real algebra $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}$ carries its own operator M, given by $M(Y) = iY$. It satisfies $M^2 = -1$ and $[MY, Z] = M[Y, Z]$. After complexifying, $M \otimes 1$ has the eigenvalues i and $-i$. Its eigenspaces are ideals, since $[(M \otimes 1)u, v] = (M \otimes 1)[u, v]$. They commute: if u and v have eigenvalues i and $-i$, then $i[u, v] = (M \otimes 1)[u, v] = -i[u, v]$. The next proposition identifies these ideals with two copies of $\mathfrak{sl}(2,\mathbb{C})$. Under it, $M \otimes 1$ becomes multiplication by i on the first copy and by $-i$ on the second.

Proposition 24.7. *The complex-linear map*

$$\iota : \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \to \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}), \qquad Y \otimes z \mapsto (zY,\ z\bar{Y}),$$

is an isomorphism of complex Lie algebras.

Proof. The pair $(zY, z\bar{Y})$ is real bilinear in (Y, z), so ι is well defined and complex linear. Complex conjugation of matrices preserves brackets, so

$$\iota([Y \otimes z, Y' \otimes z']) = \big(zz'[Y,Y'],\, zz'[\bar{Y},\bar{Y}']\big) = [\iota(Y \otimes z), \iota(Y' \otimes z')].$$

Both Lie algebras have complex dimension 6, so it suffices to prove that ι is injective. A general element $Y \otimes 1 + Y' \otimes i$ has image $(Y + iY',\ \bar{Y} + i\bar{Y}')$. If this is zero, then $Y = -iY'$ and $\bar{Y} = -i\bar{Y}'$. Conjugating the first equation gives $\bar{Y} = i\bar{Y}'$. Hence $\bar{Y}' = 0$, so $Y' = 0$ and $Y = 0$. ∎

Since $\overline{iX_k} = -i\bar{X}_k$, we have $\iota(X_k \otimes 1) = (X_k, \bar{X}_k)$ and $\iota(iX_k \otimes i) = (-X_k, \bar{X}_k)$. Hence the elements

$$X_k^L = \tfrac{1}{2}(X_k \otimes 1 - iX_k \otimes i), \qquad X_k^R = \tfrac{1}{2}(X_k \otimes 1 + iX_k \otimes i)$$

satisfy $\iota(X_k^L) = (X_k, 0)$ and $\iota(X_k^R) = (0, \bar{X}_k)$. In other words, X_k^L and X_k^R are the images of $X_k \otimes 1$ under the projections $\frac{1}{2}(1 - iM \otimes 1)$ and $\frac{1}{2}(1 + iM \otimes 1)$ onto the two ideals. We identify $\mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathfrak{so}(1,3)_{\mathbb{C}} = \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$ through the complex-linear extension of $d\varphi$. Then $X_k^L = \frac{1}{2}(J_k - iK_k)$ and $X_k^R = \frac{1}{2}(J_k + iK_k)$. These elements satisfy

$$[X_i^L, X_j^L] = \epsilon_{ijk} X_k^L, \qquad [X_i^R, X_j^R] = \epsilon_{ijk} X_k^R, \qquad [X_i^L, X_j^R] = 0.$$

Moreover $J_k = X_k^L + X_k^R$ and $K_k = i(X_k^L - X_k^R)$.

24.4 Classification of finite-dimensional representations

For $m \geq 0$ let $V_m = \mathrm{Sym}^m(\mathbb{C}^2)$, and let $S^m(A)$ be the restriction of $A^{\otimes m}$ to V_m. The entries of $S^m(A)$ are polynomials in the entries of A, so dS^m is complex linear on $\mathfrak{sl}(2,\mathbb{C})$. From the Lie algebra background, dS^m is the irreducible module of highest weight m. The element $H = \mathrm{diag}(1,-1)$ acts with eigenvalues $m, m-2, \dots, -m$. On $\mathrm{SU}(2)$, V_m is the spin-$m/2$ representation.

Definition 24.8. For $j_L, j_R \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, the representation (j_L, j_R) of $\mathrm{SL}(2,\mathbb{C})$ is

$$D^{(j_L, j_R)}(A) = S^{2j_L}(A) \otimes S^{2j_R}(\bar{A}) \quad \text{on} \quad V_{2j_L} \otimes V_{2j_R}.$$

Since $A \mapsto \bar{A}$ is a continuous automorphism of $\mathrm{SL}(2,\mathbb{C})$, this is a continuous representation. Its dimension is $(2j_L + 1)(2j_R + 1)$.

A representation Π of $\mathrm{SL}(2,\mathbb{C})$ *descends* to $\mathrm{SO}^+(1,3)$ if there is a continuous representation Π' of $\mathrm{SO}^+(1,3)$ with

$$\Pi = \Pi' \circ \varphi.$$

This is the descent of Lecture 4, now through the covering φ.

Theorem 24.9. *(a) Each (j_L, j_R) is irreducible, and distinct pairs give inequivalent representations. Every irreducible finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is equivalent to some (j_L, j_R).*

(b) $D^{(j_L, j_R)}(-I) = (-1)^{2j_L + 2j_R}$. Hence (j_L, j_R) descends to $\mathrm{SO}^+(1,3)$ if and only if $2j_L + 2j_R$ is even. Up to equivalence, these descended representations are all the irreducible finite-dimensional continuous complex representations of $\mathrm{SO}^+(1,3)$.

(c) The restriction of (j_L, j_R) to $\mathrm{SU}(2)$ is equivalent to $\bigoplus_j V_{2j}$, where $j = |j_L - j_R|, |j_L - j_R| + 1, \dots, j_L + j_R$.

Proof. (a) Write $D = D^{(j_L, j_R)}$, $m = 2j_L$ and $n = 2j_R$. Since $\overline{\exp tY} = \exp t\bar{Y}$, differentiation gives

$$dD(Y) = dS^m(Y) \otimes 1 + 1 \otimes dS^n(\bar{Y}), \qquad Y \in \mathfrak{sl}(2,\mathbb{C})_{\mathbb{R}}.$$

So the complex-linear extension is

$$dD_{\mathbb{C}}(Y \otimes z) = dS^m(zY) \otimes 1 + 1 \otimes dS^n(z\bar{Y}) = (dS^m \boxtimes dS^n)\big(\iota(Y \otimes z)\big).$$

Thus, through ι, $dD_{\mathbb{C}}$ is the module $V_m \boxtimes V_n$ of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Here $(Y_1, Y_2) \in \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ acts on $V_m \boxtimes V_n = V_m \otimes V_n$ by $dS^m(Y_1) \otimes 1 + 1 \otimes dS^n(Y_2)$. We import from the prerequisite course Lie Algebras (Lectures 13 and 26): these modules are irreducible and pairwise non-isomorphic, since their highest weights (m, n) differ. Every finite-dimensional irreducible module of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ is one of them.

Now we pass from algebra to group. Every continuous finite-dimensional representation Π is smooth (Lecture 3), with $\Pi(\exp Y) = \exp d\Pi(Y)$ (Lecture 2). By Step 2 of the proof of Theorem 24.5, $\mathrm{SL}(2,\mathbb{C})$ is connected, so it is generated by $\exp \mathfrak{sl}(2,\mathbb{C})$ (Lecture 2). Hence a subspace is Π-invariant if and only if it is $d\Pi$-invariant. A linear map intertwines two group representations if and only if it intertwines their differentials. By the extension facts in § 24.3, both statements pass to $d\Pi_{\mathbb{C}}$. So D is irreducible, and distinct pairs are inequivalent.

Finally, let Π be an irreducible finite-dimensional continuous complex representation. Then $d\Pi_{\mathbb{C}} \circ \iota^{-1} \cong V_m \boxtimes V_n$ for some (m, n). This is the complexified differential of $D^{(m/2, n/2)}$, so $\Pi \cong D^{(m/2, n/2)}$.

(b) We have $S^m(-I) = (-1)^m$ and $\overline{-I} = -I$, so $D(-I) = (-1)^{m+n}$. Covering maps are open. So by the double-cover theorem, Theorem 24.5, φ identifies $\mathrm{SO}^+(1,3)$ with $\mathrm{SL}(2,\mathbb{C})/\{\pm I\}$ as topological groups. By Lecture 4, continuous representations of $\mathrm{SO}^+(1,3)$ correspond to continuous representations of $\mathrm{SL}(2,\mathbb{C})$ that are trivial on $-I$. The correspondence preserves irreducibility and equivalence. Now combine this with (a).

(c) Let $\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. For $U \in \mathrm{SU}(2)$ one has $\bar{U} = \varepsilon U \varepsilon^{-1}$ (Exercise 24.3). So $S^n(\bar{U}) = S^n(\varepsilon) S^n(U) S^n(\varepsilon)^{-1}$, and the restriction is equivalent to $V_m \otimes V_n$. The Clebsch–Gordan rule from the prerequisite course Lie Algebras (Lectures 6 and 27) gives the decomposition. ∎

Every finite-dimensional continuous complex representation of $\mathrm{SL}(2,\mathbb{C})$ is a direct sum of representations (j_L, j_R). This follows from Weyl's complete reducibility theorem for the semisimple algebra $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$, transferred as in the proof.

The representations $(j, 0)$ are the holomorphic extensions from $\mathrm{SU}(2)$ given by Lecture 14, since the entries of $S^{2j}(A)$ are polynomials in the entries of A.

Only $(0, 0)$ admits an invariant inner product. The reason is that a boost generator has the nonzero real eigenvalue $j_L + j_R$ on every $(j_L, j_R) \neq (0, 0)$. Exercise 24.6 gives the proof. So every nontrivial continuous unitary representation of this group is infinite-dimensional. Such representations exist. They are induced from the upper triangular subgroup, as in Lecture 22, and their models use the sphere of light rays. We do not study them; see Knapp, *Representation Theory of Semisimple Groups*, Chapter II, Section 4.

In physics, let a field ψ take values in a representation space V with matrices $D(A)$. Its transformation is

$$\psi'(x) = D(A)\psi(\varphi(A)^{-1}x).$$

The label (j_L, j_R) classifies the components of ψ. It does not describe the unitary Hilbert space of particle states, which is the subject of Lecture 25.

24.5 Weyl spinors and four-vectors

Example 24.10 (Weyl spinors)**.** The representation $(\frac{1}{2}, 0)$ is $\mathbb{C}^2$ with $A \mapsto A$. The representation $(0, \frac{1}{2})$ is $\mathbb{C}^2$ with $A \mapsto \bar{A}$. Physicists call their vectors left-handed and right-handed Weyl spinors. The choice of hands is a convention. By the classification theorem, Theorem 24.9, the two representations are inequivalent. A direct check uses traces. For $A = \operatorname{diag}(2i, -\frac{i}{2}) \in \mathrm{SL}(2, \mathbb{C})$,

$$\operatorname{tr} A = \tfrac{3i}{2}, \qquad \operatorname{tr} \bar{A} = -\tfrac{3i}{2}.$$

Equivalent representations have equal traces. These traces differ, so the trace check confirms that the two representations are inequivalent. Both representations restrict to V_1 on $\mathrm{SU}(2)$. Both send $-I$ to -1, so neither descends to $\mathrm{SO}^+(1,3)$. The Dirac representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ also sends $-I$ to -1, so it does not descend either.

Example 24.11 (Four-vectors)**.** Let $\Psi : \mathbb{C}^2 \otimes \mathbb{C}^2 \to M_2(\mathbb{C})$ be the linear isomorphism given by $\Psi(\xi \otimes \zeta) = \xi\zeta^T$. The map Ψ sends $e_i \otimes e_j$ to the matrix unit E_{ij}. For $A \in \mathrm{SL}(2, \mathbb{C})$,

$$\Psi(A\xi \otimes \bar{A}\zeta) = A\xi(\bar{A}\zeta)^T = A\xi\zeta^T\bar{A}^T = A\,\Psi(\xi \otimes \zeta)\,A^*.$$

So $D^{(\frac{1}{2},\frac{1}{2})}(A)$ becomes $Z \mapsto AZA^*$ on $M_2(\mathbb{C}) = \mathcal{H} \oplus i\mathcal{H}$. This action preserves $\mathcal{H}$ and acts there by $\varphi(A)$. Hence $(\frac{1}{2}, \frac{1}{2})$ is equivalent to $\mathbb{C}^4 = \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{C}$ with $\varphi(A)$ extended complex-linearly. Since $2 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2$ is even, the representation $(\frac{1}{2}, \frac{1}{2})$ descends by the descent criterion of Theorem 24.9. The descended representation is the defining representation of $\mathrm{SO}^+(1,3)$ on $\mathbb{C}^4$.

Two checks are consistent with this identification. First, the restriction rule of Theorem 24.9(c) predicts $V_0 \oplus V_2$ on $\mathrm{SU}(2)$. Indeed $U \in \mathrm{SU}(2)$ acts by $Z \mapsto UZU^{-1}$. It fixes I, which spans the time direction. It preserves the traceless matrices, which are the spatial vectors and carry spin 1. Second, compare boost eigenvalues. In $(\frac{1}{2}, \frac{1}{2})$, the element $iX_3 = \frac{1}{2}\sigma_3$ acts by $\frac{1}{2}(H \otimes 1 + 1 \otimes H)$. Its eigenvalues are $\frac{1}{2}(q + q')$ with $q, q' \in \{\pm 1\}$:

$$1, \quad 0, \quad 0, \quad -1.$$

On $\mathbb{R}^4$, the matrix $d\varphi(iX_3) = K_3$ has eigenvectors $e_0 + e_3$, e_1, e_2, $e_0 - e_3$ with eigenvalues $1, 0, 0, -1$. The lists agree.

The table lists small cases. Exercise 24.4 identifies $(1,0) \oplus (0,1)$ with $\Lambda^2\mathbb{C}^4$, the complexified antisymmetric tensors.

(j_L, j_R)	dimension	on SU(2)	descends	name
$(0,0)$	1	V_0	yes	scalar
$(\frac{1}{2},0)$, $(0,\frac{1}{2})$	2	V_1	no	Weyl spinors
$(1,0)$, $(0,1)$	3	V_2	yes	summands of $\Lambda^2\mathbb{C}^4$
$(\frac{1}{2},\frac{1}{2})$	4	$V_0 \oplus V_2$	yes	four-vector
$(1,\frac{1}{2})$	6	$V_1 \oplus V_3$	no	–

In physics language, the two summands of $\Lambda^2\mathbb{C}^4$ are spanned by the complex combinations $\vec{E} + i\vec{B}$ and $\vec{E} - i\vec{B}$ of the electric and magnetic fields. Which one is called $(1,0)$ depends on the convention for hands.

24.6 Self-checks

(1) Compute $\varphi(\mathrm{diag}(2, \frac{1}{2}))$.
(2) Find all $A \in \mathrm{SL}(2,\mathbb{C})$ with $\varphi(A) = R_3(\pi)$.
(3) Does $(\frac{3}{2}, \frac{1}{2})$ descend to $\mathrm{SO}^+(1,3)$? Give its dimension and its restriction to $\mathrm{SU}(2)$.

Answer. (1) By the diagonal formula of Example 24.4 with $a = 2$, we get $x_0 \pm x_3 \mapsto 4^{\pm 1}(x_0 \pm x_3)$. Since $a/\bar{a} = 1$, the coordinates x_1 and x_2 are fixed. This is $B_3(t)$ with $e^t = 4$, so $\cosh t = \frac{17}{8}$ and $\sinh t = \frac{15}{8}$. Thus

$$\varphi\big(\mathrm{diag}(2, \tfrac{1}{2})\big) = \begin{pmatrix} \frac{17}{8} & 0 & 0 & \frac{15}{8} \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{15}{8} & 0 & 0 & \frac{17}{8} \end{pmatrix}.$$

Check: $17^2 - 15^2 = 64$.

(2) The element $\exp(\pi X_3) = \mathrm{diag}(-i, i)$ maps to $R_3(\pi)$. The fibre is a coset of $\{\pm I\}$, so the answer is $A = \pm\,\mathrm{diag}(-i, i)$.

(3) Here $2j_L + 2j_R = 4$ is even, so it descends. Its dimension is $4 \cdot 2 = 8$. The restriction is $V_2 \oplus V_4$, of dimension $3 + 5 = 8$.

24.7 Summary

The action $X \mapsto AXA^*$ on Hermitian matrices preserves $\det X = \langle x, x\rangle$. It gives a two-sheeted covering $\varphi : \mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ with kernel $\{\pm I\}$. Its restriction to real matrices is the double cover $\mathrm{SL}(2,\mathbb{R}) \to \mathrm{SO}^+(1,2)$. The map ι identifies the complexified Lorentz algebra with $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$. Up to equivalence, the irreducible finite-dimensional continuous representations of $\mathrm{SL}(2,\mathbb{C})$ are the (j_L, j_R), given by $S^{2j_L}(A) \otimes S^{2j_R}(\bar{A})$. Exactly those with $2j_L + 2j_R$ even descend to $\mathrm{SO}^+(1,3)$. Weyl spinors are $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, and four-vectors are $(\frac{1}{2}, \frac{1}{2})$.

24.8 Reading

- A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 2001, Chapter II, Section 3, on finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$.

Lecture 25

Poincaré representations and little groups

Objectives. *This lecture reduces the classification of positive-energy irreducible unitary representations of the Poincaré group to little groups. It makes the massive and finite-helicity sectors explicit. Translations are diagonalised by momenta. Lorentz transformations move momenta along orbits. The orbit and an irreducible unitary representation of the stabiliser of one of its points determine the representation up to unitary equivalence.*

By the end of this lecture you will be able to:

- Construct the group $\mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$ and its action on momenta.
- Classify the momentum orbits and compute the massive and massless little groups.
- State Mackey's classification with its hypotheses, apply it to the $ax+b$ group, and separate the finite-helicity, continuous-spin and zero-momentum sectors.
- Verify the massive spin model and name the steps it omits.

25.1 The Poincaré group as a semidirect product

Spacetime is $\mathbb{R}^4$ with points $x = (x_0, x_1, x_2, x_3)$, where x_0 is time. The *Minkowski pairing* is

$$x \cdot y = x_0 y_0 - x_1 y_1 - x_2 y_2 - x_3 y_3.$$

Lorentz transformations are the linear maps preserving this pairing. The proper orthochronous ones form the group $\mathrm{SO}^+(1,3)$. Adding translations gives the Poincaré group. Quantum symmetries act projectively, and Lecture 13 explains why one then passes to a cover. We prove no lifting theorem. We study unitary representations of the cover built below.

Identify $\mathbb{R}^4$ with the 2×2 Hermitian matrices by

$$\Phi(x) = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}, \qquad \det \Phi(x) = x \cdot x, \qquad \operatorname{tr} \Phi(x) = 2x_0. \tag{25.1}$$

For $g \in \mathrm{SL}(2,\mathbb{C})$ define the map $\Lambda(g)\colon \mathbb{R}^4 \to \mathbb{R}^4$ by

$$\Phi(\Lambda(g)x) = g\,\Phi(x)\,g^*.$$

This is the construction of Lecture 24, with Φ in place of X, Λ in place of φ, and $x \cdot y$ in place of $\langle x, y\rangle$.

Lemma 25.1. *The map $\Lambda(g)$ is linear, and its entries are real polynomials in the real and imaginary parts of the entries of g. Moreover $\Lambda(gg') = \Lambda(g)\Lambda(g')$ and $\Lambda(g)x \cdot \Lambda(g)y = x \cdot y$.*

Proof. The matrix $g\Phi(x)g^{*}$ is Hermitian and depends linearly on x. For a Hermitian matrix $h = (h_{ij})$, the vector $\Phi^{-1}(h)$ depends linearly on h. For example, its time coordinate is $x_0 = \tfrac{1}{2}(h_{11} + h_{22})$. Hence $\Lambda(g)$ is linear. Each entry of $g\Phi(x)g^{*}$ is a sum of products of an entry of g, an entry of $\Phi(x)$ and the complex conjugate of an entry of g. So the entries of $\Lambda(g)$ are real polynomials in the real and imaginary parts of the entries of g.

The second claim follows from $gg'\Phi(x)(gg')^{*} = g\big(g'\Phi(x)g'^{*}\big)g^{*}$. For the third, $\det(g\Phi(x)g^{*}) = |\det g|^{2}\det\Phi(x) = x\cdot x$. Polarisation, $2x\cdot y = (x+y)\cdot(x+y) - x\cdot x - y\cdot y$, gives the pairing. ∎

By Lecture 24, Λ maps $\mathrm{SL}(2,\mathbb{C})$ onto $\mathrm{SO}^{+}(1,3)$ with kernel $\{\pm I\}$. The final claim of Proposition 25.3 and the descent statements use this fact. The classification itself does not depend on it.

Definition 25.2. The group $G = \mathbb{R}^{4} \rtimes \mathrm{SL}(2,\mathbb{C})$ is the set $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ with the product

$$(a, g)(a', g') = (a + \Lambda(g)a',\ gg'). \tag{25.2}$$

Proposition 25.3. *G is a Lie group of dimension 10. Its identity is $(0, I)$ and $(a,g)^{-1} = (-\Lambda(g)^{-1}a,\ g^{-1})$. The translations $A = \{(a, I)\} \cong \mathbb{R}^{4}$ form a closed normal subgroup, and*

$$(0, g)(a, I)(0, g)^{-1} = (\Lambda(g)a,\ I). \tag{25.3}$$

The map $(a, g) \mapsto (a, \Lambda(g))$ is a homomorphism onto $\mathbb{R}^{4} \rtimes \mathrm{SO}^{+}(1,3)$ with kernel $\{(0, \pm I)\}$.

Proof. Associativity is the main point. Both bracketings of a triple product have second entry $gg'g''$. Their first entries are

$$a + \Lambda(g)a' + \Lambda(gg')a'' \qquad \text{and} \qquad a + \Lambda(g)\big(a' + \Lambda(g')a''\big).$$

These agree because $\Lambda(g)$ is linear and Λ is multiplicative, by Lemma 25.1. The stated inverse is an inverse on both sides, since

$$(a, g)(-\Lambda(g)^{-1}a, g^{-1}) = (a - a,\ I), \qquad (-\Lambda(g)^{-1}a, g^{-1})(a, g) = (-\Lambda(g)^{-1}a + \Lambda(g)^{-1}a,\ I).$$

The manifold $\mathbb{R}^{4} \times \mathrm{SL}(2,\mathbb{C})$ has dimension $4 + 6$. Multiplication is polynomial in real coordinates by Lemma 25.1. So is inversion, since g^{-1} is the adjugate of g. Hence G is a Lie group. The projection $(a, g) \mapsto g$ is a continuous homomorphism with kernel A, so A is closed and normal. Next, $(0, g)(a, I) = (\Lambda(g)a, g)$, and multiplying on the right by $(0, g^{-1})$ gives (25.3). The final map respects (25.2), because only $\Lambda(g)$ acts on translations. Its image and kernel come from those of Λ. ∎

From now on, the Poincaré group means G. The central element $z = (0, -I)$ maps to the identity.

25.2 Momentum characters and mass shells

Let U be a strongly continuous unitary representation of G on a Hilbert space. On A it gives commuting unitary operators. Their building blocks are characters. Exercise 25.2 proves the next lemma.

> **Lemma 25.4.** *Every continuous homomorphism $\mathbb{R}^4 \to \mathrm{U}(1)$ equals $\chi_p(a) = e^{-ip\cdot a}$ for a unique $p \in \mathbb{R}^4$.*

The sign follows the generator convention of Lecture 15. For the one-dimensional representation χ_p and the translation generator $X = e_0$, we have $\chi_p(te_0) = e^{-itp_0}$. Thus $A_X = p_0$ in that convention, and p_0 is the energy.

To describe all these operators together we use a projection-valued measure. We recall this object from *Linear Algebra for Physics*, Lecture 14. There the spectral theorem attaches to a self-adjoint operator a projection $P(\Omega)$ for each Borel subset Ω of its spectrum. The Borel sets are obtained from the open sets by countable unions, intersections and complements. For multiplication by x on $L^2[0,1]$, the projection $P(\Omega)$ is multiplication by the indicator function of Ω, and each single point has $P(\{\lambda\}) = 0$.

The same definition works on $\mathbb{R}^4$. A *projection-valued measure* E on $\mathbb{R}^4$ assigns an orthogonal projection $E(S)$ to every Borel set $S \subseteq \mathbb{R}^4$, with $E(\mathbb{R}^4) = I$ and countable additivity in the strong operator topology for disjoint sets. We say that E is *concentrated* on a Borel set S when $E(S) = I$.

We cite the *Stone–Naimark–Ambrose–Godement (SNAG) theorem* for vector groups; see Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2. For a strongly continuous unitary representation of $\mathbb{R}^4$, it gives a unique projection-valued measure E on $\mathbb{R}^4$ with

$$U(a, I) = \int_{\mathbb{R}^4} \chi_p(a)\, dE(p).$$

Here $E(S)$ is the projection onto the part of the space whose momentum lies in S. For a vector v, the map $S \mapsto \langle v, E(S)v\rangle$ is a positive Borel measure of total mass $\|v\|^2$. For a unit vector v, it is the probability distribution of the momentum in the state v, as in the Born rule of Lecture 14 of that course. Unlike the measures of Lecture 17, it may vanish on open sets. The operator integral has the following meaning: for every vector v,

$$\langle v, U(a, I)v\rangle = \int_{\mathbb{R}^4} \chi_p(a)\, d\langle v, E(p)v\rangle.$$

By polarisation, these scalar identities determine the bounded operator $U(a, I)$.

Example 25.5 (A multiplication model)**.** On $L^2(\mathbb{R}^4, d^4p)$ let $(U(a)\psi)(p) = e^{-ip\cdot a}\psi(p)$, and let $E(S)$ multiply ψ by the indicator function of S. Then

$$\langle \psi, E(S)\psi\rangle = \int_S |\psi(p)|^2\, d^4p, \qquad \langle \psi, U(a)\psi\rangle = \int_{\mathbb{R}^4} e^{-ip\cdot a}|\psi(p)|^2\, d^4p.$$

So this E is the measure that the SNAG theorem attaches to U. A single point has Lebesgue measure zero, so $E(\{p\}) = 0$ for every p. This representation therefore has no momentum eigenvector in the Hilbert space, although every vector has a momentum distribution.

The *support* of E is the complement of the union of open sets O with $E(O) = 0$. Since $\mathbb{R}^4$ has a countable basis, this union is a countable union of such null sets. Thus the support is the smallest closed set F with $E(F) = I$. We call it the *translation spectrum*.

We use the spectral calculus for projection-valued measures as a cited analytic input (Hall, *Quantum Theory for Mathematicians*, §10.1). It defines the self-adjoint operator

$$P_0 = \int_{\mathbb{R}^4} p_0 \, dE(p), \qquad \mathrm{Dom}(P_0) = \left\{ v : \int_{\mathbb{R}^4} p_0^2 \, d\langle v, E(p)v\rangle < \infty \right\}.$$

By Stone's theorem with the convention of Lecture 15, this is the generator of $t \mapsto U(te_0, I)$. Positive energy means $P_0 \geq 0$. By the spectral calculus this is equivalent to concentration of E on $\{p_0 \geq 0\}$. Momentum eigenvectors need not exist in the Hilbert space.

Still, a formal computation shows the mechanism. If $U(a, I)\psi = \chi_p(a)\psi$ for all a, then (25.3) gives

$$U(a, I)U(0, g)\psi = U(0, g)\, U(\Lambda(g)^{-1}a, I)\psi = \chi_{\Lambda(g)p}(a)\, U(0, g)\psi. \tag{25.4}$$

Here $p \cdot \Lambda(g)^{-1}a = \Lambda(g)p \cdot a$ by the pairing invariance in Lemma 25.1. In particular, if g fixes p, then $U(0, g)\psi$ again has momentum p. So the stabiliser of p acts on the formal eigenvectors of momentum p. Moving this action around the orbit of p gives the whole space. This is the idea of induction, and it is why stabilisers appear in the classification below.

The same calculation with E, together with the uniqueness of E in the SNAG theorem, gives

$$U(0, g)E(S)U(0, g)^{-1} = E(\Lambda(g)S).$$

Hence the translation spectrum is stable under the following action.

Definition 25.6. The *mass-shell action* of $\mathrm{SL}(2, \mathbb{C})$ on momentum space $\mathbb{R}^4$ is $g \cdot p = \Lambda(g)p$. For $m, \kappa > 0$ set

$$X_m^{\pm} = \{p \cdot p = m^2,\ \pm p_0 > 0\}, \qquad C^{\pm} = \{p \cdot p = 0,\ \pm p_0 > 0\}, \qquad Y_\kappa = \{p \cdot p = -\kappa^2\}.$$

Theorem 25.7. *The orbits of the mass-shell action are exactly the sets $X_m^{\pm}$, $C^{\pm}$, Y_κ and $\{0\}$.*

Proof. Invariants. The number $p \cdot p$ is invariant, because $\Lambda(g)$ preserves the pairing by Lemma 25.1. Write $\vec{p} = (p_1, p_2, p_3)$, with Euclidean norm $|\vec{p}|$. Suppose $p \cdot p \geq 0$ and $p \neq 0$. Then $p_0^2 \geq |\vec{p}|^2$, so $p_0 \neq 0$. The eigenvalues $\mu_1 \geq \mu_2$ of $\Phi(p)$ satisfy

$$\mu_1\mu_2 = p \cdot p \geq 0, \qquad \mu_1 + \mu_2 = 2p_0.$$

So $p_0 > 0$ exactly when $\Phi(p)$ is positive semidefinite and nonzero. The map $\Phi \mapsto g\Phi g^*$ preserves both properties, because $v^* g\Phi g^* v = (g^* v)^* \Phi(g^* v)$ and g is invertible. Hence each listed set is a union of orbits. The listed sets partition $\mathbb{R}^4$.

Transitivity. Write $\Phi(p) = u\ \mathrm{diag}(\mu_1, \mu_2)\, u^*$ with $u \in \mathrm{U}(2)$. A scalar factor $e^{i\phi}$ with $e^{2i\phi} \det u = 1$ puts u in $\mathrm{SU}(2)$ and changes nothing. Choose the base points

$$k_m = (m, 0, 0, 0), \qquad k = (1, 0, 0, 1), \qquad \ell_\kappa = (0, 0, 0, \kappa).$$

Their Φ-images are $\mathrm{diag}(m, m)$, $\mathrm{diag}(2, 0)$ and $\mathrm{diag}(\kappa, -\kappa)$. For $s > 0$ let $d_s = \mathrm{diag}(s, s^{-1})$. Then

$$d_s\ \mathrm{diag}(\alpha, \beta)\, d_s^* = \mathrm{diag}(s^2\alpha,\ s^{-2}\beta). \tag{25.5}$$

On X_m^+ we have $\mu_1, \mu_2 > 0$ and $\mu_1\mu_2 = m^2$. Take $s^2 = \mu_1/m$. Then $s^{-2}m = \mu_2$, so $p = \Lambda(ud_s)k_m$. On C^+ we have $\mu_1 = 2p_0$ and $\mu_2 = 0$; take $s^2 = \mu_1/2$. On Y_κ we have $\mu_1 > 0 > \mu_2$ and $\mu_1\mu_2 = -\kappa^2$; take $s^2 = \mu_1/\kappa$. Since $\Lambda(g)$ is linear, X_m^- and C^- are the orbits of $-k_m$ and $-k$. Finally 0 is fixed. ∎

Figure 25.1 draws these orbits with one spatial coordinate suppressed. The base points k_m, k and ℓ_κ all have $p_1 = p_2 = 0$, and their stabilisers are computed in the next section.

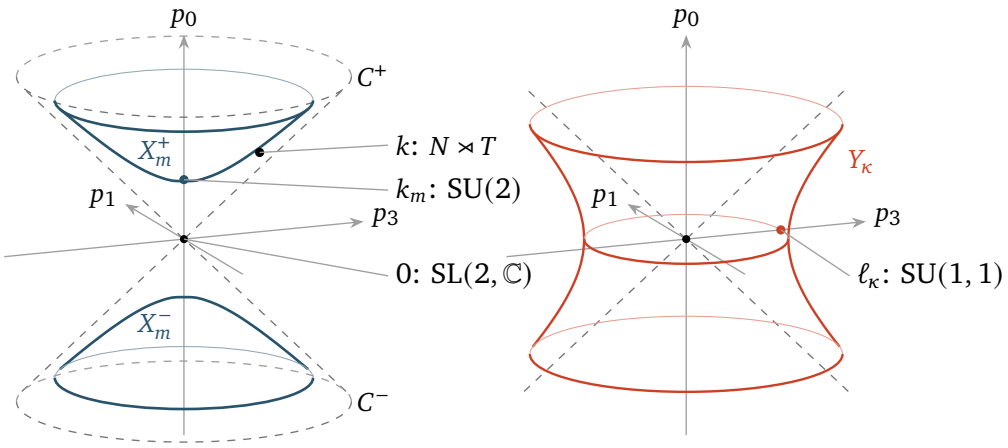


Figure 25.1: The orbits of Theorem 25.7 in the slice $p_2 = 0$. Each drawn surface is a slice of a three-dimensional orbit, since the coordinate p_2 is suppressed. In the panel with the two bowls, the bowls are $X_m^\pm$ and the dashed cone is $C^\pm$ with its vertex 0 removed. In the other panel, the connected waisted surface is Y_κ, around the same cone. Each marked base point carries the name of its stabiliser. Positive energy keeps the upper bowl, the upper cone and the origin. It excludes all of Y_κ, including the points of Y_κ with $p_0 > 0$.

By (25.5), $\Lambda(d_s)$ multiplies $p_0 + p_3$ by s^2 and $p_0 - p_3$ by s^{-2}. It is a boost along the x_3-axis.

A subset of $\mathbb{R}^4$ is *locally closed* when it is the intersection of a closed set and an open set. Each orbit is locally closed. For example, C^+ is the closed set $\{p \cdot p = 0\}$ intersected with the open set $\{p_0 > 0\}$.

The invariant Borel sets $\{p \cdot p < r\}$ with r rational, together with $\{p \cdot p \geq 0,\ p_0 > 0\}$, $\{p \cdot p \geq 0,\ p_0 < 0\}$ and $\{0\}$, separate the orbits. Local closedness and this separation property form the regularity used below.

These orbits lie in the dual of the translation subgroup. They are not the coadjoint orbits of Lecture 19, which lie in the dual of the whole Lie algebra. The two kinds are related. Restricting a functional to the translations maps a coadjoint orbit onto one orbit in the dual of the translations, because translations act trivially on that dual. With $\chi_\ell = e^{i\ell}$ in Lecture 19 and $\chi_p = e^{-ip}$ here, the restriction of ℓ is the functional $-p$. For the $ax + b$ group, this restriction sends the half-planes of Exercise 19.1 onto the frequency half-lines of Lecture 17.

25.3 Little groups and Mackey's classification

Definition 25.8. The *little group* of $p \in \mathbb{R}^4$ is its stabiliser $H_p = \{g \in \mathrm{SL}(2,\mathbb{C}) : \Lambda(g)p = p\}$.

For $b \in \mathbb{C}$ and $|\lambda| = 1$ write

$$n(b) = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}, \qquad r(\lambda) = \begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}.$$

Proposition 25.9. *(a)* $H_{k_m} = \mathrm{SU}(2)$.
(b) $H_k = \{n(b)r(\lambda) : b \in \mathbb{C},\ |\lambda| = 1\}$, *and* $r(\lambda)n(b)r(\lambda)^{-1} = n(\lambda^2 b)$.
(c) $\Lambda(r(e^{-i\theta/2}))$ *rotates the* (x_1, x_2)*-plane by the angle* $+\theta$ *and fixes* x_0 *and* x_3.
(d) $H_{\ell_\kappa} = \{g : g\sigma_3 g^* = \sigma_3\} = \mathrm{SU}(1,1)$, *where* $\sigma_3 = \mathrm{diag}(1,-1)$. *Also* $H_0 = \mathrm{SL}(2,\mathbb{C})$.

Proof. (a) The condition is $g(mI)g^* = mI$, that is, $gg^* = I$.

(b) Since $\Phi(k) = 2e_1e_1^*$, the condition is $vv^* = e_1e_1^*$ for $v = ge_1$. In entries,

$$\begin{pmatrix} |v_1|^2 & v_1\bar v_2 \\ v_2\bar v_1 & |v_2|^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $v_2 = 0$ and $v_1 = \lambda$ with $|\lambda| = 1$. Since $\det g = 1$,

$$g = \begin{pmatrix} \lambda & w \\ 0 & \bar\lambda \end{pmatrix} = n(\lambda w)\, r(\lambda).$$

Every such g fixes k. The conjugation formula is a product of three matrices:

$$\begin{pmatrix} \lambda & 0 \\ 0 & \bar\lambda \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \bar\lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 1 & \lambda^2 b \\ 0 & 1 \end{pmatrix}.$$

(c) Conjugation by $r(\lambda)$ fixes the diagonal of $\Phi(p)$. It multiplies the entry $p_1 + ip_2$ by $\bar\lambda^2$, which is $e^{i\theta}$ for $\lambda = e^{-i\theta/2}$.

(d) This restates the definition, since $\Phi(\ell_\kappa) = \kappa\sigma_3$. ∎

The group in part (d) is the group $\mathrm{SU}(1,1)$ of Lecture 23. It is also a copy of the real subgroup of Lecture 24. Every $g \in \mathrm{SL}(2,\mathbb{C})$ satisfies $g\sigma_2 g^T = \sigma_2$, so $g\sigma_2 g^* = \sigma_2$ exactly when g is real. Thus $\mathrm{SL}(2,\mathbb{R})$ is the stabiliser of the spacelike vector e_2. The Cayley matrix γ of Lecture 23 satisfies $\gamma\gamma^* = 2I$, so a scalar multiple u of γ lies in $\mathrm{SU}(2)$. It also satisfies $\gamma\sigma_2\gamma^{-1} = -\sigma_3$, so $\Lambda(u)e_2 = -e_3$. Hence $g \mapsto \gamma g \gamma^{-1}$ carries $\mathrm{SL}(2,\mathbb{R})$ onto the stabiliser of e_3, which is H_{ℓ_κ}. This does not make the spacelike orbit relevant to positive energy. It identifies the little-group data of Y_κ with the classification of Lecture 23.

Write $N = \{n(b)\} \cong \mathbb{C}$ and $T = \{r(\lambda)\} \cong \mathrm{U}(1)$. By the factorisation in Proposition 25.9(b), $H_k = N \rtimes T$. The map $n(b)r(\lambda) \mapsto (b, \lambda^2)$ is two-to-one onto the group of orientation-preserving Euclidean motions of the plane. So we call N the *little-group translations*. The maps $\Lambda(n(b))$ are Lorentz transformations fixing k, not spacetime translations. For $b = 1$ this map is the null rotation of Exercise 24.2.

We have $r(e^{-i\theta/2}) = \exp(\theta X_3)$ with $X_3 = -i\sigma_3/2$. By the rotation formula in Proposition 25.9(c), this generates positive-angle rotations about x_3, as in Lecture 24.

The next theorem uses unitary induction from Lecture 17.

Theorem 25.10 (Mackey, cited). *Let $G = \mathbb{R}^n \rtimes H$, where H is a second-countable Lie group acting linearly and continuously on $\mathbb{R}^n$. For $p \in (\mathbb{R}^n)^*$ put $\chi_p(a) = e^{-ip(a)}$. Let H act on these momenta so that $\chi_{h\cdot p}(a) = \chi_p(h^{-1}\cdot a)$. Assume the action is* regular*: orbits are locally closed, and countably many invariant Borel sets separate them. Let H_p be the stabiliser of p. For a strongly continuous unitary representation U of G, let E be the projection-valued measure on $(\mathbb{R}^n)^*$ that the SNAG theorem attaches to the translations $a \mapsto U(a, e)$.*

(a) If U is irreducible, then E is concentrated on one orbit Ω.

(b) Let $p \in \Omega$ and let σ be a strongly continuous irreducible unitary representation of H_p. Put $(\chi_p \otimes \sigma)(a, h) = \chi_p(a)\sigma(h)$. Then $\operatorname{Ind}_{\mathbb{R}^n \rtimes H_p}^{G}(\chi_p \otimes \sigma)$ is irreducible, and its measure E is concentrated on Ω.

(c) Every strongly continuous irreducible unitary representation whose measure E is concentrated on Ω is unitarily equivalent to one of the induced representations in (b). For a fixed $p \in \Omega$, two of these induced representations are unitarily equivalent exactly when their representations σ are unitarily equivalent.

This is Mackey's method for semidirect products, as presented in A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. For our G it is Wigner's classification of 1939. For our spacetime group, the Minkowski pairing identifies the functionals $p \in (\mathbb{R}^4)^*$ of the theorem with the vectors p used above. The remarks after the orbit theorem, Theorem 25.7, verify regularity. The dual action is $g \cdot p = \Lambda(g)p$ by (25.4).

Before the ten-dimensional group we run the theorem on two small groups from earlier lectures.

Example 25.11 (The $ax+b$ and Heisenberg groups). The $ax+b$ group of Lecture 17 is $\mathbb{R} \rtimes \mathbb{R}_{>0}$, where $a > 0$ acts on the translations n_c by scaling. We keep the frequency ξ of Lecture 17, with $\chi_\xi(n_c) = e^{i\xi c}$, so the momentum of the theorem is $p = -\xi$. The element a sends ξ to ξ/a. The orbits are $\xi > 0$, $\xi < 0$ and $\{0\}$. These three sets are locally closed and separate the orbits, so the action is regular.

The orbits $\xi > 0$ and $\xi < 0$ have trivial stabiliser. By part (b), each gives one irreducible representation, induced from χ_ξ on N. These are Π_+ and Π_- of Lecture 17, so both are irreducible. The orbit $\{0\}$ has stabiliser $\mathbb{R}_{>0}$. The irreducible unitary representations of $\mathbb{R}_{>0}$ are the characters $a \mapsto a^{is}$, with $s \in \mathbb{R}$. They give the one-dimensional representations $(a, b) \mapsto a^{is}$. By parts (a) and (c), this is the whole list of irreducible unitary representations. In particular, the representation Π_0 of Lecture 17 is not irreducible. Translations act trivially in it, so its measure is concentrated on $\{0\}$, and it is infinite-dimensional.

The Heisenberg group of Lecture 18 is also a semidirect product. The subgroup $A = \{(0, b, c)\}$ is a closed normal vector group, the subgroup $\{(a, 0, 0)\}$ is a copy of $\mathbb{R}$, and

$$(a, 0, 0)(0, b, c)(-a, 0, 0) = (0, b, c + ab).$$

Write a character of A as $e^{i(\mu b + \lambda c)}$. The element $(a, 0, 0)$ sends (μ, λ) to $(\mu - \lambda a, \lambda)$. For $\lambda \neq 0$, the line with this value of λ is one orbit with trivial stabiliser. So exactly one irreducible unitary representation has central character λ. By Exercise 18.5 it is the Schrödinger representation Π_λ. This is the Stone–von Neumann theorem for irreducible representations. For $\lambda = 0$, the orbits are single points $(\mu, 0)$ with stabiliser $\mathbb{R}$. The characters of this stabiliser give the characters $\chi_{\alpha,\beta}$ of Lecture 18. All these orbits are closed. Rational levels of λ, and of μ on the line $\lambda = 0$, separate them. So this action is regular too.

Corollary 25.12. *(a) For $m > 0$, up to unitary equivalence, the strongly continuous irreducible unitary representations of G with E concentrated on X_m^+ are labelled by $n \geq 0$. The label is the $\mathrm{SU}(2)$-representation V_n, of dimension $n+1$ and spin $n/2$.*

(b) Up to unitary equivalence, those with E concentrated on C^+ are labelled by the equivalence classes of strongly continuous irreducible unitary representations of $H_k = N \rtimes T$.

Proof. Combine Mackey's theorem, Theorem 25.10, with the little groups computed in Proposition 25.9. In the massive case, strongly continuous irreducible unitary representations of the compact group $\mathrm{SU}(2)$ are finite-dimensional by Lecture 10. Hence they are the V_n of Lecture 8. ∎

25.4 Massless sectors: helicity and continuous spin

Proposition 25.13. *Let σ be a finite-dimensional continuous unitary representation of H_k on V. Then $\sigma(n(b)) = I$ for all b. If σ is irreducible, then $\sigma(n(b)r(\lambda)) = \lambda^\nu$ for a unique $\nu \in \mathbb{Z}$, and*

$$\sigma(\exp \theta X_3) = e^{-i\theta\nu/2}.$$

Proof. The operators $\sigma(n(b))$ are unitary and commute. So V is the direct sum of their joint eigenspaces. On each, $b \mapsto \sigma(n(b))$ is a continuous character of $\mathbb{C} \cong \mathbb{R}^2$. As in the character lemma, Lemma 25.4, it has the form $b \mapsto e^{-i\,\mathrm{Re}(\bar\zeta b)}$ for a unique $\zeta \in \mathbb{C}$. Let V_ζ be the joint eigenspace for ζ. Let S be the finite set of ζ with $V_\zeta \neq 0$.

This is the key step. By the conjugation formula in Proposition 25.9(b), $n(b)r(\lambda) = r(\lambda)n(\bar\lambda^2 b)$. Take $v \in V_\zeta$. Then

$$\sigma(n(b))\sigma(r(\lambda))v = \sigma(r(\lambda))\sigma(n(\bar\lambda^2 b))v = e^{-i\,\mathrm{Re}(\overline{\lambda^2\zeta}\,b)}\,\sigma(r(\lambda))v.$$

So $\sigma(r(\lambda))V_\zeta = V_{\lambda^2\zeta}$. Every unit complex number is some λ^2. If some $\zeta \neq 0$ lay in S, the whole circle $|\zeta'| = |\zeta|$ would lie in S. This contradicts finiteness. So $S = \{0\}$, and N acts trivially.

Then σ factors through $n(b)r(\lambda) \mapsto \lambda$. This is a homomorphism onto $\mathrm{U}(1)$, since $n(b)r(\lambda)n(b')r(\lambda') = n(b + \lambda^2 b')r(\lambda\lambda')$. The operators $\sigma(r(\lambda))$ commute. If σ is irreducible, a common eigenline is invariant, so $\dim V = 1$. By the one-variable character argument of Exercise 25.2(a), $\theta \mapsto \sigma(r(e^{i\theta}))$ equals $e^{i\nu\theta}$ for some real ν. Periodicity in θ forces $\nu \in \mathbb{Z}$. Finally $\exp \theta X_3 = r(e^{-i\theta/2})$ maps to $e^{-i\nu\theta/2}$. ∎

Compare with the generator convention $\Pi(\exp \theta X) = \exp(-i\theta A_X)$ of Lecture 15. The self-adjoint generator of rotations about the momentum direction is $A_{X_3} = \nu/2$. This number is the *helicity*.

The *finite-helicity* massless representations are the representations induced from $\chi_k \otimes \sigma$, where σ is a finite-dimensional continuous irreducible unitary representation of H_k. By Proposition 25.13, these are the representations $\mathrm{Ind}(\chi_k \otimes \sigma_\nu)$ with $\nu \in \mathbb{Z}$ and $\sigma_\nu(n(b)r(\lambda)) = \lambda^\nu$. Each has the single helicity $\nu/2$. In these representations the little-group translations act trivially.

Remark 25.14. Mackey's theorem, Theorem 25.10, also applies to $H_k = N \rtimes T$. The T-orbits on the characters of N are $\{0\}$ and the circles $|\zeta| = \rho > 0$. These orbits are closed. The sets $\{|\zeta| < r\}$ with $r \in \mathbb{Q}$, together with $\{0\}$, separate them. Hence the action is regular.

The orbit $\{0\}$ gives the σ_ν. A circle has stabiliser $\{r(\pm 1)\}$ with two characters. These give infinite-dimensional irreducible representations of H_k that are nontrivial on N (Exercise 25.7). Inducing them to G gives the *continuous-spin* representations. Their spectral measure is concentrated on C^+, so their energy is positive. Positive energy alone does not exclude them. They form a sector separate from finite helicity.

Now suppose a nonzero representation has E concentrated on C^+. Its closed support meets C^+ and is Lorentz-invariant by covariance of E. Transitivity then puts all of C^+ in the support. Closedness adds its limit point 0. So concentration on an orbit differs from equality of closed supports.

The orbit $\{0\}$ is also separate. Translations act trivially there. Any strongly continuous irreducible unitary representation of $\mathrm{SL}(2,\mathbb{C})$ gives a representation of this zero-momentum sector by composition with the projection $G \to \mathrm{SL}(2,\mathbb{C})$. The trivial one is the usual vacuum choice. Positive energy alone does not select it.

Positive energy excludes X_m^- and C^-, where $p_0 < 0$. It also excludes Y_κ, because this orbit contains points with $p_0 < 0$ and the group moves every point of it to such points. In detail, the set $S = Y_\kappa \cap \{p_0 < 0\}$ is nonempty and relatively open. By transitivity, the Lorentz translates of S cover Y_κ. By second countability, countably many of these translates already cover Y_κ. Positive energy gives $E(S) = 0$, since $S \subset \{p_0 < 0\}$. Then covariance of E makes every translate of S an E-null set. Disjointising this countable cover and using countable additivity would give $E(Y_\kappa) = 0$. This contradicts concentration of E on that orbit in a nonzero representation.

The table collects the sectors. Here $\rho > 0$ is the radius of a T-orbit on the characters of N, and the sign is the value at $r(-1)$ of a character of its stabiliser $\{r(\pm 1)\}$.

Orbit	Base point	Stabiliser	Labels	Energy ≥ 0
X_m^+	k_m	$\mathrm{SU}(2)$	V_n, $n \geq 0$	yes
C^+, N trivial	k	$N \rtimes T$	σ_ν, $\nu \in \mathbb{Z}$	yes
C^+, N nontrivial	k	$N \rtimes T$	$\rho > 0$ and a sign	yes
$\{0\}$	0	$\mathrm{SL}(2,\mathbb{C})$	not computed here	yes
Y_κ	ℓ_κ	$\mathrm{SU}(1,1)$	the list of Lecture 23	no
X_m^-, C^-	$-k_m$, $-k$	as for X_m^+, C^+	as for X_m^+, C^+	no

In physics language, X_m^+ with $n = 0$, 1, 2 describes massive particles such as the pion, the electron and the W boson, while C^+ with $\nu = \pm 2$ describes the photon and $\nu = \pm 4$ the graviton, and no known particle has continuous spin.

25.5 The massive induced model

Fix $m > 0$ and $n \geq 0$. Let D_n be the irreducible unitary representation of $\mathrm{SU}(2)$ on V_n. Put $\omega(\vec{p}) = \sqrt{m^2 + |\vec{p}|^2}$. Define a measure μ_m on X_m^+ by

$$\int_{X_m^+} f \, d\mu_m = \int_{\mathbb{R}^3} f(\omega(\vec{p}), \vec{p}) \, \frac{d^3\vec{p}}{2\,\omega(\vec{p})}.$$

The factor $1/\omega$ makes this measure invariant. For a boost along x_3 this is visible at once. Put $p_\perp = (p_1, p_2)$ and write $p_3 = \sqrt{m^2 + |p_\perp|^2}\,\sinh\eta$. Then $\omega = \sqrt{m^2 + |p_\perp|^2}\,\cosh\eta$ and $dp_3/\omega = d\eta$. A boost along x_3 fixes $p_\perp$ and shifts the rapidity η. Exercise 25.4 proves invariance under all of $\mathrm{SL}(2,\mathbb{C})$.

For $p \in X_m^+$ the matrix $\Phi(p)$ is positive definite. Let $h_p = (\Phi(p)/m)^{1/2}$ be its positive square root. Then $\det h_p = 1$ and $h_p (mI) h_p^* = \Phi(p)$, so $\Lambda(h_p) k_m = p$. The *Wigner rotation* is

$$W(g, p) = h_p^{-1}\, g\, h_{\Lambda(g)^{-1}p}.$$

Put $q = \Lambda(g)^{-1}p$. Read from the right, the three factors move the base point k_m to q, then q to p, then p back to k_m. Figure 25.2 shows this loop for the exact data of Exercise 25.5(e).

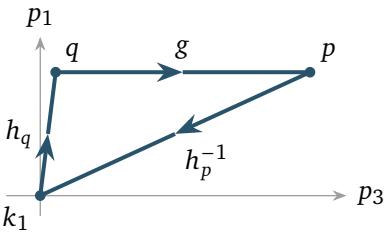


Figure 25.2: The loop of the Wigner rotation for $m = 1$ and $g = \mathrm{diag}(2, \tfrac{1}{2})$. The slice $p_2 = 0$ of X_1^+ is drawn as a graph over the (p_3, p_1)-plane, with $q = (\tfrac{713}{128}, \tfrac{87}{16}, 0, \tfrac{87}{128})$ and $p = (\tfrac{6713}{512}, \tfrac{87}{16}, 0, \tfrac{6087}{512})$. The boost h_q moves k_1 to q, the boost g moves q to p, and the boost h_p^{-1} moves p back to k_1. The product fixes k_1, so it lies in $\mathrm{SU}(2)$. It acts on the spin fibre V_n, which the drawing of the base does not show.

The model space is $\mathcal{H}_{m,n} = L^2(X_m^+, \mu_m; V_n)$. The operators are

$$(U(a,g)\psi)(p) = e^{-ip\cdot a}\, D_n\big(W(g,p)\big)\, \psi(\Lambda(g)^{-1}p). \tag{25.6}$$

Proposition 25.15. *(a)* $W(g,p) \in \mathrm{SU}(2)$.
(b) $W(gg', p) = W(g,p)\,W(g', \Lambda(g)^{-1}p)$.
(c) $U((a,g)(a',g')) = U(a,g)\,U(a',g')$.
(d) If μ_m is Λ-invariant, then each $U(a,g)$ is unitary.
(e) If $g \in \mathrm{SU}(2)$, then $W(g,p) = g$.

Proof. (a) Put $q = \Lambda(g)^{-1}p$. Then h_q maps k_m to q, g maps q to p, and h_p^{-1} maps p to k_m. So $W(g,p) \in H_{k_m} = \mathrm{SU}(2)$.

(b) With q as above, $W(g,p)W(g',q) = h_p^{-1} g h_q\, h_q^{-1} g' h_{\Lambda(g')^{-1}q}$. The middle factors cancel, and $\Lambda(g')^{-1}q = \Lambda(gg')^{-1}p$.

(c) Apply (25.6) twice:

$$U(a,g)U(a',g')\psi(p) = e^{-ip\cdot a}\, e^{-iq\cdot a'}\, D_n(W(g,p))\, D_n(W(g',q))\, \psi(\Lambda(g')^{-1}q).$$

By the pairing invariance in Lemma 25.1, $q\cdot a' = p\cdot\Lambda(g)a'$, so the phase is $e^{-ip\cdot(a+\Lambda(g)a')}$. By the cocycle identity (b), the operator is $D_n(W(gg',p))$. The argument is $\Lambda(gg')^{-1}p$. By (25.2) this is $U(a+\Lambda(g)a', gg')\psi(p)$.

(d) The phase has modulus one and D_n is unitary. Therefore

$$\|U(a,g)\psi\|^2 = \int \|\psi(\Lambda(g)^{-1}p)\|^2\, d\mu_m(p) = \|\psi\|^2.$$

By the homomorphism property (c), $U(a,g)$ has inverse $U((a,g)^{-1})$.

(e) For unitary g, the matrix gh_qg^{-1} is positive definite, and its square is $g\Phi(q)g^*/m = \Phi(p)/m$. By uniqueness of the positive square root, $gh_qg^{-1} = h_p$. Hence $W(g,p) = g$. ∎

On translations, $W(I,p) = I$, so $U(a,I)$ is multiplication by $e^{-ip\cdot a}$. Hence, as in Example 25.5, the measure E of this model has $E(S)$ equal to multiplication by the indicator function of $S\cap X_m^+$. So $E(X_m^+) = I$, and $E(\{p\}) = 0$ for every point p. Formula (25.6) also has no square root of a Jacobian, unlike the induced representations of Lecture 17, because μ_m is invariant.

By the rotation case of the Wigner rotation, Proposition 25.15(e), rotations act by $(U(0,u)\psi)(p) = D_n(u)\psi(\Lambda(u)^{-1}p)$. At the rest momentum k_m the little group acts through D_n, which is spin $n/2$. The central element acts by

$$U(z) = D_n(-I) = (-1)^n I.$$

Hence half-integer spin does not descend to $\mathbb{R}^4 \rtimes \mathrm{SO}^+(1,3)$. Time translation by t acts by multiplication by e^{-itp_0}. Its self-adjoint generator P_0 is multiplication by p_0, with domain

$$\mathrm{Dom}(P_0) = \{\psi\in\mathcal{H}_{m,n} : p_0\psi\in\mathcal{H}_{m,n}\}.$$

Since $p_0\geq m$ on X_m^+, we get $P_0\geq m$. The energy is positive.

Three steps complete the identification with a unitary induced representation. The measure and intertwining calculations are assigned to the sheet. Continuity is explained here.

1. *Invariance of μ_m.* Exercise 25.4 writes $g = u\, d_s\, v$ with $u,v\in\mathrm{SU}(2)$. Rotations fix p_0 and act orthogonally on $\vec{p}$. The boost d_s needs a Jacobian computation.
2. *Strong continuity.* For continuous compactly supported ψ, the formula is jointly continuous in (a,g,p). The positive square root h_p is continuous by matrix functional calculus. For (a,g) in a compact neighbourhood, transformed supports lie in one compact set and the integrands are uniformly bounded there. Dominated convergence gives norm continuity. Density and unitarity extend it to all ψ.
3. *Identification with induction.* The section map is

   $$f\longmapsto\psi_f, \qquad \psi_f(p) = f(0,h_p).$$

   It carries the induced representation from $A\rtimes\mathrm{SU}(2)$, with inducing data $\chi_{k_m}\otimes D_n$, onto (25.6). Exercise 25.5 checks the intertwining. Then the irreducibility of induced representations in Mackey's theorem, Theorem 25.10(b), shows that the model is irreducible.

Remark 25.16. The finite-dimensional representations of $\mathrm{SL}(2,\mathbb{C})$ admit no invariant inner product unless trivial, as proved in Lecture 24. In physics they label the components of fields. Restricted to $\mathrm{SU}(2)$, the components of a field in (j_L, j_R) carry the spins $|j_L - j_R|, \dots, j_L + j_R$ of Lecture 24. They do not classify the particle Hilbert space. For a massive particle of spin $n/2$, that space is $\mathcal{H}_{m,n}$, built from unitary $\mathrm{SU}(2)$ data on an orbit.

Self-checks

1. Is $(-1,0,0,0)$ in the orbit of k_1?
   Answer. No. $\Phi(k_1) = I$ is positive definite, but $\Phi(-1,0,0,0) = -I$ is not. Conjugation $\Phi \mapsto g\Phi g^*$ preserves positivity.
2. Do the defining matrices of H_k on $\mathbb{C}^2$ preserve some inner product?
   Answer. No. By Proposition 25.13, every finite-dimensional continuous unitary representation of H_k is trivial on N. An invariant inner product would make the defining representation such a representation. But $n(1) \neq I$.
3. In the model with $n = 2$, what are $U(z)$ and $W(u,p)$ for $u \in \mathrm{SU}(2)$?
   Answer. $U(z) = D_2(-I) = 1$, and $W(u,p) = u$ by the rotation case of the Wigner rotation, Proposition 25.15(e).

Summary

The Poincaré cover is $G = \mathbb{R}^4 \rtimes \mathrm{SL}(2,\mathbb{C})$. Translations act through momentum characters $e^{-ip\cdot a}$, and $\mathrm{SL}(2,\mathbb{C})$ moves momenta by Λ. The orbits are mass shells, light cones, spacelike hyperboloids and $\{0\}$. The projection $E(S)$ selects the part of the space with momentum in S, and in the massive model it is multiplication by an indicator function. Mackey's theorem reduces the classification to little groups. It already gives the irreducible unitary representations of the $ax+b$ group and of the Heisenberg group. Up to unitary equivalence, irreducible unitary representations with E concentrated on X_m^+ are labelled by an $\mathrm{SU}(2)$ spin $n/2$ with $n \geq 0$. Finite-helicity massless representations have trivial little-group translations. Continuous-spin sectors also have positive energy and are separate. The zero-momentum orbit $\{0\}$ gives a further separate sector.

Reading

- R. Berndt, *Representations of Linear Groups* (2007), Section 5.2, Theorem 5.2: the spectral-measure theorem for vector groups.
- E. P. Wigner, *On Unitary Representations of the Inhomogeneous Lorentz Group*, Annals of Mathematics 40 (1939), 149–204. Read the classification of momentum orbits and little groups.
- A. A. Kirillov, *Elements of the Theory of Representations* (1976), §13. Read the treatment of induced representations and Mackey's method for semidirect products.

Lecture 26

Equivariant line bundles and Borel–Weil

Objectives. *Lecture 12 induced from a character of a torus and obtained infinitely many irreducible representations. In this lecture a complex structure selects one of them. Induced representations act on equivariant functions, and the same functions can be read as sections of a vector bundle. We compute the holomorphic sections completely for the projective line. We then state the general Borel–Weil theorem, work the case* SU(3) *by hand, and prove the theorem from Peter–Weyl.*

By the end of this lecture you will be able to:

- Compute all holomorphic sections of $\mathcal{O}(m)$ on $\mathbb{CP}^1$.
- Identify the resulting SU(2) representation and its central action.
- State Borel–Weil with a specified bundle character and dual convention, and write explicit sections for SU(3).
- Prove Borel–Weil from Peter–Weyl, and name the inputs that stay cited: $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem.

26.1 From equivariant functions to sections

Let G be a Lie group, H a closed subgroup and $\sigma : H \to \mathrm{GL}(V)$ a continuous finite-dimensional complex representation. The quotient projection $G \to G/H$ is a submersion by Lecture 3. So it has a smooth local section near every point of G/H, by the local-section theorem (Lee, *Introduction to Smooth Manifolds*, second edition, Theorem 4.26). We now attach a copy of V above each coset.

Definition 26.1. The *associated vector bundle* is

$$E = G \times_H V = (G \times V)/\!\sim, \qquad (gh, v) \sim (g, \sigma(h)v).$$

Its projection is $[g, v] \mapsto gH$. A *section* is a map $s : G/H \to E$ whose value at x lies in the fibre above x. If $\dim V = 1$, it is a *line bundle*.

Each fibre is a vector space: choose g over the coset and use $v \mapsto [g, v]$. Changing g changes this identification by an invertible linear map.

Choose smooth local sections $s_i : U_i \to G$ whose domains cover G/H. The map

$$U_i \times V \longrightarrow E|_{U_i}, \qquad (x, v) \longmapsto [s_i(x), v]$$

is a bijection, because every g over x has the form $s_i(x)h$ with $h \in H$. It is linear on each fibre. On $U_i \cap U_j$, write $s_j(x) = s_i(x)h_{ij}(x)$. The same bundle vector then has coordinates

$$v_i = \sigma(h_{ij})v_j.$$

The identity $h_{ij}h_{jk} = h_{ik}$ ensures consistency on triple overlaps.

A *smooth complex vector bundle* of rank k over a manifold M is a smooth manifold E with a smooth surjection onto M and a complex vector-space structure on each fibre E_x, the preimage of x. It is required to be locally a product. Each point of M has an open neighbourhood U and a diffeomorphism $E|_U \to U \times \mathbb{C}^k$ mapping each fibre E_x complex-linearly onto $\{x\} \times \mathbb{C}^k$. These diffeomorphisms are the *smooth local trivialisations*.

These transition functions make $E = G \times_H V$ a smooth complex vector bundle, and this structure does not depend on the chosen family of local sections. The tool is the vector bundle chart lemma (Lee, Lemma 10.6). The section on sources at the end of this lecture checks its hypotheses.

A section is equivalently a function $f : G \to V$ with

$$f(gh) = \sigma(h)^{-1} f(g). \tag{26.1}$$

The correspondence is $s(gH) = [g, f(g)]$. It is well-defined because

$$[gh, f(gh)] = [g, \sigma(h)f(gh)] = [g, f(g)].$$

Conversely, a section determines a unique $f(g)$ in the coordinates supplied by g. Over U_i, the section is $x \mapsto [s_i(x), f(s_i(x))]$. The local trivialisation is a diffeomorphism, so s is smooth over U_i exactly when the coefficient $f \circ s_i$ is smooth. Moreover, s is smooth exactly when f is smooth, and s is continuous exactly when f is continuous. The section on sources gives the proof.

Left multiplication defines $a[g, v] = [ag, v]$. On sections its action is

$$(\Pi(a)s)(x) = a\, s(a^{-1}x).$$

In the function description this becomes $(\Pi(a)f)(g) = f(a^{-1}g)$. This is precisely the compact induction convention of Lecture 12.

26.2 The projective line and its two charts

We first compute everything on one example, and we start with the holomorphic language it needs. A function on an open subset of $\mathbb{C}^d$ is *holomorphic* if it is real C^∞ and its differential is $\mathbb{C}$-linear at every point. Polynomials are holomorphic, and so are quotients of polynomials where the denominator is nonzero. Composition of holomorphic maps is holomorphic by the chain rule.

A *complex manifold* is a Hausdorff, second-countable topological space covered by charts. These charts are homeomorphisms of open sets onto open subsets of $\mathbb{C}^d$. The transition maps between them have holomorphic components. A map between complex manifolds is *holomorphic* if it is continuous and its expressions in charts have holomorphic components.

Definition 26.2. The *complex projective line* $\mathbb{CP}^1$ is the set of one-dimensional complex subspaces of $\mathbb{C}^2$. A nonzero vector (v_0, v_1) determines the point $[v_0 : v_1]$, with $[cv_0 : cv_1] = [v_0 : v_1]$ for $c \neq 0$.

There are two charts. On $U_0 = \{v_0 \neq 0\}$ put $z = v_1/v_0$. On $U_1 = \{v_1 \neq 0\}$ put $w = v_0/v_1$. On the overlap, the transition between the charts is $w = 1/z$.

The charts give $\mathbb{CP}^1$ its topology. The points $[1 : 0]$ and $[0 : 1]$ have disjoint neighbourhoods $\{|z| < 1\}$ and $\{|w| < 1\}$. Every other pair lies in a common chart, so the space is Hausdorff. The two charts give a countable base. Each chart is a copy of $\mathbb{C}$, and the transition $w = 1/z$ is holomorphic on the overlap. Hence the two charts make $\mathbb{CP}^1$ a complex manifold, in the sense above.

The group $\mathrm{SU}(2)$ acts on lines by matrix multiplication. In the z chart, the action of $g = \left(\begin{smallmatrix} a & -\bar{b} \\ b & \bar{a} \end{smallmatrix}\right)$ is

$$z \longmapsto \frac{b + \bar{a}z}{a - \bar{b}z}$$

where the denominator is nonzero. The other chart expressions are similar. These rational expressions show that the action is smooth. It acts transitively. Indeed, a unit vector $v = (a, b)$ is the first column of

$$g_v = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2).$$

The stabiliser of the line $\mathbb{C}(1, 0)$ is

$$T = \{t_\theta = \mathrm{diag}(e^{i\theta}, e^{-i\theta})\}.$$

By the homogeneous-space theorem of Lecture 3, $\mathbb{CP}^1 \cong \mathrm{SU}(2)/T$ as smooth manifolds. It is compact, as the image of the compact group.

The *tautological line bundle* has fibre the line itself:

$$\mathcal{O}(-1) = \{(\ell, v) : \ell \in \mathbb{CP}^1,\ v \in \ell\}.$$

Its dual is $\mathcal{O}(1)$. For $m \geq 0$, let $\mathcal{O}(m) = \mathcal{O}(1)^{\otimes m}$. Set $\mathcal{O}(0)$ equal to the trivial line bundle. For $m < 0$, use the $(-m)$th tensor power of $\mathcal{O}(-1)$.

The local frames of $\mathcal{O}(-1)$ are $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. On the overlap, $e_1 = z^{-1}e_0$. The dual frames satisfy $e_1^* = ze_0^*$. For $m < 0$, $(e_i^*)^m$ means $e_i^{\otimes(-m)}$. Since $e_1^{\otimes(-m)} = z^m e_0^{\otimes(-m)}$, the same transition rule holds for $m < 0$. Hence a section of $\mathcal{O}(m)$ is a pair of functions f_0, f_1 with

$$f_0(z) = z^m f_1(1/z), \qquad f_1(w) = w^m f_0(1/w). \tag{26.2}$$

The frames $(e_0^*)^m$ and $(e_1^*)^m$ give fibrewise-linear bijections $\mathcal{O}(m)|_{U_i} \to U_i \times \mathbb{C}$. Their transition is multiplication by z^m, which is smooth and holomorphic on $U_0 \cap U_1$. The vector bundle chart lemma of Lee, Lemma 10.6, therefore makes $\mathcal{O}(m)$ a smooth line bundle with these trivialisations. We therefore call the section *holomorphic* when both functions are *entire*, that is, holomorphic on all of $\mathbb{C}$. The Borel–Weil discussion below shows that this definition agrees with the notion for associated bundles. We denote this space by $H^0(\mathbb{CP}^1, \mathcal{O}(m))$. The superscript 0 means ordinary global sections.

Since $w = 1/z$, the plane of z with one point added at infinity is all of $\mathbb{CP}^1$, so $\mathbb{CP}^1$ is a sphere. Write $E_0 = (e_0^*)^m$ and $E_1 = (e_1^*)^m$ for the two frames of $\mathcal{O}(m)$. They satisfy $E_1 = z^m E_0$ on the overlap. A section value over a point has one coefficient in each frame. If the frame grows by z^m, then the coefficient shrinks by z^{-m}. Figure 26.1 shows this at one point.

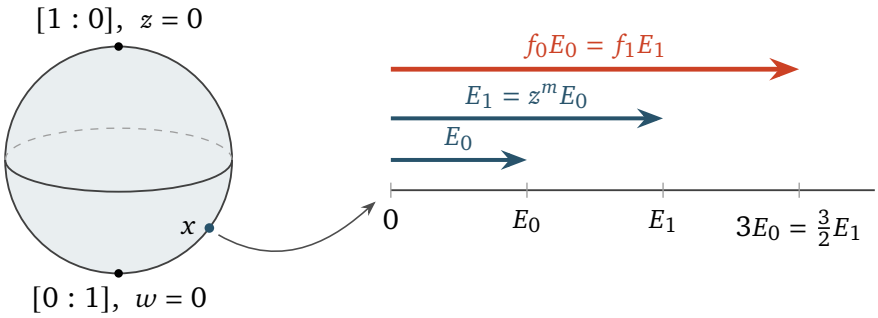


Figure 26.1: The sphere $\mathbb{CP}^1$ with the point x where $z = 2$ and $w = \frac{1}{2}$, and the fibre of $O(1)$ over x, drawn as a real line. On this sphere the chart z omits the south pole $[0 : 1]$, and the chart w omits the north pole $[1 : 0]$. Here $E_1 = 2E_0$, and the section value $3E_0$ has coefficient $f_0 = 3$ in the frame E_0 and coefficient $f_1 = \frac{3}{2}$ in the frame E_1.

26.3 Computing the holomorphic sections

We import two results from complex analysis, the power-series theorem and Cauchy's estimate. If f is entire, then

$$f(z) = \sum_{k \geq 0} a_k z^k, \qquad |a_k| \leq R^{-k} \max_{|z|=R} |f(z)| \quad (R > 0).$$

Both are imported from Stein and Shakarchi, *Complex Analysis* (2003), Chapter 2, Section 4. The section on sources records the exact statements. These results apply to our entire functions, because a $\mathbb{C}$-linear differential makes a function complex differentiable.

Theorem 26.3. *For $m \geq 0$, the holomorphic sections of $O(m)$ are naturally the homogeneous polynomials of degree m on $\mathbb{C}^2$. In particular,*

$$\dim H^0(\mathbb{CP}^1, O(m)) = m + 1.$$

For $m < 0$, the space of holomorphic sections is zero.

Proof. Let (f_0, f_1) satisfy the transition rule (26.2). Since f_1 is bounded on $|w| \leq 1$, there is C with

$$|f_0(z)| \leq C|z|^m \qquad (|z| \geq 1).$$

If $f_0(z) = \sum a_k z^k$, Cauchy's estimate on a circle of radius $R \geq 1$ gives $|a_k| \leq CR^{m-k}$. Let $R \to \infty$. For $k > m$ this forces $a_k = 0$.

If $m \geq 0$, it follows that f_0 is a polynomial of degree at most m. Conversely every such polynomial works, because

$$f_0(z) = \sum_{k=0}^{m} a_k z^k \quad \Longrightarrow \quad f_1(w) = \sum_{k=0}^{m} a_k w^{m-k}$$

is entire and satisfies the transition. The corresponding homogeneous polynomial is

$$F(v_0, v_1) = \sum_{k=0}^{m} a_k v_0^{m-k} v_1^k.$$

Its chart coefficients are $F(1,z) = f_0(z)$ and $F(w,1) = f_1(w)$. It obeys $F(cv) = c^m F(v)$ for $c \in \mathbb{C}$, so its restriction to each line is an element of that line's mth dual tensor power. Hence F is the coordinate-free description of the section.

If $m < 0$, every $k \geq 0$ satisfies $k > m$, so all a_k vanish. The transition then makes $f_1 = 0$ too. ∎

For $m = 2$, take $f_0(z) = a_0 + a_1 z + a_2 z^2$. Then $f_1(w) = a_0 w^2 + a_1 w + a_2$. Thus $F = a_0 v_0^2 + a_1 v_0 v_1 + a_2 v_1^2$.

26.4 The group action and its dual

A matrix $g \in \mathrm{SL}(2,\mathbb{C})$ sends a tautological fibre ℓ linearly to $g\ell$. The inverse transpose of this linear map sends the dual line ℓ^* to $(g\ell)^*$. Hence g acts on $O(1)$ and on all its tensor powers. Under the polynomial description, the section action is

$$(\Pi_m(g)F)(v) = F(g^{-1}v). \tag{26.3}$$

Indeed, at the line ℓ, the transformed functional evaluates a vector $v \in \ell$ by first pulling it back to $g^{-1}v \in g^{-1}\ell$. This gives the inverse in the formula.

The representation is $\mathrm{Sym}^m(\mathbb{C}^2)^*$, the dual of the symmetric tensor representation. For $\mathrm{SU}(2)$ this is isomorphic to V_m, by the prerequisite highest-weight theory. Here is also a direct weight check. On the basis $F_k = v_0^{m-k} v_1^k$,

$$\Pi_m(t_\theta) F_k = e^{i(2k-m)\theta} F_k, \qquad 0 \leq k \leq m.$$

The weights are $-m, -m+2, \ldots, m$. They match V_m, with highest weight m and dimension $m+1$. Irreducibility is the symmetric-power result from Lie Algebras.

For a concrete matrix computation, put

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

For $m = 2$, in the ordered basis $(v_0^2, v_0 v_1, v_1^2)$, expansion gives

$$D^{(2)}(g) = \begin{pmatrix} \bar{a}^2 & -\bar{a}b & b^2 \\ 2\bar{a}\bar{b} & |a|^2 - |b|^2 & -2ab \\ \bar{b}^2 & a\bar{b} & a^2 \end{pmatrix}.$$

For example, the middle column comes from

$$(\bar{a}v_0 + \bar{b}v_1)(-bv_0 + av_1) = -\bar{a}b\, v_0^2 + (|a|^2 - |b|^2) v_0 v_1 + a\bar{b}\, v_1^2.$$

The centre acts by $\Pi_m(-I) = (-1)^m I$. Thus the action on sections descends to $\mathrm{SO}(3)$ precisely for even m.

The same sections inside compact induction

For a homogeneous polynomial F of degree $m \geq 0$, a corresponding equivariant function $f_F : \mathrm{SU}(2) \to \mathbb{C}$ is given by $f_F(g) = F(g(1,0)^T)$. In the proof of Theorem 26.3, the section defined by F assigns to each line the restriction of F to that line. So $f_F(g)$ is the value of this restriction at the spanning vector $g(1,0)^T$. Since $gt_\theta(1,0)^T = e^{i\theta}g(1,0)^T$, homogeneity gives

$$f_F(gt_\theta) = e^{im\theta} f_F(g).$$

This is the covariance (26.1) for $H = T$ and $\sigma(t_\theta) = e^{-im\theta}$. Also

$$f_{\Pi_m(a)F}(g) = F(a^{-1}g(1,0)^T) = f_F(a^{-1}g),$$

so $F \mapsto f_F$ intertwines Π_m with the left-translation action $(\Pi(a)f)(g) = f(a^{-1}g)$ of the first section.

Next we identify the whole bundle. Let m be any integer. At the line $\mathbb{C}(1,0)$, the element t_θ acts on the tautological fibre by $e^{i\theta}$. It acts on the mth dual power of that fibre by $\chi_{-m}(t_\theta) = e^{-im\theta}$. Write $\mathbb{C}_{-m}$ for $\mathbb{C}$ with T acting through the character χ_{-m}.

Let ε be the value at $z = 0$ of the frame $(e_0^*)^m$. For $m < 0$, this frame means $e_0^{\otimes(-m)}$, consistently with $e_1^* = ze_0^*$ and $e_1 = z^{-1}e_0$. The map $[g,u] \mapsto g \cdot (u\varepsilon)$ from $\mathrm{SU}(2) \times_T \mathbb{C}_{-m}$ to $\mathcal{O}(m)$ is well defined, because t_θ multiplies ε by $\chi_{-m}(t_\theta)$. It is $\mathrm{SU}(2)$-equivariant and linear on each fibre. Since $\mathrm{SU}(2)$ acts transitively on $\mathbb{CP}^1$ with stabiliser T, it is bijective. We now show that its coefficients in the local trivialisations are smooth and nowhere zero.

Over U_0, use the smooth local section $s(z) = g_v$ of $\mathrm{SU}(2) \to \mathrm{SU}(2)/T$, where $v = (1,z)/\sqrt{1+|z|^2}$. Since $g_v(1,0)^T = v$, we have

$$s(z)^{-1}(1,z)^T = \sqrt{1+|z|^2}\,(1,0)^T.$$

The element $s(z)$ sends a functional ϕ on $\mathbb{C}(1,0)$ to $\phi \circ s(z)^{-1}$ on the line through $(1,z)$. Evaluating at $e_0(z) = (1,z)$ gives $s(z) \cdot e_0^*(0) = \sqrt{1+|z|^2}\, e_0^*(z)$. For vectors, the same display gives $s(z)e_0(0) = (1+|z|^2)^{-1/2}e_0(z)$. Taking mth or $(-m)$th tensor powers, we obtain for every integer m

$$s(z) \cdot (u\varepsilon) = u\,(1+|z|^2)^{m/2}\,(e_0^*)^m(z).$$

Over U_1, the section g_v with $v = (w,1)/\sqrt{1+|w|^2}$ gives the coefficient $u(1+|w|^2)^{m/2}$ with respect to $(e_1^*)^m$, by the same computation.

Thus, in both local trivialisations, the map is $u \mapsto u(1+|x|^2)^{m/2}$, where x is the chart coordinate. This coefficient is smooth. Its factor $(1+|x|^2)^{m/2}$ is nowhere zero, so the inverse $c \mapsto c(1+|x|^2)^{-m/2}$ is smooth too. Hence the map is an isomorphism of smooth line bundles

$$\mathcal{O}(m) \cong \mathrm{SU}(2) \times_T \mathbb{C}_{-m}.$$

A section in the equivariant-function description therefore obeys $f(gt_\theta) = e^{im\theta} f(g)$. The positive exponent belongs to the function covariance. The fibre itself has the negative character. For $m \geq 0$ and a homogeneous polynomial F of degree m, the vector $g \cdot (u\varepsilon)$ takes the value u at $g(1,0)^T$, by the definition of ε. With $u = f_F(g) = F(g(1,0)^T)$, it agrees with F at that vector, so the two descriptions of the section agree.

Now we can say what holomorphy does. The Hilbert completion of the continuous induced representation of Lecture 12 is the space of all L^2 sections of $\mathcal{O}(m)$. Let $m \geq 0$. By Frobenius

reciprocity, V_d occurs in this space once when $e^{-im\theta}$ is a weight of V_d, that is, for $d = m, m+2, m+4, \ldots$. Figure 26.2 shows this list. By Theorem 26.3, the holomorphic sections form exactly the first term V_m. For $m < 0$ the same reciprocity gives $V_{|m|}, V_{|m|+2}, \ldots$ while holomorphy keeps nothing.

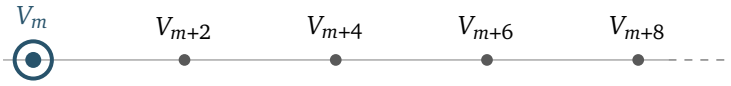


Figure 26.2: The irreducible representations of SU(2) in the L^2 sections of $O(m)$, for $m \geq 0$. Each dot occurs once. The ringed dot V_m is the space of holomorphic sections.

In physics language, $O(m)$ is the bundle of a charged particle on a sphere around a magnetic monopole, its two charts are two gauge patches, and the holomorphic sections are the lowest Landau level.

26.5 The general Borel–Weil theorem

We now replace $\mathrm{SL}(2,\mathbb{C})$ and its upper triangular subgroup by a complex group G and a Borel subgroup B. The SU(2) case at the end of this section shows that the calculation above is the first instance.

A *complex submanifold* of a complex manifold M is a subset N with the following property. Each point of N lies in a chart of M in which N is given by the vanishing of some coordinates. The remaining coordinates give charts of N. Hence a holomorphic map into M with values in N is also holomorphic as a map into N. We call this the *slice-chart property*.

A *complex Lie group* is a group with a complex manifold structure for which multiplication and inversion are holomorphic. The groups $K_{\mathbb{C}}$ of Lecture 14 are examples. A *closed complex subgroup* of a complex Lie group G is a closed subgroup that is also a complex submanifold.

Holomorphic bundles. Suppose that G is a complex Lie group, H is a closed complex subgroup and G/H is a complex manifold. Suppose also that the projection $\pi : G \to G/H$ is holomorphic, that σ is holomorphic and that π has holomorphic local sections s_i whose domains U_i cover G/H. Then $h_{ij} = s_i^{-1} s_j$ is holomorphic as a map into G, because the group operations of G are holomorphic. By the slice-chart property, h_{ij} is holomorphic as a map into H. Then $\sigma(h_{ij})$ is holomorphic, as a composition of holomorphic maps. So the local trivialisations given by the sections s_i make E a complex manifold. In this case E is a *holomorphic* vector bundle. These trivialisations are its *holomorphic bundle charts*, and the complex structure of E does not depend on the family (s_i); the section on sources checks these points. A section is *holomorphic* when each coefficient $f \circ s_i$ is holomorphic.

A section is holomorphic exactly when its function $f : G \to V$ is holomorphic. Indeed, if f is holomorphic, then so is $f \circ s_i$. Conversely, the map $q(g) = s_i(\pi(g))^{-1} g$ on $\pi^{-1}(U_i)$ is holomorphic into H by the slice-chart property. The covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big),$$

and both factors are holomorphic. Hence f is holomorphic on each set $\pi^{-1}(U_i)$, and therefore on G.

The groups. Let K be compact, connected, simply connected and semisimple. Its complexification $G = K_{\mathbb{C}}$ was introduced in Lecture 14, and it is connected because K is. Fix a maximal torus T of K and positive roots. The *positive Borel subalgebra* $\mathfrak{b}$ is the complex Cartan subalgebra plus the positive root spaces. Let B be the connected Lie subgroup of G with this Lie algebra.

We cite two facts about these groups. First, B is closed, and the inclusion of K in G induces a diffeomorphism from K/T onto the *flag manifold* G/B (Sepanski, *Compact Lie Groups*, Lemma 7.49 and Theorem 7.50). In particular,

$$G = KB.$$

Second, B is a closed complex subgroup, and G/B is a complex manifold with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections (Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 2.11 and its proof). So the paragraph on holomorphic bundles applies to G, B and G/B. The section on sources identifies Sepanski's matrix model of G with the group of Lecture 14, and it proves that B is a complex subgroup.

The character. Let λ be an integral weight. Since K is simply connected, $X^*(T) = P$ (Lecture 7). Let N_+ be the connected subgroup of G whose Lie algebra is the sum of the positive root spaces. Each $b \in B$ is uniquely a product $t\exp(iH)n$ with $t \in T$, $H \in \mathfrak{t}$ and $n \in N_+$, and these factors depend smoothly on b (Sepanski, Lemma 7.49(d)). Our character of weight λ on T is $\exp H \mapsto e^{i\lambda(H)}$. Define $\chi_\lambda : B \to \mathbb{C}^\times$ by

$$\chi_\lambda\big(\exp(H')\exp(iH)n\big) = e^{i\lambda(H')-\lambda(H)} \qquad (H, H' \in \mathfrak{t},\ n \in N_+).$$

We import from Sepanski, Definition 7.52, the fact that this formula defines a homomorphism. The formula is well defined on $T = \exp\mathfrak{t}$ because $\lambda \in P = X^*(T)$. By the smooth dependence of the factors on b, the homomorphism χ_λ is smooth. It is trivial on N_+. Its differential at e is the complex-linear extension of $i\lambda$ on $\mathfrak{t}_{\mathbb{C}}$ and zero on the positive root spaces, so it is $\mathbb{C}$-linear. Left translations of B are holomorphic, and χ_λ is multiplicative. So its differential is $\mathbb{C}$-linear at every point, and χ_λ is holomorphic.

The bundle. Define

$$L_\lambda = G \times_B \mathbb{C}_{-\lambda}, \qquad (gb, u) \sim (g, \chi_\lambda(b)^{-1}u).$$

Sections correspond to functions with

$$f(gb) = \chi_\lambda(b) f(g). \tag{26.4}$$

A section is holomorphic exactly when its function $f : G \to \mathbb{C}$ is holomorphic. Left translation acts as before. Write $H^0(G/B, L_\lambda)$ for the space of holomorphic sections of L_λ.

Theorem 26.4 (Borel–Weil). *With the preceding positive Borel and bundle convention, if λ is dominant integral, then*

$$H^0(G/B, L_\lambda) \cong V_\lambda^*$$

as a K-representation. This is finite-dimensional and irreducible. If an integral λ is not dominant, then $H^0(G/B, L_\lambda) = 0$.

The proof below uses the cited facts about B, χ_λ and G/B above, in particular $G = KB$. It also uses the holomorphic extension of Lecture 14 and the highest-weight theorem of the prerequisite course. The rest comes from the Peter–Weyl theorem.

The dual V_λ^* has highest weight $-w_0\lambda$. Here w_0 is the longest element of the Weyl group, the element that sends the positive roots to the negative roots. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels. For $\mathrm{SU}(2)$ we have $-w_0\lambda = \lambda$, so the dual is invisible there.

The case $\mathrm{SU}(3)$ by hand

Here $G = \mathrm{SL}(3,\mathbb{C})$, and B is the subgroup of upper triangular matrices. A *flag* is a line ℓ inside a plane P in $\mathbb{C}^3$. Figure 26.3 shows one. The group G moves flags, and B is the stabiliser of the standard flag $\mathbb{C}e_1 \subset \mathbb{C}e_1 \oplus \mathbb{C}e_2$. The group $K = \mathrm{SU}(3)$ already moves flags transitively, with stabiliser the diagonal torus T. So $K/T = G/B$ is the space of flags.

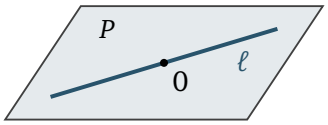


Figure 26.3: A flag $\ell \subset P$ in $\mathbb{C}^3$, drawn schematically: the complex line ℓ and the complex plane P appear as a real line in a real plane.

Example 26.5 (Sections for $\mathrm{SU}(3)$)**.** Let $\lambda = a\omega_1 + b\omega_2$ with integers $a, b \geq 0$, and write β for an upper triangular matrix in G. The character is trivial on the unipotent matrices. On the torus, ω_1 is the first diagonal entry and ω_2 is the product of the first two. So

$$\chi_\lambda(\beta) = \beta_{11}^{a+b}\beta_{22}^{b}.$$

Take a linear functional Φ on $\mathbb{C}^3$ and put $f(g) = \Phi(ge_1)$. Since $(g\beta)e_1 = \beta_{11}ge_1$, we get $f(g\beta) = \beta_{11}f(g)$. So f is a section for $(a,b) = (1,0)$. These sections form $(\mathbb{C}^3)^*$, the dual of $V_{(1,0)} = \mathbb{C}^3$, as the theorem says.

Next take a linear functional Ψ on $\Lambda^2\mathbb{C}^3$ and put $f(g) = \Psi(ge_1 \wedge ge_2)$. Since $(g\beta)e_1 \wedge (g\beta)e_2 = \beta_{11}\beta_{22}\, ge_1 \wedge ge_2$, this f is a section for $(0,1)$. These sections form the dual of $V_{(0,1)} = \Lambda^2\mathbb{C}^3$.

A product of a sections for $(1,0)$ and b sections for $(0,1)$ is a section for (a,b). These products need not be independent. For $(1,1)$, the nine products of coordinates of ge_1 with coordinates of $ge_1 \wedge ge_2$ satisfy one relation, since $ge_1 \wedge (ge_1 \wedge ge_2) = 0$. They span the eight-dimensional dual of the adjoint module.

Proof of the Borel–Weil theorem

Proof. We build sections from V_λ, and then we show that there are no others. Throughout, R_X denotes the derivative of a function φ on K along the right translation by $\exp tX$:

$$(R_X\varphi)(k) = \frac{d}{dt}\bigg|_0 \varphi(k \exp tX), \qquad X \in \mathfrak{k}.$$

We extend it complex-linearly to $Z \in \mathfrak{k}_\mathbb{C} = \mathfrak{g}$, with $R_{X+iY} = R_X + iR_Y$.

Step 1: sections from V_λ. Let λ be dominant. A module V_λ of highest weight λ exists by the highest-weight theorem of the prerequisite course and the integration theorem of Lecture 4. It extends to a holomorphic representation of G by Lecture 14. Let v_λ be a highest weight vector. Write Π_λ for this representation. On $\mathfrak{b}$, the operator $d\Pi_\lambda(Z)$ acts on v_λ by the scalar $d\chi_\lambda(Z)$: this is $i\lambda(H)$ for $H \in \mathfrak{t}$ and zero on the positive root spaces. So $b \mapsto \Pi_\lambda(b)v_\lambda$ and $b \mapsto \chi_\lambda(b)v_\lambda$ agree on $\exp \mathfrak{b}$, and hence on the connected group B:

$$\Pi_\lambda(b)v_\lambda = \chi_\lambda(b)v_\lambda.$$

For $\xi \in V_\lambda^*$ put $f_\xi(g) = \xi(\Pi_\lambda(g)v_\lambda)$. This function is holomorphic, and

$$f_\xi(gb) = \xi(\Pi_\lambda(g)\Pi_\lambda(b)v_\lambda) = \chi_\lambda(b)f_\xi(g).$$

So f_ξ is a holomorphic section of L_λ.

Step 2: the dual. Left translation by a gives

$$f_\xi(a^{-1}g) = \xi(\Pi_\lambda(a)^{-1}\Pi_\lambda(g)v_\lambda) = f_{\xi\circ\Pi_\lambda(a)^{-1}}(g).$$

So $\xi \mapsto f_\xi$ intertwines the dual representation with the section action. This is the reason for the dual: the section is linear in the functional ξ, and left translation acts on ξ. The map is injective. Indeed, if $f_\xi = 0$, then ξ vanishes on all $\Pi_\lambda(g)v_\lambda$. These vectors span V_λ because it is irreducible, so $\xi = 0$.

Step 3: the equations on K. Now let λ be any integral weight, and let f be holomorphic on G with the covariance (26.4). Let $\varphi = f|_K$, a smooth function on K. For $Z \in \mathfrak{g}$,

$$(R_Z\varphi)(k) = \left.\frac{d}{dt}\right|_0 f(k\exp tZ),$$

since the differential of the holomorphic function f is $\mathbb{C}$-linear. This is the point where holomorphy is used. A smooth function on K that is only covariant under T would not satisfy the second equation below. For $Z \in \mathfrak{b}$, the covariance gives $f(k\exp tZ) = e^{t\,d\chi_\lambda(Z)}f(k)$. Hence

$$R_H\varphi = i\lambda(H)\varphi \quad (H \in \mathfrak{t}), \qquad R_E\varphi = 0 \quad (E \text{ in a positive root space}). \tag{26.5}$$

Step 4: one block at a time. Let Π be an irreducible unitary representation of K, and let P_Π be the isotypic projector of Lecture 11. It is an integral of left translations:

$$(P_\Pi\varphi)(x) = d_\Pi \int_K \chi_\Pi(y)\,\varphi(y^{-1}x)\,dy.$$

Here χ_Π is the character of Π. The derivative $R_X\varphi$ is continuous on the compact group K, so we may differentiate under this integral, and $R_ZP_\Pi\varphi = P_\Pi R_Z\varphi$ for $Z \in \mathfrak{g}$. So $P_\Pi\varphi$ also satisfies (26.5).

The function $P_\Pi\varphi$ lies in the coefficient block M_Π. Let (ξ_i) be a basis of V_Π^*. Then

$$(P_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)v_i), \qquad v_i \in V_\Pi,$$

and differentiating gives $(R_ZP_\Pi\varphi)(k) = \sum_i \xi_i(\Pi(k)d\Pi(Z)v_i)$. By Schur orthogonality (Lecture 5), the matrix coefficients are linearly independent: a sum $\sum_i \xi_i(\Pi(k)w_i)$ vanishes for all k only if every $w_i = 0$. So (26.5) gives

$$d\Pi(H)v_i = i\lambda(H)v_i, \qquad d\Pi(E)v_i = 0.$$

Thus each nonzero v_i is a highest weight vector of weight λ in the irreducible module V_Π. By the highest-weight theorem of the prerequisite course, if some $v_i \neq 0$, then λ is dominant and $\Pi \cong V_\lambda$. Its highest weight line is one-dimensional, so $v_i = c_i v_\lambda$, and

$$(P_\Pi \varphi)(k) = \xi(\Pi(k) v_\lambda), \qquad \xi = \sum_i c_i \xi_i.$$

For every other Π, we get $P_\Pi \varphi = 0$.

Step 5: completeness. We have not assumed that φ is a finite sum of coefficients. The Peter–Weyl theorem, in the form of the Plancherel theorem of Lecture 11, says that the sums of $P_\Pi \varphi$ converge to φ in $L^2(K)$. By Step 4 at most one term is nonzero. So φ equals that term in $L^2(K)$, and both functions are continuous, so they agree on K. If λ is not dominant, then $\varphi = 0$. If λ is dominant, then $\varphi = f_\xi|_K$ for some ξ.

Finally, $G = KB$, and the covariance determines a function on G from its restriction to K. So $f = 0$ when λ is not dominant, and $f = f_\xi$ when λ is dominant. With Step 2, this gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$. The dual of an irreducible module is irreducible. ∎

The case SU(2)

For $K = \mathrm{SU}(2)$ the group G is $\mathrm{SL}(2, \mathbb{C})$ (Lecture 14), and B is the subgroup of upper triangular matrices in G. The character is

$$\chi_m(b) = c^m, \qquad b = \begin{pmatrix} c & \beta \\ 0 & c^{-1} \end{pmatrix} \in B.$$

It is trivial on the unipotent matrices, where $c = 1$. On the diagonal matrices it is $\mathrm{diag}(c, c^{-1}) \mapsto c^m$, the character of weight m. The fibre at $\mathbb{C}(1,0)$ transforms by c^{-m}. The map $gB \mapsto [g(1,0)^T]$ identifies G/B with $\mathbb{CP}^1$. Indeed, G acts transitively on lines, since $\mathrm{SU}(2)$ already does, and B is the stabiliser of the line $\mathbb{C}(1,0)$.

We now compare the two section spaces directly. Let f be holomorphic on G with $f(gb) = \chi_m(b) f(g)$. The character is trivial on unipotent b, so f depends only on the first column $v = g(1,0)^T$. Put $F(v) = f(g)$. On $\{v_0 \neq 0\}$ and on $\{v_1 \neq 0\}$, the local sections

$$v \longmapsto \begin{pmatrix} v_0 & 0 \\ v_1 & 1/v_0 \end{pmatrix}, \qquad v \longmapsto \begin{pmatrix} v_0 & -1/v_1 \\ v_1 & 0 \end{pmatrix}$$

have determinant one and holomorphic entries. So F is holomorphic on $\mathbb{C}^2 \setminus \{0\}$. Also $F(cv) = c^m F(v)$ for $c \neq 0$, by the character on $\mathrm{diag}(c, c^{-1})$. Hence $f_0(z) = F(1, z)$ and $f_1(w) = F(w, 1)$ are entire, and they satisfy the transition rule (26.2). By Theorem 26.3, for $m \geq 0$ there is a homogeneous polynomial, again called F, of degree m with these two coefficients, so that $f = f_F$. For $m < 0$, the same theorem gives $f_0 = f_1 = 0$, so $f = 0$. Conversely, each $f_F(g) = F(g(1,0)^T)$ is holomorphic and covariant.

So $F \mapsto f_F$ is an isomorphism $H^0(\mathbb{CP}^1, \mathcal{O}(m)) \cong H^0(G/B, L_m)$ of $\mathrm{SU}(2)$-representations for $m \geq 0$, and for $m < 0$ both spaces are zero. Hence our explicit calculation is exactly the case $\mathrm{SU}(2)$ of Borel–Weil. In Step 1 of the proof, $V_m = \mathrm{Sym}^m \mathbb{C}^2$ has highest weight vector $e_1^{\otimes m}$, with $e_1 = (1,0)^T$. So $f_\xi(g) = \xi((g e_1)^{\otimes m})$, which is the function f_F for the polynomial $F(v) = \xi(v^{\otimes m})$.

Self-checks

1. What is wrong with choosing an arbitrary entire f_0 as a section of $\mathcal{O}(1)$?

Answer. The coefficient $f_1(w) = w f_0(1/w)$ must also extend holomorphically to $w = 0$. This forces f_0 to have degree at most one.

2. Which representation occurs in $H^0(\mathbb{CP}^1, \mathcal{O}(3))$?

Answer. It is $\mathrm{Sym}^3(\mathbb{C}^2)^* \cong V_3$, of dimension four and spin $3/2$. The central element $-I$ acts by the scalar $(-1)^3 = -1$, so this representation does not descend to $\mathrm{SO}(3)$.

3. Does L_λ always produce highest weight λ in our general convention?

Answer. No. Its section space is V_λ^*, of highest weight $-w_0\lambda$, where w_0 is the longest element of the Weyl group. For $\mathrm{SU}(2)$ these weights coincide. For $\mathrm{SU}(n)$, the element $-w_0$ reverses the Dynkin labels, so for $\mathrm{SU}(3)$ the bundle $L_{(1,0)}$ gives highest weight $(0, 1)$.

Summary

Equivariant functions correspond to sections of associated bundles. On $\mathbb{CP}^1$ and for $m \geq 0$, the transition rule for $\mathcal{O}(m)$ identifies holomorphic sections with homogeneous polynomials of degree m. Compact induction contains $V_m, V_{m+2}, \dots$, and holomorphy keeps only V_m. For compact, connected, simply connected, semisimple K, the Borel–Weil theorem gives $H^0(G/B, L_\lambda) \cong V_\lambda^*$ for dominant integral λ, and zero otherwise. We proved it from Peter–Weyl. Covariance and holomorphy make the vectors in each coefficient block highest weight vectors of weight λ, so only one block survives. The facts $G = KB$, the complex structure of G/B, the character χ_λ and the highest-weight theorem stay cited. For $\mathrm{SU}(3)$, first-column and wedge coordinates give explicit sections.

Sources and hypotheses

This section collects what the main line cites, with the checks of hypotheses and the matching of conventions. Nothing later depends on the order in which it is read.

Smooth structure of associated bundles. We show that $E = G \times_H V$ is a smooth complex vector bundle. The local sections s_i are smooth, so the maps $h_{ij} = s_i^{-1} s_j$ are smooth into G. They take values in the embedded subgroup H, so they are smooth into H by the codomain-restriction result for embedded submanifolds [C: Lee, Corollary 5.30]. The representation σ is smooth by the automatic-smoothness theorem of Lecture 3. Hence the transition functions $\sigma(h_{ij})$ are smooth. Let $k = \dim V$, choose a basis of V, and regard $\mathbb{C}^k$ as $\mathbb{R}^{2k}$. Each fibre of E is then a real vector space of dimension $2k$, and the real matrices of $\sigma(h_{ij})$ are smooth maps into $\mathrm{GL}(2k, \mathbb{R})$. The vector bundle chart lemma [C: Lee, Lemma 10.6] applies to these real data. It gives E a unique topology and smooth structure for which the inverses of the maps above are smooth local trivialisations. These maps are complex-linear on fibres, so E is a smooth complex vector bundle.

This structure does not depend on the chosen family of local sections. For a second family (s'_j), the maps $s_i^{-1}s'_j$ are smooth into H by the same codomain-restriction argument. So the mixed transition functions $\sigma(s_i^{-1}s'_j)$ are smooth, and the chart lemma applies to the union of the two families. By its uniqueness clause, the union gives the same structure as each family alone.

Regularity of covariant functions. If f is smooth, then each $f \circ s_i$ is smooth. Conversely, suppose that each $f \circ s_i$ is smooth. Write $\pi : G \to G/H$ for the projection, and put

$$q(g) = s_i(\pi(g))^{-1}g \qquad (g \in \pi^{-1}(U_i)).$$

This map is smooth into G and takes values in H, so it is smooth into the embedded subgroup H by the same codomain-restriction result. Since $g = s_i(\pi(g))q(g)$, the covariance (26.1) gives

$$f(g) = \sigma\big(q(g)\big)^{-1} f\big(s_i(\pi(g))\big).$$

Hence f is smooth on each open set $\pi^{-1}(U_i)$, and therefore on G. The same argument with continuity in place of smoothness shows that s is continuous exactly when f is continuous.

Holomorphic inverse functions. Let Φ be a map between open subsets of $\mathbb{C}^d$ with holomorphic components and invertible differential at a point. The real inverse function theorem gives a smooth local inverse. Its differential is the inverse of a $\mathbb{C}$-linear map, hence $\mathbb{C}$-linear. So the local inverse is holomorphic. This is the *holomorphic inverse function theorem*.

Holomorphic bundles. We work in the setting of the paragraph on holomorphic bundles in the main line. If $g = s_i(x)h$, then $R_h \circ s_i$ is a holomorphic, hence smooth, local section through g, because $\pi \circ R_h = \pi$. By the local-section theorem of Lee, Theorem 4.26, π is a smooth submersion.

The topology of G/H is the quotient topology. Indeed, π is continuous, since it is holomorphic. Conversely, let $U \subset G/H$ be a set such that $\pi^{-1}(U)$ is open. Then $U \cap U_i = s_i^{-1}(\pi^{-1}(U))$ is open for each i, because $\pi \circ s_i = \mathrm{id}$. The sets U_i cover G/H, so U is open. By the uniqueness of the quotient structure in Lecture 3, the smooth structure underlying the complex manifold G/H is the quotient smooth structure.

Over the domain of a holomorphic chart φ of G/H inside U_i, the map $[s_i(x), v] \mapsto (\varphi(x), v)$ is a chart of E, with V identified with $\mathbb{C}^k$ by a basis. The change between two such charts has the form $(y, u) \mapsto (\psi(y), A(y)u)$. Here ψ is a transition map of G/H, and $A(y)$ is the matrix of $\sigma(h_{ij})$ at the point with coordinate y. The entries of A are holomorphic, so this change is holomorphic. Hence these charts make E a complex manifold. For a second family (s'_j) of holomorphic local sections, the mixed changes of chart have the same form with $\sigma(s_i^{-1}s'_j)$, which is holomorphic by the slice-chart property. Hence both families give one holomorphic atlas, and the complex structure of E does not depend on the family.

Complex analysis. [C: Stein and Shakarchi, *Complex Analysis*, 2003, Chapter 2, Section 4]. Theorem 4.4 and the paragraph after it state that an entire function has a power series about 0 that converges on all of $\mathbb{C}$, with $a_k = f^{(k)}(0)/k!$. The Cauchy inequalities, Corollary 4.3, give

$$|f^{(k)}(0)| \le k! R^{-k} \max_{|z|=R} |f(z)|.$$

Dividing by $k!$ gives the estimate for a_k.

The matrix model of $K_{\mathbb{C}}$. Sepanski, *Compact Lie Groups*, states Borel–Weil for a matrix model of G. We identify the two groups. A faithful representation ρ of K exists (Lecture 10; also [C: Bröcker and tom Dieck (BD), Section III.4, Theorem 4.1]). Averaging makes ρ unitary (Lecture 5). The entries of $r = \rho \oplus \overline{\rho} : K \to \mathrm{U}(N)$ generate $\mathscr{R}(K)$ [C: BD, Section III.4, Proposition 4.3].

Recall from Lecture 14 that each $\psi \in K_{\mathbb{C}} = \mathcal{X}(K)$ is a unital algebra homomorphism from $\mathscr{R}(K)$ to $\mathbb{C}$. The *coefficient map* $r_{\mathbb{C}}$ evaluates ψ on the matrix entries r_{ij} of r:

$$r_{\mathbb{C}}(\psi) = \big(\psi(r_{ij})\big)_{i,j=1}^{N}.$$

By [C: BD, Section III.8, Proposition 8.2 and the next paragraph], the map $r_{\mathbb{C}}$ sends $K_{\mathbb{C}}$ biholomorphically onto a closed subgroup and analytic submanifold of $\mathrm{GL}(N,\mathbb{C})$.

The resulting complex structure on $K_{\mathbb{C}}$ does not depend on r. Let r' be a second faithful unitary representation of K whose entries also generate $\mathscr{R}(K)$. Let $r'_{\mathbb{C}}$ be its coefficient map, defined by the same construction. Each entry of r' is a polynomial in the entries of r, and conversely. Every $\psi \in K_{\mathbb{C}}$ is a unital algebra homomorphism, so it respects these polynomial relations. Hence $r'_{\mathbb{C}} \circ r_{\mathbb{C}}^{-1}$ and its inverse are restrictions of polynomial maps. By the slice-chart property, they are holomorphic between the two analytic submanifolds. This common structure is the complex structure of Lecture 14.

The Lie algebra $\mathfrak{g}$ of $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complex span of that of $r(K)$ [C: BD, Proposition 8.4 and the preceding paragraph].

Since K is connected, so is $K_{\mathbb{C}}$ (Lecture 14). A connected Lie subgroup of $\mathrm{GL}(N,\mathbb{C})$ is determined by its Lie algebra [C: Kirillov Jr., *An Introduction to Lie Groups and Lie Algebras*, Theorem 3.40]. Hence $r_{\mathbb{C}}(K_{\mathbb{C}})$ is the complexification of $r(K)$ in Sepanski, Definition 7.48. We identify G with $r_{\mathbb{C}}(K_{\mathbb{C}})$ and K with $r(K)$, since $r_{\mathbb{C}}(\mathrm{ev}_g) = r(g)$.

Sepanski's construction applies to every faithful unitary r.

The Borel subgroup and the quotient. By Sepanski, Definition 7.48, B is the connected Lie subgroup of G with Lie algebra $\mathfrak{b}$. By Sepanski, Lemma 7.49, B is closed. By Sepanski, Theorem 7.50, the inclusion induces a diffeomorphism from K/T onto G/B.

Since B is closed, it is an embedded Lie subgroup of G by the closed-subgroup theorem of Lecture 3. The inclusion of B with Sepanski's Lie subgroup structure is smooth into this embedded submanifold by Lee, Corollary 5.30. It is a bijective Lie group homomorphism, so it is a Lie group isomorphism by Lee, Corollary 7.6. Hence the embedded group B has Lie algebra $\mathfrak{b}$, and exp maps some neighbourhood O of 0 in $\mathfrak{b}$ diffeomorphically onto an open subset of B.

Choose a complex complement $\mathfrak{m}$ of $\mathfrak{b}$ in $\mathfrak{g}$. For $b \in B$, define $\Phi : \mathfrak{b} \times \mathfrak{m} \to G$ by

$$\Phi(X, Y) = b \exp X \exp Y.$$

The matrix exponential is holomorphic and maps $\mathfrak{g}$ into G, so Φ is holomorphic into G by the slice-chart property. Its differential at $(0, 0)$ is the complex-linear isomorphism $(X, Y) \mapsto b(X + Y)$. By the holomorphic inverse function theorem, after shrinking O, there is a neighbourhood O' of 0 in $\mathfrak{m}$ such that Φ maps $O \times O'$ biholomorphically onto an open set $W' \subset G$. The set $\Phi(O \times \{0\}) = b \exp O$ is open in B. Because B has the subspace topology, it equals $B \cap W''$ for some open set W''. On $W = W' \cap W''$, the set B is exactly the slice $Y = 0$. Hence B is a closed complex subgroup.

The complex quotient theorem [C: Kirillov Jr., Theorem 2.11 and its proof] makes G/B a complex manifold, with holomorphic projection $\pi : G \to G/B$ and holomorphic local sections. Thus the first section applies to G, B and G/B.

Weights and the source of the theorem. Our character of weight λ is $\exp H \mapsto e^{i\lambda(H)}$, so Sepanski's weight is our $i\lambda$ on $\mathfrak{t}$ (Sepanski, Definitions 6.23 and 6.26, Theorem 6.27 and formula (6.28)). In our labels, Sepanski, Definition 7.52, defines the character χ_λ of the main line. Sepanski's Theorem 7.58 states the Borel–Weil theorem for the groups G and B above. His Definition 7.54 calls a section holomorphic when its function on G is holomorphic, and by the equivalence in the main line his holomorphic sections are ours. Our L_λ is his line bundle of index $-\lambda$, so his theorem gives the same V_λ^*. Our proof does not use his theorem.

The group $\mathrm{SL}(2, \mathbb{C})$ as $\mathrm{SU}(2)_{\mathbb{C}}$. For $\mathrm{SU}(2)$, take r to be the standard representation. Its four entries generate the representative functions, and $r_{\mathbb{C}}(K_{\mathbb{C}}) = \mathrm{SL}(2, \mathbb{C})$ (Lecture 14). We give $\mathrm{SL}(2, \mathbb{C})$ its complex structure as the subset $\{\det = 1\}$ of $\mathbb{C}^4$. This subset is a complex submanifold, with explicit slice charts. On the open set $\{g_{11} \neq 0\} \subset \mathbb{C}^4$, put

$$\varphi(g_{11}, g_{12}, g_{21}, g_{22}) = (g_{11}, g_{12}, g_{21},\ g_{11}g_{22} - g_{12}g_{21} - 1).$$

The map φ is holomorphic onto the open set $\{(a, b, c, t) : a \neq 0\}$, with holomorphic inverse

$$(a, b, c, t) \longmapsto \big(a, b, c, (t + 1 + bc)/a\big).$$

In this chart, $\mathrm{SL}(2, \mathbb{C})$ is the set $t = 0$. On $\{g_{12} \neq 0\}$, $\{g_{21} \neq 0\}$ and $\{g_{22} \neq 0\}$, solve $\det g = 1 + t$ in the same way for the entry that multiplies the nonzero one in $\det g$. These four open sets cover $\mathrm{SL}(2, \mathbb{C})$, because $\det g = 1$ forces some entry of g to be nonzero.

By the slice-chart property, a map into $\mathrm{SL}(2, \mathbb{C})$ with holomorphic entries is holomorphic. On $\mathrm{SL}(2, \mathbb{C})$ the inverse is

$$g^{-1} = \begin{pmatrix} g_{22} & -g_{12} \\ -g_{21} & g_{11} \end{pmatrix}.$$

Hence multiplication and inversion have polynomial entries, and $\mathrm{SL}(2, \mathbb{C})$ is a complex Lie group.

We now compare this structure with the complex structure of $\mathrm{SU}(2)_{\mathbb{C}} = \mathcal{X}(\mathrm{SU}(2))$ from Lecture 14. The example for $\mathrm{SU}(2)$ in Lecture 14 shows that the four matrix entries D_{ij} of

the standard representation generate the representative functions of SU(2). It also gives the bijection

$$P : \mathrm{SU}(2)_{\mathbb{C}} \longrightarrow \mathrm{SL}(2,\mathbb{C}), \qquad P(\psi) = \big(\psi(D_{ij})\big)_{i,j=1,2}.$$

The formula $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$ gives $P(\psi_1\psi_2) = P(\psi_1)P(\psi_2)$, so P is a group isomorphism. Also $P(\mathrm{ev}_g) = g$ for $g \in \mathrm{SU}(2)$. In the notation of Bröcker and tom Dieck, P is the map $r_{\mathbb{C}}$ for the standard representation r of SU(2).

Cited result. We apply [C: Bröcker and tom Dieck, *Representations of Compact Lie Groups*, Section III.8, Proposition 8.2 and the paragraph after it] to the standard representation r, whose entries generate the representative functions. Proposition 8.2 states that $r_{\mathbb{C}} = P$ maps $\mathrm{SU}(2)_{\mathbb{C}}$ bijectively onto an affine variety in $\mathbb{C}^4$. This variety is the image $\mathrm{SL}(2,\mathbb{C})$ of P. The paragraph after it states that the topology of $\mathrm{SU}(2)_{\mathbb{C}}$ is the subspace topology induced by P. It also states that the variety is an analytic submanifold of $\mathbb{C}^4$, on which multiplication and inversion are holomorphic. With this structure, P is a biholomorphic group isomorphism from $\mathrm{SU}(2)_{\mathbb{C}}$ onto $\mathrm{SL}(2,\mathbb{C})$ that is the identity on SU(2).

This analytic-submanifold structure on $\mathrm{SL}(2,\mathbb{C})$ coincides with the slice-chart structure above. Indeed, the inclusion of $\mathrm{SL}(2,\mathbb{C})$ into $\mathbb{C}^4$ is holomorphic for each structure. By the slice-chart property of the other structure, the identity map of $\mathrm{SL}(2,\mathbb{C})$ is holomorphic in each direction.

Here we use the standard representation r. We identify $\mathrm{SU}(2)_{\mathbb{C}}$ with $G = \mathrm{SL}(2,\mathbb{C})$ through P, which is the identity on SU(2). By the polynomial argument above, the complex structure of Lecture 14 does not depend on the generating representation. So it is the structure transported by P, which is $r_{\mathbb{C}}$ for the standard representation. By the cited result and the preceding paragraph, this structure coincides with the submanifold structure on G.

Consider the maps

$$s_0(z) = \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}, \qquad s_1(w) = \begin{pmatrix} w & -1 \\ 1 & 0 \end{pmatrix}.$$

Both matrices have determinant one. Their first columns are the tautological frames $e_0(z) = (1, z)$ and $e_1(w) = (w, 1)$. Identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$, as in the main line. Then s_0 and s_1 are local sections of $G \to G/B$ over U_0 and U_1. Their entries are polynomials, so they are holomorphic maps into G.

At a point of the overlap, $w = 1/z$ and

$$h_{01} = s_0(z)^{-1} s_1(1/z) = \begin{pmatrix} 1 & 0 \\ -z & 1 \end{pmatrix} \begin{pmatrix} z^{-1} & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} z^{-1} & -1 \\ 0 & z \end{pmatrix} \in B.$$

Hence $\chi_m(h_{01}) = z^{-m}$.

Let $f : G \to \mathbb{C}$ satisfy $f(gb) = \chi_m(b) f(g)$, and put $f_0 = f \circ s_0$ and $f_1 = f \circ s_1$. Since $s_1(1/z) = s_0(z) h_{01}$, covariance gives

$$f_1(1/z) = f\big(s_0(z) h_{01}\big) = z^{-m} f_0(z) \qquad (z \neq 0).$$

With $w = 1/z$, this is the transition rule $f_1(w) = w^m f_0(1/w)$ of $\mathcal{O}(m)$.

Next we compare the two complex structures on G/B. Write $\pi : G \to G/B$ for the projection, and identify G/B with $\mathbb{CP}^1$ by $gB \mapsto [g(1,0)^T]$. The vector $g(1,0)^T$ is the first column

$(g_{11}, g_{21})^T$ of g. Hence $\pi^{-1}(U_0) = \{g_{11} \neq 0\}$ and $\pi^{-1}(U_1) = \{g_{21} \neq 0\}$ are open in G. In the charts,

$$z \circ \pi(g) = \frac{g_{21}}{g_{11}} \quad \text{on } \pi^{-1}(U_0), \qquad w \circ \pi(g) = \frac{g_{11}}{g_{21}} \quad \text{on } \pi^{-1}(U_1).$$

These are quotients of holomorphic functions with nonvanishing denominators. Hence π is holomorphic for the chart structure. The sections s_0 and s_1 are holomorphic for the chart structure, because their entries are polynomials. By the comparison of the two group structures above, the cited structure on G/B also makes π holomorphic for the submanifold structure on G, and it has holomorphic local sections.

Let s be a holomorphic local section for one of the two structures. Then $\pi \circ s$ is the identity map on the domain of s. This composition is holomorphic into the other structure, because π is holomorphic for that structure. Applying this in both directions shows that the identity map of G/B is holomorphic with holomorphic inverse. Hence the two complex structures coincide.

Let s' be another holomorphic local section over an open subset of U_0. Then $q = s_0^{-1}s'$ takes values in B, and it is holomorphic into B by the slice-chart property. Covariance gives $f \circ s' = \chi_m(q) f_0$. The factor $\chi_m(q)$ is holomorphic and nowhere zero, so it does not change holomorphy. The same holds over U_1. Hence a section of L_m is holomorphic exactly when f_0 and f_1 are entire.

This is the agreement of the two notions of holomorphic section that the projective-line calculation uses.

Reading

- M. R. Sepanski, *Compact Lie Groups* (2007), Section 7.4: the Borel–Weil theorem and the groups G and B.
- J. M. Lee, *Introduction to Smooth Manifolds*, second edition (2013), Chapters 4, 5 and 10: local sections, embedded submanifolds and the vector bundle chart lemma.
- T. Bröcker and T. tom Dieck, *Representations of Compact Lie Groups* (1985), Sections III.4 and III.8: faithful representations and complexification.

Lecture 27

Compact orbits and geometric quantisation

Objectives. *A compact coadjoint orbit has a symplectic form. Not every such orbit supplies a representation of the chosen global group. We calculate the obstruction on a sphere. The symplectic area of the sphere divided by $2\pi\hbar$ must be integral. An equivariant lift imposes a further condition when the group is* $\mathrm{SO}(3)$. *Holomorphic sections then connect the orbit to the representations of the previous lecture.*

By the end of this lecture you will be able to:

- Compute the coadjoint-orbit form of $\mathrm{SU}(2)$ with its radius normalisation.
- Define a unitary connection locally and prove the sphere integrality condition.
- Distinguish an ordinary prequantum bundle from an equivariant one.
- Recover spin $m/2$ by holomorphic sections and keep this construction separate from shifted character formulas.

27.1 The orbit sphere and its area

Use the basis of Lecture 4,

$$X_j = -\frac{i}{2}\sigma_j, \qquad [X_1, X_2] = X_3$$

with cyclic versions of the bracket. The Pauli matrices satisfy

$$\sigma_i\sigma_j = \delta_{ij}I + i\sum_k \epsilon_{ijk}\sigma_k.$$

Identify $\mathfrak{su}(2)^*$ with $\mathbb{R}^3$ by $x_j = \ell(X_j)$. For $u = (u_1, u_2, u_3)$ put $X_u = \sum_j u_j X_j$. Then $[X_u, X_v] = X_{u\times v}$.

The adjoint action is the rotation cover from Lecture 4. Its dual action is the same rotation action in these orthonormal coordinates. Thus its orbits are the spheres

$$S^2_r = \{x : |x| = r\} \quad (r > 0)$$

and the origin. The infinitesimal action is $\xi_u(x) = u \times x$. Indeed, pairing it with v gives $(u \times x)\cdot v = -x \cdot (u \times v)$, which is exactly the dual derivative from Lecture 19.

Proposition 27.1. *Let $r > 0$. The orbit form at $x \in S^2_r$ is*

$$\omega_x(v, w) = \frac{x \cdot (v \times w)}{r^2}, \qquad v, w \in T_x S^2_r.$$

If Ω is the outward-oriented area form of the sphere, then $\omega = \Omega/r$. In spherical coordinates,

$$\omega = r \sin\theta \, d\theta \wedge d\phi, \qquad \int_{S^2_r} \omega = 4\pi r. \tag{27.1}$$

Proof. The KKS formula of Lecture 19 requires

$$\omega_x(u \times x, v \times x) = x \cdot (u \times v).$$

At $x = re_3$, the two tangent vectors are $r(u_2, -u_1, 0)$ and $r(v_2, -v_1, 0)$. Substitution in the stated formula gives $r(u_1 v_2 - u_2 v_1)$, the required value. Rotation invariance gives the formula at every x.

The outward area form is $\Omega_x(v, w) = x \cdot (v \times w)/r$. Thus $\omega = \Omega/r$. The usual sphere parametrisation has area form $r^2 \sin\theta \, d\theta \wedge d\phi$, giving the spherical formula (27.1). Integration over $0 \le \theta \le \pi$ and $0 \le \phi \le 2\pi$ gives $4\pi r$. ∎

Write $\xi_j = \xi_{e_j}$ for the standard basis vectors e_j. At the north pole, $\xi_1 = -r\partial_{x_2}$ and $\xi_2 = r\partial_{x_1}$, and $\omega(\xi_1, \xi_2) = r$. This is a convenient sign check. The origin must be treated separately: it is a point orbit, and none of the formulas dividing by r applies there.

Fix a positive constant $\hbar$. It sets the scale at which we compare symplectic area with phases. Keeping it explicit will make the integrality condition easier to track.

27.2 Connections described in local frames

A Hermitian line bundle is a complex line bundle with a smoothly varying positive inner product on its fibres. A local *unitary frame* is a section of norm one that never vanishes. In such a frame, a section is described by a complex coefficient function u.

Definition 27.2. A *unitary connection* differentiates local coefficients by

$$\nabla u = du + Au,$$

where A is an imaginary-valued 1-form. Let unitary frames e_i on U_i and e_j on U_j have connection forms A_i and A_j. If $e_j = g_{ij} e_i$, with $g_{ij} : U_i \cap U_j \to \mathrm{U}(1)$, the connection forms are required to satisfy

$$A_j = A_i + g_{ij}^{-1} dg_{ij}.$$

The *curvature* F_∇ is the imaginary-valued 2-form given on U_i by $F_\nabla = dA_i$.

Here du is the ordinary differential, evaluated on tangent vectors. A section s has coefficients u_i and u_j in the two frames. Since $e_j = g_{ij} e_i$, these coefficients satisfy $u_i = g_{ij} u_j$, or equivalently $u_j = g_{ij}^{-1} u_i$. Differentiating $u_i = g_{ij} u_j$ gives

$$du_i + A_i u_i = g_{ij}\big(du_j + (A_i + g_{ij}^{-1} dg_{ij}) u_j\big).$$

The left side is the coefficient of ∇s in the frame e_i. In the frame e_j this coefficient is $du_j + A_j u_j$, and coefficients change by the factor g_{ij}. Hence frame independence of ∇s for every section is exactly the rule for A_j in the definition.

Since g_{ij} is scalar, $d(g_{ij}^{-1} dg_{ij}) = 0$ locally. Indeed, write $g_{ij} = e^{i\alpha}$ with real α, so this form is $i\,d\alpha$. Hence $dA_i = dA_j$ and the curvature is global.

The same connection can be written in any nonvanishing frame. The change-of-frame formulas remain valid for $g_{ij} \in \mathbb{C}^\times$, with complex-valued A_i. Only unitary frames require imaginary-valued forms.

Along a path, the equation for parallel transport is $du + Au = 0$. In one frame its solution multiplies u by $\exp(-\int A)$. This is a phase because A is imaginary. Curvature measures the resulting phase around small loops. In physics language, A is a gauge potential and ∇ is a covariant derivative.

Definition 27.3. A *prequantum line bundle* for (M, ω) at scale $\hbar$ is a Hermitian line bundle with unitary connection satisfying

$$F_\nabla = -\frac{i}{\hbar}\omega. \tag{27.2}$$

The sign is part of our convention. A prequantum bundle is an initial geometric input. Its space of all smooth sections is usually far larger than the desired representation. A polarisation, such as holomorphicity, supplies a further selection.

We cite the general prequantisation theorem. A closed real 2-form ω on a smooth manifold admits a prequantum line bundle at scale $\hbar$ exactly when its periods divided by $2\pi\hbar$ are integers. A *period* of ω is the integral of ω over a closed oriented surface mapped smoothly into the manifold, or over an integer combination of such surfaces. Equivalently, $[\omega/(2\pi\hbar)]$ is an integral cohomology class. See Kirillov, Chapter 1, Section 2.4.

We prove the sphere case directly.

27.3 The integral sphere bundle

Put

$$m = \frac{2r}{\hbar}.$$

Take a northern chart excluding the south pole and a southern chart excluding the north pole. Define

$$A_N = -\frac{im}{2}(1 - \cos\theta)d\phi, \qquad A_S = \frac{im}{2}(1 + \cos\theta)d\phi. \tag{27.3}$$

Although $d\phi$ is singular at each pole, A_N is smooth at the north pole and A_S at the south pole. To see this we use a complex coordinate, and we recall its rules. For $z = x + iy$ put

$$dz = dx + i\,dy, \qquad d\bar{z} = dx - i\,dy, \qquad \partial_z = \tfrac{1}{2}(\partial_x - i\partial_y), \qquad \partial_{\bar{z}} = \tfrac{1}{2}(\partial_x + i\partial_y).$$

Then $df = \partial_z f\, dz + \partial_{\bar{z}} f\, d\bar{z}$ for a function f, and $dz \wedge d\bar{z} = -2i\, dx \wedge dy$. Writing the 1-form $a\,dz + b\,d\bar{z}$ in dx and dy and using the rule of Lecture 19 gives

$$d(a\,dz + b\,d\bar{z}) = (\partial_z b - \partial_{\bar{z}} a)\, dz \wedge d\bar{z}.$$

In the northern complex coordinate $z = \tan(\theta/2)e^{i\phi}$,

$$A_N = \frac{m}{2}\frac{z\,d\bar{z} - \bar{z}\,dz}{1+|z|^2}.$$

This expression is smooth at $z = 0$. The southern formula has the same property in its own coordinate. In physics language, these are the potentials of a magnetic monopole on two patches.

Differentiation gives

$$dA_N = dA_S = -\frac{im}{2}\sin\theta\,d\theta\wedge d\phi = -\frac{i}{\hbar}\omega.$$

On the overlap, $A_S - A_N = im\,d\phi$. Thus A_N and A_S satisfy the change-of-frame formula if the frame transition is

$$e_S = e^{im\phi}e_N.$$

This is a single-valued function on the overlap exactly when $m \in \mathbb{Z}$. For positive radius this means m is a positive integer. In physics language, $m \in \mathbb{Z}$ is Dirac's quantisation condition.

The idea of the necessity proof below is one loop with two caps. Transport around a loop multiplies a fibre by a phase. In a frame on a disk D bounded by the loop, Stokes' theorem makes this phase $\exp(-\int_D F_\nabla)$. In the plane of Lecture 19, a loop bounds one disk, and the Weyl relation of Lecture 18 has a phase of the same kind, a constant times an area. No condition arises there. On a sphere, a loop bounds two disks, with opposite orientations, as Fig. 27.1 shows. The two disks must give the same phase. Hence $\exp(-\int_{S^2} F_\nabla) = 1$, which says that $\int_{S^2}\omega \in 2\pi\hbar\mathbb{Z}$. We now write this with frames. In physics language, the latitude phase of Exercise 27.2 is the Berry phase of a spin.



Figure 27.1: One latitude loop, oriented by increasing ϕ, on two copies of the sphere. Its arrow points the same way in both. In the first copy, the shaded north cap D_N has this loop as its oriented boundary. In the second copy, the shaded south cap D_S has the reversed loop as its oriented boundary, so its phase has the opposite sign in the exponent.

Proposition 27.4. *The sphere S^2_r, $r > 0$, admits a prequantum line bundle if and only if $2r/\hbar \in \mathbb{Z}$. For $m = 2r/\hbar$, the preceding construction supplies one.*

Proof. Sufficiency is the explicit construction. For necessity, suppose that S^2_r carries a prequantum line bundle.

Step 1 (unitary frames on the two disks). Use the two open disk charts obtained by deleting the opposite poles. On each chart, parallel transport a unit vector at its origin along radial

segments. Cover each radial segment by finitely many local unitary frames. The change-of-frame law makes transport agree on overlaps. The resulting finite composition of ODE solutions depends smoothly on the endpoint and gives a smooth unitary frame. Each chart contains a neighbourhood of its closed hemisphere, so these frames are valid up to and beyond the equator. Call them e_N and e_S, with connection forms A_N and A_S.

Step 2 (the transition function and its lift). Write $e_S = g e_N$ near the equator, with $g : S^1 \to \mathrm{U}(1)$. Orient the equator by increasing ϕ, as the boundary of the northern disk.

Lift $g(e^{i\phi})$ to a real phase $\alpha(\phi)$ for $0 \le \phi \le 2\pi$, using the circle covering from Lecture 4. Since the endpoints give the same value, $\alpha(2\pi) - \alpha(0) = 2\pi k$ for an integer k.

Step 3 (Stokes). By the change-of-frame law, $A_S = A_N + g^{-1}dg$. Moreover $g^{-1}dg = i\,d\alpha$, since $g = e^{i\alpha}$ along the equator. Let D_N and D_S be the closed northern and southern hemispheres. We apply the calculus Stokes formula for a 1-form on each oriented disk. The southern disk induces the opposite orientation on the equator. Hence

$$\int_{S^2} F_\nabla = \int_{D_N} dA_N + \int_{D_S} dA_S = \int_{S^1} (A_N - A_S) = -\int_{S^1} g^{-1}dg = -2\pi i k.$$

By the prequantum condition (27.2) and the spherical formula (27.1),

$$k = \frac{i}{2\pi}\int F_\nabla = \frac{1}{2\pi\hbar}\int \omega = \frac{2r}{\hbar}.$$

Hence $2r/\hbar \in \mathbb{Z}$. ∎

Since $m > 0$, the curvature has no global primitive on the sphere. If $F = dA$ globally, Stokes' theorem would give $\int_{S^2} F = 0$. The period is nonzero because $m \neq 0$. The two local potentials are therefore essential.

27.4 Holomorphic sections and equivariance

First we show that $O(m)$, with one metric, carries the prequantum connection. Its holomorphic sections are then the quantisation. Finally we show that the natural SU(2) action is the equivariant lift, and that SO(3) needs even m.

The metric and the connection

Identify the sphere with $\mathbb{CP}^1$ by

$$z \longmapsto \frac{r}{1+|z|^2}\big(2\operatorname{Re} z, 2\operatorname{Im} z, 1-|z|^2\big).$$

The north pole is $z = 0$, and $z = \tan(\theta/2)e^{i\phi}$ is the northern coordinate used above. Recall from Lecture 26 that SU(2) acts on lines by $g\cdot[v] = [gv]$. The identification map is equivariant for this action and the coadjoint action. Indeed, the line of $v = (1, z)$ corresponds to the traceless Hermitian matrix proportional to

$$vv^*/\|v\|^2 - I/2.$$

Conjugation by g sends this matrix to the matrix for the line of gv, and it rotates the Pauli coordinates of the matrix. With $|z| = \tan(\theta/2)$, we have $d|z| = \frac{1}{2}(1+|z|^2)\,d\theta$ and $r\sin\theta = 2r|z|/(1+|z|^2)$. Substitution in the spherical formula (27.1) gives the sphere form

$$\omega = \frac{2ir\,dz\wedge d\bar z}{(1+|z|^2)^2}.$$

Since $i\,dz\wedge d\bar z = 2\,d(\operatorname{Re} z)\wedge d(\operatorname{Im} z)$, this form has the same positive orientation as the spherical formula (27.1).

The holomorphic frame transition of $O(m)$ in Lecture 26 is $e_S^{\mathrm{hol}} = z^m e_N^{\mathrm{hol}}$. The Hermitian metric with

$$\|e_N^{\mathrm{hol}}\|^2 = (1+|z|^2)^{-m}$$

gives unitary frames $e_N = (1+|z|^2)^{m/2}e_N^{\mathrm{hol}}$ and e_S whose transition is $(z/|z|)^m = e^{im\phi}$. This is the transition of the integral sphere bundle, so the potentials (27.3) define a unitary connection on $O(m)$.

The metric is SU(2)-invariant. To see this, we describe it fibre by fibre. A point f of the fibre over a line $[v]$ is a function on that line with $f(\lambda v) = \lambda^m f(v)$ for $\lambda\in\mathbb{C}^\times$. In this model, the holomorphic sections of Lecture 26 are the homogeneous polynomials of degree m, restricted to each line. On the fibre over $[v]$ put

$$\|f\|^2 = \frac{|f(v)|^2}{\|v\|^{2m}}.$$

Replacing v by λv multiplies the numerator and the denominator by $|\lambda|^{2m}$. Hence the norm does not depend on the chosen vector v. An element $g\in\mathrm{SU}(2)$ sends f to the point gf over $[gv]$ with $(gf)(gv) = f(v)$. Since g is unitary, $\|gv\| = \|v\|$, and so $\|gf\| = \|f\|$. The northern frame e_N^{hol} takes the value 1 at $v = (1,z)$, as the polynomial v_0^m does. Since $\|(1,z)\|^2 = 1+|z|^2$, this norm agrees with the metric displayed above.

A connection is *compatible* with the holomorphic structure when its form in a holomorphic frame has no $d\bar z$ term. Apply the change-of-frame law to $e_N^{\mathrm{hol}} = (1+|z|^2)^{-m/2}e_N$ and use the coordinate formula for A_N above. The form of this connection in the frame e_N^{hol} is then

$$A_N^{\mathrm{hol}} = A_N - \frac{m}{2}\,d\log(1+|z|^2) = -m\frac{\bar z\,dz}{1+|z|^2}.$$

Exercise 27.4 carries out this frame change in coordinates. The form A_N^{hol} has only a dz term, and its curvature is

$$dA_N^{\mathrm{hol}} = m\frac{dz\wedge d\bar z}{(1+|z|^2)^2} = -\frac{i}{\hbar}\omega.$$

Thus the prequantum bundle can be taken to be $O(m)$, with its usual holomorphic structure.

For a given metric, the unitary connection compatible with the holomorphic structure is unique. Let e be a holomorphic frame and put $h = \|e\|^2$. Let A be the connection form in the frame e. The unitary frame $h^{-1/2}e$ has an imaginary-valued connection form A_u. The frame e is $h^{1/2}$ times this unitary frame, so the change-of-frame law gives $A = A_u + \frac{1}{2}d\log h$. Moreover $\bar A_u = -A_u$, and $\log h$ is real. Hence

$$A + \bar A = d\log h.$$

For a function f, write $\partial f = (\partial f/\partial z)\,dz$ and $\bar\partial f = (\partial f/\partial \bar z)\,d\bar z$, so that $df = \partial f + \bar\partial f$. Suppose that A has no $d\bar z$ term. Then $\bar A$ has no dz term. Comparing the dz terms on the two sides gives

$$A = \partial \log h.$$

Thus the metric determines the compatible unitary connection. For $h = (1+|z|^2)^{-m}$ this formula gives $-m\bar z\,dz/(1+|z|^2)$, which is A_N^{hol} again.

The quantisation

When $m = 2r/\hbar$ is a positive integer, the *holomorphic quantisation* of S^2_r is the section space

$$Q(S^2_r) = H^0(\mathbb{CP}^1, O(m)) \cong V_m.$$

By the section computation of Lecture 26, this space has dimension $m+1$ and spin $j = m/2 = r/\hbar$. An invariant positive inner product comes from compact averaging. Explicitly, the formula

$$\|s\|^2 = \frac{1}{4\pi r}\int_{S^2_r} \|s\|^2_{\mathrm{fibre}}\,\omega$$

is the norm of Exercise 26.4. Indeed, in the coordinate z, the section z^k of $O(m)$ has squared fibre norm $|z|^{2k}(1+|z|^2)^{-m}$, and with $u = |z|^2$ this integral is

$$\int_0^\infty u^k(1+u)^{-m-2}\,du = \frac{k!(m-k)!}{(m+1)!}.$$

The point orbit uses the trivial line and gives the trivial representation, without a radius denominator.

The equivariant lift

Definition 27.5. Let K be a Lie group with Lie algebra $\mathfrak{k}$, and let $M \subset \mathfrak{k}^*$ be a coadjoint orbit with its KKS form ω. An *equivariant prequantum bundle* for K is a prequantum line bundle for (M,ω) at scale $\hbar$ with a smooth K-action on its total space. The action must cover the coadjoint action on M, be linear and unitary on fibres, preserve the connection, and satisfy the *orbit normalisation*. This normalisation requires that, at each $\ell \in M$, the stabiliser K_ℓ act on the fibre over ℓ with differential

$$X \mapsto i\ell(X)/\hbar, \qquad X \in \mathrm{Lie}(K_\ell).$$

The orbit normalisation fixes the infinitesimal lift. Without it, merely asking for a bundle action would not specify its relation to the orbit functional.

At the north pole, the stabiliser in $\mathrm{SU}(2)$ is generated by $\exp(tX_3)$, of period 4π. In an equivariant prequantum bundle, this stabiliser acts on the fibre by the character

$$\exp(tX_3) \longmapsto e^{irt/\hbar} = e^{imt/2}.$$

This is a character exactly when $m \in \mathbb{Z}$. It is the action on the fibre of $O(m)$ from Lecture 26: $t_\theta = \exp(-2\theta X_3)$ acts there by $e^{-im\theta}$. The natural $\mathrm{SU}(2)$ action is linear on fibres and preserves the invariant metric. Each $g \in \mathrm{SU}(2)$ acts by a holomorphic bundle map. It sends

holomorphic frames to holomorphic frames. Its holomorphic coordinate change sends forms with only a dz term to forms of the same kind. Hence g transports the connection to a unitary connection compatible with the holomorphic structure. By uniqueness, the transported connection is the original one. Therefore the natural action supplies the required lift.

For SO(3) the same stabiliser rotation has period 2π. Its required character is single-valued only when $e^{2\pi i r/\hbar} = 1$, or m is even. Conversely, for even m the centre $-I$ acts trivially on every fibre of $O(m)$, so the action descends. Therefore

ordinary prequantum bundle	$m \in \mathbb{Z}$
SU(2)-equivariant prequantum bundle	$m \in \mathbb{Z}$
SO(3)-equivariant prequantum bundle	$m \in 2\mathbb{Z}$

for the positive-radius sphere, with $m > 0$. An ordinary integral sphere with odd m does exist. It does not quantise equivariantly for the global rotation group SO(3). Figure 27.2 shows the first integral spheres.

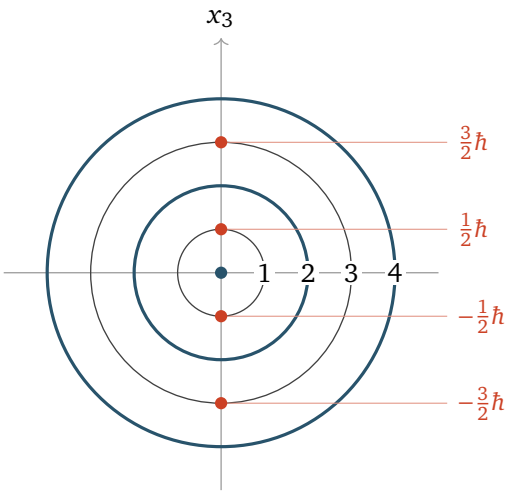


Figure 27.2: Cross-sections of the integral spheres, labelled by $m = 2r/\hbar$ with radius $r = \hbar m/2$, and the point orbit at the origin. Heavy circles have even m and admit an SO(3)-equivariant lift. The marks on the axis give the four heights $x_3 = \frac{\hbar}{2}k$, for $k = 3, 1, -1, -3$, associated with the weights of V_3, the quantisation of the sphere $m = 3$. Only the two extreme marks are points of that sphere on the axis.

Phase changes and the odd-degree obstruction

One might try to change the SU(2) lift by phases so that $-I$ acts trivially. We show that, for this sphere and its fixed connection, every such phase change is trivial.

Throughout this subsection, an action on $O(m)$ means a smooth action that is linear and unitary on fibres, as in the definition of an equivariant prequantum bundle. Suppose two such SU(2) actions cover the same sphere action and preserve the connection. For fixed g, compare their fibre maps. Their ratio is a bundle automorphism covering the identity. It is linear and unitary on each fibre, and each fibre is a complex line. Hence the ratio

is multiplication by a smooth function $a_g : S^2 \to \mathrm{U}(1)$. Since both actions preserve the connection,

$$a_g^{-1} da_g = 0.$$

The sphere is connected, so a_g is constant. Call this constant $\chi(g)$. The group-action law then implies

$$\chi(gh) = \chi(g)\chi(h).$$

Thus changing the lift while preserving the connection can only twist it by a continuous character of SU(2).

Every such character is trivial. Its differential is a real Lie algebra map from $\mathfrak{su}(2)$ to the abelian algebra $i\mathbb{R}$, so it vanishes on brackets. Since

$$[\mathfrak{su}(2), \mathfrak{su}(2)] = \mathfrak{su}(2),$$

the differential vanishes. Since the connected group SU(2) is generated by exponentials (Lecture 2), the character is identically one. Consequently a phase change of this kind cannot alter the action of the centre. Hence the natural action of Lecture 26 is the only such SU(2) action on $O(m)$ that covers the sphere action and preserves the connection.

Now suppose that SO(3) acted on $O(m)$ in this way. Composition with the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ would give such an SU(2) action on which $-I$ acts trivially. By uniqueness, that action would be the natural one, where $-I$ acts on every fibre by $(-1)^m$. Hence for odd m no SO(3) action on $O(m)$ that is linear and unitary on fibres covers the rotation action and preserves the connection. This conclusion does not use the orbit normalisation.

This is the sign of Lecture 13 on a fibre. The lift to SU(2) of the rotation through 2π about the north pole is $-I$, and it acts on the fibre by $(-1)^m$. For odd m, the calculation above shows that no change of phases removes this sign. So SO(3) acts only projectively on the sections of odd degree, and its cover SU(2) acts as a representation.

A spin-one comparison with functions on the sphere

Functions on the sphere contain every integer spin, by Lecture 12, and spin one is their linear part. Holomorphic sections of $O(2)$ contain spin one alone. We compare the two copies. For $r = \hbar$, the integer is $m = 2$. The quantisation is the three-dimensional space of homogeneous quadratic polynomials in (v_0, v_1)

$$F(v_0, v_1) = Av_0^2 + Bv_0v_1 + Cv_1^2.$$

There is another spin-one model in Lecture 12: the linear coordinate functions on the sphere. The two models are isomorphic as representations, although a bundle section and a scalar function are different geometric objects.

We now identify the two models explicitly. Write $F(v) = v^T Q v$ and set

$$Q = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}, \qquad \epsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T(Q) = \epsilon^{-1} Q = \begin{pmatrix} -B/2 & -C \\ A & B/2 \end{pmatrix}.$$

This is an invertible linear map from symmetric matrices to traceless matrices. The polynomial action sends Q to $g^{-T} Q g^{-1}$. Since $g^T \epsilon g = \epsilon$ for $g \in \mathrm{SL}(2, \mathbb{C})$,

$$T(g^{-T} Q g^{-1}) = g T(Q) g^{-1}.$$

Thus T intertwines the quadratic section representation with the complexified adjoint representation.

For $Y \in \mathfrak{su}(2)_{\mathbb{C}}$, let ℓ_x be the functional with coordinates $x_j = \ell_x(X_j)$ and define the linear function $f_Y(x) = \ell_x(Y)$ on the sphere, extending ℓ_x complex-linearly. Under pullback by g^{-1},

$$f_Y(g^{-1} \cdot x) = \ell_x(\mathrm{Ad}(g)Y) = f_{\mathrm{Ad}(g)Y}(x).$$

Hence the map sending Y to f_Y is another intertwiner. The three basis elements X_j give the independent functions x_j, so this map is an isomorphism onto their span. Since $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, the map $Q \mapsto f_{T(Q)}$ identifies the two spin-one models explicitly. The normalised period is still $m = 2$, the dimension is three, and the action descends to $\mathrm{SO}(3)$.

27.5 The compact construction and the shift

This section compares the construction with the recipe of Lecture 19. There a subalgebra $\mathfrak{m}$ with $\ell([\mathfrak{m},\mathfrak{m}]) = 0$ gave a character χ_ℓ of the group of $\mathfrak{m}$, and we induced from it. In $\mathfrak{su}(2)$ this fails for a nonzero ℓ. A real polarisation would have dimension two, and $\mathfrak{su}(2)$ has no two-dimensional subalgebra, because the bracket of two independent vectors is perpendicular to their plane. So we pass to the complexified algebra and take the positive Borel subalgebra.

The sphere illustrates the general compact picture. Let K be compact, connected, simply connected and semisimple. Let λ be a dominant integral weight in the real convention $d\chi = i\lambda$. Recall the positive Borel subalgebra fixed in Lecture 26. Our sign conventions follow that choice. Extend λ by zero on the root directions, and take the orbit through $\ell = -\hbar\lambda$. By the orbit normalisation, the stabiliser of ℓ acts on the fibre over ℓ by a character with differential $-i\lambda$. Extended by zero on the root directions, ℓ vanishes on $[\mathfrak{b},\mathfrak{b}]$, which is the sum of the positive root spaces. The character of B with differential $i\ell/\hbar$ is therefore χ_λ^{-1}. So the rule $f(gm) = \chi_\ell(m)^{-1} f(g)$ of Lecture 19 becomes $f(gb) = \chi_\lambda(b) f(g)$, which is the rule for the sections of L_λ in Lecture 26.

For strictly dominant λ, the stabiliser is a maximal torus T. The orbit is K/T and the bundle is L_λ of Lecture 26. Choose its invariant complex structure from the positive Borel subalgebra. The Borel–Weil theorem of Lecture 26 then gives V_λ^*. The sign in $\ell = -\hbar\lambda$ and the dual cancel. Let w_0 be the longest element of the Weyl group, and put $\mu = -w_0\lambda$, the highest weight of V_λ^*. Since w_0 is represented in K, the orbit through $-\hbar\lambda$ contains $\hbar\mu$. So the irreducible representation of highest weight μ belongs to the orbit through $\hbar\mu$, and the dominant integral weights label integral coadjoint orbits. For $\mathrm{SU}(2)$, the weight m has $\lambda(X_3) = -m/2$, so $-\hbar\lambda$ is the north pole of the sphere of radius $\hbar m/2$. For a weight on a chamber wall, the stabiliser K_ℓ is larger. The orbit is then the partial flag manifold K/K_ℓ. The analogous line bundle and holomorphic-section result are cited from Kirillov, Chapter 5, Sections 2–3. We do not prove that general geometric construction here.

For a different global form of K, the stabiliser character must exist for the actual stabiliser group. This is the equivariant integrality test. It reproduces the character-lattice restriction of Lecture 8. An ordinary integral period condition alone does not encode the global form.

Kirillov's character formula, which we do not prove, integrates over the orbit through $\lambda+\rho$ (Kirillov, Chapter 5). On the sphere, $r^2 = j^2\hbar^2$, while the Casimir is $j(j+1)\hbar^2 = \big((j+\frac{1}{2})^2-\frac{1}{4}\big)\hbar^2$. So the shift appears in quantum formulas and not in the orbit.

No ρ shift was used in this construction. On the sphere, the orbit label is m, and the uncorrected section space has dimension $m+1$. The character nevertheless contains $m+1$:

$$\chi_m(t_\theta) = \sum_{k=0}^{m} e^{i(m-2k)\theta} = \frac{\sin((m+1)\theta)}{\sin\theta} \qquad (\sin\theta \neq 0).$$

This is the Weyl character formula, with its familiar $\lambda+\rho$ numerator. It does not say that the prequantum orbit above had label $m+1$. Half-form corrections are another construction and also require a convention. Optional Exercise 27.6 shows one such choice on the sphere, and nothing in this lecture depends on it.

Self-checks

1. If $r = 3\hbar/2$, find the normalised period and the quantised representation.

Answer. The period divided by $2\pi\hbar$ is $2r/\hbar = 3$. The bundle is $O(3)$ and its sections give V_3, of dimension four and spin $3/2$.

2. Does that sphere admit an SO(3)-equivariant prequantum lift with the orbit normalisation?

Answer. No. Its stabiliser character would be $e^{3it/2}$ on a rotation of period 2π. It takes the value -1 at $t = 2\pi$.

3. Why can the connection have local potentials although no global potential exists?

Answer. The difference of two local potentials is the logarithmic differential of a transition function. A nonzero curvature period prevents a single global primitive but is consistent with these local potentials.

Summary

The SU(2) orbit of radius $r > 0$ has symplectic area $4\pi r$. Prequantisation requires $m = 2r/\hbar$ to be integral. Holomorphic sections of $O(m)$ give spin $m/2$. Equivariance for SO(3) further requires even m. The ordinary period, the global equivariant lift and the holomorphic selection are separate steps. The orbit through $-\hbar\lambda$ and the dual V_λ^* of Borel–Weil combine to attach V_μ to the orbit through $\hbar\mu$. For SU(2) this attaches V_m to the sphere of radius $\hbar m/2$.

Reading

Kirillov, *Lectures on the Orbit Method*, Chapter 1, Section 2.4, for integrality, and Chapter 5, Sections 2–3, for compact orbits, holomorphic constructions and character formulas. Sepanski, *Compact Lie Groups*, Section 7.4, states the Borel–Weil theorem proved in Lecture 26.

Lecture 28

Reconstruction and the boundary of quantum groups

Objectives. *The group, its coefficient algebra and its tensor representations encode the same global information in different forms. Differentiation connects these descriptions to the enveloping algebra. It loses information, however, unless the allowed group representations are retained. We make that connection explicit and identify the structures that a later course on quantum groups will deform.*

By the end of this lecture you will be able to:

- Construct the pairing between $U(\mathfrak{k}_{\mathbb{C}})$ and representative functions, with the correct multiplication order.
- For connected K, explain why representative functions form a restricted part of an algebraic dual, and show that for simply connected K they fill the finite dual.
- Recover compact and complex groups from the appropriate tensor automorphisms.
- Use the circle and the SU(2)/SO(3) pair to identify what global data survive differentiation.

28.1 The two algebras

Let K be a compact Lie group and $\mathfrak{k}$ its real Lie algebra. As in Lecture 14, $\mathcal{R}(K)$ is the span of the coefficients

$$c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)v), \qquad \lambda \in V^*,\ v \in V,$$

of finite-dimensional continuous complex representations. Every sum of coefficients is one coefficient of a direct sum. Its Hopf structure is

$$\Delta_{\mathcal{R}} f(g,h) = f(gh), \qquad \varepsilon_{\mathcal{R}} f = f(e), \qquad S_{\mathcal{R}} f(g) = f(g^{-1}), \qquad f^*(g) = \overline{f(g)}.$$

These formulas and their targets were proved in Lecture 14. We use the ordinary algebraic tensor product. Thus $\Delta_{\mathcal{R}} f$ is a finite sum of tensors, not an infinite expansion.

On the other side, put $U = U(\mathfrak{k}_{\mathbb{C}})$, the universal enveloping algebra from Lie Algebras. It is generated by $\mathfrak{k}_{\mathbb{C}}$ with $XY - YX = [X,Y]$. Its standard Hopf maps are determined by

$$\Delta_U X = X \otimes 1 + 1 \otimes X, \qquad \varepsilon_U X = 0, \qquad S_U X = -X. \tag{28.1}$$

In physics language, the first rule is the rule for adding spins. Here Δ_U and ε_U extend as algebra homomorphisms. The antipode S_U extends so that it reverses products. This

extension is well defined because reversing the defining relation sends $XY - YX - [X,Y]$ to $YX - XY + [X,Y]$, which is again zero in U. We check that Δ_U respects the defining relation:

$$[\Delta_U X, \Delta_U Y] = [X,Y] \otimes 1 + 1 \otimes [X,Y] = \Delta_U [X,Y].$$

Mixed tensor factors commute, which is why the cross terms vanish. Coassociativity and the counit identities follow by checking the generators, since the maps are algebra homomorphisms.

The antipode identities hold on X because $-X + X = 0$. They extend to products. Suppose that they hold on u and v. Then

$$\begin{aligned}\sum S_U(u_{(1)}v_{(1)})u_{(2)}v_{(2)} &= \sum S_U(v_{(1)})S_U(u_{(1)})u_{(2)}v_{(2)} \\ &= \varepsilon_U(u) \sum S_U(v_{(1)})v_{(2)} = \varepsilon_U(u)\varepsilon_U(v)1.\end{aligned}$$

The first equality uses that S_U reverses products. The second uses the identity for u, and the third uses the identity for v. The other antipode identity is proved the same way.

The two algebras have opposite kinds of commutativity. The coefficient algebra is commutative. It is cocommutative exactly when K is abelian, because coefficients separate points. The enveloping algebra is cocommutative, as the generator formulas (28.1) show, but is generally noncommutative.

The compact real form also defines an involution on U by $X^* = -X$ for real $X \in \mathfrak{k}$, extended antilinearly with $(uv)^* = v^* u^*$. For real X, Y, applying $*$ to the defining relation gives $YX - XY + [X,Y]$, which is again zero. Thus $*$ is well defined. Applying it twice fixes the generators, so it is an involution.

In every finite-dimensional unitary representation,

$$d\Pi(u^*) = d\Pi(u)^*.$$

This follows first for real generators, whose operators are skew-adjoint, then for sums and products. The involution remembers the compact real form inside its complexification.

28.2 Differentiating coefficients

For $X \in \mathfrak{k}$, let

$$(R_X f)(g) = \left.\frac{d}{dt}\right|_0 f(g \exp(tX)).$$

This is the left-invariant vector field X^L of Lecture 1, acting on functions. The letter R records multiplication on the right in the argument. It is distinct from the right regular group action, although differentiating that action gives this operator. By Lecture 1, $[R_X, R_Y] = R_{[X,Y]}$. Hence, by the universal property of U, these operators extend to R_u for $u \in U$.

Definition 28.1. The *coefficient pairing* is

$$\langle u, f \rangle = (R_u f)(e), \qquad u \in U,\ f \in \mathcal{R}(K).$$

The coefficient pairing is complex bilinear. Its angle brackets denote a pairing, not a Hermitian inner product. For real X, the value $\langle X^k, f \rangle$ is the kth derivative at 0 of $t \mapsto$

$f(\exp tX)$. These pairings record the Taylor derivatives of f at e. For a general u, the pairing is a finite linear combination of such derivatives. For a coefficient, direct differentiation gives

$$R_X c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)d\Pi_V(X)v), \qquad \langle u, c^V_{\lambda,v}\rangle = \lambda(d\Pi_V(u)v). \tag{28.2}$$

The second identity follows for products of generators and then by linearity. Since the pairing was defined on functions by differential operators, it does not depend on the chosen coefficient expression.

The order matters. For two real generators $X, Y \in \mathfrak{k}$,

$$\langle XY, f\rangle = \partial_t\partial_s|_0 f(\exp(tX)\exp(sY)).$$

Indeed, for $f = c^V_{\lambda,v}$ we have $R_X R_Y f(g) = \lambda(\Pi_V(g)d\Pi_V(X)d\Pi_V(Y)v)$. Interchanging X and Y would reverse the matrix product. Subtracting the pairing of YX with f from the pairing of XY with f gives $\langle [X,Y], f\rangle$.

Proposition 28.2 (Hopf pairing identities). *Write finite coproducts as* $\Delta_U u = \sum u_{(1)} \otimes u_{(2)}$ *and* $\Delta_{\mathcal{R}} f = \sum f_{(1)} \otimes f_{(2)}$. *Then*

$$\begin{aligned}
\langle ab, f\rangle &= \sum \langle a, f_{(1)}\rangle\langle b, f_{(2)}\rangle,\\
\langle u, fh\rangle &= \sum \langle u_{(1)}, f\rangle\langle u_{(2)}, h\rangle,\\
\langle 1, f\rangle &= \varepsilon_{\mathcal{R}} f, \qquad \langle u, 1\rangle = \varepsilon_U u,\\
\langle S_U u, f\rangle &= \langle u, S_{\mathcal{R}} f\rangle.
\end{aligned}$$

Proof. Choose a basis (e_j) of V with dual (e^j). For $f = c^V_{\lambda,v}$,

$$\Delta_{\mathcal{R}} f = \sum_j c^V_{\lambda,e_j} \otimes c^V_{e^j,v}.$$

Applying the right side of the first identity gives

$$\sum_j \lambda(d\Pi(a)e_j)\, e^j(d\Pi(b)v) = \lambda(d\Pi(a)d\Pi(b)v).$$

This is $\langle ab, f\rangle$ by the coefficient formula (28.2).

For the second identity, $R_X(fh) = (R_X f)h + f(R_X h)$ is the ordinary Leibniz rule. Iterating it gives exactly the coproduct (28.1). For instance,

$$\Delta_U(XY) = XY \otimes 1 + X \otimes Y + Y \otimes X + 1 \otimes XY.$$

Thus evaluation at e proves the claim for every product and hence for every u. The identity $\langle 1, f\rangle = \varepsilon_{\mathcal{R}} f$ follows because R_1 is the identity. Applying derivatives to constants gives $\langle u, 1\rangle = \varepsilon_U u$.

Finally, write $\Pi^\vee$ for the dual representation on V^*, given by $\Pi^\vee(g)\mu = \mu \circ \Pi_V(g)^{-1}$. The function $S_{\mathcal{R}} c^V_{\lambda,v}(g) = \lambda(\Pi_V(g)^{-1}v)$ is the coefficient of $\Pi^\vee$ with vector λ and with the functional given by evaluation at v. On V^*, a real generator acts by $d\Pi^\vee(X)\mu = -\mu \circ d\Pi(X)$. For every $u \in U$,

$$d\Pi^\vee(u)\mu = \mu \circ d\Pi(S_U u).$$

This holds for generators and extends multiplicatively because S_U reverses products. Apply the coefficient formula (28.2) to this coefficient of $\Pi^\vee$, and then to $c^V_{\lambda,v}$. Both sides of the last identity equal $\lambda(d\Pi(S_U u)v)$, which proves it. ∎

Hence the pairing makes the product of each algebra dual to the coproduct of the other. In this algebraic form, it relates the infinitesimal action to group multiplication. The proposition does not assert that one of these algebras is the entire linear dual of the other.

A noncommutative calculation

Take $K = \mathrm{SU}(2)$, so $\mathfrak{k}_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$. We test the order identity in its defining representation. Put

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These are complexified generators. The derivative is extended complex-linearly; we are not claiming that $\exp(tE)$ belongs to $\mathrm{SU}(2)$ for real t. Let $D_{ij}(g) = g_{ij}$ be coefficients of its defining representation. Then

$$EF = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad FE = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

It follows that $\langle EF, D_{11}\rangle = 1$, whereas $\langle FE, D_{11}\rangle = 0$. Their difference is $\langle H, D_{11}\rangle = 1$. The pairing therefore distinguishes EF from FE in U, even though its other argument belongs to a commutative function algebra.

The coefficient coproduct gives the same answer:

$$\Delta_{\mathcal{R}} D_{11} = D_{11} \otimes D_{11} + D_{12} \otimes D_{21}.$$

Pairing the first tensor factor with E and the second with F yields

$$E_{11}F_{11} + E_{12}F_{21} = 0 + 1 = 1.$$

Pairing the first factor with F and the second with E yields zero. Thus a commutative algebra can have a coproduct that encodes a noncommutative group law. Commutativity of the product does not by itself decide cocommutativity of the coproduct.

The regular actions on coefficients

Let $L_g f(x) = f(g^{-1}x)$ and $R_g f(x) = f(xg)$, as in the compact harmonic-analysis lectures. Here R_g denotes the group action. The symbols R_X and R_u denote differential operators. The pairings of the translates of a coefficient are

$$\langle u, L_g c_{\lambda,v}\rangle = \lambda\big(\Pi(g^{-1})d\Pi(u)v\big),$$

$$\langle u, R_g c_{\lambda,v}\rangle = \lambda\big(d\Pi(u)\Pi(g)v\big).$$

For the first formula, $L_g c$ is the coefficient with functional $\lambda \circ \Pi(g^{-1})$. For the second, $R_g c$ has vector $\Pi(g)v$. Inserting these into the coefficient formula (28.2) proves both identities without any operator-domain question.

Notice the positions of the group matrix and the derivative matrix. Left translation changes the dual variable. Right translation changes the vector variable. This is the same tensor order $V^* \otimes V$ that appeared in the Peter–Weyl theorem. The enveloping-algebra pairing continues that convention rather than introducing a second one.

28.3 What the dual contains

For any unital algebra A, its *finite dual* A° consists of the functionals that vanish on a two-sided ideal of finite codimension. Equivalently they are matrix coefficients of finite-dimensional A-modules. Here is the elementary equivalence.

A finite-dimensional module gives a homomorphism $A \to \mathrm{End}(V)$ with finite-codimensional kernel; every coefficient vanishes on that kernel. Conversely, if φ vanishes on such an ideal I, use the finite-dimensional regular module A/I, vector $1 + I$ and functional $\bar{\varphi}$. Then $\varphi(a) = \bar{\varphi}(a(1 + I))$.

By the coefficient formula (28.2), the pairing therefore maps $\mathcal{R}(K)$ into U°. Only the modules that integrate to the chosen compact group occur in this image. A finite-dimensional representation of $\mathfrak{k}_{\mathbb{C}}$ need not respect the periods of K or the restrictions imposed by a central quotient.

> **Proposition 28.3** (Injectivity for connected groups). *If K is connected, the map $\mathcal{R}(K) \to U^\circ$ from the pairing is injective.*

Proof. Suppose $\langle u, f\rangle = 0$ for every u. Since every sum of coefficients is one coefficient of a direct sum, we may write $f = c^V_{\lambda,v}$. For any finite list $X_1, \dots, X_s \in \mathfrak{k}$, expand the finite-dimensional matrix exponentials in

$$f(\exp(t_1X_1)\cdots\exp(t_sX_s)) = \lambda\big(e^{t_1 d\Pi(X_1)}\cdots e^{t_s d\Pi(X_s)}v\big).$$

These power series converge absolutely for every real or complex tuple $(t_1, \dots, t_s)$. The Taylor coefficient of $t_1^{a_1}\cdots t_s^{a_s}$ is

$$\frac{\langle X_1^{a_1}\cdots X_s^{a_s}, f\rangle}{a_1!\cdots a_s!} = 0.$$

Thus the displayed function vanishes for all real tuples. A connected Lie group is generated by its exponentials (Lecture 2), so $f = 0$ on K. ∎

For disconnected K, the same argument detects only the identity component. A coefficient may vanish on that component and be nonzero elsewhere. Thus connectedness cannot simply be removed.

Example 28.4 (The circle). Let $K = \mathrm{U}(1)$ and choose X with $\exp(tX) = e^{it}$. Then

$$U = \mathbb{C}[X], \qquad \mathcal{R}(K) = \mathbb{C}[z, z^{-1}], \qquad \langle X^k, z^n\rangle = (in)^k.$$

The image consists of finite linear combinations of the evaluation functionals $p(X) \mapsto p(in)$, $n \in \mathbb{Z}$. The finite dual is larger. Evaluation at $X = 1$ belongs to U° but is not in this image. If it were a combination supported on $n_1, \dots, n_s$, the polynomial $p(X) = \prod_j (X - in_j)$ would have zero value under that combination, but $p(1) \neq 0$.

The full linear dual is larger still. It contains the functional L given by $L(X^k) = k!$. By the equivalence above, every element of U° is a matrix coefficient of a finite-dimensional module. Suppose that X acts on that module by the operator A and that the coefficient uses the vector v and the functional λ. Then the coefficient takes the value $\lambda(A^k v)$ on X^k. Choose a norm on the module, with the corresponding operator and dual norms. This value is

then bounded in absolute value by $C\|A\|^k$, where $C = \|\lambda\|\|v\|$. The sequence $k!$ has no such exponential bound. Thus L is not even in U°.

Figure 28.1 draws these functionals on $U = \mathbb{C}[X]$ in the plane of the variable X. A dot at a point X_0 means the evaluation $p(X) \mapsto p(X_0)$.

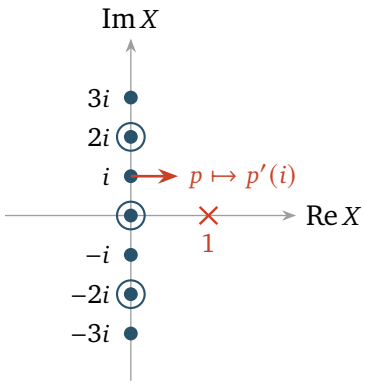


Figure 28.1: Functionals on $\mathbb{C}[X]$. The filled dots at $X_0 = in$ with $n \in \mathbb{Z}$ span the image of $\mathcal{R}(\mathrm{U}(1))$. The ringed dots at $X_0 = 2in$ span the image for the quotient of the circle by μ_2, treated later in this lecture. The cross at $X_0 = 1$ is in U° but not in the image. The arrow at i marks the functional $p \mapsto p'(i)$, which comes from the Jordan block of Exercise 28.3. It lies in U°, and it is a different functional from the evaluation at i.

28.4 Global forms and tensor reconstruction

The groups SU(2) and SO(3) have isomorphic real Lie algebras, hence the same complex enveloping algebra after this identification. Their coefficient algebras retain different representations. Pullback through $p : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives

$$p^*\mathcal{R}(\mathrm{SO}(3)) = \{f \in \mathcal{R}(\mathrm{SU}(2)) : f(-g) = f(g)\}. \tag{28.3}$$

Indeed, a representation descends exactly when $-I$ acts trivially. On V_m it acts by $(-1)^m$. The algebraic Peter–Weyl decomposition therefore identifies the right side with precisely the coefficient blocks for even m. Odd blocks have been excluded, not annihilated by differentiation. Both coefficient algebras inject into the same finite dual, with different images.

For SU(2) the image fills the finite dual. More generally, let K be compact, connected and simply connected. A finite-dimensional module of U restricts to a Lie algebra map from $\mathfrak{k}$ to $\mathfrak{gl}(V)$. By the integration theorem of Lecture 4, this map is the differential of a representation of K. So every element of U° is a coefficient of a representation of K. With Proposition 28.3, the pairing identifies $\mathcal{R}(K)$ with U°. Thus U° gives $K_{\mathbb{C}}$, whose points are the characters of $\mathcal{R}(K)$ by Lecture 14, and the involution of the first section selects K. The involution is needed: $U(\mathfrak{sl}(2,\mathbb{C}))$ is also the enveloping algebra of the complexification of $\mathfrak{su}(1,1)$. Tori and central quotients are where the coefficient algebra adds information. The circle example shows a torus, and (28.3) shows a quotient.

To recover the group, keep the category of finite-dimensional continuous complex representations of K, all its intertwiners, and its forgetful functor ω. This functor sends each representation to its underlying vector space and each intertwiner to its linear map.

A *tensor automorphism* of ω is a family of maps $\theta_V \in \mathrm{GL}(V)$, one for each representation V, such that

$$\theta_W T = T\theta_V \quad \text{for every intertwiner } T\colon V \to W, \qquad \theta_{V\otimes W} = \theta_V \otimes \theta_W, \qquad \theta_{\mathbb{C}} = 1.$$

Here $\mathbb{C}$ is the trivial representation. A tensor automorphism is *unitary* if every θ_V is unitary for every invariant inner product on V. Write $\mathrm{Aut}_{\otimes}(\omega)$ for the group of tensor automorphisms and $\mathrm{Aut}^{u}_{\otimes}(\omega)$ for its subgroup of unitary tensor automorphisms.

Each $g \in K$ supplies the unitary tensor automorphism $\theta_V = \Pi_V(g)$. A tensor automorphism θ corresponds to the algebra homomorphism $\mathcal{R}(K) \to \mathbb{C}$ given by $c^{V}_{\lambda,\upsilon} \mapsto \lambda(\theta_V \upsilon)$. Unitary tensor automorphisms correspond to $*$-homomorphisms. Exercises 14.4–14.5 prove this correspondence.

Let $K_{\mathbb{C}}$ be the complexification of K introduced in Lecture 14. It is a complex Lie group, and its identity component is reductive. The reconstruction theorem is cited from Bröcker–tom Dieck, Sections III.7–8. It identifies

$$K \cong \mathrm{Aut}^{u}_{\otimes}(\omega), \qquad K_{\mathbb{C}} \cong \mathrm{Aut}_{\otimes}(\omega).$$

The compact group has the topology of pointwise convergence on the representation spaces. In words, every unitary tensor automorphism is $\Pi_V(g)$ for a unique $g \in K$. Every tensor automorphism is given by a unique element of $K_{\mathbb{C}}$, acting through the holomorphic extensions of the representations Π_V to $K_{\mathbb{C}}$. The unitary condition distinguishes the compact target from its complexification.

For the circle, write $\mathbb{C}_n$ for the one-dimensional representation $z \mapsto z^n$ of $\mathrm{U}(1)$, where $n \in \mathbb{Z}$. Every finite-dimensional representation of $\mathrm{U}(1)$ is a direct sum of copies of these representations. Naturality makes θ act on every copy of $\mathbb{C}_n$ by one scalar $a_n \in \mathbb{C}^{\times}$. The tensor rule, the isomorphism $\mathbb{C}_m \otimes \mathbb{C}_n \cong \mathbb{C}_{m+n}$ and the unit rule give $a_{m+n} = a_m a_n$ and $a_0 = 1$. Therefore $a_n = a^n$ with $a = a_1 \in \mathbb{C}^{\times}$. Every such choice of a defines a tensor automorphism. This tensor automorphism is unitary exactly when $|a| = 1$. This recovers $\mathbb{C}^{\times}$ and $\mathrm{U}(1)$ respectively.

For $\mathrm{SU}(2)$, let $V = \mathbb{C}^2$ be the defining representation. The alternating tensor $e_1 \otimes e_2 - e_2 \otimes e_1$ is invariant. Since $\theta_{V\otimes V} = \theta_V \otimes \theta_V$ and $\theta_{\mathbb{C}} = 1$, naturality with the intertwiner from $\mathbb{C}$ given by this tensor forces $\theta_V \otimes \theta_V$ to preserve it. Because $\theta_V \otimes \theta_V$ multiplies this tensor by $\det \theta_V$, we get $\det \theta_V = 1$. Every irreducible representation is a symmetric power of V, and direct sums and intertwiners determine the rest of the tensor automorphism.

Conversely every $A \in \mathrm{SL}(2,\mathbb{C})$ acts on these tensor constructions and gives a tensor automorphism, using the complexification from Lecture 14. Hence the complex target is $\mathrm{SL}(2,\mathbb{C})$. Imposing unitarity on V gives $\mathrm{SU}(2)$. For the even-weight category the central element $-I$ acts as the identity tensor automorphism, so reconstruction instead yields the quotient global group.

A circle quotient and its representation lattice

There is a smaller example of global information in a coefficient subalgebra. Let $\mu_N \subset \mathrm{U}(1)$ be the Nth roots of unity. The quotient is again a circle, but keep the quotient map and the identification of Lie algebras fixed. A character z^n descends through this map exactly when

$$\zeta^n = 1 \quad \text{for every } \zeta \in \mu_N, \qquad \text{or equivalently } N \mid n.$$

Thus the pulled-back coefficient algebra is

$$\mathbb{C}[z^N, z^{-N}] \subset \mathbb{C}[z, z^{-1}].$$

Under the same infinitesimal pairing, its evaluation points are $iN\mathbb{Z}$, instead of $i\mathbb{Z}$. The real Lie algebras of these two circles are isomorphic. The quotient map, however, has changed which periods a representation must respect. Rescaling an abstract Lie algebra isomorphism does not erase the specified quotient map.

The subalgebra remains closed under the Hopf operations. For its generator $w = z^N$,

$$\Delta w = w \otimes w, \qquad \varepsilon w = 1, \qquad Sw = w^{-1}, \qquad w^* = w^{-1}.$$

Reconstruction from this algebra and its unitary structure recovers the quotient circle together with its representation lattice. The same calculation with even highest weights gives the SU(2) to SO(3) distinction. In both cases, keeping a Lie algebra while changing the allowed coefficient subalgebra changes the global representation problem.

28.5 What a quantum-group course changes

The structures above give a concrete starting point for deformation. The coefficient algebra encodes the multiplication of group elements in its coproduct. The enveloping algebra encodes the infinitesimal bracket in its associative product. The pairing between them relates these two operations. A deformation can change these operations while retaining the Hopf identities.

Classical object	Product	Coproduct
$\mathcal{R}(K)$	commutative	usually not cocommutative
$U(\mathfrak{k}_{\mathbb{C}})$	usually not commutative	cocommutative

A noncommutative coefficient algebra is no longer an algebra of scalar-valued functions on an ordinary group. A noncocommutative coproduct changes how tensor products of representations carry an action. The order of tensor factors can then matter. Associativity of the tensor product of representations still follows from coassociativity.

Here is one such deformation, as an algebraic calculation only.

Example 28.5 (Quantum SL_2)**.** On SU(2), the coefficients of the defining representation are $a = D_{11}$, $b = D_{12}$, $c = D_{21}$ and $d = D_{22}$. Keep the coproduct $\Delta D_{ij} = \sum_k D_{ik} \otimes D_{kj}$. Change the product. For a nonzero complex number q, let a, b, c, d generate an associative algebra with the relations

$$ab = qba, \quad ac = qca, \quad bd = qdb, \quad cd = qdc, \quad bc = cb,$$
$$ad - da = (q - q^{-1})bc, \quad ad - qbc = 1.$$

At $q = 1$ these relations say that the generators commute and that the determinant is one. For other q this algebra is not commutative, so it is not an algebra of functions on a group. The coproduct formula still respects the relations. For example, using $ab = qba$, $cd = qdc$ and $ba = q^{-1}ab$,

$$\Delta(a)\Delta(b) - q\Delta(b)\Delta(a) = ab \otimes \big(ad - da - (q - q^{-1})bc\big) = 0.$$

The other relations follow in the same way, and $ad - qbc$ is sent to $(ad - qbc) \otimes (ad - qbc)$. These relations do not give a compact quantum group for every q. The involution of the classical case does not carry over unchanged.

A braiding is additional structure, not an automatic property of a Hopf algebra. In suitable quasitriangular examples on the enveloping side, an R-matrix supplies an intertwiner between the two tensor orders. The future course will define these notions and prove their compatibility identities. They are not needed for any theorem here.

The $*$-structure also remains substantive. An arbitrary Hopf $*$-algebra is not automatically a compact quantum group; positivity and analytic completion require further hypotheses.

The useful preparation is therefore the classical bookkeeping developed in this course: distinguish an algebraic coefficient space from a completion, track coproduct order, keep the real or unitary structure, and retain the global representation lattice. These are the data a deformation must replace or preserve. The theory of deformation parameters, $U_q(\mathfrak{g})$ and roots of unity begins in the later course.

Looking back

We asked what a group determines that its Lie algebra does not. One answer runs through the course. A connected Lie group is a quotient of its simply connected covering group by a discrete central subgroup. In an irreducible representation of the cover that is finite-dimensional or unitary, a central element acts by a scalar. So a global form chooses which central characters are allowed. We used this for SU(2) and SO(3) in Lecture 4 and for lattices in Lecture 7 and Lecture 8. It separates integer and half-integer spin in Lecture 13, in the Lorentz modules of Lecture 24, in the massive states of Lecture 25, and on the sphere of Lecture 27. The coefficient algebras of this lecture record the same choice.

Part I treated compact groups. For a compact connected group, the irreducible representations are labelled by dominant characters. For every compact Lie group, the invariant integral leads to a decomposition of $L^2(K)$ as a Hilbert sum of finite blocks.

Part II removed compactness. Generators, the modular function and induction joined the invariant integral as the working tools. We then described the unitary representations

of the Heisenberg group, of $\mathrm{SL}(2,\mathbb{R})$ and of the Poincaré group. Each description has a cited theorem at its centre. For the Heisenberg group, we proved that the Schrödinger representation is irreducible, and the Stone–von Neumann theorem says that it is the only one for each nonzero central character. For $\mathrm{SL}(2,\mathbb{R})$ we constructed the principal and discrete series and derived the positivity bounds, and the full list is cited from Bargmann and Harish-Chandra. For the Poincaré group, Mackey's theorem reduces the problem to little groups. We computed the massive and finite-helicity sectors, and we only named the others.

Orbits organise much of this. In Lecture 19 a coadjoint orbit of the Heisenberg group is a plane or a point, and it gives one representation. In Lecture 27 the nonzero coadjoint orbits of $\mathrm{SU}(2)$ are spheres. For $m \geq 1$, the sphere of area $2\pi\hbar m$ gives V_m, on holomorphic sections of a line bundle over the sphere. The point orbit gives V_0. In Lecture 17 and Lecture 25 the orbits lie instead in the dual of the translations.

We did not decompose $L^2(G)$ for noncompact G, apart from the statement cited in Lecture 23, and we did not treat general semisimple groups. These are the next subjects in the representation theory of Lie groups. A second direction deforms the Hopf structures of this lecture, and a later course on quantum groups begins there.

Self-checks

1. For the circle, compute $\langle X^2, z^{-3}\rangle$.

Answer. It is $(-3i)^2 = -9$. The pairing is bilinear, so there is no complex conjugation in this calculation.

2. Does $U(\mathfrak{sl}(2,\mathbb{C}))$ alone decide whether V_1 (spin $1/2$) is allowed?

Answer. No. It is allowed for $\mathrm{SU}(2)$ and excluded for $\mathrm{SO}(3)$. The chosen coefficient subalgebra, or equivalently the allowed global representations, records that distinction.

3. For the circle tensor automorphism $a = 2$, what are $\theta_{\mathbb{C}_2}$ and $\theta_{\mathbb{C}_{-1}}$? Is it unitary?

Answer. They are 4 and $1/2$. It is a complex tensor automorphism, but it is not unitary and does not represent an element of the compact circle.

Summary

Differentiation pairs representative functions with the enveloping algebra. For connected K, the pairing embeds them in the finite dual. There the global group selects a specific subalgebra, which is the whole finite dual when K is simply connected. Tensor reconstruction recovers K when unitary structure is kept and $K_{\mathbb{C}}$ when it is forgotten. The coefficient and enveloping Hopf structures provide the classical starting point for quantum groups.

Reading

Bröcker–tom Dieck, *Representations of Compact Lie Groups*, Sections III.1 and III.7–8: coefficient algebras, reconstruction and complexification. The enveloping-algebra and tensor-product inputs are from the prerequisite Lie Algebras course. Lecture 14 contains the coefficient Hopf identities used here. The quantum-group paragraph is an outlook, not an additional classification theorem. For quantum SL_2, see Klimyk and Schmüdgen, *Quantum Groups and Their Representations* (1997), Section 4.1, whose conventions we use.